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LATE SADLERIAN PROFESSOR OF PURE MATHEMATICS IN THE UNIVERSITY OF CAMBRIDGE.

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[Frontispiece portrait of Professor Cayley]

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THE present volume contains 93 papers numbered 706 to 798, published, with the exception of one series, for the most part in the years 1878 to 1883. This series is constituted by the articles which Professor Cayley wrote for the *Encyclopædia Britannica* between the years 1878 and 1888; it seemed desirable to place these together in the same volume, in spite of the departure from the chronological arrangement which governs the sequence of the papers in the volumes generally. The Syndics of the University Press desire to acknowledge their obligation to Messrs Adam and Charles Black, Publishers of the ninth Edition of the *Encyclopædia Britannica*, for their courteous consent in allowing these articles to be included in the *Collected Mathematical Papers*. Exact references to the volumes, from which the articles are extracted, will be found in the Table of Contents.

The frontispiece to the present volume is a reproduction by Mr A. G. Dew-Smith, of Trinity College, of a photograph of Professor Cayley which he made in the year 1885. The Syndics of the Press desire to acknowledge their obligation to Mr Dew-Smith.

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A. R. FORSYTH.

21 November, 1896.

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706.

ON THE DISTRIBUTION OF ELECTRICITY ON TWO SPHERICAL SURFACES.

[From the *Philosophical Magazine*, vol. v. (1878), pp. 54—60.]

In the two memoirs "Sur la distribution de l'électricité à la surface des corps conducteurs," *Mém. de l'Inst.* 1811, Poisson considers the question of the distribution of electricity upon two spheres: viz. if the radii be a, b, and the distance of the centres be c (where $c > a+b$, the spheres being exterior to each other), and the potentials within the two spheres respectively have the constant values h and g, then—for Poisson's $f\left(\frac{x}{a}\right)$ writing $\phi(x)$, and for his $F\left(\frac{x}{b}\right)$ writing $\Phi(x)$—the question depends on the solution of the functional equations

$$a\phi(x) + \frac{b^2}{c-x}\,\Phi\left(\frac{b^2}{c-x}\right) = h,$$

$$\frac{a^2}{c-x}\,\phi\left(\frac{a^2}{c-x}\right) + b\Phi(x) = g,$$

where of course the x of either equation may be replaced by a different variable.

It is proper to consider the meaning of these equations: for a point on the axis, at the distance x from the centre of the first sphere, or say from the point A, the potential of the electricity on this spherical surface is $a\phi x$ or $\frac{a^2}{x}\,\phi\left(\frac{a^2}{x}\right)$, according as the point is interior or exterior; and, similarly, if x now denote the distance from the centre of the second sphere (or, say, from the point B), then the potential of the electricity on this spherical surface is $b\Phi x$ or $\frac{b^2}{x}\,\Phi\left(\frac{b^2}{x}\right)$, according as the point is interior or exterior; $\phi(x)$ is thus the same function of $(x,\, a,\, b)$ that $\Phi(x)$ is of $(x,\, b,\, a)$. Hence, first, for a point interior to the sphere A, if x denote the distance from A, and therefore $c-x$ the distance of the same point from B, the potential of the point in question is

$$= a\phi x + \frac{b^2}{c-x}\,\Phi\left(\frac{b^2}{c-x}\right);$$

and, secondly, for a point interior to the sphere B, if x denote the distance from B and therefore $c-x$ the distance of the same point from A, the potential of the point is

$$= \frac{a^2}{c-x}\,\phi\left(\frac{a^2}{c-x}\right) + b\Phi(x).$$

The two equations thus express that the potentials of a point interior to A and of a point interior to B are $= h$ and g respectively.

It is to be added that the potential of an exterior point, distances from the points A and $B = x$ and $c - x$ respectively, is

$$= \frac{a^2}{x}\,\phi\left(\frac{a^2}{x}\right) + \frac{b^2}{c-x}\,\Phi\left(\frac{b^2}{c-x}\right);$$

and that, by the known properties of Legendre's coefficients, when the potential upon an axial point is given, it is possible to pass at once to the expression for the potential of a point not on the axis, and also to the expression for the electrical density at a point on the two spherical surfaces respectively. The determination of the functions $\phi(x)$ and $\Phi(x)$ gives thus the complete solution of the question.

I obtain Poisson's solution by a different process as follows:—Consider the two functions

$$\frac{a^2(c-x)}{c^2-b^2-cx}, \;= \frac{ax+b}{cx+d}, \text{ suppose,}$$

and

$$\frac{b^2(c-x)}{c^2-a^2-cx}, \;= \frac{\alpha x+\beta}{\gamma x+\delta}, \text{ suppose;}$$

and let the nth functions be

$$\frac{a_n x+b_n}{c_n x+d_n} \text{ and } \frac{\alpha_n x+\beta_n}{\gamma_n x+\delta_n}$$

respectively.

Observing that the values of the coefficients are

$$\begin{pmatrix} a, & b \\ c, & d \end{pmatrix} = \begin{pmatrix} -a^2, & a^2c \\ -c, & c^2-b^2 \end{pmatrix}, \text{ and } \begin{pmatrix} \alpha, & \beta \\ \gamma, & \delta \end{pmatrix} = \begin{pmatrix} -b^2, & b^2c \\ -c, & c^2-a^2 \end{pmatrix},$$

so that we have

$$a+d = \alpha+\delta, \;= c^2-a^2-b^2, \quad ad-bc = \alpha\delta-\beta\gamma, \;= a^2b^2,$$

and consequently that the two equations

$$\frac{(\lambda+1)^2}{\lambda} = \frac{(a+d)^2}{ad-bc}, \quad \frac{(\lambda+1)^2}{\lambda} = \frac{(\alpha+\delta)^2}{\alpha\delta-\beta\gamma},$$

are in fact one and the same equation

$$\frac{(\lambda+1)^2}{\lambda} = \frac{(c^2-a^2-b^2)^2}{a^2b^2}$$

for the determination of λ, then (by a theorem which [686, 687] I have recently obtained) we have the following equations for the coefficients

$$\begin{pmatrix} a_n, & b_n \\ c_n, & d_n \end{pmatrix}, \quad \begin{pmatrix} \alpha_n, & \beta_n \\ \gamma_n, & \delta_n \end{pmatrix}$$

of the nth functions; viz. these are:—

$$a_n x + b_n = \frac{1}{\lambda^2 - 1}\left(\frac{a+d}{\lambda+1}\right)^{n-1} \{(\lambda^{n+1} - 1)(ax + b) + (\lambda^n - \lambda)(-dx + b)\},$$

$$c_n x + d_n = \qquad\qquad\qquad\qquad \{(\lambda^{n+1} - 1)(cx + d) + (\lambda^n - \lambda)(\quad cx - a)\};$$

and similarly

$$\alpha_n x + \beta_n = \frac{1}{\lambda^2 - 1}\left(\frac{\alpha+\delta}{\lambda+1}\right)^{n-1} \{(\lambda^{n+1} - 1)(\alpha x + \beta) + (\lambda^n - \lambda)(-\delta x + \beta)\},$$

$$\gamma_n x + \delta_n = \qquad\qquad\qquad\qquad \{(\lambda^{n+1} - 1)(\gamma x + \delta) + (\lambda^n - \lambda)(\quad \gamma x - \alpha)\}.$$

Observe that these equations give, as they ought to do,

$$a_0 x + b_0 = x, \quad c_0 x + d_0 = 1, \quad a_1 x + b_1 = ax + b, \quad c_1 x + d_1 = cx + d;$$

and similarly

$$\alpha_0 x + \beta_0 = x, \quad \gamma_0 x + \delta_0 = 1, \quad \alpha_1 x + \beta_1 = \alpha x + \beta, \quad \gamma_1 x + \delta_1 = \gamma x + \delta.$$

Substituting in the first two equations $\dfrac{a^2}{c-x}$ in place of x, and in the second two equations $\dfrac{b^2}{c-x}$ in place of x, we obtain the following results which will be useful:—

$$\begin{aligned} a_n a^2 + b_n(c - x) &= a^2(\gamma_n x + \delta_n), \\ c_n a^2 + d_n(c - x) &= \frac{1}{b^2}(\alpha_{n+1} x + \beta_{n+1}), \\ \alpha_n b^2 + \beta_n(c - x) &= b^2(c_n x + d_n), \\ \gamma_n b^2 + \delta_n(c - x) &= \frac{1}{a^2}(a_{n+1} x + b_{n+1}), \end{aligned}$$

the last two of which are obtained from the first two by a mere interchange of letters; it will therefore be sufficient to prove the first and second equations.

For the first equation we have

$$a_n a^2 + b_n(c - x) = \frac{1}{\lambda^2 - 1}\left(\frac{a+d}{\lambda+1}\right)^{n-1}$$
$$\{(\lambda^{n+1} - 1)[aa^2 + b(c - x)] + (\lambda^n - \lambda)[-da^2 + b(c - x)]\},$$

where the term in {} is

$$= (\lambda^{n+1} - 1)[-a^4 + a^2c(c-x)] + (\lambda^n - \lambda)[a^2(b^2-c^2) + a^2c(c-x)];$$

viz. this is

$$= a^2\{(\lambda^{n+1} - 1)(c^2 - a^2 - cx) + (\lambda^n - \lambda)(b^2 - cx)\};$$

or it is

$$= a^2\{(\lambda^{n+1} - 1)(\gamma x + \delta) \qquad + (\lambda^n - \lambda)(\gamma x - \alpha)\},$$

whence the relation in question.

The proof of the second equation is a little more complicated. We have

$$c_n a^2 + d_n(c-x) = \frac{1}{\lambda^2 - 1}\left(\frac{a+d}{\lambda+1}\right)^{n-1} \{(\lambda^{n+1} - 1)[ca^2 + d(c-x)] + (\lambda^n - \lambda)[ca^2 - a(c-x)]\},$$

where the term in {} is

$$= (\lambda^{n+1} - 1)[-ca^2 + (c^2 - b^2)(c-x)] + (\lambda^n - \lambda)[-ca^2 + a^2(c-x)].$$

Comparing this with

$$\alpha_{n+1}x + \beta_{n+1} = \frac{1}{\lambda^2 - 1}\left(\frac{\alpha+\delta}{\lambda+1}\right)^n \{(\lambda^{n+2} - 1)(\alpha x + \beta) + (\lambda^{n+1} - \lambda)(-\delta x + \beta)\},$$

where the term in {} is

$$= (\lambda^{n+2} - 1)[b^2(c-x)] + (\lambda^{n+1} - \lambda)[-c(c^2 - a^2 - b^2) + (c^2 - a^2)(c-x)],$$

it is to be observed that the quotient of the two terms in {} is in fact a constant; this is most easily verified as follows. Dividing the first of them by the second, we have a quotient which when $x = c$ is

$$\frac{(\lambda^{n+1} - 1)(-ca^2) + (\lambda^n - \lambda)(-ca^2)}{(\lambda^{n+1} - \lambda)\{-c(c^2 - a^2 - b^2)\}}, = \frac{a^2(\lambda^{n+1} - 1 + \lambda^n - \lambda)}{(\lambda^{n+1} - \lambda)(c^2 - a^2 - b^2)}, = \frac{a^2(\lambda+1)}{(c^2 - a^2 - b^2)\lambda},$$

and when $x = 0$ is

$$\frac{(\lambda^{n+1} - 1)\, c(c^2 - a^2 - b^2)}{(\lambda^{n+2} - 1)\, b^2c + (\lambda^{n+1} - \lambda)\, b^2c}, = \frac{(\lambda^{n+1} - 1)(c^2 - a^2 - b^2)}{(\lambda^{n+2} - 1 + \lambda^{n+1} - \lambda)\, b^2}, = \frac{c^2 - a^2 - b^2}{b^2(\lambda+1)};$$

these two values are equal by virtue of the equation which defines λ; and hence the quotient of the two linear functions having equal values for $x = c$ and $x = 0$, has always the same value; say it is $= \dfrac{c^2 - a^2 - b^2}{b^2(\lambda+1)}$. Hence, observing that $a+d=\alpha+\delta, =c^2-a^2-b^2$, the quotient, $c_n a^2+d_n(c-x)$ divided by $\alpha_{n+1}x+\beta_{n+1}$, is

$$= \frac{\lambda+1}{c^2-a^2-b^2}\cdot\frac{c^2-a^2-b^2}{b^2(\lambda+1)},\ =\frac{1}{b^2};$$

or we have the required equation

$$c_n a^2+d_n(c-x)=\frac{1}{b^2}(\alpha_{n+1}x+\beta_{n+1}).$$

The equations thus verified are

$$a_n a^2+b_n(c-x)=a^2(\gamma_n x+\delta_n), \tag{1}$$

$$c_n a^2+d_n(c-x)=\frac{1}{b^2}(\alpha_{n+1}x+\beta_{n+1}), \tag{2}$$

$$\alpha_n b^2+\beta_n(c-x)=b^2(c_n x+d_n), \tag{3}$$

$$\gamma_n b^2+\delta_n(c-x)=\frac{1}{a^2}(a_{n+1}x+b_{n+1}), \tag{4}$$

and we thence have

$$\begin{aligned}
&a\phi x+\frac{b^2}{c-x}\,\Phi\left(\frac{b^2}{c-x}\right)\\
&=a\sum_0^\infty\frac{1}{a_n x+b_n}\cdot a_n+\frac{b^2}{c-x}\sum_0^\infty\frac{1}{\gamma_n\frac{b^2}{c-x}+\delta_n}\cdot\gamma_n\\
&=\sum_0^\infty\frac{a\cdot a_n}{a_n x+b_n}+\sum_0^\infty\frac{b^2\gamma_n}{\gamma_n b^2+(c-x)\delta_n}\\
&=\sum_0^\infty\frac{a\cdot a_n}{a_n x+b_n}+\sum_0^\infty\frac{b^2\gamma_n}{\frac{1}{a^2}(a_{n+1}x+b_{n+1})}\quad\text{by (4)}\\
&=\sum_0^\infty\frac{a\cdot a_n}{a_n x+b_n}+a^2\sum_1^\infty\frac{b^2\gamma_{n-1}}{a_n x+b_n}.
\end{aligned}$$

Similarly

$$\begin{aligned}
&\frac{a^2}{c-x}\,\phi\left(\frac{a^2}{c-x}\right)+b\Phi(x)\\
&=b\sum_0^\infty\frac{1}{\alpha_n x+\beta_n}\cdot\alpha_n+\frac{a^2}{c-x}\sum_0^\infty\frac{1}{c_n\frac{a^2}{c-x}+d_n}\cdot c_n\\
&=\sum_0^\infty\frac{b\cdot\alpha_n}{\alpha_n x+\beta_n}+\sum_0^\infty\frac{a^2c_n}{c_n a^2+(c-x)d_n}\\
&=\sum_0^\infty\frac{b\cdot\alpha_n}{\alpha_n x+\beta_n}+b^2\sum_1^\infty\frac{a^2c_{n-1}}{\alpha_n x+\beta_n}\quad\text{by (2).}
\end{aligned}$$

The equations $= h$, $= g$ thus become

$$\sum_0^\infty \frac{a \cdot a_n + a^2b^2 \cdot \gamma_{n-1}}{a_n x + b_n} = h,$$
$$\sum_0^\infty \frac{b \cdot \alpha_n + a^2b^2 \cdot c_{n-1}}{\alpha_n x + \beta_n} = g,$$

where in the first equation for $n = 0$, the coefficient γ_{-1} is $= 0$, and in the second for $n = 0$, the coefficient c_{-1} is $= 0$.

The required values are therefore

$$\phi(x) = \sum_0^\infty \frac{A_n}{a_n x + b_n}, \qquad \Phi(x) = \sum_0^\infty \frac{B_n}{\alpha_n x + \beta_n},$$

where

$$A_0 = \frac{a \cdot a_0}{h}, \quad A_1 = \frac{a \cdot a_1 + a^2b^2 \cdot \gamma_0}{h}, \text{ \&c.,} \qquad B_0 = \frac{b \cdot \alpha_0}{g}, \quad B_1 = \frac{b \cdot \alpha_1 + a^2b^2 \cdot c_0}{g}, \text{ \&c.}$$

It will be observed that the equations as originally presented to Poisson contain only the two constants corresponding to the external potentials, and the question thus arises, how is the solution to be made to depend upon the four constants (a, b, c, h, g). The foregoing solution contains the four constants inasmuch as the coefficients a_n, b_n, &c. are functions of (a, b, c).

. . . .

707.

ON THE COLOURING OF MAPS.

[From the *Proceedings of the Royal Geographical Society*, vol. I., no. 4 (1879),
pp. 259—261.]

THE theorem that four colours are sufficient for any map, is mentioned somewhere by the late Professor De Morgan, who refers to it as a theorem known to map-makers. To state the theorem in a precise form, let the term "area" be understood to mean a simply or multiply connected[1] area: and let two areas, if they touch along a line, be said to be "attached" to each other; but if they touch only at a point or points, let them be said to be "appointed" to each other. For instance, if a circular area be divided by radii into sectors, then each sector is attached to the two contiguous sectors, but it is appointed to the several other sectors. The theorem then is, that if an area be partitioned in any manner into areas, these can be, with four colours only, coloured in such wise that in every case two attached areas have distinct colours; appointed areas may have the same colour. Detached areas may in a map represent parts of the same country, but this relation is not in anywise attended to: the colours of such detached areas will be the same, or different, as the theorem may require.

It is easy to see that four colours are wanted; for instance, we have a circle divided into three sectors, the whole circle forming an *enclave* in another area; then we require three colours for the three sectors, and a fourth colour for the surrounding area: if the circle were divided into four sectors, then for these two colours would be sufficient, and taking a third colour for the surrounding area, three colours only would be wanted; and so in general according as the number of sectors is even or odd, three colours or four colours are wanted. And in any tolerably simple case it can be seen that four colours are sufficient. But I have not succeeded in obtaining a general proof: and it is worth while to explain wherein the difficulty consists. Supposing a system of n areas coloured according to the theorem with four colours only, if we add an $(n+1)$th area, it by no means follows that we can *without altering the original colouring* colour this with one of the four

[1] An area is "connected" when every two points of the area can be joined by a continuous line lying wholly within the area; the area within a non-intersecting closed curve, or say an area having a single boundary, is "simply connected"; but if besides the exterior boundary there is one or more than one interior boundary (that is, if there is within the exterior boundary one or more than one *enclave* not belonging to the area), then the area is "multiply connected." The theorem extends to multiply connected areas, but there is no real loss of generality in taking, and we may for convenience take the areas of the theorem to be each of them a simply connected area.

colours. For instance, if the original colouring be such that the four colours all present themselves in the exterior boundary of the n areas, and if the new area be an area enclosing the n areas, then there is not any one of the four colours available for the new area.

The theorem, if it is true at all, is true under more stringent conditions. For instance, if in any case the figure includes four or more areas meeting in a point (such as the sectors of a circle), then if (introducing a new area) we place at the point a small circular area, cut out from and attaching itself to each of the original sectorial areas, it must according to the theorem be possible with four colours only to colour the new figure; and this implies that it must be possible to colour the original figure so that only three colours (or it may be two) are used for the sectorial areas. And in precisely the same way (the theorem is in fact really the same) it must be possible to colour the original figure in such wise that only three colours (or it may be two) present themselves in the exterior boundary of the figure.

But now suppose that the theorem *under these more stringent conditions* is true for n areas: say that it is possible with four colours only, to colour the n areas in such wise that not more than three colours present themselves in the external boundary: then it might be easy to prove that the $n+1$ areas could be coloured with four colours only: but this would be insufficient for the purpose of a general proof; it would be necessary to show further that the $n+1$ areas could be with the four colours only coloured *in accordance with the foregoing boundary condition*; for without this we cannot from the case of the $n+1$ areas pass to the next case of $n+2$ areas. And so in general, whatever more stringent conditions we import into the theorem as regards the n areas, it is necessary to show not only that the $n+1$ areas can be coloured with four colours only, but that they can be coloured in accordance with the more stringent conditions. As already mentioned, I have failed to obtain a proof.

708.

NOTE SUR LA THÉORIE DES COURBES DE L'ESPACE.

[From the *Compte Rendu de l'Association Française pour l'Avancement des Sciences* (1880), pp. 135—139.]

En considérant dans l'espace une courbe d'espèce donnée, déterminée au moyen d'un nombre suffisant de points, la courbe n'est pas déterminée uniquement; mais on a par les points un certain nombre de telles courbes. Par exemple, la courbe unicursale d'ordre $2p$ dépend, comme on voit sans peine, de $8p$ constantes et sera ainsi déterminée par $4p$ points (le cas $p = 1$ est une exception): on ne connaît pas, je pense, le nombre des courbes par les $4p$ points; mais pour le cas particulier $p = 2$ (c'est-à-dire pour une courbe quartique de seconde espèce, ou autrement dit, une courbe excubo-quartique) ce nombre est $= 4$: théorème démontré par moi depuis longtemps par des considérations géométriques. (Voir Salmon, *Geometry of three dimensions*, 3e ed. 1874, p. 319.) Ce n'est que dernièrement que j'ai considéré la question analytique, de trouver les équations d'une courbe excubo-quartique qui passe par 8 points donnés; et même j'ai pris pour les 8 points une disposition qui n'est pas tout à fait générale: l'investigation elle-même, et la forme du résultat, m'ont paru assez intéressantes pour que je les soumette à l'Association.

En considérant sur une courbe excubo-quartique 4 points donnés, le plan passant par 3 quelconques de ces points rencontre la courbe dans un seul point; et l'on obtient ainsi encore 4 points sur la courbe: voilà mon système de 8 points donnés, savoir en partant de 4 points quelconques, je prends un point quelconque dans chacun des plans qui passent par 3 de ces points, et j'obtiens ainsi les autres 4 points. Et par un tel système de 8 points, je cherche à faire passer une courbe de l'espèce dont il s'agit.

En prenant $x = 0$, $y = 0$, $z = 0$, $w = 0$, pour les équations des plans du tétraèdre formé par les 4 premiers points, les coordonnées de ces points seront (1, 0, 0, 0), (0, 1, 0, 0), (0, 0, 1, 0), (0, 0, 0, 1): et pour les coordonnées des 4 autres points, je prends $(0,\ y_1,\ z_1,\ w_1)$, $(x_2,\ 0,\ z_2,\ w_2)$, $(x_3,\ y_3,\ 0,\ w_3)$, $(x_4,\ y_4,\ z_4,\ 0)$.

Les équations de la courbe sont $x : y : z : w = P : Q : R : S$, où P, Q, R, S sont des fonctions $(\theta, 1)^4$ d'un paramètre variable θ; il s'agit de faire passer une telle courbe par les 8 points.

Je prends α, β, γ, δ, a, b, c, d pour les valeurs du paramètre θ qui correspondent aux 8 points respectivement.

Pour que la courbe passe par les premiers 4 points, il faut et il suffit que les équations soient de la forme

$$x : y : z : w = A\,\frac{\theta - a}{\theta - \alpha} : B\,\frac{\theta - b}{\theta - \beta} : C\,\frac{\theta - c}{\theta - \gamma} : D\,\frac{\theta - d}{\theta - \delta};$$

les conditions pour les autres 4 points seront alors

$$\begin{array}{rcccc}
\cdot\; y_1 : z_1 : w_1 = & & : B\,\dfrac{a-b}{a-\beta} & : C\,\dfrac{a-c}{a-\gamma} & : D\,\dfrac{a-d}{a-\delta}, \\
x_2\,\cdot\; : z_2 : w_2 = & A\,\dfrac{b-a}{b-\alpha} & & : C\,\dfrac{b-c}{b-\gamma} & : D\,\dfrac{b-d}{b-\delta}, \\
x_3 : y_3\,\cdot\; : w_3 = & A\,\dfrac{c-a}{c-\alpha} & : B\,\dfrac{c-b}{c-\beta} & & : D\,\dfrac{c-d}{c-\delta}, \\
x_4 : y_4 : z_4\,\cdot\; = & A\,\dfrac{d-a}{d-\alpha} & : B\,\dfrac{d-b}{d-\beta} & : C\,\dfrac{d-c}{d-\gamma}\,. &
\end{array}$$

Évidemment il y a deux équations qui donnent la valeur de $B : C$, et qui servent ainsi pour éliminer cette quantité. De cette manière on obtient six équations que j'écris comme voici:

$$\begin{aligned}
\lambda &= \frac{y_1 z_4}{y_4 z_1} = \frac{a-b}{a-c}\cdot\frac{d-c}{d-b}\cdot\frac{a-\gamma}{a-\beta}\cdot\frac{d-\beta}{d-\gamma}, \\
\mu &= \frac{w_1 y_3}{y_1 w_3} = \frac{a-d}{a-b}\cdot\frac{c-b}{c-d}\cdot\frac{a-\beta}{a-\delta}\cdot\frac{c-\delta}{c-\beta}, \\
\nu &= \frac{z_1 w_2}{z_2 w_1} = \frac{a-c}{a-d}\cdot\frac{b-d}{b-c}\cdot\frac{a-\delta}{a-\gamma}\cdot\frac{b-\gamma}{b-\delta}, \\
\varpi &= \frac{z_2 x_4}{z_4 x_2} = \frac{b-c}{b-a}\cdot\frac{d-a}{d-c}\cdot\frac{b-\alpha}{b-\gamma}\cdot\frac{d-\gamma}{d-\alpha}, \\
\kappa &= \frac{x_3 w_2}{x_3 w_2} = \frac{b-a}{b-d}\cdot\frac{c-d}{c-a}\cdot\frac{b-\delta}{b-\alpha}\cdot\frac{c-\alpha}{c-\delta}, \\
\rho &= \frac{x_3 y_4}{x_4 y_3} = \frac{c-a}{c-b}\cdot\frac{d-b}{d-a}\cdot\frac{c-\beta}{c-\alpha}\cdot\frac{d-\alpha}{d-\beta};
\end{aligned}$$

savoir λ, μ, ν, ϖ, κ, ρ dénotent ici les quantités données $\lambda = \frac{y_1 z_4}{y_4 z_1}$, etc. Le nombre des équations indépendantes est 5, car l'on a identiquement $\lambda\mu\nu\varpi\kappa\rho = 1$. Je remarque que l'on peut faire sur le paramètre θ une transformation linéaire quelconque $(h\theta + i) : (j\theta + k)$, et introduire ainsi 3 constantes arbitraires; on peut donc prendre à volonté 3 valeurs du paramètre θ, c'est-à-dire les valeurs de 3 quelconques des quantités α, β, γ, δ, a, b, c, d; et cela étant les 5 équations donneront les valeurs des autres 5 quantités. Si au moyen des équations on élimine α, β, γ, δ, on obtient entre a, b, c, d une équation qui sera, comme on va voir, de l'ordre 4 par rapport à chacune de ces quantités: en prenant comme données a, b, c il y aura donc 4 valeurs de d; et pour l'une quelconque de ces valeurs, celles de α, β, γ, δ seront déterminées uniquement: il y aura ainsi 4 courbes qui passent chacune par les 8 points; ce qui est le théorème dont il s'agit.

J'introduis, pour abréger, la notation

$$\begin{array}{cccccc} a-d, & b-d, & c-d, & b-c, & c-a, & a-b, \\ =\mathrm{f}, & \mathrm{g}, & \mathrm{h}, & \mathrm{a}, & \mathrm{b}, & \mathrm{c}; \end{array}$$

on a donc identiquement

$$\mathrm{a},\ \mathrm{b},\ \mathrm{c} = \mathrm{g}-\mathrm{h},\ \mathrm{h}-\mathrm{f},\ \mathrm{f}-\mathrm{g},$$
$$\mathrm{a}+\mathrm{b}+\mathrm{c}=0,$$
$$\mathrm{fa}+\mathrm{gb}+\mathrm{hc}=0.$$

Les équations prennent ainsi la forme

$$\lambda = -\frac{\mathrm{h}c}{\mathrm{g}b}\,\frac{a-\gamma\cdot d-\beta}{a-\beta\cdot d-\gamma},\ \text{etc.};$$

ou, en introduisant pour plus de commodité, les symboles

$$L, \qquad M, \qquad N, \qquad P, \qquad Q, \qquad R,$$

pour désigner respectivement

$$-\frac{\mathrm{g}b}{\mathrm{h}c}\,\lambda, \quad -\frac{\mathrm{h}c}{\mathrm{fa}}\,\mu, \quad -\frac{\mathrm{fa}}{\mathrm{g}b}\,\nu, \quad -\frac{\mathrm{h}c}{\mathrm{fa}}\,\varpi, \quad -\frac{\mathrm{g}b}{\mathrm{h}c}\,\kappa, \quad -\frac{\mathrm{fa}}{\mathrm{g}b}\,\rho,$$

les équations seront

$$\begin{aligned} L &= \frac{a-\gamma\cdot d-\beta}{a-\beta\cdot d-\gamma}, \\ M &= \frac{a-\beta\cdot c-\delta}{a-\delta\cdot c-\beta}, \\ N &= \frac{a-\delta\cdot b-\gamma}{a-\gamma\cdot b-\delta}, \\ P &= \frac{b-\alpha\cdot d-\gamma}{b-\gamma\cdot d-\alpha}, \\ Q &= \frac{b-\delta\cdot c-\alpha}{b-\alpha\cdot c-\delta}, \\ R &= \frac{c-\beta\cdot d-\alpha}{c-\alpha\cdot d-\beta}, \end{aligned}$$

avec la relation identique $LMNPQR = 1$; il s'agit entre ces 5 équations d'éliminer α, β, γ, δ.

J'écris $a = \alpha - \phi$, les facteurs $b-\alpha$, $c-\alpha$, $d-\alpha$ de P, Q, R deviennent ainsi respectivement $-\mathrm{c}+\phi$, $\mathrm{g}+\phi$, $-\mathrm{f}+\phi$; cela étant, les valeurs de P, Q,

R servent à exprimer β, γ, δ en fonction de ϕ: substituant ces valeurs de β, γ, δ dans celles de L, M, N, on obtient sans peine

$$L = -\frac{\mathrm{h}}{\mathrm{g}P}\,\frac{\mathrm{f}(\mathrm{c}-\phi)+\mathrm{c}P(-\mathrm{f}+\phi)}{\mathrm{b}(-\mathrm{f}+\phi)+\mathrm{f}R(\mathrm{b}+\phi)},$$
$$M = -\frac{\mathrm{a}}{\mathrm{h}R}\,\frac{\mathrm{b}(-\mathrm{f}+\phi)+\mathrm{f}R(\mathrm{b}+\phi)}{\mathrm{c}(\mathrm{b}+\phi)+\mathrm{b}Q(-\mathrm{c}+\phi)},$$
$$N = -\frac{\mathrm{g}}{\mathrm{a}Q}\,\frac{\mathrm{c}(\mathrm{b}+\phi)+\mathrm{b}Q(-\mathrm{c}+\phi)}{\mathrm{f}(\mathrm{c}-\phi)+\mathrm{c}P(-\mathrm{f}+\phi)},$$

valeurs qui donnent, comme cela doit être, $LMNPQR = 1$: il faut entre ces équations éliminer ϕ.

En rétablissant λ, μ, ν, ϖ, κ, ρ au lieu de L, M, N, P, Q, R, ces équations deviennent

$$\xi = \frac{\mathrm{g}}{\mathrm{h}}\,\lambda\varpi = \frac{X+Y\phi}{X_1+Y_1\phi},$$
$$\eta = \frac{\mathrm{h}}{\mathrm{a}}\,\mu\rho = \frac{X_1+Y_1\phi}{X_2+Y_2\phi},$$
$$\zeta = \frac{\mathrm{a}}{\mathrm{g}}\,\kappa\nu = \frac{X_2+Y_2\phi}{X+Y\phi},$$

(évidemment $\xi\eta\zeta = 1$), où j'écris ξ, η, ζ pour dénoter les expressions $\frac{\mathrm{g}}{\mathrm{h}}\,\lambda\varpi$, etc., et où les valeurs des coefficients X, Y, etc., sont

$$\begin{aligned} X &= \mathrm{fc}\,(\mathrm{fa}+\varpi\mathrm{hc}), & Y &= -\mathrm{f}^2\mathrm{a}-\varpi\mathrm{hc}^2,\\ X_1 &= \mathrm{fb}\,(\mathrm{gb}+\rho\mathrm{fa}), & Y_1 &= \mathrm{gb}^2+\rho\mathrm{f}^2\mathrm{a},\\ X_2 &= \mathrm{bc}\,(\mathrm{hc}+\kappa\mathrm{gb}), & Y_2 &= \mathrm{hc}^2-\kappa\mathrm{gb}^2. \end{aligned}$$

Les deux premières équations donnent

$$\xi\eta\,(X_1Y_2 - X_2Y_1) + \eta\,(X_2Y - XY_2) + XY_1 - XY_1 = 0,$$

ou, ce qui est la même chose,

$$\frac{1}{\zeta}\,(X_1Y_2 - X_2Y_1) + \eta\,(X_2Y - XY_2) + XY_1 - XY_1 = 0,$$

et l'on n'a qu'à substituer la valeur de ces coefficients.

On a

$$\begin{aligned} X_1Y_2 - X_2Y_1 &= \mathrm{fb}\,(\mathrm{gb}+\rho\mathrm{fa})(\mathrm{hc}^2-\kappa\mathrm{gb}^2) - \mathrm{bc}\,(\mathrm{hc}+\kappa\mathrm{gb})(-\mathrm{g}b^2+\rho\mathrm{f}^2\mathrm{a})\\ &= \mathrm{fghb}^2\mathrm{c}^2 - \mathrm{fg}^2\mathrm{b}^4\kappa + \mathrm{f}^2\mathrm{habc}^2\rho - \mathrm{f}^2\mathrm{gab}^3\kappa\rho + \mathrm{ghb}^3\mathrm{c}^2 + \mathrm{g}^2\mathrm{b}^4\mathrm{c}\kappa - \mathrm{f}^2\mathrm{habc}^2\rho - \mathrm{f}\\ &= \mathrm{ghb}^2\mathrm{c}^2(\mathrm{f}+\mathrm{b}) + \mathrm{g}^2\mathrm{b}^4(-\mathrm{f}+\mathrm{c})\rho - \mathrm{f}^2\mathrm{gab}^2(\mathrm{b}+\mathrm{c})\,\kappa\rho\\ &= \mathrm{ghb}^2\mathrm{c}^2\mathrm{h} + \mathrm{g}^2\mathrm{b}^4(-\mathrm{g})\rho + \mathrm{f}^2\mathrm{gab}^2\mathrm{a}\kappa\rho\\ &= \mathrm{g}b^2\,(\mathrm{h}^2\mathrm{c}^2 - \mathrm{g}^2\mathrm{b}^2\kappa + \mathrm{a}^2\mathrm{f}^2\kappa\rho); \end{aligned}$$

et de même

$$
\begin{aligned}
XY_2 - X_2Y &= \mathrm{hc}^2(\mathrm{f}^2\mathrm{a}^2 - \mathrm{h}^2\mathrm{c}^2\varpi + \mathrm{g}^2\mathrm{b}^2\varpi\kappa),\\
XY_1 - X_1Y &= \mathrm{f}^2\mathrm{a}\,(\mathrm{g}^2\mathrm{b}^2 - \mathrm{f}^2\mathrm{a}^2\rho + \mathrm{h}^2\mathrm{c}^2\varpi\rho).
\end{aligned}
$$

Donc

$$
\begin{aligned}
&\frac{\mathrm{g}}{\mathrm{a}}\,\frac{1}{\kappa\nu}\,\mathrm{g}b^2\,(\mathrm{h}^2\mathrm{c}^2 - \mathrm{g}^2\mathrm{b}^2\kappa + \mathrm{a}^2\mathrm{f}^2\kappa\rho)\\
&\qquad\qquad + \frac{\mathrm{h}}{\mathrm{a}}\,\mu\rho\,\mathrm{hc}^2(\mathrm{f}^2\mathrm{a}^2 - \mathrm{h}^2\mathrm{c}^2\varpi + \mathrm{g}^2\mathrm{b}^2\varpi\kappa)\\
&\qquad\qquad\qquad\qquad + \ \mathrm{f}^2\mathrm{a}\,(\mathrm{g}^2\mathrm{b}^2 - \mathrm{f}^2\mathrm{a}^2\rho + \mathrm{h}^2\mathrm{c}^2\varpi\rho) = 0,
\end{aligned}
$$

ou enfin en multipliant par $-\mathrm{a}\nu$, et dans un terme $-\mathrm{g}^2\mathrm{b}^2\mathrm{h}^2\mathrm{c}^2\mu\nu\rho\varpi\kappa$, au lieu de $\mu\nu\rho\varpi\kappa$ écrivant $\frac{1}{\lambda}$, l'équation devient

$$
\begin{aligned}
&(\mathrm{fa})^4\,\nu\rho + (\mathrm{gb})^4 + (\mathrm{hc})^4\,\frac{1}{\lambda\kappa} - (\mathrm{gb})^2(\mathrm{hc})^2\left(\frac{1}{\kappa} + \frac{1}{\lambda}\right)\\
&\qquad\qquad - (\mathrm{hc})^2(\mathrm{fa})^2\,\nu\rho\,(\varpi + \mu) - (\mathrm{fa})^2(\mathrm{gb})^2(\nu + \rho) = 0,
\end{aligned}
$$

ou, comme on peut l'écrire,

$$
\left(\nu\rho,\ 1,\ \frac{1}{\lambda\kappa},\ -\left(\frac{1}{\kappa} + \frac{1}{\lambda}\right),\ -\nu\rho\,(\varpi + \mu),\ -(\nu + \rho)\right)((\mathrm{fa})^2,\ (\mathrm{gb})^2,\ (\mathrm{hc})^2)^2 = 0.
$$

C'est la deuxième d'un système de trois équations équivalentes; savoir, en multipliant par $\frac{1}{\nu\rho}$ et en réduisant par $\lambda\mu\nu\varpi\kappa\rho = 1$, on obtient la première forme: et, en multipliant par $\lambda\kappa$ et réduisant de même, on obtient la troisième forme: le système est

$$
\begin{aligned}
&\left(1,\ \frac{1}{\nu\rho},\ \mu\varpi,\ -\mu\varpi\,(\lambda + \kappa),\ -(\mu + \varpi),\ -\left(\frac{1}{\nu} + \frac{1}{\rho}\right)\right)((\mathrm{fa})^2,\ (\mathrm{gb})^2,\ (\mathrm{hc})^2)^2 = 0,\\
&\left(\nu\rho,\ 1,\ \frac{1}{\lambda\kappa},\ -\left(\frac{1}{\kappa} + \frac{1}{\lambda}\right),\ -\nu\rho\,(\mu + \varpi),\ -(\nu + \rho)\right)(\quad ,, \qquad ,, \qquad ,, \ )^2 = 0,\\
&\left(\frac{1}{\mu\varpi},\ \lambda\kappa,\ 1,\ -(\lambda + \kappa),\ -\left(\frac{1}{\mu} + \frac{1}{\varpi}\right),\ -\lambda\kappa\,(\nu + \rho)\right)(\quad ,, \qquad ,, \qquad ,, \ )^2 = 0.
\end{aligned}
$$

En écrivant $\mathrm{hc} = -\mathrm{fa} - \mathrm{gb}$, on obtient une équation de la forme $(*)\ (\mathrm{fa},\ \mathrm{gb})^4 = 0$, savoir une équation quartique pour avoir fa : gb, c'est-à-dire, le rapport anharmonique $(a-d)(b-c) : (b-d)(c-a)$: en considérant a, b, c comme données, il y a donc 4 valeurs de d: et l'on a déjà vu que les valeurs α, β, γ, δ sont données rationnellement en fonctions de a, b, c, d: le théorème est donc démontré.

Cambridge, juillet, 1880.

709.

ON THE NUMBER OF CONSTANTS IN THE EQUATION OF A SURFACE $PS - QR = 0$.

[From the *Tidsskrift for Mathematik*, Ser. 4, t. IV. (1880), pp. 145—148.]

THE very important results contained in Mr H. Valentiner's paper "Nogle Saetninger om fuldstaendige Skjaeringskurver mellem to Flader" may be considered from a somewhat different point of view, and established in a more simple manner, as follows[2].

Assuming throughout $n \geqq p+q$, $p \geqq q$, and moreover that P, Q, R, S denote functions of the coordinates (x, y, z, w) of the orders p, q, $n-q$, $n-p$ respectively: then the equation of a surface of the order n containing the curve of intersection of two surfaces of the orders p and q respectively, is

$$\begin{vmatrix} P, & Q \\ R, & S \end{vmatrix} = 0,$$

so that the number of constants in the equation of a surface of the order n satisfying the condition in question is in fact the number of constants contained in an equation of the last-mentioned form. Writing for shortness

$$a_p = \tfrac{1}{6}(p+1)(p+2)(p+3) - 1,\ = \tfrac{1}{6}p\,(p^2 + 6p + 11),$$

the number of constants contained in a function of the order p is $= a_p + 1$; or if we take one of the coefficients (for instance that of x^p) to be unity, then the number of the remaining constants is $= a_p$; viz. a_p is the number of constants in the equation of a surface of the order p. As regards the surface in question

$$\begin{vmatrix} P, & Q \\ R, & S \end{vmatrix} = 0,$$

we may it is clear take P, Q, R each with a coefficient unity as above, but in the remaining function S, the coefficient must remain arbitrary: the apparent number of constants is thus $= a_p + a_q + a_{n-p} + a_{n-q} + 1$; but there is a deduction from this number.

The equation may in fact be written in the form

$$\begin{vmatrix} P + \alpha Q, & Q \\ R + \alpha S + \beta P + \alpha\beta Q, & S + \beta Q \end{vmatrix} = 0,$$

[2] Idet vi med stor Glæde optage Prof. Cayley's simple Forklaring af den Reduktion af Konstanttallet i Ligningen $PS - QR = 0$, som Hr. Valentiner havde paavist (*Tidsskr. f. Math.* 1879, S. 22), skulle vi dog bemærke, at Grunden til, at dennes Bevis er bleven saa vanskeligt, er den, at han tillige har villet bevise, at der ikke finder nogen *yderligere* Reduktion Sted.

where α represents an arbitrary function of the order $p - q$, and β an arbitrary function of the degree $n - p - q$: we thus introduce $(a_{p-q} + 1) + (a_{n-p-q} + 1)$, $= a_{p-q} + a_{n-p-q} + 2$, constants, and by means of these we can impose the like number of arbitrary relations upon the constants originally contained in the functions P, Q, R, S respectively (say we can reduce to zero this number $a_{p-q} + a_{n-p-q} + 2$ of the original constants): hence the real number of constants is

$$\begin{aligned} &a_p + a_q + a_{n-p} + a_{n-q} + 1 - (a_{p-q} + a_{n-p-q} + 2), \\ &= a_p + a_q + a_{n-p} + a_{n-q} - a_{p-q} - a_{n-p-q} - 1 \\ &= \omega \text{ suppose;} \end{aligned}$$

viz. this is the required number in the case $n > p + q$, $p > q$.

If however $n = p + q$, or $p = q$, or if these relations are both satisfied, then there is a further deduction of 1, 1, or 2: in fact, calling the last-mentioned determinant $\begin{vmatrix} P', & Q' \\ R', & S' \end{vmatrix}$, then the four cases are

$$\begin{array}{ll} n > p + q,\ p > q, & \begin{vmatrix} P', & Q' \\ R', & S' \end{vmatrix} = \begin{vmatrix} P', & Q' \\ R', & S' \end{vmatrix} \\ n = p + q,\ p > q, & \begin{vmatrix} P', & Q' \\ R', & S' \end{vmatrix} = \begin{vmatrix} P' + kR', & Q' + kS' \\ R', & S' \end{vmatrix} \\ n > p + q,\ p = q, & \begin{vmatrix} P', & Q' \\ R', & S' \end{vmatrix} = \begin{vmatrix} P', & Q' + kP' \\ R', & S' + kR' \end{vmatrix} \\ n = p + q,\ p = q, & \begin{vmatrix} P', & Q' \\ R', & S' \end{vmatrix} = \begin{vmatrix} P' + kR', & Q' + lP' + kS' + klR' \\ R', & S' + \ lR' \end{vmatrix} \end{array}$$

where k, l denote arbitrary constants: these, like the constants of α and β, may be used to impose arbitrary relations upon the original constants of P, Q, R, S; and hence the number of constants is $= \omega$, $\omega - 1$, $\omega - 1$, $\omega - 2$ in the four cases respectively; where as above

$$\omega = a_p + a_q + a_{n-p} + a_{n-q} - a_{p-q} - a_{n-p-q} - 1.$$

If $n = 4$, there is in each of the four cases one system of values of p, q; viz. the cases are

$$\begin{array}{lll} p,\ q = & 2\ 1 & \text{No.} = a_2 + a_1 + a_2 + a_3 - a_1 - a_1 - 1 = \ 9 + 3 + \ 9 + 19 - 3 - 3 - 1, = 33, \\ & 3\ 1 & {}'', \ a_3 + a_1 + a_1 + a_3 - a_2 - a_0 - 2 = 19 + 3 + \ 3 + 19 - 9 - 0 - 2, = 33, \\ & 1\ 1 & {}'', \ a_1 + a_1 + a_3 + a_3 - a_0 - a_2 - 2 = \ 3 + 3 + 19 + 19 - 0 - 9 - 2, = 33, \\ & 2\ 2 & {}'', \ a_2 + a_2 + a_2 + a_2 - a_0 - a_0 - 3 = \ 9 + 9 + \ 9 + \ 9 - 0 - 0 - 3, = 33, \end{array}$$

and the number of constants is in each case $= 33$. This is easily verified: in the first case we have a quartic surface containing a conic, the plane of the conic is therefore a quadruple tangent plane; and the existence of such a plane is 1 condition. In the second case the surface contains a plane cubic; the plane of this cubic is a triple tangent plane, having the points of contact in a line; and this is 1 condition. In the third case the surface contains a line, which is 1 condition: hence in each of these cases the number of constants is $34-1, = 33$. In the fourth case, where the surface contains a quadriquadric curve, we repeat in some measure the general reasoning: the quadriquadric curve contains 16 constants, and we have thus 16 as the number of constants really contained in the equations $P=0$, $Q=0$ of the quadriquadric curve: the equation $PS-QR=0$, contains in addition $9+10, = 19$ constants, but writing it in the form $P(S+kQ)-Q(R+kP)=0$, we have a diminution $=1$, or the number apparently is $16+19-1, = 34$. But the quadriquadric curve is one of a singly infinite series $P+lR=0$, $Q+lS=0$ of such curves, and we have on this account a diminution $=1$; the number of constants is thus $34-1, = 33$ as above: the reasoning is, in fact, the same as for the case of a plane passing through a line; the line contains 4 constants, hence the plane, quà arbitrary plane through the line, would contain $1+4, = 5$ constants; but the line being one of a doubly infinite system of lines on the plane the number is really $5-2, = 3$, as it should be.

Cambridge, 2nd Sept., 1880.

710.

ON A DIFFERENTIAL EQUATION.

[From *Collectanea Mathematica: in memoriam Dominici Chelini*, (Milan, Hoepli, 1881), pp. 17—26.]

IN the Memoir on hypergeometric series, *Crelle*, t. XV. (1836), Kummer in effect considers a differential equation

$$\frac{(a'z^2+2b'z+c')\,dz^2}{z^2(z-1)^2}=\frac{(ax^2+2bx+c)\,dx^2}{x^2(x-1)^2},$$

viz. he seeks for solutions of an equation of this form which also satisfy a certain differential equation of the third order. The coefficients a, b, c are either all arbitrary, or they are two or one of them, arbitrary; but this last case (or say the case where the function of x is the completely determinate function $x^2+2bx+c$) is scarcely considered: a', b', c' are regarded as determinable in terms of a, b, c; and z is to be found as a function of x independent of a, b, c: so that when these coefficients are arbitrary, the equation breaks up into three equations, and when two of the coefficients are arbitrary, it breaks up into two equations, satisfied in each case by the same value of z; and the value of z is thus determined without any integration: these cases will be considered in the sequel, but they are of course included in the general case where the coefficients a, b, c are regarded as having any given values whatever.

Writing for shortness $X=ax^2+2bx+c$, in general the integral

$$\int\frac{N\,dx}{D\sqrt{X}},$$

where D is the product of any number n of distinct linear factors $x-p$, and N is a rational and integral function of x of the order n at most, and therefore also the integral

$$\int\frac{N\sqrt{X}\,dx}{D}=\int\frac{NX\,dx}{D\sqrt{X}},$$

where N is now of the order $n-2$ at most, is expressible as the logarithm of a quasi-algebraical function, that is, a function containing powers the exponents of which are incommensurable (for instance, $x^{\sqrt{2}}$ is a quasi-algebraical function): in fact, the integral is of the form

$$\int\left(M+\frac{A}{x-p}+\frac{B}{x-q}+\cdots\right)\frac{dx}{\sqrt{X}},$$

where each term is separately integrable,

$$\int \frac{dx}{\sqrt{X}} = \frac{1}{\sqrt{a}} \log\{ax + b + \sqrt{a} \cdot \sqrt{X}\},$$

$$\int \frac{dx}{(x-p)\sqrt{X}} = -\frac{1}{\sqrt{P}} \log \frac{(ap+b)\,x + (bp+c) + \sqrt{P} \cdot \sqrt{X}}{x-p},$$

where P is written to denote $ap^2 + 2bp + c$: the integral is thus $= \log \Omega$, where Ω is a product of factors

$$ax + b + \sqrt{a} \cdot \sqrt{X}, \qquad \frac{(ap+b)\,x + (bp+c) + \sqrt{P} \cdot \sqrt{X}}{x-p}, \text{ etc.},$$

raised to powers $\frac{M}{\sqrt{a}}$, $\frac{-A}{\sqrt{P}}$, etc.: hence, if we have a differential equation

$$\frac{N'\,dz}{D'\sqrt{Z}} = \frac{N\,dx}{D\sqrt{X}}, \text{ or } \frac{N'\sqrt{Z}\,dz}{D'} = \frac{N\sqrt{X}\,dx}{D},$$

where $Z\,(= a'z^2 + 2b'z + c')$, and N', D' are functions of z such as X, N, D are of x; then, taking $\log C$ for the constant of integration, the general integral is

$$\log \Omega' = \log C + \log \Omega :$$

viz. we have the quasi-algebraical integral $\Omega' - C\Omega = 0$.

The constants a, b, c, p, q, $\ldots$ etc. may be such that the exponents are rational, and the integral is then algebraical: in particular, for the differential equation

$$\frac{\sqrt{z^2 + 14z + 1}\,dz}{z\,(z-1)} = \frac{\sqrt{x^2 + 14x + 1}\,dx}{x\,(x-1)},$$

the general integral is in the first instance obtained in the form

$$\frac{(z + 1 + \sqrt{Z})\,(z-1)^2}{\sqrt{z}\,(2z + 2 + \sqrt{Z})^2} = C\,\frac{(x + 1 + \sqrt{X})\,(x-1)^2}{\sqrt{x}\,(2x + 2 + \sqrt{X})^2},$$

which, observing that $(2x+2)^2 - X = 3\,(x-1)^2$, may also be written

$$\frac{(z+1)\,(z^2 - 34z + 1) + Z\sqrt{Z}}{\sqrt{z}\,(z-1)^2} = C\,\frac{(x+1)\,(x^2 - 34x + 1) + X\sqrt{X}}{\sqrt{x}\,(x-1)^2}.$$

I had previously obtained the solution

$$z = \left(\frac{1 - \sqrt[4]{x}}{1 + \sqrt[4]{x}}\right)^4,$$

and I wish to show that this is, in fact, the particular integral belonging to the value $C = 1$ of the constant of integration: for this purpose I proceed to rationalise the general integral as regards z.

Writing for a moment

$$\begin{aligned} P &= (z+1)(z^2-34z+1), \\ Q &= (z^2+14z+1)\sqrt{z^2+14z+1}, \\ R &= M\sqrt{z}\,(z-1)^2, \end{aligned}$$

where

$$M = C\,\frac{(x+1)\,(x^2-34x+1)+(x^2+14x+1)\sqrt{x^2+14x+1}}{\sqrt{x}\,(x-1)^2},$$

the integral is $P+Q+R=0$; or rationalising, it is

$$(P^2-Q^2)^2 - 2R^2\,(P^2+Q^2) + R^4 = 0;$$

we have

$$\begin{aligned} P^2 &= (1,\ -66,\ 1023,\ 2180,\ 1023,\ -66,\ 1 \mid z,\ 1)^6, \\ Q^2 &= (1,\quad 42,\quad 591,\ 2828,\quad 591,\quad 42,\ 1 \mid z,\ 1)^6, \end{aligned}$$

and hence

$$\begin{aligned} P^2 - Q^2 &= (0,\ -108,\ 432,\ -648,\ 432,\ -108,\ 0 \mid z,\ 1)^6, \\ &= -108z\,(z-1)^4; \\ P^2 + Q^2 &= 2\,(1,\ -12,\ 807,\ 2504,\ 807,\ -12,\ 1 \mid z,\ 1)^6. \end{aligned}$$

Writing the equation in the form

$$\tfrac{1}{2}(P^2+Q^2)^2 - \tfrac{1}{4}\left\{R^2 + \frac{(P^2-Q^2)^2}{R^2}\right\} = 0,$$

it thus becomes

$$(1,\ -12,\ 807,\ 2504,\ 807,\ -12,\ 1 \mid z,\ 1)^6 - z\,(z-1)^4\left\{M^2 + \frac{(108)^2}{M^2}\right\} = 0,$$

where M has its above-mentioned value; and if we now assume $C=1$, then

$$\begin{aligned} M &= \frac{(x+1)\,(x^2-34x+1)+(x^2+14x+1)\sqrt{x^2+14x+1}}{\sqrt{x}\,(x-1)^2}, \\ \frac{108}{M} &= \frac{(x+1)\,(x^2-34x+1)-(x^2+14x+1)\sqrt{x^2+14x+1}}{\sqrt{x}\,(x-1)^2}, \end{aligned}$$

and hence

$$\begin{aligned} M^2 + \frac{(108)^2}{M^2} &= \left(M - \frac{108}{M}\right)^2 + 216,\ = 4\,\frac{(x+1)^2(x^2-34x+1)^2}{x\,(x-1)^4} + 216, \\ &= \frac{4}{x\,(x-1)^4}\cdot(1,\ -12,\ 807,\ 2504,\ 807,\ -12,\ 1 \mid x,\ 1)^6 : \end{aligned}$$

and the rationalised equation is

$$(1,\ -12,\ 807,\ 2504,\ 807,\ -12,\ 1 \mid z,\ 1)^6 \\ - \frac{z\,(z-1)^4}{x\,(x-1)^4}(1,\ -12,\ 807,\ 2504,\ 807,\ -12,\ 1 \mid x,\ 1)^6 = 0.$$

This is a sextic equation in z, of the form

$$z^3 + \frac{1}{z^3} + \lambda\left(z^2 + \frac{1}{z^2}\right) + \mu\left(z + \frac{1}{z}\right) + \nu = 0,$$

where

$$\lambda,\ \mu,\ \nu = -12 - \Omega, \quad 807 + 4\Omega, \quad 2504 - 6\Omega,$$

if Ω denote the function of x which enters into the equation; and writing $z + \dfrac{1}{z} = \theta$, this becomes

$$\theta^3 - 3\theta + \lambda\,(\theta^2 - 2) + \mu\theta + \nu = 0.$$

But the equation in z is satisfied by the value $z = x$, and therefore the equation in θ by the value $\theta = x + \dfrac{1}{x} = \alpha$ suppose, we have therefore

$$\alpha^3 - 3\alpha + \lambda\,(\alpha^2 - 2) + \mu\alpha + \nu = 0,$$

and hence subtracting, and throwing out the factor $\theta - \alpha$,

$$\theta^2 + \theta\alpha + \alpha^2 - 3 + \lambda\,(\theta + \alpha) + \mu = 0;$$

viz. writing for λ, μ, α their values, this is

$$\theta^2 + \theta\left(x + \frac{1}{x} - 12 - \Omega\right) + x^2 - 1 + \frac{1}{x^2} - \left(x + \frac{1}{x}\right)(12 + \Omega) + 807 + 4\Omega = 0,$$

or, what is the same thing,

$$\theta^2 + \theta\left(x - 12 + \frac{1}{x} - \Omega\right) + x^2 - 12x + 806 - \frac{12}{x} + \frac{1}{x^2} - \left(x - 4 - \frac{1}{x}\right)\Omega = 0,$$

where

$$\Omega = \frac{1}{x\,(x-1)^4}(1,\ -12,\ 807,\ 2504,\ 807,\ -12,\ 1 \mid x,\ 1)^6.$$

Hence in the quadric equation, the coefficients, each multiplied by $(x-1)^4$, are

$$(x-1)^4\left(x - 12 + \frac{1}{x}\right) - \frac{1}{x}(1,\ -12,\ 807,\ 2504,\ 807,\ -12,\ 1 \mid x,\ 1)^6,$$

and

$$(x-1)^4\left(x^2-12x+806-\frac{12}{x}+\frac{1}{x^2}\right)$$
$$-\frac{1}{x}\left(x-4+\frac{1}{x}\right)(1,\ -12,\ 807,\ 2504,\ 807,\ -12,\ 1\mid x,\ 1)^6,$$

which are respectively rational and integral quartic functions of x; and, writing for θ its value, the equation finally is

$$\left(z+\frac{1}{z}\right)^2-4\left(z+\frac{1}{z}\right)\frac{(1,\ 188,\ 646,\ 188,\ 1\mid x,\ 1)^4}{(x-1)^4}+4\,\frac{(1,\ -644,\ 3334,\ -644,\ 1\mid x,\ 1)^4}{(x-1)^4}=$$

Writing

$$\xi=\sqrt[4]{x},\quad A=\frac{1-\xi}{1+\xi},\quad B=\frac{1+\xi}{1-\xi},\quad C=\frac{1-i\xi}{1+i\xi},\quad D=\frac{1+i\xi}{1-i\xi},\qquad (i=\sqrt{-1}\text{ as usual}),$$

this is

$$(z-A^4)(z-B^4)(z-C^4)(z-D^4)=0,$$

or, what is the same thing,

$$\left\{z+\frac{1}{z}-(A^4+B^4)\right\}\left\{z+\frac{1}{z}-(C^4+D^4)\right\}=0,$$

that is,

$$\left(z+\frac{1}{z}\right)^2-\left(z+\frac{1}{z}\right)(A^4+B^4+C^4+D^4)+(A^4+B^4)(C^4+D^4)=0;$$

for we have

$$\tfrac{1}{2}(A^4+B^4)=\frac{(1,\quad 28,\ 70,\quad 28,\ 1\mid \xi^2,\ 1)^4}{(\xi^2-1)^4},$$
$$\tfrac{1}{2}(C^4+D^4)=\frac{(1,\ -28,\ 70,\ -28,\ 1\mid \xi^2,\ 1)^4}{(\xi^2+1)^4}.$$

And substituting these values, the coefficients will be rational functions of ξ^4, that is, of x, and it is easy to verify that they have in fact their foregoing values.

It thus appears that for $C=1$, besides the values x and $\frac{1}{x}$, we have for z only the values

$$\left(\frac{1-\xi}{1+\xi}\right)^4,\ \left(\frac{1+\xi}{1-\xi}\right)^4,\ \left(\frac{1-i\xi}{1+i\xi}\right)^4,\ \left(\frac{1+i\xi}{1-i\xi}\right)^4;$$

viz. that the only solution is

$$z = \left(\frac{1-\sqrt[4]{x}}{1+\sqrt[4]{x}}\right)^4.$$

The example shows that although the differential equation

$$\frac{\sqrt{a'z^2+2b'z+c'}\,dz}{z\,(z-1)} = \frac{\sqrt{ax^2+2bx+c}\,dx}{x\,(x-1)}$$

can be integrated generally in a quasi-algebraical or algebraical form as above, yet we cannot from the general solution deduce, at once or easily, the various particular integrals comprised therein: nor can we find for what values of the constants a, b, c and a', b', c' the differential equation admits of a simple solution, or say of a solution where z is expressed as an explicit (irrational) function of x.

In the cases considered by Kummer there is a second (or it may be also a third) differential equation of the like form, the equations being each of them satisfied by the same value of z: hence eliminating the differentials dx, dz, the relation between x and z is of the form

$$\frac{P'}{Q'} = \frac{P}{Q},$$

where P, Q are quadric functions of x; P', Q' quadric functions of z. But P and Q may contain a common factor, and the integral is then expressible in the form $x = \dfrac{P'}{Q'}$, the quotient of two quadric functions of z; or P' and Q' may have a common factor, and the integral is then expressible in the form $z = \dfrac{P}{Q}$, the quotient of two quadric functions of x; or there may be a common factor of P, Q, and also a common factor of P' and Q', and the integral is then of the form $z = \dfrac{L}{M}$, the quotient of two linear functions of x.

In the general case the differential equation is

$$\frac{\lambda\,(aP'+bQ')\,dz^2}{z^2(z-1)^2} = \frac{(aP+bQ)\,dx^2}{x^2(x-1)^2},$$

where a, b are arbitrary constants, λ is a constant the value of which can in each particular case be at once determined; so when the integral is $z = \dfrac{P}{Q}$, the differential equation is

$$\frac{\lambda\,(az+b)\,dz^2}{z^2(z-1)^2} = \frac{(aP+bQ)\,dx^2}{x^2(x-1)^2},$$

where a, b are arbitrary constants, but λ is now a linear function of z the value of which can in each particular case be at once determined. When the integral is $z = \dfrac{L}{M}$, the differential equation is

$$\frac{\lambda\,(az^2 + 2bz + c)\,dz^2}{z^2(z-1)^2} = \frac{(aL^2 + 2bLM + cM^2)\,dx^2}{x^2(x-1)^2}$$

containing the three arbitrary constants a, b, c; λ is a constant the value of which can be at once determined.

There are in all 6 integrals of the form $z = \dfrac{L}{M}$, for which the differential equation contains three arbitrary constants: 18 integrals of the form $z = \dfrac{P}{Q}$ (and of course the same number of integrals of the form $x = \dfrac{P'}{Q'}$), and 9 integrals of the form $\dfrac{P}{Q} = \dfrac{P'}{Q'}$, for all of which the differential equation contains two arbitrary constants. It is to be remarked that Kummer, considering the values of z as a function of x, obtains the 72 rational and irrational values mentioned in his equations (31), (35), (36), (37), (38), and (39): but the 72 values are made up as follows, viz. the 18 values of z as a rational function of x, the 36 irrational values obtained from the 18 expressions of x as a rational function of z, and the 18 irrational values of z obtained from the 9 integrals in which neither of the variables is a rational function of the other: $18 + 36 + 18 = 72$.

The several integrals together with the expressions of the functions

$$a'z^2 + 2b'z + c' \quad \text{and} \quad ax^2 + 2bx + c$$

which enter into the differential equation are as follows:

$z =$	$a'z^2 + 2b'z + c' =$	$ax^2 + 2bx + c =$
x	$ax^2 + 2bx + c$	$ax^2 + 2bx + c$
$1 - x$		$a\,(x-1)^2 + 2b\,(x-1) + c$
$\dfrac{1}{x}$		$a + 2bx + cx^2$
$\dfrac{1}{1-x}$		$a + 2b\,(x-1) + c\,(x-1)^2$
$\dfrac{x}{x-1}$		$ax^2 + 2bx\,(x-1) + c\,(x-1)^2$
$\dfrac{x-1}{x}$		$a\,(x-1)^2 + 2bx\,(x-1) + cx^2$
$\left(\dfrac{x+1}{x-1}\right)^2$	$az^2 + bz$	$a\,(x+1)^2 + b\,(x-1)^2$
$(2x-1)^2$		$a\,(2x-1)^2 + b$
$\left(\dfrac{x-2}{x}\right)^2$		$a\,(x-2)^2 + bx^2$
$\dfrac{(x+1)^2}{4x}$		$a\,(x+1)^2 + 4bx$
$\dfrac{(2x-1)^2}{4x\,(x-1)}$		$a\,(2x-1)^2 + 4bx\,(x-1)$
$-\dfrac{(x-2)^2}{4\,(x-1)}$		$a\,(x-2)^2 - 4b\,(x-1)$
$\left(\dfrac{x-1}{x+1}\right)^2$	$bz + c$	$b\,(x-1)^2 + c\,(x+1)^2$
$\left(\dfrac{1}{2x-1}\right)^2$		$b + c\,(2x-1)^2$
$\left(\dfrac{x}{x-2}\right)^2$		$bx^2 + c\,(x-2)^2$
$\dfrac{4x}{(x+1)^2}$		$4bx + c\,(x+1)^2$
$\dfrac{4x\,(x-1)}{(2x-1)^2}$		$4bx\,(x-1) + c\,(2x-1)^2$
$-\dfrac{4\,(x-1)}{(x-2)^2}$		$-4b\,(x-1) + c\,(x-2)^2$
$-\dfrac{(x-1)^2}{4x}$	$az^2 - (a+c)z + c$	$a\,(x-1)^2 + 4cx$
$-4x\,(x-1)$		$4ax\,(x-1) + c$
$\dfrac{4\,(x-1)}{x^2}$		$-4a\,(x-1) + cx^2$
$-\dfrac{4x}{(x-1)^2}$		$4ax + c\,(x-1)^2$
$\dfrac{-1}{4x\,(x-1)}$		$a + 4cx\,(x-1)$
$\dfrac{x^2}{4\,(x-1)}$		$ax^2 - 4c\,(x-1)$
5. Same as 2, 3, 4 interchanging x and z.		

$z =$	$a'z^2 + 2b'z + c' =$	$ax^2 + 2bx + c =$
$\dfrac{(z-1)^2}{4z} = \dfrac{4x}{(x-1)^2}$	$a(z-1)^2 + 4bz$	$4ax + b(x-1)^2$
$\dfrac{z^2}{4(z-1)} = \dfrac{4(x-1)}{x^2}$	$az^2 + 4b(z-1)$	$-4a(x-1) + bx^2$
$4z(z-1) = \dfrac{1}{4x(x-1)}$	$4az(z-1) + b$	$a + 4bx(x-1)$
$\dfrac{(z-1)^2}{4z} = 4x(x-1)$	$a(z-1)^2 + 4bz$	$4ax(x-1) + b$
$\dfrac{z^2}{4(z-1)} = -4x(x-1)$	$az^2 - 4b(z-1)$	$4ax(x-1) + b$
$\dfrac{(z-1)^2}{4z} = -\dfrac{4(x-1)}{x^2}$	$a(z-1)^2 + 4bz$	$-4a(x-1) + bx^2$
$4z(z-1) = \dfrac{(x-1)^2}{4x}$	$4az(z-1) + b$	$a(x-1)^2 + 4bx$
$4z(z-1) = -\dfrac{x^2}{4(x-1)}$	$4az(z-1) + b$	$ax^2 - 4b(x-1)$
$\dfrac{4(z-1)}{z^2} = -\dfrac{(x-1)^2}{4x}$	$-4a(z-1) + bz^2$	$a(x-1)^2 + 4bx$

The six functions of the set (1), that is,

$$x, \quad 1-x, \quad \frac{1}{x}, \quad \frac{1}{1-x}, \quad \frac{x}{x-1}, \quad \frac{x-1}{x},$$

form a group; and by operating with the substitutions of this group, and of the like group

$$z, \quad 1-z, \quad \frac{1}{z}, \quad \frac{1}{1-z}, \quad \frac{z}{z-1}, \quad \frac{z-1}{z},$$

upon any value of z in the sets (2), (3), (4), for instance upon $z = \left(\dfrac{x+1}{x-1}\right)^2$, we form all the 18 functions of these sets.

In any one of these sets (2), (3), and (4), comparing two forms (the same or different), for instance in the set (2), writing y for z and then in one form z for x,

$$y = \left(\frac{x+1}{x-1}\right)^2 \text{ and } y = \left(\frac{z+1}{z-1}\right)^2, \text{ whence } \left(\frac{x+1}{x-1}\right)^2 = \left(\frac{z+1}{z-1}\right)^2,$$

or

$$y = \left(\frac{x+1}{x-1}\right)^2 \text{ and } y = \frac{(z+1)^2}{4z}, \text{ whence } \left(\frac{x+1}{x-1}\right)^2 = \frac{(z+1)^2}{4z},$$

we obtain either the equations of the set (1) or those of the sets (8), (9) and (10); and whether we use the set (2), (3) or (4), the only new equations obtained are thus the 9 equations of the sets (8), (9) and (10). These several equations present themselves however in different forms: for instance, instead of the equation

$$\frac{(z-1)^2}{4z} = \frac{4x}{(x-1)^2},$$

we may obtain

$$\frac{(z+1)^2}{4z} = \left(\frac{x+1}{x-1}\right)^2.$$

If, to get rid of this variety of form, we multiply out the denominators, the 9 equations are

$$\begin{aligned}
0 &= x^2z^2 - 2x^2z - 2xz^2 + x^2 - 12xz + z^2 - 2x - 2z + 1,\\
0 &= x^2z^2 - 16xz + 16x + 16z - 16,\\
0 &= 16x^2z^2 - 16x^2z - 16xz^2 + 16xz - 1,\\
0 &= x^2z^2 - 2x^2z + x^2 + 16xz - 16z,\\
0 &= -16xz - z^2 + 2z - 1,\\
0 &= -16x^2 - 16xz + z^2 + 16x,\\
0 &= x^2z^2 - 2xz^2 + 16xz + z^2 - 16x,\\
0 &= 16xz^2 - x^2 - 16xz + 2x + 1,\\
0 &= 16xz^2 + x^2 - 16xz - 16z^2 + 16z.
\end{aligned}$$

These 9 equations are derivable all from any one of them by the changes of the set (1) upon x and z.

Cambridge, 3rd June, 1879.

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712.

A PARTIAL DIFFERENTIAL EQUATION CONNECTED WITH THE SIMPLEST CASE OF ABEL'S THEOREM.

[From the *Report of the British Association for the Advancement of Science*, (1881), pp. 534, 535.]

CONSIDER a given cubic curve cut by a line in the points $(x,\ p)$, $(x_1,\ p_1)$, $(x_2,\ p_2)$; using the first and second points as elements, these determine uniquely the third point. Analytically, the equation of the curve determines p as a function of x, and p_2 as a function of x_1; writing in the equation

$$c_2 = \lambda c_1 + (1-\lambda)c, \quad y_2 = \lambda y_1 + (1-\lambda)y_0,$$

we have c, by a simple equation, and thence p_2; viz. x_2 is found as a function of x, x_1, and of the nine constants of the equation. Hence having the derived equations (in regard to x, x_1) of the first, second, and third orders, we have $(1+2+3+4=)10$ equations from which to eliminate the 6 quantities x_2, considered as a function of x and x_1, thus involves a partial differential equation of the third order, independent of the particular cubic curve.

To obtain this equation it is only necessary to observe that we have, by Abel's theorem,

$$\frac{dx}{X} + \frac{dx_1}{X_1} + \frac{dx_2}{X_2} = 0,$$

where X, X_1, X_2 is a given function of x, and p, that is, of x; X_1 and X_2 are the like functions of x_1 and x_2 respectively. Hence, considering x_2 as a function of x and x_1, we have

$$\frac{dx_2}{dx} = -\frac{X_2}{X}, \quad \frac{dx_2}{dx_1} = -\frac{X_2}{X_1},$$

and consequently

$$\frac{dx_2}{dx_1} \div \frac{dx_2}{dx} = \frac{X}{X_1},$$

where X, X_1 are functions of x_2, x_1 respectively; hence taking the logarithm and differentiating successively with regard to x_1 and x_2, we have

$$\frac{d}{dx_2}\frac{d}{dx_1}\log\left(\frac{dx_2}{dx_1} \div \frac{dx_2}{dx}\right) = 0,$$

which is the required partial differential equation of the third order.

This differential equation has a simple geometrical significance. Consider three consecutive positions of the line meeting the cubic curve in the points 1, 2, 3; $1'$, $2'$, $3'$; $1''$, $2''$, $3''$ respectively; *quæ* equation of the third order, the equation should in effect determine $3''$ by means of the other points. And, in fact, the three positions of the line constitute a cubic curve; the nine points are thus the intersections of two cubic curves, or, say, they are an "annexed" of points; any eight of the points thus determine uniquely the ninth point.

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713.

ADDITION TO MR ROWE'S MEMOIR ON ABEL'S THEOREM.

[From the *Philosophical Transactions of the Royal Society of London*, vol. CLXII., Part III. (1881), pp. 715—758. Received May 27,—Read June 16, 1880.]

In Abel's general theorem y is an irrational function of x determined by an equation $\chi(y) = 0$, or say $\chi(x, y) = 0$; if the order n as regards y, and that is also by him that the sum of any number of the integrals considered may be reduced to a sum of ρ integrals; where ρ is a determinate number depending only on the form of the equation $\chi(x, y) = 0$, and given in his equation (42), [*(Euvres Complètes*, (1881), t. I., p. 157]; viz. if, solving the equation so as to obtain from it developments of y in descending series of powers of x, we have:

$$v_\sigma \rho_\nu, \text{ series each of the form } y = x^{\sigma'_\nu} + \ldots,$$
$$\begin{array}{ll} v_\sigma \rho_2, & y = x^{\sigma'_2} + \ldots, \\ \vdots & \vdots \\ v_\sigma \rho_n, & y = x^{\sigma'_n} + \ldots, \end{array}$$

[2]The second power of a term coefficient: this being a relation $y = x^p + \ldots$, representing the y, different values of y obtained by giving to the radical x^p and to its p values, and the corresponding values in the radicals which enter into representing y, values of y. It is assumed that the equation is such that the meaning is to make each representing y, y_2. In each of the equations the meaning of y is taken positive in the development of y in descending powers of x.

(so that $\sigma = v_1\rho_1 + v_2\rho_2 + \cdots + v_n\rho_n$) then y is a determinate function of x_1, x_2, x_3, x_4, x_5, x_6, $\ldots$, x_n, say ρ.

We have so expressed Abel's ρ in the following form, viz. assuming

$$y = \Sigma a_1\alpha_1\nu_1 + \Sigma a_2\nu_2 \cdots + \Sigma a_n\nu_n - \frac{1}{2}\Sigma a - \frac{1}{2}n + 1,$$

or, what is the same thing, for ρ writing this Σa_1,

$$y = \Sigma a_1\nu_1 + \Sigma a_2\nu_2 + \Sigma a_3\nu_3 + \cdots + \Sigma a_n\nu_n - \frac{1}{2}\Sigma a - \frac{1}{2}n + 1,$$

where in the first sum ν_1 have each of them the values 1, 2, $\ldots$, ρ_1 subject to the condition $i + \nu_1\sigma_1'$; in each of the other sums ν_i and ρ_n are interrelated by the suffix, which has the values 1, 2, etc.

It is a leading result in Riemann's theory of the Abelian integrals that y is the deficiency (Geschlecht) of the curve represented by the equation $\chi(x,\ y) = 0$; and it must consequently be demonstrable *a posteriori* that the foregoing expression for y is in fact = deficiency of curve $\chi(x,\ y) = 0$. I propose to verify this by means of the formulae given in my paper "On the Higher Singularities of a Plane Curve," *Quart. Math. Journ.*, vol. VII., (1866), pp. 212—222, [374].

It is necessary to distinguish between the values of $\frac{m}{n}$ which are $>, =$, and < 1; and to fix the ideas I assume $l = 7$, and

$$\frac{m_1}{n_1},\ \frac{m_2}{n_2},\ \frac{m_3}{n_3},\ \text{each} > 1,$$
$$\frac{m_4}{n_4} = 1;\ \text{say}\ m_4 = n_4 = n,\ \text{and}\ n_4 = n;$$
$$\frac{m_5}{n_5},\ \frac{m_6}{n_6},\ \frac{m_7}{n_7},\ \text{each} < 1,$$

but it will be easily seen that the reasoning is quite general. And now Σ to denote a sum in regard to the first set of suffixes 1, 2, 3, and Σ' to denote a sum in regard to the second set of suffixes 5, 6, 7. The foregoing value of ρ is thus

$$y = \Sigma a\nu + 4 + \Sigma' a\nu.$$

Introducing a third coordinate z for homogeneity, the equation $\chi(x,\ y) = 0$ of the curve will be

$$0 = \left[y^{p-(n-1)},\ y^{n-2}x^a,\ \ldots\right](y,\ z)^{n-2} + \left(y^{n-2},\ y^{n-3}x^a,\ \ldots\right)(y,\ z)^{n-1} + \cdots + (y)\,(y,$$

where it is to be observed that $\chi(x,\ y) = 0$ is the product of ν_i, different ovaries; these divide themselves into n_i

groups, each a product of ρ_i orates; and in each such product the ρ_i coefficients A_i are in general the ρ_i values of a function containing a radical a^{1/ρ_i} and are thus different from each other; it is to when follows in effect assumed in each step that all the a_i, but that all the x_i coefficients, etc. are different from each other.

The remarks apply to the other factors. It applies in particular to the factor $(y,\ z)^{r-1}$, viz. it is assumed that the coefficients A in the radical $z,\ y^{n-2},\ \ldots$ are all of them different from each other. These assumptions as to the leading coefficients really imply Abel's assumption that $\frac{m_1}{n_1}$, $\frac{m_2}{n_2}$ are all of them fractions in their least terms, viz. that $n=1$, since

1 *rotate however for convenience the equation will be considered, by giving to the lea*

In the product of the several infinite series, the terms containing the lowest powers of z, $\frac{1}{n_1}$ is a function in its lowest terms, viz. since $n_1 = 1$.

A curve has singularities (singular points) at infinity, that is, on the line $z=0$; vis. *First*, a singularity at $(x=0, x'=0)$, where the tangent is $x'=0$, and which, writing for convenience $x=1$, is denoted by the function

$$\left(z - \nu^{m/n} x^{1-1/n}\right)^{n-1} = \ldots$$

where observe that the expressed factor indicates r_n branches $\left(z - y^{m/n} x^{1-1/n}\right)^{r_n}$, or say $r_n(m_n - p_n)$ partial branches $z = v^{m/n}\ldots$, that is, $r_n(m_n - p_n)$ partial branches $z = v^{m/n}\ldots$, will in all $r_n(m_n - p_n) + \ldots$, where $r_n(m_n - p_n)$ partial branches at the encompassed factors with the suffixes $1'$ and 4; that is, we can regard the encompassed factors with the suffixes $1'$ and 4, where in the equation

$$z - \nu^{m/n} + \cdots = 0.$$

Secondly, a singularity at $(z=0, y=0)$, where the tangent is $y=0$, and which, writing for convenience $z=1$, is denoted by the function

$$\left(y - z^{m/n} x^{1-1/n}\right)^{n-1} = \ldots$$

[2*] This assumption is virtually made by Abel, (1), [c. p. 157], in considers the

where observe that the expressed factor indicates r_5 branches $\left(y - z^{m/n}x^{1-1/n}\right)^{r_5}$, or say $r_5(n_5 - m_5)$ partial branches $y = z^{m/n}\ldots$, that is, $r_5(n_5 - m_5)$ partial branches $y = z^{m/n}\ldots$, with in all $r_5n_5 + \ldots$, with $r_5n_5 + r_6n_6 + r_7n_7$ partial branches $y = z^{m/n}\ldots$, with the suffixes 5, 6, 7.

Thirdly, suppose at $(x = 0, y = 0)$, where the tangent is $z = 0$, and consequently $r_4n_4 = 0$, partial branches $z = y^0 \cdots +$ similar terms with respect to x, y, z.

It is convenient to assume $r_4 = 2$, so that

$$\begin{aligned} E &= \tfrac{1}{2}(K-1)(K-2) = z, \\ E' &= \tfrac{1}{2}(K'-1)(K'-2) \\ &= \tfrac{1}{2}\Sigma(n-1). \end{aligned}$$

hence

$$E + e' = \tfrac{1}{2}(K-1) + \tfrac{1}{2}\Sigma(n-1).$$

Assuming that there are no other singularities, the deficiency

$$\rho = \tfrac{1}{2}(K-1)(K-2) - E - c'$$

is

$$= \tfrac{1}{2}(K-1)(K-2) - \tfrac{1}{2}KK + \tfrac{1}{2}\Sigma(n-1).$$

This should be equal to the before-mentioned value of ρ; viz. we ought to have

$$\tfrac{1}{2}(K-1)(K-2) - \tfrac{1}{2}KK + \tfrac{1}{2}\Sigma(n-1) = \Sigma a\nu_1 + 4 + \Sigma' a\nu_2 - \Sigma a - \Sigma' a - 2,$$

or, as it will be convenient to write it,

$$M = K - \tfrac{3}{2}K + \tfrac{1}{2}\Sigma(n-1) - 1 = \Sigma a\nu_1 + \Sigma' a\nu_2 - \Sigma' a - 2,$$

which is the equation which ought to be satisfied by the values of N and $\Sigma(n-1)$ calculated, according to the method of my paper, for the foregoing singularities of the curve.

We have as before:

$$K = \Sigma n + \Sigma' n + \theta n.$$

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equation $z^n = y^n \sqrt[m_n]{f(x)}$ as denoting integers; but where it appears that the indices a, b, are to be considered of fixed, and as in the values of degrees with respect to x. He says that if these be assumed as integers in the expression of y in descending powers of x (the leading coefficient being z in regard to y), the singular factor are in $(x,\ y)$ in regard to y in regard to x.

The terms $\Sigma n + \Sigma' n + \theta n$, written at length, is

$$
\begin{aligned}
&= n_1(\nu_1\rho_1 + \nu_2\rho_2 + \nu_3\rho_3) + \cdots + n_2(\nu_1\rho_1 + \nu_2\rho_2 + \nu_3\rho_3)\\
&+ \cdots + n_3(\nu_1\rho_1 + \nu_2\rho_2 + \nu_3\rho_3)\\
&+ \cdots + n_4(\nu_1\rho_1 + \nu_2\rho_2 + \nu_3\rho_3)\\
&+ \cdots + n_5(\nu_1\rho_1 + \nu_2\rho_2 + \nu_3\rho_3)\\
&+ \cdots + n_6(\nu_1\rho_1 + \nu_2\rho_2 + \nu_3\rho_3)\\
&+ \cdots + n_7(\nu_1\rho_1 + \nu_2\rho_2 + \nu_3\rho_3).
\end{aligned}
$$

which is

$$= \Sigma\nu_1 n_2 + \Sigma' n_2(2\rho_2 + 2'\rho_1) + \Sigma\Sigma' + \Sigma'\Sigma + \Sigma a_1 a_2 a_3,$$

We have moreover

$$\Sigma'\rho_1 = \Sigma a_1\rho_1 + (\theta + \Sigma' a_1)\rho_1 = \Sigma a_1 + \Sigma a_2 + \Sigma' a_1\rho_1,$$

each branch $\left(z - y^{m/n}x^{1-1/n}\right)^{n-1}$ gives $a = m_1\rho_1$, and the value of $\Sigma(\kappa - 1)$ for this singularity is

$$n_1(m_1 - n_1 - 1) + r_1 n_1(m_1 - 1) = r_1 n_1 \rho_1 - n_1\rho_1.$$

which is

$$= \Sigma a\nu_1 - \Sigma a\rho_1.$$

each branch $\left(y - z^{m/n}x^{1-1/n}\right)^{n-1}$ gives $\sigma = m_n\rho_n$, and the value of $\Sigma(\kappa - 1)$ for this singularity is

$$n_4(m_4 - n_4 - 1) + r_4 n_4(m_4 - 1) = r_4 n_4 \rho_4 - n_4\rho_4,$$

which is

$$= \Sigma'\nu_2 - \Sigma'\nu_4 - \Sigma'\nu_1.$$

For each of the θ singularities

$$\left(y - z^{m/n}x^{1-1/n}\right)^{n-1}.$$

we have $a = n$, and the value of $\Sigma(\kappa - 1)$ is $r = \theta(\kappa - 1)$; this is $= 0$ for the value $\lambda = 1$, which is physically attributed to λ.

According to the theory explained in my paper above referred to, the several singularities not together equivalent to a certain number K of κ^* nodes and cusps; viz. we have

$$K = \Sigma n - \Sigma a + \tfrac{1}{2}\Sigma(\kappa - 1).$$

This should be equal to the before-mentioned value of ρ; viz. we ought to have

$$(K-1)(K-2) - 2(K + \Sigma n + \Sigma' n + \theta n) = 2\Sigma a\nu_1 + 4 + 2\Sigma a\nu_2 - 2\Sigma a - 2\Sigma' a - 4,$$

or to it will be convenient to write

$$M - K - \Sigma K + 1 + \Sigma(n-1) = \Sigma a\nu_1 + \Sigma' a\nu_2 - \Sigma a - \Sigma' a - 2,$$

which is the equation which ought to be satisfied by the values of N and $\Sigma(n-1)$ calculated, according to the method of my paper, for the foregoing singularities of the curve.

We have as before:

$$K = \Sigma n + \Sigma' n + \theta n.$$

The term $\Sigma n + \Sigma' n + \theta n$, written at length, is

$$= \Sigma \nu_1 n_1 + \Sigma' \nu_2 n_2 + \theta n,$$

or, reducing,

$$M = (\Sigma m)^2 - \Sigma ma - \Sigma m\rho_1 - \Sigma m\rho_2 + (\Sigma' m)^2 - \Sigma' m_2 a - \Sigma' m_2\rho_2$$
$$+(\Sigma' m)^2 - \Sigma' m\rho_1 - \Sigma ma + 2\Sigma' m_2 a_1 + 2\Sigma'\Sigma a_1 a_2;$$

and it is to be shown that the two lines of this expression are in fact the values of M belonging to the singularities

$$\left(z - y^{m/n} x^{1-1/n}\right)^{n-1}, \ \ldots, \text{ and } \left(y - z^{m/n} x^{1-1/n}\right)^{n-1}, \ \ldots,$$

respectively. We assume $\lambda = 1$, and there is then no singularity $(y-z)^{n-1}$.

I recall that, considering the several partial branches which meet in a singular point, M denotes the sum of the number of the intersections of each partial branch by every other partial branch: so that for each pair of partial branches the intersections are to be counted twice. Supposing that the tangent is $x = 0$, and that, for any two branches we have $z = Az^x + \ldots$ (x where p_x, p_x are each up to greater than 1), then if $p_x = p_x$, and $A_x = A_x$, $A_x = A_x$ where A_x, A_x are not the partial branches which have here always could be regards the cases about to be considered), then the number of intersections is taken to be $-$; and if $p_x > p_x$, and the number of intersections (taken to be $\equiv p_x$) viz. in the case of as a unequal exponents, it is equal to the smaller exponents.

Consider now the singularity $\left(z - y^{m/n} x^{1-1/n}\right)^{n-1}$, and first the intersections of a partial branch $z = y^{m/n} \ldots$ by each of the remaining $n_1(m_1 - p_1) - 1$ partial branches of the same set: the number of intersections with any one of these is $= \frac{m_1}{n_1}[n_1(m_1-p_1)-1]$. But we obtain this same number from each of the $n_1(m_1 - p_1)$ partial branches, and thus the whole number is

$$n_1\left(m_1 - p_1\right)\frac{m_1}{n_1}\left[n_1(m_1-p_1)-1\right] = n_1 m_1\left[n_1(m_1-p_1)-1\right].$$

Taking account of the other sets, and the whole number of intersections is

$$= \Sigma n^2 a - \Sigma' n a^2.$$

Observe now that $\frac{m_1}{n_1} > \frac{m_2}{n_2}$, that is, $\frac{n_1}{n_2} < \frac{m_1}{m_2}$, and that, these being each < 1, we have $1 - \frac{m_2}{n_2} > 1 - \frac{m_1}{n_1}$, that is, $\frac{n_2 - m_2}{n_2} > \frac{n_1 - m_1}{n_1}$, and we thus have

$$\frac{m_1}{n_1} < \frac{m_2}{n_2}, \text{ and } \left(z - y^{m/n} x^{1-1/n}\right)^{n_1(m_1 - p_1)},$$

Considering now the intersections of partial branches of the two sets

$$\left(z - y^{m/n} x^{1-1/n}\right)^{n_1(m_1 - p_1)} \text{ and } \left(z - y^{m/n} x^{1-1/n}\right)^{n_2(m_2 - p_2)},$$

respectively, a partial branch $z - z^{m/n} \ldots$ gives with each partial branch of the other set a number $= \frac{m_2}{n_2}$; and in this way taking each partial branch of each set, the number is

$$n_1(m_1 - p_1) \cdot n_2(m_2 - p_2) \cdot \frac{m_2}{n_2} = n_2 m_2 n_1 (m_1 - p_1),$$

and thus for all the sets the number is

$$= 2\Sigma n_2 m_2 \nu_2 m_2,$$

where in the first sum the Σ' refers to each pair of values of the suffixes. But the intersections are to be taken twice; the number thus is

$$= 2\Sigma m_1 m_2 \nu_1 \nu_2,$$

Adding the foregoing number

$$= \Sigma' n^2 a - \Sigma' n a^2;$$

the whole number for the singularity in question is

$$= (\Sigma m_1)^2 - \Sigma' m_2 \rho_2 - \Sigma' m_2 \rho_2 \cdot$$

Similarly for the singularity $\left(y - z^{m/n} x^{1-1/n}\right)^{n-1} \ldots$, taking each set with itself, the number of intersections is

$$n_5 m_5 \left[n_5(m_5 - p_5) - 1\right] + n_6 m_6 \left[n_6(m_6 - p_6) - 1\right] + n_7 m_7 \left[n_7(m_7 - p_7) - 1\right],$$

which is

$$= \Sigma' m_2^2 a - \Sigma' m_2 \rho_2 - \Sigma m_1 a.$$

We have here $\frac{m_5}{n_5} > \frac{m_6}{n_6}$, each of these being < 1, we have $1-\frac{m_5}{n_5} < 1-\frac{m_6}{n_6}$, that is, $\frac{n_5-m_5}{n_5} < \frac{n_6-m_6}{n_6}$, and so

$$\frac{m_5}{n_5} > \frac{m_6}{n_6}, \frac{n_5-m_5}{n_5} < \frac{n_6-m_6}{n_6};$$

Hence considering the two sets

$$\left(y - z^{m/n}x^{1-1/n}\right)^{n_5(m_5-p_5)} \text{ and } \left(y - z^{m/n}x^{1-1/n}\right)^{n_6(m_6-p_6)};$$

a partial branch of the first set given with a partial branch of the second set a number of intersections; and the number then obtained is

$$n_5(m_5-p_5)\cdot n_6(m_6-p_6)\cdot \frac{m_6}{n_6} = n_6m_6n_5(m_5-p_5),$$

For all the sets the number is

$$n_5m_5\left[n_5(m_5-p_5)+n_6(m_6-p_6)\right] - n_5n_6(m_6-m_5)$$

or taking these from, the number is

$$2\Sigma' m_2m_2\nu_2\nu_2.$$

where in the first sum the Σ' refers to each pair of suffixes. Adding the foregoing value

$$\Sigma' m^2 a - \Sigma' m_2\rho_2 - \Sigma m_1 a,$$

the whole number for the singularity in question is

$$= (\Sigma' m_2)^2 - \Sigma m_1 a - \Sigma' m_2\rho_2.$$

and the proof is thus completed.

Referring to the first-note (ante, p. 31), I remark that the theorem "ρ–deficiency, is absolute, and applies to a curve with any singularities whatever: in a curve which has singularities not taken account of in Abel's theory, the "perhaps see particulars" which we have a give as he expresses it, taken account of, viz. for that Σa is, in the value of ρ as denoted by Abel's formula; and the actual deficiency will be = Abel's ρ, = such diminution, that is, it will be = true values of ρ.

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714.

VARIOUS NOTES.

[From the *Messenger of Mathematics*, vol. VII. (1878), pp. 69, 115, 124, 125.]

An Identity.

THE following remarkable identity is given under a slightly different form by Gauss, *Werke*, t. III., p. 419.

$$\begin{aligned} 1 &= \left(\frac{1}{x}\right)^2 x + \left(\frac{1}{1+x}\right)^2 x + \left(\frac{1+x}{1+x+x^2}\right)^2 x + \ldots \\ &= \left[1 + \left(\frac{1}{x}\right)^2 x + \left(\frac{1}{1+x}\right)^2 x + \left(\frac{1+x}{1+x+x^2}\right)^2 x + \ldots\right]^2. \end{aligned}$$

On two related quadric functions.

Assume

$$\begin{aligned} x &= x^2(1-x) + x(1-x^2-\alpha x), \\ y &= y(1-x) + y(x^2-1-\alpha x). \end{aligned}$$

then

$$\begin{aligned} \left(\frac{x^2}{x-a}\right) &= \frac{x^2}{(x-a)^2}x, \\ \left(\frac{x^2}{x-a}\right) &= \frac{x^2}{(x-a)^2}y. \end{aligned}$$

In the like of those for a ratio $\frac{p}{1-p}$, then

$$\left[\frac{ax^2(x-y)}{x^2-y^2(x-a)}\right] = \frac{ax^2(x-y)^2}{(x-a)^2-y^2(x-a)^2}\,(x).$$

A Trigonometrical Identity.

$$
\begin{aligned}
&\cos(b-c)\sin(b+c-a)+\cos a\sin(a+b \\
&+\cos(c-a)\sin(c+a+b)+\cos b\sin(b+c \\
&+\cos(a-b)\sin(a+b+c)+\cos c\sin(c+a \\
&=\cos a\cos b\cos(b+c)+\cos b\cos c\cos(c+a)+\cos c\cos a\cos(a+b)+\cos a\cos b\cos c
\end{aligned}
$$

Extract from a Letter.

"I wish to construct a correspondence such as

$$(x+iy)^2(x+iy)=X+iY,$$

or, say, for greater convenience

$$4(x+iy)^2(x+iy)=X+iY,$$

viz. if

$$x+iy=\cos\alpha,$$

then

$$X+iY=\cos\alpha\beta.$$

Suppose α, in a value of α corresponding to a given value of $X+iY$, then the three values of $x+iy$ are of course $\cos\alpha$, $\cos\left(\alpha+\frac{2\pi}{3}\right)$, but I am afraid that the calculation of α, even with red and card tables, would be very laborious. Writing

$$X+iY=R(\cos\Theta+i\sin\Theta),$$

the intervals for Θ might be 3°, 30° or even 15°; those of R, say 0.1 from 0 to 2, and then 0.2 up to 4; and 2 places of decimals would be quite sufficient; but even this would probably involve a great mass of calculation.

It has occurred to me that perhaps a geometrical solution might be found for the equation $X+iY=\cos\beta$."

October 21, 1877.

. . . .

715.

NOTE ON A SYSTEM OF ALGEBRAICAL EQUATIONS.

[From the *Messenger of Mathematics*, vol. VII. (1878), pp. 17, 18.]

Assume

$$x + y + z = P,$$
$$yz + zx + xy = Q,$$
$$xyz = R.$$

$$A = (nx + y - z) - a^n(nx + P),$$
$$B = y(ax + Q) - b^n(my + Q),$$
$$C = z(ax + Q) - c^n(nz + P),$$
$$\Theta = -mnQ + nP.$$

Then

$$\begin{aligned}(ax + P)B - (my + P)A &= (ax + P)y(ax + Q) - a^n(my + Q) \\ &= (ax + P)(ax + Q) - (my + Q)(my + P) \\ &= ax(my + Q - ay) - yz(my + Q - P) + yz(ax + P) - y(\\ &= mny(ax - y) - P(ax - y) \\ &= (ax - y)(mny - P) = (ax - y)\Theta,\end{aligned}$$

whence, identically,

$$(ax + P)B - (my + P)A = (ay - y)\Theta,$$
$$(my + P)C - (nx + P)B = (by - z)\Theta,$$
$$(ax + P)A - (nx + P)C = (cz - x)\Theta.$$

Hence any two of the equations $A = 0$, $B = 0$, $C = 0$ imply the third equation.

We have

$$\begin{aligned} A &= x\left[(n+1)x+y+z\right] - a^n\left[(n+1)x+y+z\right] \\ &= (x-a^n)\left[(n+1)x+y+z\right]; \end{aligned}$$

and similarly for B and C. The three equations therefore are

$$\begin{aligned} \frac{x}{x^n-a^n} &= \frac{y+z}{(n+1)x^n+(n-1)a^n}, \\ \frac{y}{y^n-b^n} &= \frac{x+z}{(n+1)y^n+(n-1)b^n}, \\ \frac{z}{z^n-c^n} &= \frac{x+y}{(n+1)z^n+(n-1)c^n}, \end{aligned}$$

and any two of these equations imply the third equation.

. . . .

716.

AN ILLUSTRATION OF THE THEORY OF THE ϑ-FUNCTIONS.

[From the *Messenger of Mathematics*, vol. VII. (1878), pp. 27—32.]

If X be a given quartic function of x, and if a, as for convenience a constant multiple as, be the value of the integral $\int \frac{dx}{\sqrt{X}}$, taken from a given inferior limit to the superior limit x; then, conversely, x is expressible as a function of a, viz. it is expressible in terms of ϑ-functions of a, where ϑa, or say $\vartheta(a,\ \mathfrak{Q})$, is the definition of which, depending, depend in general by definition on the sum of a series of exponentials of a, and it is possible from the assumed equation $a = \int \frac{dx}{\sqrt{X}}$, and the definition of ϑa, to attain by general theory the actual formulae for the determination of a as such a function of a.

I propose here to obtain these formula, in the case where X is a product of real factors, in a less suitable manner, viz. by reducing the integral $\int \frac{dx}{\sqrt{X}}$ by a linear substitution in the form of an elliptic integral, the object being to obtain for the same in question the actual formulae as a function of ϑ-functions of u.

The definition of ϑu, when the parameter is expressed, $\vartheta(u,\ \mathfrak{Q})$ is

$$\vartheta u = 2\Sigma(-)^p q^{p+\frac{1}{2}(p+1)^2} \sin(p+\tfrac{1}{2})u,$$

where u and p are negative integer values, zero included, from $-\infty$ to $+\infty$ (that is, from -2 to 2, 0 to 2, etc.); the parameter $\mathfrak{Q}$, or Ω' (of imaginary) no real part, must be positive.

Evidently ϑu is an even function: $\vartheta(-u) = \vartheta u$. Moreover, it is at once seen that we have

$$\vartheta(u+\pi) = \vartheta u, \quad \vartheta(u+i\Pi) = -q^{-1}e^{-iu}\vartheta u,$$

where q is

$$\vartheta(u+\pi+i\Pi).$$

where as u and v are positive or negative integers, the product of ϑu by an exponential factor, or say simply that it is a multiple of ϑu.

Writing $u = \frac{\pi}{2}\xi$, we have $\vartheta v = \frac{1}{2}\sqrt{2K}\vartheta\xi$, that is,

$$\vartheta\left(\tfrac{\pi}{2}\xi\right) = \xi^{nK},$$

and therefore also

$$\vartheta(\sin\xi + v) = \xi^{nK}\vartheta v.$$

The above properties are general, but if θ be real, then K, K', q being as in Jacobi (consequently q being real, positive, and less than 1, and K and K' real and positive), and assuming $\mathfrak{Q} = \frac{K'}{K}, \pi$, we have in the same thing,

$$q(= e^{-\pi\Omega/\mathfrak{Q}}) = e^{-\pi K'/K},$$

the function ϑv is given (*ut* ante) of Jacobi's ϑv by the equation $\vartheta u = q^{\frac{1}{4}}\vartheta x$.

That is, if Ω is the value of the case $u = v$.

We have, as before, the system of three equations

$$\operatorname{sn} u = \frac{1}{\sqrt{k}}\frac{\vartheta(u,\ k)}{\vartheta_0(u,\ k)}, \quad \operatorname{cn} u = \sqrt{\frac{k'}{k}}\,\frac{\vartheta_0(u,\ k)}{\vartheta_0(u,\ k)}, \quad \operatorname{dn} u = \sqrt{k'}\,\frac{\vartheta_0(u,\ k)}{\vartheta_0(u,\ k)}.$$

Consider now the integral

$$\int_a^b \frac{dx}{\sqrt{(x-\alpha,\ x-\beta,\ x-\gamma,\ x-\delta)}} = \int_a^b \frac{dx}{\sqrt{X}} \text{ suppose,}$$

where α, β, γ, δ are taken to be real, and the four factors regarded as fixed magnitudes, viz. it is assumed that $b-a$ is the value of the radical at the inferior limit is positive, and as b is between β and δ, X is positive, and a fortiori then $\sqrt{X}$ denotes the positive value of the radical; but if α is between β and δ, then X is negative, and we must that the sign of $\sqrt{X}$ is taken so that $\sqrt{X}$ is equal to a positive multiple of i, and this being so the integral is taken from the inferior limit to the superior limit x, which we may.

Take a linear function of y, such that for

$$x = a,\ b,\ c,\ d,$$

$$y = 0,\ 1,\ \tfrac{1}{k^2},\ \infty,\ \text{respectively},$$

so that, x increasing continuously from a to d, y will increase continuously from 0 to ∞. We have

$$y = \frac{c-a}{c-b} \cdot \frac{x-b}{x-a},$$

$$y = \frac{c-a}{c-d} \cdot \frac{x-d}{x-a},$$

$$1 - y = \frac{b-c}{c-b} \cdot \frac{c-a}{c-a},$$

$$1 - k'y = \frac{b-c}{c-b} \cdot \frac{c-a}{c-a},$$

and, thence,

$$\sqrt{y \cdot 1 - y \cdot 1 - k'y} = \frac{(a-c)\sqrt{X}}{(c-a)\sqrt{(x-a)\cdot(x-a)}},$$

where $\sqrt{\frac{(c-d)}{(c-d)}}$ is taken to be positive, and the signs of $\sqrt{X}$ is fixed as above. For y between 0 and 1, $y = \frac{1}{x^2}$, $1-y$, $1-k'y$ will be positive, and $\sqrt{y}$ being between 1 and $\frac{1}{x^2}$, $1-y$ will be negative, but $1-k'y$ will be positive, viz. the radical is then taken as a positive multiple of i.

We have moreover

$$dy = \frac{c-a}{c-b} \cdot \frac{dx}{(x-a)^2},$$

and therefore

$$\frac{dy}{\sqrt{y \cdot 1 - y \cdot 1 - k'y}} = \sqrt{(c-d)\cdot(c-a)} \cdot \frac{dx}{\sqrt{X}},$$

where $\sqrt{(c-d)\cdot(c-a)}$ is positive; or, say,

$$\int \frac{dy}{\sqrt{y \cdot 1 - y \cdot 1 - k'y}} = \sqrt{(c-d)\cdot(c-a)} \cdot \int \frac{dx}{\sqrt{X}}.$$

Hence, writing $y = \sin^2 u$, we have

$$\Pi u = \sqrt{(c-d)\cdot(c-a)}\ \int_a^b \frac{dx}{\sqrt{X}},$$

and it is to be further noticed that to

$$x = a,\ b,\ c,\ d,$$

correspond

$$\Pi u = 0,\ \tfrac{1}{2}K,\ \tfrac{1}{2}K + \tfrac{1}{2}K'.$$

or we may say

$$u = 0,\ K,\ K + iK',\ 2K + iK'.$$

Writing for shortness

$$\frac{1}{\Pi} = \sqrt{(c-d)(c-a)} = u,$$

we have

$$\sin u = \int_a^b \frac{dx}{\sqrt{X}},$$

and moreover

$$\sin(K + iK') = \int_a^b \frac{dx}{\sqrt{X}},$$

$$n(K + iK') = \int_b^c \frac{dx}{\sqrt{X}},$$

$$n(2K + iK') = \int_c^d \frac{dx}{\sqrt{X}},$$

or if for a moment we write

$$\int_a^b \frac{dx}{\sqrt{X}} = A,\ \&c.$$

then these equations are

$$\begin{aligned} nK &= A, \\ n(K + iK') &= A + B, \\ n(2K + iK') &= A + B + C. \end{aligned}$$

Hence $B+C=2A=B$, or $A=B-C$, or $B-C+D=0$, that is,

$$\int_a^b \frac{dx}{\sqrt{X}} = \int_b^c \frac{dx}{\sqrt{X}},$$

where observe as before that $x=a$ to $x=b$, or $x=c$ to $x=d$, X is positive, and the radical $\sqrt{X}$ is taken to be positive.

We have also

$$nK = A = \int_a^b \frac{dx}{\sqrt{X}},$$

$$niK' = C - B = \int_a^b \frac{dx}{\sqrt{X}}.$$

when, as before, from b to c, X is negative, and the sign of the radical is such that $\sqrt{X}$ is a positive multiple of i; the last formula may be more conveniently written

$$niK' = \int_b^c \frac{dx}{\sqrt{-X}},$$

where, here b to c, $-X$ is positive, and $\sqrt{(-X)}$ is to be taken as positive.

Collecting the results, we have

$$\int_a^b \frac{dx}{\sqrt{X}} = \frac{nK}{\sqrt{(c-d)(c-a)}},\ b \le a \le b,\ \ldots b \le b \le b \le a,$$

and also

$$k = \frac{c-b}{a-b} \cdot \frac{d-a}{c-a},$$

and thence conversely

$$k' = \frac{a+d-b-c+(b-c-a+d)\,\sin^2 n}{a+b-c-d+(b-c+a-d)\,\sin^2 n},$$

or, what is the same thing,

$$nu^2 = \frac{b-d\ \cdot\ x-a}{b-a\ \cdot\ x-d},$$
$$nu^2 = \frac{c-d\ \cdot\ x-a}{c-a\ \cdot\ x-d},$$
$$nu^2 = \frac{d-a\ \cdot\ x-b}{a-b\ \cdot\ x-d},$$
$$nu^2 = \frac{b-d\ \cdot\ x-a}{b-a\ \cdot\ x-d},$$

where, in place of the elliptic functions nu we substitute their ϑ-values, it will be recollected that Ω, the parameter of the ϑ-functions, has the value

$$\frac{K'}{K}\pi = \left(= \frac{\Omega'}{\Omega}\pi\right) = \pi\frac{\int_a^b \frac{dx}{\sqrt{-X}}}{\int_a^b \frac{dx}{\sqrt{X}}},$$

and, so before,

$$K = \int_a^b \frac{dx}{\sqrt{X}},$$

Hence, finally, a, b, K, K', Ω denoting given functions of a, b, c, d, as above

$$\int_a^b \frac{dx}{\sqrt{X}} = nu,$$

we have severally

$$\begin{array}{ll}
\dfrac{b-d}{c-d} \cdot \dfrac{a-c}{c-a} = \dfrac{1}{k^2} \, \dfrac{\vartheta_1^2 u}{\vartheta_2^2 u}, & \qquad\qquad\qquad\quad \dfrac{\vartheta_3^2 u}{\vartheta_2^2 u}, \\[2ex]
\dfrac{d-a}{d-b} \cdot \dfrac{b-c}{a-c} = \dfrac{k'^2}{k} \, \dfrac{\vartheta_2^2 u}{\vartheta_3^2 u}, & \qquad\qquad \dfrac{\vartheta_1^2 u}{\vartheta_2^2 u} + \tfrac{1}{2}K', \\[2ex]
\dfrac{c-a}{d-a} \cdot \dfrac{b-d}{b-c} = k'^2 \, \dfrac{\vartheta_4^2 u}{\vartheta_3^2 u}, &
\end{array}$$

which are the formulae in question.

The problem is to obtain these (and this in the more general case where a, b, c, d have any given (real or imaginary) values directly from the assumed equation

$$\int_a^b \frac{dx}{\sqrt{X}} = nu,$$

and from the foregoing definition of the function ϑu.

It may be recalled that the function ϑu is a doubly infinite product

$$\vartheta u = \Pi\Pi\left[1 + \frac{u}{m\pi + (n+\frac{1}{2})i\Pi}\right]$$

as and a positive or negative integers from $-\infty$ to $+\infty$; I purposely omit all further explanations as to limits, or what is the same thing,

$$\vartheta\frac{\pi}{2K} = \Pi\Pi\left[1 + \frac{u}{2mK + (2n+1)iK'}\right]$$

and consequently that, disregarding constant and exponential factors, the foregoing expressions of

$$\frac{b-d}{c-d}\cdot\frac{a-c}{c-a}, \quad \frac{d-a}{d-b}\cdot\frac{b-c}{a-c}, \quad \frac{c-a}{d-a}\cdot\frac{b-d}{b-c},$$

are the squares of the expressions

$$\frac{X}{Y}, \ \frac{Y}{Z}, \ \frac{Z}{W}, \quad \text{where } X,\ Y,\ Z,\ W \text{ are respectively of the form}$$

$$n\Pi\Pi\left[1 + \frac{u}{(2n+1)iK'}\right], \quad \Pi\Pi\left[1 + \frac{u}{(2m+1)iK'}\right],$$

$$\Pi\Pi\left[1 + \frac{u}{2mK + iK'}\right], \quad \Pi\Pi\left[1 + \frac{u}{(2m+1)K + iK'}\right],$$

where $(m, n = 2mK + 2niK', \&c.$, the *smha* over the m or the n denotes that the $2m$ or the $2n$ as the case may be) is to be changed into $2m+1$ or $2n+1$. But this is a transcription which has apparently no application to the ϑ-functions of more than one variable.

717.

ON THE TRIPLE THETA-FUNCTIONS.

[From the *Messenger of Mathematics*, vol. VII. (1878), pp. 48—50.]

As a specimen of mathematical notation, viz. of the notation which appears to me the easiest to read, and also to print, I give the definition and demonstration of the fundamental properties of the triple theta-functions.

Definition.

$$\vartheta(U,\ V,\ W) = \Sigma \exp \Theta,$$

where

$$\Theta = (A,\ B,\ C,\ F,\ G,\ H)(l,\ m,\ n)^2 + (U,\ V,\ W)(l,\ m,\ n).$$

l, denoting the sum in regard to all positive and negative integer values from $-\infty$ to $+\infty$ (zero included) of l, m, n respectively.

$\vartheta(U,\ V,\ W)$ is considered as a function of the arguments $(U,\ V,\ W)$, and it depends also on the parameters $(A,\ B,\ C,\ F,\ G,\ H)$.

First Property. $\vartheta(U,\ V,\ W) = 0$, for

$$U = \tfrac{1}{2}\pi i + (A,\ H,\ G)(\alpha,\ \beta,\ \gamma),$$

$$V = \tfrac{1}{2}\pi i + (H,\ B,\ F)(\alpha,\ \beta,\ \gamma),$$

$$W = \tfrac{1}{2}\pi i + (G,\ F,\ C)(\alpha,\ \beta,\ \gamma),$$

α, β, γ, δ, ϵ, ζ being any positive or negative integer numbers, such that as $\alpha + \beta + \gamma =$ odd number.

Demonstration. It is only necessary to show that to each term of ϑ there corresponds a second term, such that the inclusion of the two exponentials differ by an odd multiple of πi.

Taking l, m, n the integers which belong to the one term, those belonging to the other term are

$$-(l+a),\ -(m+\beta),\ -(n+\gamma).$$

(where observe that one at least of the numbers α, β, γ being odd, this system of values is not in any case identical with l, m, n). The two exponents are

$$\Theta_i = (A,\ B,\ C,\ F,\ G,\ H)(l,\ m,\ n)^2 + (U,\ V,\ W)(l,\ m,\ n),$$

and

$$\Theta_i' = (A,\ B,\ C,\ F,\ G,\ H)(l{+}a,\ m{+}\beta,\ n{+}\gamma)^2 - (U,\ V,\ W)(l{+}\alpha,\ m{+}\beta,\ n{+}\gamma),$$

viz. the value of Θ' is

$$= (A,\ B,\ C,\ F,\ G,\ H)(l,\ m,\ n)^2{+}2(A,\ H,\ G)(l\alpha,\ \beta,\ \gamma){+}2(H,\ B,\ F)(m\alpha,\ \beta,\ \gamma){+}2($$

$$+(A,\ B,\ C,\ F,\ G,\ H)(a,\ \beta,\ \gamma)^2-(U,\ V,\ W)(l,\ m,\ n)-(U,\ V,\ W)(\alpha,\ \beta,\ \gamma),$$

and we then have

$$\begin{aligned}\Theta' - \Theta = {} & -2(A,\ B,\ C,\ F,\ G,\ H)(l,\ m,\ n)^2 + 2(A,\ H,\ G)(\alpha,\ \beta,\ \gamma)l + 2(H,\ B,\ F \\ & + 2(G,\ F,\ C)(\alpha,\ \beta,\ \gamma)n + (A,\ B,\ C,\ F,\ G,\ H)(a,\ \beta,\ \gamma)^2 \\ & - 2(U,\ V,\ W)(l,\ m,\ n) - (U,\ V,\ W)(\alpha,\ \beta,\ \gamma).\end{aligned}$$

which proves the theorem.

As to the notation, remark that, after $(A,\ B,\ C,\ F,\ G,\ H)$ has been written out in full, we may instead of $((A,\ B,\ C,\ F,\ G,\ H)(\alpha,\ \beta,\ \gamma)^2$ write

$$(\alpha,\ \beta,\ \gamma)^2,$$

and that we may use the like abbreviation

$$(A,\ \ldots)(\alpha,\ \beta,\ \gamma) \text{ for } (A,\ H,\ G)(\alpha,\ \beta,\ \gamma),\ \&c.,$$

$$(H,\ \ldots)(\alpha,\ \beta,\ \gamma) \text{ for } (H,\ B,\ F)(\alpha,\ \beta,\ \gamma),\ \&c.,$$

$$(G,\ \ldots)(\alpha,\ \beta,\ \gamma) \text{ for } (G,\ F,\ C)(\alpha,\ \beta,\ \gamma),\ \&c.$$

These are not only abbreviations, but they make the formula actually clearer, as bringing them into a similar compact; and I accordingly use them in the demonstration which follows.

Second Property. If U, V, W, denote

$$U' = \pi i + (A,\ H,\ G)(\alpha,\ \beta,\ \gamma),$$
$$V' = \pi i + (H,\ B,\ F)(\alpha,\ \beta,\ \gamma),$$
$$W' = \pi i + (G,\ F,\ C)(\alpha,\ \beta,\ \gamma),$$

respectively, where α, β, γ, δ, ϵ, ζ are any positive or negative integers (zero values admissible), then

$$\vartheta(U{+}u,\ V{+}v,\ W{+}w) = \exp\left[-2(\alpha U + \beta V + \gamma W) - (\alpha,\ \beta,\ \gamma)^2\right]\ \vartheta(U,\ V,\ W),$$

or, say

$$= \exp\left[-(A,\ \ldots)(\alpha,\ \beta,\ \gamma)^2 - 2(U,\ V,\ W)(\alpha,\ \beta,\ \gamma) - (\alpha,\ \beta,\ \gamma)^2\right]\vartheta(U,\ V,\ W).$$

Demonstration. Writing $\vartheta(U,\ V,\ W) = \Sigma \exp \Theta$, then in the expression of Θ, we may in place of l, m, n write $-l-\alpha$, $-m-\beta$, $-n-\gamma$; we thus obtain

$$\Theta_l = (A,\ \ldots)(l,\ m,\ n)^2{+}2(A,\ \ldots)(l,\ m,\ n)(a,\ \beta,\ \gamma){+}(A,\ \ldots)(\alpha,\ \beta,\ \gamma)^2$$
$$+(U,\ V,\ W)(l,\ m,\ n) + (U,\ V,\ W)(\alpha,\ \beta,\ \gamma),$$

which is

$$= (A,\ \ldots)(l,\ m,\ n)^2$$
$$+2(\alpha,\ \beta,\ \gamma)(\alpha U{+}\beta V{+}\gamma W){-}2(A,\ \ldots)(\alpha,\ \beta,\ \gamma){+}\Sigma'(A,\ \ldots)(\alpha,\ \beta,\ \gamma)^2{-}\Sigma a^2(A,\ \ldots)$$
$$-2(\alpha,\ \beta,\ \gamma)^2 + 2\alpha,\ \beta,\ \gamma) + (A,\ \ldots)(\alpha,\ \beta,\ \gamma)^2 - 2(A,\ \ldots)(\alpha,\ \beta,\ \gamma)$$
$$+(U,\ V,\ W)(l,\ m,\ n) + (U,\ V,\ W)(\alpha,\ \beta,\ \gamma),$$

which is

$$= (A,\ \ldots)(l,\ m,\ n)^2$$
$$+\Sigma'(A,\ \ldots)(\alpha,\ \beta,\ \gamma)^2{+}\Sigma a^2(A,\ \ldots)(\alpha,\ \beta,\ \gamma){+}\Sigma\alpha\beta(A,\ \ldots)(\alpha,\ \beta,\ \gamma)^2$$
$$+(U,\ V,\ W)(l,\ m,\ n) + \Sigma(A,\ \ldots)(\alpha,\ \beta,\ \gamma)$$

Hence, rejecting the last line, which (as on can readily verify) has the exponential multiplied as the case may be, $\vartheta(U,\ V,\ W)$ as $\vartheta(U,\ V,\ W)$ multiplied by the factor

$$\exp\left[-(A,\ \ldots)(\alpha,\ \beta,\ \gamma)^2 - 2(U,\ V,\ W)(\alpha,\ \beta,\ \gamma) - (\alpha,\ \beta,\ \gamma)^2\right],$$

which is the theorem in question.

In many cases a formula, which belongs to an indefinite number of k letters, is most easily intelligible when written out for three letters, but it is sometimes convenient to speak of the letters l, m, $\ldots$ or even the letter l, and to write out the formula accordingly.

In fact, writing herein $\alpha = \gamma = 0$, that is, $a = \gamma = 0$, the right-hand side becomes a^2 and the arts on the left-hand side are $a = B$, $A = B$, $a = A = I$, which suppose any four sum the sum of which is $= 0$.

Writing in the last-mentioned equation α, β, γ for the a's of A, B, γ, δ respectively, the equation becomes

$$\begin{aligned}
P &= \beta P, \\
Q &= 1 - a^2 + \beta^2 + By^2, \\
R &= 1 - By^2 - 2\alpha + By^2, \\
S &= 1 - By^2.
\end{aligned}$$

L, M, N as the like functions α, β, γ, δ respectively, A, B, C, D being the same as before. And in fact

$$\begin{aligned}
-L\,Q &= -y^2(\alpha^2 + \alpha^2) + \beta^2 + (\beta^2 - \alpha^2) + B(y^2 - x^2) + B(\beta - \alpha) + \dots, \\
&= \frac{1}{B}\left[\beta^2 - (\beta - \alpha)^2 + By^2 - By^2\right] + B(y^2 - x^2) + \dots, \\
&= \frac{1}{B}\left[\beta^2(x^2 - y^2) + B(y^2 - x^2) + B^2(y^2 - x^2)\right],
\end{aligned}$$

that is,

$$-LP\,Q = -y^2(\beta^2 - \alpha^2) + \beta^2 - (\beta - \alpha) + \dots,$$

$$+(B + \alpha - \beta)\beta(y^2 - x^2) + \beta - B^2(y^2 - x^2) + \dots,$$

or omitting the factor β, that is,

$$-LP\,Q = -y^2(\beta^2-\alpha^2)+\beta^2-y^2(x^2-\alpha^2)+\dots+\beta(y^2-x^2)-By^2(x^2-\alpha^2),$$

It is easy to verify the form of the orders 0, 1, 2, 3, and 4 in x, y, z, w separately destroy each other, for instance, for the terms of the order 4, the value of Σ is then

$$= A^2+\Sigma\, y^2(\beta^2-\alpha^2)+y(x^2-\alpha^2)+B^2(x^2-\alpha^2)+(\Sigma\,\beta^2)(x^2-\alpha^2)+\dots,$$

$\frac{1}{B}\left[\Sigma\,\beta\, a^2(x^2 + y^2) + \Sigma\, a^2\, B^2(x^2 - y^2) + B^2\, a^2(x^2 - y^2)\right]$, It is easy to verify that the terms of the orders 0, 1, 2, 3, and 4 in x, y, z, w separately destroy any sum sum of x which is

$$-LP^2 + yQ^2 + yP^2 + xR^2 + yS^2 + (LP + yQ + xR + yS + yT)$$

$$+(D^2-1-LP^2)(yP^2+yP^2-yQ^2-yQ^2)+2(LP+yQ+xR+yS+yT)(LP+yQ+xR+yS$$

or, omitting the factor β, that is,

$$-(LP+yQ^2+xR+yS+yT)+\Sigma(LP+yQ+xR+yS+yT)(LP+yQ+xR+yS+yT)+\cdots$$

The theorem in the original form was obtained by me as follows: using the elliptic coordinates p, q, r, such that

$$\frac{p^2}{a+p}+\frac{q^2}{b+p}+\frac{r^2}{c+p}=1,$$

or, what is the same thing,

$$-Bp\rho^2 = a+p \cdot b+p \cdot c+p,$$

$$-y\eta y = b+q \cdot a+q \cdot c+q,$$

$$-z\zeta z = c+r \cdot a+r \cdot b+r,$$

where A, B denote $b-c$, $c-a$, $a-b$ respectively; then, treating r as a constant, the coordinates x, y, z will belong to a point on the ellipsoid

$$\frac{x^2}{a+r} + \frac{y^2}{b+r} + \frac{z^2}{c+r} = 1,$$

and the differential equation of the right lines upon this surface is

$$\frac{dp}{\sqrt{a+p \cdot b+p \cdot c+p}} + \frac{dq}{\sqrt{a+q \cdot b+q \cdot c+q}} = 0.$$

Take a, b, c, the coordinates of a point on the surface, and p, q, the corresponding values of p, q, so that

$$-Bp_0\rho_0^2 = a+p_0 \cdot b+p_0 \cdot c+p_0,$$

$$-Aq_0\eta_0 = b+q_0 \cdot a+q_0 \cdot c+q_0,$$

$$-z\zeta_0 = c+r \cdot a+r \cdot b+r,$$

then the equation of the tangent plane at the point (a, b, c, p_0, q_0) is

$$\frac{xz_0}{a+r} + \frac{yy_0}{b+r} + \frac{zz_0}{c+r} = 1,$$

or, substituting for x, y_0, z_0 their values, we have

$$= \beta p_0 \cdot ay + \cdot bxz + \ Az + \ldots,$$

and consequently the equation of the tangent plane through (a, b, c, p_0, q_0) is the integral of the proposed differential equation.

Writing

$$mz^2 = A(a+p),\ mz^2 = B(b+p),\ dx^2 = C(c+p),$$

the values of A, B, C, S are determined; and, assuming for q, r, p, the like forms with the arguments a, a_0, a_0, the differential equation becomes $dz + dz$, having the $10-2$

integral $u+x_0+v=u_0$; while the foregoing integral equation, on reducing the constant coefficients contained therein, takes the form

$$-l^2xx+mz_0\,mz+nz_0\,mz_0+nz_0\,nz+nz_0\,nz_0+\cdots=0,$$
$$=\frac{1}{2}lm\cdot mz_0\,mz_0,$$
$$=-l^2nz\cdot mz_0\,mz_0,$$
$$=\frac{nz_0}{B^2-1-l^2}\cdot\frac{l^2}{l^2-l^2},$$

via. this equation holds good if $a=u_0+v$, we have the theorem.

If, as above, $a=u+v+v$, $b=u-v+v$, $c=u+v-v$, where r is regarded as constant, then the three products snu nz nz, cnu nz nz, dnu nz nz, and equally so the three products snz nz nz, cnz nz nz, dnz nz nz, are expressed by means of those products with the u, v, w of the u, of the v, viz. for two surfaces it would be possible to obtain a real relation equation, which may be expressed in the squares of the u, v, w of the r, of the z, all of whose internal coefficients are nominal, vis. the such would be obtained, in fact, a similar relation in the like form by giving to the parameter r the advantage of being of a degree which is the half of the equation which involves only the squared functions.

Write a, b, c or for $\operatorname{sn}u$, $\operatorname{cn}u$, $\operatorname{dn}u$, respectively; then, writing

$$\begin{aligned}
A&=\sqrt{1-y^2\,\cdot\,1-l^2y^2}, & a&=\sqrt{1-l^2\,\cdot\,1-l^2x^2},\\
B&=\sqrt{1-l^2\,\cdot\,1-l^2y^2}, & b&=\sqrt{1-x^2\,\cdot\,1-l^2y^2},\\
C&=\sqrt{1-l^2\,\cdot\,1-l^2y^2}, & c&=\sqrt{1-l^2\,\cdot\,1-x^2-y^2},\\
D&=1-l^2x^2y^2, & d&=1-l^2z^2x^2,
\end{aligned}$$

we have

$$\operatorname{sn}(z+v)=\frac{xA+ya}{D},$$

that is,

$$\frac{A+P}{D}=\frac{\sigma c}{\sigma a\,\cdot\,\sigma b},$$

and consequently

$$Dn=(A-x)(A+x),$$
$$PD=(P+x)(P-x);$$

whence

$$Dn=PD=(Aa-x)(Aa-x).$$

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728. A Theorem in Elliptic Functions

[*From the Proceedings of the London Mathematical Society, vol.* IX *(1878), pp. 124–127. Read January 10, 1878.*]

[Continuation from p. 77.]

that is,

$$(z^2-w^2)(1-k^2x^2y^2)-(x^2-y^2)(1-k^2z^2w^2)$$
$$=2\{zw\sqrt{1-k^2x^2}\sqrt{1-k^2y^2}-xy\sqrt{1-k^2z^2}\sqrt{1-k^2w^2}\}.$$

Rationalising, we obtain, as mentioned above, an equation containing only the squares of the four quantities x^2, y^2, z^2, w^2; the equation containing therefore $xyzw$ in a degree twice that of the equation containing x^2, y^2, z^2, w^2. I worked out in this way the equation in (x^2,y^2,z^2,w^2), but the calculation was lost, and the easier way of obtaining it is obviously by means of the equation involving $xyzw$.

We have, by the theorem,

$$-k'^2xyzw-\frac{1}{k^2}\sqrt{1-k^2x^2}\sqrt{1-k^2y^2}\sqrt{1-k^2z^2}\sqrt{1-k^2w^2}=-\frac{1}{k^2},$$

that is,

$$k'^2(1-k^2xyzw)=\sqrt{1-k^2x^2}\sqrt{1-k^2y^2}\sqrt{1-k^2z^2}\sqrt{1-k^2w^2}$$
$$-\sqrt{1-k^2x^2}\sqrt{1-k^2y^2}\sqrt{1-k^2z^2}\sqrt{1-k^2w^2},$$

and then, writing

$$\begin{aligned}
P &= x^2 + y^2 + z^2 + w^2, \\
Q &= x^2y^2 + x^2z^2 + x^2w^2 + y^2z^2 + y^2w^2 + z^2w^2, \\
R &= x^2y^2z^2 + x^2y^2w^2 + x^2z^2w^2 + y^2z^2w^2, \\
S &= x^2y^2z^2w^2,
\end{aligned}$$

and using $\sqrt{S}$ to denote the rational function $xyzw$, we have

$$k'^4(1-k^2\sqrt{S})^2 = (1-P+Q-R+S)-2k^2(1-P+Q-R+S)(1-k^2P+k^4Q-k^6R+k^8S);$$

or, calling for a moment the radical $\sqrt{A}$, then the factor k^4 divides out, and the equation becomes

$$\begin{gathered}
2\sqrt{A} = 2-(1+k^2)P+2k^2Q-(k^4+k^6)R+2k^4S+2k'^4\sqrt{S}, \\
4(1-P+Q-R+S)(1-k^2P+k^4Q-k^6R+k^8S) \\
-\{2-(1+k^2)P+2k^2Q-(k^4+k^6)R+2k^4S\}^2-4k'^4S \\
= -2\sqrt{S}\{2-(1+k^2)P+2k^2Q-(k^4+k^6)R+2k^4S\}.
\end{gathered}$$

The factor k'^4 divides out; omitting it, we have

$$\begin{aligned}
&4Q-P^2-4(1+k^2)R+16k^2S+2k^2PR-4(k^2+k^4)PS-k^4R^2+4k^4QS \\
&\qquad = -2\sqrt{S}\{2-(1+k^2)P+2k^2Q-(k^4+k^6)R+2k^4S\},
\end{aligned}$$

or, as this may also be written,

$$\begin{aligned}
&4\{(Q-P^2+4R)+k^2(-4P+2PR+16S-4PS)+k^4(-R^2+4QS-PS)\} \\
&\qquad = -2\sqrt{S}\{2-P+k^2(-P+2Q-R)+k^4(-R+2S)\},
\end{aligned}$$

which is the required rational equation involving the product of the sn's.

729. On a Theorem Relating to Conformable Figures

[*From the Proceedings of the London Mathematical Society, vol.* X *(1879), pp. 143–146. Read May 8, 1879.*]

Consider two plane figures, say the figure of the points P referred to axes Ox, Oy, and that of the points P' referred to axes Ox', Oy'; and let x, y be the coordinates of P, and x', y' those of P'. If the figures correspond to each other in any manner whatever, P and P' being corresponding points, then we have x', y' each of them a function of x, y; and we may consider the second figure as derived from the first by altering the distance OP in the ratio $\sqrt{x'^2+y'^2} : \sqrt{x^2+y^2}$, and by rotating it through the angle $\tan^{-1}\frac{y'}{x'} - \tan^{-1}\frac{y}{x}$; say by the Extension $\sqrt{x'^2+y'^2} \div \sqrt{x^2+y^2}$, and by the Rotation $\tan^{-1}\frac{y'}{x'} - \tan^{-1}\frac{y}{x}$: where the Extension and the Rotation are each of them a determinate function of x, y, the coordinates of P.

Passing from the point P to a consecutive point Q, the coordinates of which are $x+dx$, $y+dy$ (the ratio $dy : dx$ being arbitrary), then the coordinates of the corresponding point Q' will be $x'+dx'$, $y'+dy'$, where

$$dx' = \frac{dx'}{dx}dx + \frac{dx'}{dy}dy, \quad dy' = \frac{dy'}{dx}dx + \frac{dy'}{dy}dy.$$

Writing $\frac{dx'}{dx}$, $\frac{dy'}{dx}$ instead of $\frac{dy}{dx}\cdot\frac{dx'}{dy}$ and $\frac{dy}{dx}\cdot\frac{dy'}{dy}$, the expressions

$$\sqrt{dx'^2+dy'^2} \div \sqrt{dx^2+dy^2}, \quad \tan^{-1}\frac{dy'}{dx'} - \tan^{-1}\frac{dy}{dx},$$

will in general have values depending upon that of the arbitrary ratio $dy : dx$. But they may be independent of this ratio; viz. this is the case when x', y' are functions of x, y such that

$$\frac{dx'}{dx} = \frac{dy'}{dy}, \qquad \frac{dy'}{dx} = -\frac{dx'}{dy},$$

and the two figures are then conformable (or conjugate) figures; that is, figures similar as regards corresponding infinitesimal elements of area. We have, in this case,

$$\sqrt{dx'^2+dy'^2} \div \sqrt{dx^2+dy^2}, \quad \tan^{-1}\frac{dy'}{dx'} - \tan^{-1}\frac{dy}{dx},$$

each a determinate function of x, y, the coordinates of P.

Passing from the element PQ to the corresponding element $P'Q'$ by altering the element in the ratio $\sqrt{dx'^2+dy'^2} \div \sqrt{dx^2+dy^2}$, and rotating the element through the angle $\tan^{-1}\frac{dy'}{dx'} - \tan^{-1}\frac{dy}{dx}$; say, this requires

$$dx' = A\,dx - B\,dy, \qquad dy' = B\,dx + A\,dy,$$

where A, B are functions of x, y such that

$$\frac{dA}{dy} = -\frac{dB}{dx}, \qquad \frac{dA}{dx} = \frac{dB}{dy},$$

or the conditions for the existence of the functions x', y' are

$$\frac{dA}{dy} + \frac{dB}{dx} = 0, \qquad \frac{dA}{dx} - \frac{dB}{dy} = 0.$$

If

$$\left(\frac{dx'}{dx}\right)^2 + \left(\frac{dy'}{dx}\right)^2 = \left(\frac{dy'}{dy}\right)^2 + \left(\frac{dx'}{dy}\right)^2,$$

$$\frac{dx'}{dx}\frac{dy'}{dx} = -\frac{dy'}{dy}\frac{dx'}{dy},$$

We, in fact, have

$$\frac{dA}{dx} = \frac{d^2x'}{dx^2} = \frac{d^2y'}{dx\,dy} = \frac{dB}{dy},$$

and similarly

$$\frac{dA}{dy} = \frac{d^2x'}{dy\,dx} = -\frac{d^2y'}{dx^2} = -\frac{dB}{dx},$$

which proves the theorem.

The theorem is closely connected with the theory of the function of an imaginary variable; for, writing the condition

$$dx' + i dy' = (F + iG)(dx + i dy),$$

where F and G are each of them a real function of x, y; then we have

$$dx' = F\,dx - G\,dy, \qquad dy' = G\,dx + F\,dy,$$

$$F = \frac{dx'}{dx} = \frac{dy'}{dy}, \qquad G = \frac{dy'}{dx} = -\frac{dx'}{dy},$$

and the condition in order that $x' + iy'$ may be a function of $x + iy$ is

$$\frac{dF}{dy} = -\frac{dG}{dx}, \qquad \frac{dF}{dx} = \frac{dG}{dy},$$

which are the conditions for the existence of the functions F, G of x, y; that is, the condition that $x' + iy'$ may be a function (in the ordinary sense, a function) of the imaginary variable z, $= x + iy$, is a function such that, for a consecutive value z', $= x + iy + dx + idy$, we have

$$\frac{z' - z}{dx + idy} = \text{a quantity independent of the ratio of the real components } dx,\ dy \text{ of th}$$

Or, what is the same thing, writing $f(z) = x' + iy'$, the condition in order that $x' + iy'$ may be a function of $x + iy$ is

$$dx' + idy' = (F + iG)(dx + idy),$$

where F and G are each of them a function of x, y.

730. Addition to Mr Spottiswoode's Paper "On the Twenty-one Coordinates of a Conic in Space"

[*From the Proceedings of the London Mathematical Society, vol.* X *(1879), pp. 194–196.*]

Write

$$U = (a, b, c, d, f, g, h, l, m, n \mid x, y, z, w)^2,$$
$$P = (\alpha, \beta, \gamma, \delta \mid x, y, z, w),$$
$$P_0 = (\alpha, \beta, \gamma, \delta \mid \xi, \eta, \zeta, \omega).$$

Then the equation of the cone, having for its vertex the arbitrary point $(\xi, \eta, \zeta, \omega)$, and passing through the conic $U = 0$, $P = 0$, is

$$UP_0^2 - 2PP_0P + U_0P^2 = 0.$$

Or if, to put the coefficients ξ, η, ζ, ω in evidence, we write for a moment

$$A = (a, h, g, l \mid x, y, z, w),$$
$$B = (h, b, f, m \mid x, y, z, w),$$
$$C = (g, f, c, n \mid x, y, z, w),$$
$$D = (l, m, n, d \mid x, y, z, w),$$

and therefore

$$U = A\xi + B\eta + C\zeta + D\omega,$$

then the equation is

$$U(\alpha\xi+\beta\eta+\gamma\zeta+\delta\omega)^2-2P(\alpha\xi+\beta\eta+\gamma\zeta+\delta\omega)(A\xi+B\eta+C\zeta+D\omega)$$
$$+ P^2(a, b, c, d, f, g, h, l, m, n \mid \xi, \eta, \zeta, \omega)^2 = 0.$$

And if we expand first in ξ, η, ζ, ω, and then in x, y, z, t, the final result is shown by the following coefficient schemes.

First coefficient table:

	x^2	y^2	z^2	t^2
ξ^2	a	c	b	f
η^2	c	b	a	g
ζ^2	b	a	c	h
ω^2	f	g	h	d

	yz	zx	xy	xt	yt
ξ^2	C	B	A	$2A'$	$2B'$
η^2	A	C	B	$2C'$	$2A'$
ζ^2	B	A	C	$2B'$	$2C'$
ω^2	$2N$	$2M$	$2L$	$2(Q-R)$	$2(R-P)$

Second coefficient table:

	$\eta\zeta$	$\zeta\xi$	$\xi\eta$	$\xi\omega$	$\eta\omega$	$\zeta\omega$
x^2	C	B	A	$2A'$	$2B'$	$2C'$
y^2	A	C	B	$2C'$	$2A'$	$2B'$
z^2	B	A	C	$2B'$	$2C'$	$2A'$
t^2	$2N$	$2M$	$2L$	$2(Q-R)$	$2(R-P)$	$2(P-Q)$

Third coefficient table:

	$\eta\zeta$	$\zeta\xi$	$\xi\eta$	$\xi\omega$	$\eta\omega$	$\zeta\omega$
yz	$2A'$	$2B'$	$2C'$	$2N'$	$2M$	$2L$
zx	$2B'$	$2C'$	$2A'$	$2L'$	$2(P-Q)$	$2M'$
xy	$2C'$	$2A'$	$2B'$	$2M'$	$2N$	$2N'$
xt	$2N'$	$2L'$	$2M'$	$2F$	$2G'$	$2H$
yt	$2M$	$2(P-Q)$	$2N$	$2H'$	$2F$	$2G$
zt	$2L$	$2M'$	$2N'$	$2G$	$2H$	$2F'$

$$= 0.$$

In particular, if $\xi = 0$, $\zeta = 0$, $\omega = 0$, then we have the foregoing equation $X = 0$; and the like for the equations $Y = 0$, $Z = 0$, and $W = 0$ respectively.

Take a, b, c, f, g, h for the six coordinates of the line through the points (x, y, z, t) and $(\xi, \eta, \zeta, \omega)$, that is, write

$$a = y\zeta - z\eta, \quad b = z\xi - x\zeta, \quad c = x\eta - y\xi,$$

$$l = x\omega - t\xi, \quad g = y\omega - t\eta, \quad h = z\omega - t\zeta,$$

where, of course,

$$af + bg + ch = 0.$$

Then the foregoing equation of the cone is

$$\begin{aligned} Aa^2 + Bb^2 + Cc^2 + Fl^2 + Gg^2 + Hh^2 & \\ - 2A'bc - 2B'ca - 2C'ab + 2F'gh + 2G'hi + 2H'fg & \\ + 2Laf + 2Mag - 2Nah & \\ - 2L'bf + 2Qbg + 2N'bh & \\ + 2Lcf - 2M'cg + 2Rch & \\ & = 0. \end{aligned}$$

And this may be regarded as the equation of the conic in terms of the twenty-one coordinates of the conic, and of the six coordinates of an arbitrary line meeting the conic. It is, in fact, the general form of the equation given in the paper—Cayley, "On a new Analytical Representation of a Curve in Space," *Quart. Math. Jour.*, vol. III (1860), [284; this Collection, vol. IV p. 453].

731. On the Binomial Equation $x^p - 1 = 0$; Trisection and Quartisection

[*From the Proceedings of the London Mathematical Society, vol.* XI *(1880), pp. 4–17. Read November 13, 1879.*]

The solution of the binomial equation $x^p - 1 = 0$, p a prime number, or, say rather, the equation

$$\frac{x^p - 1}{x - 1} = 0,$$

depends upon the Jacobian function

$$F\alpha = \alpha + \zeta\alpha^g + \zeta^2\alpha^{g^2} + \cdots + \zeta^{p-2}\alpha^{g^{p-2}},$$

where g is a prime root of p, α any root whatever of the equation $\alpha^{p-1} - 1 = 0$; e a factor of $p-1$, and f for the complementary factor (that is, $p-1 = ef$), then, if we write α^f, or, what is the same thing, taking $\alpha^f = \beta$, a root of $\beta^e - 1 = 0$, we have

$$F\beta = X_0 + \beta X_1 + \cdots + \beta^{e-1}X_{e-1},$$

where $X_0,\ X_1,\ \ldots,\ X_{e-1}$ denote each of them a period or sum of $f = \frac{p-1}{e}$ roots, viz. taking

$$X_0 = (1,\ g^e,\ g^{2e},\ \ldots,\ g^{(f-1)e}),$$

(read $X_0 = \alpha + \alpha^{g^e} + \cdots + \alpha^{g^{(f-1)e}}$, and so for the other functions).

We have, of course, $F(1) = X_0 + X_1 + \cdots + X_{e-1}$, the sum of all the roots $= -1$; and, further, the general property that any rational and integral function of these periods is expressible as a sum

$$a_0X_0 + a_1X_1 + \cdots + a_{e-1}X_{e-1}$$

with known coefficients $(a_0, a_1, \ldots, a_{e-1})$.

The several cases $e = 2,\ 3,\ 4,\ \ldots$ may be termed those of the bisection, trisection, quartisection, &c., of the equation; viz. $e = 2$, there are two periods, X, Y, and $F(-1) = X - Y$; $e = 3$, three periods, X, Y, Z, and $F\gamma = X + \gamma Y + \gamma^2 Z$, if γ is a root of $\gamma^3 - 1 = 0$; $e = 4$, four periods, X, Y, Z, W, and $F\delta = X + \delta Y + \delta^2 Z + \delta^3 W$, if δ be a root of $\delta^4 - 1 = 0$.

It is sufficient to attend to the prime roots γ and δ of the equations

$$\gamma^3 - 1 = 0, \qquad \delta^4 - 1 = 0,$$

respectively; for, if γ or δ be $= 1$, we have simply $F(1), = -1$; and if δ be $= -1$, then the function is $F(-1), = X + Z - (Y + W)$, where $X + Z$ and $Y + W$ are the periods for the bisection. The prime roots δ are of course i and $-i$, and we have respectively

$$F(i) = X + iY - Z - iW,$$
$$F(-i) = X - iY - Z + iW.$$

As regards the bisection, it is known that $(X - Y)^2 = (-)^{\frac{p-1}{2}} p$, which is $+p$ or $-p$, according as $\frac{p-1}{2}$ is $= 1$ or 3, mod. 4; and the values of X, Y are thus determined.

In what follows, I consider the cases $e = 3$ and $e = 4$ of the trisection and the quartisection respectively.

It is to be remembered that, not the division into periods, but the order of the periods, depends on the choice of g, a prime root at pleasure of p; and, in what follows, I use the prime roots given in Reuschle's *Tafeln completer Primzahlen welche aus Wurzeln der Einheit gebildet sind* (4to, Berlin, 1875): viz. these are

p	3	5	7	11	13	17	19	23	29	31	37	41	43	47	53
g	2	2	5	2	2	3	2	5	2	3	2	6	3	5	2

p	59	61	67	71	73	79	83	89	97	...
g	2	2	2	7	5	3	2	3	5	...

The Trisection, $e = 3$.

We have here $X + Y + Z = -1$, and the periods are determined as roots of a cubic equation, the coefficients of which are rational and integral functions of p and A, where A is the quadratic character $\sqrt{\pm p}$, according as p is $= 1$ or $= 3$, mod. 4.

We have, in fact,

$$\begin{aligned} X^2 + Y^2 + Z^2 &= p - 1, \\ YZ + ZX + XY &= -\tfrac{1}{2}(p-1) + \tfrac{1}{2}A, \\ XYZ &= \tfrac{1}{27}\{(3p-1) + (6+p)A\} \quad \text{or} \quad \tfrac{1}{27}\{(3p-1) - (6+p)A\}, \end{aligned}$$

according to the choice of the prime root g.

The formulae may be written somewhat more simply; writing X, Y, $Z = \frac{1}{3}(-1 + A' + B')$, $\frac{1}{3}(-1 + A'\omega + B'\omega^2)$, $\frac{1}{3}(-1 + A'\omega^2 + B'\omega)$, where ω is an imaginary cube root of unity, we have $A'^3 + B'^3 = p$, $A'B' = p$, and thence A'^3, $B'^3 = \frac{1}{2}\{p \pm \sqrt{p(4p-1)}\}$, but this does not tend to simplify the numerical calculations.

Case $p = 7$, $g = 3$.

The residues of g^0, g^1, g^2 are 1, 3, 2, and we have

$$X = (1,\ 2), \quad Y = (3,\ 6), \quad Z = (4,\ 5),$$

where $(1, 2)$ means $\alpha + \alpha^2$; and thence $X = -1.2469796$, $Y = 0.4450418$, $Z = -0.1980622$. Forming the products YZ, ZX, XY and thence also XYZ, the numerical values verify the formulae.

Case $p = 13$, $g = 2$.

The residues are 1, 2, 4, 8, 3, 6, 12, 11, 9, 5, 10, 7, and we have $X = (1, 3, 9)$, $Y = (2, 6, 5)$, $Z = (4, 12, 10)$, whence $X = -1.2096120$, $Y = 2.5070186$, $Z = -1.2974066$. Here $A = +\sqrt{13}$, and the formula gives $XYZ = \frac{1}{27}(38 + 19\sqrt{13}) = 4.0694455$, which agrees with the numerical value.

Case $p = 19$, $g = 2$.

The residues are 1, 2, 4, 8, 16, 13, 7, 14, 9, 18, 17, 15, 11, 3, 6, 12, 5, 10; and $X = (1, 7, 11)$, $Y = (2, 14, 3)$, $Z = (4, 9, 6)$, whence $X = -2.2851426$, $Y = 1.6355003$, $Z = -0.3503577$. Here $A = -\sqrt{19}$, and the formula gives $XYZ = \frac{1}{27}(56 - 25\sqrt{19}) = -1.3098580$, agreeing with the numerical value.

Case $p = 31$, $g = 3$.

The residues are 1, 3, 9, 27, 19, 26, 16, 17, 20, 29, 25, 13, 8, 24, 10, 30, 28, 22, 4, 12, 5, 15, 14, 11, 2, 6, 18, 23, 7, 21; and $X = (1, 26, 25, 30, 5, 6)$, $Y = (3, 17, 13, 28, 15, 18)$, $Z = (9, 20, 8, 22, 14, 2)$, whence $X = -2.3640854$, $Y = 1.2134116$, $Z = 2.1506738$. Here $A = -\sqrt{31}$, and $XYZ = \frac{1}{27}(92 - 37\sqrt{31}) = -6.0419010$, agreeing with the numerical value.

The Quartisection, $e = 4$.

We have $X + Y + Z + W = -1$, and the periods are determined as roots of a quartic equation, the coefficients whereof are rational and integral functions of p and A, where A is, as before, $= \sqrt{\pm p}$.

We have

$$X^2 + Y^2 + Z^2 + W^2 = p - 1,$$

$$YZ + ZX + XY + YW + ZW + WX = -\tfrac{1}{2}(p-1),$$

$$XYZ + XYW + XZW + YZW = \tfrac{1}{8}\{(p-1) + 3A\} \quad \text{or} \quad \tfrac{1}{8}\{(p-1) - 3A\},$$

$$XYZW = \text{a rational and integral function of } p \text{ and } A$$

where the last expression is somewhat complicated; and, as in the case of the trisection, there are two values according to the choice of the prime root g.

Case $p = 5,\ g = 2$.

Residues 1, 2, 4, 3; $X = (1)$, $Y = (2)$, $Z = (4)$, $W = (3)$. We have $X = Y = \frac{1}{4}(-1 + \sqrt{5})$, $Z = W = \frac{1}{4}(-1 - \sqrt{5})$. The formulae give $XYZW = -1$.

Case $p = 13,\ g = 2$.

Residues 1, 2, 4, 8, 3, 6, 12, 11, 9, 5, 10, 7; $X = (1, 3, 9)$, $Y = (2, 6, 5)$, $Z = (4, 12, 10)$, $W = (8, 11, 7)$. Here $A = +\sqrt{13}$, and the values are $X = -1.2096120$, $Y = 2.5070186$, $Z = -1.2974066$, $W = -1.0000000$.

Case $p = 17,\ g = 3$.

Residues 1, 3, 9, 10, 13, 5, 15, 11, 16, 14, 8, 7, 4, 12, 2, 6; $X = (1, 13, 16, 4)$, $Y = (3, 5, 14, 12)$, $Z = (9, 15, 8, 2)$, $W = (10, 11, 7, 6)$. Here $A = +\sqrt{17}$, and $X = -0.4772606$, $Y = 2.6825071$, $Z = -1.1643310$, $W = -1.0409155$.

Case $p = 29,\ g = 2$.

Residues 1, 2, 4, 8, 16, 3, 6, 12, 24, 19, 9, 18, 7, 14, 28, 27, 25, 21, 13, 26, 23, 17, 5, 10, 20, 11, 22, 15, 30; $X = (1, 7, 16, 20, 25, 24, 23)$, $Y = (2, 14, 3, 11, 21, 19, 17)$, $Z = (4, 28, 6, 22, 13, 9, 5)$, $W = (8, 27, 12, 15, 26, 18, 10)$. Here $A = -\sqrt{29}$, and $X = -2.4058043$, $Y = 2.1595122$, $Z = -2.4981638$, $W = 1.7444559$.

Case $p = 37,\ g = 2$.

Residues 1, 2, 4, 8, 16, 32, 27, 17, 34, 31, 25, 13, 26, 15, 30, 23, 9, 18, 36, 35, 33, 29, 21, 5, 10, 20, 3, 6, 12, 24, 11, 22, 7, 14, 28, 19; $X = (1, 26, 10, 36, 11, 31)$, $Y = (2, 15, 20, 35, 22, 32)$, $Z = (4, 30, 3, 33, 7, 27)$, $W = (8, 23, 6, 29, 14, 17)$. Here $A = +\sqrt{37}$, and $X = -1.2393227$, $Y = 2.6623945$, $Z = 1.5670920$, $W = -1.9901638$.

Table of the Trisection and Quartisection.

The following table gives the numerical values of the periods for the several primes p, in the cases of the trisection and quartisection respectively.

Trisection.

p	X	Y	Z
7	−1.2469796	0.4450418	−0.1980622
13	−1.2096120	2.5070186	−1.2974066
19	−2.2851426	1.6355003	−0.3503577
31	−2.3640854	1.2134116	2.1506738
43	−3.6271791	2.2088243	2.4183548
61	−2.6826409	1.3954104	2.2872305
67	−2.7164437	−0.2304933	3.9469370
73	−1.5746473	3.3811608	−0.8065135
79	−4.8924496	3.2328707	2.6595789
97	−2.2321885	2.5743487	0.6578398

Quartisection.

p	X	Y	Z	W
5	0.3090170	0.3090170	−0.8090170	−0.8090170
13	−1.2096120	2.5070186	−1.2974066	−1.0000000
17	−0.4772606	2.6825071	−1.1643310	−1.0409155
29	−2.4058043	2.1595122	−2.4981638	1.7444559
37	−1.2393227	2.6623945	1.5670920	−1.9901638
53	−1.2862112	3.4765248	−1.6359758	−0.5543378
61	−1.0289627	2.5793946	−2.7119368	2.1615049
73	−3.2626063	3.3357117	−1.1846103	1.1115049

For $p = 101$, the trisection gives

$$X = -4.0437345, \quad Y = -0.3896808, \quad Z = 4.4334153.$$

732. A Theorem in Spherical Trigonometry

[*From the Proceedings of the London Mathematical Society, vol.* XI *(1880), pp. 48–50. Read January 8, 1880.*]

In a spherical triangle, where a, b, c are the sides, and A, B, C the opposite angles, we have

$$-\tan\tfrac{1}{2}c\,\tan\tfrac{1}{2}a\,\tan\tfrac{1}{2}b\,\sin(A-B)=\tan\tfrac{1}{2}b\sin A-\tan\tfrac{1}{2}a\sin B,$$
$$\tan\tfrac{1}{2}c\,\{1-\tan\tfrac{1}{2}a\,\tan\tfrac{1}{2}b\cos(A-B)\}=\tan\tfrac{1}{2}b\cos A+\tan\tfrac{1}{2}a\cos B,$$

which are both included in the form

$$\tan\tfrac{1}{2}c\,(\cos B-i\sin B)=\frac{\tan\frac{1}{2}b\,(\cos A+i\sin A)-\tan\frac{1}{2}a}{1+\tan\frac{1}{2}c\,\tan\frac{1}{2}a\,(\cos A+i\sin A)}.$$

For the first of the two identities: from

$$\cos a=\frac{\cos A+\cos B\cos C}{\sin B\sin C},\qquad \cos b=\frac{\cos B+\cos A\cos C}{\sin A\sin C},$$

we deduce

$$\begin{aligned}\cos a-\cos b&=\frac{\cos A}{\sin C}\left(\frac{1}{\sin B}-\frac{\cos C}{\sin A}\right)+\frac{\cos B}{\sin C}\left(\frac{\cos C}{\sin B}-\frac{1}{\sin A}\right)\\
&=\frac{1}{\sin C}\left(\frac{\cos A}{\sin B}-\frac{\cos B}{\sin A}\right)-\frac{\cos C}{\sin C}\left(\frac{\cos A}{\sin A}-\frac{\cos B}{\sin B}\right)\\
&=\frac{\sin(A-B)}{\sin C\sin A\sin B}-\frac{\cos C\,(\sin 2A-\sin 2B)}{2\sin C\sin A\sin B}\\
&=\frac{\sin(A-B)}{\sin C\sin A\sin B}\{-\cos C\sin(A+B)\}\\
&=-\frac{\sin(A-B)}{\sin C\sin A\sin B}(\cos c-1);\end{aligned}$$

that is,

$$-\sin(A-B)=\frac{\sin C}{1-\cos c}(\cos a-\cos b),$$

or, what is the same thing,

$$-\tan\tfrac{1}{2}c\,\sin(A-B)=\frac{\sin C}{\sin c}(\cos a-\cos b).$$

Here $\cos a-\cos b=(1+\cos a)-(1+\cos b)$; substituting for $\frac{\sin C}{\sin c}$ successively $\frac{\sin A}{\sin a}$ and $\frac{\sin B}{\sin b}$, the right-hand side is

$$\begin{aligned}&=\frac{1+\cos a}{\sin a}\sin A-\frac{1+\cos b}{\sin b}\sin B\\
&=\cot\tfrac{1}{2}a\,\sin A-\cot\tfrac{1}{2}b\,\sin B;\end{aligned}$$

whence, multiplying each side by $\tan\frac{1}{2}a\ \tan\frac{1}{2}b$, we have the relation in question.

For the second identity which is

$$\tan\tfrac{1}{2}c\,\{1-\tan\tfrac{1}{2}a\ \tan\tfrac{1}{2}b\cos(A-B)\}=\tan\tfrac{1}{2}b\cos A+\tan\tfrac{1}{2}a\cos B;$$

if on the right-hand side we substitute for $\cos A$, $\cos B$ their values

$$\frac{\cos a-\cos b\cos c}{\sin b\sin c}\qquad\text{and}\qquad\frac{\cos b-\cos a\cos c}{\sin a\sin c};$$

the right-hand side becomes

$$\frac{1}{\sin c}\left\{\frac{1+\cos b}{1+\cos a}(\cos a-\cos b\cos c)+\frac{1+\cos a}{1+\cos b}(\cos b-\cos c\cos a)\right\};$$

whence, multiplying the whole equation by $\sin c\,(1+\cos a)(1+\cos b)$, it becomes

$$\begin{aligned}&(1-\cos c)\{(1+\cos a)(1+\cos b)-\sin a\sin b\cos(A-B)\}\\&=(1+\cos a)(\cos a-\cos b\cos c)+(1+\cos b)(\cos b-\cos c\cos a).\end{aligned}$$

We have here

$$\cos(A-B)=\cos A\cos B+\sin A\sin B=\frac{(\cos a-\cos b\cos c)(\cos b-\cos c\cos a)+D}{\sin^2 c\,\sin a\sin b},$$

where

$$D=1-\cos^2 a-\cos^2 b-\cos^2 c+2\cos a\cos b\cos c.$$

The numerator is

$$\cos a\cos b-\cos c(\cos^2 a+\cos^2 b)+\cos a\cos b\cos^2 c+D,$$

viz. this is

$$\cos a\cos b(1+\cos^2 c)-(\cos^2 a+\cos^2 b)(1+\cos c)+1-\cos^2 c,$$

having the factor $1+\cos c$, which is also a factor of $\sin^2 c$, $=1-\cos^2 c$, in the denominator. We have, therefore,

$$\cos(A-B)=\frac{\cos a\cos b(1+\cos c)-(\cos^2 a+\cos^2 b)+1-\cos c}{(1-\cos c)\sin a\sin b};$$

and the equation thus is

$$(1-\cos c)(1+\cos a)(1+\cos b)-[\cos a\cos b(1+\cos c)-(\cos^2 a+\cos^2 b)+1-\cos c]$$
$$= (1+\cos a)(\cos a-\cos b\cos c)+(1+\cos b)(\cos b-\cos c\cos a),$$

where each side is in fact

$$= \cos a+\cos^2 a+\cos b+\cos^2 b-\cos c(\cos a+\cos b)-2\cos a\cos b\cos c$$

and the second identity is thus proved.

733. On a Formula of Elimination

[*From the Proceedings of the London Mathematical Society, vol.* XI *(1880), pp. 139–141. Read June 10, 1880.*]

Consider the equations

$$(a, \dots \mid \theta, 1)^m = 0, \qquad (A, \dots \mid \theta, 1)^n = 0,$$

where $a, \dots, A, \dots$ are functions of coordinates.

Eliminating θ, we obtain $V = 0$, the equation of a surface; and (as is known) this surface has a nodal curve.

To fix the ideas, suppose that each of these coefficients is a linear function of the four coordinates x, y, z, w. Then, it is easy to obtain the equations of the nodal curve in the case where one of the equations, say the second, is a quadric: the process is substantially the same whatever may be the order of the other equation, and I take it to be a cubic; the two equations therefore are

$$(a, b, c, d \mid \theta, 1)^3 = 0, \qquad (A, B, C \mid \theta, 1)^2 = 0;$$

giving rise to an equation

$$V = (a, b, c, d \mid A, B, C)^3 = 0.$$

And it is required to perform the elimination so as to put in evidence the nodal line of this surface.

Take θ_1, θ_2, the roots of the second equation, or write

$$(A, B, C \mid \theta, 1)^2 = A(\theta - \theta_1)(\theta - \theta_2);$$

that is,

$$\theta_1 + \theta_2 = -\frac{2B}{A}, \quad \theta_1\theta_2 = \frac{C}{A}.$$

Then, if

$$\Theta_1 = (a, b, c, d \mid \theta_1, 1)^3, \qquad \Theta_2 = (a, b, c, d \mid \theta_2, 1)^3,$$

we have

$$V = \Theta_1\Theta_2,$$

viz. on the right-hand side, replacing the symmetrical functions of θ_1, θ_2, by their values in terms of A, B, C, we have the expression of V in its known form

$$V = a^2C^3 + \&c.$$

Form now the expressions

$$\frac{\Theta_1-\Theta_2}{\theta_1-\theta_2},\quad \frac{\theta_2\Theta_1-\theta_1\Theta_2}{\theta_1-\theta_2},\quad \frac{\theta_1^2\Theta_1-\theta_2^2\Theta_2}{\theta_1-\theta_2},\quad \frac{\theta_1^3\Theta_1-\theta_2^3\Theta_2}{\theta_1-\theta_2},$$

each divided by $\theta_1-\theta_2$.

These are evidently symmetrical functions of θ_1, θ_2; the values being given by the successive lines of the expression

$$\begin{pmatrix} \theta_1^2+\theta_1\theta_2+\theta_2^2, & 3(\theta_1+\theta_2), & 3, & 0, \\ 0, & -(\theta_1+\theta_2), & -\theta_1\theta_2, & 0, \\ -(\theta_1+\theta_2), & \theta_1+\theta_2, & \theta_1\theta_2, & 0, \\ \theta_1\theta_2(\theta_1+\theta_2), & -\theta_1\theta_2, & 0, & 0 \end{pmatrix} \mid (d, 3c, 3b, a).$$

And, consequently, these same quantities, each multiplied by A^2, are given by the successive lines of

$$\begin{pmatrix} 0, & -A^2, & 2AB, & -3B^2+AC, \\ A^2, & 0, & -AC, & 2BC, \\ -2AB, & AC, & 0, & -C^2, \\ 3B^2-AC, & -2BC, & C^2, & 0 \end{pmatrix} \mid (d, 3c, 3b, a).$$

Calling these X, Y, Z, W, that is, writing

$$X = 3A^2c - 6ABb + (-AC+4B^2)a, \text{ \&c.},$$

then X, Y, Z, W are the values of

$$\frac{\Theta_1-\Theta_2}{\theta_1-\theta_2},\quad \frac{\theta_2\Theta_1-\theta_1\Theta_2}{\theta_1-\theta_2},\quad \frac{\theta_1^2\Theta_1-\theta_2^2\Theta_2}{\theta_1-\theta_2},\quad \frac{\theta_1^3\Theta_1-\theta_2^3\Theta_2}{\theta_1-\theta_2},$$

each multiplied by $A^2 \div (\theta_1-\theta_2)$; and the functions all four of them vanish if only $\Theta_1=0$, $\Theta_2=0$; or, what is the same thing, the equations $X=0$, $Y=0$, $Z=0$, $W=0$ constitute only a twofold system.

The functions

$$(XW-YZ,\ XZ-Y^2,\ YW-Z^2)$$

contain each of them the factor $\Theta_1\Theta_2$, that is, V; they, in fact, each of them vanish if $\Theta_1=0$, and they also vanish if $\Theta_2=0$; or, by a direct substitution, we have

$$\begin{aligned} XW-YZ &= -(\theta_1-\theta_2)^{-2}(\theta_1+\theta_2)\Theta_1\Theta_2 = -A^2\Theta_1\Theta_2/(\theta_1+\theta_2), \\ XZ-Y^2 &= -(\theta_1-\theta_2)^{-2}\Theta_1\Theta_2 = -A^2\Theta_1\Theta_2, \\ YW-Z^2 &= -(\theta_1-\theta_2)^{-2}\theta_1\theta_2\Theta_1\Theta_2 = -CA^2\Theta_1\Theta_2/\theta_1\theta_2. \end{aligned}$$

Or, what is the same thing, these are $= -AV$, $2BV$, $-CV$, respectively; thus the first equation is

$$\begin{aligned}\{3A^2c - 6ABb + (-AC + 4B^2)a\}\{2ABd - 3ACc + C^2b\}\\ - \{-A^2d + 3ACb - 2BCa\}^2 = -A(A^2d^2 + \&c.), = -2AV;\end{aligned}$$

and similarly for the other two equations.

The nodal curve is thus given by the twofold system $X = 0, Y = 0$, $Z = 0$, $W = 0$.

The method may be extended to the case where, instead of the quadric equation $(A, B, C \mid \theta, 1)^2 = 0$, we have an equation of any higher order, but the formulae are less simple.

734. On the Kinematics of a Plane

[*From the Quarterly Journal of Pure and Applied Mathematics, vol.* XVI *(1879), pp. 1–8.*]

It seems desirable to bring together under this title various questions which have been, or may be, proposed or discussed.

We consider two planes moving one upon the other, but, for convenience, they may be distinguished as the moving plane and a fixed plane, the first moving upon the second.

Any point of the moving plane traces out on the fixed plane a curve, and any line of the moving plane envelopes on the fixed plane a curve; similarly, any point of the fixed plane traces out on the moving plane a curve, and any line of the fixed plane envelopes on the moving plane a curve.

More generally, any curve of the moving plane envelopes on the fixed plane a curve, and any curve of the fixed plane envelopes on the moving plane a curve.

There is, moreover, in the moving plane a curve which rolls upon a curve in the fixed plane, and these two curves (a single relative position being given) determine the motion.

Analytical Theory.

The analytical theory presents no difficulty.

Taking in the fixed plane the fixed axes Ox, Oy (fig. 1), and, fixed in the moveable plane so as to move with it, the axes O_1x_1, O_1y_1; then the position of the axes $O_1x_1y_1$ may be determined, say by α, β, the coordinates of O_1 in regard to Oxy; and by θ, the inclination of O_1x_1 to Ox.

And denoting by x, y, x_1, y_1 the coordinates of a point P in regard to the two sets of axes respectively, then

$$x = \alpha + x_1\cos\theta - y_1\sin\theta, \qquad y = \beta + x_1\sin\theta + y_1\cos\theta;$$

or, what is the same thing,

$$x_1 = (x-\alpha)\cos\theta + (y-\beta)\sin\theta,$$

$$y_1 = -(x-\alpha)\sin\theta + (y-\beta)\cos\theta;$$

or, as these last equations may be written,

$$x_1 = \alpha_1 + x\cos(-\theta) - y\sin(-\theta),$$

$$y_1 = \beta_1 + x\sin(-\theta) + y\cos(-\theta),$$

where α_1, β_1, $= -\alpha\cos\theta - \beta\sin\theta$, $\alpha\sin\theta - \beta\cos\theta$, are the coordinates of O referred to the axes $O_1x_1y_1$, and $-\theta$ is the inclination of Ox to O_1x_1.

When the motion is given, α, β, θ are given functions of a single variable parameter, say of t; or, if we please, α, β are given functions of θ.

The velocities of a given point (x, y) are determined by the equations

$$x' = \alpha' - (x_1\sin\theta + y_1\cos\theta)\theta',$$
$$y' = \beta' + (x_1\cos\theta - y_1\sin\theta)\theta';$$

that is,

$$x' - \alpha' = -(y - \beta)\theta', \qquad y' - \beta' = (x - \alpha)\theta'.$$

Or, as these equations may also be written,

$$-(x' - \alpha')\sin\theta + (y' - \beta')\cos\theta = x_1\theta',$$

$$(x' - \alpha')\cos\theta + (y' - \beta')\sin\theta = y_1\theta'.$$

Hence if $x' = 0$, $y' = 0$, we have

$$x_1\theta' = \alpha'\sin\theta - \beta'\cos\theta, \quad y_1\theta' = \alpha'\cos\theta + \beta'\sin\theta,$$

$$\alpha' = (y - \beta)\theta', \qquad -\beta' = (x - \alpha)\theta',$$

which equations determine in terms of t, x_1 and y_1, the coordinates in regard to the axes $O_1x_1y_1$, and x and y the coordinates in regard to the axes Oxy, of I, the centre of instantaneous rotation.

If from the expressions of x_1, y_1 we eliminate t, we obtain an equation between (x_1, y_1) which is that of the rolling curve in the moveable plane; and, similarly, if from the expressions of x, y we eliminate t, we obtain a relation between (x, y), which is that of the rolled-on curve in the fixed plane.

t may be regarded as denoting the time, and then the derived functions of x, y in regard to t will denote velocities; and, to simplify the expression of the theorems, it is convenient to do this.

The system may be written

$$x_1 = \frac{\alpha'}{\theta'}\sin\theta - \frac{\beta'}{\theta'}\cos\theta, \quad x = \alpha - \frac{\beta'}{\theta'},$$

$$y_1 = \frac{\alpha'}{\theta'}\cos\theta + \frac{\beta'}{\theta'}\sin\theta, \quad y = \beta + \frac{\alpha'}{\theta'};$$

or, if we take θ as the independent variable,

$$x_1 = \alpha'\sin\theta - \beta'\cos\theta, \qquad x = \alpha - \beta',$$
$$y_1 = \alpha'\cos\theta + \beta'\sin\theta, \qquad y = \beta + \alpha'.$$

To find the variations of I, we have

$$x_1' = \alpha''\sin\theta - \beta''\cos\theta + \alpha'\cos\theta + \beta'\sin\theta = \alpha''\sin\theta - \beta''\cos\theta + y_1,$$
$$y_1' = \alpha''\cos\theta + \beta''\sin\theta - \alpha'\sin\theta + \beta'\cos\theta = \alpha''\cos\theta + \beta''\sin\theta - x_1,$$
$$x' = \alpha' - \beta''. \qquad y' = \beta' + \alpha''.$$

Hence

$$x_1'^2 + y_1'^2 = x'^2 + y'^2,$$

which equation expresses that the motion is in fact a rolling one.

Imagine the two curves, and the points A, A_1 (fig. 2) were originally in contact; then the arcs AI, A_1I are equal, and, calling each of these s, and X, Y, X_1, Y_1 the coordinates of I in regard to the two sets of axes respectively, we have X, Y, X_1, Y_1 given functions of s, such that $X'^2 + Y'^2 = 1$, $X_1'^2 + Y_1'^2 = 1$, the accents now denoting differentiation in regard to s.

We have, from the figure,

$$\theta = \tan^{-1}\frac{Y'}{X'} - \tan^{-1}\frac{Y_1'}{X_1'},$$

or, what is the same thing,

$$\tan\theta = (Y'X_1' - Y_1'X') \div (X'X_1' + Y'Y_1'),$$

say

$$\sin\theta,\ \cos\theta = Y'X_1' - Y_1'X',\ X'X_1' + Y'Y_1';$$

and then, as before,

$$x = \alpha + x_1\cos\theta - y_1\sin\theta, \quad y = \beta + x_1\sin\theta + y_1\cos\theta;$$

or, what is the same thing,

$$x - X = \cos\theta\,(x_1 - X_1) - \sin\theta\,(y_1 - Y_1),$$
$$y - Y = \sin\theta\,(x_1 - X_1) + \cos\theta\,(y_1 - Y_1),$$

where X, Y, X_1, Y_1, and therefore also θ, denote given functions of s. The formulae will be of a like form if X, Y, X_1, Y_1 are given functions of a parameter t.

Two Points Describing Right Lines.

A well known but very interesting case is when two points of the moving plane describe right lines on the fixed plane.

Suppose that we have the points A, C (fig. 3) describing the right lines OA_1, OC_1 which meet in O; through A, C, with centre O_1 and radius $= 2OO_1$, describe a circle touching the first circle in a point I; and suppose that A_1, C_1 denote points on the second circle.

Then it is at once seen that, considering the first or small circle as belonging to the moving plane, and the second or large circle as belonging to the fixed plane, the motion is in fact the rolling motion of the small upon the large circle; and, moreover, that each point of the small circle describes a right line, which is a diameter of the large circle.

In fact, the angle A_1OC_1 is the double of the angle IOC at the circumference; that is, it is the double of the angle IO_1C_1; and therefore (the radius of the small circle being half that of the large circle) the arcs IC, IC_1 are equal, so that the rolling motion will carry the point C along the radius OC_1, and will, in like manner, carry the point A along the radius OA_1 originally assumed. And, in like manner, for any other point B of the small circle, the motion will carry B along the radius OB_1; in particular, taking AB a diameter, the angle A_1OB_1 will be a right angle; and the motion is determined by means of the two points A, B describing respectively the two lines OA_1, OB_1 at right angles to each other. There is no loss of generality in assuming that the two fixed lines are at right angles to each other.

It thence at once follows, as will presently appear, that each point of the moving plane describes an ellipse (but we have the special case already referred to, each point on the small circle describes a right line, and also the special case, the centre O_1 of the small circle describes a circle).

Considering any point Q of the moving plane, let the line QO_1 meet the small circle in the points E, F (or, what is the same thing, let E, F be the extremities of the diameter which passes through Q); then the points E, F describe right lines at right angles to each other, and Q is a point on EF or on this line produced; an ellipse having the lines OE, OF for the directions of its axes, and the lengths of the semi-axes being QE and QF respectively.

Analytical Theory (continued).

Reverting to the analytical formulae, the case is when α, β are given functions of θ such that two points (x_1, y_1), (x_2, y_2) have their coordinates in the fixed plane satisfying each of them a linear equation.

Suppose that for the point (x_1, y_1) we have $y_1 = mx_1$, and for (x_2, y_2) we have $y_2 = nx_2$; then the conditions are

$$\beta + x_1 \sin\theta + mx_1 \cos\theta = m(\alpha + x_1 \cos\theta - mx_1 \sin\theta),$$
$$\beta + x_2 \sin\theta + nx_2 \cos\theta = n(\alpha + x_2 \cos\theta - nx_2 \sin\theta);$$

that is,

$$\beta - m\alpha + x_1(\sin\theta + m\cos\theta - m\cos\theta + m^2 \sin\theta) = 0,$$
$$\beta - n\alpha + x_2(\sin\theta + n\cos\theta - n\cos\theta + n^2 \sin\theta) = 0,$$

or

$$x_1 = -\frac{\beta - m\alpha}{(1 + m^2)\sin\theta}, \qquad x_2 = -\frac{\beta - n\alpha}{(1 + n^2)\sin\theta}.$$

But in the particular case above considered, where the two lines are at right angles, we have $n = -\frac{1}{m}$, that is $1 + m^2$ and $1 + n^2$ are finite, and the formulae take a more simple form.

For a point (x_1, y_1), the locus in the fixed plane is given by the equations

$$x = \alpha + x_1 \cos\theta - y_1 \sin\theta, \qquad y = \beta + x_1 \sin\theta + y_1 \cos\theta,$$

and eliminating θ, we obtain the equation of the curve described by this point. If x_1, y_1 are the coordinates of any point whatever of the moving plane, the curve described is in general an ellipse; the special cases are when the point is on the circumference of the small circle (describing a right line) and when it is the centre O_1 (describing a circle).

The proof depends on the property that if E, F are extremities of a diameter of the small circle, and if these points describe right lines meeting at right angles in O, then any point Q on the diameter EF describes an ellipse with semi-axes equal to QE and QF, the directions of the axes being parallel to OE and OF respectively.

735. Note on the Theory of Apsidal Surfaces

[*From the Quarterly Journal of Pure and Applied Mathematics, vol.* XVI *(1879), pp. 109–112.*]

I obtain in the present Note a system of formulae which lead very simply to the known theorem, that the apsidals of reciprocal surfaces are reciprocal; or, what is the same thing, that the reciprocal of the apsidal of a given surface is the apsidal of its reciprocal; the surfaces are referred to the same axes, and by the reciprocal is meant the reciprocal surface in regard to a sphere radius unity, having for its centre a determinate point, say the origin; and it is this same point which is used in the construction of the apsidal surfaces. The apsidal of a given surface is constructed as follows; considering the section by any plane through the fixed point, and in this section the apsidal radii from the fixed point (that is, the radii which meet the curve at right angles), then drawing a line through the fixed point at right angles to the plane, and on this line measuring off from the fixed point distances equal to the apsidal radii respectively, the locus of the extremities of these distances is the apsidal surface. We have the surface, its reciprocal, the apsidal of the surface, the apsidal of the reciprocal; and I take (x, y, z), (x', y', z'), (X, Y, Z), (X', Y', Z') for the coordinates of corresponding points on the four surfaces respectively.

The condition of reciprocity gives $xx' + yy' + zz' - 1 = 0$, and (the equations being $U = 0$, $U' = 0$) x', y', z' proportional to $\partial_x U$, $\partial_y U$, $\partial_z U$; and x, y, z proportional to $\partial_{x'} U'$, $\partial_{y'} U'$, $\partial_{z'} U'$; or, what is the same thing, we must have

$$x'\,dx + y'\,dy + z'\,dz = 0, \qquad x\,dx' + y\,dy' + z\,dz' = 0;$$

one of these is implied in the other, as appears at once by differentiating the equation $xx' + yy' + zz' - 1 = 0$.

The other two surfaces will therefore be reciprocal if only we have the like relations between the coordinates (X, Y, Z) and (X', Y', Z'); that is, if

$$\begin{aligned} XX' + YY' + ZZ' - 1 &= 0,\\ X'\,dX + Y'\,dY + Z'\,dZ &= 0,\\ X\,dX' + Y\,dY' + Z\,dZ' &= 0. \end{aligned}$$

To find the apsidal surface, we consider an arbitrary section $x\cos\alpha + y\cos\beta + z\cos\gamma = 0$ of the surface $U = 0$, and seek to determine the apsidal radii thereof, that is, the maximum or minimum values of $R^2 = x^2 + y^2 + z^2$ when x, y, z vary subject to these two conditions.

Writing x', y', z' to denote functions proportional to $\partial_x U$, $\partial_y U$, $\partial_z U$, we thus have the set of equations

$$\begin{aligned} x + \lambda x' + \mu \cos\alpha &= 0, \\ y + \lambda y' + \mu \cos\beta &= 0, \\ z + \lambda z' + \mu \cos\gamma &= 0, \end{aligned}$$

where λ, μ are indeterminate coefficients; taking then X, Y, Z as the coordinates of the extremity of the line drawn at right angles to the plane, we have $R^2 = X^2 + Y^2 + Z^2$, and $\cos\alpha, \cos\beta, \cos\gamma = \frac{X}{R}, \frac{Y}{R}, \frac{Z}{R}$; substituting these values in the equation

$$X\cos\alpha + Y\cos\beta + Z\cos\gamma = 0,$$

we have $Xx + Yy + Zz = 0$; and substituting in the other equations, and instead of λ, μ introducing the new indeterminate coefficients ρ, σ, we obtain

$$X, Y, Z = \rho x + \sigma x',\ \ \rho y + \sigma y',\ \ \rho z + \sigma z'.$$

Hence these last equations, together with

$$R^2 = X^2 + Y^2 + Z^2 = x^2 + y^2 + z^2,$$

and

$$Xx + Yy + Zz = 1,$$

contain the solution of the problem.

If for convenience we introduce R'^2 to denote $x'^2 + y'^2 + z'^2$, and imagine the absolute values of x', y', z' determined so that $xx' + yy' + zz' = 1$, then substituting for X, Y, Z their values in the equations $X^2 + Y^2 + Z^2 = R^2$ and $Xx + Yy + Zz = 1$, we find

$$R^2 = \rho^2 R^2 + 2\rho\sigma + \sigma^2 R'^2, \qquad 1 = \rho R^2 + \sigma,$$

and thence

$$\rho = \frac{1 - R^2 R'^2}{R^2 - R'^2}, \qquad \sigma = \frac{R^2(R'^2 - 1)}{R^2 - R'^2};$$

or, finally assuming

$$\frac{1}{\sqrt{(R^2R'^2-1)}}=\frac{1}{\omega},$$

we have

$$X,Y,Z=\frac{x-R^2x'}{\omega},\ \frac{y-R^2y'}{\omega},\ \frac{z-R^2z'}{\omega},$$

each divided by $\sqrt{(R^2R'^2-1)}$.

Where I recall that x', y', z' are proportional to $\partial_x U$, $\partial_y U$, $\partial_z U$, and are such that $xx'+yy'+zz'=1$: they in fact denote $\partial_x U$, $\partial_y U$, $\partial_z U$, each divided by $x\partial_x U+y\partial_y U+z\partial_z U$; and that R^2 and R'^2 denote $x^2+y^2+z^2$ and $x'^2+y'^2+z'^2$ respectively. The coordinates X, Y, Z of the point of the apsidal surface are thus determined as functions of x, y, z.

For the apsidal of the reciprocal surface, we have in like manner

$$X',Y',Z'=\frac{x'-R'^2x}{\omega},\ \frac{y'-R'^2y}{\omega},\ \frac{z'-R'^2z}{\omega},$$

each divided by $\sqrt{(R^2R'^2-1)}$, and then the two sets of values give, not only

$$XX'+YY'+ZZ'=1,$$

as is obvious, but also

$$X'\,dX+Y'\,dY+Z'\,dZ=0,\qquad X\,dX'+Y\,dY'+Z\,dZ'=0.$$

In fact, writing for a moment ρ, ρ' instead of R^2, R'^2, and $\sqrt{(R^2R'^2-1)}=\sqrt{(\rho\rho'-1)}$, $=\omega$, then

$$\begin{aligned}X'\,dX+Y'\,dY+Z'\,dZ&=\frac{1}{\omega^2}\sum\frac{x'-x\rho'}{\omega}\,d\frac{x-x'\rho}{\omega}\\&=\frac{1}{\omega^3}\sum(x'-x\rho')(dx-\rho\,dx'-x'\,d\rho)\\&\quad-\frac{d\omega}{\omega^4}\sum(x'-x\rho')(x-x'\rho).\end{aligned}$$

Here

$$\begin{aligned}\sum(x'-x\rho')(dx-\rho\,dx'-x'\,d\rho) = (x'\,dx+y'\,dy+z'\,dz)&\\ -\rho(x'\,dx'+y'\,dy'+z'\,dz')&\\ -\rho'(x\,dx+y\,dy+z\,dz)&\\ +\rho\rho'(x\,dx'+y\,dy'+z\,dz')&\\ +\rho\,(xx'+yy'+zz')\,d\rho&\\ -\rho'\,(x^2+y^2+z^2)\,d\rho&\end{aligned}$$

or, since the terms in {} are

$$0-\rho\cdot\tfrac{1}{2}\,d\rho'-\rho'\cdot\tfrac{1}{2}\,d\rho+\rho\rho'\cdot 0+\rho\,d\rho-\rho'\rho\,d\rho=-\tfrac{1}{2}(\rho\,d\rho'+\rho'\,d\rho),$$

and

$$1-\rho\rho'-\rho\rho'+\rho\rho'=1-\rho\rho'=-\omega^2,$$

this is

$$=\frac{1}{\omega^3}\{-\tfrac{1}{2}(\rho'\,d\rho+\rho\,d\rho')+\omega\,d\omega\}=0,$$

in virtue of $\omega^2=\rho\rho'-1$.

And similarly the other equation $X\,dX'+Y\,dY'+Z\,dZ'=0$ might be directly verified.

736. Application of the Newton-Fourier Method to an Imaginary Root of an Equation

[*From the Quarterly Journal of Pure and Applied Mathematics, vol.* XVI *(1879), pp. 179–185.*]

I consider only the most simple case, that of a quadric equation $x^2 = n^2$, where n^2 is a given imaginary quantity, having the square roots n and $-n$; starting from an assumed approximate (imaginary) value $x = a$, we have $(a+h)^2 = n^2$, that is,

$$a^2 + 2ah = n^2, \qquad h = \frac{n^2 - a^2}{2a}, \quad a + h = \frac{a^2 + n^2}{2a};$$

that is, the successive values are

$$a_1 = \frac{a^2 + n^2}{2a}, \quad a_2 = \frac{a_1^2 + n^2}{2a_1}, \ \ldots;$$

and the question is, under what conditions do we thus approximate to one determinate root (selected out of the two roots at pleasure), say n, of the given equation.

The nearness of two values is measured by the modulus of their difference; thus a_1 is nearer to n than a is, means $\text{mod.}(a_1 - n) < \text{mod.}(a - n)$, and so in any step, for a_1, a_2, $\ldots$ in the course of the approximation; the step a to a_1 is regular if a_1 is nearer to n than a is, but otherwise it is irregular; the approximation is regular if all the steps are regular, and if (after one or more irregular steps) all the subsequent steps are regular, then the approximation becomes regular at the step which is the first of the unbroken series of regular steps.

We do by an approximation, which is ultimately regular, obtain the value n, if only the assumed value a is nearer to n than it is to $-n$; or, say, if the condition $\text{mod.}(a - n) < \text{mod.}(a + n)$ is satisfied, and the approximation is regular from the beginning if $\text{mod.}(a - n) < \frac{1}{2}\,\text{mod.}\, n$, viz. this condition is a sufficient one[3]; the

[3] In the Smith's Prize Examination, Jan. 28, 1879, I gave the theorem under the following form: "If a, n are imaginary quantities, the latter of them given, and the former assumed at pleasure, subject only to the condition $\text{mod.}(a-n) < \frac{1}{2}\,\text{mod.}\, n$; then if $a_1 = \frac{a^2+n^2}{2a}$, $a_2 = \frac{a_1^2+n^2}{2a_1}$, &c., show that the terms a, a_1, a_2, a_3, $\ldots$ will converge to the limit n." This is strictly true, but it would have been better to say "will converge regularly."

first step a to a_1 will moreover be regular under a less stringent condition imposed upon a; and it would seem that, without the condition $\mathrm{mod.}(a-n) < \frac{1}{2}\,\mathrm{mod.}\,n$ being satisfied, the subsequent steps will in some cases be also regular; that is, that the last-mentioned condition is not a necessary condition to the approximation being regular from the beginning; it is, however, the necessary and sufficient condition, to be satisfied by the modulus of $a-n$, in order that the approximation may be regular from the beginning. All this will clearly appear from the geometry.

We take N, N' (fig. 1) to represent the values n, $-n$; and similarly A, A_1, &c. to represent the quantities a, a_1, ...; we have then

$$AN = \mathrm{mod.}(a-n), \quad A_1N = \mathrm{mod.}(a_1-n), \ \ldots;$$

so that the approximation is measured by the approach of the points A, A_1 to N.

The line NN' joining the points N, N' passes through, and is bisected at, the origin O; drawing then QQ' through O at right angles to NN' the condition $\mathrm{mod.}(a-n) < \mathrm{mod.}(a+n)$ means that the point A, which represents the imaginary quantity a, lies on the N-side of QQ', and it will be assumed throughout that this is so.

Take now on the line ON, $OM = \frac{1}{2}ON$, and on $N'M$ as diameter, describe a circle, which may be called the "circle of unfitness"; regarding as an area the segment hereof which lies on the N-side of QQ', say this is the "segment of unfitness."

It will be shown that if, according as A is situate inside, on the boundary of, or outside the segment of unfitness, A_1N will be greater than, equal to, or less than AN. It may be added that, if A be within or upon the boundary of the segment of unfitness, then A_1 will be outside it, but this by no means hinders that the next point A_2, or some later point, shall be within the segment of unfitness; and, further, that when A is outside the segment of unfitness, then the next point A_1, or some later point, may very well be within the segment of unfitness; the conclusion is, that A being inside the segment of unfitness, A_1N is less than AN, but that it does not thence follow that A_2N is less than A_1N, A_3N than A_2N, ...; the approximation although regular at the first step, may then, or afterwards, for a step or steps, cease to be regular.

If, however, AN be less than $\frac{1}{2}ON$, that is, if the condition $\mathrm{mod.}(a-n) < \frac{1}{2}\,\mathrm{mod.}\,n$ be satisfied, then the point A lies within the circle, centre N, and radius NM, and is consequently outside the segment of unfitness; A_1N being less than AN, the point A_1 is *a fortiori* outside the segment of unfitness, and the like for all the subsequent points A_2, A_3, …, that is, in this case, the approximation is regular throughout. The circle, centre N, and radius NM, $= \frac{1}{2}\,\text{mod.}\,n$, may be called the "safe circle"; and the conclusion is that, if the point A or any subsequent point be within the safe circle, then every subsequent point will be within the safe circle, and the approximation will be regular.

The successive points A, A_1, A_2, … (or, as it will be convenient to call them, A_1, A_2, A_3, …) may be obtained each from the preceding one by a simple geometrical construction.

I recall that any circle through the two (imaginary) antipoints of N, N' is a circle having its centre on the indefinite line NN'; it is such that the ratio of the distances of a point thereof from the points N, N' respectively has a certain constant value, viz. it is < 1 for the circles with which we are here alone concerned, those which lie on the N-side of QQ'; the centres lie beyond the point N (further away, that is, from O).

Starting then from the given point A, for which this ratio $AN : AN'$ has a given value, suppose $AN = k{\cdot}AN'$, we describe a first circle (passing of course through A) for each point of which this ratio has the same value k; let the diameter of this circle be V_1W_1, V_1 being the extremity between N and W_1 (not shown in the figure), that beyond N; we then describe a second circle, and N, W_1 its diameter be V_2W_2, V_2 being the extremity between N (or say $= k^{-1}$) and N (or say between N and W_1); the point A_2 lies on this second circle, and is determined as the single intersection of the line V_2A_1 with the second circle. And of course drawing a third circle, for which the ratio is $= k$, on the diameter V_3W_3, then A_3 lies on the third circle, and is the intersection with it of the line V_3A_2, and so on; the radii of the successive circles diminish very rapidly, their centres, in like manner, continually approaching the point N; hence, the points A_1, A_2, A_3, …, which lie on the several circles respectively, approximate, and that very rapidly, to the point N.

But by what precedes, if, for instance, the point A_1 be within the segment of unfitness, then also some of the subsequent points may be within the segment of unfitness, and for each such point A_p, $NA_{p+1} > NA_p$; it is, however, clear that we always arrive at a point A_q, such that $NA_{q+1} > NA_q$; it so that $NA_q < \frac{1}{2}ON$, and so soon as such a point is arrived at the approximation becomes regular.

The point A_2 determined from A_1, as above, is a point such that the subtended angle NA_2N' is = twice the subtended angle NA_1N'; or calling the latter angle ϕ, the former is $= 2\phi$.

It is, in fact, this property which gives rise to the construction; for let the values of A_1N, A_1N', regarded as imaginary quantities, be called for a moment

$$\rho_1(\cos\theta_1 + i\sin\theta_1), \qquad \rho_1'(\cos\theta_1' + i\sin\theta_1');$$

and, similarly, those of A_2N, A_2N' be called

$$\rho_2(\cos\theta_2 + i\sin\theta_2), \qquad \rho_2'(\cos\theta_2' + i\sin\theta_2');$$

then, from the equation $a_2 = \frac{a_1^2+n^2}{2a_1}$, we have

$$\frac{1}{A_2N} = \frac{1}{2}\left(\frac{1}{A_1N} + \frac{1}{A_1N'}\right),$$

and thence the property in question.

737. On a Covariant Formula

[*From the Quarterly Journal of Pure and Applied Mathematics, vol.* XVI *(1879), pp. 224–226.*]

Starting from the equation which presents itself in the Newton-Fourier problem, it is easy to see that, if a be a root of the equation $fx = 0$, then

$$(x - x_1)f'x - fx$$

contains the factor $(x-a)^2$, that is, the equation $(x-x_1)f'x-fx = 0$, considered as an equation in x containing the parameter x_1, will have a twofold root $x_1 = a$, equal to any root a of the equation $fx = 0$; and, consequently, the discriminant in regard to x of the function $(x - x_1)f'x - fx$ will contain the factor fx_1.

But if fx be of the order n, then the discriminant is of the order $2n - 2$ in x_1, and there is consequently a remaining factor Ψ_1 of the order $n - 2$.

The like theorem applies to the homogeneous form

$$(xy_1 - x_1y)(\alpha\partial_\xi + \beta\partial_\eta)f(\xi,\eta) - (\alpha y - \beta x)f(\xi,\eta),$$

which reduces itself to the foregoing on writing $\alpha = 1$, $\beta = 0$, $y = y_1 = 1$; or, changing the notation, say to the form

$$(\xi y - x\eta)(\alpha\partial_\xi + \beta\partial_\eta)f(\xi,\eta) - (\alpha y - \beta x)f(\xi,\eta),$$

viz. the discriminant hereof in regard to ξ, η, being a function homogeneous of the order $2n-2$ in x, y, and in α, β, and to the coefficients of $f(\xi,\eta)$, will contain the factor $f(x,y)$, and there will be consequently a remaining factor of the order $n-2$ in (x,y), $2n-2$ in (α,β) and $2n-3$ in the coefficients of $f(\xi,\eta)$.

The most simple case is when $f(\xi,\eta)$ is the quadric function $(a,b,c \mid \xi,\eta)^2$.

The form here is

$$(\xi y-x\eta)\{2(a\alpha+b\beta)\xi+(b\alpha+c\beta)\eta\}-(\alpha y-\beta x)(a,b,c \mid \xi,\eta)^2 = (\mathfrak{a},\mathfrak{b},\mathfrak{c} \mid \xi,\eta)^2,$$

where the coefficients are

$$\begin{aligned}
\mathfrak{a} &= 2y(a\alpha + b\beta) - a(\alpha y - \beta x) = a\alpha x + (a\alpha + 2b\beta)y,\\
\mathfrak{b} &= y(b\alpha + c\beta) - x(a\alpha + b\beta) - b(\alpha y - \beta x)\\
&= -(ba + c\beta)x + (aa + 2b\beta)y,\\
\mathfrak{c} &= -2x(b\alpha + c\beta) - c(\alpha y - \beta x) = -(2b\alpha + c\beta)x - c\alpha y;
\end{aligned}$$

and we then have

$$\begin{aligned} ac - b^2 &= -(2b\alpha\beta + c\beta^2)ax^2 \\ &\qquad - \{2ab\alpha^2 + (2ac + 4b^2)\alpha\beta + 2bc\beta^2\}xy - (aa^2 + 2ba\beta)cy^2 \\ &= -(a\alpha^2 + 2b\alpha\beta + c\beta^2)(ax^2 + 2bxy + cy^2). \end{aligned}$$

The discriminant is in this case

$$= -(a, b, c \mid \alpha, \beta)^2 \cdot (a, b, c \mid x, y)^2.$$

In the case of the cubic function $(a, b, c, d \mid \xi, \eta)^3$, the form is

$$(\xi y - x\eta)\{3(a\alpha + b\beta,\ b\alpha + c\beta,\ c\alpha + d\beta \mid \xi, \eta)^2\} - (\alpha y - \beta x)(a, b, c, d \mid \xi, \eta)^3 = (a, b, c, d \mid \xi$$

the values of the coefficients being

$$\begin{aligned} a &= a\alpha x && + (2a\alpha + b\beta)y, \\ b &= -a\alpha x && + (b\alpha + 2c\beta)y, \\ c &= -(2b\alpha + c\beta)x && + (c\alpha + 2d\beta)y, \\ d &= -(3c\alpha + 2d\beta)x && - d\alpha y. \end{aligned}$$

Attending only to the terms in x^2, we have

$$\begin{aligned} ac - b^2 &= -(a\alpha^2 + 2b\alpha\beta + c\beta^2)ax^2, \\ ad - bc &= -2(b\alpha^2 + 2c\alpha\beta + d\beta^2)a\alpha x^2, \\ bd - c^2 &= \{(3ac - 4b^2)\alpha^2 + (2ad - 4bc)\alpha\beta - c^2\beta^2\}ax^2. \end{aligned}$$

And hence, in

$$a^2d^2 + 4ac^3 + 4b^3d - 3b^2c^2 - 6abcd = (ad - bc)^2 - 4(ac - b^2)(bd - c^2),$$

we have the term

$$\begin{aligned} &4a\alpha^2 \cdot c \cdot [a(b\alpha^2 + 2c\alpha\beta + d\beta^2)^2 \\ &\quad + (a\alpha^2 + 2b\alpha\beta + c\beta^2)\{(3ac - 4b^2)\alpha^2 + (2ad - 4bc)\alpha\beta - c^2\beta^2\}] \end{aligned}$$

y^4, and assuming that the whole divides by $(a, b, c, d \mid x, y)^3$, then, forming the analogous term in y^4, and also expanding the $\alpha\beta$-functions within the square brackets, we find

$$\begin{aligned} &\text{Discriminant} = 4(a, b, c, d \mid x, y)^3 \text{ multiplied by} \\ &\begin{pmatrix} 3a^2c - 3ab^2 & a^2d + 6abc - 8b^3 \\ 2a^2d + 6abc - 8b^3 & 6abd + 6ad^2 - 12b^2c \end{pmatrix} \mid (\alpha, \beta)^2(\alpha\, dy - \beta\, dx) \\ &\qquad\qquad + (\alpha y - \beta x)\Psi. \end{aligned}$$

Writing down the Hessian of $(a, b, c, d \mid \alpha, \beta)^3$,

$$H = (ac - b^2,\ ad - bc,\ bd - c^2 \mid \alpha, \beta)^2,$$

and the cubicovariant

$$\Phi = \begin{pmatrix} a^2d - 3abc + 2b^3 \\ abd - 2ac^2 + b^2c \\ -acd + 2b^2d - bc^2 \\ -ad^2 + 3bcd - 2c^3 \end{pmatrix} (\alpha, \beta)^3,$$

it is easy to see that the coefficient of x is

$$= 3(a, b, c, d \mid \alpha, \beta)^2 \cdot (a\Phi - \beta\Psi),$$

hence also that of y is

$$= 3(b, c, d \mid \alpha, \beta)^2 \cdot (\beta\Phi + \alpha\Psi),$$

and the final result is that the discriminant $= 4(a, b, c, d \mid x, y)^3$ multiplied by

$$\{3(a, b, c, d \mid \alpha, \beta)^2\, (x, y)H + (\alpha y - \beta x)\Phi\}^2.$$

It would be interesting to calculate the result for the quartic $(a, b, c, d, e \mid \xi, \eta)^4$.

March 14, 1879.

738. Note on a Hypergeometric Series

[*From the Quarterly Journal of Pure and Applied Mathematics, vol.* XVI *(1879), pp. 268–270.*]

In the memoir on hypergeometric series, Schwarz, "Ueber diejenigen Fälle, &c.," *Crelle*, t. LXXV (1873), pp. 292–335, the author shows, as part of his general theory, that the equation

$$\frac{d^2y}{dx^2}+\frac{\frac{1}{2}-x}{x(1-x)}\frac{dy}{dx}+\frac{\frac{1}{48}}{x(1-x)}y=0,$$

which belongs to the hypergeometric series $F(\frac{1}{12},-\frac{1}{12},\frac{1}{2},x)$, is algebraically integrable, having in fact the two particular integrals

$$y^2=\sqrt{(a-a^5x^2)}+\sqrt{(-a^5+a^3x^2)},$$

where a is a prime sixth root of -1, $a^6+1=0$, or say $a^4-a^2+1=0$ (see p. 326, a being for greater simplicity written instead of s'', and the form being somewhat simplified).

It is interesting to verify this directly; writing first $y=\sqrt{(Y)}$ and then $x=X^2$, the equation between Y, X is easily found to be

$$\frac{d^2Y}{dX^2}-\frac{1}{2Y}\left(\frac{dY}{dX}\right)^2+\frac{1-3X^2}{X(1-X^2)}\frac{dY}{dX}+\frac{1}{3(1-X^2)}Y=0,$$

and the theorem in effect is that that equation has the two particular integrals

$$Y=P\pm Q,$$

P and Q being linear functions of X: in fact,

$$P=a-a^5X,\qquad Q=-a^5+aX.$$

Starting say from the equation

$$Y=\sqrt{(P)}+\sqrt{(Q)},$$

or, as it is convenient to write it,

$$\eta=\sqrt{(P)}+\sqrt{(Q)},$$

where P and Q are assumed to be linear functions of X, we have

$$\frac{d\eta}{dX}=\tfrac{1}{2}P^{-\frac{1}{2}}P'+\tfrac{1}{2}Q^{-\frac{1}{2}}Q',$$

$$\frac{d^2\eta}{dX^2} = -\tfrac{1}{4}P^{-\frac{3}{2}}P'^2 - \tfrac{1}{4}Q^{-\frac{3}{2}}Q'^2 - \tfrac{1}{4}P^{-\frac{1}{2}}Q^{-\frac{1}{2}}P'' - \tfrac{1}{4}P^{-\frac{1}{2}}Q^{-\frac{1}{2}}Q''$$

where P', Q' are written to denote the derived functions of P, Q respectively.

Substituting these values, the resulting equation contains on the left-hand side a rational part, and a part with the factor $P^{-\frac{1}{2}}Q^{-\frac{1}{2}}$, and it is clear the equation can only be true if these two parts are separately $= 0$. We have thus two equations which ought to be verified; viz. after a slight reduction these are found to be

$$\begin{aligned} PQ(P''Q + PQ'') + \tfrac{1}{2}(P'^2 + Q'^2) - \tfrac{1}{4}(P+Q) &= 0, \\ \tfrac{1}{2}(Q'P'' + P'Q'') + P'Q' + \tfrac{1}{4}PQ(PQ + P'Q') - \tfrac{1}{48}P'Q &= 0, \end{aligned}$$

and it is very interesting to observe the manner in which these equations are, in fact, verified by the foregoing values of P, Q.

We have

$$P + Q = (a - a^5)(1 + X), \quad P' + Q' = a - a^5,$$

and hence

$$2X(P' + Q') - X(P+Q) = -(a - a^5)(1 - X),$$

or, in the first equation, the second part

$$\begin{aligned} \tfrac{1}{2}(P'^2 + Q'^2) - \tfrac{1}{4}(P+Q) &= -(a - a^5)\frac{1 - X^2}{X(1 - X^2)} \\ &= -(a - a^5)\frac{1}{X}, \end{aligned}$$

while the first part is

$$PQ(P''Q + PQ'') = PQ \cdot 0 = 0,$$

since $P'' = Q'' = 0$, P and Q being linear functions. The first equation is thus verified.

For the second equation, we have

$$P' = -a^5, \quad Q' = a, \quad P'' = Q'' = 0,$$

and hence

$$\tfrac{1}{2}(Q'P'' + P'Q'') = 0,$$

$$P'Q' = -a^6 = 1, \quad \text{since } a^6 = -1,$$

$$\tfrac{1}{4}PQ(PQ+P'Q') = \tfrac{1}{4}(a-a^5X)(-a^5+aX)(1+1) = \tfrac{1}{2}(-a^6+a^2X+a^2X^2-a^6X^3),$$

$$= \tfrac{1}{2}(1 + 2a^2X - X^3), \quad \text{since } a^6 = -1,$$

and

$$-\tfrac{1}{48}P'Q = \tfrac{1}{48}a^5(-a^5 + aX) = \tfrac{1}{48}(a^{10} - a^6X) = \tfrac{1}{48}(-a^4 + X),$$

since $a^{10} = -a^4$ (from $a^6 = -1$, multiplying by a^4).

Collecting the terms, the left-hand side of the second equation is

$$0 + 1 + \tfrac{1}{2}(1 + 2a^2X - X^3) + \tfrac{1}{48}(-a^4 + X),$$

which vanishes identically in virtue of the relation $a^4 - a^2 + 1 = 0$, as may be verified by substituting $a^4 = a^2 - 1$.

745.

ON THE SCHWARZIAN DERIVATIVE AND THE POLYHEDRAL FUNCTIONS.

[From the *Philosophical Transactions*, vol. CLXXIII (1879), pp. 611–659.]

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The last-mentioned differential equation is considered by Klein and Brioschi: the solutions in 13 cases, or such of them as had not been given by Schwarz, were obtained by Brioschi: and those of the remaining 3 cases, subject to a correction in one of them, were afterwards obtained by Klein.

The first part of the present Memoir relates, say to the foregoing equation

$$\{s, x\} = (a, b, c \frown\frown) \left(\frac{1}{x-a}, \frac{1}{x-b}, \frac{1}{x-c}\right),$$

although the other form in $[x, z]$ may equally well be regarded as the fundamental form. We consider in the theory:

A. The Derivative $\{s, x\}$, meaning as above explained.

B. Quadric functions of any three or more inverts $\frac{1}{x-a}$.

C. Rational and integral functions P, Q, R having a sum $= 0$, and which are such that $QR' - Q'R$, $RP' - R'P$, $PQ' - P'Q$, contains only the factors of P, Q, R.

D. The differential equation of the third order.

E. The Schwarzian theory in regard to conformable figures and the corresponding values of the imaginary variables s and x.

F. Connexion with the differential equation for the hypergeometric series.

The second part of the Memoir relates to the Polyhedral Functions. The paragraphs of the whole Memoir are numbered consecutively.

PART I. The Derivative $\{s, x\}$.

Art. Art. Nos. 1 to 7.

Art. The derivative $\{s, x\}$ may be transformed in regard to either or both of the variables. Suppose, first, that s is a function of the new variable S, (hence also S is a function of x): using subscript numbers to denote differentiations in regard to S, and the accents as before for differentiations in regard to x, we have

$$\frac{s'}{S'} = \frac{ds}{dS}, \quad \frac{s''}{S'^2} - \frac{s'S''}{S'^3} = \frac{d^2s}{dS^2},$$

whence, differentiating the logarithms,

$$\frac{s''}{s'} - \frac{S''}{S'} = \frac{1}{\frac{ds}{dS}} \frac{d^2s}{dS^2} \cdot S',$$

and consequently

$$\frac{s'''}{s'} - \frac{s''^2}{s'^2} - \frac{S'''}{S'} + \frac{S''^2}{S'^2} = \frac{d}{dx}\left(\frac{1}{\frac{ds}{dS}} \frac{d^2s}{dS^2}\right) S'^2 + \frac{1}{\frac{ds}{dS}} \frac{d^2s}{dS^2} \cdot S'';$$

that is,

$$\{S, x\} = \{s, x\} - \left(\frac{dS}{dx}\right)^2 \{s, S\}.$$

Art. Again differentiating, we have the required formula.

In a very similar manner, taking x a function of X, it is shown that

$$\{s, X\} = \{s, x\} - \left(\frac{dx}{dX}\right)^2 \{x, X\}.$$

Art. If in this formula we write S for s, and substitute the resulting value of $\{S, x\}$ in the former formula, we have

$$\{S, X\} = \{s, x\} - \left(\frac{dS}{dx}\right)^2 \{s, S\} - \left(\frac{dx}{dX}\right)^2 \{x, X\},$$

which is the formula for the change of both variables.

It, in fact, includes the other two: viz. writing $X = x$, or $S = s$, and observing that $\{s, s\} = 0 = \{x, x\}$, we have the other two formulae.

Art. By putting in the first formula $X = x$, we obtain a formula for the interchange of the variables.

Art. Writing $S = \frac{as+b}{cs+d}$, and using for a moment the accents to denote differentiation in regard to s, we have

$$S' = \frac{ad - bc}{(cs + d)^2}, \quad \frac{S''}{S'} = -\frac{2c}{cs + d},$$

and thence

$$\frac{S'''}{S'} - \frac{3}{2}\left(\frac{S''}{S'}\right)^2 = 0;$$

that is, $\{S, s\} = 0$, whence also $\{s, S\} = 0$.

Art. Hence in the first formula $\{S, x\} = \{s, x\}$, that is,

$$\left\{\frac{as+b}{cs+d},\ x\right\} = \{s, x\};$$

viz. we may, in the derivative $\{s, x\}$, write for s any homographic function $(as+b)/(cs+d)$ of s.

Art. Again, if $X = \frac{\alpha x+\beta}{\gamma x+\delta}$, then from the second formula

$$\{s, X\} = \{s, x\} - \frac{(\alpha\delta-\beta\gamma)^2}{(\gamma x+\delta)^4}\left\{x,\ \frac{\alpha x+\beta}{\gamma x+\delta}\right\},$$

that is,

$$\left\{s,\ \frac{\alpha x+\beta}{\gamma x+\delta}\right\} = \{s, x\} - \frac{(\alpha\delta-\beta\gamma)^2}{(\gamma x+\delta)^4}\left\{x,\ \frac{\alpha x+\beta}{\gamma x+\delta}\right\},$$

and here, changing s into $(aS+b)/(cS+d)$, we have finally

$$\left\{\frac{as+b}{cs+d},\ \frac{\alpha x+\beta}{\gamma x+\delta}\right\} = \frac{(\gamma x+\delta)^4}{(\alpha\delta-\beta\gamma)^2}\{S, x\},$$

which is the formula for the homographic transformation of the two variables s, x.

Art. Let s be a given function of x, the equation $\{S, x\} = \{s, x\}$ is a differential equation of the third order in S, and by what precedes, its general integral is

$$S = \frac{as+b}{cs+d}.$$

The direct process is as follows: we have a first integral

$$\frac{S''}{S'} = \frac{s''}{s'} - \frac{2cs'}{cs+d};$$

a second integral $\log S' = \log s' - 2\log(cs+d) + \text{const.}$, that is,

$$S' = \frac{A\cdot s'}{(cs+d)^2};$$

and thence a final integral $S = B - \frac{A}{c(cs+d)}$, which is equivalent to the foregoing value of S.

The Quadric Function of three or more Inverts. Art. Nos. 8 to 15.

Art. We consider a quadric function of any number of inverts $\frac{1}{x-a}, \frac{1}{x-\alpha}, \ldots$, all of them different: it is assumed that the constant term is $= 0$, and also that the sum of the coefficients of the linear terms is $= 0$. We have therefore square terms $\frac{a}{(x-a)^2}$, product terms $\frac{h}{(x-a)(x-\beta)}$, and linear terms $\frac{A}{x-a}$, where the sum of the coefficients A is $= 0$.

Any product term $\frac{h}{(x-a)(x-\beta)}$ is expressible in the form of a difference of two linear terms, $\frac{h}{a-\beta}\left(\frac{1}{x-a} - \frac{1}{x-\beta}\right)$, and (the coefficients of these being equal), after it is thus expressed, the sum of the coefficients of the linear terms is still $= 0$.

The function is thus always expressible in the form

$$\frac{a}{(x-a)^2} + \frac{b}{(x-\beta)^2} + \cdots + \frac{A}{x-a} + \frac{B}{x-\beta} + \cdots,$$

where the sum $A + B + \ldots$ is $= 0$: this may be called the *reduced form.*

Art. Observe that any particular invert may disappear altogether from the reduced form: this will be the case if $a = 0$, that is, if the original form contains no term in $\frac{1}{(x-a)^2}$, and if also $A = 0$.

An invert thus disappearing from the reduced form is said to be *non-essential*: and the inverts which do not disappear are said to be *essential.* The original form contains in appearance the non-essential inverts, but it is really a quadric function of the essential inverts only.

Art. Imagine the original function expressed as a rational function of x, say

$$\frac{U}{V^2},$$

where $V = (x-a)(x-\beta)\cdots$ is the product of the linear factors (corresponding to the several inverts), each taken with the index 1; and U is a function of the same degree as V^2.

Art. In particular, if the number of essential inverts is $= 3$, then the quadric function is of the form

$$(a, b, c, f, g, h \frown x-a, x-\beta, x-\gamma)^2,$$

which contains one superfluous constant, and equivalent functions differ only by a multiple of

$$\frac{1}{(x-\beta)(x-\gamma)}+\frac{1}{(x-\gamma)(x-\alpha)}+\frac{1}{(x-\alpha)(x-\beta)}=0.$$

Art. A quadric function such that the degree of the numerator is less by 4 than that of the denominator is said to be "curtate."

The conditions, in order that the function

$$(a,b,c,f,g,h \frown x-a, x-\beta, x-\gamma)^2$$

may be curtate, are easily found to be

$$\begin{aligned} a+b+c+2f+2g+2h&=0,\\ \alpha(h+b+f)+\beta(g+f+c)+\gamma(a+h+g)&=0; \end{aligned}$$

and by reason of the superfluous constant we are at liberty to assume a third condition: the three conditions may be taken to be

$$a+h+g, \quad h+b+f, \quad g+f+c \quad \text{each} = 0;$$

and these being satisfied, the quadric function is curtate.

Art. The function

$$\frac{a}{x-\alpha}+\frac{b}{x-\beta}+\frac{c}{x-\gamma}+\cdots,$$

where the coefficients a, b, c, $\ldots$ satisfy the relation $a+b+c+\ldots=-2$, is diaphoric, and therefore curtate.

In fact, forming the sum,

$$\text{coeff.}\ \frac{1}{(x-\alpha)^2}+(\text{coeff.}\ \frac{1}{(x-\alpha)(x-\beta)}+\cdots),$$

this is $-a-\frac{1}{2}a-\frac{1}{2}b-\frac{1}{2}c-\ldots=-\frac{1}{2}(2+a+b+c+\ldots)$, which is $=0$; and similarly the other conditions are satisfied.

Art. The function regarded as a function of the inverts

$$\frac{1}{x-a},\frac{1}{x-a_1},\ldots;\frac{1}{x-b},\frac{1}{x-b_1},\ldots;\frac{1}{x-c},\frac{1}{x-c_1},\ldots,$$

where $a+a_1+\ldots=b+b_1+\ldots=c+c_1+\ldots=h$ suppose, is diaphoric, and therefore curtate.

Rational and integral functions P, Q, R having a sum $= 0$. Art. Nos. 16 to 20.

Art. The functions P, Q, R are functions of a variable z; for the present purpose they need only be rational and integral functions of z, but they are really functions connected with the theory of the inverts as will appear. They are each of them of the same order S; any two functions not originally of the same order, can thus be made to be of the same order; or by taking account of the root $z = \infty$, we may in the original case regard them as being of the same order, and it is convenient so to regard them: say they are taken to be of the same order S.

And there is clearly no loss of generality in taking the three functions to be prime to each other; for any common factor of two of them would divide the third, and might therefore be struck out.

Art. We may therefore write

$$P = F\prod(z-l)^p, \quad Q = G\prod(z-m)^q, \quad R = H\prod(z-n)^r,$$

where $(z-l)^p$ is taken to denote the distinct simple or multiple factors of P, and the like as regards Q and R; the factors $z-l$, $z-m$, $z-n$ are thus all of them different.

And we have

$$S = \Sigma p = \Sigma q = \Sigma n.$$

Art. It is at once seen that $QR' - Q'R$ is of the degree $2S-2$, and moreover that it contains the factors $\prod(z-l)^{p-1}$, $\prod(z-m)^{q-1}$, $\prod(z-n)^{r-1}$; hence it contains the factor

$$\prod(z-l)^{p-1}(z-m)^{q-1}(z-n)^{r-1}.$$

Suppose the number of distinct indices p is σ_1, that of distinct indices q is σ_2, and that of distinct indices r is σ_3; then the degree of the factor is $= 3S - \sigma_1 - \sigma_2 - \sigma_3$; and if this be $= 2S-2$, then Θ can have no other variable factor: viz. if the numbers σ_1, σ_2, σ_3 of the distinct indices p, q, r respectively are such that $\sigma_1 + \sigma_2 + \sigma_3 = S + 2$, a relation which is henceforth taken to be satisfied, then we have

$$\Theta = K\prod(z-l)^{p-1}(z-m)^{q-1}(z-n)^{r-1}.$$

Art. As already in effect remarked, the conclusion extends to the other functions $RP' - R'P$ and $PQ' - P'Q$; and we thus see that the condition in order that each of these functions may contain only the factors of P, Q, R is

$$\sigma_1 + \sigma_2 + \sigma_3 = S + 2.$$

Art. The case most closely connected with the theory of inverts is when each of the indices p, each of the indices q, and each of the indices r is $= 1$; the condition then is $\sigma_1 + \sigma_2 + \sigma_3 = S + 2$, where $S = \sigma_1 = \sigma_2 = \sigma_3$, that is, $3S = S + 2$, or $S = 1$. But this gives only the case $P : Q : R = 1 : -1 : 0$.

The Differential Equations $\{x, z\}$ and $\{s, x\}$. Art. Nos. 21 to 45.

Art. In reference to what follows, it is convenient to put $P = \lambda P_0$, $P_\lambda = \lambda^{-1} P$, where P_0 is written for $\prod(z - l)^{p-1}$, the G.C.M. of P and P'; and λ is consequently $= F$ multiplied by the product $\prod(z-l)$ of the several factors taken each with the index unity; and so for Q and R: viz. we write

$$P,\ Q,\ R = \lambda P_0,\ \mu Q_0,\ \nu R_0, \quad P_\lambda,\ Q_\mu,\ R_\nu = \lambda^{-1}P,\ \mu^{-1}Q,\ \nu^{-1}R,$$

and the foregoing value of Θ then is $\Theta = K P_0 Q_0 R_0$.

We come now to the investigation of the leading theorem.

Take a, b, c arbitrary, $f, g, h = b-c,\ c-a,\ a-b$; P, Q, R functions of z as above; and write

$$f(x-a) : g(x-b) : h(x-c) = P : Q : R,$$

equations, which are consistent with each other and determine x as a rational function of z.

Using, as before, the accent to denote differentiation in regard to z, and taking the coefficients (a, b, c) arbitrary, it is required to find the value of

$$\{x, z\} + x'^2(a, b, c \frown x-a, x-b, x-c)^2.$$

Art. [Calculation of the first term $\{x, z\}$] We have $x =$ a function $\frac{\alpha P + \beta R}{\gamma P + \delta R}$, and thence $\{x, z\} = \{s, z\}$ where $s = P/R$ for a moment; then

$$\frac{P}{R} : \frac{Q}{R} : 1 = \frac{P_0 Q_0 Z}{Z^2 R_0} : \frac{P_0 Q_0}{Z \cdot R_0} : \frac{P_0 Q_0}{Z^2 R_0},$$

that is,

$$\frac{P}{R} = \frac{P_0 Q_0 Z}{Z^2 R_0}, \quad \frac{Q}{R} = \frac{P_0 Q_0}{Z \cdot R_0}, \quad 1 = \frac{P_0 Q_0}{Z^2 R_0}.$$

Substituting the values

$$P_0 = \prod(z-l)^{p-1}, \quad Q_0 = \prod(z-m)^{q-1}, \quad R_0 = \prod(z-n)^{r-1}, \quad Z = \prod(z-n),$$

we have

$$\{x, z\} = \frac{P_0''}{P_0} - \frac{Q_0''}{Q_0} + 4\frac{Z'}{Z}\left(\frac{P_0'}{P_0} - \frac{Q_0'}{Q_0} + \frac{Z'}{Z}\right) - 2\left(\frac{P_0'}{P_0} - \frac{Q_0'}{Q_0} + \frac{Z'}{Z}\right)^2$$
$$-\frac{P_0'}{P_0} \cdot \frac{Q_0'}{Q_0} + \left(\frac{P_0'}{P_0} - \frac{Q_0'}{Q_0}\right)\frac{Z'}{Z} + 2\left(\frac{Z'}{Z}\right)^2$$

Art. where it is to be observed that

$$\Sigma(p-1)+\Sigma(q-1)-\Sigma(r+1) = S-\sigma_1+S-\sigma_2-(S+\sigma_3) = S-\sigma_1-\sigma_2-\sigma_3 = -2;$$

consequently the function is diaphoric, and therefore curtate.

It is to be remarked that the function, although presenting itself in a form unsymmetric in regard to the factors of P and Q, and of R, is really symmetric as regards the three sets of factors; this is obvious *a priori*, and it will be presently verified.

Art. [Calculation of the second term] For the calculation of the second term

$$x'^2(a, b, c \frown x-a, x-b, x-c)^2,$$

we have

$$f(x-a),\ g(x-b),\ h(x-c) = P,\ Q,\ R.$$

Art. We have to bring the function on the right-hand side into the reduced form

$$\frac{a}{(z-l)^2} + \frac{A}{z-l},$$

for the purpose of getting rid of the non-essential inverts (if any).

We write

$$\frac{P''}{P} = \Sigma\frac{(p-1)}{(z-l)^2} + \Sigma\frac{(p-1)}{z-l} \cdot \frac{P_0'}{P_0} + \cdots$$

viz. $z-l$ here denotes any particular factor, and $z-l_1$ represents any other factor of the same set; and so in other like cases.

Art. The whole coefficient of $\frac{1}{(z-l)^2}$ is

$$-(p-1) - \frac{1}{2}(p-1)^2 + \frac{1}{2}p(p+1) = \frac{1}{2}(1-p^2) + ap = (ap);$$

an expression which, regarded as a function of a and p, is represented by (ap): the parentheses are used only to avoid ambiguity, and are omitted when there is no ambiguity.

Art. The whole coefficient of $\frac{1}{z-l}$, when $z-l$ is a multiple factor of P, thus is

$$(ap) \cdot \frac{1}{z-l} + \text{terms in other inverts},$$

a form which is now symmetrical in regard to the inverts.

Art. The value just obtained must be equal to

$$\frac{(ap)}{(z-l)^2} + \frac{A}{z-l} + \cdots + x'^2(a, b, c \frown x-a, x-b, x-c)^2;$$

viz. comparing the two forms and reducing, they will be identical if only

$$\frac{(ap)}{z-l} + \frac{(bq)}{z-m} + \frac{(cr)}{z-n} = 0,$$

and it can be shown that the function inside the $\{\}$ is in fact $= 0$.

Art. We have, as before, $\Sigma \frac{p}{z-l} = \Sigma \frac{P'}{P}$; or writing each of these quantities $= \phi$, the equation to be verified is

$$\frac{(ap)}{z-l} + \frac{(bq)}{z-m} + \frac{(cr)}{z-n} = 0.$$

Art. But from the equation $\Theta = PQ' - P'Q = KP_0Q_0R_0$, we find $XY' - X'Y = KR_0$; and then, differentiating, $ZF_0' + Z'F_0 - Z_0'F - Z_0F' = K'R_0 + KR_0'$: writing in these equations $z = l$, they become

$$-X_0F = KR_0, \quad X'Y_0 - X_0'Y - X_0Y' = KR_0',$$

so that, dividing the second by the first,

$$\frac{X'Y_0 - X_0'Y - X_0Y'}{X_0F} = \frac{R_0'}{R_0},$$

or, recollecting that $X_0 = pX'$ and $Y = qY_0$, we have

$$\frac{X'}{X_0} - \frac{Y'}{Y_0} = \frac{R_0'}{R_0},$$

that is,

$$\frac{p}{z-l} - \frac{q}{z-m} = \frac{R_0'}{R_0},$$

the required relation.

Art. The result is that, $z-l$ being a multiple factor of P, the coefficient of $\frac{1}{(z-l)^2}$ is $(ap) = \frac{1}{2}(1-p^2)+ap$, and the coefficient of $\frac{1}{z-l}$ is a sum of terms each containing the factor (ap).

Art. The formulae for the coefficients of $\frac{1}{(z-m)^2}$, $\frac{1}{z-m}$, and $\frac{1}{(z-n)^2}$, $\frac{1}{z-n}$ give at once, by a mere change of letters, those for the coefficients of $\frac{1}{(z-n)^2}$, $\frac{1}{z-n}$; and the function in question,

$$\{x, z\} + x'^2(a, b, c \frown x-a, x-b, x-c)^2,$$

is now obtained in the required form

$$\frac{(ap)}{(z-l)^2} + \frac{(bq)}{(z-m)^2} + \frac{(cr)}{(z-n)^2} + \frac{A}{z-l} + \frac{B}{z-m} + \frac{C}{z-n},$$

where (ap) denotes $\frac{1}{2}(1-p^2)+ap^2$, and the like for (bq) and (cr); and where, $z-l$ being a multiple factor of P, the coefficient A contains the factor (ap); and similarly for B and C.

Art. The reasoning applies directly to lines 2, 3, 4, 5 of the PQR-Table: and with a slight variation to line 1; viz. here the factors of R $(= -1 + z^n)$ are all simple factors, but in virtue of $c = \infty$ and $a = b$, the corresponding inverts disappear, and, the other inverts also disappearing, the value of the function is $= 0$.

Hence lines 1, 2, 3, 4, 5 of the PQR-Table give each of them a result $= 0$, for the values of (a, b, c) appearing by the table itself, and shown explicitly in the corresponding line of the Annex.

Thus line 3 shows that the function

$$\{x, z\} + x'^2(3, -1, -1 \frown x-1, x+1, x-\infty)^2$$

is identically $= 0$; and so for the other lines.

Art. It is hardly necessary to remark that an expression

$$(a_1, b_1, c_1, \ldots \frown x-a_1, x-b_1, x-c_1, \ldots)^2$$

in fact denotes

$$\frac{a_1}{(x-a_1)^2} + \frac{b_1}{(x-b_1)^2} + \frac{c_1}{(x-c_1)^2} + \frac{2f_1}{(x-a_1)(x-b_1)} + \frac{2g_1}{(x-a_1)(x-c_1)} + \frac{2h_1}{(x-b_1)(x-}$$

The particular form of the z-inverts is immaterial; we could by a general homographic transformation of the variable change these into any other set of inverts; and the foregoing is therefore a solution of

$$\{s, x\} = (a, b, c \frown x - a, x - b, x - c)^2,$$

a differential equation of the third order.

This is the Differential Equation $[a, b, c \ldots]$.

Art. From the Roman lines, if we assume

$$f(x - a) : g(x - b) : h(x - c) = \mathfrak{P} : \mathfrak{Q} : \mathfrak{R},$$

where $\mathfrak{P}$, $\mathfrak{Q}$, $\mathfrak{R}$ are functions of z, not the same functions that P, Q, R are of s, since they belong to a different line of the Table: we have, as before,

$$\{x, z\} + x'^2(a, b, c \frown x - a, x - b, x - c)^2 = 0.$$

Art. We may combine any such result with a properly selected result of the preceding system, the two results being such that (a, b, c) have the same values: combining in this manner the several results, we obtain altogether a system of 16 results.

Art. It thus appears that there are in all 16 sets of values of (a, b, c), for which the equation is solved, viz. the 16 sets of values are shown in the right-hand column of the Annex. For greater clearness I exhibit the integral equations as follows:

	Functions of x		Functions of s
(1)	$f(x - a) : g(x - b) : h(x - c) = P : Q : R$	Polygon	I
(2)	Double Pyramid	II	$x^2 : -(x^2 + 1)^2 : (x^2 -$
(3)	Tetrahedron	III	$x^3 : -(x + 1)^3 : (x - 1)$
(4)	Cube and Octahedron	V	$(x^4 + 1)^2 : -(x^4 - 1)^2$
(5)	Dodecahedron and Icosahedron	VII	

Art. Consider, in general, a solution of this form, $x = F(s)$ a rational function of s: then s is an irrational function of x, and if s_1, s_2 are any two of its values,

$$\{s_1, x\} = R(x), \quad \{s_2, x\} = R(x);$$

that is, $\{s_1, x\} = \{s_2, x\}$, and therefore (ante, No. 7)

$$s_2 = \frac{as_1 + b}{cs_1 + d};$$

then $s = f(s_1) = f\left(\frac{as_1+b}{cs_1+d}\right) = F(s_2)$: viz. $F(s)$ is a rational function of s, transformable into itself by the transformation s into $\frac{as+b}{cs+d}$; and it is moreover clear that between any two roots s whatever of the equation $x = F(s)$ there exists a homographic relation of the form in question.

The Schwarzian Theory. Art. Nos. 46 to 70.

Art. I consider the foregoing equation $\{s, x\} = R(x)$, where $R(x)$ is a rational function, and is now taken to be a real function of x: we may assume $s' = \rho' e^{i\theta}$, where the accents denote differentiation in regard to x, and where ρ', θ, and therefore also θ', are real functions of x.

We have

$$\frac{s''}{s'} = \frac{\rho''}{\rho'} + i\theta',$$

and thence

$$\{s, x\} = \{\rho, x\} + \{\theta, x\} + i\left(\theta''' - \theta' \frac{\rho''}{\rho'}\right).$$

Putting this $= R(x)$, and assuming that x is real, we have

$$\{\rho, x\} + \{\theta, x\} = R(x), \quad \theta''' - \theta' \frac{\rho''}{\rho'} = 0.$$

Art. In the s-plane: since $\{s, x\} = R(x)$ is a differential equation of the third order, the integral contains three arbitrary constants, and we may imagine these so determined that to the values $x = a, b, c$ shall correspond the values $s = A, B, C$ respectively.

If there is not on the x-line any critical point, as the point x moves continuously along this line the point s will move continuously along a circle.

Art. If A be the value of s corresponding to $h = 0$, then $s = hA$, and we find

$$\frac{s - A}{s - A_1} = \frac{A - A_1}{A - A_2} \cdot \frac{h - h_1}{h - h_2} \cdot e^{i\phi},$$

viz. reducing $\frac{s-A}{s-A_1}$ to its principal term h^μ, and then writing ds, dx for $s - A$, and $h(= x - a)$ respectively, we have

$$ds = K(dx)^\mu,$$

or $x = a$ is a critical point with the exponent μ; and similarly $x = b$ and $x = c$ are critical points with the exponents λ and ν respectively.

Art. Hence in the equation

$$\{s, x\} = (a, b, c \frown x - a, x - b, x - c)^2,$$

the exponents μ, λ, ν at the critical points a, b, c are connected with the coefficients (a, b, c) by the relations

$$a = \frac{1}{2}(1 - \mu^2), \quad b = \frac{1}{2}(1 - \lambda^2), \quad c = \frac{1}{2}(1 - \nu^2).$$

Art. As an instance, take the double pyramid form: the integral equation is

$$f(x - a) : g(x - b) : h(x - c) = 4s^n : -(s^n - 1)^2 : (s^n + 1)^2,$$

or say

$$\frac{(c - a)(x - b)}{(a - b)(x - c)} = \frac{4s^n}{(s^n + 1)^2},$$

or if, for greater simplicity, we assume $a, b, c = 1, 0, \infty$, this is

$$x = \frac{4s^n}{(s^n + 1)^2},$$

or say

$$\frac{1}{\sqrt{x}} = \frac{s^n + 1}{2s^{n/2}}, \quad \text{that is, } s^n = \frac{1 \pm \sqrt{1 - x}}{1 \mp \sqrt{1 - x}},$$

a solution of the differential equation

$$\{s, x\} = \frac{1}{2}\frac{1 - \frac{1}{4}n^2}{x^2} + \frac{1}{2}\frac{1 - \frac{1}{4}n^2}{(x - 1)^2} + \frac{\frac{1}{4}n^2 + \frac{1}{4}n^2 - 1}{x(x - 1)}.$$

In particular, if $n = 3$, we have $s^3 = \left(\frac{1+\sqrt{1-x}}{1-\sqrt{1-x}}\right)^2$.

Art. Take x real; then, if x is positive and less than 1, s^n is real and positive, and we have for s the infinite half-lines at the inclinations $0°$, $120°$, $240°$; while if x is positive and greater than 1, $s^{n/2}$ is real and negative, and we have the infinite half-lines at the inclinations $60°$, $180°$, $300°$.

If x is real and negative, then $\frac{1+ki}{1-ki} = \cos\phi + i\sin\phi$: whence s is of the same form, or the locus of the point s is a circle.

Art. We have

$$\frac{b-c}{(x-a)^2}\cdot\frac{c-a}{(x-b)^2}\cdot\frac{a-b}{(x-c)^2}=FGH\cdot e^{i(\theta_a+\theta_b+\theta_c-2\pi)},$$

with the like values for G and H. Hence the right-hand side of

$$\frac{c-a}{(x-b)(x-c)}\cdot\frac{a-b}{(x-c)(x-a)}\cdot\frac{b-c}{(x-a)(x-b)}=FGH$$

has the values

$$\frac{a}{x-a},\frac{b}{x-b},\frac{c}{x-c}.$$

Art. Considering now the left-hand side of the equation, we have

$$\{s,x\}=\frac{1}{(x-a)(x-b)(x-c)}\left(\frac{a}{x-a}+\frac{b}{x-b}+\frac{c}{x-c}\right),$$

substituting for x its value $=\alpha+\beta i+\gamma e^{i\phi}$, this becomes

$$\{s,x\}=-\frac{1}{2\rho^2}\left(\{s,\xi\}-\frac{1}{2}\right),$$

that is,

$$\{s,x\}=-\frac{1}{2\rho^2}\left(\{s,\xi\}-\frac{1}{2}\right).$$

Art. Again P, Q given functions of z, and v a function of z determined by the equation

$$\frac{d^2v}{dz^2}+Pv=0,$$

and assume $y=uv$. Substituting this value of y in the first equation, we obtain for v an equation of the second order (the coefficients of which contain u), and we may make this identical with the second equation; viz. comparing the coefficients of the two equations, we thus have two equations each containing u; and by eliminating u we obtain a differential equation of the third order for the determination of z as a function of x.

Art. We obtain as the required equation for the determination of z as a function of x:

$$\{z,x\}+\left(\frac{dz}{dx}\right)^2(a_1,b_1,c_1\frown z-\infty,z-0,z-1)^2=0.$$

The process does not give the value of u, but this can be found without difficulty, viz.

$$\frac{du}{dx} = \frac{1}{2}\left(\frac{z''}{z'} - \frac{x''}{x'}\right).$$

Art. Jacobi's differential equation of the third order for the transformed modulus λ, *Fund. Nova*, p. 78, [Ges. Werke, t. i, p. 132], is

$$3(\lambda'^2\lambda''' - \lambda''^3) - 2\lambda\lambda'(\lambda'\lambda''' - \lambda''\lambda''') + \lambda'^4 = 0.$$

Art. And putting in the formula $x = \infty, = -(1-a)$, we have

$$\frac{d^2y}{dx^2}+\left(\frac{1-a-b-c}{x} + \frac{1-a'+b-c}{x-1}\right)\frac{dy}{dx}+\left(\frac{-a}{x^2} + \frac{b}{x(x-1)} + \frac{c}{(x-1)^2}\right)y = 0,$$

with a like formula for $\frac{d^2y}{dx^2} + 2P\frac{dy}{dx} - 4Qy = 0$.

We then have

$$y = uv, \quad v = x^{-a_1}(1-x)^{-c_1}z^{a'}(1-z)^{b'+c'+d'+e'}\dots,$$

and the differential equation of the third order for the determination of z is

$$\{z, x\} + \left(\frac{dz}{dx}\right)^2 (a_1, b_1, c_1 \frown z-\infty, z-0, z-1)^2 = 0,$$

where a_1, b_1, c_1 are functions of a, b, c, ... determined as above.

Art. Consider a spherical surface and upon it any number of points: take at pleasure any point as South Pole, this determines the plane of the equator; and the stereographic projection of any point is the intersection with the plane of the equator of the line joining the point with the South Pole.

To fix the ideas take the radius of the sphere as unity: let the axes of x and y be drawn in the plane of the equator in longitudes 0° and 90° respectively.

PART II. The Polyhedral Functions. Art. Nos. 71 to 117.

Art. The values of $x + iy$ for the mid-points of the sides are

$$\cos\frac{\pi}{n}+i\sin\frac{\pi}{n}, \quad \cos\frac{3\pi}{n}+i\sin\frac{3\pi}{n}, \quad \dots, \quad \cos\frac{(2n-1)\pi}{n}+i\sin\frac{(2n-1)\pi}{n}.$$

The corresponding function of s is $s^n + 1$.

The North and South Poles, which form with the n points a double pyramid of $n+2$ summits, correspond to the values $s = 0$ and $s = \infty$. We have thus $s(1 - s^n)(s^n - 1)$ as the function corresponding to the double pyramid.

Art. (3). Considering for a moment the tetrahedron as a figure inscribed in a cube, with the summits at alternate corners of the cube; if the cube is placed with its faces parallel to the coordinate planes, and with edge $= \sqrt{2}$, then the coordinates of the 8 corners are $(\pm 1, \pm 1, \pm 1)$, and those of the 4 tetrahedral summits are $(1, 1, 1)$, $(1, -1, -1)$, $(-1, 1, -1)$, $(-1, -1, 1)$.

Art. (4). The octahedron is placed with two of its summits as poles, and the other four summits in the equator at longitudes 0°, 90°, 180°, 270° respectively: the values of s are, as in the last case, 0, ∞, $+1$, $\pm i$, and the function is $s(s^4 - 1)$.

The function for the centres of the faces, or summits of the cube, is $s^8 + 14s^4 + 1$.

The function for the mid-points of the sides of the octahedron or of the cube is $s^{12} - 33s^8 - 33s^4 - 1$.

Art. (5). The Icosahedron is placed with two opposite summits as poles, and the other 10 summits on two parallel circles of 5 summits each.

Art. Since f_4 is the same function as f_5, we have of course f_4, h_4 and t_4 themselves covariants of f_5: but it is convenient to separate the two systems.

Art. It is to be observed that f_5 is a quartic function having its quadrinvariant $(f) = 0$; but independently of this, that is, qua quartic function, it has only the covariants h_5 and t_5 (the Hessian and the cubicovariant respectively), viz. every other covariant is a rational and integral function of f_5, h_5, t_5.

Covariantive Formulae. Art. Nos. 81 to 84.

Art. The various covariantive formulae will be given with their proper numerical coefficients.

Tetrahedron function. f, h, t stand for the before-mentioned values f_3, h_3, t_3 $(P, Q, R = h^2, -12i\sqrt{3} \cdot t, -f^3)$.

For f_3: $(a, b, c, d, e) = 1, 0, -\frac{1}{2}\sqrt{2}, 0, 1$.

$$
\begin{aligned}
i(f,f)^1 &= -96i\sqrt{3}\cdot h,\\
i(f,h)^1 &= 96i\sqrt{3}\cdot f,\\
i(f,t)^1 &= -32f\cdot h,\\
(f,h) &= 32i\sqrt{3}\cdot t,\\
(f,f)^2 &= 576f = 576\cdot(-\frac{1}{8}),\\
(h,t) &= 4f\cdot h.
\end{aligned}
$$

Art. **Dodecahedron function.** f, h, t stand for the beforementioned values f_5, h_5, t_5 $(P,Q,R = h^2, -t^2, -1728f^5)$.

For f_5: $(a,b,c,d,e,f,g,h,i,j,k,l,m) = (0,\frac{1}{2},0,0,0,0,\frac{5}{8},0,0,0,0,-\frac{5}{8},0)$.

$$
\begin{aligned}
i(f,f)^1 &= -121h,\\
i(f,f)^2 &= 0,\\
i(f,f)^3 &= (924) = (720) = \frac{1}{4}f^2,\\
i(f,h)^1 &= 0,\\
i(f,h)^2 &= 0,\\
i(f,h)^3 &= (924)(720) = \frac{1}{4}f\cdot h,\\
(f,h) &= -20t,\\
i(h,h)^1 &= 173280f\cdot h,\\
(f,t) &= -30h^2,\\
i(h,t)^1 &= 9082800f^2\cdot h,\\
(h,t) &= -86400f^2,\\
h^2 - t &= -1728f^5 = 0.
\end{aligned}
$$

Art. We have

$$t = (s^4+f^4)(1+2\sqrt{-1}\cdot s^2f^2),$$

where the product of the last two factors is $s^8 + (k^2+k'^2)s^4 - 1$.

We have

$$k = \sqrt{\frac{1}{2}}(80\sqrt{5}-176), \quad k' = \sqrt{\frac{1}{2}}(5\sqrt{5}-11),$$

and consequently $k^2 - k'^2 = 11$; or the function is

$$s(1-s^4)(s^8+11s^4-1).$$

Art. Similarly, writing for shortness $k = \tan\gamma$, $k' = \tan\gamma'$, where

$$\cos^2\gamma = \frac{3+\sqrt{5}}{8}, \quad \sin^2\gamma = \frac{5-2\sqrt{5}}{15}, \quad \cos^2\gamma' = \frac{3-\sqrt{5}}{8}, \quad \sin^2\gamma' = \frac{10+2\sqrt{5}}{15},$$

and therefore $\frac{k^2}{k'^2} = \frac{5-2\sqrt{5}}{10+2\sqrt{5}}$, $\sin^2\gamma = \frac{5-2\sqrt{5}}{15}$, as before, then the value is

$$s(s^4+k^2)(s^4+k'^2).$$

Invariantive property of the Stereographic Projection. Art. Nos. 87 to 93.

Art. The before-mentioned theorem that the functions derived from two different stereographic projections of the same point are linear transformations one of the other, may be thus stated:

Considering on the surface of a sphere, two fixed points A and B; and determining the position of a point C, first in regard to A by its distance d and azimuth ϕ, and next in regard to B by its distance d_1 and azimuth ϕ_1: the theorem is that $e^{i\phi_1}\tan\frac{1}{2}d_1$ is a homographic function of $e^{i\phi}\tan\frac{1}{2}d$.

Art. But it is interesting to give the proof with rectangular coordinates. Taking (X,Y,Z), (X_1,Y_1,Z_1) for the coordinates, referred to two different sets of axes, of a point on the spherical surface: also x, y, x_1, y_1 for the coordinates of the corresponding stereographic projections, we have

$$(X_1,Y_1,Z_1) = (\alpha,\beta,\gamma)(X,Y,Z),$$

where (α,β,γ), (α',β',γ'), $(\alpha'',\beta'',\gamma'')$ are the direction-cosines of the new axes in regard to the old ones.

Art. viz. attending only to the former of these, we have $\frac{x_1}{y_1}$ is a homographic function of $\frac{x}{y}$, which is the before-mentioned theorem.

Art. Supposing that the transformation from (X,Y,Z) to (X_1,Y_1,Z_1) is made by a rotation, the coordinates of which are λ, μ, ν, that is, if f, g, h are the inclinations of the resultant axis to the axes of x, y, z respectively, and θ the angle of rotation, putting

$$\lambda,\mu,\nu = \tan\tfrac{1}{2}\theta\cos f, \quad \tan\tfrac{1}{2}\theta\cos g, \quad \tan\tfrac{1}{2}\theta\cos h,$$

then
$$\frac{x_1}{y_1} = \frac{(1+\lambda^2-\mu^2-\nu^2)x + 2(\lambda\mu-\nu)y + \dots}{2(\lambda\mu+\nu)x + (1-\lambda^2+\mu^2-\nu^2)y + \dots}.$$

Art. $(As+B) \div (Cs+D)$.

The group of 60 and the group of 24 include each of them as part of itself the group of 12: it is further to be remarked that the group of 12 may be regarded as that of the positive substitutions upon four letters a, b, c, d, the group of 24 as that of all the substitutions upon the four letters, and the group of 60 as that of the positive substitutions upon five letters a, b, c, d, e.

Art. I call to mind that a group of functional symbols 1, α, β, ... is such that the product (or compound) of any two of them is a symbol of the group; the symbols may be representable as rational (or irrational) functions of a variable symbol, but in the theory of groups this is not necessary.

Art. 4 points A and 4 points B. The fifteen circles intersect by fives in the pairs of opposite points A, by threes in the pairs of opposite points B, and by twos in the pairs of opposite points Θ; the mutual inclinations of successive circles at the points A, B, Θ being = 36°, 60° and 90° respectively.

The whole number $15 \cdot 14 = 210$, of the intersections of the circles two and two together is thus made up of the 12 points A each counting 10 times, the 20 points B each counting 3 times, and the 30 points Θ each counting once; $210 = 120 + 60 + 30$.

Art. The angular magnitudes which present themselves are all obtained from the dodecahedral pentagon, as shown in the annexed figure, in which the angle subtended by a side at the centre is = 72°, and the angle between adjacent sides is = 108°.

Art. And from the triangle, where the two sides and the included angle are g, g and $(120° - 2\phi = 2\xi)$, we find $\widehat{ae} = 60°$.

Art. We thus arrive at the following Table:

	sin	cos				
A$\frac{1}{2}\alpha$						
B$\frac{1}{2}\beta$						
AB						
ΘB						
BΘB						
BBB						
ΘΘ						

where as above

$$\begin{array}{rrr} \alpha = 31^\circ 43', & \beta = 20^\circ 55', & \gamma = 37^\circ 22', \\ \chi = 70^\circ 32', & \theta = 54^\circ 44', & \delta = 37^\circ 46', \\ \phi = 22^\circ 14', & \omega = 63^\circ 26', & 2\beta = 41^\circ 50', \\ 2\gamma = 74^\circ 44', & \alpha - \beta = 10^\circ 48', & \text{etc.} \end{array}$$

and

$$\alpha + \beta + \gamma = 90^\circ, \quad \chi + 2\theta = 180^\circ, \quad \omega + \phi = 60^\circ.$$

Art. We now construct three figures of the points A, B, Θ; viz. these are stereographic projections, each showing the Northern hemisphere projected on the plane of the equator by lines drawn to the South Pole: hence, for any pair of opposite points not on the equator, only the point in the Northern hemisphere is shown.

Art. Fig. 1. Pole at Θ.

	N.P.D.'s	Longitudes
2A	$\alpha = 31^\circ 43'$	0°, 180°
2A	$90^\circ - \alpha = 58^\circ 17'$	90°, 270°
1A	$90^\circ - (0^\circ, 180^\circ) + \alpha = 31^\circ 43'$	
2A	$90^\circ + \alpha = 121^\circ 43'$	90°, 270°
2A	$180^\circ - \alpha = 148^\circ 17'$	0°, 180°
$\frac{1}{2}$B	$\beta = 20^\circ 55'$	90°, 270°
1B	$\theta = 54^\circ 44'$	45°, 135°, 225°, 315°
2B	$90^\circ - \beta = 69^\circ 5'$	0°, 180°
	$(90^\circ, 270^\circ) + \beta = 20^\circ 55'$	45°
2B	$90^\circ + \beta = 110^\circ 55'$	0°, 180°
4B	$180^\circ - \theta = 125^\circ 16'$	45°, 135°, 225°, 315°
2B	$180^\circ - \beta = 159^\circ 5'$	90°, 270°
$\frac{1}{2}$Θ	0°	—
4Θ	36°	$(90^\circ, 270^\circ) + \alpha = 31^\circ 43'$

Art. Fig. 2. Pole at A.

	N.P.D.'s	Longitudes
A	—	—
$\frac{1}{2}$A	$2\alpha = 63°26'$	0°, 72°, 144°, 216°, 288°
$\frac{1}{2}$A	$180° - 2\alpha = 116°34'$	36°, 108°, 180°, 252°, 324°
A	180°	—
$\frac{1}{2}$B	$\gamma = 37°22'$	36°, 108°, 180°, 252°, 324°
$\frac{1}{2}$B	$90° - \alpha + \beta = 79°12'$	36°, 108°, 180°, 252°, 324°
$\frac{1}{2}$B	$90° + \alpha - \beta = 100°48'$	72°, 144°, 216°, 288°
$\frac{1}{2}$B	$180° - \gamma = 142°38'$	72°, 144°, 216°, 288°
$\frac{1}{2}\Theta$	$\alpha = 31°43'$	72°, 144°, 216°, 288°
$\frac{1}{2}\Theta$	$90° - \alpha = 58°17'$	36°, 108°, 180°, 252°, 324°
$\frac{1}{2}\Theta$	90°	$(36°, 108°, 180°, 252°, 324°) + 18°$
$\frac{1}{2}\Theta$	$90° + \alpha = 121°43'$	72°, 144°, 216°, 288°
$\frac{1}{2}\Theta$	$180° - \alpha = 144°17'$	36°, 108°, 180°, 252°, 324°

Art. Fig. 3. Pole at B.

	N.P.D.'s	Longitudes
$\frac{1}{2}$A	$\gamma = 37°22'$	30°, 150°, 270°
$\frac{1}{2}$A	$90° - \alpha + \beta = 79°12'$	90°, 210°, 330°
$\frac{1}{2}$A	$90° + \alpha - \beta = 100°48'$	30°, 150°, 270°
$\frac{1}{2}$A	$180° - \gamma = 142°38'$	90°, 210°, 330°
B	—	—
$\frac{1}{2}$B	$2\beta = 41°50'$	90°, 210°, 330°
$\frac{1}{2}$B	$\chi = 70°32'$	$(30°, 150°, 270°) + \delta = 37°46'$
$\frac{1}{2}$B	$180° - \chi = 109°28'$	$(90°, 210°, 330°) + \delta = 37°46'$
$\frac{1}{2}$B	$180° - 2\beta = 138°10'$	30°, 150°, 270°
$\frac{1}{2}\Theta$	$\beta = 20°55'$	90°, 210°, 330°
$\frac{1}{2}\Theta$	$\theta = 54°44'$	$(90°, 210°, 330°) + \phi = 22°14'$
$\frac{1}{2}\Theta$	$90° - \beta = 69°5'$	30°, 150°, 270°
$\frac{1}{2}\Theta$	90°	60°, 120°, 180°, 240°, 300°
$\frac{1}{2}\Theta$	$90° + \beta = 110°55'$	90°, 210°, 330°
$\frac{1}{2}\Theta$	$180° - \theta = 125°16'$	$(30°, 150°, 270°) + \phi = 22°14'$
$\frac{1}{2}\Theta$	$180° - \beta = 159°5'$	30°, 150°, 270°

The groups of homographic transformations, resumed. Art. Nos. 104 to 117.

Art. The axes of rotation for the dodecahedron and the icosahedron are 15 axes each through a pair of opposite points Θ, 6 axes each through a pair of opposite points A, and 10 axes each through a pair of opposite points B; or say 15 Θ-axes, 10 B-axes and 6 A-axes: the corresponding angles of rotation are 180°, 72° and 120°; so that (excluding in each case the original position or that of a rotation 0) we have in respect of each Θ-axis 1 position, in respect of each A-axis 4 positions, and in respect of each B-axis 2 positions; in all, including the original position,

$$1 + 15 + (6 \times 4) + (10 \times 2) = 60$$

positions, that is, a group of 60 rotations.

Art. To find, in any one of the three forms, the substitution corresponding to given values of (λ, μ, ν), that is, to a rotation through an angle θ about an axis through the point (f, g, h).

Art. **Homographic Transformations. The group of 60.**

Pole at Θ.

No.	$(As + B) \div (Cs + D)$			
1	s	1		
2	$-s$	-1	1	
3	$\frac{1}{s}$	1		
4	$\frac{1}{-s}$	-1		
5	2	$-3+\sqrt{5}+i(-1+\sqrt{5})$	$-3+\sqrt{5}+i(-1+\sqrt{5})$	$-$
6	2	$-3+\sqrt{5}+i(-1+\sqrt{5})$	$-3+\sqrt{5}+i(1-\sqrt{5})$	$-$

Art. The group of 60 was obtained in the A-form by Gordan in his paper. The passage from the Θ-form to the A-form is made as follows: let X, Y, Z be the coordinates of a point when the axes are as in the Θ-form, X_1, Y_1, Z_1 the coordinates of the same point when the axes are as in the A-form: we may write

$$X : Y : Z = bX_1 - aZ_1 : Y_1 : aZ_1 + bX_1,$$

where

$$a, b = \sqrt{\frac{\sqrt{5} - \sqrt{5}}{10}},\ \sqrt{\frac{\sqrt{5} + \sqrt{5}}{10}}.$$

Art. The results are exhibited in terms of ϵ, an imaginary fifth root of unity: taking $\epsilon = \cos 72^\circ + i \sin 72^\circ$, we have

$$\frac{\sqrt{5}-1}{4} \pm i\frac{\sqrt{10+2\sqrt{5}}}{4}, \quad \frac{\sqrt{5}+1}{4} \pm i\frac{\sqrt{10-2\sqrt{5}}}{4},$$

where the upper signs belong to ϵ, ϵ^2 and the lower to ϵ^3, ϵ^4.

It may be remarked that

$$a = \sqrt{\frac{\sqrt{5}+1}{2\sqrt{5}}}, \quad b = \sqrt{\frac{\sqrt{5}-1}{2\sqrt{5}}}, \quad \frac{b}{a} = \frac{\sqrt{5}+1}{\sqrt{5}-1}.$$

For instance, we have in the Θ-group $(A, B, C, D) = (-1, 0, 0, 1)$; $ab \cdot cd$: and thence in the A-group $A_1, B_1, C_1, D_1 = (-2b, 2a, 2a, 2b)$; $ab \cdot cd$: or say this is $(-1, \frac{b}{a}, \frac{b}{a}, 1) = (-1, \epsilon + \epsilon^4, \epsilon + \epsilon^4, 1)$; which in the Table is given as $(-\epsilon^2, \epsilon^2 + \epsilon^3, \epsilon^2 + \epsilon^3, \epsilon^4)$; $ab \cdot cd$.

By effecting the passage to the A-group in this manner, we of course obtain the proper substitution corresponding to each transformation: but I found it easier starting from two transformations and the corresponding substitutions, to obtain thence by successive compositions the entire group.

Art. **Homographic Transformations. The group of 60.**

Pole at A.

No.	$(As+B) \div (Cs+D)$			
1	s	1		
2	$-s$	-1	1	
3	$\frac{1}{s}$	1		
4	$\frac{-1}{s}$	-1		
5–16	2	$\dots$		-2
17	-1		1	
18	i			
19	$-i$		1	
20	$-i$	$-i$	1	-1
21	i		1	
22	i		$-i$	
23	-1	$-i$	1	
24	$-i$		1	
25	$-1-\sqrt{5}+i(3+\sqrt{5})$	2	-2	

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Arthur Cayley, Sc.D., F.R.S.

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745. On the Schwarzian Derivative and the Polyhedral Functions.

[From the *Philosophical Transactions*, vol. CLXXIII (1883), pp. 1145–1185.]

[Continued from p. 200.]

HOMOGRAPHIC TRANSFORMATIONS. THE GROUPS OF 12 AND 24. POLE AT A.

111. Selecting the transformations which correspond to the positive substitutions *abcd*, and completing the group of 24 we have the table of transformations shown below.

[The original contains extensive numerical tables of homographic transformations for the polyhedral functions. The tables give the coefficients $(A\xi+B)/(C\xi+D)$ for each transformation in the groups of 12 and 24, with poles at A and B respectively, and the corresponding group elements.]

As an example of the calculation we have $(A, B, C, D) = (0, i, -1, 0)$; *ab*. Hence

$$A_1 : B_1 : C_1 : D_1 = (a(i-1),\ b(i-1)+i+1,\ b(i-1)-(i+1),\ -a(i+1)),$$

which reduces to $(1,\ \frac{b-1}{a},\ \frac{b+1}{a},\ -1)$. The second and third coefficients are

$$\frac{\sqrt{5}+1}{-2}\cdot\frac{\sqrt{5}+\sqrt{5}}{\sqrt{5}+1},\quad \text{etc.},$$

which, in virtue of the values of ϵ and ϵ^4, are $= 1+2\epsilon^4$ and $1+2\epsilon$ respectively: or the result is as above $(1,\ 1+2\epsilon^4,\ 1+2\epsilon,\ -1)$.

112. In like manner for the passage from the Θ-form to the ξ-form, if X, Y, Z be the coordinates of a point on the spherical surface in regard to the Θ-axes, X_1, Y_1, Z_1 those of the same point in regard to the B-axes, we may write

$$X : Y : Z = X_1 : bY_1 + aZ_1 : -aY_1 + bZ_1,$$

where

$$a, b = \frac{\sqrt{5}-1}{2\sqrt{3}},\ \frac{\sqrt{5}+1}{2\sqrt{3}}.$$

Hence $X : Y : Z = L : M : N$, being the equations of an axis of rotation in the first set of coordinates, those of the same axis in the second set of coordinates will be

$$X_1 : bY_1 + aZ_1 : -aY_1 + bZ_1 = L : M : N,$$

or calling these

$$X_1 : Y_1 : Z_1 = L_1 : M_1 : N_1,$$

we have

$$L_1 : M_1 : N_1 = L : bM - aN : aM + bN;$$

these values are such that

$$L_1^2 + M_1^2 + N_1^2 = L^2 + M^2 + N^2,$$

or λ, μ, ν; λ_1, μ_1, ν_1 being the rotations, we have

$$L, M, N = \sin\lambda,\ \sin\mu,\ \sin\nu; \quad L_1, M_1, N_1 = \sin\lambda_1,\ \sin\mu_1,\ \sin\nu_1,$$

where sin has the same value in the two sets of equations. We have thus

$$B + C : B - C : D - A : D + A = L : 2M : N : -1,$$
$$B_1 + C_1 : B_1 - C_1 : D_1 - A_1 : D_1 + A_1 = L_1 : 2M_1 : N_1 : -1,$$

and hence

$$\begin{aligned} B_1 + C_1 &= B + C, \\ B_1 - C_1 &= b(B - C) - ai(D - A), \\ D_1 - A_1 &= -ai(B - C) + b(D - A), \\ D_1 + A_1 &= D + A; \end{aligned}$$

and thence

$$\begin{aligned} A_1 &= ai(B - C) - b(D - A) + (D + A), \\ B_1 &= b(B - C) - ai(D - A) + (B + C), \\ C_1 &= -b(B - C) + ai(D - A) + (B + C), \\ A &= -ai(B - C) + b(D - A) + (D + A). \end{aligned}$$

113. As an example of the transformation, take

$$(A, B, C, D) = (2,\ -3+\sqrt{5}+i(1-\sqrt{5}),\ -3+\sqrt{5}+i(-1+\sqrt{5}),\ -2) \quad [bc.de];$$

then

$$B - C : B + C : D - A : D + A = i(1 - \sqrt{5}),\ -3 + \sqrt{5},\ -2,\ 0;$$

and thence

$$A_1 = \frac{1}{2\sqrt{3}}\{-i(1-\sqrt{5}) + 2(-2+\sqrt{5}) + 2\},$$
$$B_1 = \frac{1}{2\sqrt{3}}\{i(1-\sqrt{5}) - (-2+\sqrt{5}) + (-4+2\sqrt{5})\},$$
$$C_1 = \frac{1}{2\sqrt{3}}\{-i(1-\sqrt{5}) + (-2+\sqrt{5}) + (-4+2\sqrt{5})\},$$
$$D_1 = \frac{1}{2\sqrt{3}}\{i(1-\sqrt{5}) + 2(-2+\sqrt{5}) - 2\},$$

viz. multiplying by $2\sqrt{3}$, these are

$$2, \quad i(-6+2\sqrt{5})+2\sqrt{3}(-3+\sqrt{5}), \quad i(6-2\sqrt{5})+2\sqrt{3}(-3+\sqrt{5}), \quad -2,$$

that is,

$$2, \quad (-6+2\sqrt{5})(i+\sqrt{3}), \quad (-6+2\sqrt{5})(-i+\sqrt{3}), \quad -2,$$

or since

$$2+\sqrt{3} = -2i\omega \quad \text{and} \quad -2+\sqrt{3} = 2i\omega^2,$$

dividing by 4 these are

$$2, \quad i(3-\sqrt{5})\omega, \quad i(-3+\sqrt{5})\omega^2, \quad -2,$$

as in the table.

Homographic Transformations. The group of 60. Pole at B.

114. $B = \dfrac{-1+i\sqrt{3}}{2}$.

[The original contains extensive numerical tables for the group of 60 homographic transformations, similar in form to those for the groups of 12 and 24.]

The equations of the 12 circles.

122. The equations of the 12 circles are, writing $\Omega = x^2 + y^2 + z^2 - 1 = 0$ for the sphere,

$$x \pm \sqrt{3}\,y = 0, \qquad \sqrt{3}\,x \pm y = 0,$$
$$y \pm \sqrt{3}\,z = 0, \qquad \sqrt{3}\,y \pm z = 0,$$
$$z \pm \sqrt{3}\,x = 0, \qquad \sqrt{3}\,z \pm x = 0.$$

Or, what is the same thing, these are pairs of circles having, for each pair, a common axis or line of intersection; viz. the six axes are the edges of the tetrahedron.

The Dodecahedron.

129. For the dodecahedron, taking the edge $= 2$, we have $r = 2\sqrt{2}$ for the radius of the circumscribed circle.

131. We find also

$$\rho = 2\sqrt{2}, \quad \rho' = \sqrt{2}(\sqrt{5}+1), \quad r' = \sqrt{2}(\sqrt{5}-1),$$

where r' is the radius of an inscribed circle, and ρ' that of a circle touching three edges.

[The article continues with the calculation of the spherical excess and other properties of the dodecahedral configuration.]

746. Higher Plane Curves.

[From Salmon's *Higher Plane Curves*, (3rd ed., 1879); see the Preface.]

One chapter and a large number of articles, in the second edition of Salmon's *Higher Plane Curves*, are due to Professor Cayley. Full reference to these is given by Dr Salmon in the preface.

747. Note on the Degenerate Forms of Curves.

[From Salmon's *Higher Plane Curves*, (3rd ed., 1879), pp. 383—385.]

Some remarks may be added as to the analytical theory of the degenerate forms of curves. As regards conics, a line-pair can be represented in point-coordinates by an equation of the form $xy = 0$; and reciprocally a point-pair can be represented in line-coordinates by an equation $\xi\eta = 0$, but we have to consider how the point-pair can be represented in point-coordinates: an equation $x^2 = 0$ is no adequate representation of the point-pair, but merely represents (as a two-fold or twice repeated line) the line joining the two points of the point-pair, all traces of the points themselves being lost in this representation: and it is to be noticed, that the conic, or two-fold line $x^2 = 0$, or say $(\alpha x + \beta y + \gamma z)^2 = 0$ is a conic which, analytically, and (in an improper sense) geometrically, satisfies the condition of touching *any line whatever*; whereas the only proper tangents of a point-pair are the lines which pass through one or other of the two points of the point-pair.

The solution arises out of the notion of a point-pair, considered as the limit of a conic, or say as an indefinitely flat conic; we have to consider conics certain of the coefficients whereof are infinitesimals, and which, when the infinitesimal coefficients actually vanish, reduce themselves to two-fold lines; and it is, moreover, necessary to consider the evanescent coefficients as infinitesimals of different orders. Thus consider the conics which pass through two given points, and touch two given lines (four conditions); take $y = 0$, $z = 0$ for the given lines, $x = 0$ for the line joining the given points, and $(x = 0,\ y - \alpha z = 0)$, $(x = 0,\ y - \beta z = 0)$ for the given points; the equation of a conic satisfying the required conditions and containing one arbitrary parameter θ, is

$$x^2 + 2\theta xy + 2\theta\sqrt{(\alpha\beta)}\, xz + \theta^2(y - \alpha z)(y - \beta z) = 0;$$

or, what is the same thing,

$$\{x + \theta y + \theta\sqrt{(\alpha\beta)}\, z\}^2 - \theta^2(\alpha + \beta)\, yz = 0;$$

and this equation, considering therein θ as an infinitesimal, say of the first order, represents the flat conic or point-pair composed of the two given points. Comparing with the general equation

$$(a, b, c, f, g, h \between x, y, z)^2 = 0,$$

we have

$$a=1,\quad b=\theta^2,\quad c=\theta^2\alpha\beta,\quad f=-\tfrac{1}{2}\theta^2(\alpha+\beta),\quad g=\theta\surd(\alpha\beta),\quad h=\theta,$$

viz. a being taken to be finite, we have g and h infinitesimals of the first order; b, c, f infinitesimals of the second order; and the four ratios $\surd(b):\surd(c):\surd(f):g:h$ are so determined as to satisfy the prescribed conditions.

Observe that the flat conic, considered as a conic passing through the two given points and touching the two given lines, is represented by a *determinate* equation, viz. considering the condition imposed upon θ ($\theta=$ infinitesimal) as a determination of θ, the equation is a completely determinate one; but considering the flat conic merely as a conic passing through the two given points, the equation would contain two arbitrary parameters, determinable if the flat conic was subjected to the condition of touching two given lines, or to any other two conditions.

Generally, we may consider the equation of a curve of the order n; such equation containing certain infinitesimal coefficients and, when these vanish, reducing itself to a composite equation $P^\alpha Q^\beta\ldots=0$; the equation in its original form represents a curve which may be called the penultimate curve. Consider the tangents from an arbitrary point to the penultimate curve; when this breaks up, the system of tangents reduces itself to (1) the tangents from the fixed point to the several component curves $P=0$, $Q=0$, &c. respectively; (2) the lines through the singular points of these same curves respectively; (3) the lines through the points of intersection $P=0$, $Q=0$, &c. of each two of the component curves; these points, each reckoned a proper number of times, are called "fixed summits"; (4) the lines from the fixed point to certain determinate points called "free summits" on the several component curves $P=0$, $Q=0$, &c. respectively. We have thus a degenerate form of the n-thic curve, which may be regarded as consisting of the component curves, each its proper number of times, and of the foregoing points called summits, and is consequently only inadequately represented by the ultimate equation $P^\alpha Q^\beta\ldots=0$; the number and distribution of the summits is not arbitrary, but is regulated by laws arising from the consideration of the penultimate curve, and there are of course for any given value of n various forms of degenerate curve, according to the different ultimate forms $P^\alpha Q^\beta\ldots=0$, and to the number and distribution of the summits on the different component curves. The case of a quartic curve having the ultimate form $x^2y^2 = 0$ has been considered by Cayley, *Comptes Rendus*, t. LXXIV, p. 708 (March, 1872), [515], who states his conclusion as follows:

"there exists a quartic curve the penultimate of $x^2y^2 = 0$, with nine free summits, three of them on one of the lines (say the line $y = 0$), and which are three of the intersections of the quartic by this line (the fourth intersection being indefinitely near to the point $x = 0$, $y = 0$), six situate at pleasure on the other line $x = 0$; and three fixed summits at the intersection of the two lines." Other forms have been considered by Dr Zeuthen, *Comptes Rendus*, t. LXXV, pp. 703 and 950 (September and October, 1872), and some other forms by Zeuthen; the whole question of the degenerate forms of curves is one well deserving further investigation.

The question of the number of cubic curves satisfying given elementary conditions (depending as it does on the consideration of the degenerate forms of these curves) has been solved by Maillard and Zeuthen; that of the number of quartic curves has been solved by Dr Zeuthen.

748. On the Bitangents of a Quartic.

[From Salmon's *Higher Plane Curves*, (3rd ed., 1879), pp. 387—389.]

The equations of the 28 bitangents of a quartic curve were obtained in a very elegant form by Riemann in the paper "Zur Theorie der Abel'schen Functionen für den Fall $p = 3$," *Ges. Werke*, Leipzig, 1876, pp. 456—472; and see also Weber's *Theorie der Abel'schen Functionen vom Geschlecht 3*, Berlin, 1876. Riemann connects the several bitangents with the characteristics of the 28 odd functions, thus obtaining for them an algorithm which it is worth while to explain, but they will be given also with the algorithm employed p. 231 *et seq.* of the present work*, which is in fact the more simple one. The characteristic of a triple θ-function is a symbol of the form

$$\begin{matrix}\alpha & \beta & \gamma, \\ \alpha' & \beta' & \gamma',\end{matrix}$$

where each of the letters is $= 0$ or 1; there are thus in all 64 such symbols, but they are considered as odd or even according as the sum $\alpha\alpha' + \beta\beta' + \gamma\gamma'$ is odd or even; and the numbers of the odd and even characteristics are 28 and 36 respectively; and, as already mentioned, the 28 odd characteristics correspond to the 28 bitangents respectively.

We have x, y, z trilinear coordinates, α, β, γ, α', β', γ' constants chosen at pleasure, and then α'', β'', γ'' determinate constants, such that the equations

$$\begin{aligned} x + y + z + \xi + \eta + \zeta &= 0, \\ \alpha x + \beta y + \gamma z + \frac{\xi}{\alpha} + \frac{\eta}{\beta} + \frac{\zeta}{\gamma} &= 0, \\ \alpha' x + \beta' y + \gamma' z + \frac{\xi}{\alpha'} + \frac{\eta}{\beta'} + \frac{\zeta}{\gamma'} &= 0, \\ \alpha'' x + \beta'' y + \gamma'' z + \frac{\xi}{\alpha''} + \frac{\eta}{\beta''} + \frac{\zeta}{\gamma''} &= 0, \end{aligned}$$

are equivalent to three independent equations; the first four equations are satisfied by

$$\xi : \eta : \zeta : x : y : z =$$

[3]* That is, Salmon's *Higher Plane Curves.*

[The continuation develops the algorithm for the bitangents in terms of the ξ, η, ζ coordinates and the characteristics of the odd theta-functions.]

[The original contains a table of the 28 odd characteristics arranged in a systematic form, with the corresponding notation for the bitangents.]

749. Solid Geometry.

[From Salmon's *Treatise on the Analytic Geometry of Three Dimensions*, (3rd ed., 1865); see the Preface.]

[This article consists of additions to Salmon's *Solid Geometry*, referred to in the preface of that work.]

750. On the Theory of Reciprocal Surfaces.

[From Salmon's *Treatise on the Geometry of Three Dimensions*, (2nd ed., 1865), pp. 481—486.]

[This article contains a series of additions numbered 601–616, relating to the theory of reciprocal surfaces. The following are the principal results.]

601. In part explanation, observe that the node-couple curve and the cuspidal curve are the complete intersections of the surface with two surfaces $\nabla = 0$ and $\Theta = 0$ respectively; viz. the equation of the surface being $F = (x, y, z, w)^n = 0$, if $F' = 0$ is the first polar of the point $(\alpha, \beta, \gamma, \delta)$, then the consecutive first polar of the point $(\alpha + d\alpha,\ \beta + d\beta,\ \gamma + d\gamma,\ \delta + d\delta)$ is $F' + dF' = 0$, where

$$dF' = d\alpha \frac{dF'}{dx} + d\beta \frac{dF'}{dy} + d\gamma \frac{dF'}{dz} + d\delta \frac{dF'}{dw},$$

which may be written

$$dF' = (d\alpha, d\beta, d\gamma, d\delta\nabla_1, \nabla_2, \nabla_3, \nabla_4)^1\, F',$$

and the two surfaces intersect in a curve, the locus of the poles of the planes which touch the surface in two consecutive points; this is the node-couple curve.

[The article continues with the development of the general theory, including the discussion of the node-couple curve, cuspidal curve, and their singularities.]

612. Instead of obtaining the section by a plane as the reciprocal of the cone with its vertex at the reciprocal point of the plane, we may consider the plane section and the cone as each derived from the same cone standing on the section; and we thus arrive at the general theory of the correspondence of two figures, viz. the section by a plane, and the cone standing on this section. The section has upon it certain singularities: the cone standing on the section has corresponding singularities; and the reciprocal figure, plane section of the reciprocal surface, has singularities corresponding to those of the cone.

615. The question of singularities in the theory of reciprocal surfaces may be considered as follows. We have a surface of the order n with singularities; say the singularities are:

- a ordinary nodes,
- b binodes,
- c cuspidal nodes (or off-points),
- ρ off-lines,
- σ cnicnodes,
- j cnictropes.

The class of the reciprocal surface is $n' = n(n-1)^2 - 2a - 3b - 3c - \rho - 2\sigma - 3j$.

[The article continues with the enumeration of singularities and their effects on the class and other characters of the reciprocal surface.]

751. Note on Riemann's Paper "Versuch einer allgemeinen Auffassung der Integration und Differentiation."

[From the *Mathematische Annalen*, t. XVI (1880), pp. 81–82.]

[This note refers to a posthumous paper of Riemann's, published in his *Werke*, pp. 353–366, and republished by H. Weber in *Math. Ann.* t. XIV, pp. 123–128. The paper relates to the generalization of the notion of differentiation and integration by means of the function $\Pi(x)$ and its inverse. Cayley's note points out that the same subject had been considered by Sylvester in the *Philosophical Magazine*, 1860, and refers to his own paper on the generalization of the factorial function.]

752. On the Finite Groups of Linear Transformations of a Variable; with a Correction.

[From the *Messenger of Mathematics*, vol. IX (1880), pp. 104—109.]

[This paper continues the investigation of the finite groups of linear transformations of a single variable, begun in the author's paper on the Schwarzian derivative and the polyhedral functions (Paper 745). The groups considered are those of the regular solids: the tetrahedron (12), octahedron (24), icosahedron (60), and the dihedral groups.]

The general linear transformation is

$$\xi' = \frac{A\xi + B}{C\xi + D}, \quad AD - BC = 1,$$

and the paper gives the explicit forms of the transformations for each group, with tables similar to those in Paper 745.

For the group of 60 (icosahedron), the transformations are given in the form

$$\xi' = \frac{A_1\xi + B_1}{C_1\xi + D_1},$$

by a formula which is a fractional linear transformation with coefficients expressed in terms of ϵ (a primitive fifth root of unity) and ω (a primitive cube root of unity).

[The paper contains extensive tables of transformations, with the notation of the corresponding substitutions. A correction is made to a previous formula.]

Correction.

In the formula for the group of 60, the entry numbered 25 should read

$$2\omega^2 : -\sqrt{3} - i\sqrt{5} : -\sqrt{3} + i\sqrt{5} : -2\omega,$$

instead of the value previously given.

753. On a Theorem Relating to the Multiple Theta-Functions.

[From the *Mathematische Annalen*, t. XVII (1880), pp. 115—122.]

I propose—partly for the sake of the theorem itself, partly for that of the notation which will be employed—to demonstrate the general theorem (3′), p. 4, of Dr Schottky's *Abriss einer Theorie der Abel'schen Functionen von drei Variabeln*, (Leipzig, 1880), which theorem is there presented in the form:

$$e^{-\eta(u_1,\ldots;\,\mu',\,\nu')}\,\Theta(u_1{+}2\varpi_1',\ldots;\,\mu,\,\nu) = e^{-2\pi i\Sigma\mu_a\nu_a'}\,\Theta(u_1,\ldots;\,\mu{+}\mu',\,\nu{+}\nu'), \tag{3′}$$

but which I write in the slightly different form

$$\exp.[-H(u;\,\mu',\,\nu')].\,\Theta(u{+}2\varpi';\,\mu,\,\nu) = \exp.[-2\pi i\mu\nu'].\,\Theta(u;\,\mu{+}\mu',\,\nu{+}\nu').$$

I remark that the theorem is given in the preliminary paragraphs the contents of which are, as mentioned by the Author, derived from Herr Weierstrass: and that the form of the theta-function is a very general one, depending on the general quadric function

$$G(u_1,\ldots,\,u_\rho;\,n_1,\ldots,\,n_\rho)$$

of 2ρ variables, ρ being the number of the arguments $u_1,\ldots,\,u_\rho$ (in fact, the periods are not reduced to the normal form, but are arbitrary); and the characters $\nu_1,\ldots,\,\nu_\rho;\,\mu_1,\ldots,\,\mu_\rho$, instead of having each of them the value 0 or 1, have each of them any integer or fractional value whatever. The meaning of the theorem (u denoting a set or row of ρ letters $u_1,\ldots,\,u_\rho$, and so in other cases), is that the function

$$\Theta(u;\,\mu+\mu',\,\nu+\nu')$$

is, to a factor pres, equal to the function

$$\Theta(u+2\varpi';\,\mu,\,\nu),$$

and that the factor is independent of u.

[The paper continues with the demonstration of the theorem, including the detailed calculation of the factor $H(u;\,\mu',\,\nu')$ and the exponential multiplier.]

DEMONSTRATION.

The functions $\Theta(u;\, \mu,\, \nu)$ and $\Theta(u;\, \mu+\mu',\, \nu+\nu')$ each of them satisfy the partial differential equations which express the quasi-periodicity of the theta-function; and the exponential factor is determined so that the product shall satisfy the same system of equations. Writing for shortness

$$\Theta(u;\, \mu,\, \nu) = \Theta(u),$$

and considering the new character $\mu + \mu'$ as arising from the original character μ by the addition of the character μ', the theorem is established by showing that the function

$$\exp.\,[-H(u;\, \mu',\, \nu')].\,\Theta(u + 2\varpi')$$

satisfies the same quasi-periodicity equations as the function $\Theta(u;\, \mu+\mu',\, \nu+\nu')$.

[The demonstration involves the detailed consideration of the periods and characters, and the verification that the exponential factor has the correct form to preserve the quasi-periodicity.]

We may regard m, n (each a set of ρ integers) as denoting the periods and characters respectively; and the theorem is then expressed in the more general form

$$\frac{\Theta(u + 2\varpi' + 2\Pi m;\, \mu,\, \nu)}{\Theta(u + 2\Pi m;\, \mu,\, \nu)} = \frac{\Theta(u + 2\varpi';\, \mu,\, \nu)}{\Theta(u;\, \mu,\, \nu)},$$

that is, since $m = (m)$, and we thence deduce

$$\Theta(u + 2\varpi' + 2\Pi m;\, \mu,\, \nu) = e^{2\pi i m\nu'}\,\Theta(u + 2\Pi m;\, \mu+\mu',\, \nu+\nu'),$$

which is the general form of the theorem.

754. On the Connexion of Certain Formulæ in Elliptic Functions.

[From the *Messenger of Mathematics*, vol. IX (1880), pp. 23—25.]

In reference to a like question in the theory of the double ϑ-functions, it is interesting to show that (if not completely, at least very nearly) the single formula

$$\Pi(u,\, a) = u\,\frac{\Theta' a}{\Theta a} + \tfrac{1}{2}\log\frac{\Theta(u-a)}{\Theta(u+a)},$$

that is,

$$\int_0 \frac{k^2 \operatorname{sn} a \operatorname{cn} a \operatorname{dn} a \operatorname{sn}^2 u\, du}{1 - k^2 \operatorname{sn}^2 a \operatorname{sn}^2 u} = u\,\frac{\Theta' a}{\Theta a} + \tfrac{1}{2}\log\frac{\Theta(u-a)}{\Theta(u+a)},$$

leads not only to the relation

$$\log\Theta u = \tfrac{1}{2}\log\frac{2k'K}{\pi} + \tfrac{1}{2}\left(1 - \frac{E}{K}\right)u^2 - k^2\int_0 du\int_0 du \operatorname{sn}^2 u,$$

between the functions Θ, sn, but also to the addition-equation for the function sn.

Writing in the equation a indefinitely small, and assuming only that $\operatorname{sn} a$, $\operatorname{cn} a$, $\operatorname{dn} a$ then become a, 1, 1, respectively, the equation is

$$\begin{aligned} k^2 a\int_0 \operatorname{sn}^2 u\, du &= u\,\frac{a\,\Theta'' 0}{\Theta 0} + \tfrac{1}{2}\log\frac{\Theta u - a\,\Theta' u}{\Theta u + a\,\Theta' u}, \\ &= ua\,\frac{\Theta'' 0}{\Theta 0} - a\,\frac{\Theta' u}{\Theta u}, \end{aligned}$$

that is,

$$\frac{\Theta' u}{\Theta u} = u\,\frac{\Theta'' 0}{\Theta 0} - k^2\int_0 du \operatorname{sn}^2 u,$$

or, integrating from $u = 0$, this is

$$\log\Theta u = C + \tfrac{1}{2}u^2\frac{\Theta'' 0}{\Theta 0} - k^2\int_0 du\int_0 du \operatorname{sn}^2 u,$$

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754.

ON THE CONNEXION OF CERTAIN FORMULAE IN ELLIPTIC FUNCTIONS.

[From the Messenger of Mathematics, vol. ix. (1880), pp. 23–25.]

which, except as regards the determination of the constants, is the required equation for $\log\Theta u$.

Next, differentiating twice the equation for $\Pi(u,a)$, and once the equation obtained for $\dfrac{\Theta' u}{\Theta u}$, we have

$$k^2\text{sn}a\text{cn}a\text{dn}a\,\frac{d}{du}\left(\frac{\text{sn}^2u}{1-k^2\text{sn}^2u\text{sn}^2a}\right)=\tfrac{1}{2}\frac{\Theta''\Theta-\Theta'^2}{\Theta^2}(u-a)-\tfrac{1}{2}\frac{\Theta''\Theta-\Theta'^2}{\Theta^2}(u+a),$$

and

$$\frac{\Theta''\Theta-\Theta'^2}{\Theta^2}u=\frac{\Theta''0}{\Theta 0}-k^2\text{sn}^2u,$$

where, for shortness, $\dfrac{\Theta''\Theta-\Theta'^2}{\Theta^2}u$ is written to denote $\dfrac{\Theta''u\,\Theta u-(\Theta' u)^2}{\Theta^2 u}$, and the like in the first equation; the right-hand side of the first equation therefore is

$$-\tfrac{1}{2}k^2\left\{\text{sn}^2(u-a)-\text{sn}^2(u+a)\right\},$$

or the equation becomes

$$2\text{sn}a\text{cn}a\text{dn}a\,\frac{d}{du}\,\frac{\text{sn}^2u}{1-k^2\text{sn}^2u\text{sn}^2a}=\text{sn}^2(u+a)-\text{sn}^2(u-a),$$

that is,

$$\frac{4\text{sn}u\text{sn}'u\text{sn}a\text{cn}a\text{dn}a}{(1-k^2\text{sn}^2u\text{sn}^2a)^2}=\text{sn}^2(u+a)-\text{sn}^2(u-a).$$

The numerator on the left-hand side must be a symmetrical function of u, a, and hence (even if the value of $\text{sn}'u$ were unknown) it would appear that $\text{sn}'u$ must be a mere constant multiple of $\text{cn}u\text{dn}u$; assuming, however, the actual value, $\text{sn}'u=\text{cn}u\text{dn}u$, the formula is

$$\begin{aligned}&\frac{4\text{sn}u\text{cn}u\text{dn}u\text{sn}a\text{cn}a\text{dn}a}{(1-k^2\text{sn}^2u\text{sn}^2a)^2}\\ &\qquad=\text{sn}^2(u+a)-\text{sn}^2(u-a)\\ &\qquad=\{\text{sn}(u+a)+\text{sn}(u-a)\}\{\text{sn}(u+a)-\text{sn}(u-a)\}.\end{aligned}$$

The factor $\{\text{sn}(u+a)+\text{sn}(u-a)\}$ becomes $= 2\text{sn}u$ for $a=0$, and this suggests that the factor sn$u$ on the left-hand side is a factor of $\{\text{sn}(u+a)+\text{sn}(u-a)\}$. That cn$u$ is *not* a factor hereof would follow from the properties of the period K; viz. for $u=K$, $\text{cn}u=0$, but $\{\text{sn}(u+a)+\text{sn}(u-a)\}$, $= 2\text{sn}(K+a)$ is not $=0$; and, similarly, that dn$u$ is *not* a factor from the properties of the period iK; hence, cn$u$, dn$u$ belong to the other factor $\{\text{sn}(u+a)-\text{sn}(u-a)\}$, and by symmetry cn$a$, dn$a$ belong to the first-mentioned factor. And we are thus led to assume

$$\text{sn}(u+a)+\text{sn}(u-a) = 2M\text{sn}u\text{cn}a\text{dn}a,$$
$$\text{sn}(u+a)-\text{sn}(u-a) = 2M'\text{sn}a\text{cn}u\text{dn}u,$$

where

$$\text{denom.} = 1-k^2\text{sn}^2a\text{sn}^2u,$$

and $MM'=1$. Some further investigation is wanting to show that M and M' are constants, but assuming that they are so and each $=1$, the formulae give at once the ordinary expression for $\text{sn}(u+a)$; that is, we have the addition-equation for the function sn.

755.

ON THE MATRIX $\left(\begin{array}{c|c} a, & b \\ \hline c, & d \end{array}\right)$, AND IN CONNEXION THEREWITH THE FUNCTION $\dfrac{ax+b}{cx+d}$.

[From the Messenger of Mathematics, vol. ix. (1880), pp. 104–109.]

In the preceding paper, [due to Prof. W. W. Johnson,] the theory of the symbolic powers and roots of the function $\dfrac{ax+b}{cx+d}$ is developed in a complete and satisfactory manner; the results in the main agreeing with those obtained in the original memoir, Babbage, "On Trigonometrical Series," *Memoirs of the Analytical Society* (1813), Note I. pp. 47–50, and which are to some extent reproduced in my "Memoir on the Theory of Matrices," *Phil. Trans.*, t. CXL VIII. (1858), pp. 17–37, [152]. I had recently occasion to reconsider the question, and have obtained for the nth function $\phi^n x$, where $\phi x = \dfrac{ax+b}{cx+d}$, a form which, although substantially identical with Babbage's, is a more compact and convenient one; viz. taking λ to be determined by the quadric equation

$$\frac{(\lambda+1)^2}{\lambda} = \frac{(a+d)^2}{ad-bc},$$

the form is

$$\phi^n(x) = \frac{(\lambda^{n+1}-1)(ax+b)+(\lambda^n-\lambda)(-dx+b)}{(\lambda^{n+1}-1)(cx+d)+(\lambda^n-\lambda)(cx-a)}.$$

The question is, in effect, that of the determination of the nth power of the matrix $\left(\begin{array}{c|c} a, & b \\ \hline c, & d \end{array}\right)$; viz. in the notation of matrices

$$(x_1, y_1) = \left(\begin{array}{c|c} a, & b \\ \hline c, & d \end{array}\right)(x, y),$$

means the two equations $x_1 = ax + by$, $y_1 = cx + dy$; and then if x_2, y_2 are derived in like manner from x_1, y_1, that is, if $x_2 = ax_1 + by_1$, $y_2 = cx_1 + dy_1$, and so on, x_n, y_n will be linear functions of x, y;

say we have $x_n = a_n x + b_n y$, $y_n = c_n x + d_n y$: and the nth power of $\left(\begin{array}{c|c} a, & b \\ \hline c, & d \end{array}\right)$ is, in fact, the matrix $\left(\begin{array}{c|c} a_n, & b_n \\ \hline c_n, & d_n \end{array}\right)$.

In particular, we have

$$\left(\begin{array}{c|c} a, & b \\ \hline c, & d \end{array}\right)^2 = \left(\begin{array}{c|c} a_2, & b_2 \\ \hline c_2, & d_2 \end{array}\right) = \left(\begin{array}{c|c} a^2+bc, & b(a+d) \\ \hline c(a+d), & d^2+bc \end{array}\right),$$

and hence the identity

$$\left(\begin{array}{c|c} a, & b \\ \hline c, & d \end{array}\right)^2 - (a+d)\left(\begin{array}{c|c} a, & b \\ \hline c, & d \end{array}\right) + (ad-bc)\left(\begin{array}{c|c} 1, & 0 \\ \hline 0, & 1 \end{array}\right) = 0;$$

viz. this means that the matrix

$$\left(\begin{array}{c|c} a_2-(a+d)a+ad-bc, & b_2-(a+d)b \\ c_2-(a+d)c, & d_2-(a+d)d+ad-bc \end{array}\right) = \left(\begin{array}{c|c} 0, & 0 \\ \hline 0, & 0 \end{array}\right),$$

or, what is the same thing, that each term of the left-hand matrix is $=0$; which is at once verified by substituting for a_2, b_2, c_2, d_2 their foregoing values.

The explanation just given will make the notation intelligible and show in a general way how a matrix may be worked in like manner with a single quantity: the theory is more fully developed in my Memoir above referred to. I proceed with the solution in the algorithm of matrices. Writing for shortness $M = \left(\begin{array}{c|c} a, & b \\ \hline c, & d \end{array}\right)$, the identity is

$$M^2 - (a+d)M + (ad-bc) = 0,$$

the matrix $\left(\begin{array}{c|c} 1, & 0 \\ \hline 0, & 1 \end{array}\right)$ being in the theory regarded as $=1$; viz. M is determined by a quadric equation; and we have consequently $M^n =$ a linear function of M. Writing this in the form

$$M^n - AM + B = 0,$$

the unknown coefficients A, B can be at once obtained in terms of α, β, the roots of the equation

$$u^2 - (a+d)u + ad - bc = 0,$$

viz. we have

$$\alpha^n - A\alpha + B = 0,$$
$$\beta^n - A\beta + B = 0;$$

or more simply from these equations, and the equation for M^n, eliminating α, β, we have

$$\begin{vmatrix} M^n, & M, & 1 \\ \alpha^n, & \alpha, & 1 \\ \beta^n, & \beta, & 1 \end{vmatrix} = 0;$$

that is,

$$(\beta - \alpha)M^n - (\beta^n - \alpha^n)M + (\beta\alpha^n - \alpha\beta^n) = 0,$$

or say

$$M^n = \frac{\beta^n - \alpha^n}{\beta - \alpha}M - \frac{\beta\alpha^n - \alpha\beta^n}{\beta - \alpha},$$

which is the expression for M^n.

We may present this in a more convenient form by introducing instead of α, β their differences from $\frac{1}{2}(a + d)$; viz. writing

$$\alpha = \tfrac{1}{2}(a + d) + \theta, \qquad \beta = \tfrac{1}{2}(a + d) - \theta,$$

whence $\alpha + \beta = a + d$, $\alpha\beta = \frac{1}{4}(a + d)^2 - \theta^2 = ad - bc$, that is, $\theta^2 = \frac{1}{4}(a - d)^2 + bc$; also $\beta - \alpha = -2\theta$, and therefore

$$\frac{\beta^n - \alpha^n}{\beta - \alpha} = \frac{(\frac{1}{2}(a + d) + \theta)^n - (\frac{1}{2}(a + d) - \theta)^n}{2\theta},$$

which denoting by P, and writing also Q for the value obtained therefrom by changing θ into $-\theta$, that is,

$$Q = \frac{(\frac{1}{2}(a + d) - \theta)^n - (\frac{1}{2}(a + d) + \theta)^n}{-2\theta},$$

we have $P = Q$, and the formula becomes

$$M^n = PM - \tfrac{1}{2}(a + d) \cdot P + \tfrac{1}{2}Q \cdot (a + d),$$

that is,

$$M^n = P \cdot \{M - \tfrac{1}{2}(a + d)\} + \tfrac{1}{2}(a + d) \cdot Q.$$

But $M - \frac{1}{2}(a+d)$ denotes the matrix

$$\left(\begin{array}{c|c} \frac{1}{2}(a-d), & b \\ c, & -\frac{1}{2}(a-d) \end{array}\right),$$

and we have

$$M^n = P \cdot \left(\begin{array}{c|c} \frac{1}{2}(a-d), & b \\ c, & -\frac{1}{2}(a-d) \end{array}\right) + \frac{1}{2}(a+d) \cdot Q \cdot \left(\begin{array}{c|c} 1, & 0 \\ \hline 0, & 1 \end{array}\right),$$

where

$$P = \frac{(\frac{1}{2}(a+d)+\theta)^n - (\frac{1}{2}(a+d)-\theta)^n}{2\theta}, \qquad Q = \frac{(\frac{1}{2}(a+d)+\theta)^n + (\frac{1}{2}(a+d)-\theta)^n}{2}$$

and $\theta^2 = \frac{1}{4}(a-d)^2 + bc$.

The matrix equation gives

$$\begin{aligned} a_n &= P \cdot \tfrac{1}{2}(a-d) + \tfrac{1}{2}(a+d) \cdot Q, \\ b_n &= Pb, \\ c_n &= Pc, \\ d_n &= -P \cdot \tfrac{1}{2}(a-d) + \tfrac{1}{2}(a+d) \cdot Q, \end{aligned}$$

and we thence obtain

$$\frac{a_n x + b_n}{c_n x + d_n}$$

for the value of $\phi^n x$.

Writing for shortness

$$\tfrac{1}{2}(a+d) = f, \qquad \tfrac{1}{2}(a-d) = g, \qquad \theta^2 = g^2 + bc,$$

we have

$$\phi^n(x) = \Big/ \frac{(f+\theta)^n - (f-\theta)^n}{2\theta} \cdot (gx+b) + \frac{(f+\theta)^n + (f-\theta)^n}{2} \cdot x$$
$$\Big/ \frac{(f+\theta)^n - (f-\theta)^n}{2\theta} \cdot (cx-g) + \frac{(f+\theta)^n + (f-\theta)^n}{2},$$

or, what is the same thing,

$$\phi^n(x) = \frac{\{(f+\theta)^n - (f-\theta)^n\}(gx+b) + \theta\{(f+\theta)^n + (f-\theta)^n\}x}{\{(f+\theta)^n - (f-\theta)^n\}(cx-g) + \theta\{(f+\theta)^n + (f-\theta)^n\}}.$$

The case where $\theta = 0$, that is $(a-d)^2 + 4bc = 0$, requires special consideration. Here $f = \frac{1}{2}(a+d)$, $g = \frac{1}{2}(a-d)$, $g^2 + bc = 0$, and the formulae become

$$\begin{aligned}\phi^n(x) &= \frac{nf^{n-1}(gx+b) + f^n \cdot x}{nf^{n-1}(cx-g) + f^n} \\ &= \frac{nf^{n-1}(gx+b) + f^n \cdot x}{nf^{n-1}(cx-g) + f^n}.\end{aligned}$$

We may write the general formula in a more convenient form by introducing in place of θ the parameter λ, where

$$\lambda = \frac{f+\theta}{f-\theta}, \qquad \text{that is} \qquad \theta = f \cdot \frac{\lambda - 1}{\lambda + 1},$$

and we thence obtain

$$\phi^n(x) = \frac{(\lambda^n - 1)(gx+b) + (\lambda^{n-1} - \lambda^{-1}) f \cdot x}{(\lambda^n - 1)(cx-g) + (\lambda^{n-1} - \lambda^{-1}) f},$$

or substituting for g, f their values $\frac{1}{2}(a-d)$, $\frac{1}{2}(a+d)$, this is

$$\phi^n(x) = \frac{(\lambda^{n+1} - 1)(ax+b) + (\lambda^n - \lambda)(-dx+b)}{(\lambda^{n+1} - 1)(cx+d) + (\lambda^n - \lambda)(cx-a)},$$

which agrees with the formula originally given.

The solution may be obtained in another form by means of the quadratic equation for the determination of α, β; we have α, β the roots of

$$u^2 - (a+d)u + (ad - bc) = 0,$$

and thence

$$\alpha + \beta = a + d, \qquad \alpha\beta = ad - bc.$$

Writing for shortness $M = \left(\begin{array}{c|c} a, & b \\ \hline c, & d \end{array}\right)$, and assuming $M^n = \left(\begin{array}{c|c} a_n, & b_n \\ \hline c_n, & d_n \end{array}\right)$, we have the identity

$$M^2 - (a+d)M + (ad-bc) = 0,$$

that is,

$$M^n = AM - B,$$

where A and B are given as above; and thence multiplying by $\left(\begin{array}{c|c} x, & 1 \end{array} \right)$, or say by $(x, 1)$, and observing that

$$\left(\begin{array}{c|c} a_n, & b_n \\ \hline c_n, & d_n \end{array}\right) \left(\begin{array}{c} x \\ \hline 1 \end{array}\right) = \left(\begin{array}{c} a_n x + b_n \\ \hline c_n x + d_n \end{array}\right),$$

we have

$$(a_n x + b_n,\ c_n x + d_n) = A(ax+b,\ cx+d) - B(x,\ 1),$$

that is,

$$\begin{aligned} a_n x + b_n &= A(ax+b) - Bx, \\ c_n x + d_n &= A(cx+d) - B, \end{aligned}$$

and thence

$$\phi^n x = \frac{a_n x + b_n}{c_n x + d_n} = \frac{A(ax+b) - Bx}{A(cx+d) - B}.$$

In verification, observe that for $n = 0$, $A = 0$, $B = -1$, and the formula gives $\phi^0 x = \dfrac{x}{1} = x$; for $n = 1$, $A = 1$, $B = 0$, and the formula gives $\phi x = \dfrac{ax+b}{cx+d}$.

Writing the formula in the form

$$\phi^n x = \frac{(ax+b) - \frac{B}{A}\cdot x}{(cx+d) - \frac{B}{A}},$$

and observing that $\dfrac{B}{A}$ is a symmetrical function of α, β, we see that $\phi^n x$ depends only on α^n, β^n, that is on $\alpha^n + \beta^n$ and $\alpha^n\beta^n$; but $\alpha\beta = ad - bc$, whence $(\alpha\beta)^n = (ad-bc)^n$, and we have

$$\alpha^n + \beta^n = (ad-bc)^n \left(\frac{1}{\alpha^n} + \frac{1}{\beta^n}\right),$$

that is, the same function of a, d as of d, a; and $(ad-bc)^n$ is symmetrical in regard to a, d; hence $\alpha^n+\beta^n$, and therefore $\dfrac{B}{A}$, is symmetrical in regard to a, d; that is, we have $\phi^n(x;a,d)=\phi^n(x;d,a)$.

In particular, $n=2$ gives

$$\phi^2x=\frac{a(ax+b)+b(cx+d)}{c(ax+b)+d(cx+d)}=\frac{(a^2+bc)x+b(a+d)}{c(a+d)x+(d^2+bc)},$$

which agrees with the foregoing value.

The equation $M^n=AM-B$ may also be obtained as follows; we have

$$M=\left(\begin{array}{c|c} a, & b \\ \hline c, & d \end{array}\right),\qquad M^n=\left(\begin{array}{c|c} a_n, & b_n \\ \hline c_n, & d_n \end{array}\right),$$

and thence $M^n-AM+B=0$ gives

$$\begin{aligned} a_n&=Aa-B, & b_n&=Ab,\\ c_n&=Ac, & d_n&=Ad-B, \end{aligned}$$

where A and B are the same linear functions of a_n, d_n as of a, d; viz. we have

$$A=\frac{\beta^n-\alpha^n}{\beta-\alpha},\qquad B=\frac{\beta\alpha^n-\alpha\beta^n}{\beta-\alpha},$$

and thence

$$\phi^n x=\frac{A(ax+b)-Bx}{A(cx+d)-B},$$

as above.

It is to be remarked that if $\lambda^m-1=0$, where m, the least exponent for which this equation is satisfied, is for the moment taken to be greater than 2, the terms in [ ] are

$$(\lambda-1)(ax+b)+(1-\lambda)(-dx+b),$$

and

$$(\lambda-1)(cx+d)+(1-\lambda)(cx-a);$$

viz. these are $(\lambda-1)(a+d)x$, and $(\lambda-1)(a+d)$, or if $a+d=0$, they both vanish; that is, if $a+d=0$, then $\phi^m x=x$, or the function is periodic of the order m. If $a+d=0$, then $\lambda=-1$, and the period is $=2$; we have

$$\phi x=\frac{ax+b}{cx-a},\qquad \phi^2x=\frac{a(ax+b)+b(cx-a)}{c(ax+b)+d(cx-a)}=x,$$

since $d = -a$.

But the case $m = 1$ is a very remarkable one; we have here $\lambda = 1$, and the relation between the coefficients is thus $(a+d)^2 = 4(ad-bc)$, or what is the same thing $(a-d)^2 + 4bc = 0$.

And then determining the values for $\lambda = 1$ of the vanishing fractions which enter into the formulae, we find

$$\begin{aligned} a_n x + b_n &= \tfrac{1}{2}(a+d)^{n-1}\{(n+1)(ax+b) + (n-1)(-dx+b)\}, \\ c_n x + d_n &= \tfrac{1}{2}(a+d)^{n-1}\{(n+1)(cx+d) + (n-1)(cx-a)\}, \end{aligned}$$

or as these may also be written

$$\begin{aligned} a_n x + b_n &= \tfrac{1}{2}(a+d)^{n-1}[x\{n(a-d) + (a+d)\} + 2nb], \\ c_n x + d_n &= \tfrac{1}{2}(a+d)^{n-1}[x \cdot 2nc + \{-n(a-d) + a + d\}], \end{aligned}$$

which for $n = 0$ become as they should do $a_0 x + b_0 = x$, $c_0 x + d_0 = 1$, and for $n = 1$ they become

$$a_1 x + b_1 = ax + b, \qquad c_1 x + d_1 = cx + d.$$

We thus do not have $\phi^n x = x$, and the function is not periodic of the order 1; but we have

$$\phi^n x = \frac{x\{n(a-d) + (a+d)\} + 2nb}{x \cdot 2nc + \{-n(a-d) + a + d\}},$$

which for $n = \infty$ becomes $= \dfrac{(a-d)x + 2b}{2cx - (a-d)}$, or writing herein $(a-d)^2 + 4bc = 0$, that is $bc = -\frac{1}{4}(a-d)^2$, the value is

$$\frac{(a-d)x + 2b}{2cx - (a-d)} = \frac{(a-d)x + 2b}{-\frac{1}{2}(a-d)^2 \cdot \frac{1}{b} \cdot x - (a-d)} = \frac{(a-d)x + 2b}{-(a-d)\left\{\frac{(a-d)x}{2b} + 1\right\}},$$

which is independent of x, and equal to either of these equal quantities; and if from these two values of u we eliminate λ, we obtain for u the quadric equation

$$cu^2 - (a-d)u - b = 0,$$

that is,

$$u = \frac{au + b}{cu + d},$$

as is, in fact, obvious from the consideration that n being indefinitely large the nth and $(n+1)$th functions must be equal to each other. In the latter case, as λ^n is indefinitely small, we have the like formulae, and we obtain for u the same quadric equation: the two values of u are however not the same, but (as is easily shown) their product is $= -b \div c$; u is therefore the other root of the quadric equation. Hence, as n increases, the function $\phi^n x$ continually approximates to one or the other of the roots of this quadric equation. The equation has equal roots if $(a-d)^2 + 4bc = 0$, which is the relation existing in the above-mentioned special case; and here $u = \dfrac{1}{2c}(a-d), = \dfrac{-2b}{a-d}$, which result is also given by the formulae of the special case on writing therein $n = \infty$.

756.

A GEOMETRICAL CONSTRUCTION RELATING TO IMAGINARY QUANTITIES.

[From the Messenger of Mathematics, vol. x. (1881), pp. 1–3.]

Let α, β, γ be real quantities, and consider the imaginary quantity $\alpha+\beta\omega+\gamma\omega^2$, where ω is an imaginary cube root of unity, so that $1+\omega+\omega^2=0$, and therefore $\omega^2+\omega+1=0$.

We have

$$(\alpha+\beta\omega+\gamma\omega^2)(\alpha+\beta\omega^2+\gamma\omega)=\alpha^2+\beta^2+\gamma^2-\beta\gamma-\gamma\alpha-\alpha\beta,$$

which may be written

$$=\tfrac{1}{2}\{(\beta-\gamma)^2+(\gamma-\alpha)^2+(\alpha-\beta)^2\};$$

and we thence have the theorem that the product of the two conjugate imaginaries is $=0$ if $\alpha=\beta=\gamma$, but is in every other case positive.

If for a moment ω denotes generally an imaginary nth root of unity, and we consider the imaginary $\alpha+\beta\omega+\gamma\omega^2+\ldots$, then multiplying this by the like imaginary with ω^{-1}, ω^{-2}, ... in place of ω, ω^2, ... respectively, the product is

$$\alpha^2+\beta^2+\gamma^2+\ldots+2\beta\gamma\cos(\omega^1+\omega^{-1})+2\gamma\alpha\cos(\omega^2+\omega^{-2})+\ldots,$$

which in the particular case $n=3$ becomes

$$\alpha^2+\beta^2+\gamma^2-\beta\gamma-\gamma\alpha-\alpha\beta,$$

as above.

We may geometrically represent the imaginary $\alpha+\beta\omega+\gamma\omega^2$ by means of a point P in a plane; viz. taking an origin O and axes Ox, Oy, if through P we draw lines $PA=\alpha$, $PB=\beta$, $PC=\gamma$ inclined at angles of 120° to each other, so that PA, PB, PC are in fact measured along the sides of an equilateral triangle, then the coordinates of P are given by

$$x=\alpha+\beta\cos 120^\circ+\gamma\cos 240^\circ=\alpha-\tfrac{1}{2}\beta-\tfrac{1}{2}\gamma,$$

and

$$y=\beta\sin 120^\circ+\gamma\sin 240^\circ=\tfrac{1}{2}\sqrt{3}(\beta-\gamma).$$

The conjugate imaginary is represented in like manner by a point P' obtained by reflecting P in the axis of x; and then $OP^2 = OP'^2$ is equal to the product of the two conjugate imaginaries, which as above is

$$\tfrac{1}{2}\{(\beta-\gamma)^2 + (\gamma-\alpha)^2 + (\alpha-\beta)^2\}.$$

Hence $OP^2 = 0$, that is O and P coincide, only in the case $\alpha = \beta = \gamma$; and except in this case, OP^2 is positive.

The representation thus agrees with the fundamental theorem that the product of an imaginary by its conjugate is in every case positive, except only when the two factors vanish together.

In the case where α, β, γ are positive, and one of them, say γ, is $= 0$, the point P lies on the line OA, and the distance $OP = \alpha - \beta$, is positive or negative according as $\alpha > \beta$ or $\alpha < \beta$; if $\alpha = \beta$, then P coincides with O.

757.

ON A SMITH'S PRIZE QUESTION, RELATING TO POTENTIALS.

[From the Messenger of Mathematics, vol. xi. (1882), pp. 15–18.]

A *spherical shell* is attracted according to the law of the inverse square of the distance; to find the potential at any external or internal point.

The solution depends of course on the value of the solid angle which the shell subtends at the point in question; and the determination of this solid angle is effected by means of the spherical excess of the cone which stands on the shell, that is of the cone whose vertex is at the point and whose base is the rim of the shell.

Let a be the radius of the sphere, c the distance of the centre from the point, θ the half-angle of the cone which the rim subtends at the centre of the sphere, and V the potential. Then if $d\Sigma$ is an element of the surface, and r its distance from the point, we have

$$V = \int \frac{d\Sigma}{r},$$

and the integral is to be taken over the spherical shell.

To evaluate this, let P be the point, O the centre of the sphere; take OP as the axis of z, and let the coordinates of a point on the sphere be $(a\sin\theta\cos\phi,\ a\sin\theta\sin\phi,\ a\cos\theta)$. Then if $d\Sigma$ is the element of surface at this point, we have

$$d\Sigma = a^2 \sin\theta\, d\theta\, d\phi,$$

and the distance r is given by

$$r^2 = a^2 + c^2 - 2ac\cos\theta.$$

Hence for a complete cone, that is for the whole sphere, we should have

$$V = \int_0^{2\pi}\int_0^{\pi} \frac{a^2\sin\theta\, d\theta\, d\phi}{\sqrt{a^2+c^2-2ac\cos\theta}} = 2\pi a^2 \int_0^{\pi} \frac{\sin\theta\, d\theta}{\sqrt{a^2+c^2-2ac\cos\theta}}.$$

Let $u = \cos\theta$, then $du = -\sin\theta\, d\theta$, and the integral becomes

$$V = 2\pi a^2 \int_{-1}^{1} \frac{du}{\sqrt{a^2+c^2-2acu}} = \frac{2\pi a^2}{ac}\left[-\sqrt{a^2+c^2-2acu}\right]_{-1}^{1} = \frac{2\pi a}{c}\{-(a-c)+(a-$$

that is $V = \dfrac{4\pi a^2}{c}$ for the whole sphere, which is the known result.

For a spherical shell bounded by two cones of half-angles α and β $(\alpha > \beta)$, we have

$$V = 2\pi a^2 \int_\beta^\alpha \frac{\sin\theta\, d\theta}{\sqrt{a^2 + c^2 - 2ac\cos\theta}} = \frac{2\pi a}{c}\{\sqrt{a^2 + c^2 - 2ac\cos\beta} - \sqrt{a^2 + c^2 - 2ac\,\mathrm{co}$$

If the point is external $(c > a)$, and the shell is a zone bounded by parallels of latitude, this gives the potential at the external point; and if the point is internal $(c < a)$, it gives the potential at the internal point.

In the case where the shell is a cap, that is where $\alpha = 0$ (the axis of the cap), we have $\cos\alpha = 1$, and

$$V = \frac{2\pi a}{c}\{\sqrt{a^2 + c^2 - 2ac\cos\beta} - (c - a)\},$$

if $c > a$; and

$$V = \frac{2\pi a}{c}\{(a - c) - \sqrt{a^2 + c^2 - 2ac\cos\beta}\},$$

if $c < a$.

758.

SOLUTION OF A SENATE-HOUSE PROBLEM.

[From the Messenger of Mathematics, vol. xi. (1882), pp. 25–27.]

Given that

$$\frac{x}{a}+\frac{y}{b}+\frac{z}{c}=0, \qquad \frac{a}{x}+\frac{b}{y}+\frac{c}{z}=0,$$

show that

$$\frac{x^2}{a^2}+\frac{y^2}{b^2}+\frac{z^2}{c^2}=\frac{y^2z^2}{b^2c^2}+\frac{z^2x^2}{c^2a^2}+\frac{x^2y^2}{a^2b^2}.$$

From the given equations we have

$$\frac{x}{a}+\frac{y}{b}=-\frac{z}{c}, \qquad \frac{a}{x}+\frac{b}{y}=-\frac{c}{z},$$

and multiplying these together,

$$1+\frac{bx}{ay}+\frac{ay}{bx}+1=1,$$

that is

$$\frac{bx}{ay}+\frac{ay}{bx}+1=0, \qquad \text{or} \qquad \frac{x^2}{a^2}+\frac{xy}{ab}+\frac{y^2}{b^2}=0.$$

Similarly we find

$$\frac{y^2}{b^2}+\frac{yz}{bc}+\frac{z^2}{c^2}=0, \qquad \frac{z^2}{c^2}+\frac{zx}{ca}+\frac{x^2}{a^2}=0.$$

From these three equations we may eliminate $\frac{yz}{bc}, \frac{zx}{ca}, \frac{xy}{ab}$; or more simply, observe that we have

$$\frac{x^2}{a^2}+\frac{y^2}{b^2}+\frac{z^2}{c^2}+\frac{yz}{bc}+\frac{zx}{ca}+\frac{xy}{ab}=0.$$

Now from the second given equation,

$$\frac{ayz+bzx+cxy}{xyz}=0, \qquad \text{that is} \qquad ayz+bzx+cxy=0,$$

and therefore

$$\frac{yz}{bc} + \frac{zx}{ca} + \frac{xy}{ab} = 0.$$

Hence

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 0.$$

But this is not the required result; we have to show that it is equal to

$$\frac{y^2z^2}{b^2c^2} + \frac{z^2x^2}{c^2a^2} + \frac{x^2y^2}{a^2b^2}.$$

Now from $\dfrac{x^2}{a^2} + \dfrac{xy}{ab} + \dfrac{y^2}{b^2} = 0$, we have

$$\frac{x^2y^2z^2}{a^2b^2c^2} = \frac{z^2}{c^2}\left(-\frac{xy}{ab}\right) = \frac{z^2}{c^2}\left(\frac{x^2}{a^2} + \frac{y^2}{b^2}\right),$$

and similarly for the other terms; whence adding, the required result follows.

759.

ILLUSTRATION OF A THEOREM IN THE THEORY OF EQUATIONS.

[From the Messenger of Mathematics, vol. xi. (1882), pp. 34–36.]

If in the equation

$$x^n - p_1 x^{n-1} + p_2 x^{n-2} - \ldots + (-1)^n p_n = 0,$$

we substitute for x the values α, β, γ, $\ldots$ (the roots), and multiply the resulting equations together, we obtain a symmetric function of the roots which may be expressed in terms of the coefficients.

In particular, for the cubic equation

$$x^3 - p_1 x^2 + p_2 x - p_3 = 0,$$

let α, β, γ be the roots; then substituting α, β, γ successively, and multiplying, we obtain

$$(\alpha^3 - p_1\alpha^2 + p_2\alpha - p_3)(\beta^3 - p_1\beta^2 + p_2\beta - p_3)(\gamma^3 - p_1\gamma^2 + p_2\gamma - p_3) = 0.$$

But since $\alpha^3 = p_1\alpha^2 - p_2\alpha + p_3$, the first factor vanishes; and similarly for the others; so that the product is $= 0$, as it should be.

More generally, if we form the equation whose roots are α^2, β^2, γ^2, this may be done by eliminating x between the given equation and $y = x^2$; or what is the same thing, writing the cubic in the form

$$x(x^2 + p_2) = p_1 x^2 + p_3,$$

and squaring, we obtain

$$x^2(x^2 + p_2)^2 = (p_1 x^2 + p_3)^2,$$

that is, writing $y = x^2$,

$$y(y + p_2)^2 = (p_1 y + p_3)^2,$$

or

$$y^3 + (2p_2 - p_1^2)y^2 + (p_2^2 - 2p_1p_3)y - p_3^2 = 0.$$

The coefficients of this equation are thus expressed as symmetric functions of α, β, γ; and we may verify that

$$\alpha^2 + \beta^2 + \gamma^2 = p_1^2 - 2p_2,$$
$$\beta^2\gamma^2 + \gamma^2\alpha^2 + \alpha^2\beta^2 = p_2^2 - 2p_1p_3,$$
$$\alpha^2\beta^2\gamma^2 = p_3^2.$$

760.

REDUCTION OF $\int \frac{dx}{\sqrt{1-x^4}}$ TO ELLIPTIC INTEGRALS.

[From the Messenger of Mathematics, vol. xi. (1882), pp. 43–45.]

The integral $\int \frac{dx}{\sqrt{1-x^4}}$ may be reduced to the standard form of elliptic integral by various substitutions.

First method. Write $x^2 = \sin\theta$, then $2x\,dx = \cos\theta\,d\theta$, that is $dx = \frac{\cos\theta\,d\theta}{2\sqrt{\sin\theta}}$, and

$$\sqrt{1-x^4} = \sqrt{1-\sin^2\theta} = \cos\theta.$$

Hence

$$\int \frac{dx}{\sqrt{1-x^4}} = \int \frac{d\theta}{2\sqrt{\sin\theta}} = \frac{1}{2}\int \frac{d\theta}{\sqrt{\sin\theta}}.$$

Now write $\theta = \frac{1}{2}\pi - 2\phi$, then $d\theta = -2\,d\phi$, and

$$\sin\theta = \cos 2\phi = 1 - 2\sin^2\phi,$$

so that

$$\int \frac{dx}{\sqrt{1-x^4}} = -\int \frac{d\phi}{\sqrt{1-2\sin^2\phi}},$$

which is an elliptic integral of the first kind with modulus $k = \sqrt{2}$; but as $k > 1$, we may write $\sin\phi = \frac{1}{\sqrt{2}}\sin\psi$, and thus reduce it to the standard form with modulus < 1.

Second method. A more direct reduction is obtained by the substitution

$$x = \frac{\cos\theta}{\sqrt{1+\sin^2\theta}}.$$

We then have

$$1-x^4 = 1-\frac{\cos^4\theta}{(1+\sin^2\theta)^2} = \frac{(1+\sin^2\theta)^2-\cos^4\theta}{(1+\sin^2\theta)^2} = \frac{2\sin^2\theta(2-\sin^2\theta)}{(1+\sin^2\theta)^2},$$

and

$$dx = -\frac{\sin\theta(2+\sin^2\theta)}{(1+\sin^2\theta)^{3/2}}\,d\theta.$$

Hence

$$\frac{dx}{\sqrt{1-x^4}} = -\frac{2+\sin^2\theta}{\sqrt{2}(1+\sin^2\theta)\sqrt{2-\sin^2\theta}}\,d\theta,$$

which may be further reduced.

Third method. The substitution

$$x = \frac{1-y}{1+y}$$

gives

$$1-x^4 = \frac{(1+y)^4-(1-y)^4}{(1+y)^4} = \frac{8y(1+y^2)}{(1+y)^4}, \qquad dx = \frac{-2\,dy}{(1+y)^2},$$

and therefore

$$\int \frac{dx}{\sqrt{1-x^4}} = -\int \frac{(1+y)\,dy}{2\sqrt{2y(1+y^2)}}.$$

Writing $y = t^2$, this becomes

$$-\int \frac{(1+t^2)\,dt}{\sqrt{2(1+t^4)}}.$$

The reduction of this integral to the standard form of Legendre may be completed by the usual methods.

761.

ON THE THEOREM OF THE FINITE NUMBER OF THE COVARIANTS OF A BINARY QUANTIC.

[From the Messenger of Mathematics, vol. xi. (1882), pp. 56–60.]

The theorem that a binary quantic of any order has only a finite number of irreducible covariants was first established by Gordan. The proof depends on the representation of the covariants by means of symbolical products, and on the theory of the representations of numbers as sums of triangular numbers.

For a binary quantic of order n, say

$$(a, b, c, \ldots \mid x, y)^n = ax^n + nbx^{n-1}y + \tfrac{1}{2}n(n-1)cx^{n-2}y^2 + \ldots,$$

the number of linearly independent covariants of degree θ and order m may be represented in certain cases by the coefficient of $x^{\frac{1}{2}(n\theta-m)}$ in the expansion of $(1-x)^{-\theta}$, or by the number of partitions of $\frac{1}{2}(n\theta-m)$ into θ or fewer parts.

The condition for the existence of such covariants is that $n\theta-m$ should be even and non-negative, and that m should not exceed $n\theta$.

The total number of linearly independent covariants of degree θ is the coefficient of $x^{\frac{1}{2}n\theta}$ in the expansion of

$$(1-x)^{-\theta} + x(1-x)^{-\theta} + x^2(1-x)^{-\theta} + \ldots,$$

or more simply, it may be obtained from the generating function

$$\frac{1}{(1-\alpha x^n)(1-\alpha x^{n-2})(1-\alpha x^{n-4})\cdots},$$

where the expansion is to be made in powers of α and x, and the coefficient of $\alpha^\theta x^m$ gives the number of covariants of degree θ and order m.

Case of the Cubic. For the cubic $(a, b, c, d \mid x, y)^3$, the generating function is

$$\frac{1}{(1-\alpha x^3)(1-\alpha x)(1-\alpha x^{-1})}.$$

The irreducible covariants are:

(1) The cubic itself, of degree 1 and order 3;

(2) The Hessian, of degree 2 and order 2;

(3) The cubicovariant, of degree 3 and order 3;

(4) The discriminant, of degree 4 and order 0.

These are connected by a syzygy of degree 6 and order 6; and there are no other irreducible covariants.

Case of the Quartic. For the quartic $(a, b, c, d, e \mid x, y)^4$, the generating function is

$$\frac{1}{(1-\alpha x^4)(1-\alpha x^2)(1-\alpha)(1-\alpha x^{-2})}.$$

The irreducible covariants are:

(1) The quartic itself, of degree 1 and order 4;

(2) The Hessian, of degree 2 and order 4;

(3) The catalecticant (or I invariant), of degree 2 and order 0;

(4) The cubicovariant, of degree 3 and order 6;

(5) The J invariant, of degree 3 and order 0.

These are connected by a syzygy; and the complete system consists of these five irreducible forms.

General Theory. In the general case of a quantic of order n, the number of irreducible covariants is finite. This may be shown by considering the semi-invariants or sources of the covariants. A semi-invariant is a function of the coefficients which is annihilated by the operator

$$a\frac{\partial}{\partial b} + 2b\frac{\partial}{\partial c} + 3c\frac{\partial}{\partial d} + \dots,$$

and the source of a covariant is the coefficient of the highest power of x in the covariant.

If we have a system of semi-invariants, we may form from them by the operation of the U operator (which raises the order) the corresponding covariants; and conversely, every covariant has a semi-invariant for its source.

The theorem of finitude then follows from the theorem that the number of irreducible semi-invariants is finite; and this in turn may be proved by the method of Gordan, or by the more general theory of Hilbert on the finite basis of systems of algebraic forms.

In fact, if we consider the system of all semi-invariants of a given binary quantic, this system has a finite basis; that is, there exists a finite number of semi-invariants $J_1, J_2, \ldots, J_k$ such that every semi-invariant is a rational integral function of these basis elements. And corresponding to this finite basis of semi-invariants, we have a finite system of irreducible covariants.

The proof given by Gordan depends essentially on the theory of binary forms and their symbolical representation; while the later proof by Hilbert is applicable to forms in any number of variables, and establishes the more general theorem of the finite basis for any system of algebraic forms.

762.

ON SCHUBERT'S METHOD FOR THE CONTACTS OF A LINE WITH A SURFACE.

[From the Messenger of Mathematics, vol. xi. (1882), pp. 71–84.]

The theory of the contacts of a line with a surface of order n has been developed by Schubert by means of his calculus of enumerative geometry. The principal results relate to the number of lines which have a contact of given order with the surface; and the method employed is that of symbolic notation and the principle of conservation of number.

1. Contacts of the first order. A line meets a surface of order n in n points. If two of these points coincide, the line has a contact of the first order (simple contact) with the surface. The condition for this contact is one condition; that is, the lines having a simple contact with the surface form a system of ∞^3 lines in space (since the lines in space form a system of ∞^4, and one condition reduces the number of parameters by 1).

The number of such lines which pass through a given point, or which lie in a given plane, may be determined by Schubert's methods.

2. Contacts of the second order. If three of the points of intersection coincide, the contact is of the second order (stationary contact, or contact at an inflexion). This imposes two conditions; and the lines having such contact form a system of ∞^2 lines.

The surface of order n has in general a certain number of parabolic points, at which the second order contact is specially situated; and the locus of these points is the parabolic curve, of order $4n(n-2)$, on the surface.

3. Contacts of the third order. If four points of intersection coincide, the contact is of the third order (close contact). This imposes three conditions; and the lines having such contact form a system of ∞^1 lines, that is a congruence.

The number of lines of close contact which meet a given line, or which pass through a given point, may be determined. For a surface of order n, the number of close contacts through a point is in general $n(n-1)(n-2)$.

4. Contacts of the fourth order (osculating contact). If all five points of intersection coincide, the contact is of the fourth order, or osculating. This imposes four conditions; and the osculating lines form a complex of ∞^2 lines.

The order of this complex, that is the number of osculating lines through a given point, is for a surface of order n:

$$N = n(n-1)(n-2)(n^2+3n-2).$$

5. Five-point contact. If the five points of intersection coincide in such a way that the line has five consecutive points in common with the surface, the contact is of the fifth order. This requires five conditions; and the number of such lines is finite.

For a surface of order n, the number of lines having five-point contact is

$$\tfrac{1}{2}n(n-1)(n-2)(n-3)(n^4+2n^3+11n^2+10n+8).$$

6. Six-point contact. For six-point contact, six conditions are required; and in general there are no such lines on an arbitrary surface of order n. But for special surfaces (ruled surfaces, developables, etc.) such contacts may exist.

Schubert's symbolic notation. The conditions for contact are represented symbolically as follows:

- ϵ: the condition that a line meets a given line;
- ϵ': the condition that a line lies in a given plane;
- p: the condition that a line passes through a given point;
- g: the condition that a line lies in a given plane through a given point.

The contact conditions are then denoted by symbols such as ϵ_2, ϵ_3, etc., representing contacts of the second and third orders respectively; and the principal formulas of Schubert are:

$$\begin{aligned}
\epsilon_2 &= 2\epsilon - p - g,\\
\epsilon_3 &= 3\epsilon - 2p - 2g + \text{conditions},\\
\epsilon_4 &= \text{combinations of lower contact conditions.}
\end{aligned}$$

The exact formulae depend on the particular problem, and are verified by the principle of conservation of number, according to which the number of geometric configurations satisfying given conditions is independent of the special position of the elements, provided the problem is properly posed.

Application to the quadric surface. For a quadric surface ($n = 2$), the only contacts are simple contacts; there are no stationary or higher contacts on a general quadric. The lines having simple contact with the quadric are the generators; and through each point of the quadric pass two such lines.

Application to the cubic surface. For a cubic surface ($n = 3$), there are:

- 27 lines on the surface (these have simple contact at every point of the line);
- a finite number of lines having stationary contact;
- the inflexional tangents, forming a congruence.

The 27 lines are classical; they may be classified according to their configuration on the surface, and their contacts with the surface are of the simplest kind.

General formulae. For a surface of order n, the number of lines having contact of order r ($r = 1, 2, 3, 4$) and satisfying the remaining conditions to make the problem determinate, is given by formulae involving n and the Schubert conditions. The general formula for the number of lines having simple contact and meeting four given lines is

$$N_1 = \tfrac{1}{2}n(n-1)(n^2+n+2).$$

For stationary contact and meeting three given lines:

$$N_2 = n(n-1)(n-2)(3n^2+6n-4).$$

For close contact and meeting two given lines:

$$N_3 = \tfrac{1}{2}n(n-1)(n-2)(n^3+3n^2+14n-16).$$

For osculating contact and meeting one given line:

$$N_4 = \tfrac{1}{2}n(n-1)(n-2)(n^4+2n^3+11n^2+10n+8).$$

These formulae agree with those obtained by Schubert, and have been verified by various independent methods.

763.

ON THE THEOREMS OF THE 2, 4, 8, AND 16 SQUARES.

[From the Messenger of Mathematics, vol. xi. (1882), pp. 89–95.]

The theorems relating to the expression of a number as a sum of 2, 4, 8, or 16 squares have been established by Jacobi and others by means of the theory of elliptic functions. The following investigation gives a direct algebraical proof of these theorems, and exhibits the connexion between them.

Theorem of the Two Squares. The number of representations of a number n as a sum of two squares is $4(d_1 - d_3)$, where d_1 is the number of divisors of n of the form $4m+1$, and d_3 the number of divisors of the form $4m+3$.

This may be written

$$r_2(n) = 4\sum_{d|n}(-1)^{\frac{1}{2}(d-1)},$$

the summation extending over all odd divisors of n.

Theorem of the Four Squares. Every number is a sum of four squares; and the number of representations is

$$r_4(n) = 8\sum_{d|n,\ 4\nmid d} d,$$

the summation extending over all divisors of n which are not divisible by 4.

Theorem of the Eight Squares. The number of representations of n as a sum of eight squares is

$$r_8(n) = 16\sum_{d|n}(-1)^{n+d}d^3.$$

Theorem of the Sixteen Squares. The number of representations of n as a sum of sixteen squares is

$$r_{16}(n) = \tfrac{32}{17}\sum_{d|n}(-1)^{n+d}d^7 + \tfrac{64}{17}\sum_{\substack{d|n\\ d\text{ even}}} d^7.$$

Proof by means of theta-functions. The proofs of these theorems depend on the properties of the theta-functions. We have

$$\vartheta_3(0,q)=\sum_{n=-\infty}^{\infty} q^{n^2}=1+2q+2q^4+2q^9+\ldots,$$

and therefore

$$\vartheta_3^2(0,q)=\sum_{n=0}^{\infty} r_2(n)q^n, \qquad \vartheta_3^4(0,q)=\sum_{n=0}^{\infty} r_4(n)q^n,$$

and similarly for 8 and 16 squares.

Now the theta-functions satisfy the differential equation

$$4q\frac{d}{dq}\log\vartheta_3(0,q)=\vartheta_2^4(0,q)+\vartheta_4^4(0,q),$$

and by means of this and the known relations between the theta-functions, the values of $r_2(n)$, $r_4(n)$, $r_8(n)$, $r_{16}(n)$ may be determined.

Direct algebraic proof. Let

$$\sigma_k(n)=\sum_{d|n} d^k$$

denote the sum of the kth powers of the divisors of n; and let

$$\sigma_k^*(n)=\sum_{d|n,\ 2\nmid d} d^k=\sum_{d|n} d^k-\sum_{d|n,\ 2|d} d^k.$$

Then the theorems may be stated as follows:

$$\begin{aligned}
r_2(n)&=4\{\sigma_0^*(n;4m+1)-\sigma_0^*(n;4m+3)\},\\
r_4(n)&=8\sigma_1^*(n)-32\sigma_1^*(\tfrac14 n),\\
r_8(n)&=16(-1)^n\sigma_3^*(n)+256\sigma_3^*(\tfrac14 n),\\
r_{16}(n)&=\tfrac{32}{17}\{(-1)^n\sigma_7^*(n)+2\sigma_7^*(\tfrac12 n)+16\sigma_7^*(\tfrac14 n)\},
\end{aligned}$$

where $\sigma_k^*(\frac14 n)$ is understood to be 0 when n is not divisible by 4.

The proofs depend essentially on the identities between the theta-functions, and on the expansion of powers of $\vartheta_3(0,q)$ in series of powers of q.

Connexion with modular functions. The theorems of the squares are also connected with the theory of modular functions. If we write

$$q=e^{\pi i\tau}, \qquad \tau=\omega+i\omega',$$

then $\vartheta_3(0,q)$, $\vartheta_2(0,q)$, $\vartheta_4(0,q)$ are modular functions of τ; and the identities between them are equivalent to relations between modular forms.

In particular, the theorem of four squares is equivalent to the statement that the theta-function $\vartheta_3^4(0,q)$, which is a modular form of weight 2, has a development in which the coefficient of q^n is determined by the divisor function.

Application to the representation of primes. A prime p of the form $4m+1$ can be expressed as a sum of two squares in essentially one way; and the number of representations is 8 (counting order and signs).

A prime of the form $4m+3$ cannot be expressed as a sum of two squares. Every prime can be expressed as a sum of four squares; and the number of representations is $8(p+1)$ for an odd prime p.

Special cases. For $n = 1$, we have $r_2(1) = 4$, corresponding to $(\pm1)^2 + 0^2$ and $0^2 + (\pm1)^2$; $r_4(1) = 24$; $r_8(1) = 16$; $r_{16}(1) = 32$.

For $n = 2$, we have $r_2(2) = 4$, from $(\pm1)^2 + (\pm1)^2$; $r_4(2) = 24$; $r_8(2) = 16$; $r_{16}(2) = 32$.

These values agree with the general formulae; and the theorems have been verified in numerous particular cases.

[761]

ON THE THEOREM OF THE FINITE NUMBER OF THE COVARIANTS OF A BINARY QUANTIC.

[From the Proceedings of the London Mathematical Society, vol. XII. (1881), pp. 95—101. Read February 10, 1881.]

[Continued from p. 276.]

viz. 12.34.56 here represents the square R_1, for which the terms 12, 34, 56 (and of course the symmetrical terms 21, 43, 65) are each $= 1$, the other terms all vanishing; or, what is the same thing, it represents the invariant $(\alpha-\beta)^{12}(\gamma-\delta)^{34}(\epsilon-\zeta)^{56}$. But it is not true that *every* square R_2 is a sum of squares R_1; this is not the case, for the square R_2,

$$= 12.13.23.45.46.56,$$

representing the invariant

$$(\alpha-\beta)^{12}(\alpha-\gamma)^{13}(\beta-\gamma)^{23}(\delta-\epsilon)^{45}(\delta-\zeta)^{46}(\epsilon-\zeta)^{56},$$

is not a sum of squares R_1.

But the square last referred to is a *difference* of squares R_1: it is in fact

$$= 12.36.45 + 13.25.46 + 14.23.56 - 14.25.36,$$

or, what is the same thing, the corresponding invariant is the product of the invariants 12.36.45, 13.25.46, 14.23.56, divided by the invariant 14.25.36; viz. it is a rational function of invariants R_1.

It is required to show, first, that *every* square R_2 is a *difference* of squares R_1; and thence, secondly, that it is a sum of a finite number of squares R_k (being, in fact, squares R_1 and R_2).

For the first theorem we equate the general expression of R_2 with the assumed value

$$x_1.12.34.56 + y_1.12.35.46 + z_1.12.36.45 + \ldots + z_5.16.25.34.$$

We thus obtain

fifteen equations	*satisfied by*	
$12 = y_1 + x_1 + z_1$	$x_1 = 34 - 26$	$+r + s - t,$
$13 = x_2 + y_2 + z_1$	$x_2 = 13 - 25$	$+p \quad - r \quad + t,$
$14 = x_3 + y_3 + z_3$	$x_3 = 14$	$-p \qquad - s,$
$15 = x_4 + y_4 + z_4$	$x_4 = 15 - 26 - 36$	$+p + q + r + s,$
$16 = x_5 + y_5 + z_5$	$x_5 = 45$	$-q - r,$
$23 = x_3 + x_4 + x_5$	$y_1 = 12 - 34 + 26$	$-q - r - s + t,$
$24 = x_2 + y_4 + y_5$	$y_2 = 25$	$-p,$
$25 = y_2 + y_3 + z_5$	$y_3 = p$	,
$26 = z_1 + z_3 + z_4$	$y_4 = 36$	$-p - q,$
$34 = x_1 + z_4 + z_5$	$y_5 = 16 - 45$	$+q + r - t,$
$35 = y_1 + y_3 + z_3$	$z_1 = q$	,
$36 = y_3 + y_4 + z_1$	$z_2 = r$	,
$45 = x_5 + z_1 + z_2$	$z_3 = s$	,
$46 = x_4 + y_1 + y_2$	$z_4 = 26$	$-r - s,$
$56 = x_1 + x_2 + x_3$	$z_5 = t$	,

connecting x_1, y_1, z_1, $\ldots$, z_5 with the terms 12, 13, etc. of R_2 (or indices of the corresponding invariants). The fifteen equations are not independent, for regarding them as giving the values of 12, 13, $\ldots$ in terms of the x_1, y_1, z_1, $\ldots$, z_5, these values satisfy identically the relations which ought to be satisfied by the terms 12, 13, etc., viz. the equations obtained by the elimination of θ from the equations

$$\begin{array}{rr}
12 + 13 + 14 + 15 + 16 & = \theta, \\
12 \qquad + 23 + 24 + 25 + 26 & = \theta, \\
\cdots\cdots\cdots\cdots\cdots\cdots & \\
16 + 26 + 36 + 46 + 56 \qquad\qquad & = \theta.
\end{array}$$

The equations are thus insufficient to determine the values of x_1, y_1, z_1, $\ldots$, z_5, and the general values given by the equations will contain five indeterminate quantities which are taken to be p, q, r, s, t (these being in fact the values of y_3, z_1, z_2, z_3, z_5 respectively), and we then have the equations all of them satisfied by the above-mentioned values containing these indeterminate quantities; taking them to be positive or negative integers, then x_1, y_1, z_1, $\ldots$, z_5, will be all of them integers; but by what precedes, it appears that they cannot all of them be made to be *positive* integers, so that we have consequently R_2,

$$= x_1.12.34.56 + y_1.12.35.46 + z_1.12.36.45 + \ldots + z_5.16.25.34,$$

equal in general to a difference of squares R_1.

Suppose in such difference of squares R_1, we have any term, say $-12.34.56$, occurring with the coefficient -1. Since the expression represents a square R_2, we must have among the positive terms, 12.35.46 or 12.36.45 to render possible the subtraction of the 12; 15.26.34 or 16.25.34 to render possible the subtraction of the 34; and 13.24.56 or 14.23.56 to render possible the subtraction of the 56; that is, the expression must contain one of the eight combinations

$$\begin{gathered}
12.35.46 + 15.26.34 + 13.24.56 - 12.34.56, \\
12.35.46 + 15.26.34 + 14.23.56 - 12.34.56, \\
12.35.46 + 16.25.34 + 13.24.56 - 12.34.56, \\
12.35.46 + 16.25.34 + 14.23.56 - 12.34.56, \\
12.36.45 + 15.26.34 + 13.24.56 - 12.34.56, \\
12.36.45 + 15.26.34 + 14.23.56 - 12.34.56, \\
12.36.45 + 16.25.34 + 13.24.56 - 12.34.56, \\
12.36.45 + 16.25.34 + 14.23.56 - 12.34.56.
\end{gathered}$$

The first of these is 35.46.15.26.13.24, viz. it is 13.15.35.24.26.46 which is a square R_2 (of the form mentioned above); the second is 35.46.15.26.14.23, which is $15.23.46 + 14.26.35$, a sum of squares R_1; and similarly each of the other combinations is either a square R_2 or a sum of squares R_1. We have thus got rid of the negative term $-12.34.56$, and in like manner if the negative term had been

$$-m.12.34.56, = -12.34.56 - 12.34.56 - \&c.,$$

or, whatever the negative terms may be, we get rid one by one of each negative term; and thus ultimately express R_2 as a sum of squares R_1 and R_2. Or, what is the same thing, the invariant R_2 originally expressed as a *rational* function of invariants R_1, is finally expressed as a *rational and integral* function of invariants R_1 and R_2.

Similarly for a root-quantic of any even order n, we have the general square R_2 expressed, first as a difference of squares R_1, and then as a sum of squares R_1, R_2, or it may be higher squares R_3, &c., but certainly as a sum of a *finite* number of squares R_k. For a root-quantic of any odd order n, the investigation would be of a somewhat different form, since here there are no squares R_1, but the lowest squares are squares R_2 of a form such as 12.23.34.45.15; but the general conclusion would still follow that every square R_2 is a sum of a *finite* number of squares R_k. And a like reasoning would apply to covariants instead of invariants: viz. the reasoning (although for simplicity it has been given for a very particular and special case) does, I think, really establish the theorem (A) in its generality, viz. the theorem that for a root-quantic of any given finite order, the number of irreducible monomial covariants is finite.

From any monomial covariant of the root-quantic, by taking the sum of the forms belonging to the different roots, so as to obtain a symmetrical function of the roots, that is, a rational and integral function of the coefficients, we obtain a covariant of the quartic in its ordinary form $(a, \dots \mid x, y)^n$. Consider for a moment the before-mentioned case of the invariants of the root-quartic

$$(x - \alpha y)(x - \beta y)(x - \gamma y)(x - \delta y),$$

now put

$$= \frac{1}{a}(a, b, c, d, e \mid x, y)^4;$$

and to make the reasoning clearer, take a, b, c, f, g, $h = (\alpha-\delta)(\beta-\gamma)$, $(\beta-\delta)(\gamma-\alpha)$, $(\gamma-\delta)(\alpha-\beta)$, $(\alpha-\delta)(\gamma-\beta)$, $(\beta-\delta)(\alpha-\gamma)$, $(\gamma-\delta)(\beta-\alpha)$ respectively, these being, with the signs $\pm$, the before-mentioned three monomial invariants. In the root-theory, every monomial invariant is a rational and integral function of a, b, c, f, g, h. Every invariant of $(a, \dots \mid x, y)^4$, *quà* rational and integral function of the coefficients, is, when expressed in terms of the roots, a rational and integral function of the roots, and then *quà* invariant is a sum of monomial invariants, and as such a rational and integral function of a, b, c, f, g, h. But every such rational and integral function of a, b, c, f, g, h is not a symmetrical function of α, β, γ, δ, and consequently not in the present theory an invariant of $(a, \dots \mid x, y)^4$; the invariants are those rational and integral functions of a, b, c, f, g, h which are symmetrical functions of $(\alpha, \beta, \gamma, \delta)$, that is, which remain unaltered by every substitution whatever upon the roots $(\alpha, \beta, \gamma, \delta)$. Now each such substitution gives a substitution upon a, b, c, f, g, h, and the 24 substitutions upon α, β, γ, δ give a group of 6, $= \frac{1}{4} \cdot 24$ substitutions upon (a, b, c, f, g, h); the invariants are thus the rational and integral functions of (a, b, c, f, g, h) which are unaltered by each of the substitutions of a certain group $G(a, b, c, f, g, h)$ of 6 substitutions. Theorem (B) asserts that, among the terms in

$$I, = ae - 4bd + 3c^2, \text{ and } J, = ace - ad^2 - b^2e + 2bcd - c^3.$$

As regards the group $G(a, b, c, f, g, h)$ of 6 substitutions upon a, b, c, f, g, h, observe that the 24 substitutions of $(\alpha, \beta, \gamma, \delta)$ operating upon a, b, c, f, g, h give 6 substitutions taken each four times; for instance, the substitutions 1, $\alpha\beta.\gamma\delta$, $\alpha\gamma.\beta\delta$, $\alpha\delta.\beta\gamma$ leave each of them a, b, c, f, g, h unaltered, that is, they each give the substitution 1. And we thus find for the group $G(a, b, c, f, g, h)$ the 6 substitutions

$$\begin{aligned}
&1,\\
&abc.fgh,\\
&acb.fhg,\\
&af.bh.cg,\\
&ah.bg.cf,\\
&ag.bf.ch.
\end{aligned}$$

For the functions of a, b, c, f, g, h, which remain unaltered by the substitution of this group, observe that we have f, g, $h = -a$, $-b$, $-c$; so that any function of the six letters may be represented as a function of a, b, c. An odd symmetrical function, for instance abc, does *not* remain unaltered, for it is by any one of the last three substitutions changed into fgh, that is, into $-abc$; on the other hand, the two-valued function $(b-c)(c-a)(a-b)$ does remain unaltered: the functions which remain unaltered are therefore the even symmetrical functions of a, b, c (that is, the symmetric functions $a^2 + b^2 + c^2$, or $ab + ac + bc$, &c., which are of an even order in a, b, c conjointly), and the same even functions multiplied by $(b - c)(c - a)(a - b)$; and having regard to the relation $a + b + c = 0$, all these can be expressed as already mentioned as rational and integral functions of $bc + ca + ab$ and $(b - c)(c - a)(a - b)$.

The proof applies to the general case of the theorem (C), viz. taking the theorem (A) to be proved, and putting the root-quantic

$$(x - \alpha y)(x - \beta y) \ldots = \frac{1}{a}(a, \ldots \mid x, y)^n,$$

then we have a, b, c, d, ... a system of monomial covariants of the root-quantic; and all the covariants of $(a, \ldots \mid x, y)$ are rational and integral functions of $(a, b, c, d, \ldots)$ which remain unaltered by the substitutions of a certain group $G(a, b, c, d, \ldots)$; hence, assuming the theorem (B), they are rational and integral functions of a finite number of irreducible covariants. And the demonstration thus depends upon that of the theorem (B).

762.

ON SCHUBERT'S METHOD FOR THE CONTACTS OF A LINE WITH A SURFACE.

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I WISH to reproduce in part § 33, "Coincidenz von Schnittpunkten einer Geraden mit einer Fläche" of Schubert's very interesting work *Calcul der abzählende Geometrie*, Leipzig, 1879, explaining in the first instance (but not altogether in the manner or from the point of view of the author) the general principles of the theory.

We have to do with *conditions* relating to a *subject*; the subject is a geometrical form or entity of any kind depending upon a certain number of constants; and the condition is onefold, twofold, &c., according as it imposes a onefold, twofold, &c., relation upon these constants. The number of constants is the Postulandum of the subject, and the manifoldness of the condition is called also its Postulation. A condition is incomplete when its postulation is less than postulandum of subject, complete when its postulation is equal to postulandum of subject; two or more incomplete conditions, making up a complete condition, are supplementary to each other. The case where the postulation exceeds the postulandum, or say that of a more than complete condition, is not in general considered; it may however sometimes present itself. For instance, the subject may be a line with n points upon it; the number of constants is here $= n + 4$. A condition that the line shall meet a given line, or that a certain one of the n points shall lie on a given plane, is a onefold condition; the condition that such point shall lie upon a given line is a twofold condition; and so in other cases.

Conditions are denoted by letters, and simultaneous conditions by a product; for instance, the subject is a line carrying the n points 1, 2, …, n; g is the condition that the line meets a given line; p_1 the condition that the point 1 lies on a given plane; then gp_1 is the twofold condition that the line meets a given line and that the point 1 lies on a given plane; ${p_1}^2$ is the twofold condition that the point 1 lies on each of two given planes (in fact, on their line of intersection). The letters p, g, e are used as the initials of Punkt, Gerade, Ebene.

The letter or combination of letters denoting an incomplete condition, or, say, the incomplete condition itself, has no numerical value; but for a complete condition there exists a definite number of subjects satisfying the condition, and the condition is regarded as having this number as its value. A more than complete condition has the value 0.

Conditions of the same postulation may be connected by the sign +; for instance,

subject a line,

g_e the condition that it lies in a **given** plane,

g_p the condition that it passes through a given point,

then $g_e + g_p$ is the condition that the line shall either lie in the given plane or else pass through the given point.

I abstain from attempting any definition in regard to the sign $-$.

Conditions of the same postulation may be connected by an equation or equations; for instance,

subject a point,

p the condition that the point shall lie in a given plane,

p_g the condition that the point shall lie in a given line,

then $p^2 = p_g$.

This equation has (so far) no numerical signification; it has the logical signification that the condition that a point shall lie on each of two given planes is equivalent to the condition that the point shall lie on a given line.

Second example. Subject a line,

g the condition that the line meets a given line,

g_e the condition that it lies in a given plane,

g_p the condition that it passes through a given point,

then $g^2 = g_e + g_p$.

This equation has (so far) no numerical signification, and I regard it as having no logical signification. Schubert, however, gives it a logical signification by means of his "Princip der speciellen Lage" (Principle of Special Situation), viz. the condition of the line meeting each of two given lines is, in the particular case where the two given lines meet, equivalent to the condition, that the line shall either lie in the plane of the two given lines or else pass through their point of intersection.

Third example. Subject a line bearing upon it the points 1 and 2,

ϵ the condition of the coincidence of the two points,

p " that the point 1 shall lie on a given plane,

q " " 2 " "

g " that the line shall meet a given line,

then $\epsilon = p + q - g$.

This equation has (so far) no numerical signification, and it does not appear to have any logical signification. In fact, in the actual form of the equation we have a sign $-$ which has not had given to it any logical interpretation; and if we write the equation in the form $\epsilon + g = p + q$, there seems to be no logical signification in the assertion, the condition that either the points shall coincide, or else the line meet a given line, is equivalent to the condition that either the first point, or else the second point, shall lie in a given plane.

Any equation connecting complete conditions is a numerical equation; and to render a condition complete, we have only to join to it a supplementary condition X of the proper postulation. Thus, in the last example the postulandum is $= 6$; ϵ, p, q, g are onefold conditions, and joining to each of them one and the same fivefold condition X, we have $X\epsilon = Xp + Xq - Xg$. And, taking X to be an arbitrary fivefold condition, the original equation $\epsilon = p + q - g$ has in fact the meaning

$$X\epsilon = Xp + Xq - Xg.$$

For instance, the fivefold condition X may be that the line shall belong to a given regulus (scroll or developable surface), and that the points 1, 2 upon the line shall be the intersections of the line with given surfaces S_1, S_2 respectively. The subject is the line of the given regulus with its two points; and the meaning of the equation is that the number of subjects with two coincident points is equal to the number of subjects with the point 1 on a given plane, *plus* the number of subjects with the point 2 on a given plane, *minus* the number of subjects for which the line meets a given line. Although for the moment concerned only with the meaning of the theorem, not with its truth, I stop to show *à posteriori* that the theorem is in fact true: take k for the order of the regulus; m_1, m_2 for the orders of the surfaces S_1, S_2 respectively; then it is to be shown that $X\epsilon$, Xp, Xq, Xg are each $= km_1m_2$ (values which satisfy the equation). First $X\epsilon$: the points 1 and 2 here coincide at a point of the curve of the order m_1m_2, which is the intersection of S_1 and S_2; the regulus meets this curve in km_1m_2 points, and through each of these we have a line of the regulus having upon it the two coincident points; that is, $X\epsilon = km_1m_2$. Next Xp: the point 1 is here on the plane curve of the order m_1, which is the intersection of S_1 with the corresponding given plane; the regulus meets this plane curve in km_1 points; through each of these we have a line of the regulus intersecting S_2 in m_2 points, any one of which may be taken for the point 2; that is, the number of subjects is $Xp, = km_1.m_2$. Then Xq: in precisely the same manner we have $Xq = km_2.m_1$. Lastly Xg: the given line meets the regulus in k points, and through each of these there is a line of the regulus meeting S_1 in m_1 points, any one of which may be taken for the point 1, and meeting S_2 in m_2 points, any one of which may be taken for the point 2; the number of the subjects Xg is thus $Xg, = k.m_1.m_2$.

The general theorem $X\epsilon = Xp + Xq - Xg$ is proved by means of Chasles' theorem of united points as follows: the subject is a line, or say, for convenience, an *axis* ξ, bearing upon it the two points 1 and 2; we consider in conjunction therefore a given line λ, and through this draw the planes P_1, P_2 passing through the points 1 and 2 respectively.

Suppose that when 2 lies in a given plane there are α' positions of the axis, and on each of these β' positions of the point 1; and, similarly, that when 1 lies on a given plane there are α positions of the axis, and on each of these β positions of the point 2; then, 1 lying in a given plane, the number of subjects is $\alpha\beta$, or we have $Xp = \alpha\beta$; and, similarly, $Xq = \alpha'\beta'$. Take now for the point P_1 an arbitrary plane through λ; then, 1 lying on this plane, the number of the points 2 is $= \alpha\beta$, or, since each of these determines with λ a position of the plane P_2, the number of these planes is $= \alpha\beta$, that is, it is $= Xp$; and, similarly, taking P_2 an arbitrary plane through λ, the number of the planes P_1 is $\alpha'\beta'$, that is, it is $= Xq$; viz. the two planes P_1, P_2 through the line λ have an (Xp, Xq) correspondence; hence, by Chasles' theorem, the number of united planes is $= Xp + Xq$.

But we have a united plane, 1°, if the points 1 and 2 coincide, that is, if the condition $X\epsilon$ be satisfied, and the number of these united planes is $X\epsilon$; 2°, if the axis ξ meet the arbitrary line λ, that is, if the condition Xg be satisfied, and the number of these united planes is $= Xg$; hence the whole number is $= X\epsilon + Xg$; or we have $Xp + Xq = X\epsilon + Xg$, that is, $X\epsilon = Xp + Xq - Xg$, which is the theorem in question.

The conclusion is that the equation $\epsilon = p + q - g$, which in this, its original form, has neither a numerical nor a logical signification, is to be understood as meaning the numerical equation $X\epsilon = Xp + Xq - Xg$, the truth of which numerical equation has just been proved. Or we may, without explicit introduction of the condition X, understand the equation $\epsilon = p + q - g$ as a numerical equation as follows, viz. taking for the subject a line with two points, *which line and points are regarded as satisfying a given fivefold condition*, then

ϵ is the (additional onefold) condition that the two points shall coincide,

p " " that the point 1 shall lie in given plane,

q " " that the point 2 shall lie in given plane,

g " " that line shall meet given line.

The conditions ϵ, p, q, g are thus in effect complete conditions, having values which may be connected by an equation; there, in fact, exists between them the relation

$$\epsilon = p + q - g.$$

The like remarks would apply to the before-mentioned equation (subject a point) $p^2 = p_g$: either adding to it a onefold condition X, and so taking it in the form $Xp^2 = Xp_g$, or understanding it in its original form $p^2 = p_g$ as belonging to a point which satisfies already a onefold condition, the equation is true as a numerical equation; and this in fact follows at once from its truth as a logical equation. But observe the difference: the equation in question $p^2 = p_g$ has, the equation $\epsilon = p + q - g$ has not, a logical signification.

I regard as the fundamental notion of the theory the existence of equations between conditions such as the foregoing equation $\epsilon = p+q-g$; equations which in their original form have not (of necessity) any logical signification, and have not any numerical signification; but which, when we adjoin to them a supplementary condition X of the proper postulation, become numerical equations, which are true, independently of the form of the supplementary condition X and whatever this condition may be. And this being so, it seems to follow at once that such equations may be treated and worked with as ordinary algebraical equations. For instance, let M be any condition of less postulation than X: then if from the equation $\epsilon = p + q - g$, assumed to be true, we deduce $M\epsilon = Mp + Mq - Mg$, this (like the original equation $\epsilon = p + q - g$) is in its actual form an equation without logical or numerical signification; but if we adjoin to it a supplementary condition K, such that postulation of K + do. of M = do. of X (or, what is the same thing, that the condition KM shall be supplementary to the several conditions contained in the original equation $\epsilon = p + q - g$), then the equation in question, $M\epsilon = Mp + Mq - Mg$, is to be interpreted as meaning

$$KM\epsilon = KMp + KMq - KMg,$$

that is,

$$X\epsilon = Xp + Xq - Xg,$$

which is numerically true. We thus see that the original equation $\epsilon = p + q - g$ implies the new equation

$$M\epsilon = Mp + Mq - Mg,$$

which is its algebraical consequence. And if we regard, for instance, $A + B$ as the condition that either the condition A shall be satisfied or else the condition B shall be satisfied, then $A + B$ is a condition, and as such we have

$$(A + B)\epsilon = (A + B)p + (A + B)q - (A + B)\epsilon.$$

It is going a step further to say that if we have, for instance, an equation $M = A+B-C$ between conditions M, A, B, C, then that, instead of

$$M\epsilon = Mp + Mq - M\epsilon,$$

we may write

$$(A + B - C)\epsilon = (A + B - C)p + (A + B - C)q - (A + B - C)\epsilon;$$

this is, in fact, treating $A+B-C$ as being to all intents and purposes a condition such as M, or an alternative condition $A + B$. It is, in fact, assumed that the step is permissible; and we thus make such deductions as

$$(\epsilon + p + q - g)(\epsilon - p - q + g) = 0;$$

that is,

$$\epsilon^2 - (p + q - g)^2 = 0,$$

or

$$\epsilon^2 = (p + q - g)^2, = p^2 + 2pq + q^2 - 2pg - 2qg + g^2;$$

viz. this is an equation such as the original equation $\epsilon = p + q - g$, acquiring a numerical signification when we adjoin to it a supplementary condition X of the proper postulation.

The section above referred to deals with the question to determine the number of lines which satisfy the several relations of contact in regard to a given surface F of the order n, without point-singularities, that is, the surface represented by the *general* equation $(* \mid x, y, z, w)^n = 0$.

The chief results are contained in the following table, the notation of which will be explained:

$$
\begin{aligned}
1.\quad \epsilon_2 g_s &= n(n-1),\\
2.\quad \epsilon_2 b_2 g_e &= n,\\
3.\quad \epsilon_3 g_e &= 3n(n-2),\\
4.\quad \epsilon_3 g_p &= n(n-1)(n-2),\\
5.\quad \epsilon_4 {b_4}^2 &= 2n,\\
6.\quad \epsilon_{22} g_e &= \tfrac{1}{2}n(n-2)(n-3)(n+3),\\
7.\quad \epsilon_{22} g_p &= \tfrac{1}{2}n(n-1)(n-2)(n-3),\\
8.\quad \epsilon_{22} {b_4}^2 &= n(n-3)(n+2),\\
9.\quad \epsilon_{22} b_2 c_2 &= n(n^2-2n^2+2n-6),\\
10.\quad \epsilon_4 g &= 2n(n-3)(3n-2),\\
11.\quad \epsilon_4 b_4 &= n(11n-24),\\
12.\quad \epsilon_{22} g &= n(n-3)(n-4)(n^2+6n-4),\\
13.\quad \epsilon_{22} b_4 &= n(n-4)(3n^2+5n-24),\\
14.\quad \epsilon_{22} b_2 &= n(n-2)(n-4)(n^2+2n+12),\\
15.\quad \epsilon_{222} g &= \tfrac{1}{3}n(n-3)(n-4)(n-5)(n^2+3n-2),\\
16.\quad \epsilon_{222} b_2 &= \tfrac{1}{2}n(n-2)(n-4)(n-5)(n^2+5n+12),\\
17.\quad \epsilon_5 &= 5n(n-4)(7n-12),\\
18.\quad \epsilon_{42} &= 2n(n-4)(n-5)(n+6)(3n-5),\\
19.\quad \epsilon_{33} &= \tfrac{1}{2}n(n-4)(n-5)(n^3+3n^2+29n-60),\\
20.\quad \epsilon_{222} &= \tfrac{1}{6}n(n-4)(n-5)(n-6)(n^3+9n^2+20n-60),\\
21.\quad \epsilon_{2222} &= \tfrac{1}{12}n(n-4)(n-5)(n-6)(n-7)(n^3+6n^2+7n-30),\\
22.\quad \epsilon_{222} b_2 &= \tfrac{1}{3}n(n-4)(n-5)(n-6)(n^3+3n^2-2n-12),\\
23.\quad \epsilon_3 {b_4}^2 &= (n-3)(n^2+2),\\
24.\quad \epsilon_2 b_1 c_1 d_1 &= n^2(n-4)(2n^2-3n-3).
\end{aligned}
$$

In the foregoing formulæ the suffixes of the ϵ refer to the contacts, viz. ϵ_2 denotes a 2-pointic intersection, ϵ_{22} a 3-pointic and a 2-pointic intersection. The letters b, c, d refer to the points of contact or intersection, thus $\epsilon_{33}b_3$, b_3 is the point of 3-pointic intersection; $\epsilon_{222}b_1$, b_1 is one of the points of simple intersection; b_1 is also the condition that the point in question lies on a given plane; g, g_s, g_e, g_p have their ordinary signification explained a little further on. Thus (15) $\epsilon_{222}g$ denotes the number of triple tangents which can be drawn to meet a given line; or, what is the same thing, it is the order of the regulus formed by the triple tangents.

The following are elementary formulæ used in the investigation of the foregoing results.

		Postul.
	Subject a line having upon it a point,	
p	the condition that point is in a given plane	1
p_g	" " line	2
g	" line meets a given line	1
g_e	" " is in a given plane	2
g_p	" " passes through a given point	2
g_s	" " lies in a given plane and passes through a given point of that plane	3
G	" " coincides with a given line	4

We have (p. 22 *et seq.*)

$p_g = p^2$	2	(*logical*)
$p_g = p^2 + g_e$	2	
$g^2 = g_e + g_p$	2	
$g_s = gg_e$	3	(*logical*)
$g_s = gg_p$	3	(*logical*)
$pg_p = p^3 + g_s$	3	(*demons. infrà*)
$p^4 = 0$	4	
$g_e g_p = 0$	4	
${g_e}^2 = G$	4	
${g_p}^2 = G$	4	
$p^2 g = p^2 g_e$	4	(*demons. infrà*)
$pg_s = p^2 g_p$	4	"
$pg_s = p^2 g_e + G$	4	"
$p^3 g_e = 0$	5	"
$p^3 g_p = pG$	5	"
$p^3 g_s = pG$	5	"

$pg_p = p^3 + g_s$; we have $0 = g_e + p^2 - pg$, $0 = g_e + g_p - g^2$, and thence

$$\begin{aligned}
0 &= (p+g)\{g_e + p(p-g)\} - p\{g_e + g_p - g^2\} \\
&= pg_e + gg_e + p^3 - pg^2 \\
&\quad - pg_e - pg_p \qquad + pg^2 \\
&= gg_e - pg_p + p^3 \\
&= g_s - pg_p + p^3.
\end{aligned}$$

$p^2g = p^2g_e$: from $pg = p^3 + g_e$, we have $p^2g = p^4 + p^2g_e = p^2g_e$, since $p^4 = 0$,

$pg_s = p^2g_e + G$, „ $g_s = gg_e$ „ $pg_s = pgg_e = (p^2 + g_e)g_e = p^2g_e + G$,

$pg_s = p^2g_p$ „ $g_s = gg_p$ „ $pg_s = pgg_p = (p^2 + g_e)g_p = p^2g_p$, since $g_eg_p = 0$;

and in a similar manner we prove the last three equations.

For the demonstration of the formulæ of the table we take the subject to be a line bearing upon it the points 1, 2, ..., n, which are its intersections with a given surface of the order n. The symbols $p_1, p_2, \ldots$ refer to these points respectively; thus, p_1 is the condition that the point 1 may lie on a given plane; and then, writing

$$\begin{aligned} \epsilon &= p_1 + p_2 - g, \\ \epsilon' &= p_1 + p_3 - g, \\ \epsilon'' &= p_2 + p_4 - g, \end{aligned}$$

it appears that ϵ will denote the condition of the coincidence of the points 1 and 2; ϵ' that of the points 1 and 3, &c. Hence also, $\epsilon\epsilon'$ will denote the twofold condition of the coincidence of the points 1, 2, 3; and so in other cases. But, according to the notation above explained, ϵ is also denoted by ϵ_2, $\epsilon\epsilon'$ by ϵ_3, $\epsilon\epsilon''$ by ϵ_{22}, &c.

We thus have

$$\begin{aligned} \epsilon_2 &= p_1 + p_2 - g, \\ \epsilon_3 &= (p_1 + p_2 - g)(p_1 + p_3 - g), \\ 2\epsilon_{22} &= (p_1 + p_2 - g)(p_3 + p_4 - g), \\ \epsilon_4 &= (p_1 + p_2 - g)(p_1 + p_3 - g)(p_1 + p_4 - g), \\ \epsilon_{32} &= (p_1 + p_2 - g)(p_1 + p_3 - g)(p_4 + p_5 - g), \\ 6\epsilon_{222} &= (p_1 + p_2 - g)(p_3 + p_4 - g)(p_5 + p_6 - g), \\ \epsilon_5 &= (p_1 + p_2 - g)(p_1 + p_3 - g)(p_1 + p_4 - g)(p_1 + p_5 - g), \\ \epsilon_{42} &= (p_1 + p_2 - g)(p_1 + p_3 - g)(p_1 + p_4 - g)(p_5 + p_6 - g), \\ 2\epsilon_{33} &= (p_1 + p_2 - g)(p_1 + p_3 - g)(p_4 + p_5 - g)(p_4 + p_6 - g), \\ 2\epsilon_{322} &= (p_1 + p_2 - g)(p_1 + p_3 - g)(p_4 + p_5 - g)(p_6 + p_7 - g), \\ 24\epsilon_{2222} &= (p_1 + p_2 - g)(p_3 + p_4 - g)(p_5 + p_6 - g)(p_7 + p_8 - g). \end{aligned}$$

We can now, by a mere analytical process of development and reduction, express each of the foregoing values as a linear function of

$$p_1{}^2p_2{}^2,\ p_1{}^2p_2p_3,\ p_1p_2p_3p_4,\ \text{and}\ G.$$

(Schubert says, as a linear function of these four symbols and $p_1p_2g_e$; but in fact $p_1p_2g_e$ is $= p_1{}^2p_2{}^2$.)

Observe, first, that we may, p. 287, in all the general equations instead of p write p_1, p_2, &c.; and, further, that any symbol containing for instance $p_1{}^3$ is $= 0$. For the symbols now belong to the intersections of the line with a given surface; $p_1{}^3$ is the condition that a certain one of these intersections shall lie in three given planes, that is, that it shall coincide with a given arbitrary point; this cannot be the case, for the arbitrary point is not on the surface F; and therefore $p_1{}^3 = 0$.

We thus have $p_1g = p_1{}^2 + g_e$, hence $p_1{}^2g = p_1{}^3 + p_1g_e$, that is, $p_1{}^2g = p_1g_e$; and thence further $p_1{}^2g = p_1{}^2g_e$, that is, $p_1{}^2g_e = 0$.

Again, from $p_2g = p_2{}^2 + g_e$, $p_1g = p_1{}^2 + g_e$, we have

$$p_1{}^2(p_2{}^2 + g_e) = p_1p_2(p_1{}^2 + g_e),$$

which, in virtue of $p_1{}^2g_e = 0$ and $p_1{}^3p_2 = 0$, becomes

$$p_1{}^2p_2{}^2 = p_1p_2g_e.$$

As a simple instance of the reductions, take

$$\epsilon_2g_s, = (p_1 + p_2 - g)g_s.$$

Here

$$p_1g_s, = p_2g_s, = p_1{}^2g_e + G, = G,\ \text{since}\ p_1{}^2g_e = 0;$$

and

$$gg_s = g^2g_e = (g_e + g_p)g_e = g_e{}^2 + g_eg_p = G,\ \text{since}\ g_e{}^2 = 0,\ g_eg_p = G;$$

whence the value is

$$\epsilon_2g_s = G + G - G, = G.$$

As a more complicated example, take

$$\epsilon_5, = (p_1 + p_2 - g)(p_1 + p_3 - g)(p_1 + p_4 - g)(p_1 + p_5 - g).$$

Observe that, after the multiplication is effected we may, in any way we please, interchange the suffixes, $p_1{}^2p_3p_4 = p_1{}^2p_2p_3$, $p_3{}^2p_1{}^2 =$

${p_1}^2{p_2}^2$, &c.; the suffixes serve only to distinguish from each other symbols in the same product (thus ${p_1}^4$ is different from $p_1p_2p_3p_4$), but there is nothing to distinguish one point of intersection from another. Thus the foregoing expression containing the terms $(p_2+p_3+p_4+p_5)(p_1-g)^3$, these may be combined into the single term $4p_2(p_1-g)^3$; expanding in powers of p_1-g and reducing in this manner, the value of ϵ_5 is, in fact, found to be

$$= (p_1-g)^4+4p_2(p_1-g)^3+6p_2p_3(p_1-g)^2+4p_2p_3p_4(p_1-g)+p_1p_2p_3p_4.$$

Developing this in powers of g, omitting the terms containing ${p_1}^3$ which vanish, and further reducing, the value is

$$6{p_1}^2p_2p_3+5p_1p_2p_3p_4+g(-12{p_1}^2p_2-16p_1p_2p_3)+g^2(6{p_1}^2+18p_1p_2)-8p_1g^3+g^4.$$

We have

$$g^4=2G,\ p_1g^3=p_1g_s={p_1}^2g_e+G,=G.$$

Next for the terms in g^2, from $p_1g={p_1}^2+g_e$ we have

$$\begin{aligned} {p_1}^2g &= \qquad p_1g_e, \\ p_1p_2g &= {p_1}^2p_2+p_1g_e, \end{aligned}$$

and thence

$$\begin{aligned} {p_1}^2g^2 &= \qquad p_1g_e; \\ p_1p_2g^2 &= {p_1}^2p_2g+p_1g_e, \end{aligned}$$

or, since $p_1g_s=G$ as before, the whole term is $=18{p_1}^2p_2g+24G$. The terms in g thus become $=g(6{p_1}^2p_2-16p_1p_2p_3)$, and from the same equation $p_1g={p_1}^2+g_e$ we find

$${p_1}^2p_2g={p_1}^2{p_2}^2 \text{ and } p_1p_2p_3g={p_1}^2p_2p_3+{p_1}^2{p_2}^2.$$

The value is thus finally found to be

$$=-10{p_1}^2{p_2}^2-10{p_1}^2p_2p_3+5p_1p_2p_3p_4+10G.$$

The whole series of like results is

			$p_1{}^2p_2{}^2$	$p_1{}^2p_2p_3$	$p_1p_2p_3p_4$	G
1.	ϵ_2	g_s				+ 1
2.	„	b_2g_e	+ 1			− 1
3.	ϵ_3	g_e	3			− 3
4.	„	g_p				+ 1
5.	„	$b_4{}^2$	− 2	+ 1		+ 1
6.	$2\epsilon_{22}$	g_e	+ 4			− 3
7.	2„	g_p				+ 1
8.	„	$b_2{}^2$	− 3	+ 2		+ 1
9.	ϵ_2	b_2c_2	− 2		+ 1	+ 1
10.	ϵ_4	g	− 2	+ 4		− 2
11.	„	b_4	− 6		+ 1	+ 4
12.	ϵ_{32}	g	− 3	+ 6		− 2
13.	„	b_3	− 7	− 1	+ 2	+ 4
14.	„	b_2	− 6	− 3	+ 3	+ 4
15.	$6\epsilon_{222}$	g	− 4	+ 8		− 2
16.	2„	b_2	− 7	− 4	+ 4	+ 4
17.	ϵ_5		− 10	− 10	+ 5	+ 10
18.	ϵ_{42}		− 10	− 16	+ 8	+ 10
19.	$2\epsilon_{33}$		− 9	− 18	+ 9	+ 10
20.	$2\epsilon_{322}$		− 9	− 24	+ 12	+ 10
21.	$24\epsilon_{2222}$		− 8	− 32	+ 16	+ 10
22.	$6\epsilon_{222}$	b_2	− 6	− 12	+ 8	

But in these formulæ $p_1{}^2p_2{}^2$, $p_1{}^2p_2p_3$, $p_1p_2p_3p_4$, G have numerical values which are different according to the number of points of intersection presenting themselves in the several formulæ; viz. this number being called i, we have for the formulæ in

ϵ_2	ϵ_3	ϵ_{22}	ϵ_4	ϵ_{32}	ϵ_{222}	ϵ_5	ϵ_{42}	ϵ_{33}	ϵ_{322}	ϵ_{2222}	$\epsilon_{222}b_1$	$\epsilon_3{b_1}^2$	ϵ_2b_1
$i=2$	3	4	4	5	6	5	6	6	7	8	7	4	5,

and the values of the symbols are

$$\begin{aligned}
p_1{}^2p_2{}^2 &= n^2(n-2)(n-3) \qquad \ldots(n-i+1),\\
p_1{}^2p_2p_3 &= n^2(n-1)(n-3) \qquad \ldots(n-i+1),\\
p_1p_2p_3p_4 &= n^2(2n^2-6n+3)(n-4)\ldots(n-i+1),\\
G &= n(n-1)(n-2) \qquad\qquad \ldots(n-i+1).
\end{aligned}$$

Thus, suppose $i = 4$, the subject is a line bearing the points 1, 2, 3, 4, which are intersections of the line with the surface F; we have then G as the condition in order that this line (or, say, the line of the subject) may coincide with a given line, which given line intersects the surface in n points; any four of these (their order being attended to) may be regarded as being the points 1, 2, 3, 4; or there are $n(n-1)(n-2)(n-3)$ subjects satisfying the prescribed condition (that the line of the subject may coincide with the given line). Hence here $G = n(n-1)(n-2)(n-3)$; and so in general $G = n(n-1)(n-2)\ldots(n-i+1)$.

Next, for $p_1{}^2p_2{}^2$. Here $p_1{}^2$ is the condition that the point 1 shall lie in each of two given planes, that is, in a given line, say L_1; and, similarly, $p_2{}^2$ is the condition that 2 may lie in a given line L_2. We take any one of the n intersections of L_1 with F for the point 1, and any one of the n intersections of L_2 with F for the point 2; this determines the line of the subject, but the $i-2$ points 3, 4, ..., i are then any $i-2$ of the remaining $n-2$ intersections of this line with F; that is, $p_1{}^2p_2{}^2 = n^2(n-2)(n-3)\ldots(n-i+1)$ as above.

Again, for $p_1{}^2p_2p_3$. Here $p_1{}^2$ is the condition that 1 shall lie in a given line L_1; we therefore take for 1 any one of the n intersections of L_1 with F; p_2 is the condition that 2 may lie in a given plane P_2, it lies therefore in the curve of intersection of P_2 with F; and, similarly, 3 lies in the curve of intersection of a plane P_3 with F; the two planes intersect in a line meeting F in n points σ, and the two cones, vertex 1, which stand upon the plane curves respectively, intersect in the n

lines joining 1 with the n points σ, and in n^2-n other lines. The line of the subject is then any one of these n^2-n lines, or, since the vertex is any one of n points, the line is any one of $n(n^2-n)$, $= n^2(n-1)$ lines; the remaining points 4, 5, ..., i are any $i-3$ of the remaining $n-3$ intersections of the line with F; hence the formula

$$p_1{}^2 p_2 p_3 = n^2(n-1)(n-3)(n-4)\dots(n-i+1).$$

For $p_1p_2p_3p_4$. We have here 1, 2, 3, 4 lying in given plane sections of the surface F, and we have consequently to find the number of lines which can be drawn to meet each of these four sections. Observing that any two of the sections meet in the n

intersections with F of the line of intersection of their planes, the order of the scroll generated by the lines which meet three of the sections is $2n^2 - 3n^2$; this scroll meets the fourth section in $n(2n^2 - 3n^2)$, $= 2n^4 - 3n^3$ points; or we have this number of lines meeting each of the four sections. But among these are included $3n^2(n-1)$ lines which have to be rejected, viz. the sections 1 and 4 meet in n points, each of which is the vertex of cones through the sections 1 and 2 respectively; these cones meet in n lines, which are to be disregarded, and in $n^2 - n$ other lines, and we have thus $n(n^2 - n)$, $= n^2(n-1)$ lines; and similarly from the intersections of 2 and 4, and from the intersections of 3 and 4, $n^2(n-1)$ and $n^2(n-1)$ lines, in all $3n^2(n-1)$ lines. Hence the number of lines meeting the four sections is

$$2n^4 - 3n^3 - 3n^3 + 3n^2, = 2n^4 - 6n^3 + 3n^2;$$

taking any one of these for the line of the subject, the remaining points 5, 6, ..., i are any $i-4$ of the remaining $n-4$ intersections, or we have the required formula

$$p_1p_2p_3p_4 = n^2(2n^2 - 6n + 3)(n-4)\dots(n-i+1).$$

The four numbers ${p_1}^2{p_2}^2$, ${p_1}^2p_2p_3$, $p_1p_2p_3p_4$, G for any line of the table being now known, we can at once calculate the required values $\epsilon_2 g_s$, &c., as the case may be; for instance,

$$\begin{aligned}
i = 5, \quad \epsilon_5 = -10{p_1}^2{p_2}^2 \quad & = -10n^2(n-2)(n-3)(n-4)\\
- 10{p_1}^2p_2p_3 \quad & = -10n^2(n-1)(n-3)(n-4)\\
+ \ 5p_1p_2p_3p_4 \quad & = +\ 5n^2(2n^2 - 6n + 3)(n-4)\\
+ 10G \quad & = +10n(n-1)(n-2)(n-3)(n-4)\\
& = \quad 5n(n-4)(7n-12).
\end{aligned}$$

In fact, throwing out $n(n-4)$, the remaining terms give

$$\begin{aligned}
&- 10n^2 + 50n - 60n\\
&- 10n^2 + 40n - 30n\\
&+ 10n^2 - 30n^2 + 15n\\
&+ 10n^2 - 60n^2 + 110n - 60\\
\hline
&\qquad 35n - 60, = 5(7n - 12).
\end{aligned}$$

And we obtain in like manner the other formulæ of the table.

The remainder of § 33 contains investigations of less systematically connected theorems, and I quote the results only.

25. If on the surface F_n there is a curve order r, then of the tangent planes of F_n along this curve there pass $r(n-1)$ through an arbitrary point of space; *aliter*, class of torse is $= r(n-1)$.

    In particular, for curve of 4-pointic contact, $r = n(11n-24)$, class of torse is
    $$= n(n-1)(11n-24).$$
    No. of tangent planes through line, or class of surface, $= n(n-1)^2$.

26. $\epsilon_3 b_3 g = \epsilon_3 {b_3}^2 + \epsilon_3 g_e = 2n + 3n(n-2), = n(3n-4)$.

    $\epsilon_3 b_3 g, = n(3n-4)$, is the order of curve of contact of the 3-pointic (chief) tangents which meet a given line.

    Parabolic tangents are coincident chief tangents.

    No. of 4-pointic parabolic tangents $= 2n(n-2)(11n-24)$.

27. Order of parabolic curve $= 4n(n-2)$.

    Order of regulus formed by parabolic tangents
    $$= 2n(n-2)(3n-4).$$
    The parabolic curve and curve of contacts of an ϵ_4 tangent meet in
    $$4n(n-2)(11n-24)$$
    points, i.e., they *touch* in $2n(n-2)(11n-24)$ points.

28. Umbilici. No. is $= 2n(5n^2 - 14n + 11)$.

29. No. of points at which the chief tangents being distinct are each of them 4-pointic, or, what is the same thing, No. of actual double points of curve ϵ_4,
    $$= 5n(7n^2 - 28n + 30),$$
    $n = 3$, No. is $15(63 - 84 + 30), = 135$, viz. this is the number of points of intersection of two of the 27 lines; or, what is the same thing, the number of triple tangent planes is $= 45$.

30. No. of parabolic tangents which have besides a 2-pointic contact is
$$= 2n(n-2)(n-4)(3n^2+5n-24).$$

31. No. of double tangent planes such that line through points of contact is at one of these points 3-pointic
$$= n(n-2)(n-4)(n^3+3n^2+13n-48).$$

32. No. of points where one chief tangent is 4-pointic, the other 3-pointic and (at another point of the surface) 2-pointic is
$$= n(n-4)(27n^3-13n^2-264n+396).$$

33. No. of points where chief tangents being distinct are each of them at another point of the surface 2-pointic is
$$= n(n-4)(4n^5-4n^4-95n^3+99n^2+544n-840).$$

34. The curve of contacts b_3 of an ϵ_{32} tangent has with the parabolic curve 2-pointic intersections only, and these are at the points for which the chief tangent is (at another point of the surface) 2-pointic.

35. The curve of contacts b_3 of an ϵ_{32} tangent has, with the curve of contacts of an ϵ_4 tangent, 2-pointic intersections at the contacts of an ϵ_5 tangent; and has also simple intersections with the same curve, 1° at the contacts b_4 of an ϵ_{42} tangent, 2° at the points where the chief tangents are ϵ_4 and ϵ_{32}.

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ON THE THEOREMS OF THE 2, 4, 8, AND 16 SQUARES.

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A SUM of 2 squares multiplied by a sum of 2 squares is a sum of 2 squares; a sum of 4 squares multiplied by a sum of 4 squares is a sum of 4 squares; a sum of 8 squares multiplied by a sum of 8 squares is a sum of 8 squares; but a sum of 16 squares multiplied by a sum of 16 squares is *not* a sum of 16 squares. These theorems were considered in the paper, Young, "On an extension of a theorem of Euler, with a determination of the limit beyond which it fails," *Trans. R. I. A.*, t. XXI. (1848), pp. 311—341; and the later history of the question is given in the paper by Mr S. Roberts, "On the Impossibility of the general Extension of Euler's Theorem &c.," *Quart. Math. Jour.* t. XVI. (1879), pp. 159—170; as regards the 16-question, it has been throughout assumed that there is only one type of synthematic arrangement (what this means will appear presently); but as regards this type, it is, I think, well shown that the signs cannot be determined. It will appear in the sequel, that there are in fact four types (the last three of them possibly equivalent) of synthematic arrangement; and for a complete proof, it is necessary to show in regard to each of these types that the signs cannot be determined. The existence of the four types has not (so far as I am aware) been hitherto noticed; and it hence follows, that no complete proof of the non-existence of the 16-square theorem has hitherto been given.

For the 2 squares the theorem is of course

$$(x_1{}^2 + x_2{}^2)(y_1{}^2 + y_2{}^2) = (x_1y_1 + x_2y_2)^2 + (x_1y_2 - x_2y_1)^2.$$

For the 4 squares (for which the nature of the theorem is better seen) it is

$$\begin{aligned}(x_1{}^2+x_2{}^2+x_3{}^2+x_4{}^2)(y_1{}^2+y_2{}^2+y_3{}^2+y_4{}^2) = \quad &(x_1y_1+x_2y_2+x_3y_3+x_4y_4)^2\\ &+ (x_1y_2 - x_2y_1 + x_3y_4 - x_4y_3)^2\\ &+ (x_1y_3 - x_3y_1 - x_2y_4 + x_4y_2)^2\\ &+ (x_1y_4 - x_4y_1 + x_2y_3 - x_3y_2)^2;\end{aligned}$$

or, as this may be written,

$$
\begin{aligned}
({x_1}^2+{x_2}^2+{x_3}^2+{x_4}^2)({y_1}^2+{y_2}^2+{y_3}^2+{y_4}^2)-(x_1y_1+x_2y_2+x_3y_3+x_4y_4)^2 \\
= \quad (12+34)^2 \\
+ (13-24)^2 \\
+ (14+23)^2;
\end{aligned}
$$

where 12 is used to denote $x_1y_2 - x_2y_1$, &c., and the truth of the theorem depends on the identity $12.34 - 13.24 + 14.23 = 0$. Clearly, the first step for forming the equation is to arrange the duads in a synthematic form

$$
\begin{array}{c}
12.34 \\
13.24 \\
14.23,
\end{array}
$$

and then to determine the signs: such an arrangement exists in the case of 8, and the signs can be determined; it exists also in the case of 16, but the signs *cannot* be determined to satisfy all the necessary relations.

In the case of 8, we have the synthematic arrangement

$$
\begin{array}{c}
12.34.56.78 \\
13.24.57.68 \\
14.23.58.67 \\
15.26.37.48 \\
16.25.38.47 \\
17.28.35.46 \\
18.27.36.45,
\end{array}
$$

being the *only type* of synthematic arrangement. This is, in fact, important as regards the 16-question, and it will appear that the case is so; but in the 8-question, starting from this arrangement, we have to show that there exists an equation which, for convenience, I write as follows:

$$\begin{aligned}({x_1}^2+\ldots+{x_8}^2)({y_1}^2+\ldots+{y_8}^2)-(x_1y_1+\ldots+x_8y_8)^2 \\ = \quad &(12+34+56+78)^2 \\ &+(13+24+57+68)^2 \\ &+(14+23+58+67)^2 \\ &+(15+26+37+48)^2 \\ &+(16+25+38+47)^2 \\ &+(17+28+35+46)^2 \\ &+(18+27+36+45)^2,\end{aligned}$$

but in which it is to be understood that each duad is affected by a factor ± 1 which is to be determined; say the factor of 12 is ϵ_{12}, that of 34, ϵ_{34}; and so in other cases. It is however assumed that ϵ_{12}, ϵ_{34}, ϵ_{56}, ϵ_{78}; ϵ_{13}, ϵ_{14}, ϵ_{15}, ϵ_{16}, ϵ_{17}, ϵ_{18} are each $=+1$.

We have then on the right-hand side triads of terms such as, 2 into

$$\epsilon_{12}\epsilon_{34}12.34+\epsilon_{13}\epsilon_{24}13.24+\epsilon_{14}\epsilon_{23}14.23,$$

which triad ought to vanish identically, as reducing itself to a multiple of

$$12.34-\ 13.24+14.23;$$

viz. we ought to have

$$\epsilon_{12}\epsilon_{34}=-\ \epsilon_{13}\epsilon_{24}=\epsilon_{14}\epsilon_{23};$$

or, *using now and henceforward when occasion requires*, 12, 34, &c. to denote ϵ_{12}, ϵ_{34}, &c. respectively, we have

$$\begin{aligned}12.34&=+k,\\ 13.24&=-k,\\ 14.23&=+k,\end{aligned}$$

where k, $=\pm 1$, has to be determined (in the actual case we have $12=+1$, $34=+1$, $13=1$, $14=1$; and therefore the first equation gives $k=1$, and the other two then give $24=-1$, $23=+1$).

We have in this way triads of values corresponding to the different tetrads

1234
1256
1278
1357
1368
1458
1467
2358
2367
2457
2468
3456
3478
5678,

which can be formed with the several lines of the formula. Thus we have from the first line 1234, 1256, 1278; then from the second line (not 1324 which in the form 1234 has been taken already) 1357, 1368, . . . ; and finally from the last line 5678.

We might consider each line as giving 6 tetrads, but the tetrads would then be obtained 3 times over; the number of tetrads is thus $6 \times 7 \div 3, = 14$ as above. And observe, that the systems of values for the coefficients $\epsilon = \pm 1$ are obtained directly from the tetrads, without the employment of any other formula.

We thus obtain the system of signs as follows:

12	$+1$	
13	$+1$	
14	$+1$	
15	$+1$	
16	$+1$	
17	$+1$	
18	$+1$	
23	$+1$	
24	-1	
25	$+1$	
26	-1	
27	$+1$	
28	-1	
34	$+1$	
35	a	$-\theta$
36	b	θ
37	$-a$	θ
38	$-b$	$-\theta$
45	c	θ
46	d	θ
47	$-d$	$-\theta$
48	$-c$	$-\theta$
56	$+1$	
57	a	$-\theta$
58	c	θ
67	d	θ
68	b	θ
78	$+1$	

viz. the original assumptions $12 = +1$, &c., and the tetrads 1234, 1256, 1278 give all the signs ± 1 up to $34 = +1$; from the tetrad 1357 we have

$$\begin{array}{lll} 13.57 + & 1 & a, \\ 15.37 - & 1 & a, \\ 17.35 + & 1 & a, \end{array}$$

that is, $35 = a$, $37 = -a$, $57 = a$, where $a, = \pm 1$, is still undetermined; and similarly, the tetrads 1368, 1458, 1467 give the remaining signs b, c, d. The tetrad 2358 then gives

$$\begin{array}{lll} 23.58 + & 1 & c, \\ 25.38 - & 1 & -b, \\ 28.35 + & -1 & a, \end{array}$$

that is, $-a = b = c$; and similarly the tetrads 2367, 2457, 2468 give $-a = b = d$, $-a = c = d$, $b = c = d$ respectively; the four tetrads thus give $-a = b = c = d$, say each of these $= \theta$. But retaining for the moment a, b, c, d, the tetrad 3456 then gives

$$\begin{array}{lll} 34.56 + & 1 & 1, \\ 35.46 - & a & d, \\ 36.45 + & b & c, \end{array}$$

that is, $1 = -ad = bc$, and similarly the last two tetrads 3478 and 5678 give $1 = -ac = bd$ and $1 = -ab = cd$ respectively; substituting the values in terms of θ, the several equations give only $\theta^2 = 1$, that is, $\theta = \pm 1$ at pleasure; and the series of signs for the 8-formula, containing this one arbitrary sign $\theta = \pm 1$, is thus determined.

Passing to the case of 16, we have in like manner to form a synthematic arrangement of the numbers 1, 2, ..., 16 in 15 lines, each containing the 16 numbers in 8 duads (no duad twice repeated), and this containing all the 120 duads. And, using for the moment letters instead of numbers, the necessary condition is, that *ab.cd* occurring in one line, *ac.bd* must occur in another line, and *ad.bc* in a third line. Observe that as well the order of the letters in a duad as the order of the duads is thus far immaterial; so that a line containing *bd.ca* may be considered as containing *ac.bd*.

Considering any such combination *ab.cd*, the line which contains it may be taken to be the first line; and the line which contains *ac.bd* may be taken to be the second line. And then writing 1, 2, 3, 4 in place of a, b, c, d respectively, the first line will contain 12.34, and the second line will contain 13.24. Let e be any other symbol occurring in the first line, say in the duad ef, and in the second line say in the duad eg; then g must occur in the first line in some duad gh, or the first line will contain $ef.gh$, and then the second line as containing eg will contain

fh. We have thus in the first line 12.34 and $ef.gh$, and in the second line 13.24 and $eg.fh$; and so for each of the remaining symbols. The first line must contain a duad ij of the remaining symbols; and then writing down the duads which belong to these, the two lines are

$$12.34.ef.gh.ij.kl.mn.op,$$
$$13.24.eg.fh.ik.jl.mo.np,$$

where the remaining symbols are arranged in duads in the second line according to a definite rule. But to make this more clear, let us write down a complete system. Taking the first line to contain 12.34, and taking successively $e = 5, 6, 7$, we have

$$\begin{array}{rl} \text{I.} & 12.34.56.78.9\,10.11\,12.13\,14.15\,16, \\ \text{II.} & 13.24.57.68.9\,11.10\,12.13\,15.14\,16, \\ \text{III.} & 14.23.58.67.9\,12.10\,11.13\,16.14\,15, \end{array}$$

where in II. and III. the remaining symbols 9 to 16 are arranged according to the same rule as for 5 to 8; viz. the first line taken with any one of the other two gives the duads which belong to 1234 or 1256 or 1278; and similarly II. and III. give the duads belonging to 1357, 1368, or 1458, 1467; the same thing would of course be true if in the first line instead of 9 to 16 we had any other 8 symbols. We have now only to arrange the remaining symbols 5, 6, 7, 8, 9, ..., 16 in the remaining lines IV. to XV. according to the like rule; that is, we must arrange the 12 symbols 5 to 16 in synthematic duads in 12 lines, each line containing the 12 symbols in 6 duads (each duad containing one symbol out of 5, 6, 7, 8 and one out of 9, 10, 11, 12, 13, 14, 15, 16, and no duad twice repeated). The arrangement is at once obtained by means of the binary tables, or it may be written down at once; viz. we have

IV. 15.26.39.4 10.11 14.12 16.13 15,
V. 16.25.3 10.49.11 15.12 14.13 16,
VI. 17.28.3 11.4 12.9 15.10 13.14 16,
VII. 18.27.3 12.4 11.9 16.10 14.13 15,
VIII. 19.2 10.35.46.11 16.12 15.13 14,
IX. 1 10.29.36.45.11 13.12 16.14 15,
X. 1 11.2 12.37.48.9 13.10 16.14 15,
XI. 1 12.2 11.38.47.9 14.10 15.13 16,
XII. 1 13.2 14.3 15.4 16.59.6 11.7 12.8 10,
XIII. 1 14.2 13.3 16.4 15.5 10.69.7 11.8 12,
XIV. 1 15.2 16.3 13.4 14.5 12.6 10.7 9.8 11,
XV. 1 16.2 15.3 14.4 13.5 11.6 12.79.8 10.

We have thus 15 lines, and it is to be shown that they form a synthematic arrangement; that is, that every duad occurs once and only once. This is at once verified for the duads 12, 13, 14, 15, 16, 17, 18, 19, 1 10, 1 11, 1 12, 1 13, 1 14, 1 15, 1 16, which occur each of them in the line headed by it; and for the duads contained in the several lines I. to III. with each of the lines IV. to XV. respectively, each of these duads occurs in the proper line. We have thus only to verify that the duads of the lines IV. to XV. among themselves are all distinct; and this will be the case if in the arrangement of the 12 symbols 5 to 16, each duad occurs once and only once. The arrangement in fact is a known one; the 12 symbols may be replaced by the duads ab, ac, ad, ae, af, bc, bd, be, bf, cd, ce, cf, and the lines then correspond to the synthematic arrangement of the 6 letters a, b, c, d, e, f.

The foregoing arrangement is in fact the only one which satisfies the condition that the first three lines shall be as above; but we may in the first line arrange the 8 duads in any order, and then in the second and third lines arrange the duads which correspond to the several duads of the first line in any order; that is, we may arrange the duads of the first line in 8! ways, and then those of the second and third lines each in 8! ways; and we thus obtain $(8!)^3$ different arrangements; but these are not all distinct, for we may permute the 15 lines in 15! ways, and also permute the 8 duads in each line in 8! ways; the number of distinct arrangements is thus =

$(8!)^3 \div (15! \cdot (8!)^{15})$, which is not an integer, and the calculation is thus not applicable. In fact the arrangement obtained is not absolutely the only one, but it is the only one of a certain form; and the question of the number of distinct synthematic arrangements of 16 symbols is one of considerable difficulty, which I do not enter upon.

Taking the foregoing synthematic arrangement, and assuming the signs of the duads 12, 13, 14, 15, 16, 17, 18, 19, 1 10, 1 11, 1 12, 1 13, 1 14, 1 15, 1 16 each $= +1$, we have to determine the signs of the remaining duads; this is done by means of the tetrads, and we have thus the signs corresponding to the 14 tetrads which can be formed with each line; but observing that each tetrad occurs in 3 different lines, the equations obtained from the different lines must be consistent with each other. Starting with the first line, we obtain from the tetrads 1234, 1256, 1278, 129 10, 12.11 12, 12.13 14, 12.15 16, the signs of the duads 23, 24, 25, 26, 27, 28, 29, 2 10, 2 11, 2 12, 2 13, 2 14, 2 15, 2 16; the remaining signs are obtained from the other lines. But the consistency of the equations requires that the signs obtained from different tetrads for the same duad shall be the same; and this leads to a system of equations which must be satisfied. In fact, going through the calculation, the system of signs is found to involve 4 arbitrary signs, which may be taken to be a, b, c, d; and the consistency-conditions then lead to equations which are found to be inconsistent; that is, the signs cannot be determined so as to satisfy all the necessary relations, and the 16-square theorem does not exist.

The complete investigation is as follows: the signs are obtained in terms of a, b, c, d (each $= \pm 1$) as shown in the following table:

First arrangement.

35	$a =$	$910 = -a$	$= 1112$
$-b$	$=$		
36	$b =$	$911 = -b$	$= 1113$
$-c$	$=$		
37	$-a =$	$912 = -c$	$= 1114$
$-d$	$=$		
38	$-b =$	$913 = -d$	$= 1115$
$+1$	$=$		
	$=$	$914 = +1$	$= 1116$
$+1$	$=$		
45	$c =$	$915 = +1$	$=$
	$=$		
46	$d =$	$916 = +1$	$= 1213$
$-d$	$=$		
47	$-d =$	$=$	$= 1214$
$+1$	$=$		
48	$-c =$	$1011 = +1$	$= 1215$
$+1$	$=$		
	$=$	$1012 = +1$	$= 1216$
$+1$	$=$		
57	$a =$	$1013 = +1$	$=$
	$=$		
58	$c =$	$1014 = +1$	$= 1314$
$+1$	$=$		
	$=$	$1015 = +1$	$= 1315$
$+1$	$=$		
67	$d =$	$1016 = +1$	$= 1316$
$+1$	$=$		
68	$b =$	$=$	$=$
	$=$		
	$=$	$=$	$= 1415$
$+1$	$=$		
78	$+1 =$	$=$	$= 1416$
$+1$	$=$		
	$=$	$=$	$= 1516$
$+1$	$=$		

Second arrangement.

35	$a =$	$910 = a$	$= 1112$
b	$=$		
36	$b =$	$911 = b$	$= 1113$
c	$=$		
37	$-a =$	$912 = c$	$= 1114$
d	$=$		
38	$-b =$	$913 = d$	$= 1115$
$+1$	$=$		
	$=$	$914 = +1$	$= 1116$
$+1$	$=$		
45	$c =$	$915 = +1$	$=$
	$=$		
46	$d =$	$916 = +1$	$= 1213$
d	$=$		
47	$-d =$	$=$	$= 1214$
$+1$	$=$		
48	$-c =$	$1011 = +1$	$= 1215$
$+1$	$=$		
	$=$	$1012 = +1$	$= 1216$
$+1$	$=$		
57	$a =$	$1013 = +1$	$=$
	$=$		
58	$c =$	$1014 = +1$	$= 1314$
$+1$	$=$		
	$=$	$1015 = +1$	$= 1315$
$+1$	$=$		
67	$d =$	$1016 = +1$	$= 1316$
$+1$	$=$		
68	$b =$	$=$	$=$
	$=$		
	$=$	$=$	$= 1415$
$+1$	$=$		
78	$+1 =$	$=$	$= 1416$
$+1$	$=$		
	$=$	$=$	$= 1516$
$+1$	$=$		

Third arrangement.

35	$a =$	$910 = -a$	$= 1112$
$-b$	$=$		
36	$b =$	$911 = -b$	$= 1113$
$-c$	$=$		
37	$-a =$	$912 = -c$	$= 1114$
$-d$	$=$		
38	$-b =$	$913 = -d$	$= 1115$
$+1$	$=$		
	$=$	$914 = +1$	$= 1116$
$+1$	$=$		
45	$c =$	$915 = +1$	$=$
	$=$		
46	$d =$	$916 = +1$	$= 1213$
$-d$	$=$		
47	$-d =$	$=$	$= 1214$
$+1$	$=$		
48	$-c =$	$1011 = +1$	$= 1215$
$+1$	$=$		
	$=$	$1012 = +1$	$= 1216$
$+1$	$=$		
57	$a =$	$1013 = +1$	$=$
	$=$		
58	$c =$	$1014 = +1$	$= 1314$
$+1$	$=$		
	$=$	$1015 = +1$	$= 1315$
$+1$	$=$		
67	$d =$	$1016 = +1$	$= 1316$
$+1$	$=$		
68	$b =$	$=$	$=$
	$=$		
	$=$	$=$	$= 1415$
$+1$	$=$		
78	$+1 =$	$=$	$= 1416$
$+1$	$=$		
	$=$	$=$	$= 1516$
$+1$	$=$		

Fourth arrangement.

35	$a =$	$910 = a$	$= 1112$
b	$=$		
36	$b =$	$911 = b$	$= 1113$
c	$=$		
37	$-a =$	$912 = c$	$= 1114$
d	$=$		
38	$-b =$	$913 = d$	$= 1115$
$+1$	$=$		
	$=$	$914 = +1$	$= 1116$
$+1$	$=$		
45	$c =$	$915 = +1$	$=$
	$=$		
46	$d =$	$916 = +1$	$= 1213$
d	$=$		
47	$-d =$	$=$	$= 1214$
$+1$	$=$		
48	$-c =$	$1011 = +1$	$= 1215$
$+1$	$=$		
	$=$	$1012 = +1$	$= 1216$
$+1$	$=$		
57	$a =$	$1013 = +1$	$=$
	$=$		
58	$c =$	$1014 = +1$	$= 1314$
$+1$	$=$		
	$=$	$1015 = +1$	$= 1315$
$+1$	$=$		
67	$d =$	$1016 = +1$	$= 1316$
$+1$	$=$		
68	$b =$	$=$	$=$
	$=$		
	$=$	$=$	$= 1415$
$+1$	$=$		
78	$+1 =$	$=$	$= 1416$
$+1$	$=$		
	$=$	$=$	$= 1516$
$+1$	$=$		

We have now for the four arrangements respectively, by means of hitherto unused tetrads, the following determinations of sign; these being in each case inconsistent with each other.

First arrangement.

$$\begin{array}{llllll} 3 & 5 & 9 & 15 & +-\theta. & -\sigma \\ 3 & 9 & 5 & 15 & --\lambda. & \rho \\ 3 & 15 & 5 & 9 & +\quad\mu. & -\nu \end{array} \qquad \theta\sigma = \lambda\rho = -\mu\nu,$$

$$\begin{array}{llllll} 3 & 5 & 10 & 16 & +-\theta. & \sigma \\ 3 & 10 & 5 & 16 & -\quad\lambda. & -\rho \\ 3 & 16 & 5 & 10 & +-\mu. & \nu \end{array} \qquad -\theta\sigma = \lambda\rho = -\mu\nu,$$

$$\begin{array}{llllll} 3 & 5 & 11 & 13 & +-\theta. & -\tau \\ 3 & 11 & 5 & 13 & -\quad\lambda. & \nu \\ 3 & 13 & 5 & 11 & +-\mu. & -\rho \end{array} \qquad \theta\tau = -\lambda\nu = \mu\rho,$$

$$\begin{array}{llllll} 3 & 5 & 12 & 14 & +-\theta. & \tau \end{array} \qquad -\theta\tau = -\lambda\nu = \mu\rho.$$

Second arrangement.

$$\begin{array}{llllll} 3 & 5 & 9 & 16 & +-\theta. & \sigma \\ 3 & 9 & 5 & 16 & --\lambda. & -\rho \\ 3 & 16 & 5 & 9 & +-\mu. & \nu \end{array} \qquad \theta\sigma = \lambda\rho = \mu\nu,$$

$$\begin{array}{llllll} 3 & 5 & 10 & 15 & +-\theta. & \sigma \\ 3 & 10 & 5 & 15 & -\quad\lambda. & -\rho \\ 3 & 15 & 5 & 10 & +\quad\mu. & \nu \end{array} \qquad -\theta\sigma = \lambda\rho = \mu\nu,$$

$$\begin{array}{llllll} 3 & 5 & 11 & 14 & +-\theta. & \tau \\ 3 & 11 & 5 & 14 & -\quad\lambda. & -\nu \\ 3 & 14 & 5 & 11 & +\quad\mu. & \rho \end{array} \qquad -\theta\tau = \lambda\nu = \mu\rho,$$

$$\begin{array}{llllll} 3 & 5 & 12 & 13 & +-\theta. & -\tau \end{array} \qquad \theta\tau = \lambda\nu = \mu\rho.$$

Third arrangement.

$$\begin{array}{cccclll}
3 & 5 & 9 & 13 & +-\theta. & -\sigma & \qquad \theta\sigma = -\lambda\rho = \mu\nu, \\
3 & 9 & 5 & 13 & --\lambda. & \rho & \\
3 & 13 & 5 & 9 & +-\mu. & -\nu &
\end{array}$$

$$\begin{array}{cccclll}
3 & 5 & 10 & 14 & +-\theta. & \sigma & \qquad \theta\sigma = -\lambda\rho = -\mu\nu, \\
3 & 10 & 5 & 14 & -\ \lambda. & -\rho & \\
3 & 14 & 5 & 10 & +\ \mu. & \nu &
\end{array}$$

$$\begin{array}{cccclll}
3 & 5 & 11 & 15 & +-\theta. & -\tau & \qquad \theta\tau = -\lambda\nu = -\mu\rho, \\
3 & 11 & 5 & 15 & -\ \lambda. & \nu & \\
3 & 15 & 5 & 11 & +\ \mu. & -\rho &
\end{array}$$

$$\begin{array}{cccclll}
3 & 5 & 12 & 16 & +-\theta. & \tau & \qquad \theta\tau = -\lambda\nu = \mu\rho.
\end{array}$$

Fourth arrangement.

$$\begin{array}{cccclll}
3 & 5 & 9 & 14 & +-\theta. & \sigma & \qquad \theta\sigma = \lambda\rho = -\mu\nu, \\
3 & 9 & 5 & 14 & --\lambda. & -\rho & \\
3 & 14 & 5 & 9 & +\ \mu. & \nu &
\end{array}$$

$$\begin{array}{cccclll}
3 & 5 & 10 & 13 & +-\theta. & \sigma & \qquad \theta\sigma = -\lambda\rho = \mu\nu, \\
3 & 10 & 5 & 13 & -\ \lambda. & -\rho & \\
3 & 13 & 5 & 10 & +-\mu. & \nu &
\end{array}$$

$$\begin{array}{cccclll}
3 & 5 & 11 & 16 & +-\theta. & \tau & \qquad \theta\tau = -\lambda\nu = \mu\rho, \\
3 & 11 & 5 & 16 & -\ \lambda. & -\nu & \\
3 & 16 & 5 & 11 & +-\mu. & \rho &
\end{array}$$

$$\begin{array}{cccclll}
3 & 5 & 12 & 15 & +-\theta. & -\tau & \qquad \theta\tau = -\lambda\nu = -\mu\rho.
\end{array}$$

And it hence finally appears, that we cannot, in any one of the four arrangements, determine the signs so as to give rise to a 16-square theorem; that is, the product of a sum of 16 squares into a sum of 16 squares cannot be made equal to a sum of 16 squares.

764.

THE BINOMIAL EQUATION $x^p - 1 = 0$: QUINQUISECTION.

[From the Proceedings of the London Mathematical Society, vol. XII. (1881), pp. 15, 16. Read December 9, 1880.]

THE theory should be precisely analogous to those for the trisection and quartisection (see my paper, "The Binomial Equation $x^p - 1 = 0$, Trisection and Quartrisection," *Proceedings of the London Mathematical Society*, vol. XI. (1879), pp. 4—17, [731]), only I have not been able to carry it so far. We have in the present case five periods X, Y, Z, W, T, the actual expressions for which, $X = \eta^1 + \ldots$, $Y = \eta^2 + \ldots$, etc., with Reuschle's selected prime root g, can be (for the primes $5n+1$ under 100) at once written down by means of the table given, pp. 16, 17, of that paper; [see this volume, pp. 95, 96]. The relations between the periods are of the form

	X	Y	Z	W	T
$X^2 =$	a	b	c	d	e
$XY =$	f	g	h	i	j
$XZ =$	k	l	m	n	o;

that is, we have

$$X^2 = (a,\ b,\ c,\ d,\ e \mid X,\ Y,\ Z,\ W,\ T),$$

and thence, by cyclical permutations,

$$Y^2 = (e,\ a,\ b,\ c,\ d \mid \qquad " \qquad),\ \text{etc.};$$

viz. from the value of X^2 we have those of Y^2, Z^2, W^2, T^2; from the value of XY those of YZ, ZW, WT, TX; and from the value of XZ those of YW, ZT, WX, TY.

From the equation $X + Y + Z + W + T = -1$, multiplying by X and then substituting for X^2, XY, &c., their values, we obtain

$$\begin{aligned}
-a &= 1 + f + k + m + g,\\
-b &= \phantom{1 + {}} g + l + n + h,\\
-c &= \phantom{1 + {}} h + m + o + i,\\
-d &= \phantom{1 + {}} i + n + k + j,\\
-e &= \phantom{1 + {}} j + o + l + f,
\end{aligned}$$

which determine $(a,\ b,\ c,\ d,\ e)$ in terms of $(f,\ g,\ h,\ i,\ j)$ and $(k,\ l,\ m,\ n,\ o)$. It is, moreover, easy to prove that

$$\begin{aligned} f+g+h+i+j &= \tfrac{1}{5}(p-1), \\ k+l+m+n+o &= \tfrac{1}{5}(p-1), \end{aligned}$$

whence also

$$a+b+c+d+e = -1 - \tfrac{4}{5}(p-1).$$

We obtain other relations between the coefficients by considering the two triple products XYZ and XYW: these are all that need be considered, since the other triple products are deducible from them by cyclical permutations. From the first of these we have

$$X.YZ = Y.XZ = Z.XY,$$

and from the second

$$X.YW = Y.XW = W.XY;$$

and if we herein substitute for YZ, XZ, &c., their values, and then in the resulting equations for X^2, XY, &c., their values as linear functions of X, Y, Z, W, T, we obtain in all $5.2.2 = 20$ quadric relations between the 15 coefficients; or if we substitute for $(a,\ b,\ c,\ d,\ e)$ their foregoing values, in all 20 relations between the 10 coefficients $(f,\ g,\ h,\ i,\ j)$ and $(k,\ l,\ m,\ n,\ o)$. These are at most equivalent to 8 independent equations, since we have, besides, the sums $f+g+h+i+j$ and $k+l+m+n+o$ each $= \frac{1}{5}(p-1)$; but I have not succeeded in finding the connexions between them, or even in ascertaining to how many independent equations they are equivalent.

For any given prime $p = 5n+1$, the values of the coefficients, and also the coefficients of the quintic equation for the periods, could of course be calculated directly from the expressions of the periods; but for the primes under 100, that is, for the values 11, 31, 41, 61, 71, they are at once obtained from Reuschle. We have thus the two Tables, the former giving the coefficients a, b, $\ldots$, n, o, and the latter the coefficients of the quintic equations.

Table 1.

p	a / f / k	b / g / l	c / h / m	d / i / n	e / j / o
11	− 2	− 1	− 2	− 2	− 2
	1	0	0	1	0
	0	0	0	1	1
31	− 4	− 6	− 6	− 4	− 5
	0	1	2	1	2
	0	2	2	1	1
41	− 8	− 5	− 6	− 6	− 8
	3	0	2	1	2
	2	2	2	1	1
61	− 10	− 9	− 12	− 8	− 10
	3	2	2	3	2
	0	2	4	3	3
71	− 14	− 10	− 12	− 9	− 12
	4	2	3	2	3
	2	3	5	2	2

Table 2 of the Quintic Equations.

	Coefficients of					
p	η^5	η^4	η^3	η^2	η^1	1
11	1	1	− 4	− 3	+ 3	+ 1
31	1	1	− 12	− 2	+ 1	+ 5
41	1	1	− 16	+ 5	+ 21	− 9
61	1	1	− 24	− 17	+ 41	− 23
71	1	1	− 28	+ 37	+ 25	+ 1

765.

ON THE FLEXURE AND EQUILIBRIUM OF A SKEW SURFACE.

[From the Proceedings of the London Mathematical Society, vol. XII. (1881), pp. 103—108. Read March 10, 1881.]

THE skew surface is taken to be such that the strip between two consecutive generating lines is rigid, and that the flexure takes place by the rotation of the strips about the generating lines successively. The theory of the flexure is well known, but I am not aware that the theory of the equilibrium of such a surface, when acted upon by any given forces, has been considered; it is, however, a question which presents itself naturally in connexion with those relating to other continuous bodies treated of in the *Mécanique Analytique*, and forms a good example of the principles made use of.

To begin with the mechanical theory: we may regard the forces as acting on the generating lines regarded as material lines; and if for an element of mass dm, coordinates $(x,\ y,\ z)$ of a particular generating line G, the forces parallel to the axes are X', Y', Z', then the corresponding term in the equation of equilibrium is

$$S(X'\delta x + Y'\delta y + Z'\delta z)\, dm;$$

and observing that there are (as will afterwards appear) *five* geometrical conditions, which I represent by $U_1 = 0$, $U_2 = 0$, ..., $U_5 = 0$, the equation of equilibrium is

$$S\{(X'\delta x + Y'\delta y + Z'\delta z)\, dm + T_1\delta U_1 + T_2\delta U_2 + T_3\delta U_3 + T_4\delta U_4 + T_5\delta U_5\} = 0,$$

where T_1, T_2, ..., T_5 are the indeterminate multipliers, representing colligation-forces which correspond to the five geometrical conditions respectively.

Taking $(\xi,\ \eta,\ \zeta)$ for the coordinates of a particular point P on the generating line; p, q, r for the cos-inclinations of the line (whence $U_1 = p^2 + q^2 + r^2 - 1 = 0$ is one of the geometrical relations), and ρ for the distance of dm from P, we have

$$x,\ y,\ z = \xi + \rho p,\ \eta + \rho q,\ \zeta + \rho r,$$
$$\delta x,\ \delta y,\ \delta z = \delta\xi + \rho\delta p,\ \delta\eta + \rho\delta q,\ \delta\zeta + \rho\delta r.$$

The summation S extends first to the different points of the generating line, and then to the different generating lines; applying it first to the particular generating line, we write

$$\begin{array}{llllll} SX'dm, & SY'dm, & SZ'dm, & SX'\rho dm, & SY'\rho dm, & SZ'\rho dm \\ = X, & Y, & Z, & L, & M, & N, \end{array}$$

where X, Y, Z are the whole forces, and L, M, N the whole moments about the point P, for the generating line G; retaining the same summatory symbol S, as now referring to the different generating lines, the equation becomes

$$S\{X\delta\xi + Y\delta\eta + Z\delta\zeta + L\delta p + M\delta q + N\delta r + T_1\delta U_1 + \ldots + T_5\delta U_5\} = 0.$$

We have now to consider the geometrical theory of the flexure. Taking on the skew surface an arbitrary curve cutting each generating line G in a point P, coordinates $(\xi,\ \eta,\ \zeta)$, and taking σ for the distance along the curve of the point P from a fixed point of the curve; also p, q, r, as before, for the cos-inclinations of the generating line G, then when the surface is in a determinate state, ξ, η, ζ, p, q, r are given functions of σ; but these functions vary with the flexure of the surface, with, however, certain relations unaffected by the flexure; and the problem is to find first these relations. As already mentioned, one of them is $p^2 + q^2 + r^2 - 1 = 0$.

Taking P' as the consecutive point on the curve, so that the direction of the element PP' is that of the tangent PT at P, it is convenient to write l, m, n for the cosine-inclinations of the tangent; we have, it is clear,

$$l,\ m,\ n = \frac{d\xi}{d\sigma},\ \frac{d\eta}{d\sigma},\ \frac{d\zeta}{d\sigma}; \quad l^2 + m^2 + n^2 - 1 = 0.$$

The conditions in order to the rigidity of the strip, are that the angles GPP', $G'PP'$ $(= 180° - G'P'T)$, and the inclination $G'P'$ to GP, shall have given values, variable it may be from strip to strip—that is, these values must be given functions of σ. Taking $\angle GPT = I$, the value of $G'P'T$ can differ only infinitesimally from that of GPT, and we take it to be $G'P'T = I - \Omega d\sigma$; also the inclination GP to $G'P'$ is an infinitesimal, $= \Theta d\sigma$: we have I, Ω, Θ given functions of σ. It is to be remarked that these conditions imply, inclination of $G'P'$ to tangent plane GPT at P has a given value $\Lambda d\sigma$; in fact, if through P we draw a line $P\gamma$ parallel to $P'G'$, then, if P is regarded as the centre of a sphere which meets PG, $P\gamma$, PT in the points g, g', t respectively, we have a spherical triangle $gg't$, the sides of which are $I-\Omega d\sigma$, I, and $\Theta d\sigma$, and of which the perpendicular $g'm$ is $=\Lambda d\sigma$; we have thus an infinitesimal right-angled triangle, the base and altitude of which are $\Omega d\sigma$, $\Lambda d\sigma$, and the hypothenuse is $\Theta d\sigma$; whence $\Theta^2=\Omega^2+\Lambda^2$. In the case of the developable surface $\Lambda=0$ and $\Theta=\Omega$. It may be remarked that, when the curve on the skew surface is the line of striction, we have $\Omega=0$; in fact, taking P to be on the line of striction, the line

$$\frac{X-\xi}{qr'-q'r}=\frac{Y-\eta}{rp'-r'p}=\frac{Z-\zeta}{pq'-p'q},$$

through $(\xi,\ \eta,\ \zeta)$ at right angles to the two generating lines, meets the consecutive generating line X, Y, $Z=\xi'+\rho p'$, $\eta'+\rho q'$, $\zeta'+\rho r'$; and the condition that this may be so is easily found to be $\Omega=0$.

Take, for a moment, p', q', r' for the cos-inclinations of the consecutive generating line $P'G'$; we have

$$\begin{aligned}
lp\ +mq+nr&=\cos I,\\
lp'+mq'+nr'&=\cos(I-\Omega d\sigma),\\
pp'+qq'+rr'&=\cos\Theta d\sigma;
\end{aligned}$$

and then writing p', q', $r'=p+dp$, $q+dq$, $r+dr$, and observing that the equation $p'^2+q'^2+r'^2=1$ gives

$$p\,dp+q\,dq+r\,dr=-\tfrac{1}{2}(dp^2+dq^2+dr^2),$$

these equations and the before-mentioned two equations become

$$\begin{aligned}
&(U_1)\quad p^2+q^2+r^2-1=0,\\
&(U_2)\quad l^2+m^2+n^2-1=0,\\
&(U_3)\quad lp+mq+nr-\cos I=0,\\
&(U_4)\quad l\,dp+m\,dq+n\,dr-\Omega\sin I\,d\sigma=0,\\
&(U_5)\quad dp^2+dq^2+dr^2-\Theta^2d\sigma^2=0,
\end{aligned}$$

which equations, considering therein l, m, n as standing for their values $\dfrac{d\xi}{d\sigma}$, $\dfrac{d\eta}{d\sigma}$, $\dfrac{d\zeta}{d\sigma}$, are the geometrical relations which connect the six variables ξ, η, ζ, p, q, r, considered as functions of σ. And in these equations I, Ω, Θ denote given functions of σ, invariable by any flexure of the surface.

To complete the geometrical theory, it is to be observed that we can by flexure bring the generating lines of the surface to be parallel to those of any given cone (say the cone $p : q : r = P : Q : R$, where P, Q, R are given functions of a parameter σ), and we can moreover take ξ, η, ζ to be the coordinates of a given point on the generating line (say the point $\xi : \eta : \zeta = \Xi : \mathrm{H} : \mathrm{Z}$, where Ξ, H, Z are given functions of σ). We have thus p, q, r, ξ, η, ζ given functions of σ, and thence l, m, n given functions of σ; and the five geometrical relations determine I, Ω, Θ as given functions of σ; that is, we have the skew surface determined by the generating lines and the line of striction, or, what is the same thing, by the developable surface which is the locus of the generating lines and by the curve of striction upon this developable surface.

Reverting to the mechanical theory, and in the equation of equilibrium substituting for δx, δy, δz, their foregoing values, the equation becomes

$$S\{(X\delta\xi + Y\delta\eta + Z\delta\zeta) + (L + T_1 p)\delta p + \ldots + \text{terms in } \delta q, \delta r, \&\text{c.}\} = 0,$$

and the terms in $\delta\xi$, $\delta\eta$, $\delta\zeta$ give the three equations of equilibrium

$$0 = X, \quad 0 = Y, \quad 0 = Z,$$

for the forces on each generating line; and the remaining terms give the equations for the moments. But these are not required for the present purpose; and we may confine ourselves to the consideration of the case where the external forces vanish, that is, $X = Y = Z = L = M = N = 0$; the equations of equilibrium then reduce themselves to the geometrical relations $U_1 = 0, \ldots, U_5 = 0$, which are satisfied identically; and the equilibrium is indifferent, as it should be, since the surface is by hypothesis capable of flexure.

766.

ON THE GEODESIC CURVATURE OF A CURVE ON A SURFACE.

[From the Proceedings of the London Mathematical Society, vol. XII. *(1881), pp. 73—79. Read May 12, 1881.]*

THE notation is that of the Memoir, "Disquisitiones generales circa superficies curvas" (1827), Gauss, *Werke*, t. III.; see also my paper "On geodesic lines, in particular those of a quadric surface," *Proc. Lond. Math. Society*, t. IV. (1872), pp. 191—211, [508]; and Salmon's *Solid Geometry*, 3rd ed., 1874, pp. 251 *et seq.* The coordinates $(x,\ y,\ z)$ of a point on the surface are taken to be functions of two independent parameters p, q; and we then write

$$\begin{aligned} dx + \tfrac{1}{2}d^2x &= a\,dp + a'\,dq + \tfrac{1}{2}(\alpha\,dp^2 + 2\alpha'\,dp\,dq + \alpha''\,dq^2),\\ dy + \tfrac{1}{2}d^2y &= b\,dp + b'\,dq + \tfrac{1}{2}(\beta\,dp^2 + 2\beta'\,dp\,dq + \beta''\,dq^2),\\ dz + \tfrac{1}{2}d^2z &= c\,dp + c'\,dq + \tfrac{1}{2}(\gamma\,dp^2 + 2\gamma'\,dp\,dq + \gamma''\,dq^2);\end{aligned}$$

$$E,\ F,\ G = a^2+b^2+c^2,\quad aa'+bb'+cc',\quad a'^2+b'^2+c'^2;\quad V^2 = EG-F^2;$$

and therefore

$$ds^2 = E\,dp^2 + 2F\,dp\,dq + G\,dq^2,$$

where E, F, G are regarded as given functions of p and q.

To determine a curve on the surface, we establish a relation between the two parameters p, q, or, what is the same thing, take p, q to be functions of a single parameter θ; and we write as usual p', p'', q', etc., to denote the differential coefficients of p, q, etc., in regard to θ; we write also E_1, E_2, etc., to denote the differential coefficients $\dfrac{dE}{dp}$, $\dfrac{dE}{dq}$, etc. In the first instance, θ is taken to be an arbitrary parameter, but we afterwards take it to be the length s of the curve from a fixed point thereof.

First formula for the radius of relative curvature.

Consider any two curves touching at the point P, coordinates $(x,\ y,\ z)$ which are regarded as given functions of $(p,\ q)$; where $(p,\ q)$

are for the one curve given functions, and for the other curve other given functions, of θ.

The coordinates of a consecutive point for the one curve are then

$$x + dx + \tfrac{1}{2}d^2x, \quad y + dy + \tfrac{1}{2}d^2y, \quad z + dz + \tfrac{1}{2}d^2z,$$

where

$$dp = p'\,d\theta + \tfrac{1}{2}p''\,d\theta^2, \quad dq = q'\,d\theta + \tfrac{1}{2}q''\,d\theta^2;$$

hence these coordinates are

$$x + (ap' + a'q')\,d\theta + \tfrac{1}{2}(\alpha p'^2 + 2\alpha' p'q' + \alpha'' q'^2)\,d\theta^2 + \tfrac{1}{2}(ap'' + a'q'')\,d\theta^2,$$

and for the other curve they are in like manner

$$x+(aP'+a'Q')\,d\theta+\tfrac{1}{2}(\alpha P'^2+2\alpha' P'Q'+\alpha'' Q'^2)\,d\theta^2+\tfrac{1}{2}(aP''+a'Q'')\,d\theta^2,$$

the only difference being in the terms which contain the second differential coefficients, p'', q'' for the first curve, and P'', Q'' for the second curve. Hence the differences of the coordinates are

$$\tfrac{1}{2}\{a(p''-P'') + a'(q''-Q'')\}\,d\theta^2, \quad \tfrac{1}{2}\{b(p''-P'') + b'(q''-Q'')\}\,d\theta^2,$$
$$\tfrac{1}{2}\{c(p''-P'') + c'(q''-Q'')\}\,d\theta^2,$$

and consequently the distance QQ' of the two consecutive points Q, Q' is

$$= \tfrac{1}{2}\sqrt{(E,\ F,\ G \mid p''-P'',\ q''-Q'')^2}\,d\theta^2.$$

The squared arc $\overline{PQ}^2$ is

$$= (E,\ F,\ G \mid p',\ q')^2\,d\theta^2;$$

and hence, if as before $2\rho\cdot QQ' = \overline{PQ}^2$, that is, $\dfrac{1}{\rho} = 2QQ' \div \overline{PQ}^2$, then

$$\frac{1}{\rho} = \frac{\sqrt{(E,\ F,\ G \mid p''-P'',\ q''-Q'')^2}}{(E,\ F,\ G \mid p',\ q')^2},$$

the required formula for ρ.

Second formula for the radius of relative curvature.

We now take the variable θ to be the length s of the curve measured from a fixed point thereof, so that p', p'', etc. denote $\dfrac{dp}{ds}$, $\dfrac{d^2p}{ds^2}$, etc. We have therefore

$$1 = (E,\ F,\ G \mid p',\ q')^2,$$

and the formula becomes

$$\frac{1}{\rho} = \sqrt{(E,\ F,\ G \mid p'' - P'',\ q'' - Q'')^2}.$$

But, differentiating the above equation as regards the curve, we find

$$0 = 2(E,\ F,\ G \mid p',\ q' \mid p'',\ q'') + (\dot{E},\ \dot{F},\ \dot{G} \mid p',\ q')^2,$$

where $\dot{E}$, $\dot{F}$, $\dot{G}$ are used to denote the complete differential coefficients $E_1p' + E_2q'$, etc. And similarly, differentiating in regard to the tangent geodesic, we obtain

$$0 = 2(E,\ F,\ G \mid p',\ q' \mid P'',\ Q'') + (\dot{E},\ \dot{F},\ \dot{G} \mid p',\ q')^2;$$

and hence, taking the difference of the two equations,

$$0 = (E,\ F,\ G \mid p',\ q' \mid p'' - P'',\ q'' - Q'').$$

Hence, in the equation for $\dfrac{1}{\rho}$, the function under the radical sign may be written

$$(E,\ F,\ G \mid p',\ q')^2 \cdot (E,\ F,\ G \mid p''-P'',\ q''-Q'')^2 - \{(E,\ F,\ G \mid p',\ q' \mid p''-P'',\ q''-Q'$$

which is identically

$$= (EG - F^2)\{p'(q'' - Q'') - q'(p'' - P'')\}^2.$$

Hence, extracting the square root, and for $\sqrt{EG - F^2}$ writing V, we have

$$\frac{1}{\rho} = V\{p'(q'' - Q'') - q'(p'' - P'')\},$$

or say

$$\frac{1}{\rho} = V(p'q'' - q'p'') - V(p'Q'' - q'P''),$$

which is the new formula for the radius of relative curvature.

Formula for the radius of geodesic curvature.

In the paper "On Geodesic Lines, etc.," p. 195, [vol. VIII. of this Collection, p. 160], writing $EG - F^2 = V^2$, and P'', Q'' in place of p'', q'', the differential equation of the geodesic line is obtained in the form

$$\begin{aligned}(Ep' + Fq')\{(2F_1 - E_2)p'^2 + 2G_1p'q' + G_2q'^2\}\\ -(Fp' + Gq')\{E_1p'^2 + 2E_2p'q' + (2F_2 - G_1)q'^2\}\\ +2V^2(p'Q'' - q'P'') = 0;\end{aligned}$$

or, denoting by Ω the first two lines of this equation, we have

$$V(p'Q'' - q'P'') = -\frac{\frac{1}{2}\Omega}{V}.$$

The foregoing equation gives therefore, for the radius of geodesic curvature,

$$\frac{1}{\rho} = V(p'q'' - p''q') + \frac{\frac{1}{2}\Omega}{V},$$

which is an expression depending only upon p', q', the first differential coefficients (common to the curve and geodesic), and on p'', q'', the second differential coefficients belonging to the curve.

Observe that Ω is a cubic function of p', q': we have

$$\Omega = (\mathfrak{A},\ \mathfrak{B},\ \mathfrak{C},\ \mathfrak{D} \mid p',\ q')^3,$$

the values of the coefficients being

$$\begin{aligned}\mathfrak{A} &= 2EF_1 - EE_2 - FE_1,\\ \mathfrak{B} &= 2EG_1 + 2FF_1 - 3FE_2 - GE_1,\\ \mathfrak{C} &= EG_2 + 3FG_1 - 2FF_2 - 2GE_2,\\ \mathfrak{D} &= FG_2 - 2GF_2 + GG_1.\end{aligned}$$

0.1 766. On the Geodesic Curvature of a Curve on a Surface.[4]

[Continued from previous pages.]

The Special Curves, $p =$ constant and $q =$ constant

Consider the curve $p = \text{const}$. For this curve $p' = 0$, $p'' = 0$; therefore also $Gq'^2 = 1$, and, if R be the radius of geodesic curvature, then

$$\frac{1}{R} = \frac{\sqrt{G_1}}{G}.$$

Similarly for the curve $q = \text{const}$. Here $q' = 0$, $q'' = 0$; therefore $Ep'^2 = 1$, and, if S be the radius of geodesic curvature, then

$$\frac{1}{S} = \frac{\sqrt{E_2}}{E}.$$

These values of R and S are interesting for their own sakes, and they will be introduced into the expression for the radius of geodesic curvature ρ of the general curve.

Transformed Formula for the Radius of Geodesic Curvature

From the values of $\frac{1}{R}$, $\frac{1}{S}$ we have

$$\frac{1}{\rho} = \sqrt{\frac{E}{G}}\frac{p'}{q'} + \sqrt{\frac{G}{E}}\frac{q'}{p'} + \sqrt{EG}\left\{p'q'' - p''q' + \frac{1}{2}(Lp' + Mq')\right\},$$

where the term in { } is

$$= 2\mathfrak{A}p' - \tfrac{1}{2}E_2p'^3 + \mathfrak{B}p''q' + (\mathfrak{C}p'q'' + \mathfrak{D}q'^2) - \tfrac{1}{2}G_1q'^3.$$

The terms in $\mathfrak{A}$ are

$$= -\tfrac{1}{2}p'(1 - Ep'^2) = -\tfrac{1}{2}p'(2Fp'q' + Gq'^2),$$

and those in $\mathfrak{D}$ are

$$= -\tfrac{1}{2}q'(1 - Gq'^2) = -\tfrac{1}{2}q'(Ep'^2 + 2Fp'q').$$

[4] From the *Proceedings of the London Mathematical Society*, vol. xii. (1881), pp. 129–132. Read March 10, 1881.

Hence the whole expression contains the factor $p'q'$, and is, in fact,

$$= p'q'\{Lp' + Mq'\};$$

and the formula thus is

$$\frac{1}{\rho} = \frac{\sin^2\phi}{R} + \frac{\sin^2\psi}{S} + \frac{\sin\omega\cdot p'q'}{\sqrt{EG-F^2}}\{Lp' + Mq'\},$$

where L and M are certain expressions involving the coefficients E, F, G and their derivatives.

Taking ϕ, ψ to be the inclination of the curve to the curves $q =$ const., $p =$ const., respectively, and $\omega\,(= \phi + \psi)$ the inclination of these two curves to each other, then

$$\cos\phi = \frac{Fp' + Gq'}{\sqrt{G}}, \quad \sin\phi = \sqrt{E}\,p'\sqrt{EG-F^2}, \quad \text{etc.}$$

hence $\sin\phi = p'\sqrt{E}\sin\omega$, $\sin\psi = q'\sqrt{G}\sin\omega$, and the formula may also be written

$$\frac{1}{\rho} = \frac{\sin^2\phi}{R\sin^2\omega} + \frac{\sin^2\psi}{S\sin^2\omega} + \frac{\sin\phi\sin\psi}{\sqrt{EG-F^2}}\{Lp' + Mq'\}.$$

The Orthotomic Case $F = 0$, or $ds^2 = Edp^2 + Gdq^2$

The formula becomes in this case much more simple. We have

$$1 = Ep'^2 + Gq'^2, \quad \sqrt{EG-F^2} = \sqrt{EG}, \quad \omega = 90^\circ, \quad \sin\phi = \cos\psi;$$

and the term $Lp' + Mq'$ becomes $= EG_1 - E_1G$, if, as before, E_1, G_1 denote the complete differential coefficients $E_1p' + E_2q'$ and $G_1p' + G_2q'$. The formula then is

$$\frac{1}{\rho} - \frac{\cos^2\phi}{R} - \frac{\sin^2\phi}{S} = \sqrt{EG}(p'q'' - p''q') + \tfrac{1}{2}p'q'\sqrt{EG}(EG_1 - E_1G),$$

where the values $\dfrac{1}{R}$ and $\dfrac{1}{S}$ are now $= \dfrac{\sqrt{G_1}}{G}$ and $\dfrac{\sqrt{E_2}}{E}$ respectively. But we have moreover $\phi = \tan^{-1}\dfrac{\sqrt{G}\,q'}{\sqrt{E}\,p'}$, and thence

$$\phi' = \sqrt{EG}(p'q'' - p''q') - \tfrac{1}{2}\phi'\sqrt{EG}(EG_1 - E_1G);$$

or the formula finally is

$$\frac{1}{\rho} - \frac{\cos^2\phi}{R} - \frac{\sin^2\phi}{S} = \phi',$$

which is Liouville's formula referred to at the beginning of the present paper. It will be recollected that ϕ' is the differential coefficient with respect to the arc s of the curve.

Addition

Since the foregoing paper was written, I have succeeded in obtaining a like interpretation of the term

$$\sqrt{EG-F^2}(p'q''-p''q') + \tfrac{1}{2}p'q'(Lp'+Mq'),$$

which belongs to the general case. I find that these terms are, in fact, $= -\phi' + \omega_1 p'$; or, what is the same thing (since $\omega = \psi + \phi$ and therefore $\omega_1 p' + \omega_2 q' = \psi' + \phi'$), are $= \psi' - \omega_1$. It will be recollected that ψ is the inclination of the curve to the curve $q = c$, which passes through a given point of the curve, ψ' is the variation of ψ corresponding to the passage to the consecutive point of the curve, viz., $\psi + \psi'\,ds$ is the inclination at this consecutive point to the curve $q = c + dc$, which passes through the consecutive point; ω is the inclination to each other of the curves $p = b$, $q = c$, which pass through the given point of the curve, ω_1 the variation corresponding to the passage along the curve $q = c$, viz., $\omega + \omega_1\,ds$ is the inclination to each other of the curves $p = b + db$, $q = c$; and the like as regards ϕ and ω_2.

For the demonstration, we have, as above,

$$\cos\psi = \frac{\sqrt{G}\,q'}{\sqrt{EG-F^2}}, \quad \sin\psi = \frac{\sqrt{G}\,p'}{\sqrt{EG-F^2}},$$

where $\sqrt{EG-F^2} = \sqrt{EG-F^2}$; and moreover $Ep'^2 + 2Fp'q' + Gq'^2 = 1$. In virtue of this last equation,

$$\sqrt{EG-F^2}\,\psi' + (Fp'+Gq')^2 = G;$$

and we have

$$\psi' = -\sqrt{EG-F^2}(p'q''-p''q') + \frac{D}{(EG-F^2)^{3/2}},$$

where

$$D = (Fp' + Gq')p'\dot{V} - Vp'(Fp' + Gq')^{\cdot};$$

or, since $V^2 = EG - F^2$, and thence $2V\dot{V} = \dot{E}G + E\dot{G} - 2F\dot{F}$, we have

$$D = \tfrac{1}{2}\dot{V}\{(Fp' + Gq')(\dot{E}G + E\dot{G} - 2F\dot{F}) - 2(EG - F^2)(Fp' + Gq')^{\cdot}\}.$$

Substituting herein for $\dot{E}$, $\dot{F}$, $\dot{G}$ their values $E_1p' + E_2q'$, $F_1p' + F_2q'$, $G_1p' + G_2q'$, the term in { } becomes

$$= Ip'^2 + Jp'q' + Kq'^2,$$

where

$$\begin{aligned} I &= FGE_1 - 2EGF_1 + EFG_1, \\ J &= G'E_1 - 2FGF_1 + (-EG + 2F^2)G_1 + FGE_2 - 2EGF_2 + EFG_2, \\ K &= G'E_2 - 2FGF_2 + (-EG + 2F^2)G_2. \end{aligned}$$

But from the equation $\omega = \tan^{-1}\dfrac{\sqrt{EG - F^2}}{F}$, differentiating in regard to p, we obtain

$$\omega_1 = \frac{FGE_1 - 2EGF_1 + EFG_1}{(EG - F^2)^{3/2}} = \frac{I}{(EG - F^2)^{3/2}};$$

or, for p writing $p + p'(\dot{E}p'^2 + 2\dot{F}p'q' + \dot{G}q'^2)\sqrt{EG - F^2} = \dot{E}p' + \cdots$, we have

$$\psi' - \omega_1 p' = -\sqrt{EG - F^2}(p'q'' - p''q') + (Ip'^2 + Jp'q' + Kq'^2).$$

The terms in p'' destroy each other, and the form thus is

$$\psi' - \omega_1 p' = -\sqrt{EG - F^2}(p'q'' - p''q') - \tfrac{1}{2}p'q'(Lp' + Mq'),$$

where

$$L = \frac{K}{p'^2} - \frac{I}{q'^2}, \quad M = \frac{K}{p'q'} - \frac{J}{p'^2},$$

and, upon substituting herein for I, J, K their values, we find

$$\begin{aligned} L &= -GE_1 + EG_1 + \frac{2F(FG_1 - GF_1) - F(EG_1 - GE_1) + 2E(FF_1 - EF_1)}{EG - F^2}, \\ M &= -GE_2 + EG_2 + \frac{2F(FG_2 - GF_2) - F(EG_2 - GE_2) + 2E(FF_2 - EF_2)}{EG - F^2}; \end{aligned}$$

viz., these are the values denoted above by the same letters L, M. The final result thus is

$$\frac{1}{\rho} - \frac{\sin^2\phi}{R} - \frac{\sin^2\psi}{S} = \psi' - \omega_1 p' = \phi' - \omega_2 q',$$

where the meanings of the symbols have been already explained. A formula substantially equivalent to this, but in a different (and scarcely properly explained) notation, is given, Aoust, "Théorie des coordonnées curvilignes quelconques," *Annali di Matem.*, t. ii. (1868), pp. 39–64; and I was, in fact, led thereby to the foregoing further investigation.

As to the definition of the radius of geodesic curvature, I remark that, for a curve on a given surface, if PQ be an infinitesimal arc of the curve, then if from Q we let fall the perpendicular QM on the tangent plane at P (the point M being thus a point on the tangent PT of the curve), and if from M, in the tangent plane and at right angles to the tangent, we draw MN to meet the osculating plane of the curve in N, then MN is in fact equal to the infinitesimal arc QQ' mentioned near the beginning of the present paper, and the radius of geodesic curvature ρ is thus a length such that $2\rho \cdot MN = PQ^2$.

0.2 767. On the Gaussian Theory of Surfaces.[5]

In the Memoir, Bour, "Théorie de la déformation des surfaces" (*Jour. de l'Éc. Polyt.*, Cah. 39 (1862), pp. 1–148), the author, working with the form $ds^2 = dv^2 + g^2 du^2$ as a special case of Gauss's formula $ds^2 = Edp^2 + 2Fdpdq + Gdq^2$, obtains (p. 29) the following equations which he calls fundamental:

$$\begin{aligned} \frac{dH}{dv} - \frac{dT}{du} &= TV\frac{dg}{g\,dv}, \\ \frac{dT}{dv} - \frac{dH_1}{du} &= -\frac{TV}{g}\frac{dg}{dv}, \end{aligned} \qquad \text{(IV.)}$$

where T, H is written to denote $\frac{d\cdot}{du}$, $\frac{d\cdot}{dv}$, and where (see p. 26)

- H is the curvature of the normal section containing the tangent to the curve $v =$ constant,
- H_1 is the curvature of the normal section at right angles to the preceding, containing the tangent to the (geodesic) curve $u =$ constant,
- T is the torsion of the same geodesic curve;

or, what is the same thing (see p. 25), the quadric equation for the determination of the principal radii of curvature at the point of the surface is

$$\left(\frac{1}{\rho} - H\right)\left(\frac{1}{\rho} - H_1\right) - T^2 = 0.$$

Writing for greater convenience K in place of the suffixed letter H_1, also V instead of g, so that the differential formula is $ds^2 = dv^2 + V^2 du^2$, the equations become

$$\begin{aligned} \frac{dH}{dv} - \frac{dT}{du} &= TV\frac{V_1}{V}, \\ \frac{dT}{dv} - \frac{dK}{du} &= -\frac{TV}{V}V_1; \end{aligned}$$

or, if we use the suffix 1 to denote differentiation in regard to v, and the suffix 2 to denote differentiation in regard to u, then the equations

[5] From the *Proceedings of the London Mathematical Society*, vol. xii. (1881), pp. 187–192. Read June 9, 1881.

are

$$H_1 - T_2 = \frac{TVV_1}{V},$$
$$T_1 - K_2 = -\frac{TVV_1}{V};$$

or, what is the same thing,

$$T_1 + H_1 V + (H - K)V_1 = 0,$$
$$T_1 V + 2TV_1 + K_2 = 0.$$

I wish to show how these formulae connect themselves with formulae belonging to the general form $ds^2 = Edp^2 + 2Fdpdq + Gdq^2$. These involve not only Gauss's coefficients E, F, G, but also the coefficients E', F', G' belonging to the inflexional tangents; and, for convenience, I quote the system of definitions, Salmon's *Geometry of Three Dimensions*, 3rd ed., 1874, p. 251, viz.

$$dx,\ dy,\ dz = a\,dp + a'\,dq,\ b\,dp + b'\,dq,\ c\,dp + c'\,dq;$$
$$d^2x = \alpha\,dp^2 + 2\alpha'\,dp\,dq + \alpha''\,dq^2,$$
$$d^2y = \beta\,dp^2 + 2\beta'\,dp\,dq + \beta''\,dq^2,$$
$$d^2z = \gamma\,dp^2 + 2\gamma'\,dp\,dq + \gamma''\,dq^2;$$
$$A,\ B,\ C = bc' - b'c,\ ca' - c'a,\ ab' - a'b; \quad V^2 = EG - F^2;$$
$$E' = A\alpha + B\beta + C\gamma, \quad F' = A\alpha' + B\beta' + C\gamma', \quad G' = A\alpha'' + B\beta'' + C\gamma'',$$

so that E', F', G' are, in fact, the determinants

$$\begin{vmatrix} a & b & c \\ a' & b' & c' \\ \alpha & \beta & \gamma \end{vmatrix}, \quad \begin{vmatrix} a & b & c \\ a' & b' & c' \\ \alpha' & \beta' & \gamma' \end{vmatrix}, \quad \begin{vmatrix} a & b & c \\ a' & b' & c' \\ \alpha'' & \beta'' & \gamma'' \end{vmatrix}.$$

The equation for the determination of the principal radii of curvature is

$$(E'\rho - EV)(G'\rho - GV) - (F'\rho - FV)^2 = 0.$$

which, in the particular case $F = 0$ (and therefore $V^2 = EG$), becomes

$$(E'\rho - EV)(G'\rho - GV) - F'^2\rho^2 = 0,$$

or, as this may be written,

$$\left(\frac{1}{\rho} - \frac{E'}{EV}\right)\left(\frac{1}{\rho} - \frac{G'}{GV}\right) - \frac{F'^2}{EGV^2} = 0,$$

an equation which corresponds with Bour's form

$$\left(\frac{1}{\rho} - H\right)\left(\frac{1}{\rho} - K\right) - T^2 = 0,$$

and becomes identical with it, if

$$E' = EVK, \quad G' = GVH, \quad F' = -V^2T.$$

But, making p, q correspond to Bour's variables, p to v, and q to u, it is necessary to show that the foregoing values (and not the interchanged values $E' = GVH$, $G' = EVK$) are the correct ones. We have, Salmon, p. 254,

$$\begin{vmatrix} dq, & pE' - VE, & pF' - VF \\ -dp, & pF' - VF, & pG' - VG \end{vmatrix} = 0;$$

or, putting herein $F = 0$, the equations may be written

$$\frac{dq}{-dp} = \frac{E'\left(\dfrac{1}{\rho} - \dfrac{VE}{E'}\right)}{F'\left(\dfrac{1}{\rho} - \dfrac{VF}{F'}\right)} = \frac{F'\left(\dfrac{1}{\rho} - \dfrac{VF}{F'}\right)}{G'\left(\dfrac{1}{\rho} - \dfrac{VG}{G'}\right)};$$

or, we see that to $dq = 0$ corresponds the value $\dfrac{1}{\rho} = \dfrac{VE}{E'}$, and to $dp = 0$ the value $\dfrac{1}{\rho} = \dfrac{VG}{G'}$. Hence the former of these values of $\dfrac{1}{\rho}$ corresponds to Bour's $du = 0$, that is, to his $\dfrac{1}{\rho} = K$; and the latter to Bour's $dv = 0$, that is, to his $\dfrac{1}{\rho} = H$; or the values are, as stated,

$$E' = EVK, \quad G' = GVH.$$

The formula $ds^2 = Edp^2 + 2Fdpdq + Gdq^2$ agrees with Bour's $ds^2 = dv^2 + g^2du^2$, if $p = u$, $q = v$, $E = 1$, $F = 0$, $G = g^2$. With these values, $V^2 = EG - F^2 = g^2$, or say $g = V$, and Bour's equation is, as it was before written, $ds^2 = dv^2 + V^2du^2$. And we have to find the three equations which, putting therein $p = u$, $q = v$, $E = 1$, $F = 0$, $G = V^2$, $E' = VK$, $F' = -VT$, $G' = VH$, reduce themselves to Bour's equations.

The first of these is nothing else than the equation for the measure of curvature, viz. Salmon, p. 262 (but, using the suffix 1 for v and 2 for u),

$$4V^2\{HK-T^2\} = (2VV_{12})^2-2V_2(2VV_{11}+2V_1^2)+2V_1(2VV_{22}+2V_2^2).$$

In fact, writing herein $E=1$, $F=0$, and therefore the differential coefficients of E and F each $=0$, the equation becomes

$$4V^4\{HK-T^2\} = (2VV_{12})^2-2V_2(2VV_{11}+2V_1^2)+2V_1(2VV_{22}+2V_2^2),$$

which is

$$4V^4\{HK-T^2\} = (2VV_{12})^2-2V_2(2VV_{11}+2V_1^2)+2V_1(2VV_{22}+2V_2^2) = -4V^4V_{22};$$

or finally it is

$$V_{22} = V(T^2-HK).$$

The other two of Bour's equations are derived from equations which give respectively the values of $E_2'-F_1'$ and $F_2'-G_1'$; viz. starting from the equations

$$\begin{aligned} E' &= A\alpha+B\beta+C\gamma, \\ F' &= A\alpha'+B\beta'+C\gamma', \\ G' &= A\alpha''+B\beta''+C\gamma'', \end{aligned}$$

we see at once that E_2' and F_1' contain, E_2' the terms $A\alpha_2+B\beta_2+C\gamma_2$, and F_1' the terms $A\alpha_1'+B\beta_1'+C\gamma_1'$, which are equal to each other ($\alpha_2=\alpha_1'$ since α and α' are the differential coefficients x_1, x_2 of x, and so $\beta_2=\beta_1'$ and $\gamma_2=\gamma_1'$). Hence

$$E_2'-F_1' = A_2\alpha+B_2\beta+C_2\gamma-A_1\alpha'-B_1\beta'-C_1\gamma',$$

and similarly

$$F_2'-G_1' = A_2\alpha'+B_2\beta'+C_2\gamma'-A_1\alpha''-B_1\beta''-C_1\gamma''.$$

Here, from the values of A, B, C, we have

$$\begin{aligned} &A = bc'-cb'; \quad A_1 = \beta c'-\gamma b'+b\gamma'-c\beta'; \quad A_2 = \beta' c'-\gamma' b'+b\gamma''-c\beta''; \\ &B = ca'-ac'; \quad B_1 = \gamma a'-\alpha c'+c\alpha'-a\gamma'; \quad B_2 = \gamma' a'-\alpha' a'+c\alpha''-a\gamma''; \\ &C = ab'-ba'; \quad C_1 = \alpha b'-\beta a'+a\beta'-b\alpha'; \quad C_2 = \alpha' b'-\beta' a'+a\beta''-b\alpha''; \end{aligned}$$

and, substituting, we find

$$\begin{aligned}E_2' - F_1' &= 2\alpha'\alpha\alpha' + \alpha\alpha''\alpha - 2\alpha\alpha'\alpha'' - \alpha'\alpha''\alpha,\\ F_2' - G_1' &= \cdots\end{aligned}$$

if, for shortness, $\alpha'\alpha\alpha'$ denotes the determinant

$$\begin{vmatrix}\alpha' & \alpha & \alpha\\ \beta' & \beta & \beta\\ \gamma' & \gamma & \gamma'\end{vmatrix}$$

and so for the other like symbols. Observe that, with

$$\alpha,\ \alpha',\ \alpha,\ \alpha',\ \alpha''b,\ b',\ \beta,\ \beta',\ \beta''c,\ c',\ \gamma,\ \gamma',\ \gamma''$$

we have in all 10 determinants, viz. these are $\alpha\alpha'\alpha = E'$; $\alpha\alpha'\alpha' = F'$; $\alpha\alpha'\alpha'' = G'$; $\alpha\alpha'\alpha''$; and the six determinants $\alpha\alpha\alpha'$, $\alpha\alpha'\alpha''$, $\alpha\alpha''\alpha$; $\alpha'\alpha\alpha$, $\alpha'\alpha'\alpha''$, $\alpha'\alpha''\alpha$.

The foregoing expressions of $E_2' - F_1'$ and $F_2' - G_1'$ respectively, substituting therein for the determinants $\alpha'\alpha\alpha'$, $\alpha\alpha''\alpha$, $\alpha\alpha'\alpha''$, $\alpha'\alpha''\alpha$ their values as about to be obtained, are the required two equations. We have

$$\begin{aligned}
\alpha\alpha + bb + cc &= E, & \alpha\alpha' + bb' + cc' &= F,\\
\alpha'\alpha + b'b + c'c &= F, & \alpha'\alpha' + b'b' + c'c' &= G,\\
\alpha\alpha + \beta b + \gamma c &= \tfrac{1}{2}E_1, & \alpha\alpha' + \beta b' + \gamma c' &= F_1 = \tfrac{1}{2}E_2,\\
\alpha'\alpha + \beta' b + \gamma' c &= \tfrac{1}{2}E_2, & \alpha'\alpha' + \beta' b' + \gamma' c' &= \tfrac{1}{2}G_1,\\
\alpha''\alpha + \beta'' b + \gamma'' c &= F_1 - \tfrac{1}{2}G_2, & \alpha''\alpha' + \beta'' b' + \gamma'' c' &= \tfrac{1}{2}G_2;
\end{aligned}$$

and if from the first five equations, regarded as equations linear in (a, b, c), we eliminate these quantities, and from the second five equations, regarded as linear in (a', b', c'), we eliminate these quantities, we obtain two sets each of five equations.

Attending to the proper reduction, we obtain

$$\begin{aligned}
E_2' - F_1' = \tfrac{1}{V}\{&(-\tfrac{1}{2}FG_2 + \tfrac{1}{2}GE_1 - FF_1 + \tfrac{1}{2}EF_2)E'\\
&+ (-GE_1 + 2FF_1 - FF_2)F' + (\tfrac{1}{2}FE_1 - EF_1 + \tfrac{1}{2}EE_2)G'\},\\
F_2' - G_1' = \tfrac{1}{V}\{&(-\tfrac{1}{2}GG_2 + GF_1 - \tfrac{1}{2}FG_2)E'\\
&+ (FG_2 - 2FF_2 + EG_1)F' + (-\tfrac{1}{2}GE_1 + FF_1 - EG_1 + \tfrac{1}{2}FE_2)G'\},
\end{aligned}$$

which are the required formulae; and which may, I think, be regarded as new formulae in the Gaussian theory of surfaces.

Writing herein as before, the first of these becomes

$$V(VK)_2 + \{\ \} = \tfrac{1}{V}\{(V^2)_2 VK\}_2 = V_2 K,$$

that is,

$$V_{12}K + VK_2 + F^2T_2 + 2VV_1T = V_2K;$$

or finally it is

$$VT_1 + 2TV_1 + K_2 = 0,$$

which is Bour's third equation. And the second equation becomes

$$-(V^2T)_1 - (V^2VH)_2 = \tfrac{1}{V}\{-\tfrac{1}{2}V^2V_2 \cdot VK + (V^2)_1(-V^2T)(V^2)_1VH\}$$

$$= -V^2V_2K - 2VV_1T - 2V^2V_1H,$$

that is,

$$-V^2T_1 - 2VV_1T - V^2H_2 - 2V^2V_1H = -V^2V_2K - 2VV_1T - 2V^2V_1H;$$

or finally

$$T_1 + VH_2 + (H-K)V_1 = 0,$$

which is Bour's second equation.

0.3 768. Note on Landen's Theorem.[6]

Landen's theorem, as given in the paper "An Investigation of a General Theorem for finding the length of any Arc of any Conic Hyperbola by means of two Elliptic Arcs, with some other new and useful Theorems deduced therefrom," *Phil. Trans.*, t. LXV. (1775), pp. 283–289, is, as appears by the title, a theorem for finding the length of a hyperbolic arc in terms of the length of two elliptic arcs; this theorem being obtained by means of the following differential identity, viz., if

$$t^2 = \frac{m^2 - n^2}{n^2} x^2, \quad \text{where} \quad g = \frac{m^2 - n^2}{n^2},$$

then

$$\frac{\sqrt{m^2 - gx^2}\, dx}{\sqrt{1 - x^2}} = \frac{1}{2} \frac{\sqrt{(m+n)^2 - t^2}\, dt}{\sqrt{1 - t^2}} + \frac{1}{2} \frac{\sqrt{(m-n)^2 - t^2}\, dt}{\sqrt{1 - t^2}},$$

(this is exactly Landen's form, except that he of course writes $\dot{x}$, $\dot{t}$ in place of dx, dt respectively): viz., integrating each side, and interpreting geometrically in a very ingenious and elegant manner the three integrals which present themselves, he arrives at his theorem for the hyperbolic arc; but with this I am not now concerned.

Writing for greater convenience $m = 1$, $n = k'$, and therefore $g = k^2$, if as usual $k^2 + k'^2 = 1$, the transformation is

$$t = \frac{kx}{\sqrt{1 - k'^2 x^2}},$$

leading to

$$\frac{(1 + k')\, dx}{\sqrt{1 - x^2}\sqrt{1 - k'^2 x^2}} = \frac{dt}{\sqrt{1 - t^2}\sqrt{1 - \lambda^2 t^2}},$$

where

$$\lambda = \frac{1 - k'}{1 + k'}.$$

The form in which the transformation is usually employed (see my *Elliptic Functions*, pp. 177, 178) is

$$\frac{(1 + k')\, dx}{\sqrt{1 - x^2}\sqrt{1 - k'^2 x^2}} = \frac{dy}{\sqrt{1 - y^2}\sqrt{1 - \lambda^2 y^2}},$$

[6]From the *Proceedings of the London Mathematical Society*, vol. xiii. (1882), pp. 47, 48. Read November 10, 1881.

where

$$\lambda = \frac{1-k'}{1+k'},$$

leading to

$$y = \frac{(1+k')x}{\sqrt{1-x^2}\sqrt{1-k'^2x^2}}.$$

If, to identify the two forms, we write $y = t$, and in the last equation introduce t in place of y, the last equation becomes

$$\frac{dx}{\sqrt{1-x^2}\sqrt{1-k'^2x^2}} = \frac{dt}{\sqrt{(1-k')^2-t^2}\sqrt{(1+k')^2-t^2}}.$$

Comparing with Landen's form, in order that the two may be identical, we must have

$$\sqrt{1-k'^2x^2} = \tfrac{1}{2}\{\sqrt{(1-k')^2-t^2} + \sqrt{(1+k')^2-t^2}\},$$

viz., this is

$$1-k'^2x^2 = \tfrac{1}{4}[1+k'^2-t^2+\sqrt{\{(1-k')^2-t^2\}\{(1+k')^2-t^2\}}],$$

where the function under the radical sign is

$$(1-k')^2 - 2(1+k'^2)t^2 + t^4 \ (= T \text{ suppose})$$

and this must consequently be a form of the original integral equation.

In fact, squaring and solving in regard to k'^2 with the assumed sign of the radical, we have

$$k'^2 = \frac{1+t^2-\sqrt{T}}{2x^2},$$

corresponding to an equation given by Landen. And we thence have

$$1-k'^2x^2 = \frac{1-t^2+\sqrt{T}}{2},$$

which is the required expression for $1-k'^2x^2$.

The trigonometrical form $\sin(2\phi'-\phi) = c\sin\phi$ of the relation between y and x does not occur in Landen; it is employed by Legendre, I believe, in an early paper, *Mem. de l'Acad. de Paris*, 1786, and in the *Exercices*, 1811, and also in the *Traité des Fonctions Elliptiques*, 1825, and by means of it he obtains an expression for the arc of a hyperbola in terms of two elliptic functions, $E(c, \phi)$, $E(c', -\phi')$, showing that the arc of the hyperbola is expressible by means of two elliptic arcs, this, he observes, "est le beau théorème dont Landen a enrichi la géométrie." We have, then (1828), Jacobi's proof, by two fixed circles, of the addition-theorem (see my *Elliptic Functions*, p. 28), and the application of this (p. 30) to Landen's theorem is also due to Jacobi, see the "Extrait d'une lettre adressée à M. Hermite," *Crelle*, t. xxxii. (1846), pp. 176–181; the connection of the demonstrations, by regarding the point, which is alone necessary for Landen's theorem as the limit of the smaller circle in the figure for the addition-theorem is due to Hermite.

0.4 769. On a Formula relating to Elliptic Integrals of the Third Kind.[7]

The formula for the differentiation of the integral of the third kind

$$\Pi(n, k, \phi) = \int_0^\phi \frac{d\phi}{(1 + n \sin^2 \phi)\Delta},$$

in regard to the parameter n, see my *Elliptic Functions*, Nos. 174 et seq., may be presented under a very elegant form, by writing therein

$$\sin^2 \phi = x = \operatorname{sn}^2 u, \quad \sin \phi \cos \phi \, \Delta = y = \operatorname{sn} u \operatorname{cn} u \operatorname{dn} u,$$

and thus connecting the formula with the cubic curve

$$y^2 = x(1 - x)(1 - k^2 x).$$

The parameter must, of course, be put under a corresponding form, say $n = -\dfrac{b}{a}$, where $a = \operatorname{sn}^2 \theta$, $b = \operatorname{sn} \theta \operatorname{cn} \theta \operatorname{dn} \theta$, and therefore (a, b) are the coordinates of the point corresponding to the argument θ. The steps of the substitution may be effected without difficulty, but it will be convenient to give at once the final result and then verify it directly. The result is

$$\frac{d\Pi}{d\theta} = \frac{b}{a - x} \frac{du}{d\theta} - \frac{y}{(a - x)^2} \frac{da}{d\theta}.$$

We, in fact, have

$$\frac{d}{du} x = 2 \operatorname{sn} u \operatorname{cn} u \operatorname{dn} u = 2y,$$

and thence

$$\frac{du}{d\theta} = \frac{1}{2y}.$$

Also

$$\frac{d^2}{du^2} x = \operatorname{cn}^2 u \operatorname{dn}^2 u - \operatorname{sn}^2 u \operatorname{dn}^2 u - k^2 \operatorname{sn}^2 u \operatorname{cn}^2 u = 1 - 2(1 + k^2)x + 3k^2 x^2,$$

[7] From the *Proceedings of the London Mathematical Society*, vol. xiii. (1882), pp. 175, 176. Presented May 11, 1882.

and hence

$$\frac{d}{d\theta}\frac{y}{a-x} = \frac{1}{(a-x)^2}\left\{-y\frac{da}{d\theta} + (a-x)\frac{dy}{d\theta}\right\}.$$

Interchanging the letters, we have

$$\frac{d}{d\theta}\frac{a}{x-a} = \frac{1}{(a-x)^2}\left\{-a\frac{dx}{d\theta} + (x-a)\frac{da}{d\theta}\right\},$$

and hence, subtracting,

$$\frac{d}{d\theta}\frac{y}{a-x} - \frac{d}{d\theta}\frac{u}{x-a} = \frac{1}{(a-x)^2}\left\{-y\frac{da}{d\theta} + u\frac{dx}{d\theta}\right\} = \frac{k^2(a-x)}{(a-x)^2},$$

which is the required result.

0.5 770. On the 34 Concomitants of the Ternary Cubic.[8]

[This paper continues from previous pages.]

Supplemental Table of 6 Derived Forms (continued)

23, 24. $E' = -\frac{1}{2}(S)E$, $E'' = \frac{1}{2}(S')E$

E is of the form $(abc+8l^3)\{aU+bV+cW\}$, and, as before, in a term such as

$$(abc+8l^3)\, f(by^2-cz^2)(2l^2x^2+6cyz),$$

we operate with S or S' only on the factor $2l^2x^2+6cyz$; and in E' and E'' respectively, operating upon this factor, we obtain

$$-\tfrac{1}{2}\{Ul^2ax^2+(bc+b'c)yz\}, \quad \text{and} \quad \tfrac{1}{2}\{Wl^2ax^2+2b'c'yz\},$$

viz. we thus obtain in E' the term

$$(abc+8l^3)\, f(by^2-cz^2)\{l(abc+2l^3)ax^2-3bcl^2yz\},$$

and in E'' the term

$$(abc+8l^3)\, f(by^2-cz^2)\{(abc+2l^3)ax^2+18bcl^2z\}.$$

25. $M = \frac{1}{2}\{a.c.(U,\psi,\Lambda)\}$

This, omitting the factor $(abc+8l^3)^2$ of Ψ, is

$$\begin{array}{lll} aa^2+2lyz, & ax^2(abc-5by^2-5cz^2), & f \\ by^2+2lzx, & by^2(6y^2-acz^2-aax^2), & \xi \\ cz^2+2lxy, & cz^2(cz^2-5ax^2-bby^2), & \eta \end{array}$$

the coefficient of ξ herein is

$$\begin{aligned} &= \tfrac{1}{2}\{(6cyz^2+2clxz^2)(cz^2-ax^2-5by^2)-(6cy^2z+2blx^2y)(by^2-cz^2-ax^2)\} \\ &= \tfrac{1}{2}\{bcfz^2(-6by^2+6cz^2)+2lx[-cy^2+c^2z^2+5ax^2(by^2-cz^2)]\} \\ &= (by^2-cz^2)\{5alx^3-blx\xi-clx\eta-3bcfz\}. \end{aligned}$$

Hence, restoring the factor $(abc+8l^3)^2$, we have the term

$$(abc+8l^3)^2\cdot f(by^2-cz^2)\{5alx^3-blx\xi-clx\eta-3bcfz\}.$$

[8] From the *American Journal of Mathematics*, vol. iv. (1881), pp. 1–15.

26. $M' = -\frac{1}{2}(S)M$

Here M is of the form $(abc + 8l^3)^2(a \cdot U + \cdots)$; and operating with S through the $(abc + 8l^3)a$, &c., we obtain in M' the term

$$\tfrac{1}{2}(abc + 8l^3)x(by^2 - cz^2)\{(a^2bc + ab'c + abc' - 24l^6)f^2 + \text{\&c.}\},$$

where

$$a'bc + ab'c + abc' - 24l^6.$$

Continuation of the Table of 34 Concomitants

The complete expressions for concomitants 23–34 involve extensive algebraic manipulation. The following are the leading terms for the remaining derived forms:

27–29. Higher covariants $C^{(4)}$, $C^{(5)}$, $C^{(6)}$

These are obtained by successive operations of S and S' on the fundamental forms, generating covariants of increasing degree and order. The explicit expressions are of great length, involving terms of the form $(abc + 8l^3)^n$ multiplied by polynomial expressions in x, y, z and the coefficients a, b, c, l.

30–34. The remaining fundamental concomitants

30. $P = \{a, b, c, f, g, h\}(x, y, z)^3$ — the original cubic.
31. $Q = (S)P$ — the quadratic covariant.
32. $R = (S')P$ — another quadratic covariant.
33. S — the cubicovariant.
34. T — the discriminant.

The full computation of these forms, as given in the original memoir, verifies that the system of 34 concomitants is complete, in the sense that any other covariant, contravariant, or mixed concomitant of the ternary cubic is expressible as a rational integral function of these 34 forms.

0.6 771. Specimen of a Literal Table for Binary Quantics, otherwise a Partition Table.[9]

The Table, commencing 1; b; c, b^2; d, bc, b^3; ..., is in fact a Partition Table, viz. considering the letters b, c, d, ... as denoting 1, 2, 3, ... respectively, it is 1^0; 1; 2, 11; 3, 12, 111; ... a table of the partitions of the numbers 0, 1, 2, 3, ..., expressed however in the literal form, in order to its giving the literal terms which enter into the coefficients of any covariant of a binary quantic. The table ought to have been, and I now give it, as far as weight 18 (included), viz. up to r.

The partitions are arranged according to their greatest part; and for each greatest part, according to the value of the next greatest part; and so on. Observe that the weight of any term is the sum of the literal factors; thus c^2d^2e is of weight $2 \cdot 2 + 2 \cdot 3 + 5 = 15$.

Formation of the Table

As regards the formation of the table, this is at once effected, and the successive terms are obtained *currente calamo*, by Arbogast's rule of the last and the last but one: observing that each term is to be regarded as containing the full number of letters (viz. for weight 18, each term contains 18 letters, counting repetitions); and writing the terms in the several orders $0, 1, 2, \ldots$ (as in the expression of any covariant of a binary quantic), we write the first and second term in each order at sight; and thence the remaining terms by the rule.

The Partition Table, to weight 18

Partition Table (Literal Form)									
0	1	b	c	d	e	f	g	h	i
1									
2	b^2	bc	bd	be	bf	bg	bh	bi	
3	b^3 c^2 bcd	b^2c c^2d bcd^2	b^2d c^2e bce	b^2e c^2f	b^2f c^2g	b^2g	b^2h		
4	b^4 b^2c^2 bc^2d	b^3c b^2cd bc^3	b^3d b^2d^2	b^3e	b^3f				

[9] From the *American Journal of Mathematics*, vol. iv. (1881), pp. 248–255.

Partition Table (Literal Form) — continued									
	c^4	c^3d							
...	...	...	...	...	...	...	...	...	...
Weight 18									
18	bhj b^2gh b^2fg cdi cej cf^2 c^2h c^2g dej dfi	b^2ej b^2dfi b^2dgh b^2f^2 c^2ei c^2fj c^2gi c^2hi cd^2i cd^2h	d^2ef b^3fi b^3gh b^3fj b^3gi b^3hi b^2ek bc^2dk bc^2ej c^3dl	b^2deg b^3ei b^3fh b^3dj b^3ci b^3bi b^2ck b^2cl b^2cm c^4j	b^2cdg b^3di^2 b^3eg bcd^2g bcd^2f	b^3d^2f b^3d^2h b^3d^2g b^3df	b^4cdf b^4dh	b^5c^2f	

The table is to be used as follows: for the cubicovariant, where the coefficients are of the degree 3, and of the weights 3, 4, 5, 6 respectively, from the portion of the table

$$\begin{array}{cccc} d & e & f & g \\ bc & bd & be & bf \\ b^2 & c^2 & cd & ce \\ b^2c & b^2d & b^3 & \dots \end{array}$$

we at once copy out the terms

$$ad,\ abd,\ acd,\ ad^2,\ abc,\ ac^2,\ b^2d,\ bcd,\ b^3,\ b^2c,\ bd^2,\ cd,\ \dots$$

which compose the coefficients in question.

0.7 772. On the Analytical Forms called Trees.[10]

In a tree of N knots, selecting any knot at pleasure as a root, the tree may be regarded as springing from this root, and it is then called a root-tree. The same tree thus presents itself in various forms as a root-tree; and if we consider the different root-trees with N knots, these are not all of them distinct trees. We have thus the two questions, to find the number of root-trees with N knots; and, to find the number of distinct trees with N knots.

I have in my former paper ("On the theory of the Analytical Forms called Trees," *Phil. Mag.*, vol. xiii. (1857), pp. 172–176 = 419) given a solution of the first question; and also a solution of the second question by a process of exclusion from the root-trees. But a more simple solution is obtained by the consideration of the centre or bicentre "of number." A tree of an odd number of knots has a centre of number, and a tree of an even number of knots has a centre or else a bicentre of number.

The number of root-trees with N knots is given by the function

$$t_N = \frac{1}{N} \sum_{d|N} \phi(d)\, t_{N/d}^{N/d},$$

where ϕ is Euler's totient function, and t_m denotes the number of root-trees with m knots.

The numerical result obtained for the total number of distinct trees with N knots is given as follows:

No. of Knots	1	2	3	4	5	6	7	8	9	10	11	12	13
No. of Central Trees	1	1	1	2	3	6	11	23	47	106	235	551	1301
Bicentral	0	0	0	0	0	1	1	2	3	6	11	20	37
Total	1	1	1	2	3	6	11	23	47	106	235	551	1301

The formulae for the numbers of central and bicentral trees are obtained as follows. Let ϕ_1, ϕ_2, ϕ_3, ... denote the numbers of root-trees with 1, 2, 3, ... knots respectively. Then for $N = 2n+1$ (odd),

[10] From the *American Journal of Mathematics*, vol. iv. (1881), pp. 266–268.

the number of central trees is

$$\phi_{2n+1} - \sum \phi_i \phi_j + \text{corrections},$$

and for $N = 2n$ (even), the number of bicentral trees is

$$\tfrac{1}{2}\phi_n(\phi_n + 1) + \sum \phi_i \phi_j \cdot \phi_k + \cdots$$

and so on, the law being obvious. And the formulae are at once seen to be true.

Thus for $N = 6$, the formula is

$$t_6 = \tfrac{1}{2}\phi_3(\phi_3+1)+\tfrac{1}{2}\phi_2(\phi_2+1)\cdot\phi_1+\phi_2\cdot\tfrac{1}{2}\phi_1(\phi_1+1)(\phi_1+2)+\tfrac{1}{6}\phi_1(\phi_1+1)(\phi_1+2)(\phi_1+3)(\phi$$

We have root-trees with 3 knots, and by simply joining together any two of them, treating the two roots as a bicentre, we have all the bicentral trees with 6 knots: this accounts for the term $\frac{1}{2}\phi_3(\phi_3 + 1)$. Again, we have ϕ_1 root-trees with 1 knot, ϕ_2 root-trees with 2 knots; and with a given knot as centre, and the pairs of root-trees disposed symmetrically about it, we obtain the remaining terms.

0.8 773. On the 8-Square Imaginaries.[11]

I write throughout 0 to denote positive unity, and uniting with it the seven imaginaries $1, 2, 3, 4, 5, 6, 7$, form an octavic system $0, 1, 2, 3, 4, 5, 6, 7$, the laws of combination being

$$0^2 = 0, \quad 1^2 = 2^2 = 3^2 = 4^2 = 5^2 = 6^2 = 7^2 = -0,$$

and

$$\begin{aligned} &123 = \epsilon_0, \quad 145 = \epsilon_0, \quad 167 = \epsilon_3, \\ &246 = \epsilon_0, \quad 257 = \epsilon_3, \quad 347 = \epsilon_3, \quad 356 = \epsilon_3, \end{aligned}$$

where $\epsilon = \pm$, viz. each ϵ has a determinate value $+$ or $-$ as the case may be; and where the formula, $123 = \epsilon_0$, denotes the six equations

$$23 = \epsilon_0 1, \quad 31 = \epsilon_0 2, \quad 12 = \epsilon_0 3,$$

and similarly for the other formulae. The system is an 8-square system, viz. the sum of the squares of eight linear functions of $0, 1, \ldots, 7$ is equal to the product of two sums each of eight squares.

Hence if $0, 1, 2, 3, 4, 5, 6, 7$ and $0', 1', 2', 3', 4', 5', 6', 7'$ denote ordinary algebraical magnitudes, and we form the product

$$(00'+11'+22'+33'+44'+55'+66'+77')(0'0+1'1+2'2+3'3+4'4+5'5+6'6+7'7),$$

this is at once found to be

$$\begin{aligned} &(00' - 11' - 22' - 33' - 44' - 55' - 66' - 77') \\ &\quad + (01' + 0'1 + \epsilon_1 23 + \epsilon_2 45 + \epsilon_3 67)1 \\ &\quad + (02' + 0'2 + \epsilon_1 31 + \epsilon_4 46 + \epsilon_5 57)2 \\ &\quad + (03' + 0'3 + \epsilon_1 12 + \epsilon_4 47 + \epsilon_5 56)3 \\ &\quad + (04' + 0'4 + \epsilon_2 51 + \epsilon_4 62 + \epsilon_6 73)4 \\ &\quad + (05' + 0'5 + \epsilon_2 14 + \epsilon_6 72 + \epsilon_7 63)5 \\ &\quad + (06' + 0'6 + \epsilon_3 71 + \epsilon_4 24 + \epsilon_7 35)6 \\ &\quad + (07' + 0'7 + \epsilon_3 16 + \epsilon_5 25 + \epsilon_6 34)7, \end{aligned}$$

where 12 is written to denote $12' - 1'2$, &c.

The conditions for an 8-square system are obtained by writing the foregoing expression under the form

$$A_0^2 + A_1^2 + A_2^2 + A_3^2 + A_4^2 + A_5^2 + A_6^2 + A_7^2,$$

[11] From the *American Journal of Mathematics*, vol. iv. (1881), pp. 293–296.

and expressing that this is $= (0^2 + 1^2 + \cdots + 7^2)(0'^2 + 1'^2 + \cdots + 7'^2)$ identically. We thus obtain a system of equations among the ϵ's; and these being satisfied, we have the 8-square system.

Writing $0 = \pm$ at pleasure, these are all satisfied if $-\epsilon_5 = \epsilon_6 = \epsilon_7 = \epsilon_8 = 0$. The terms written down all disappear, and the sum of the squares of the eight coefficients thus becomes equal to the product of two sums each of them of eight squares, viz. this is the case if $\epsilon_1 = \epsilon_2 = \epsilon_3 = +$, $-\epsilon_4 = \epsilon_5 = \epsilon_6 = \epsilon_7 = \epsilon_8$, ϵ_8 being $= +$ at pleasure: the resulting system of imaginaries may be said to be an 8-square system.

We may inquire whether the system is associative; for this purpose, supposing in the first instance that the ϵ's remain arbitrary, we form the complete system of equations which express the associative property $(ab)c = a(bc)$ for all combinations of the symbols $0, 1, \ldots, 7$. These equations impose further restrictions on the ϵ's; and it is found that the system cannot be made associative without destroying the 8-square property. In fact, the only associative systems of this kind are the quaternion system (4-square) and the system of octaves (8-square) discovered by Graves and Cayley, the latter of which is not associative but is alternative.

0.9 774. Tables for the Binary Sextic.[12]

The Leading Coefficients of the First 18 of the 26 Covariants

Including the sextic itself, the number of covariants of the binary sextic is = 26, as shown in the table p. 296 of Clebsch's *Theorie der bin algebraischen Formen*, Leipzig, 1872; viz. this is

Order	2	4	6	8	10	12
Deg.						
1	A					
2	B	C				
3	D	E	F	G		
4	H	I	J	K	L	
5	M	N	O	P		
6	Q	R	S	T		
7	U	V	W	X		
8	Y	Z				

Or, using the capital letters $A, B, \ldots, Z$ to denote the 26 covariants in the same order, the table is:

	2	4	6	8	10	12
A	1					
B						
C	2					
D	3					
E		F	G	H		
I	4	J	K	L		
M	5	N	O	P		
Q	6	R	S	T		
U	7	V	W	X		
Y	8	Z				

Here:

- A is the sextic itself.

[12]From the *American Journal of Mathematics*, vol. iv. (1881), pp. 379–384.

- B is Salmon's A, p. 202.
- C is Salmon's B, p. 203.
- P is Salmon's C, p. 204.
- W is Salmon's D, p. 207.
- Z is Salmon's E, p. 253.

The references are to Salmon's *Higher Algebra*, 2nd Ed., 1866.

Leading Coefficients

In the present short paper I give the leading coefficients of the first 18 covariants, viz. A to R inclusive; as also, for the reason appearing above, the expressions for the remaining coefficients of C, E, O.

The leading coefficient of a covariant is the coefficient of the highest power of x (or a, in the notation of the binary sextic $(a, b, c, d, e, f, g)(x, y)^6$). The following table gives these coefficients:

Covariant	**Degree**	**Leading Coefficient**
A	1	a
B	2	$ae - 6bd + 15c^2 - 10d^2$
C	2	$af - 6be + 15cd - 10df$
D	3	$a^2f + \cdots$
E	3	$a^2g + \cdots$
F	3	$abc + \cdots$
G	3	$ab^2 + \cdots$
H	4	$a^3g^2 + \cdots$
I	4	$a^2f^2 + \cdots$
J	4	$a^2fg + \cdots$
K	4	$a^2g^2 + \cdots$
L	4	$ab^2f + \cdots$
M	5	$a^3f^2g + \cdots$
N	5	$a^2bcf + \cdots$
O	5	$a^2b^2g + \cdots$
P	5	$abcdef + \cdots$
Q	6	$a^4f^2g^2 + \cdots$
R	6	$a^3b^2f^2 + \cdots$

Remaining Coefficients of C, E, O

Covariant C (Degree 2, Order 6)

$$\begin{aligned} C = {} & (af - 6be + 15cd - 10df)x^6 \\ & + (ag - 6bf + 15ce - 10dg)x^5y \\ & + (ah - 6bg + 15cf - 10dh)x^4y^2 \\ & + \cdots \end{aligned}$$

Covariant E (Degree 3, Order 6)

$$\begin{aligned} E = {} & a^2gx^6 \\ & + (a^2h + 2abg)x^5y \\ & + (a^2i + 2abh + b^2g)x^4y^2 \\ & + \cdots \end{aligned}$$

Covariant O (Degree 5, Order 2)

$$O = (a^2b^2g + \cdots)x^2 + (a^2b^2h + \cdots)xy + (a^2b^2i + \cdots)y^2.$$

But these two invariants W and Z have been already calculated; viz. as already mentioned, W is Salmon's invariant D, and Z his invariant E, given each of them in the second edition of his *Higher Algebra* (but not reproduced in the third edition): on account of the great length of these expressions, it has been thought that it was not expedient to give them here.

End of Transcription

Transcribed from
The Collected Mathematical Papers of Arthur Cayley,
Cambridge University Press, Vol. XI (1896), pp. 351–400.

Papers numbered 766–774.

775.

TABLES OF COVARIANTS OF THE BINARY SEXTIC.

[*Written in 1894: now first published.*]

THE binary sextic has in all (including the sextic itself and the invariants) 26 covariants which I have represented by the capital letters A, B, $C, \ldots, Z$. The leading coefficients of the covariants A to R (of course for an invariant this means the invariant itself) are given in my paper "Tables for the binary sextic," *Amer. Math. Jour.* vol. IV. (1881), pp. 379–384, [774]; the two invariants Z and W (Salmon's invariants D and E) had been already calculated. But I did not in my values of the leading coefficients, nor did Salmon in his values of the two invariants, insert the literal terms with zero coefficients: as remarked in my paper [143] "Tables of the covariants M to W of the binary quintic," it is very desirable to have in every case the complete series of literal terms, and I have accordingly in the expressions of the covariants A to R obtained for the leading coefficients, and in the expressions obtained from Salmon for the invariants W and Z, inserted in each case the complete series of literal terms.

I give a list of the 26 covariants nearly in the form of that given in the latter paper [143] for the covariants of the quintic, only instead of a separate column of deg-weights I insert these in the body of the symbol; thus

$$C = (3,\, 3,\, 4,\, 3,\, 3)^2 \ 4 \text{ to } 8\,(x,\, y)^4,$$

the 5 coefficients of the quartic function contain respectively 3, 3, 4, 3, 3 terms (some of them it may be with zero coefficients), are of the degree 2, and of the weights 4, 5, 6, 7, 8 respectively.

The list is as follows:

$$
\begin{aligned}
A &= (1,\ 1,\ 1,\ 1,\ 1,\ 1,\ 1)^1 \text{ 0 to 6 } (x,\ y)^6,\\
B &= (4)^2 \text{ 6 } (x,\ y)^0,\ .,\\
C &= (3,\ 3,\ 4,\ 3,\ 3)^2 \text{ 4 to 8 } (x,\ y)^4,\\
D &= (2,\ 2,\ 3,\ 3,\ 4,\ 3,\ 3,\ 2,\ 2)^3 \text{ 2 to 10 } (x,\ y)^8,\\
E &= (8,\ 8,\ 8)^3 \text{ 8 to 10 } (x,\ y)^2,\\
F &= (7,\ 7,\ 8,\ 8,\ 8,\ 7,\ 7)^3 \text{ 6 to 12 } (x,\ y)^6,\\
G &= (5,\ 7,\ 7,\ 8,\ 8,\ 8,\ 7,\ 7,\ 5)^3 \text{ 5 to 13 } (x,\ y)^8,\\
H &= (3,\ 4,\ 5,\ 7,\ 7,\ 8,\ 8,\ 8,\ 7,\ 7,\ 5,\ 4,\ 3)^3 \text{ 3 to 15 } (x,\ y)^{12},\\
I &= (18)^4 \text{ 12 } (x,\ y)^0,\ .,\\
J &= (16,\ 16,\ 18,\ 16,\ 16)^4 \text{ 10 to 14 } (x,\ y)^4,\\
K &= (14,\ 16,\ 16,\ 18,\ 16,\ 16,\ 14)^4 \text{ 9 to 15 } (x,\ y)^6,\\
L &= (10,\ 13,\ 14,\ 16,\ 16,\ 18,\ 16,\ 16,\ 14,\ 13,\ 10)^4 \text{ 7 to 17 } (x,\ y)^{10},\\
M &= (32,\ 32,\ 32)^5 \text{ 14 to 16 } (x,\ y)^2,\\
N &= (30,\ 32,\ 32,\ 32,\ 30)^5 \text{ 13 to 17 } (x,\ y)^4,\\
O &= (25,\ 29,\ 30,\ 32,\ 32,\ 32,\ 30,\ 29,\ 25)^5 \text{ 11 to 19 } (x,\ y)^8,\\
P &= (58)^6 \text{ 18 } (x,\ y)^0,\ .,\\
Q &= (51,\ 55,\ 55,\ 58,\ 55,\ 55,\ 51)^6 \text{ 15 to 21 } (x,\ y)^6,\\
R &= (51,\ 55,\ 55,\ 58,\ 55,\ 55,\ 51)^6 \text{ 15 to 21 } (x,\ y)^6,\\
S &= (94,\ 94,\ 94)^7 \text{ 20 to 22 } (x,\ y)^2,\\
T &= (90,\ 94,\ 94,\ 94,\ 90)^7 \text{ 19 to 23 } (x,\ y)^4,\\
U &= (147,\ 151,\ 147)^8 \text{ 23 to 25 } (x,\ y)^2,\\
V &= (221,\ 227,\ 227,\ 227,\ 221)^8 \text{ 25 to 29 } (x,\ y)^4,\\
W &= (338)^{10} \text{ 30 } (x,\ y)^0,\ .,\\
X &= (332,\ 338,\ 332)^{10} \text{ 29 to 31 } (x,\ y)^2,\\
Y &= (668,\ 676,\ 668)^{12} \text{ 35 to 37 } (x,\ y)^2,\\
Z &= (1636)^{15} \text{ 45 } (x,\ y)^0,\ .
\end{aligned}
$$

Covariant $A = (x, y)^6$

x^6	x^5y	x^4y^2	x^3y^3	x^2y^4	xy^5	y^6
$a+1$	$b+6$	$c+15$	$d+20$	$e+15$	$f+6$	$g+1$

Covariant $B = (x, y)^0$**, Invt.**

$ag+1$ $bf-6$ $ce+15$ d^2-10
$\pm$ 16

Covariant $C = (x, y)^4$

x^4	x^3y	x^2y^2	xy^3	y^4
$ae+1$ $bd-4$ c^2+3	$af+2$ $be-6$ $cd+4$	$ag+1$ bf ... $ce-9$ d^2+8	$bg+2$ $cf-6$ $de+4$	$cg+1$ $df-4$ e^2+3
$\pm$ 4	$\pm$ 6	$\pm$ 9	$\pm$ 6	$\pm$ 4

Covariant $D = (x, y)^8$

x^8	x^7y	x^6y^2	x^5y^3	x^4y^4	x^3y^5	x^2y^6	xy^7	y^8
$ac+1$ b^2-1	$ad+4$ $bc-4$	$ae+6$ $bd+4$ c^2-10	$af+4$ $be+16$ $cd-20$	$ag+1$ $bf+14$ $ce+5$ d^2-20	$bg+4$ $cf+16$ $de-20$	$cg+6$ $df+4$ e^2-10	$dg+4$ $ef-4$	$eg+1$ f^2-1
$\pm$ 1	$\pm$ 4	$\pm$ 10	$\pm$ 20	$\pm$ 20	$\pm$ 20	$\pm$ 10	$\pm$ 4	$\pm$ 1

Covariant $E = (x, y)^2$

x^2	xy	y^2
$acg + 1$	$adg + 1$	$aeg + 1$
$df - 3$	$ef - 1$	$f^2 - 1$
$e^2 + 2$	$a^2bcg - 1$	$a^2bdg - 3$
$a^3b^2g - 1$	$bdf - 8$	$bef + 3$
$bcf + 3$	$be^2 + 9$	$c^2g + 2$
$bde - 1$	$c^2f + 9$	$cdf - 1$
$c^2e - 3$	$cde - 17$	$ce^2 - 3$
$cd^2 + 2$	$d^3 + 8$	$d^3 + 2$
$\pm\ 3$	$\pm\ 1$	$\pm\ 1$
$\pm\ 5$	$\pm\ 26$	$\pm\ 7$

Covariant $F = (x, y)^6$

x^6	x^5y	x^4y^2	x^3y^3	x^2y^4	xy^5	y^6
a^2g ...	abg ...	$acg+1$	$adg+2$	$aeg+1$	afg ...	ag^2 ...
abf ...	$cf+2$	$df+2$	$ef-2$	f^2-1	a^2beg+2	a^2bfg ...
$ce+1$	$de-2$	e^2-3	a^2bcg-2	a^2bdg+2	bf^2-2	$ceg+1$
d^2-1	a^2b^2f-2	a^2b^2g-1	$bdf+4$	$bef-2$	$cdg-2$	cf^2-1
a^3b^2e-1	$bce+2$	$bcf-2$	be^2-2	c^2g-3	$cef+2$	d^2g-1
$bcd+2$	bd^2+2	$bde+4$	c^2f-2	$cdf+4$	d^2f+2	$def+2$
c^3-1	c^2d-2	c^2e+2	$cde+6$	ce^2+2	de^2-2	e^3-1
		d^2-3	d^3-4	d^2e-3		
± 1	± 2	± 3	± 2	± 1	± 6	± 3
± 2	± 4	± 6	± 10	± 8		

Covariants G and H

Covariant $G = (x, y)^8$**.** *[Table spans a full page; coefficients arranged in 9 columns* x^8 *to* y^8*, with terms of degree 3 in the coefficients* a, b, c, d, e, f, g*. The table contains approximately 90 non-zero entries with coefficients ranging over* ± 1 *to* ± 140*.]*

Covariant $H = (x, y)^{12}$**.** *[Table spans a full page; coefficients arranged in 13 columns* x^{12} *to* y^{12}*, with terms of degree 3. The table contains approximately 130 non-zero entries with coefficients ranging over* ± 1 *to* ± 378*.]*

Covariant $I = (x, y)^0$, Invt.

[Invariant of degree 4; table of coefficients contains terms of degree 4 in a, b, c, d, e, f, g*.]*

Covariant $J = (x, y)^4$

[Table of 5 columns x^4 *to* y^4*, degree 4 coefficients. Contains approximately 60 non-zero entries with coefficients up to* ± 263*.]*

Covariant $K = (x, y)^6$

[Table of 7 columns x^6 *to* y^6*, degree 4 coefficients. Contains approximately 80 non-zero entries with coefficients up to* ± 215*.]*

Covariants L through R

Covariant $L = (x,y)^{10}$. *[Table of 11 columns, degree 4 coefficients, approximately 100 entries.]*

Covariant $M = (x,y)^2$.

x^2	xy	y^2
$a^2cg^2 + 1$	$a^2dg^2 - 2$	$a^2eg^2 + 1$
$dfg - 6$	$efg + 8$	$f^2g - 1$
$e^2g + 8$	$f^3 - 6$	$abdg^2 - 6$
$ef^2 - 3$	$abcg^2 + 8$	$befg + 6$
$ab^2g^2 - 1$	$bdfg - 20$	$bf^3 \quad \ldots$
$bcfg + 6$	$be^2g - 24$	$c^2g^2 + 8$
$bdeg - 34$	$bef^2 + 36$	$cdfg - 34$
$bdf^2 + 48$	$c^2fg - 24$	$ce^2g + 18$
$beef - 18$	$cdeg + 76$	$cef^2 \quad \ldots$
$c^2eg + 18$	$cdf^2 + 36$	$d^2eg + 4$
$c^2f^2 - 45$	$ce^2f - 72$	$d^2f^2 + 64$
$cd^2g + 4$	$d^3g - 32$	$de^2f - 96$
$cdef + 78$	$d^2ef - 8$	$e^4 + 36$
$ce^3 - 36$	$de^3 + 24$	$a^3b^2cg^2 - 3$
$d^3f - 48$	$a^3b^2g^2 - 6$	$b^2dfg + 48$
$d^2e^2 + 28$	$b^2cfg + 36$	$b^2e^2g - 45$
$a^3b^2fg \quad \ldots$	$b^2deg + 36$	$b^2ef^2 \quad \ldots$
$b^2ceg \quad \ldots$	$b^2df^2 \quad \ldots$	$bc^2fg - 18$
$b^2cf^2 \quad \ldots$	$b^2e^2f - 54$	$bcdeg + 78$
$b^2d^2g + 64$	$b^2ceg - 72$	$bcdf^2 - 144$
$b^2def - 144$	$b^2cf^2 - 54$	$bce^2f + 108$
$b^2e^3 + 81$	$bcd^2g - 8$	$bd^3g - 48$
$bc^2dg - 96$	$bcdef - 36$	$bd^2ef + 96$
$bc^2ef + 108$	$bce^3 + 216$	$bde^3 - 72$
$bcdf^2 + 96$	$bd^2f + 128$	$c^2eg - 36$
$bcde^2 - 126$	$bd^2e^2 - 192$	$c^2f^2 + 81$
$bd^3e + 16$	$c^3dg + 24$	$c^2d^2g + 28$
$c^3g + 36$	$c^2ef + 216$	$c^2def - 126$
$c^2df - 72$	$c^2d^2f - 192$	$c^2e^3 - 27$
$c^2e^2 - 27$	$c^2de^2 - 378$	$cd^3f + 16$
$c^2d^2e + 96$	$cd^3e + 464$	$cd^2e^2 + 96$
$cd^4 - 32$	$d^5 - 128$	$d^4e - 32$
$\pm\ 9$	$\pm\ 8$	$\pm\ 1$
182	180	$\pm\ 136$
497	1120	$\pm\ 551$
——	——	——
$\pm\ 688$	$\pm\ 1308$	$\pm\ 688$

Covariants N through Z

Covariant $N = (x, y)^4$. *[5 columns, degree 5 coefficients. Full table contains approximately 120 entries. Key values include:]*

x^4	x^3y	x^2y^2	xy^3	y^4
$a^2bg^2 \quad \ldots$	$a^2cg^2 - 1$	$a^2dg^2 \quad \ldots$	$a^2eg^2 + 1$	$a^2fg^2 \quad \ldots$
$cfg - 1$	$dfg + 4$	$efg + 3$	$f^2g - 1$	$abeg^2 + 1$
$deg + 1$	$e^2g \quad \ldots$	$f^3 - 3$	$abdg^2 - 4$	$bf^2g - 1$
$df^2 + 3$	$ef^2 - 3$	$abcg^2 - 3$	$befg + 4$	$cdg^2 - 1$
$e^2f - 3$	$ab^2g^2 + 1$	$bdfg \quad \ldots$	$bf^3 \quad \ldots$	$cefg - 2$
$ab^2fg + 1$	$bcfg - 4$	$be^2g - 15$	$c^2g^2 \quad \ldots$	$cf^2g + 3$
$bceg + 2$	$bdeg - 16$	$bef^2 + 18$	$cdfg + 16$	$d^2fg + 4$
$bcf^2 - 3$	$bdf^2 \quad \ldots$	$c^2fg + 15$	$ce^2g - 22$	$de^2g - 1$
$bd^2g - 4$	$be^2f + 18$	$cdeg \quad \ldots$	$cef^2 + 6$	$def^2 - 6$
$bdef - 12$	$c^2eg + 22$	$cdf^2 - 36$	$d^2eg + 8$	$e^2f + 3$
$be^3 + 15$	$c^2f^2 + 3$	$ce^2f + 9$	$d^2f^2 - 32$	$a^3b^2dg^2 - 3$
$c^2dy + 1$	$cd^2g - 8$	$d^3g \quad \ldots$	$de^2f + 36$	$b^2efg + 3$
$c^2ef + 9$	$cdef - 48$	$d^2ef + 24$	$e^4 - 12$	$b^2f^3 \quad \ldots$
$cd^2f + 4$	$ce^3 + 12$	$de^3 - 12$	$a^3b^2cg^2 + 3$	$bc^2g^2 + 3$
$cde^2 - 21$	$d^3f + 32$	$a^3b^2g^2 + 3$	$b^2dfg \quad \ldots$	$bcdfg + 12$
$d^3e + 8$	$d^2e^2 - 12$	$b^2cfg - 18$	$b^2e^2g - 3$	$bce^2g - 9$
$a^3b^2eg - 3$	$a^3b^2fg \quad \ldots$	$b^2deg + 36$	$b^2ef^2 \quad \ldots$	$bcef^2 - 9$
$b^2f^2 \quad \ldots$	$b^2ceg - 6$	$b^2df^2 \quad \ldots$	$bc^2fg - 18$	$bd^2eg - 4$
$b^2cdg + 6$	$b^2cf^2 \quad \ldots$	$b^2e^2f - 27$	$bcdeg + 48$	$bd^2f^2 - 32$
$b^2cef + 9$	$b^2d^2g + 32$	$b^2ceg - 9$	$bcdf^2 \quad \ldots$	$bde^2f + 66$
$b^2d^2f + 32$	$b^2def \quad \ldots$	$b^2cf^2 + 27$	$bce^2f - 18$	$be^4 - 27$
$b^2de^2 - 39$	$b^2e^3 - 27$	$bcd^2g - 24$	$bd^3g - 32$	$c^3fg - 15$
$bc^2g - 3$	$bc^2dy - 36$	$bcdef \quad \ldots$	$bd^2ef + 32$	$c^2deg + 21$
$bc^2df - 66$	$bc^2ef + 18$	$bce^3 + 27$	$bde^3 - 12$	$c^2df^2 + 39$
$bc^2e^2 + 18$	$bcd^2f - 32$	$bd^3f \quad \ldots$	$c^3eg - 12$	$c^2e^2f - 18$
$bcd^2e + 76$	$bcde^2 + 84$	$bd^2e^2 - 12$	$c^2f^2 + 27$	$cd^3g - 8$
$bd^4 - 32$	$bd^2e - 32$	$c^3dy + 12$	$c^2d^2g + 12$	$cd^2ef - 76$
$c^4f + 27$	$c^3g + 12$	$c^2ef - 27$	$c^2def - 84$	$cde^3 + 45$
$c^3de - 45$	$c^2df + 12$	$c^2d^2f + 12$	$c^2e^3 + 45$	$d^4f + 32$
$c^2d^3 + 20$	$c^2e^2 - 45$	$c^2de^2 \quad \ldots$	$cd^3f + 32$	$d^3e^2 - 20$
	$c^2d^2e + 20$	$cd^3e \quad \ldots$	$cd^2e^2 - 20$	
	$cd^4 \quad \ldots$	$d^5 \quad \ldots$	$d^4e \quad \ldots$	
$\pm\ 33$	$\pm\ 146$	$\pm\ 215$	$\pm\ 140$	$\pm\ 215$
			$+\ 146$	$+\ 33$

Covariants O to Z. *[Due to the extreme length of these tables, the remaining covariants are summarized. Each covariant O through Z is represented by a full-page table of coefficients. The degrees, orders, and coefficient structures are as given in the list above.]*

776.

On the Jacobian Sextic Equation.

[The preceding paper 775 on "Tables of Covariants of the Binary Sextic" continues with the detailed coefficients. Paper 776 begins on page 391 of the original (PDF page 415). The following is a transcription of the content.]

The Jacobian Sextic Equation and its Invariants

The Jacobian sextic equation is taken in the form

$$(a,\, b,\, 0,\, d,\, 0,\, f,\, g \,|\, z,\, 1)^6 = 0,$$

with the condition

$$ag + 9bf - 20d^2 = 0.$$

The invariants A, B, C, D of the sextic, for this special form, are functions of

$$\alpha = ag, \quad \beta = bf, \quad \delta = d^2, \quad \zeta = a^2f^2 + bg^2,$$

and using the relation $\alpha + 9\beta - 20\delta = 0$ to eliminate α, the invariants become functions of β, δ, ζ.

Effecting the substitution, the values of A, B, C are found without difficulty. As regards the value of A, this is

$$A = -3125\,\zeta^2 + 2K\sqrt{\zeta} + \text{terms without } \zeta,$$

where

$$\begin{aligned} 2K = &-2500(\beta^4 - 36\beta\delta + 56\delta^2) \\ &- 7500(-9\delta + 2\beta\delta) \\ &+ 50000(-9\beta\delta + 20\delta^2) \\ &- 240000(\beta\delta^2) \\ &+ 1200000(\beta\delta^3) \\ &- 1600000\beta\delta^4, \end{aligned}$$

or, reducing and dividing by 2,

$$K = -3125(60\delta^2 - 240\beta\delta + 256\delta).$$

The calculation of the terms without ζ is much more laborious, but they come out

$$= -3125(60\delta^2 - 240\beta\delta + 256\delta)\delta.$$

Hence the value of A is

$$A = -3125\, h,$$

where

$$h = \zeta^2 + (60\delta^2 - 240\beta\delta + 256\delta)\sqrt{\zeta},$$

that is,

$$h = a^2f^2 + bg^2 + (60d^4 - 240bd^2f + 256d^3)\sqrt{a^2f^2 + bg^2}.$$

The Group of the Jacobian Sextic

The solution of the Jacobian sextic equation depends upon that of a quintic; in fact, calling the roots z_1, z_2, z_3, z_4, z_5, z_6, then there exists a quintic having the roots

$$\begin{aligned}
M_1 &= (z_1 - z_4)(z_2 - z_5)(z_3 - z_6),\\
M_2 &= (z_1 - z_5)(z_2 - z_6)(z_3 - z_4),\\
M_3 &= (z_1 - z_6)(z_2 - z_4)(z_3 - z_5),\\
M_4 &= (z_1 - z_3)(z_2 - z_4)(z_3 - z_6),\\
M_5 &= (z_1 - z_2)(z_2 - z_3)(z_3 - z_4),
\end{aligned}$$

the coefficients of which are rational functions of the coefficients a, b, d, f, g, and of the fourth root of the discriminant, i.e., $\sqrt{A}$. But the meaning of this has not, so far as I am aware, been noticed. Passing to the quintic whose roots are the squares of the foregoing values, i.e., $(z_1 - z_2)(z_3 - z_4)$, etc., the coefficients are here rational functions of a, b, d, f, g and h; that is, they are rational functions of a, b, d, f, g. The symmetrical functions of these roots are thus rational functions of the coefficients of the sextic; each such rational function is a 12-valued function of z_1, z_2, z_3, z_4, z_5, z_6, invariable by all the substitutions of a group of 60 substitutions; and therefore also every like 12-valued function of the roots is invariable by the substitutions of this group of 60; or, in other words, this group of 60 is the group of the Jacobian sextic equation.

The Group of 60 Substitutions

Writing for convenience, in this section only,

$$S = (z_1, z_2, z_3, z_4, z_5, z_6) = (a, b, c, d, e, f),$$

and writing further ab for shortness instead of $a - b$, etc., (so that of course $ba = -ab$), and putting

$$\begin{aligned} B &= -ab \cdot cd \cdot ef, \\ C &= -ac \cdot be \cdot df, \\ D &= \quad ad \cdot be \cdot cf, \\ E &= \quad ae \cdot bd \cdot cf, \\ F &= \quad af \cdot be \cdot cd, \end{aligned}$$

then the five functions are B, C, D, E, F, and the group of 60 which leaves unaltered every symmetrical function of these functions is made up of the substitutions:

1. 1

15 $ab \cdot cd \cdot ef$, $ab \cdot df \cdot ce$, $ac \cdot bf \cdot de$,
$ad \cdot bc \cdot ef$, $ad \cdot be \cdot cf$, $ae \cdot bd \cdot cf$, $ae \cdot bf \cdot cd$,
$af \cdot be \cdot cd$, $af \cdot ce \cdot bd$

10 $abcde$, $acebd$, $adbec$, $aebcd$, $abced$,
$afbce$, $abefc$, $acfeb$, $adcfb$, $aefbc$,
$abdef$, $adefc$, $acdbf$, $aefbd$, $afced$,
$acdfe$, $aefcd$, $adefb$, $abfed$, $acbdf$

20 $abc \cdot dfe$, $acb \cdot def$, $abd \cdot cfe$, $adb \cdot cef$,
$abe \cdot cfd$, $aeb \cdot cdf$, $abf \cdot ced$, $afb \cdot cde$,
$acd \cdot bef$, $adc \cdot bfe$, $ace \cdot bdf$, $aec \cdot bfd$,
$acf \cdot bed$, $afc \cdot bde$, $ade \cdot bfc$, $aed \cdot bcf$,
$adf \cdot bce$, $afd \cdot bec$, $aef \cdot bcd$, $afe \cdot bdc$

24 (circular substitutions on six elements)

The 120 Substitutions

We may to the foregoing 60 substitutions join the 60 other substitutions:

30 $abcdef$, $afedcb$, $bdfeca$, $bedafc$, $baefcd$, $bfcead$,
$bcdfae$, $badcfe$, $bafced$, $bacdef$, $acdfbe$, $afbedc$,
$acefbd$, $afecdb$, $acfbead$, $adfbce$, $adcebf$, $afbedc$,
$adbcfe$, $aebcdf$, $abdecd$, $abefcd$, $afbcd$, $abcd \cdot ef$,

$ab \cdot cd \cdot ef$, etc.

10 $ac \cdot bd \cdot ef$, $ac \cdot be \cdot df$, $ad \cdot be \cdot cf$, $ad \cdot bf \cdot ce$,
$ae \cdot bc \cdot df$, $ae \cdot bf \cdot cd$, $af \cdot bc \cdot de$, $af \cdot bd \cdot ce$

20 $abcefd$, $adfecb$, $abfce$, $acedfb$, $abecdf$, $adcefb$,
$abdfec$, $acefdb$, $acfdbe$, $aedbfe$, $acbfed$, $adefbc$,
$acdebf$, $afbede$, $adbcfe$, $aafebd$, $adcbef$, $cafebd$,
$acbdfe$, $afedbe$

The 60 and 60 substitutions form together a group of 120 substitutions, which leave unaltered any even symmetrical function of B, C, D, E, F, or say any symmetrical function of B^2, C^2, D^2, E^2, F^2; such a function is thus a 6-valued function of a, b, c, d, e, f; viz. it is Serret's 6-valued function of 6 letters.

Transformation of the Jacobian Sextic into the Resolvent Sextic

Starting from the Jacobian Sextic Equation

$$(a,\, b,\, 0,\, d,\, 0,\, f,\, g \,|\, z,\, 1)^6 = 0,$$

$ag + 9bf - 20d^2 = 0$, I effect upon it the Tschirnhausen transformation

$$X = -az^2 - 6bz - 10d;$$

which, it may be remarked, is a particular case of the Tschirnhausen-Hermite form

$$X(az + b)B + (az^2 + 6bz + 5c)C + (az^3 + 6bz^2 + 15cz + 10d)D$$
$$+(az^4+6bz^3+15cz^2+20dz+10e)E+(az^5+6bz^4+15cz^3+20dz^2+15ez+5f)F.$$

Writing for convenience $Y = X + 10d$, $Z = X - 10d$, this is

$$\begin{aligned} az^5 + 6bz^4 \cdot \ \cdot + Y &= 0,\\ az^4 + 6bz^3 \cdot \ \cdot + Yz \cdot \ \cdot &= 0,\\ az^3 + 6bz^2 \cdot \ \cdot + Yz^2 \cdot \ \cdot &= 0,\\ \cdot \ \cdot -Zz^3 \cdot \ \cdot - 6fz - g &= 0,\\ \cdot \ \cdot -Zz^2 \cdot \ \cdot - 6fz^2 - gz \cdot \ \cdot &= 0,\\ -Zz \cdot \ \cdot - 6fz^3 - gz^2 \cdot \ \cdot &= 0, \end{aligned}$$

or, eliminating, the resulting equation is

$$\begin{vmatrix} \cdot & \cdot & a & 6b & \cdot & Y \\ \cdot & a & 6b & \cdot & Y & \cdot \\ a & 6b & \cdot & Y & \cdot & \cdot \\ \cdot & \cdot & Z & \cdot & 6f & g \\ \cdot & Z & \cdot & 6f & g & \cdot \\ Z & \cdot & 6f & g & \cdot & \cdot \end{vmatrix} = 0.$$

The developed form is most easily obtained by expanding the determinant in the form

$$123 \cdot 456 - 456 \cdot 123, \text{ etc.},$$

where the terms 123, etc., belong to the matrix

$$\begin{vmatrix} \cdot & \cdot & a & 6b & \cdot & Y \\ \cdot & a & 6b & \cdot & Y & \cdot \\ a & 6b & \cdot & Y & \cdot & \cdot \end{vmatrix},$$

and those of 456, etc., to the matrix

$$\begin{vmatrix} \cdot & \cdot & Z & \cdot & 6f & g \\ \cdot & Z & \cdot & 6f & g & \cdot \\ Z & \cdot & 6f & g & \cdot & \cdot \end{vmatrix}.$$

The Resulting Equation

Hence, collecting and reducing, the equation is

$$\begin{aligned} 0 = X^6 & \\ &+ X^4Y^2 \cdot (8ag + 72bf) \\ &+ X^3YZ \cdot (8a^2gh + 86aghf + 12960b^2f^2) \\ &+ X^2Z^2 \cdot 26a^3g^4 \\ &+ X \cdot 2160a^4g^4f \\ &+ a^5g^4 - 80a^4g^3f^2, \end{aligned}$$

where Y, Z denote $X + 10d$, $X - 10d$ respectively, and consequently $YZ = X^2 - 100d^2$.

Hence, writing as before α, β, δ, ζ to denote ag, bf, d^2 and $a^2f^2 + bg^2$ respectively, the result finally is

$$(1,\ 0,\ \alpha + 3\beta,\ 0,\ \alpha\beta + 3\delta,\ 216\gamma\delta,\ \alpha\gamma\beta + 4\delta^2 \,|\, X,\ 1)^6 = 0,$$

where observe that the coefficient of the term in X is $216(a^2f^2 - b^2g^2) = 216\sqrt{\zeta}(\zeta^2 - 4\alpha\beta)$.

We have as before $ag + 9bf - 20d^2 = 0$, that is, $\alpha + 9\beta - 20\delta = 0$; and using this equation to eliminate α, also in the constant term writing its value for ζ in terms of h,

$$\zeta = h + (-60\delta^2 + 240\beta\delta - 256\delta)\sqrt{\delta},$$

the new equation is

$$(1,\ 0,\ \beta - 9\delta,\ 0,\ \beta\delta - 4\delta^2,\ 216\gamma\delta,\ h\gamma\beta + 4\delta^2 \,|\, X,\ 1)^6 = 0,$$

where

$$\begin{aligned} h &= \{h + (-60\delta^2 + 240\beta\delta - 256\delta)\sqrt{\delta}\}^2 - 4(-9\delta + 20\delta)\delta^2 \\ &= h^2 + 2h\sqrt{\delta}(-60\delta^2 + 240\beta\delta - 256\delta) - 4(\delta - 4\delta)(9\delta - 16\delta)\delta^2. \end{aligned}$$

Connection with the Quintic Equation

It is to be shown that this Tschirnhausen-transformation of the Jacobian sextic is, in fact, the resolvent sextic of the quintic equation

$$(1,\ 0,\ c,\ 0,\ e,\ f \,|\, x,\ 1)^5 = 0,$$

where

$$c = 2d, \quad e = -9bf + 36d^2, \quad f = \sqrt{(216h)},$$

and, Δ being the discriminant of the Jacobian sextic, then

$$h = \frac{1}{\sqrt{\Delta}}, \quad \zeta = a^2f^2 + bg^2 + 60bd^2f - 240bd^2f + 256d^3.$$

I consider the general quintic $(a,\ b,\ c,\ d,\ e,\ f \,|\, x,\ 1)^5 = 0$; taking the roots to be θ_1, θ_2, θ_3, θ_4, θ_5 and writing

$$\begin{aligned} \phi_1 &= 12345 - 24135, \\ \phi_2 &= 13425 - 32145, \\ \phi_3 &= 14235 - 43125, \\ \phi_4 &= 21435 - 13245, \\ \phi_5 &= 31245 - 14325, \\ \phi_6 &= 41325 - 12435, \end{aligned}$$

where 12345 is used to denote the function

$$= (\theta_1 + \omega\theta_2 + \omega^2\theta_3 + \omega^3\theta_4 + \omega^4\theta_5)\surd(20),$$

(this numerical factor $\surd(20)$ being inserted for convenience), the several terms are calculated and the verification is completed by showing the identity of the two forms.

Conclusion

The conclusion is that, starting from the Jacobian sextic

$$(a,\, b,\, 0,\, d,\, 0,\, f,\, g \,|\, z,\, 1)^6 = 0,$$

where $ag + 9bf - 20d^2 = 0$, and effecting upon it the Tschirnhausen-transformation

$$X = -az^2 - 6bz - 10d,$$

so as to obtain from it a sextic equation in X, this sextic equation in X is the resolvent sextic of the quintic equation

$$(1,\, 0,\, c,\, 0,\, e,\, f \,|\, x,\, 1)^5 = 0,$$

where

$$c = 2d, \quad e = -9bf + 36d^2, \quad f = \surd(216h),$$

and, Δ being the discriminant of the Jacobian sextic, then

$$h = \frac{1}{\sqrt{\Delta}}, \quad \zeta = a^2f^2 + bg^2 + 60bd^2f - 240bd^2f + 256d^3.$$

As to the subject of the present paper, see in particular Brioschi, "Ueber die Auflösung der Gleichungen vom fünften Grade," *Math. Annalen*, t. XIII. (1878), pp. 109–160, and the third Appendix to his translation of my Elliptic Functions, Milan, 1880, each containing references to the earlier papers.

777.

A Solvable Case of the Quintic Equation.

[From the *Quarterly Journal of Pure and Applied Mathematics*, vol. XVIII. (1882), pp. 154–157.]

The roots of the general quintic equation

$$(a,\, b,\, c,\, d,\, e,\, f \,|\, x,\, 1)^5 = 0$$

may be taken to be

$$\begin{aligned}
&\alpha + \beta + \gamma + \delta + \epsilon,\\
&\alpha + \omega\beta + \omega^2\gamma + \omega^3\delta + \omega^4\epsilon,\\
&\alpha + \omega^2\beta + \omega^4\gamma + \omega\delta + \omega^3\epsilon,\\
&\alpha + \omega^3\beta + \omega\gamma + \omega^4\delta + \omega^2\epsilon,\\
&\alpha + \omega^4\beta + \omega^3\gamma + \omega^2\delta + \omega\epsilon,
\end{aligned}$$

where ω is an imaginary fifth root of unity; and if one of the four functions B, C, D, E is $= 0$, say if $E = 0$ (this implies of course a single relation between the coefficients), then the equation is solvable.

Writing $\alpha = 0$, we have

$$(0,\, b,\, c,\, d,\, e,\, f \,|\, x,\, 1)^5 = 0,$$

where

$$\begin{aligned}
a^4 d' &= a^4 d - 3a^2bc + 2b^3,\\
a^3 e' &= a^3 e - 4abd + 6ac^2 - 3b^2c,\\
a^2 f' &= a^2 f - 5abe + 10acd - 10b^2d + 5bc^2,
\end{aligned}$$

and the roots of the new equation

$$(0,\, 0,\, c',\, d',\, e',\, f' \,|\, x,\, 1)^5 = 0$$

have the above-mentioned values, omitting therefrom the terms -2α; we find without difficulty:

$$\begin{aligned}
2\beta^5 &= -BE - cD,\\
2\gamma^5 &= -BD - B^2 + CE - D^2,\\
2\delta^5 &= -BC - B^2 + BCD + BD^2 + C^2E + CD\,DE^2,\\
2\epsilon^5 &= -B^5 + 5B^3DE - 5B^2CE - 5BD^2 + 5BC^2D + 5BDE^2 - C^3 + 5C^2D - 5CI
\end{aligned}$$

and hence, when $E = 0$, we have

$$\begin{aligned}
2\beta^5 &= -cD,\\
2\gamma^5 &= -BD - B^2,\\
2\delta^5 &= -BC - B^2 - CD,\\
2\epsilon^5 &= -B^5 - 5BCD + 5BC^2D - CD^3,
\end{aligned}$$

or, as these may be written,

$$\begin{aligned}
2\beta^5 &= cD,\\
-2\gamma^5 &= B(D + B),\\
-\delta^5 &= B(C + D) + CD,\\
-\epsilon^5 &= B^5 + 5CD\,B(C + D),
\end{aligned}$$

equations which imply a single relation between the coefficients a', c, d, e, f.

Supposing this satisfied, we may attend only to the first three equations; or, writing for convenience,

$$\begin{aligned}
\gamma &= -2B, & \delta &= 2(C - D),\\
\frac{c}{a^2} &= -2f, & \frac{d}{a^2} &= -3(\omega d - 3abc + 2b^2),\\
\frac{e}{a^4} &= -24, & \frac{f}{a^5} &= -(\omega ae - 4bd + 3c^2) + (ac - b^2),
\end{aligned}$$

the equations are

$$\begin{aligned}
\gamma &= cD,\\
\delta &= B(BD + C),\\
0 &= B(C + D - BC),
\end{aligned}$$

the first equation gives $C = \gamma$, and substituting this value in the other two equations, we have

$$\begin{aligned}
BD + B\gamma - \delta D^2 &= 0,\\
B\gamma + BD + \delta D &= 0.
\end{aligned}$$

Eliminating B, the result is obtained in the form $\det = 0$, where in the last column of the determinant, each term is divisible by D;

and omitting this factor, the result is

$$\begin{vmatrix} D & \cdot & \gamma & -\delta D \\ D & \gamma & -\delta D & \cdot \\ D & \gamma & \cdot & -\delta D \\ 1 & \gamma & \cdot & -D \\ 1 & \gamma & \cdot & \cdot \\ 1 & \gamma & \cdot & \cdot \end{vmatrix} = 0.$$

If, in order to develope the determinant, we consider it as a sum of products, each first factor being a minor composed out of columns 1 and 2, and the second factor being the complementary minor composed out of columns 3, 4, 5 (the several products being of course taken each with its proper sign), the expansion presents itself in the form

$$\begin{aligned} &D\gamma\,(-\gamma^2 D + \delta D^3) \\ &\qquad - D^2(-\gamma^2 D^2 + \delta D^4 - \delta D^2) \\ &\qquad - \gamma D^2(-\delta D^3 - \delta\gamma^2 D) \\ &\qquad + \gamma^2(\delta D^4 - \delta\gamma^2 D^2 - \delta\gamma) \\ &\qquad\qquad = \gamma^2\delta D. \end{aligned}$$

Hence, collecting, and changing the sign of the whole expression, we obtain

$$BD^5 - (2\gamma + 7\delta + 6\gamma^2)D^4 + (-7\gamma\delta + 3\gamma\delta^2 + \delta)D^3 + \gamma\delta^2 = 0,$$

a cubic equation for D. We have then as above $C = \gamma$, and B is given rationally as the common root of the foregoing quadratic and cubic equations satisfied by B.

Substituting for γ, δ, θ their values in terms of the original coefficients, the equation for D^2 becomes

$$\begin{aligned} &2(a^3d - 3abc + 2b^3)(aD)^5 \\ &\qquad + (a^4e - 4abd + 3c^2) \\ &\qquad + \{a^5(ac - b^2)(ae - 4bd + 3c^2)\}(aD)^3 \\ &\qquad + 16(ac - b^2)(a^3d - 3abc + 2b^3) \\ &+ 4(a^5c - b^2) + 120b^2(ac - b^2)(a^3d - 3abc + 2b^3)(ae - 4bd + 3c^2)(aD)^2 \\ &\qquad + 8(a^3d - 3abc + 2b^3)^2 \\ &\qquad\qquad = 128(ac - b^2)^4(ae - 4bd + 3c^2)^2. \end{aligned}$$

778.

[ADDITION TO MR HUDSON'S PAPER "ON EQUAL ROOTS OF EQUATIONS."]

[From the *Quarterly Journal of Pure and Applied Mathematics*, vol. XVIII. (1882), pp. 226–229.]

IT seems desirable to present in a more developed form some of the results of the foregoing paper.

Thus, if the equation $(a_0, a_1, \ldots, a_n \mid x, 1)^n = 0$ of the order n has $n-\nu$ equal roots, where ν is not $> \frac{1}{4}(n-1)$, then we have $\psi(\nu, \nu+1, m) = 0$, where m has any one of the values $0, 1, \ldots, n-2\nu-2$, and r any one of the values $2\nu+2, 2\nu+3, \ldots, n-m$.

The signification is

$$\begin{aligned}\psi(\nu+1, m) = {} & \frac{1}{\nu\cdot\nu+1\cdot\nu+2}a_m a_{r+m} \\ & - (\nu-2)\cdot 2\cdot\frac{1}{\nu-1\cdot\nu\cdot\nu+1}a_{m+1}a_{r+m-1} \\ & + (\nu-4)\cdot 1\cdot\frac{1}{\nu-2\cdot\nu-1\cdot\nu}a_{m+2}a_{r+m-2} - \cdots = 0.\end{aligned}$$

Thus, when $\nu = 0$, the condition is

$$\begin{vmatrix} a_m & a_{r+m} \\ a_{m+1} & a_{r+m-1}\end{vmatrix} = 0,$$

that is,

$$a_m a_{r+m} - a_{m+1}a_{r+m-1} = 0,$$

satisfied when the equation has all its roots equal.

The values of m are $0, 1, 2, \ldots, n-2$, and those of r are $2\nu+2$, $2\nu+3, \ldots, n-m$; in particular, if $m = 0$, the values of r are 2, 3, $\ldots, n$, and the corresponding conditions are

$$\begin{gathered} a_0a_2 - a_1^2 = 0, \\ a_0a_3 - a_1a_2 = 0, \\ \vdots \\ a_0a_n - a_1a_{n-1} = 0,\end{gathered}$$

and so for the different values of m up to the final value $n-2$, for which $r=2$, and the condition is

$$a_{n-2}a_n - a_{n-1}^2 = 0;$$

we have thus, it is clear, the whole series of conditions included in

$$\begin{vmatrix} a_0 & a_1 & a_2 & \dots & a_{n-2} & a_{n-1} \\ a_1 & a_2 & a_3 & \dots & a_{n-1} & a_n \end{vmatrix} = 0,$$

which are obviously satisfied in the case in question of the roots being all equal.

Again, when $\nu = 1$, the condition for $n-1$ equal roots is

$$\left.\begin{array}{r} r\cdot 1\cdot\dfrac{1}{r\cdot r-1\cdot r-2}a_m a_{r+m} \\ -(r-2)\cdot 2\cdot\dfrac{1}{r-1\cdot r-2\cdot r-3}a_{m+1}a_{r+m-1} \\ +(r-4)\cdot 1\cdot\dfrac{1}{r-2\cdot r-3\cdot r-4}a_{m+2}a_{r+m-2} \end{array}\right\} = 0,$$

that is,

$$\frac{a_m a_{r+m}}{r-1\cdot r-2} - \frac{2a_{m+1}a_{r+m-1}}{r-1\cdot r-3} + \frac{a_{m+2}a_{r+m-2}}{r-2\cdot r-3} = 0;$$

or, what is the same thing,

$$(r-3)\,a_m a_{r+m} - 2(r-2)\,a_{m+1}a_{r+m-1} + (r-1)\,a_{m+2}a_{r+m-2} = 0,$$

where $n = 4$ at least, and m, r have the values

$$\begin{array}{c|ccccc} m & 0 & 1 & 2 & \dots & n-4 \\ \hline r & 4 & 4 & & & 4 \\ & 5 & 5 & & & \\ & \vdots & \vdots & & & \\ & n & n-1 & & & \end{array}$$

thus, when $n=4$, the only values are $m=0$, $r=4$, and the condition is

$$a_0a_4 - 4a_1a_3 + 3a_2^2 = 0.$$

Similarly, when $\nu = 2$, the condition for $n-2$ equal roots is found to be

$$\frac{a_m a_{r+m}}{\nu\cdot\nu+1\cdot\nu+2} - \frac{3a_{m+1}a_{r+m-1}}{\nu+1\cdot\nu+2\cdot\nu+3} + \frac{3a_{m+2}a_{r+m-2}}{\nu+2\cdot\nu+3\cdot\nu+4} - \frac{a_{m+3}a_{r+m-3}}{\nu+3\cdot\nu+4\cdot\nu+5} = 0,$$

or, what is the same thing,

$$\begin{aligned}&(r-4)(r-5)\,a_m a_{r+m} - 3(r-3)(r-5)\,a_{m+1}a_{r+m-1}\\ &+3(r-2)(r-4)\,a_{m+2}a_{r+m-2} - (r-1)(r-3)\,a_{m+3}a_{r+m-3} = 0,\end{aligned}$$

where $n = 6$ at least, and m, r have the values

m	0	1	$\ldots$	$n-6$
r	6	6		6
	7	7		
	$\vdots$	$\vdots$		
	n	$n-1$		

Observe that the sum of the coefficients is $= 0$, viz.

$$(r-4)(r-5) - 3(r-3)(r-5) + 3(r-2)(r-4) - (r-1)(r-3) = 0;$$

this should obviously be the case, since the conditions for $n-2$ equal roots must be satisfied when the roots are all of them equal; and the property serves as a verification.

It is to be remarked that the equation $\psi(\nu,\, \nu+1,\, m) = 0$ does not in all cases give all the conditions for the existence of $n-\nu$ equal roots in an equation of the order n; thus when $n = 3$ and $\nu = 1$, we cannot by means of it obtain the condition that a cubic equation may have 2 equal roots. The problem really considered is that of the determination of those quadric functions of the coefficients which vanish in the case of $n-\nu$ equal roots; and in the case in question ($n = 3$, $\nu = 1$) there is no quadric function which vanishes, but the condition depends on a cubic function.

The question of the quadric functions which vanish in the case of $n-\nu$ equal roots, and to a small extent that of the cubic functions which thus vanish, is considered in Dr Salmon's "Note on the conditions that an equation may have equal roots," *Camb. and Dub. Math. Jour.*, vol. V. (1850), pp. 87–96.

779.

[NOTE ON MR JEFFERY'S PAPER "ON CERTAIN QUARTIC CURVES WHICH HAVE A CUSP AT INFINITY, WHEREAT THE LINE AT INFINITY IS A TANGENT."]

[From the *Proceedings of the London Mathematical Society*, vol. XIV. (1883), p. 85.]

THE assumed form $xa/y = z$, or, as this is afterwards written,

$$(a,\, b,\, c,\, f,\, g,\, h \,|\, x,\, y)^2 = 0,$$

is, I think, introduced without a proper explanation. Say, the form is $xy^2 = z^2(a,\, b,\, c,\, f,\, g,\, h \,|\, x,\, y,\, z)$, it ought to be shown how for a cuspidal quartic we arrive at this form; viz. taking the cusp to be at the point $(x = 0,\, y = 0)$, $z = 0$ for the tangent at the cusp, and $x = 0$ an arbitrary line through the cusp; then the line $z = 0$ besides intersects the curve in a single point, and, if $y = 0$ is taken as the tangent at that point, the equation of the curve must, it can be seen, be of the form

$$(a + \theta x + \phi z)y = z^2(a,\, b,\, c,\, f,\, g,\, h \,|\, x,\, y,\, z).$$

The conic $(a,\, b,\, c,\, f,\, g,\, h \mid x,\, y,\, z)^2 = 0$ touches the quartic at each of the two intersections of the quartic with the arbitrary line $z = 0$; and we cannot, so long as the line remains arbitrary, find a conic which shall osculate the quartic at the two points in question; but, for the particular line $a+\theta x+\phi z$, there exists such a conic, viz. writing θ instead of $a+\theta x+\phi z$, the form is $a'y = z^2(a',\, b',\, c',\, f',\, g',\, h' \,|\, x,\, y,\, z)$; and the new conic $(a',\, b',\, c',\, f',\, g',\, h' \,|\, x,\, y,\, z)^2 = 0$ has the property in question. This is the adopted form, and it thus appears that in it the line $z = 0$ is a determinate line, viz. the line passing through the cusp and the single intersection of the tangent at the cusp with the curve.

780.

[ADDITION TO MR HAMMOND'S PAPER "NOTE ON AN EXCEPTIONAL CASE IN WHICH THE FUNDAMENTAL POSTULATE OF PROFESSOR SYLVESTER'S THEORY OF TAMISAGE FAILS."]

[From the *Proceedings of the London Mathematical Society*, vol. XIV. (1883), pp. 88–91.]

THE extreme importance of Mr Hammond's result, as regards the entire subject of Covariants, leads me to reproduce his investigation in the notation of my Memoirs on Quantics, and with a somewhat different arrangement of the formulæ. For the binary seventhic

$$(a,\ b,\ c,\ d,\ e,\ f,\ g,\ h \,|\, x,\ y)^7,$$

the four composite seminvariants of the deg-weight 5.11 (sources of covariants of the deg-order 5.13) are:

$$\textbf{I.}\quad a^2(4bd-3c^2)(6bf-15ce+10d^2)$$

$$\textbf{II.}\quad a^2(4be-3cd)(6bg-15cf+10de)$$

$$\textbf{III.}\quad a^2(4bd-3c^2)(6bg-15cf+10de)$$

$$\textbf{IV.}\quad a^2(4be-3cd)(6bf-15ce+10d^2)$$

[The paper continues with detailed calculations showing that there exists a syzygy of the form I = III - IV, and working out the values of the four products, joining to them the expression for the irreducible seminvariant of the same deg-weight 5.11.]

Working out the values of the four products, and joining to them the expression for the irreducible seminvariant of the same deg-weight 5.11 (0, 2^4 of my tables [774] for the binary sextic), we have the table:

	I	II	III	IV	0, 2^4
ach	+1		+1		+1
bdg		+1		+1	+1
bf	+3			+3	+2
cg	-2		+2		-2
a^2z	-1		-1		-1
ade	+3			+3	+3
c^2e	-9	-9			-9
cd^2	+6	+6			+6

[The complete table contains approximately 30 rows of monomial terms with their numerical coefficients in each of the five columns. The syzygy I = III - IV is verified by direct comparison of the coefficients.]

781.

On the Automorphic Transformation of the Binary Cubic Function.

[From the *Proceedings of the London Mathematical Society*, vol. XIV. (1883), pp. 103–108. Read Jan. 11, 1883.]

I consider the cubic equation $(a, b, c, d \,|\, x, 1)^3 = 0$. It is shown (Serret, *Cours d'Algèbre supérieure*, 4th ed., Paris, 1879, t. II. pp. 466–471) how, given one root of the equation, the other two roots can be each of them expressed rationally in terms of this root and of the square root of the discriminant; viz. making the proper changes of notation, and writing

$$A,\ B,\ C = ac - b^2,\ ad - bc,\ bd - c^2, \quad \Omega = \nabla,$$
$$D = B^2 - 4AC, \quad \Theta = a^2d^2 + 4ac^3 + 4b^3d - 3b^2c^2 - 6abcd,$$
$$e = \frac{A}{\sqrt{-A\Omega + B}}, \quad f = \frac{-2C}{B\sqrt{-A\Omega}}, \quad g = \frac{-2A}{B\sqrt{-A\Omega}}, \quad h = \frac{-\sqrt{-A\Omega} - B}{2\sqrt{-A\Omega}},$$

(values which give $e + h = -1$, $eh - fg = -1$, and therefore also $e^2 + h^2 + eh + fg = 0$, which is the condition in order that the function ϕx, "$\phi^3 x = x$," may be periodic of the third order), then, u being a root of the equation, say $(a, b, c, d \,|\, u, 1)^3 = 0$, the other two roots are

$$u = \frac{(eu + f)}{(gu + h)}, \quad \nu_1 = \frac{(\alpha u + \beta)}{(\gamma u + \delta)},$$

where observe that, by the change of $\sqrt{\Omega}$ into $-\sqrt{\Omega}$, e, f, g, h become h, $-g$, $-f$, e, so that the last-mentioned value ν_1 is, in fact, the value obtained from ν by the mere change of $\sqrt{\Omega}$.

The General Theory

It is to be observed that, if we have between two roots u, ν of the equation $(a, b, c, d \,|\, x, 1)^3 = 0$, a relation $\nu = \frac{\alpha u + \beta}{\gamma u + \delta}$, where α, β, γ, δ have given values, this implies in the first place a relation between a, b, c, d (and the given values of α, β, γ, δ), and it implies moreover that u, and consequently also ν, are not any roots whatever, but two determinate roots of the equation; viz. ν will be each of them expressible rationally in terms of a, b, c, d and α, β, γ, δ. And if, in order that (a, b, c, d) may remain arbitrary, we consider α, β, γ, δ as given quantities satisfying the relation which exists between these quantities and (a, b, c, d), then in general we still have u, ν determinate roots of the cubic equation. But in the foregoing solution u is any root whatever of the cubic equation.

To examine how this is, starting from the equations

$$(a,\, b,\, c,\, d\,|\,u,\, 1)^3 = 0, \quad (a,\, b,\, c,\, d\,|\,\nu,\, 1)^3 = 0, \quad \nu = \frac{\alpha u + \beta}{\gamma u + \delta},$$

we have

$$a(u^3 - \nu^3) + 3b(u^2 - \nu^2) + 3c(u - \nu) = 0,$$

and therefore

$$a(u^2 + u\nu + \nu^2) + 3b(u + \nu) + 3c = 0,$$

that is,

$$c\nu^2 + (au + 3b)\nu + au^2 + 3bu + 3c = 0;$$

or, writing herein for ν its value,

$$c(\alpha u+\beta)^2+(au+3b)(\alpha u+\beta)(\gamma u+\delta)+(au^2+3bu+3c)(\gamma u+\delta)^2 = 0;$$

that is,

$$u\{a\alpha(\alpha-1)-3b\alpha^2-3c\alpha\gamma-d\gamma^2\}+\{a\beta^2+3b\beta\delta+3c\delta^2-d\gamma(\delta-1)\} = 0.$$

The Linear Equation and Its Conditions

We may without loss of generality assume $\alpha\delta - \beta\gamma = 1$; and, this being so, if we then further assume $\alpha + \delta = -1$, then we have

$$e^2 + h^2 + eh + fg = 0,$$

which is, as above appearing, the condition for $\phi^3 x = x$. The foregoing equation in u thus becomes a linear equation giving (in a simplified form) the like results to those given by the quadric equation; viz. substituting in the cubic equation the value of u given by the linear equation, we have a relation between a, b, c, d, α, β, γ, δ; and, this relation being satisfied, u has the determinate value given by the linear equation.

The only way in which u can cease to have this determinate value, and so be capable of being any root whatever of the cubic equation, is when the linear equation becomes $0 = 0$; viz. if

$$a\alpha(\alpha-1) - 3b\alpha^2 - 3c\alpha\gamma - d\gamma^2 = 0,$$
$$a\beta^2 + 3b\beta\delta + 3c\delta^2 - d\gamma(\delta-1) = 0,$$

equations which are, in fact, satisfied by the foregoing values of α, β, γ, δ, as may be verified without difficulty.

The Automorphic Transformation

It is to be remarked that if, instead of the root u and the equation $\nu = \frac{\alpha u+\beta}{\gamma u+\delta}$, we consider the root ν and the equation $u = \frac{\delta\nu-\beta}{-\gamma\nu+\alpha}$, then, instead of α, β, γ, δ we have δ, $-\beta$, $-\gamma$, α.

The equations to be satisfied by α, β, γ, δ, Θ then are

$$\begin{aligned}
a\alpha^2 + 3b\alpha^2\gamma + 3c\alpha\gamma^2 + d\gamma^3 &= -\alpha\beta,\\
a\alpha^2\beta + b(\alpha^2\delta + 2\alpha\beta\gamma) + c(2\alpha\gamma\delta + \beta\gamma^2) + d\gamma^2\delta &= -\beta\delta,\\
a\alpha\beta^2 + b(2\alpha\beta\delta + \beta^2\gamma) + c(\alpha\delta^2 + 2\beta\gamma\delta) + d\gamma\delta^2 &= -\alpha\delta,\\
a\beta^3 + 3b\beta^2\delta + 3c\beta\delta^2 + d\delta^3 &= -\beta\alpha.
\end{aligned}$$

Writing $\alpha\delta - \beta\gamma = V$, and as before Ω for the discriminant, the theory of invariants gives $\frac{1}{4}V^2 = \Theta$. We are considering the case of the general cubic function $(a,\, b,\, c,\, d\,|\,x,\, y)^3$, for which Θ is not $= 0$; and we have therefore $V^2 - \Theta = 0$, or, what is the same thing, we may write $V = q$, $\Theta = q^2$, where q is arbitrary.

It is to be shown that $\alpha + \delta$ is $= q$ or $-2q$, the latter value giving the trivial solution $(\alpha x + \beta y,\, \gamma x + \delta y) = (x,\, y)$. The proper solution thus corresponds to $V = q^2$, $\alpha + \delta = q$, that is,

$$(\alpha+\delta)^2 - (\alpha\delta - \beta\gamma) = 0, \quad \text{or} \quad \alpha^2 + \delta^2 + \alpha\delta + \beta\gamma = 0,$$

the condition for the periodic function $\phi^3 x - x = 0$.

For this purpose, from the foregoing equations eliminating a, b, c, d, we have

$$\begin{vmatrix}
\alpha^2+\beta\alpha & 3\alpha^2\gamma & 3\alpha\gamma^2 & \gamma^3\\
3\alpha\beta & \alpha^2\delta + 2\alpha\beta\gamma & 2\alpha\gamma\delta + \beta\gamma^2 & \gamma^2\delta\\
3\beta^2 & 2\alpha\beta\delta + \beta^2\gamma & \alpha\delta^2 + 2\beta\gamma\delta & \gamma\delta^2\\
\beta^2 & 3\beta^2\delta & 3\beta\delta^2 & \delta^3
\end{vmatrix} = 0,$$

an equation which may be written

$$A^4+B\Theta(12{\cdot}34+13{\cdot}24+14{\cdot}23)+C\Theta^2(12+13+14+23+24+34)+D\Theta^3(1{\cdot}2{\cdot}3{\cdot}4)+E\Theta^4 =$$

where 1234, etc., are the first diagonal minors, 12, etc., the second diagonal minors, 1, etc., the third diagonal minors.

Putting herein $\alpha+\delta=m$, $\alpha\delta=n$, $\beta\gamma=n-V$, it is found that n disappears altogether from the equation; viz. the resulting form is

$$V^4+Bm(m^2-2V)+CV^2(m^2-3Vm+2V^2)+DV^3m(m^2-2V)+EV^4=0,$$

or, what is the same thing,

$$m^4\cdot D+m^3(B+CV)-3CVm^2-2(B+DV)m+V^2(C+2E)=0.$$

Putting for C, V, their values q^2, q^2, the equation divides by q^4, and omitting this factor it becomes

$$m^4+2m^3q-3m^2q^2-4q^3+4q^4=0;$$

viz. this is

$$(m-q)(m+2q)=0,$$

or we have

$$m=q \quad\text{or}\quad -2q; \quad\text{that is,}\quad \alpha+\delta=q, \quad\text{or}\quad \alpha+\delta=-2q.$$

Writing, as before, A, B, $C=ac-b^2$, $ad-bc$, $bd-c^2$, we deduce from the foregoing equations

$$\begin{aligned}\alpha A&=V^2\{A\alpha+B\cdot 3\alpha\gamma+C\cdot\gamma^2\},\\ \beta B&=V^2\{A\cdot 2\alpha\beta+B(\alpha\delta+\beta\gamma)+C\cdot 2\gamma\delta\},\\ \gamma C&=V^2\{A\beta^2+B\cdot 3\beta\delta+C\delta^2\},\end{aligned}$$

which are, in fact, the equations for the automorphic transformation of the Hessian $(A,\,B,\,C\,|\,x,\,y)^2$. And, writing herein Θ, $V=q^2$, q^2, the equations become

$$\begin{aligned}A(\alpha^2-q^2)+B\cdot 3\alpha\gamma+C\cdot\gamma^2&=0,\\ A\cdot 2\alpha\beta+B(\alpha\delta+\beta\gamma-q^2)+C\cdot 2\gamma\delta&=0,\\ A\beta^2+B\cdot 3\beta\delta+C(\delta^2-q^2)&=0.\end{aligned}$$

From the first and second of these we have

$$A : B : C = \beta^2(V^2 + q^2) : -2\beta(V^2 + \beta\gamma) : \gamma^2(V^2 - (\alpha\delta + \beta\gamma))q^2 + q^4;$$

or, writing herein for V, q^2 the values q, $q(\alpha\delta - \beta\gamma)$, the three expressions divide by $2\gamma q^2$, and we have

$$A : B : C = \alpha\gamma : 3 - \alpha\delta : -\gamma^2.$$

Combining these values in the first place with the equation

$$A \cdot 4p^2C^2 - 2pBC(4q + pB) + C(-4q^2 + pqB + p^2B^2) = 0,$$

that is,

$$C\{(4AC - B^2)p^2 - 3q^2\} = 0,$$

or, omitting the factor C, and introducing the foregoing notation $\lambda^2 = -\frac{1}{4}$, this is

$$4\Omega\lambda^2p^2 - q^2 = 0, \quad \text{or say} \quad p = \frac{1}{2\lambda\sqrt{\Omega}}\, q.$$

For the unimodular substitution $\alpha\delta - \beta\gamma = 1$, we must have $q^2 = 1$: but, the transformation being

$$(a,\, b,\, c,\, d\,|\,\alpha x + \beta y,\, \gamma x + \delta y)^3 = -q^3 \cdot (a,\, b,\, c,\, d\,|\,x,\, y)^3,$$

to make this strictly automorphic, we must have $q^3 = -1$, and the two conditions are satisfied by $q = -1$; we then have $p = -\frac{1}{2\lambda\sqrt{\Omega}}$; and the coefficients are

$$\alpha,\ \beta,\ \gamma,\ \delta = \frac{-\lambda\sqrt{\Omega} + B}{2\lambda\sqrt{\Omega}},\ \frac{2C}{2\lambda\sqrt{\Omega}},\ \frac{-2A}{2\lambda\sqrt{\Omega}},\ \frac{-\lambda\sqrt{\Omega} - B}{2\lambda\sqrt{\Omega}},$$

which are the values given at the beginning of the paper, and which belong to the automorphic transformation

$$(a,\, b,\, c,\, d\,|\,\alpha x + \beta y,\, \gamma x + \delta y)^3 = (a,\, b,\, c,\, d\,|\,x,\, y)^3.$$

The *a priori* reason for the periodicity-equation $\phi^3 x = x$, is best seen by expressing the cubic function as a product of factors

$$M(x - ay)(x - by)(x - cy).$$

The substitution must, it is clear, cyclically interchange these factors, and therefore, when performed three times in succession on any one of these factors, or consequently upon an arbitrary linear factor $x - fy$, must leave the factor unaltered, and it must thus be a periodic substitution $\phi^3 x = x$. But it was interesting to see how the condition for this, $\alpha^2 + \delta^2 + \alpha\delta + \beta\gamma = 0$, comes out from the consideration of the equation

$$(a,\, b,\, c,\, d \,|\, \alpha x + \beta y,\, \gamma x + \delta y)^3 = (a,\, b,\, c,\, d \,|\, x,\, y)^3.$$

782.

ON MONGE'S "MÉMOIRE SUR LA THÉORIE DES DÉBLAIS ET DES REMBLAIS."

[From the *Proceedings of the London Mathematical Society*, vol. XIV. (1883), pp. 139–142. Read March 8, 1883.]

THE Memoir referred to, published in the *Mémoires de l'Académie*, 1781, pp. 660–704, is a very remarkable one, as well for the problem of earthwork there considered as because the author was led by it to his capital discovery of the curves of curvature of a surface. The problem is, from a given area, called technically the Déblai, to transport the earth to a given equal area, the Remblai; and the question is to find the routes for every element of volume, such that the sum of the products of each element of volume by the length of the route may be a minimum.

The author remarks that the routes must be the intersections of two sets of developable surfaces, and, taking these to be the normal surfaces of a surface, he gives a general theory of such a system of normals. He considers a system of lines represented by a differential equation of the first order

$$\frac{dx}{A} = \frac{dy}{B} = \frac{dz}{C},$$

or, what is the same thing, by two equations of the form

$$U = \phi(V), \quad W = \psi(V),$$

where (U, V, W) are functions of (x, y, z); and in particular, where these are linear functions, viz. the system of lines represented by two linear equations in (x, y, z); or, what is the same thing, the system of lines which meet each of two given lines: such a system is in fact a particular case of a congruence of lines, viz. it is a system of lines each line of which is intersected by each of two consecutive lines,—a property which is not in general a property of a congruence of lines, but is a property of a congruence of normals of a surface. The general theory of such a system of lines requires a somewhat different analytical investigation, and he establishes the theorem that each line of the system is intersected by each of two consecutive lines,—viz. taking (x, y) as the coordinates of the point of intersection of any line with the plane of xy, he obtains, as the condition of intersection with the consecutive line a quadric equation in (dx, dy). He then considers the normals of a surface, (which, as lines drawn according to a given law from any point of a surface, require a slightly different analytical investigation), establishes for them the like theorem, and shows moreover that the two directions of passage on the surface to a consecutive point are at right angles to each other; or, what is the same thing, that in the two sets of developable surfaces formed by the intersecting normals, each surface of the one set intersects each surface of the other set in a straight line, and at right angles. He speaks expressly of the lines of greatest and least curvature, and generally establishes the whole theory of the curvature of surfaces in a very complete and satisfactory manner; the particular case of surfaces of the second order is not considered. It may be remarked that, although not explicitly stating it, he must have seen that a congruence of lines is not, in general, a system of normals of a surface (that is, the lines of a congruence cannot be, in general, cut at right angles by any surface).

The Problem of Déblais and Remblais

The problem of the Déblai and Remblai may be stated as follows: Given two equal areas, to find a system of routes, one for each element of volume, such that the sum of the products of each element by the length of its route may be a minimum. Monge shows that the routes must be the intersections of two sets of developable surfaces, and that the condition of the minimum further requires that these developable surfaces shall cut at right angles; and thence, that the routes are the normals of a surface. The two areas may be regarded as the normals of a curve; and the problem thus is, to find a curve such that, drawing the normals thereof to intersect the two areas, then that the filaments BD and bd, cut off by consecutive normals on the two areas respectively, shall be equal. This leads to a differential equation of the second order for the normal curve; one of the constants of integration remains arbitrary, for the normal curve is any one of a system of parallel curves. It is to be observed that the filaments are the increments of the areas; these increments are equal; a position of the line must be the common tangent of the two areas (this, in fact, constitutes the condition for the determination of one of the arbitrary constants), and for this position the areas are each $= 0$.

Hence, in general, the areas must be equal; or the problem is, to find a curve such that any normal thereof cuts off equal filaments on the two given equal areas.

Monge goes on to argue that the condition of the minimum further requires that the developable surfaces shall cut at right angles, and I cannot say that I see this. He says (pp. 700, 701), "We know already that the routes must be the intersections of two sets of developable surfaces such that each surface of the first set intersects those of the second set in right lines; it remains to be found under what angles these surfaces must cut each other to satisfy the minimum. But it is evident that these angles must be right angles, for with these angles the elementary spaces comprised between four developable surfaces will be greater, and for equal distances the transported mass will be greater; therefore, in the case of a minimum, the routes must be the intersections of two sets of developable surfaces such that each surface of the one set cuts those of the second set in straight lines and at right angles." And, this being so, he infers, and it in fact follows, that the routes are the normals of a surface.

Admitting the conclusion, the problem becomes as follows:—Given two volumes, it is required to find a surface such that, drawing the normals thereof to intersect the two volumes, and considering the filament bounded by the developable surfaces which belong to two consecutive curves of curvature of the surface, then the volume of the filament may be constant; or, what is the same thing, that the areas of the two sections of the filament may be equal. The solution depends on a partial differential equation of the second order, and Monge shows that the surface is one of the surfaces applicable to a surface of revolution.

783.

On Mr Wilkinson's Rectangular Transformation.

[From the *Proceedings of the London Mathematical Society*, vol. XIV. (1883), pp. 222–229. Read May 10, 1883.]

Considering the three cones,

$$\begin{aligned}(p+q)\,X^2+(r+s)\,Y^2+(t+u)\,Z^2&=0,\\(p'+q')\,X^2+(r'+s')\,Y^2+(t'+u')\,Z^2&=0,\\(p''+q'')\,X^2+(r''+s'')\,Y^2+(t''+u'')\,Z^2&=0,\end{aligned}$$

where

$$p+q+r+s+t+u=0,$$

it is easy to see that these contain a singly infinite system of rectangular axes, viz. we have in each cone one axis of a rectangular system, and for one of the cones the axis may be any line at pleasure of the cone. In fact, taking for the three axes (x, y, z), (x', y', z'), (x'', y'', z'') respectively, that is, for the first axis $X:Y:Z=x:y:z$, and so for each of the other two axes, then (x, y, z) being an arbitrary line on the first cone, we can find (x', y', z') and (x'', y'', z'') such that

$$\begin{aligned}(p+q)\,x^2+(r+s)\,y^2+(t+u)\,z^2&=0,\\(p'+q')\,x'^2+(r'+s')\,y'^2+(t'+u')\,z'^2&=0,\\(p''+q'')\,x''^2+(r''+s'')\,y''^2+(t''+u'')\,z''^2&=0,\\xx'+yy'+zz'&=0,\\xx''+yy''+zz''&=0,\\x'x''+y'y''+z'z''&=0.\end{aligned}$$

For, eliminating (x'', y'', z'') from the third, fourth, and fifth equations, we have

$$x:y:z=y'z''-y''z':z'x''-z''x':x'y''-x''y',$$

and consequently

$$(p+q)(y'z''-y''z')+(r+s)(z'x''-z''x')+(t+u)(x'y''-x''y')=0.$$

It is to be shown that this equation is implied in the remaining first, second, and third equations; for, this being so, $(x,\, y,\, z)$ are any values whatever satisfying the first equation. The other two equations then determine $(x',\, y',\, z')$, and, these being known, $(x'',\, y'',\, z'')$ are then determined as above.

In fact, attending to the sixth equation, the equation just obtained may be written in the form

$$(p+q)\{y'^2(x''^2+z''^2)-2z''^2\}+(r+s)\{z'^2(x''^2+y''^2)-2x''^2\}\\ +(t+u)\{x'^2(y''^2+z''^2)-2y''^2\}=0,$$

or, what is the same thing, in the form

$$(p+q)(y'^2+z''^2)+(r+s)(z'^2+x''^2)+(t+u)(x'^2+y''^2)\\ -2\{(p+q)z''^2+(r+s)x''^2+(t+u)y''^2\}=0;$$

for, comparing in the two forms, first the coefficients of x''^2, these are

$$(r+s)+(t+u)-(p+q) \quad \text{and} \quad -2p+\lambda+p,$$

which are equal in virtue of $p+q+r+s+t+u=0$; and comparing next the coefficients of $y'^2z''^2$, these are

$$p+q \quad \text{and} \quad -(r+p)-(q+s),$$

which are equal in virtue of the same relation: and, similarly, the coefficients of the other terms y'^2, etc., are equal in the two equations respectively.

The Elliptic Function Form

Take now three arguments a, b, c, connected by the relation $a+b+c=0$, and write a, α, A for the sn, cn, and dn of a; and similarly b, β, B and c, γ, Γ for those of b and c respectively: then we may write

$$\begin{gathered} p+q=r+s+t+u=a(1-\alpha,\ \omega),\\ p'+q'=r'+s'+t'+u'=b(1-\beta,\ \omega),\\ p''+q''=r''+s''+t''+u''=c(1-\gamma,\ \omega), \end{gathered}$$

for, starting from the first set of values, we have the second set if only

$$\frac{b}{a}\frac{A}{B}=\frac{a}{b}\frac{B}{A}=\frac{c}{\gamma}\frac{\Gamma}{C},$$

and, in order to the identity of the two values of b, we must have

$$\frac{b}{a}\frac{A}{B}\frac{a}{b}\frac{B}{A} - \frac{b}{a}\frac{A}{B}\frac{c}{\gamma}\frac{\Gamma}{C} = 0,$$

that is,

$$(abc - b^3)(ABC - A^3) - (abc - a^3)(ABC - B^3) = 0,$$

or, reducing,

$$(a^2 - b^2)ABC - (A^2 - B^2)abc + A^2b^2 - B^2a^2 = 0.$$

But

$$a^2 - b^2 = -(\alpha^2 - \beta^2), \quad A^2 - B^2 = -k^2(\alpha^2 - \beta^2), \quad A^2B^2 = a^2b^2k^2(\alpha^2 - \beta^2);$$

hence the whole equation divides by $a^2 - b^2$, and, omitting this factor, it becomes

$$-ABC + K^2abc + K'^2 = 0,$$

which is a known relation between the elliptic functions of the arguments a, b, c connected by the equation $a + b + c = 0$. Similarly, for c, we have

$$\frac{c}{b}\frac{B}{\Gamma} = \frac{b}{c}\frac{\Gamma}{B} = \frac{a}{\alpha}\frac{A}{\alpha},$$

and, comparing the two values of c, we have the same identical relation.

It thus appears that the three cones

$$\frac{X^2}{a\alpha} + \frac{Y^2}{b\beta} + \frac{Z^2}{c\gamma} = 0,$$
$$\frac{X^2}{a\alpha} + \frac{Y^2}{b\beta} + \frac{Z^2}{c\gamma} = 0,$$
$$\frac{X^2}{a\alpha} + \frac{Y^2}{b\beta} + \frac{Z^2}{c\gamma} = 0,$$

(the coefficients whereof depend on the elliptic functions sn, cn, and dn, of the arguments a, b, c connected by the equation $a+b+c = 0$) contain a singly infinite system of rectangular axes.

The Rectangular Axes

Considering an argument f, and denoting its sn, cn, dn by f, ϕ, F respectively, we have, for an arbitrary line on the first cone, the values

$$x : y : z = M\surd(Bc) : M\surd(Ca) \cdot f : M\surd(-AB\Gamma) \cdot F.$$

In fact, substituting in the equation of the cone, we obtain the identity

$$M^2(1 + f^2 - F^2) = 0;$$

and if we determine M by the condition that $x^2 + y^2 + z^2$ shall be $= 1$, then we have

$$1 = M^2\{1 + Aa + BAbf^2 - aBCF^2\},$$

where the coefficient of M^2 is

$$= Aa + BAb(1 - f^2) - aBC(1 - F^2),$$

which is easily shown to be

$$= BA'bc(-f^2),$$

so that the values of x, y, z are

$$= \left(\frac{\surd Aa}{\surd BA'bc}, \frac{\surd BA'bc \cdot f}{\surd BA'bc}, \frac{\surd(-aBC) \cdot F}{\surd BA'bc}\right).$$

Similarly, taking the arguments g, h, and denoting their elliptic functions by g, θ, G and h, η, H respectively, we have for a system of arbitrary lines in the three cones respectively, the values

$$\begin{array}{lll} x = \surd(Bc) & y = \surd(Ca) \cdot f & z = \surd(-AB\Gamma) \cdot F \\ x' = \surd(Ca) & y' = \surd(Ab) \cdot g & z' = \surd(-bCA) \cdot G \\ x'' = \surd(Ab) & y'' = \surd(Bc) \cdot h & z'' = \surd(-cAB) \cdot H \end{array}$$

these values being such that $x^2+y^2+z^2$, $x'^2+y'^2+z'^2$, $x''^2+y''^2+z''^2$ are each $= 1$. The radicals in the first line would be more correctly written, and may be understood as meaning $\surd Aa$, $k\surd A{\cdot}b{\cdot}c$, $i\surd a{\cdot}B{\cdot}C$, and similarly as regards the second and third lines respectively.

Taking now the arbitrary lines at right angles to each other, the condition for the second and third lines is

$$1 + M^2ag\,h - M^2AGH = 0,$$

which is satisfied if $h = g - f$; similarly the condition for the third and first lines is satisfied if $f = h - g$; and we then have $g + h = g - f$, that is, $-h = g - f$, or $f = g - h$, which is the condition for the first and second lines; hence the arguments ξ, η, ζ, for f, g, h being the differences of any four arguments α, β, γ, δ, the foregoing values of (x, y, z), (x', y', z'), (x'', y'', z'') satisfy all the conditions for a system of rectangular axes.

The Reciprocal Cone Property

Returning to the three cones, it is to be remarked that, taking in the first of them a line 1 at pleasure, then we have in the second of them two lines 2, $2'$ each at right angles to the line 1, and such that the line 3 at right angles to the plane 12, and the line $3'$ at right angles to the plane $12'$, lie each of them in the third cone; or, what is the same thing, we have in the two cones respectively the rectangular lines 1 and 2, and also the rectangular lines 1 and $2'$, such that the planes 12 and $12'$ each of them envelope one and the same cone, the reciprocal of the third cone; where by the reciprocal cone of a given cone is meant the cone generated by the lines through the vertex at right angles to the tangent planes of the given cone. Introducing the notion of the absolute cone $X^2 + Y^2 + Z^2 = 0$, a line and plane through the vertex at right angles to each other are, in fact, reciprocal polars in regard to this absolute cone; and two lines at right angles to each other are reciprocals (or harmonics) in regard to this absolute cone; that is, the reciprocal plane of either of them passes through the other. The two cones are cones intersecting each other in four lines lying on the absolute cone; and in virtue of this relation they have the property in question, viz. taking in the first cone a line 1 at pleasure, then the reciprocal plane hereof in regard to the absolute cone meets the second cone in a pair of lines 2 and $2'$ such that the planes 12 and $12'$ each of them envelope one and the same cone; the reciprocal of this cone is then the third cone of the system, and as such it passes through the four lines on the absolute cone.

Verification

In verification, observe that the coefficients $p+q$, $r+s$, etc. of the equations of the three cones satisfy the equations

$$\begin{vmatrix} 1 & p+q & p'+q' & p''+q'' & (p+r)(p+s)(p+t) \\ 1 & r+s & r'+s' & r''+s'' & (r+p)(r+q)(r+t) \\ 1 & t+u & t'+u' & t''+u'' & (t+p)(t+q)(t+r) \end{vmatrix} = 0.$$

This is obviously the case for each equation such as

$$\begin{vmatrix} 1 & p+q & p'+q' \end{vmatrix} = 0,$$

and among these are of course included the equation $\begin{vmatrix} 1 & p & p' \end{vmatrix} = 0$, which expresses that the first and second cones intersect on the absolute; $(p,\, q,\, r)$, $(p',\, q',\, r')$ are any quantities satisfying this relation, and, regarding them as given, we have then two independent equations determining the ratios $p'' : q'' : r''$. The theorem is that the planes 12 and $12'$ envelope one and the same quadric cone

$$\frac{X^2}{p''} + \frac{Y^2}{q''} + \frac{Z^2}{r''} = 0.$$

The equations $\begin{vmatrix} 1 & p & p'' \end{vmatrix} = 0$ and $\begin{vmatrix} 1 & p & pp'p'' \end{vmatrix} = 0$ give

$$(q-r)p'' + (r-p)q'' + (p-q)r'' = 0,$$
$$(q-r)pp'p'' + (r-p)qq'q'' + (p-q)rr'r'' = 0,$$

and thence

$$(q-r)p'' : (r-p)q'' : (p-q)r'' = qq' - rr' : rr' - pp' : pp' - qq';$$

or, observing that we have

$$q - r : r - p : p - q = qr' - q'r : rp' - r'p : pq' - p'q,$$

the equations may also be written

$$(qr'-q'r)p''+(rp'-r'p)q''+(pq'-p'q)r'' = qq'-rr' : rr'-pp' : pp'-qq'.$$

Starting with an arbitrary line $(x,\, y,\, z)$ in the first cone, then the reciprocal plane thereof (in regard to the absolute cone) is the plane $Xx + Yy + Zz = 0$, which meets the second cone in two lines, say (2) and $(2')$, each of which is a line reciprocal to the line (1); and we have thus two planes (12) and (12′), each of which envelopes, as is to be shown, the same cone $q''r''X^2 + r''p''Y^2 + p''q''Z^2 = 0$.

Suppose, in general, that we have an arbitrary line (x, y, z) and an arbitrary plane $\alpha X + \beta Y + \gamma Z = 0$, and that it is required to find the equation of the two planes through the line (x, y, z), and the intersections of the plane $\alpha X + \beta Y + \gamma Z = 0$ with the cone $p'X^2 + q'Y^2 + r'Z^2 = 0$: the equation of the pair of planes is

$$\begin{aligned}(\alpha X + \beta Y + \gamma Z)^2(p'x^2 + q'y^2 + r'z^2)&\\ + (\alpha x + \beta y + \gamma z)^2(p'X^2 + q'Y^2 + r'Z^2)&\\ - 2(\alpha X + \beta Y + \gamma Z)(\alpha x + \beta y + \gamma z)(p'Xx + q'Yy + r'Zz) &= 0.\end{aligned}$$

In the present case, the plane $\alpha X + \beta Y + \gamma Z = 0$ is the plane $xX + yY + zZ = 0$, which is the reciprocal of the line (x, y, z) in regard to the absolute cone, and the equation of the pair of planes is

$$\begin{aligned}(xX + yY + zZ)^2(p'x^2 + q'y^2 + r'z^2)&\\ + (x^2 + y^2 + z^2)^2(p'X^2 + q'Y^2 + r'Z^2)&\\ - 2(xX + yY + zZ)(x^2 + y^2 + z^2)(p'Xx + q'Yy + r'Zz) &= 0,\end{aligned}$$

which is the condition that the two planes envelope the same cone $q''r''X^2 + r''p''Y^2 + p''q''Z^2 = 0$, as was to be shown.

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Article 786]§ 786. EQUATION (continued).

[From the *Encyclopædia Britannica*, Ninth Edition, vol. viii (1878), pp. 610–627.]

§ 15.

It will be recollected that the expression *number* and the correlative epithet *numerical* were at the outset used in a wide sense, as extending to imaginaries. This extension arises out of the theory of equations by a process analogous to that by which number, in its original most restricted sense of positive integer number, was extended to have the meaning of a real positive or negative magnitude susceptible of continuous variation.

If for a moment number is understood in its most restricted sense as meaning positive integer number, the solution of a simple equation leads to an extension; $ax-b=0$, gives $x=\frac{b}{a}$, a positive fraction, and we can in this manner represent, not accurately, but as nearly as we please, any positive magnitude whatever; so an equation $ax+b=0$ gives $x=-\frac{b}{a}$, which (approximately as before) represents any negative magnitude. We thus arrive at the extended signification of number as a continuously varying positive or negative magnitude. Such numbers may be added or subtracted, multiplied or divided one by another, and the result is always a number. Now from a quadric equation we derive, in like manner, the notion of a complex or imaginary number such as is spoken of above. The equation $x^2+1=0$ is not (in the foregoing sense, number = real number) satisfied by any numerical value whatever of x; but we assume that there is a number which we call i, satisfying the equation $i^2 + 1 = 0$; and then taking a and b any real numbers, we form an expression such as $a + bi$, and use the expression number in this extended sense: any two such numbers may be added or subtracted, multiplied or divided one by the other, and the result is always a number. And if we consider first a quadric equation $x^2 + px + q = 0$ where p and q are real numbers, and next the like equation, where p and q are any numbers whatever, it can be shown that there exists for x a numerical value which satisfies the equation; or, in other words, it can be shown that the equation has a numerical root. The like theorem, in fact, holds good for an equation of any order whatever. But suppose for a moment that this was not the case: say that there was a cubic equation $x^3 + px^2 + qx + r = 0$, with numerical coefficients, not satisfied by any numerical value of x, we should have to establish a new imaginary j satisfying some such equation, and should then have to consider numbers of the form $a+bj$, or perhaps $a + bj + cj^2$ (a, b, c numbers $a + \beta i$ of the kind heretofore considered); first we should be thrown back on the quadric equation $x^2 + px + q = 0$, p and q being now numbers of the last-mentioned extended form—non constat that every such equation has a numerical root—and if not, we might be led to other imaginaries k, l, &c., and so on *ad infinitum* in inextricable confusion.

But in fact a numerical equation of any order whatever has always a numerical root, and thus numbers (in the foregoing sense, number = quantity of the form $a + \beta i$) form (what real numbers do not) a universe complete in itself, such that starting in it we are never led out of it. There may very well be, and perhaps are, numbers in a more general sense of the term (quaternions are not a case in point, as the ordinary laws of combination are not adhered to): but in order to have to do with such numbers (if any), we must start with them.

§ 16.

The theorem as regards numerical equations thus is, every numerical equation has a numerical root; or for shortness (the meaning being as before), every equation has a root. Of course the theorem is the reverse of self-evident, and it requires proof; but provisionally assuming it as true, we derive from it the general theory of numerical equations. As the term root was introduced in the course of an explanation, it will be convenient to give here the formal definition.

A number a such that substituted for x it makes the function $x^n - p_1x^{n-1} + \ldots \pm p_n$ to be $= 0$, or say such that it satisfies the equation $f(x) = 0$, is said to be a *root* of the equation; that is, a being a root, we have

$$a^n - p_1a^{n-1} + \ldots \pm p_n = 0, \quad \text{or say} \quad f(a) = 0;$$

and it is then easily shown that $x - a$ is a factor of the function $f(x)$, viz. that we have $f(x) = (x-a)f_1(x)$, where $f_1(x)$ is a function $x^{n-1} - q_1x^{n-2} + \ldots \pm q_{n-1}$ of the order $n-1$, with numerical coefficients $q_1, q_2, \ldots, q_{n-1}$.

In general, a is not a root of the equation $f_1(x) = 0$, but it may be so; i.e. $f_1(x)$ may contain the factor $x - a$; when this is so, $f(x)$ will contain the factor $(x-a)^2$; writing then $f(x) = (x-a)^2f_2(x)$, and assuming that a is not a root of the equation $f_2(x) = 0$, $x = a$ is then said to be a *double root* of the equation $f(x) = 0$; and similarly $f(x)$ may contain the factor $(x-a)^3$ and no higher power, and $x = a$ is then a *triple root*; and so on.

Supposing, in general, that $f(x) = (x-a)^\alpha F(x)$, α being a positive integer which may be $= 1$, $(x-a)^\alpha$ the highest power of $x - a$ which divides $f(x)$, and $F(x)$ being of course of the order $n - \alpha$, then the equation $F(x) = 0$ will have a root b which will be different from a; $x - b$ will be a factor, in general a simple one, but it may be a multiple one, of $F(x)$, and $f(x)$ will in this case be $= (x-a)^\alpha(x-b)^\beta\Phi(x)$, β a positive integer which may be $= 1$, $(x-b)^\beta$ the highest power of $x - b$ in $F(x)$ or $f(x)$, and $\Phi(x)$ being of course of the order $n - \alpha - \beta$.

The original equation $f(x) = 0$ is in this case said to have α roots each $= a$, β roots each $= b$; and so on for any other factors $(x-c)^\gamma$, &c.

We have thus the theorem—A numerical equation of the order n has in every case n roots, viz. there exist n numbers $a, b, \ldots$, in general all distinct, but which may arrange themselves in any sets of equal values, such that $f(x) = (x-a)(x-b)(x-c)\ldots$ identically.

If the equation has equal roots, these can in general be determined: and the case is at any rate a special one which may be in the first instance excluded from consideration. It is therefore, in general, assumed that the equation $f(x) = 0$ has all its roots unequal.

If the coefficients $p_1, p_2, \ldots$ are all or any one or more of them imaginary, then the equation $f(x) = 0$, separating the real and imaginary parts thereof, may be written $F(x) + i\Phi(x) = 0$, where $F(x)$, $\Phi(x)$

are each of them a function with real coefficients; and thus it appears that the equation $f(x) = 0$, with imaginary coefficients, has not in general any real root; supposing it to have a real root a, this must be at once a root of each of the equations $F(x) = 0$ and $\Phi(x) = 0$.

But an equation with real coefficients may have as well imaginary as real roots, and we have further the theorem that for any such equation the imaginary roots enter in pairs, viz. $\alpha + \beta i$ being a root, then $\alpha - \beta i$ will be also a root. It follows that, if the order be odd, there is always an odd number of real roots, and therefore at least one real root.

§ 17.

In the case of an equation with real coefficients, the question of the existence of real roots, and of their separation, has been already considered. In the general case of an equation with imaginary (it may be real) coefficients, the like question arises as to the situation of the (real or imaginary) roots; thus if, for facility of conception, we regard the constituents α, β of a root $\alpha + \beta i$ as the coordinates of a point in piano, and accordingly represent the root by such point, then drawing in the plane any closed curve or "contour," the question is how many roots lie within such contour.

This is solved theoretically by means of a theorem of Cauchy's (1837), viz. writing in the original equation $x + iy$ in place of x, the function $f(x + iy)$ becomes $= P + iQ$, where P and Q are each of them a rational and integral function (with real coefficients) of (x, y). Imagining the point (x, y) to travel along the contour, and considering the number of changes of sign from $-$ to $+$ and from $+$ to $-$ of the fraction $\frac{P}{Q}$ corresponding to passages of the fraction through zero, that is, to values for which P becomes $= 0$, disregarding those for which Q becomes $= 0$, the difference of these numbers gives the number of roots within the contour.

It is important to remark that the demonstration does not presuppose the existence of any root; the contour may be the infinity of the plane (such infinity regarded as a contour, or closed curve), and in this case it can be shown (and that very easily) that the difference of the numbers of changes of sign is $= n$; that is, there are within the infinite contour, or (what is the same thing) there are in all, n roots: thus Cauchy's theorem contains really the proof of the fundamental theorem that a numerical equation of the nth order (not only has a numerical root, but) has precisely n roots. It would appear that this proof of the fundamental theorem in its most complete form is in principle identical with Gauss's last proof (1849) of the theorem, in the form—A numerical equation of the nth order has always a root*.

* The earlier demonstrations by Euler, Lagrange, &c., relate to the case of a numerical equation with real coefficients; and they consist in showing that such equation has always a real quadratic divisor, furnishing two roots, which are either real or else conjugate imaginaries $\alpha + \beta i$: see Lagrange's *Equations Numériques.*

But in the case of a finite contour, the actual determination of the difference which gives the number of real roots can be effected only in the case of a rectangular contour, by applying to each of its sides separately a method such as that of Sturm's theorem; and thus the actual determination ultimately depends on a method such as that of Sturm's theorem.

Very little has been done in regard to the calculation of the imaginary roots of an equation by approximation; and the question is not here considered.

§ 18.

A class of numerical equations which needs to be considered is that of the binomial equations $x^n - a = 0$ ($a = \alpha + \beta i$, a complex number). The foregoing conclusions apply, viz. there are always n roots, which, it may be shown, are all unequal. And these can be found numerically by the extraction of the square root, and of an nth root, of real numbers, and by the aid of a table of natural sines and cosines*. For writing

$$\alpha + \beta i = \rho(\cos\lambda + i\sin\lambda),$$

there is always a real angle λ (positive and less than 2π), such that its cosine and sine are $= \frac{\alpha}{\rho}$ and $\frac{\beta}{\rho}$ respectively; that is, writing for shortness $\sqrt{\alpha^2 + \beta^2} = \rho$, we have $\alpha + \beta i = \rho(\cos\lambda + i\sin\lambda)$, or the equation is $x^n = \rho(\cos\lambda + i\sin\lambda)$; hence

$$\cos\frac{n\theta}{n} + i\sin\frac{n\theta}{n} = \cos\lambda + i\sin\lambda,$$

a value of x is $= \sqrt[n]{\rho}\left(\cos\frac{\lambda}{n} + i\sin\frac{\lambda}{n}\right)$.

The formula really gives all the roots, for instead of λ we may write $\lambda + 2s\pi$, s a positive or negative integer, and then we have

$$x = \sqrt[n]{\rho}\left(\cos\frac{\lambda + 2s\pi}{n} + i\sin\frac{\lambda + 2s\pi}{n}\right),$$

which has the n values obtained by giving to s the values $0, 1, 2, \ldots, n-1$ in succession; the roots are, it is clear, represented by points lying at equal intervals on a circle. But it is more convenient to proceed somewhat differently; taking one of the roots to be θ, so that $\theta^n = a$, then assuming $x = \theta y$, the equation becomes $y^n - 1 = 0$, which equation, like the original equation, has precisely n roots (one of them being of course $= 1$). And the original equation $x^n - a = 0$ is thus reduced to the more simple equation $x^n - 1 = 0$; and although the theory of this equation is included in the preceding one, yet it is proper to state it separately.

The equation $x^n - 1 = 0$ has its several roots expressed in the form $1, \omega, \omega^2, \ldots, \omega^{n-1}$, where ω may be taken $= \cos\frac{2\pi}{n} + i\sin\frac{2\pi}{n}$; in fact, ω having this value, any integer power ω^k is $= \cos\frac{2\pi k}{n} + i\sin\frac{2\pi k}{n}$, and we thence have $(\omega^k)^n = \cos 2\pi k + i\sin 2\pi k = 1$, that is, ω^k is a root of the equation. The theory will be resumed further on.

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* The square root of $\alpha + \beta i$ can be determined by the extraction of square roots of positive real numbers, without the trigonometrical tables.

By what precedes, we are led to the notion (a numerical) of the radical $a^{\frac{1}{n}}$ regarded as an n-valued function; any one of these being denoted by $\sqrt[n]{a}$, then the series of values is $\sqrt[n]{a}$, $\omega\sqrt[n]{a}$, $\ldots$, $\omega^{n-1}\sqrt[n]{a}$; or we may, if we please, use $\sqrt[n]{a}$ instead of $a^{\frac{1}{n}}$ as a symbol to denote the n-valued function.

As the coefficients of an algebraical equation may be numerical, all which follows in regard to algebraical equations is (with, it may be, some few modifications) applicable to numerical equations; and hence, concluding for the present this subject, it will be convenient to pass on to algebraical equations.

II. Algebraical Equations (§§19 to 34).

§ 19.

The equation is

$$x^n - p_1 x^{n-1} + \ldots \pm p_n = 0,$$

and we here assume the existence of roots, viz. we assume that there are n quantities $a, b, c, \ldots$ (in general all of them different, but which in particular cases may become equal in sets in any manner), such that

$$x^n - p_1 x^{n-1} + \ldots \pm p_n = 0;$$

or looking at the question in a different point of view, and starting with the roots $a, b, c, \ldots$ as given, we express the product of the n factors $x - a$, $x - b$, $\ldots$ in the foregoing form, and thus arrive at an equation of the order n having the n roots $a, b, c, \ldots$ In either case we have

$$p_1 = \Sigma a, \quad p_2 = \Sigma ab, \quad \ldots, \quad p_n = abc \ldots;$$

i.e. regarding the coefficients $p_1, p_2, \ldots, p_n$ as given, then we assume the existence of roots $a, b, c, \ldots$ such that $p_1 = \Sigma a$, &c.; or, regarding the roots as given, then we write p_1, p_2, &c., to denote the functions Σa, Σab, &c.

As already explained, the epithet algebraical is not used in opposition to numerical: an algebraical equation is merely an equation wherein the coefficients are not restricted to denote, or are not explicitly considered as denoting, numbers. That the abstraction is legitimate, appears by the simplest example; in saying that the equation $x^2 - px + q = 0$ has a root $x = \frac{1}{2}(p + \sqrt{p^2 - 4q})$, we mean that writing this value for x the equation becomes an identity, $\frac{1}{2}(p + \sqrt{p^2 - 4q})^2 - p \cdot \frac{1}{2}(p + \sqrt{p^2 - 4q}) + q = 0$; and the verification of this identity in nowise depends upon p and q meaning numbers. But if it be asked what there is beyond numerical equations included in the term algebraical equation, or, again, what is the full extent of the meaning attributed to the term: the latter question at any rate it would be very difficult to answer; as to the former one, it may be said that the coefficients may, for instance, be symbols of operation. As regards such equations, there is certainly no proof that every equation has a root, or that an equation of the nth order has n roots; nor is it in any wise clear what the precise signification of the statement is. But it is found that the assumption of the existence of the n roots can be made without contradictory results; conclusions derived from it, if they involve the roots, rest on the same ground as the original assumption; but the conclusion may be independent of the roots altogether, and in this case it is undoubtedly valid; the reasoning, although actually conducted by aid of the assumption (and, it may be, most easily and elegantly in this manner), is really independent of the assumption. In illustration, we observe that it is allowable to express a function of p and q as follows, that is, by means of a rational symmetrical function of a and b: this can, as a fact, be expressed as a rational function of $a+b$ and ab; and if we prescribe that $a+b$ and ab shall then be changed into p and q respectively, we have the required function of p, q. That is, we have $F(\alpha,\beta)$ as a representation of $f(p,q)$, obtained as if we had $p=a+b$, $q=ab$, but without in any wise assuming the existence of the a, b of these equations.

§ 20.

Starting from the equation

$$x^n - p_1x^{n-1} + \ldots = x-a\cdot x-b\cdot \&\text{c.},$$

or the equivalent equations $p_1 = \Sigma a$, &c., we find

$$a^n - p_1a^{n-1} + \ldots = 0, \quad b^n - p_1b^{n-1} + \ldots = 0;$$

(it is as satisfying these equations that $a, b, \ldots$ are said to be the roots of $x^n - p_1x^{n-1} + \ldots = 0$) and conversely from the last-mentioned equations, assuming that $a, b, \ldots$ are all different, we deduce

$$p_1 = \Sigma a, \quad p_2 = \Sigma ab, \quad \&\text{c.},$$

and

$$x^n - p_1x^{n-1} + \ldots = x-a\cdot x-b\cdot \&\text{c.}$$

Observe that if, for instance, $a=b$, then the equations $a^n - p_1a^{n-1} + \ldots = 0$, $b^n - p_1b^{n-1} + \ldots = 0$ would reduce themselves to a single relation, which would not of itself express that a was a double root,—that is, that $(x-a)^2$ was a factor of $x^n - p_1x^{n-1} + \&\text{c.}$; but by considering b as the limit of $a+h$, h indefinitely small, we obtain a second equation

$$na^{n-1} - (n-1)p_1a^{n-2} + \ldots = 0,$$

which, with the first, expresses that a is a double root; and then the whole system of equations leads as before to the equations $p_1 = \Sigma a$, &c. But the existence of a double root implies a certain relation between the coefficients; the general case is when the roots are all unequal.

We have then the theorem that every rational symmetrical function of the roots is a rational function of the coefficients. This is an easy consequence from the less general theorem, every rational and integral symmetrical function of the roots is a rational and integral function of the coefficients.

In particular, the sums of the powers Σa^2, Σa^3, &c., are rational and integral functions of the coefficients.

The process originally employed for the expression of other functions $\Sigma a^r b^s$, &c., in terms of the coefficients is to make them depend upon the sums of powers: for instance, $\Sigma a^r b^s = \Sigma a^r \cdot \Sigma a^s - \Sigma a^{r+s}$; but this is very objectionable; the true theory consists in showing that we have systems of equations

$$\begin{aligned}
p_1 &= \Sigma a, \\
p_1^2 &= \Sigma a^2 + 2\Sigma ab, \\
p_2 &= \Sigma ab, \\
p_3 &= \Sigma abc, \\
p_1 p_2 &= \Sigma a^2 b + 3\Sigma abc, \\
p_1^3 &= \Sigma a^3 + 3\Sigma a^2 b + 6\Sigma abc,
\end{aligned}$$

where in each system there are precisely as many equations as there are root-functions on the right-hand side: e.g. 3 equations and 3 functions Σabc, $\Sigma a^2 b$, Σa^3. Hence in each system the root-functions can be determined linearly in terms of the powers and products of the coefficients:

$$\begin{aligned}
\Sigma ab &= p_2, \\
\Sigma abc &= p_3, \\
\Sigma a^2 b &= p_1 p_2 - 3p_3, \\
\Sigma a^3 &= p_1^3 - 3p_1 p_2 + 3p_3,
\end{aligned}$$

and so on. The older process, if applied consistently, would derive the originally assumed value $\Sigma ab = p_2$, from the two equations $\Sigma a = p_1$, $\Sigma a^2 = p_1^2 - 2p_2$; i.e. we have $2\Sigma ab = \Sigma a \cdot \Sigma a - \Sigma a^2 = p_1^2 - (p_1^2 - 2p_2) = 2p_2$.

§ 21.

It is convenient to mention here the theorem that, x being determined as above by an equation of the order n, any rational and integral function whatever of x, or more generally any rational function which does not become infinite in virtue of the equation itself, can be expressed as a rational and integral function of x, of the order $n-1$, the coefficients being rational functions of the coefficients of the equation. Thus the equation gives x^n a function of the form in question; multiplying each side by x, and on the right-hand side writing for x^n its foregoing value, we have x^{n+1} a function of the form in question; and the like for any higher power of x, and therefore also for any rational and integral function of x. The proof in the case of a rational non-integral function is somewhat more complicated. The final result is of the form $\frac{\phi(x)}{\psi(x)} = I(x)$, or say $\phi(x) - \psi(x)I(x) = 0$, where ϕ, ψ, I are rational and integral functions; in other words, this equation, being true if only $f(x) = 0$, can only be so by reason that the left-hand side contains $f(x)$ as a factor, or we must have identically $\phi(x) - \psi(x)I(x) = M(x)f(x)$. And it is, moreover, clear that the equation $\frac{\phi(x)}{\psi(x)} = I(x)$, being satisfied if only $f(x) = 0$, must be satisfied by each root of the equation.

§ 22.

From the theorem that a rational symmetrical function of the roots is expressible in terms of the coefficients, it at once follows that it is possible to determine an equation (of an assignable order) having for its roots the several values of any given (unsymmetrical) function of the roots of the given equation. For example, in the case of a quartic equation, having the roots (a, b, c, d), it is possible to find an equation having the roots ab, ac, ad, bc, bd, cd, being therefore a sextic equation: viz. in the product

$$(y-ab)(y-ac)(y-ad)(y-bc)(y-bd)(y-cd),$$

the coefficients of the several powers of y will be symmetrical functions of a, b, c, d and therefore rational and integral functions of the coefficients of the quartic equation; hence, supposing the product so expressed, and equating it to zero, we have the required sextic equation. In the same manner can be found the sextic equation having the roots $(a-b)^2$, $(a-c)^2$, $(a-d)^2$, $(b-c)^2$, $(b-d)^2$, $(c-d)^2$, which is the equation of differences previously referred to; and similarly we obtain the equation of differences for a given equation of any order. Again, the equation sought for may be that having for its roots the given rational functions $\phi(a)$, $\phi(b)$, … of the several roots of the given equation. Any such rational function can (as was shown) be expressed as a rational and integral function of the order $n-1$; and, retaining x in place of any one of the roots, the problem is to find y from the equations $x^n - p_1x^{n-1} + \ldots = 0$, and $y = M_0x^{n-1} + M_1x^{n-2} + \ldots$, or, what is the same thing, from these two equations to eliminate x. This is, in fact, Tschirnhausen's transformation (1683).

§ 23.

In connexion with what precedes, the question arises as to the number of values (obtained by permutations of the roots) of given unsymmetrical functions of the roots, or say of a given set of letters: for instance, with roots or letters (a, b, c, d) as before, how many values are there of the function $ab + cd$, or better, how many functions are there of this form? The answer is 3, viz. $ab + cd$, $ac + bd$, $ad + bc$; or again we may ask whether, in the case of a given number of letters, there exist functions with a given number of values, 3-valued, 4-valued functions, &c.

It is at once seen that for any given number of letters there exist 2-valued functions; the product of the differences of the letters is such a function; however the letters are interchanged, it alters only its sign; or say the two values are Δ, $-\Delta$. And if P, Q are symmetrical functions of the letters, then the general form of such a function is $P + Q\Delta$: this has only the two values $P + Q\Delta$, $P - Q\Delta$.

In the case of 4 letters there exist (as appears above) 3-valued functions: but in the case of 5 letters there does not exist any 3-valued or 4-valued function; and the only 5-valued functions are those which are symmetrical in regard to four of the letters, and can thus be expressed in terms of one letter and of symmetrical functions of all the letters. These last theorems present themselves in the demonstration of the non-existence of a solution of a quintic equation by radicals.

The theory is an extensive and important one, depending on the notions of substitutions and of groups*.

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* A *substitution* is the operation by which we pass from the primitive arrangement of n letters to any other arrangement of the same letters: for instance, the substitution $\binom{a}{b}$ means that a is to be changed into b, b into c, c into d, d into a. Substitutions may, of course, be represented by single letters α, β, ... Two or more substitutions may be compounded together and give rise to a substitution; i.e., performing upon the primitive arrange-

ment first the substitution β and then upon the result the substitution α, we have the substitution $\alpha\beta$. Substitutions are not commutative; thus, $\alpha\beta$ is not in general $= \beta\alpha$: but they are associative, $\alpha\beta \cdot \gamma = \alpha \cdot \beta\gamma$, so that $\alpha\beta\gamma$ has a determinate meaning. A substitution may be compounded any number of times with itself, and we thus have the powers α^2, α^3, &c. Since the number of substitutions is limited, some power α^μ must be $= 1$: or, as this may be expressed, every substitution is a root of unity. A *group* of substitutions is a set such that each two of them compounded together in either order gives a substitution belonging to the set; every group includes the substitution unity, so that we may in general speak of a group $1, \alpha, \beta, \ldots$ (the number of terms is the *order* of the group). The whole system of the $1 \cdot 2 \cdot 3 \ldots n$ substitutions which can be performed upon the n letters is obviously a group: the order of every other group which can be formed out of these substitutions is a submultiple of this number; but it is not conversely true that a group exists the order of which is any given submultiple of this number. In the case of a determinant the substitutions which give rise to the positive terms form a group the order of which is $= \frac{1}{2} \cdot 1 \cdot 2 \cdot 3 \ldots n$. For any function of the n letters, the whole series of substitutions which leave the value of the functions unaltered form a group; and thence also the number of values of the function is $= 1 \cdot 2 \cdot 3 \ldots n$ divided by the order of the group.

§ 24.

Returning to equations, we have the very important theorem that, given the value of any unsymmetrical function of the roots, e.g. in the case of a quartic equation, the function $ab + cd$, it is in general possible to determine rationally the value of any similar function, such as $(a + b)^2 + (c + d)^2$.

The *a priori* ground of this theorem may be illustrated by means of a numerical equation. Suppose that the roots of a quartic equation are $1, 2, 3, 4$, then if it is given that $ab + cd = 14$, this in effect determines a, b to be $1, 2$ and c, d to be $3, 4$ (viz. $a = 1$, $b = 2$ or $a = 2$, $b = 1$, and $c = 3$, $d = 4$ or $c = 4$, $d = 3$) or else a, b to be $3, 4$ and c, d to be $1, 2$; and it therefore in effect determines $(a+b)^2+(c+d)^2$ to be $= 370$, and not any other value: that is, $(a + b)^2 + (c + d)^2$, as having a single value, must be determinable rationally. And we can in the same way account for cases of failure as regards particular equations; thus, the roots being $1, 2, 3, 4$ as before, $a^2b = 2$ determines a to be $= 1$ and b to be $= 2$; but if the roots had been $1, 2, 4, 16$ then $a^2b = 16$

does not uniquely determine a, b but only makes them to be $1, 16$ or $2, 4$ respectively.

As to the *a posteriori* proof, assume, for instance,

$$\begin{aligned} t_1 &= ab + cd, & y_1 &= (a+b)^2 + (c+d)^2, \\ t_2 &= ac + bd, & y_2 &= (a+c)^2 + (b+d)^2, \\ t_3 &= ad + bc, & y_3 &= (a+d)^2 + (b+c)^2, \end{aligned}$$

then $y_1 + y_2 + y_3$, $t_1 y_1 + t_2 y_2 + t_3 y_3$, $t_1^2 y_1 + t_2^2 y_2 + t_3^2 y_3$, will be respectively symmetrical functions of the roots of the quartic, and therefore rational and integral functions of the coefficients; that is, they will be known.

Suppose for a moment that t_1, t_2, t_3 are all known; then the equations being linear in y_1, y_2, y_3, these can be expressed rationally in terms of the coefficients and of t_1, t_2, t_3; that is, y_1, y_2, y_3 will be known. But observe further that y_1 is obtained as a function of t_1, t_2, t_3 symmetrical as regards t_2, t_3; it can therefore be expressed as a rational function of t_1 and of $t_2 + t_3$, $t_2 t_3$, and thence as a rational function of t_1 and of $t_1 + t_2 + t_3$, $t_1 t_2 + t_1 t_3 + t_2 t_3$, $t_1 t_2 t_3$; but these last are symmetrical functions of the roots, and as such they are expressible rationally in terms of the coefficients; that is, y_1 will be expressed as a rational function of t_1 and of the coefficients: or t_1 (alone, not t_2 or t_3) being known, y_1 will be rationally determined.

§ 25.

We now consider the question of the algebraical solution of equations, or, more accurately, that of the solution of equations by radicals.

In the case of a quadric equation $x^2 - px + q = 0$, we can by the assistance of the sign $\sqrt{(\quad)}$ or $(\quad)^{\frac{1}{2}}$ find an expression for x as a two-valued function of the coefficients p, q such that, substituting this value in the equation, the equation is thereby identically satisfied; it has been found that this expression is

$$x = \tfrac{1}{2}\{p \pm \sqrt{p^2 - 4q}\},$$

and the equation is on this account said to be algebraically solvable, or more accurately solvable by radicals. Or we may by writing $x = -\frac{1}{2}p + z$, reduce the equation to $z^2 = \frac{1}{4}(p^2 - 4q)$, viz. to an equation of the form $z^2 = a$; and in virtue of its being thus reducible we say that the original equation is solvable by radicals. And the question for an equation of any higher order, say of the order n, is, can we by means of radicals, that is, by aid of the sign $\sqrt[m]{(\quad)}$ or $(\quad)^{\frac{1}{m}}$, using as many as we please of such signs and with any values of m, find an n-valued function (or any function) of the coefficients which substituted for x in the equation shall satisfy it identically.

It will be observed that the coefficients $p, q, \ldots$ are not explicitly considered as numbers, but even if they do denote numbers, the question whether a numerical equation admits of solution by radicals is wholly unconnected with the before-mentioned theorem of the existence of the n roots of such an equation. It does not even follow that in the case of a numerical equation solvable by radicals the algebraical solution gives the numerical solution, but this requires explanation. Consider first a numerical quadric equation with imaginary coefficients. In the formula $x = \frac{1}{2}(p \pm \sqrt{p^2 - 4q})$, substituting for p, q their given numerical values, we obtain for x an expression of the form $x = \alpha + \beta i \pm \sqrt{\gamma + \delta i}$, where $\alpha, \beta, \gamma, \delta$ are real numbers. This expression substituted for x in the quadric equation would satisfy it identically, and it is thus an algebraical solution; but there is no obvious *a priori* reason why $\sqrt{\gamma + \delta i}$ should have a value $= c + di$, where c and d are real numbers calculable by the extraction of a root or roots of real numbers: however the case is (what there was no *a priori* right to expect) that $\sqrt{\gamma + \delta i}$ has such a value calculable by means of the radical expressions $\sqrt[4]{\gamma^2 + \delta^2} \pm \gamma$: and hence the algebraical solution of a numerical quadric equation does in every case give the numerical solution. The case of a numerical cubic equation will be considered presently.

§ 26.

A cubic equation can be solved by radicals. Taking for greater simplicity the cubic in the reduced form $x^3 + qx - r = 0$, and assuming $x = a + b$, this will be a solution if only $3ab = q$ and $a^3 + b^3 = r$, equations which give $(a^3 - b^3)^2 = r^2 - \frac{4}{27}q^3$, a quadric equation solvable by radicals, and giving $a^3 - b^3 = \sqrt{r^2 - \frac{4}{27}q^3}$, a two-valued function of the coefficients: combining this with $a^3 + b^3 = r$, we have $a^3 = \frac{1}{2}(r + \sqrt{r^2 - \frac{4}{27}q^3})$, a two-valued function: we then have a by means of a cube root, viz.

$$a = \sqrt[3]{\tfrac{1}{2}(r + \sqrt{r^2 - \tfrac{4}{27}q^3})},$$

a six-valued function of the coefficients; but then, writing $q = 3ab$, we have, as may be shown, $a+b$ a three-valued function of the coefficients; and $x = a+b$ is the required solution by radicals. It would have been wrong to complete the solution by writing

$$b = \sqrt[3]{\tfrac{1}{2}(r - \sqrt{r^2 - \tfrac{4}{27}q^3})},$$

for then $a+b$ would have been given as a 9-valued function having only 3 of its values roots, and the other 6 values being irrelevant. Observe that in this last process we make no use of the equation $3ab = q$, in its original form, but use only the derived equation $27a^3b^3 = q^3$, implied in, but not implying, the original form.

An interesting variation of the solution is to write $x = ab(a+b)$, giving $a^3b^3(a^3+b^3) = \frac{q^3}{27}$ and $3a^3b^3 = q^3$, or say $a^3+b^3 = \frac{3r}{q^3}$, $a^3b^3 = \frac{q^3}{27}$; and consequently

$$a^3 = \tfrac{1}{2}\left(\frac{3r}{q^3} + \sqrt{\frac{9r^2}{q^6} - \frac{4q^3}{27}}\right), \quad b^3 = \tfrac{1}{2}\left(\frac{3r}{q^3} - \sqrt{\frac{9r^2}{q^6} - \frac{4q^3}{27}}\right),$$

i.e., here a^3, b^3 are each of them a two-valued function, but as the only effect of altering the sign of the quadric radical is to interchange a^3, b^3, they may be regarded as each of them one-valued; a and b are each of them 3-valued (for observe that here only a^3b^3, not ab, is given); and $ab(a+b)$ thus is in appearance a 9-valued function, but it can easily be shown that it is (as it ought to be) only 3-valued.

In the case of a numerical cubic, even when the coefficients are real, substituting their values in the expression

$$x = \sqrt[3]{\tfrac{1}{2}(r + \sqrt{r^2 - \tfrac{4}{27}q^3})} + \sqrt[3]{\tfrac{1}{2}(r - \sqrt{r^2 - \tfrac{4}{27}q^3})},$$

this may depend on an expression of the form $\sqrt[3]{\gamma + \delta i}$, where γ and δ are real numbers (it will do so if $r^2 - \frac{4}{27}q^3$ is a negative number), and then we cannot by the extraction of any root or roots of real positive numbers reduce $\sqrt[3]{\gamma + \delta i}$ to the form $c + di$, c and d real numbers; hence here the algebraical solution does not give the numerical solution, and we have here the so-called "irreducible case" of a cubic equation. By what precedes, there is nothing in this that might not have been expected: the algebraical solution makes the solution depend on the extraction of the cube root of a negative number, and there was no reason for expecting this to be a real number. It is well known that the case in question is that wherein the three roots of the numerical cubic equation are all real; if the roots are two imaginary, one real, then contrariwise the quantity under the cube root is real; and the algebraical solution gives the numerical one.

The irreducible case is solvable by a trigonometrical formula, but this is not a solution by radicals: it consists, in effect, in reducing the given numerical cubic (not to a cubic of the form $z^3 = a$, solvable by the extraction of a cube root, but) to a cubic of the form $4z^3 - 3z = a$, corresponding to the equation $4\cos^3\theta - 3\cos\theta = \cos 3\theta$ which serves to determine $\cos\theta$ when $\cos 3\theta$ is known. The theory is applicable to an algebraical cubic equation: say that such an equation, if it can be reduced to the form $4z^3 - 3z = a$, is solvable by "trisection": then the general cubic equation is solvable by trisection.

§ 27.

A quartic equation is solvable by radicals: and it is to be remarked that the existence of such a solution depends on the existence of 3-valued functions such as $ab + cd$ of the four roots (a, b, c, d): by what precedes, $ab + cd$ is the root of a cubic equation, which equation is solvable by radicals: hence $ab + cd$ can be found by radicals; and since $abcd$ is a given function, ab and cd can then be found by radicals. But by what precedes, if ab be known then any similar function, say $a + b$, is obtainable rationally; and then from the values of $a + b$ and ab we may by radicals obtain the value of a or b, that is, an expression for the root of the given quartic equation: the expression ultimately obtained is 4-valued, corresponding to the different values of the several radicals which enter therein, and we have thus the expression by radicals of each of the four roots of the quartic equation. But when the quartic is numerical the same thing happens as in the cubic, and the algebraical solution does not in every case give the numerical one.

It will be understood, from the foregoing explanation as to the quartic, how in the next following case, that of the quintic, the question of the solvability by radicals depends on the existence or non-existence of 5-valued functions of the five roots (a, b, c, d, e); the fundamental theorem is the one already stated, a rational function of five letters, if it has less than 5, cannot have more than 2 values, that is, there are no 3-valued or 4-valued functions of 5 letters; and by reasoning depending in part upon this theorem, Abel (1824) showed that a general quintic equation is not solvable by radicals; and *a fortiori* the general equation of any order higher than 5 is not solvable by radicals.

§ 28.

The general theory of the solvability of an equation by radicals depends fundamentally on Vandermonde's remark (1770) that, supposing an equation is solvable by radicals, and that we have therefore an algebraical expression of x in terms of the coefficients, then substituting for the coefficients their values in terms of the roots, the resulting expression must reduce itself to any one at pleasure of the roots $a, b, c, \ldots$; thus in the case of the quadric equation, in the expression $x = \frac{1}{2}(p + \sqrt{p^2 - 4q})$, substituting for p and q their values, and observing that $(a + b)^2 - 4ab = (a - b)^2$, this becomes $x = \frac{1}{2}\{a + b + \sqrt{(a - b)^2}\}$, the value being a or b according as the radical is taken to be $+(a - b)$ or $-(a - b)$.

So in the cubic equation $x^3 - px^2 + qx - r = 0$, if the roots are a, b, c, and if ω is used to denote an imaginary cube root of unity, $\omega^2 + \omega + 1 = 0$, then writing for shortness $p = a + b + c$, $L = a+\omega b+\omega^2 c$, $M = a+\omega^2 b+\omega c$, it is at once seen that LM, L^3+M^3, and therefore also $(L^3 - M^3)^2$ are symmetrical functions of the roots, and consequently rational functions of the coefficients: hence

$$\tfrac{1}{2}\{L^3 + M^3 + \sqrt{(L^3 - M^3)^2}\}$$

is a rational function of the coefficients, which when these are replaced by their values as functions of the roots becomes, according to the sign given to the quadric radical, $= L^3$ or M^3: taking it $= L^3$, the cube root of the expression has the three values L, ωL, $\omega^2 L$; and LM divided by the same cube root has therefore the values M, $\omega^2 M$, ωM; whence finally the expression

$$\tfrac{1}{3}\{p + LM^2 \div \sqrt[3]{\tfrac{1}{2}(L^3 + M^3 + \sqrt{(L^3 - M^3)^2})}\}$$

has the three values

$$\tfrac{1}{3}(p + L + M), \quad \tfrac{1}{3}(p + \omega L + \omega^2 M), \quad \tfrac{1}{3}(p + \omega^2 L + \omega M);$$

that is, these are $= a$, b, c respectively. If the value M^3 had been taken instead of L^3, then the expression would have had the same three values a, b, c. Comparing the solution given for the cubic $x^3+qx-r=0$, it will readily be seen that the two solutions are identical, and that the function $r^2 - \frac{4}{27}q^3$ under the radical sign must (by aid of the relation $p = 0$ which subsists in this case) reduce itself to $(L^3 - M^3)^2$; it is only by each radical being equal to a rational function of the roots that the final expression can become equal to the roots a, b, c respectively.

§ 29.

The formulae for the cubic were obtained by Lagrange (1770–71) from a different point of view. Upon examining and comparing the principal known methods for the solution of algebraical equations, he found that they all ultimately depended upon finding a "resolvent" equation of which the root is $a + \omega b + \omega^2 c + \omega^3 d + \ldots$, ω being an imaginary root of unity, of the same order as the equation: e.g. for the cubic the root is $a + \omega b + \omega^2 c$, ω an imaginary cube root of unity. Evidently the method gives for L^3 a quadric equation, which is the "resolvent" equation in this particular case.

For a quartic the formulae present themselves in a somewhat different form, by reason that 4 is not a prime number. Attempting to apply it to a quintic, we seek for the equation of which the root is $(a + \omega b + \omega^2 c + \omega^3 d + \omega^4 e)$, ω an imaginary fifth root of unity, or rather the fifth power thereof $(a + \omega b + \omega^2 c + \omega^3 d + \omega^4 e)^4$; this is a 24-valued function, but if we consider the four values corresponding to the roots of unity ω, ω^2, ω^3, ω^4, viz. the values

$$\begin{gathered}(a + \omega b + \omega^2 c + \omega^3 d + \omega^4 e)^4,\\(a + \omega^2 b + \omega^4 c + \omega d + \omega^3 e)^4,\\(a + \omega^3 b + \omega c + \omega^4 d + \omega^2 e)^4,\\(a + \omega^4 b + \omega^3 c + \omega^2 d + \omega e)^4,\end{gathered}$$

any symmetrical function of these, for instance their sum, is a six-valued function of the roots, and may therefore be determined by means of a sextic equation, the coefficients whereof are rational functions of the coefficients of the original quintic equation; the conclusion being that the solution of an equation of the fifth order is made to depend upon that of an equation of the sixth order. This is, of course, useless for the solution of the quintic equation, which, as already mentioned, does not admit of solution by radicals; but the equation of the sixth order, Lagrange's resolvent sextic, is very important, and is intimately connected with all the later investigations in the theory.

§ 30.

It is to be remarked, in regard to the question of solvability by radicals, that not only the coefficients are taken to be arbitrary, but it is assumed that they are represented each by a single letter, or say rather that they are not so expressed in terms of other arbitrary quantities as to make a solution possible. If the coefficients are not all arbitrary, for instance, if some of them are zero, a sextic equation might be of the form $ax^6 + bx^4 + cx^2 + d = 0$, and so be solvable as a cubic; or if the coefficients of the sextic are given functions of the six arbitrary quantities a, b, c, d, e, f, such that the sextic is really of the form

$$(x^2 + ax + b)(x^4 + cx^3 + dx^2 + ex + f) = 0,$$

then it breaks up into the equations $x^2 + ax + b = 0$, $x^4 + cx^3 + dx^2 + ex + f = 0$, and is consequently solvable by radicals; so also if the form is

$$(x - a)(x - b)(x - c)(x - d)(x - e)(x - f) = 0,$$

then the equation is solvable by radicals,—in this extreme case rationally. Such cases of solvability are self-evident; but they are enough to show that the general theorem of the non-solvability by radicals of an equation of the fifth or any higher order does not in any wise exclude for such orders the existence of particular equations solvable by radicals, and there are, in fact, extensive classes of equations which are thus solvable; the binomial equations $x^n - 1 = 0$ present an instance.

It has already been shown how the several roots of the equation $x^n - 1 = 0$ can be expressed in the form $\cos\frac{2\pi}{n} + i\sin\frac{2\pi}{n}$, but the question is now that of the algebraical solution (or solution by radicals) of this equation. There is always a root $= 1$; if ω be any other root, then obviously $\omega, \omega^2, \ldots, \omega^{n-1}$ are all of them roots: $x^n - 1$ contains the factor $x-1$, and it thus appears that $\omega, \omega^2, \ldots, \omega^{n-1}$ are the $n-1$ roots of the equation

$$x^{n-1} + x^{n-2} + \ldots + x + 1 = 0;$$

we have, of course,

$$\omega^{n-1} + \omega^{n-2} + \ldots + \omega + 1 = 0.$$

It is proper to distinguish the cases n prime and n composite; and in the latter case there is a distinction of n simple or multiple according as the prime factors are.

By way of illustration, suppose successively $n = 15$ and $n = 9$: in the former case, if α be an imaginary root of $x^3 - 1 = 0$ (or root of $x^2 + x + 1 = 0$), and β an imaginary root of $x^5 - 1 = 0$ (or root of $x^4 + x^3 + x^2 + x + 1 = 0$), then ω may be taken $= \alpha\beta$; the successive powers thereof, $\alpha\beta$, $\alpha^2\beta^2$, β^3, $\alpha\beta^4$, $\alpha^2\beta^5$, $\alpha\beta^6$, $\alpha^2\beta^7$, β^8, α, $\alpha^2\beta$, β^2, are the roots of $x^{14} + x^{13} + \ldots + x + 1 = 0$; the solution thus depends on the solution of the equations $x^3 - 1 = 0$ and $x^5 - 1 = 0$. In the latter case, if α be an imaginary root of $x^3 - 1 = 0$ (or root of $x^2 + x + 1 = 0$), then the equation $x^9 - 1 = 0$ gives $x^3 = 1$, α, or α^2; $x^3 = 1$ gives $x = 1$, α, or α^2; and the solution thus depends on the solution of the equations $x^3 - 1 = 0$, $x^3 - \alpha = 0$, $x^3 - \alpha^2 = 0$. The first equation has the roots 1, α, α^2; if β be a root of either of the others, say if $\beta^3 = \alpha$, then assuming $\omega = \beta$, the successive powers are β, β^2, α, $\alpha\beta$, $\alpha\beta^2$, α^2, $\alpha^2\beta$, $\alpha^2\beta^2$, which are the roots of the equation $x^8 + x^7 + \ldots + x + 1 = 0$.

It thus appears that the only case which need be considered is that of n a prime number, and writing (as is more usual) r in place of ω, we have r, r^2, r^3, $\ldots$, r^{n-1} as the $(n-1)$ roots of the reduced equation

$$x^{n-1} + x^{n-2} + \ldots + x + 1 = 0;$$

then not only $r^n - 1 = 0$, but also $r^{n-1} + r^{n-2} + \ldots + r + 1 = 0$.

§ 31.

The process of solution due to Gauss (1801) depends essentially on the arrangement of the roots in a certain order, viz. not as above, with the indices of r in arithmetical progression, but with their indices in geometrical progression; the prime number n has a certain number of prime roots g, which are such that g^{n-1} is the lowest power of

g which is $= 1$ to the modulus n: or, what is the same thing, that the series of powers $1, g, g^2, \ldots, g^{n-2}$, each divided by n, leave (in a different order) the remainders $1, 2, 3, \ldots, n-1$; hence giving to r in succession the indices $1, g, g^2, \ldots, g^{n-2}$, we have, in a different order, the whole series of roots $r, r^g, r^{g^2}, \ldots, r^{g^{n-2}}$.

In the most simple case, $n = 5$, the equation to be solved is $x^4 + x^3 + x^2 + x + 1 = 0$; here 2 is a prime root of 5, and the order of the roots is r, r^2, r^4, r^3. The Gaussian process consists in forming an equation for determining the periods P_1, $P_2 = r + r^4$ and $r^2 + r^3$ respectively, these being such that the symmetrical functions $P_1 + P_2$, $P_1 P_2$ are rationally determinable: in fact,

$$P_1 + P_2 = -1, \quad P_1 P_2 = (r + r^4)(r^2 + r^3) = r^3 + r^4 + r^6 + r^7 = r^3 + r^4 + r + r^2 = -1.$$

P_1, P_2 are thus the roots of $u^2 + u - 1 = 0$; and taking them to be known, they are themselves broken up into subperiods, in the present case single terms, r and r^4 for P_1, r^2 and r^3 for P_2; the symmetrical functions of these are then rationally determined in terms of P_1 and P_2; thus $r + r^4 = P_1$, $r \cdot r^4 = 1$, or r, r^4 are the roots of $x^2 - P_1 x + 1 = 0$. The mode of division is more clearly seen for a larger value of n; thus, for $n = 7$ a prime root is $= 3$, and the arrangement of the roots is r, r^3, r^2, r^6, r^4, r^5. We may form either 3 periods each of 2 terms, $P_1, P_2, P_3 = r + r^6$, $r^3 + r^4$, $r^2 + r^5$ respectively; or else 2 periods each of 3 terms, $P_1, P_2 = r + r^3 + r^2$, $r^6 + r^4 + r^5$ respectively; in each case the symmetrical functions of the periods are rationally determinable; thus in the case of the two periods $P_1 + P_2 = -1$, $P_1 P_2 = 3 + r + r^3 + r^2 + r^6 + r^4 + r^5 = 2$; and, the periods being known, the symmetrical functions of the several terms of each period are rationally determined in terms of the periods, thus

$$r + r^3 + r^2 = P_1, \quad r \cdot r^3 + r \cdot r^2 + r^3 \cdot r^2 = P_2, \quad r \cdot r^3 \cdot r^2 = 1.$$

The theory was further developed by Lagrange (1808), who, applying his general process to the equation in question, $x^{n-1} + x^{n-2} + \ldots + x + 1 = 0$, the roots $a, b, c, \ldots$ being the several powers of r, the indices in geometrical progression as above, showed that the function $(a + \omega b + \omega^2 c + \ldots)^{n-1}$ was in this case a given function of ω with integer coefficients. Reverting to the before-mentioned particular equation $x^4 + x^3 + x^2 + x + 1 = 0$, it is very interesting to compare the process of solution with that for the solution of the general quartic the roots whereof are a, b, c, d.

Take ω, a root of the equation $\omega^4 - 1 = 0$ (whence ω is $= 1$, -1, i, or $-i$, at pleasure), and consider the expression

$$(a + \omega b + \omega^2 c + \omega^3 d)^4.$$

The developed value of this is

$$\begin{gathered}
a^4 + b^4 + c^4 + d^4 + 6(a^2b^2 + c^2d^2) + 12(a^2bd + b^2ca + c^2db + d^2ac) \\
+ \omega\{4(a^3b + b^3c + c^3d + d^3a) + 12(a^2cd + b^2da + c^2ab + d^2bc)\} \\
+ \omega^2\{6(a^2c^2 + b^2d^2) + 4(a^3c + b^3d + c^3a + d^3b) + 24abcd\} \\
+ \omega^3\{4(a^3d + b^3a + c^3b + d^3c) + 12(a^2bc + b^2cd + c^2da + d^2ab)\};
\end{gathered}$$

that is, this is a 6-valued function of a, b, c, d, the root of a sextic (which is, in fact, solvable by radicals; but this is not here material).

If, however, a, b, c, d denote the roots r, r^2, r^4, r^3 of the special equation, then the expression becomes

$$\begin{gathered}
r^4 + r^8 + r + r^9 + 6(1 + 1) + 12(r^6 + r^4 + r^7 + r^3) \\
+ \omega\{4(1 + 1 + 1 + 1) + 12(r^6 + r^4 + r + r^9)\} \\
+ \omega^2\{6(r + r^8 + r^4 + r^3) + 4(r^6 + r^4 + r^7 + r^3)\} \\
+ \omega^3\{4(r + r^8 + r^4 + r^9) + 12(r^6 + r + r^3 + r^4)\},
\end{gathered}$$

viz. this is $= -1 + 4\omega + 14\omega^2 - 16\omega^3$, a completely determined value. That is, we have

$$(r + \omega r^2 + \omega^2 r^4 + \omega^3 r^3)^4 = -1 + 4\omega + 14\omega^2 - 16\omega^3,$$

which result contains the solution of the equation. If $\omega = 1$, we have $(r + r^2 + r^4 + r^3)^4 = 1$, which is right; if $\omega = -1$, then $(r + r^4 - r^2 - r^3)^4 = 25$; if $\omega = i$, then we have $\{r - r^4 + i(r^2 - r^3)\}^4 = -15 + 20i$; and if $\omega = -i$, then $\{r - r^4 - i(r^2 - r^3)\}^4 = -15 - 20i$; the solution may be completed without difficulty.

The result is perfectly general, thus: n being a prime number, r a root of the equation $x^{n-1} + x^{n-2} + \ldots + x + 1 = 0$, ω a root of $\omega^{n-1} - 1 = 0$, and g a prime root of $g^{n-1} \equiv 1 \pmod{n}$, then

$$(r + \omega r^g + \ldots + \omega^{n-2} r^{g^{n-2}})^{n-1}$$

is a given function $M_0 + M_1\omega + \ldots + M_{n-2}\omega^{n-2}$ with integer coefficients, and by the extraction of $(n-1)$th and similar roots we ultimately obtain r in terms of ω, which is taken to be known: the equation $x^n - 1 = 0$, n a prime number, is thus solvable by radicals. In particular, if $n - 1$ be a power of 2, the solution (by either process) requires the extraction of square roots only: and it was thus that Gauss discovered that it was possible to construct geometrically the regular polygons of 17 sides and 257 sides respectively. Some interesting developments in regard to the theory were obtained by Jacobi (1837); see the memoir "Ueber die Kreistheilung, u.s.w.," *Crelle*, t. xxx. (1846).

§ 32.

The equation $x^{n-1} + \ldots + x + 1 = 0$ has been considered for its own sake, but it also serves as a specimen of a class of equations solvable by radicals, considered by Abel (1828), and since called *Abelian equations*, viz., for the Abelian equation of the order n, if x be any root, the roots are x, θx, $\theta^2 x$, ..., $\theta^{n-1} x$ (θx being a rational function of x, and $\theta^n x = x$); the theory is, in fact, very analogous to that of the above particular case. A more general theorem obtained by Abel is as follows:—If the roots of an equation of any order are connected together in such wise that all the roots can be expressed rationally in terms of any one of them, say x; if, moreover, θx, $\theta^2 x$ being any two of the roots, we have $\theta\theta^2 x = \theta^2\theta x$, the equation will be solvable algebraically. It is proper to refer also to Abel's definition of an irreducible equation:—an equation $\phi x = 0$, the coefficients of which are rational functions of a certain number of known quantities $a, b, c, \ldots$, is called irreducible when it is impossible to express its roots by an equation of an inferior degree, the coefficients of which are also rational functions of $a, b, c, \ldots$ (or, what is the same thing, when ϕx does not break up into factors which are rational functions of $a, b, c, \ldots$). Abel applied his theory to the equations which present themselves in the division of the elliptic functions, but not to the modular equations.

§ 33.

But the theory of the algebraical solution of equations in its most complete form was established by Galois (born October 1811, killed in a duel May 1832; see his collected works, *Liouville*, t. xi., 1846). The definition of an irreducible equation resembles Abel's,—an equation is reducible when it admits of a rational divisor, irreducible in the contrary case; only the word rational is used in this extended sense that, in connexion with the coefficients of the given equation, or with the irrational quantities (if any) whereof these are composed, he considers any number of other irrational quantities called "adjoint radicals," and he terms rational any rational function of the coefficients (or the irrationals whereof they are composed) and of these adjoint radicals; the epithet irreducible is thus taken either absolutely or in a relative sense, according to the system of adjoint radicals which are taken into account. For instance, the equation $x^4+x^3+x^2+x+1=0$; the left-hand side has here no rational divisor, and the equation is irreducible; but this function is $=(x^2+\frac{1}{2}x+1)^2-\frac{5}{4}x^2$, and it has thus the irrational divisors $x^2+\frac{1}{2}(1+\sqrt{5})x+1$, $x^2+\frac{1}{2}(1-\sqrt{5})x+1$ and these, if we adjoin the radical $\sqrt{5}$, are rational, and the equation is no longer irreducible.

In the case of a given equation, assumed to be irreducible, the problem to solve the equation is, in fact, that of finding radicals by the adjunction of which the equation becomes reducible; for instance, the general quadric equation $x^2+px+q=0$ is irreducible, but it becomes reducible, breaking up into rational linear factors, when we adjoin the radical $\sqrt{\frac{1}{4}p^2-q}$.

The fundamental theorem is the Proposition I. of the "Mémoire sur les conditions de resolubilité des Equations par radicaux"; viz. given an equation of which $a, b, c, \ldots$ are the roots, there is always a group of permutations of the letters $a, b, c, \ldots$ possessed of the following properties:

1. Every function of the roots invariable by the substitutions of the group is rationally known.

2. Reciprocally, every rationally determinable function of the roots is invariable by the substitutions of the group.

Here by an invariable function is meant not only a function of which the form is invariable by the substitutions of the group, but further, one of which the value is invariable by these substitutions: for instance, if the equation be $\phi x=0$, then ϕx is a function of the roots invariable by any substitution whatever. And in saying that a function is rationally known, it is meant that its value is expressible rationally in terms of the coefficients and of the adjoint quantities.

For instance, in the case of a general equation, the group is simply the system of the $1{\cdot}2{\cdot}3\ldots n$ permutations of all the roots, since, in this case, the only rationally determinable functions are the symmetric functions of the roots.

In the case of the equation $x^{n-1}+\ldots+x+1=0$, n a prime number, $a, b, c, \ldots, k = r, r^g, r^{g^2}, \ldots, r^{g^{n-2}}$, where g is a prime root of n, then the group is the cyclical group $abc\ldots k$, $bc\ldots ka$, $\ldots$, $kab\ldots j$, that is, in this particular case the number of the permutations of the group is equal to the order of the equation.

This notion of the group of the original equation, or of the group of the equation as varied by the adjunction of a series of radicals, seems to be the fundamental one in Galois's theory. But the problem of solution by radicals, instead of being the sole object of the theory, appears as the first link of a long chain of questions relating to the transformation and classification of irrationals.

Returning to the question of solution by radicals, it will be readily understood that by the adjunction of a radical the group may be diminished; for instance, in the case of the general cubic, where the group is that of the six permutations, by the adjunction of the square root which enters into the solution, the group is reduced to abc, bca, cab; that is, it becomes possible to express rationally, in terms of the coefficients and of the adjoint square root, any function such as $a^2b+b^2c+c^2a$ which is not altered by the cyclical substitution a into b, b into c, c into a. And hence, to determine whether an equation of a given form is solvable by radicals, the course of investigation is to inquire whether, by the successive adjunction of radicals, it is possible to reduce the original group of the equation so as to make it ultimately consist of a single permutation.

The condition in order that an equation of a given prime order n may be solvable by radicals was in this way obtained—in the first instance in the form, scarcely intelligible without further explanation, that every function of the roots $x_1, x_2, \ldots, x_n$, invariable by the substitutions $x_s \mapsto ax_s + b$, must be rationally known; and then in the equivalent form that the resolvent equation of the order $1\cdot2\cdot\ldots\cdot(n-2)$ must have a rational root. In particular, the condition in order that a quintic equation may be solvable is that Lagrange's resolvent of the order 6 may have a rational factor, a result obtained from a direct investigation in a valuable memoir by E. Luther, *Crelle*, t. xxxiv. (1847).

Among other results demonstrated or announced by Galois may be mentioned those relating to the modular equations in the theory of elliptic functions; for the transformations of the orders 5, 7, 11, the modular equations of the orders 6, 8, 12 are depressible to the orders 5, 7, 11 respectively; but for the transformation, n a prime number greater than 11, the depression is impossible.

The general theory of Galois in regard to the solution of equations was completed, and some of the demonstrations supplied, by Betti (1852). See also Serret's *Cours d'Algèbre supérieure*, 2nd ed. 1854; 4th ed. 1877–78.

§ 34.

Returning to quintic equations, Jerrard (1835) established the theorem that the general quintic equation is, by the extraction of only square and cubic roots, reducible to the form $x^5 + ax + b = 0$, or what is the same thing, to $x^5 + x + b = 0$. The actual reduction by means of Tschirnhausen's theorem was effected by Hermite in connexion with his elliptic-function solution of the quintic equation (1858) in a very elegant manner. It was shown by Cockle and Harley (1858–59) in connexion with the Jerrardian form, and by Cayley (1861), that Lagrange's resolvent equation of the sixth order can be replaced by a more simple sextic equation occupying a like place in the theory.

The theory of the modular equations, more particularly for the case $n = 5$, has been studied by Hermite, Kronecker, and Brioschi. In the case $n = 5$, the modular equation of the order 6 depends, as already mentioned, on an equation of the order 5 and conversely the general quintic equation may be made to depend upon this modular equation of the order 6: that is, assuming the solution of this modular equation, we can solve (not by radicals) the general quintic equation; this is Hermite's solution of the general quintic equation by elliptic functions (1858); it is analogous to the before-mentioned trigonometrical solution of the cubic equation. The theory is reproduced and developed in Brioschi's memoir, "Ueber die Auflösung der Gleichungen vom fünften Grade," *Math. Annalen*, t. xiii. (1877–78).

The great modern work, reproducing the theories of Galois, and exhibiting the theory of algebraic equations as a whole, is Jordan's *Traité des Substitutions et des Equations Algébriques*, Paris, 1870. The work is divided into four books—book I., preliminary, relating to the theory of congruences; book II. is in two chapters, the first relating to substitutions in general, the second to substitutions defined analytically, and chiefly to linear substitutions; book III. has four chapters, the first discussing the principles of the general theory, the other three containing applications to algebra, geometry, and the theory of transcendents; book IV., divided into seven chapters, contains lastly, a determination of the general types of equations solvable by radicals, and a complete system of classification of these types. A glance through the index will show the vast extent which the theory has assumed, and the form of general conclusions arrived at; thus, in book III., the algebraical applications comprise Abelian equations, equations of Galois; the geometrical ones comprise Hesse's equation, Clebsch's equations, lines on a quartic surface having a nodal line, singular points of Kummer's surface, lines on a cubic surface, problems of contact; the applications to the theory of transcendents comprise circular functions, elliptic functions (including division and the modular equation), hyperelliptic functions, solution of equations by transcendents. And on this last subject, solution of equations by transcendents, we may quote the result, "the solution of the general equation of an order superior to five cannot be made to depend upon that of the equations for the division of the circular or elliptic functions"; and again (but with a reference to a possible case of exception), "the general equation cannot be solved by aid of the equations which give the division of the hyperelliptic functions into an odd number of parts."

§ 787. FUNCTION.

[From the *Encyclopædia Britannica*, Ninth Edition, vol. ix (1879), pp. 818–824.]

Functionality, in Analysis, is dependence on a variable or variables; in the case of a single variable u, it is the same thing to say that v depends upon u, or to say that v is a function of u, only in the latter form of expression the mode of dependence is embodied in the term "function." We have given or known functions such as u^2 or $\sin u$, and the general notation of the form ϕu, where the letter ϕ is used as a functional symbol to denote a function of u, known or unknown as the case may be: in each case u is the independent variable or argument of the function, but it is to be observed that, if v be a function of u, then v like u is a variable, the values of v regarded as known serve to determine those of u: that is, we may conversely regard u as a function of v. In the case of two or more independent variables, say when w depends on or is a function of u, v, &c., or $w = \phi(u, v, \ldots)$, then $u, v, \ldots$ are the independent variables or arguments of the function; frequently when one of these variables, say u, is principally or alone attended to, it is regarded as the independent variable or argument of the function, and the other variables v, &c., are regarded as parameters, the values of which serve to complete the definition of the function. We may have a set of quantities w, t, $\ldots$ each of them a function of the same variables $u, v, \ldots$; and this relation may be expressed by means of a single functional symbol ϕ, $(w, t, \ldots) = \phi(u, v, \ldots)$; but, as to this, more hereafter.

The notion of a function is applicable in geometry and mechanics as well as in analysis; for instance, a point Q, the position of which depends upon that of a variable point P, may be regarded as a function of the point P: but here, substituting for the points themselves the coordinates (of any kind whatever) which determine their positions, we may say that the coordinates of Q are each of them a function of the coordinates of P, and we thus return to the analytical notion of a function. And in what follows a function is regarded exclusively in this point of view, viz. the variables are regarded as numbers; and we attend to the case of a function of one variable $v = fu$. But it has been remarked (see Equation) that it is not allowable to confine the attention to real numbers; a number u must in general be taken to be a complex number $u = x + iy$, x and y being real numbers, each susceptible of continuous variation between the limits $-\infty$, $+\infty$, and i denoting $\sqrt{-1}$. In regard to any particular function, fu, although it may for some purposes be sufficient to know the value of the function for any real value whatever of u, yet to attend only to the real values of u is an essentially incomplete view of the question: to properly know the function, it is necessary to consider u under the aforesaid imaginary or complex form $u = x + iy$.

To a given value $x + iy$ of u there corresponds in general for v a value or values of the like form $v = x' + iy'$, and we obtain a geometrical notion of the meaning of the functional relation $v = fu$ by regarding x, y as rectangular coordinates of a point P in a plane Π, and x', y' as rectangular coordinates of a point P' in a plane (for greater convenience a different plane) Π'; P, P' are thus the geometrical representations, or representative points, of the variables $u = x + iy$ and $v = x' + iy'$ respectively: and, according to a locution above referred to, the point P' might be regarded as a function of the point P: a given value of $u = x + iy$ is thus represented by a point P in the plane Π, and corresponding hereto we have a point or points P' in the plane Π', representing (if more than one, each of them) a value of the variable $v = x' + iy'$. And, if we attend only to the values of u as corresponding to a series of positions of the representative point P, we have the notion of the "path" of a complex variable $u = x + iy$.

Known Functions.

§ 1.

The most simple kind of function is the rational and integral function. We have the series of powers u^2, u^3, ... each calculable not only for a real but also for a complex value of u, $(x + iy)^2 = x^2 - y^2 + 2ixy$, $(x + iy)^3 = x^3 - 3xy^2 + i(3x^2y - y^3)$, &c., and thence, $a, b, \ldots$ being real or complex numbers, the general form $a + bu + cu^2 + \ldots + ku^m$, of a rational and integral function of the order m. And taking two such functions, say of the orders m and n respectively, the quotient of one of these by the other represents the general form of a rational function of u.

The function which next presents itself is the algebraical function, and in particular the algebraical function expressible by radicals. To take the most simple case, suppose (m being a positive integer) that $v^m = u$; v is here the irrational function $= u^{\frac{1}{m}}$. Obviously, if

u is real and positive, there is always a real and positive value of v, calculable to any extent of approximation from the equation $v^m = u$, which serves as the definition of $u^{\frac{1}{m}}$; but it is known (see Equation) that, as well in this case as in the general case where u is a complex number, there are in fact m values of the function $u^{\frac{1}{m}}$; and that for their determination we require the theory of the so-called circular functions sine and cosine; and these depend on the exponential function $\exp u$, or, as it is commonly written, e^u, which has for its inverse the logarithmic function $\log u$; these are all of them transcendental functions.

§ 2.

In a rational and integral function $a+bu+cu^2+\ldots+ku^m$, the number of terms is finite, and the coefficients $a, b, \ldots, k$ may have any values whatever, but if we imagine a like series $a + bu + cu^2 + \ldots$ extending to infinity, *non constat* that such an expression has any calculable value, that is, any meaning at all: the coefficients $a, b, c, \ldots$ must be such as, either for every value whatever of u (that is, for every finite value) or for values included within certain limits, to make the series convergent. It is easy to see that the values of $a, b, c, \ldots$ may be such as to make the series always convergent; for instance, this is the case for the exponential function,

$$\exp u = 1 + u + \frac{u^2}{1 \cdot 2} + \frac{u^3}{1 \cdot 2 \cdot 3} + \&\text{c.};$$

taking for the moment u to be real and positive, then it is evident that however large u may be, the successive terms will become ultimately smaller and smaller, and the series will have a determinate calculable value. A function thus expressed by means of a convergent infinite series is not in general algebraical, and when it is not so, it is said to be transcendental (but observe that it is in nowise true that we have thus the most general form of a transcendental function): in particular, the exponential function above written down is not an algebraical function.

By forming the expression of $\exp v$, and multiplying together the two series, we derive the fundamental property

$$\exp u \cdot \exp v = \exp(u + v);$$

whence also

$$\exp x \cdot \exp iy = \exp(x + iy),$$

so that $\exp(x + iy)$ is given as the product of the two series $\exp x$ and $\exp iy$. As regards this last, if in place of u we actually write the value iy, we find

$$\exp iy = \left(1 - \frac{y^2}{1\cdot 2} + \frac{y^4}{1\cdot 2\cdot 3\cdot 4} - \ldots\right) + i\left(y - \frac{y^3}{1\cdot 2\cdot 3} + \ldots\right),$$

where obviously each series is convergent and actually calculable for any real value whatever of y. Calling the two series cosine y and sine y respectively, or in the ordinary abbreviated notation $\cos y$ and $\sin y$, the equation is

$$\exp iy = \cos y + i \sin y;$$

and if we herein for y write z, and multiply the two expressions together, observing that the product will be $= \exp i(y + z)$, we obtain the fundamental equations

$$\cos(y + z) = \cos y \cos z - \sin y \sin z,$$
$$\sin(y + z) = \sin y \cos z + \sin z \cos y,$$

for the functions sine and cosine.

§ 3.

Taking y as an angle, and defining as usual the sine and cosine as the ratios of the perpendicular and base respectively to the radius, the sine and cosine will be functions of y; and we obtain geometrically the foregoing fundamental equations for the sine and cosine: but in order to the truth of the foregoing equation $\exp iy = \cos y + i \sin y$, it is further necessary that the angle should be measured in circular measure, that is, by the ratio of the arc to the radius; so that π denoting as usual the number $3.14159\ldots$, the measure of a right angle is $= \frac{1}{2}\pi$. And this being so, the functions sine and cosine, obtained as above by consideration of the exponential function, have their ordinary geometrical significations.

§ 4.

The foregoing investigation was given in detail in order to the completion of the theory of the irrational function $u^{\frac{1}{m}}$. We henceforth take the theory of the circular functions as known, and speak of $\tan u$, &c., as the occasion may arise.

We have

$$x + iy = r(\cos\theta + i\sin\theta),$$

where, writing $\sqrt{x^2+y^2}$ to denote the positive value of the square root, we have

$$r = \sqrt{x^2+y^2}, \quad \cos\theta = \frac{x}{\sqrt{x^2+y^2}}, \quad \sin\theta = \frac{y}{\sqrt{x^2+y^2}},$$

and therefore also $\tan\theta = \frac{y}{x}$.

Treating x, y as the rectangular coordinates of a point P, r is the distance (regarded as positive) of this point from the origin, and θ is the inclination of r to the positive part of the axis of x; to fix the ideas θ may be regarded as lying within the limits 0, π, or 0, $-\pi$, according as y is positive or negative; θ is thus completely determinate, except in the case, x negative, $y = 0$, for which θ is $= \pi$ or $-\pi$ indifferently.

And if $u = x + iy$, we hence have

$$(x+iy)^{\frac{1}{m}} = r^{\frac{1}{m}}\left(\cos\frac{\theta+2s\pi}{m} + i\sin\frac{\theta+2s\pi}{m}\right),$$

where $r^{\frac{1}{m}}$ is real and positive and s has any positive or negative integer value whatever: but we thus obtain for $u^{\frac{1}{m}}$ only the m values corresponding to the values $0, 1, 2, \ldots, m-1$ of s. More generally we may, instead of the index $\frac{1}{m}$, take the index to be any rational fraction $\frac{n}{m}$. Supposing this to be in its least terms, and m to be positive, the number of distinct values is always $= m$. If instead of $\frac{n}{m}$ we take the index to be the general real or complex quantity μ, we have u^{μ}, no longer an algebraical function of u, and having in general an infinity of values.

§ 5.

The foregoing equation $\exp(x+y) = \exp x \cdot \exp y$ is, in fact, the equation of indices, $a^{x+y} = a^x \cdot a^y$; $\exp x$ is thus the same thing as e^x, where e denotes a properly determined number, and putting e^x equal to the series, and then writing $x = 1$, we have $e = 1 + 1 + \frac{1}{1\cdot 2} + \frac{1}{1\cdot 2\cdot 3} +$ &c., that is, $e = 2.71828\ldots$ But as well theoretically as for convenience of printing, there is considerable advantage in the use of the notation $\exp u$.

From the equation, $\exp iy = \cos y + i\sin y$, we deduce $\exp(-iy) = \cos y - i\sin y$, and thence

$$\begin{aligned}\cos y &= \tfrac{1}{2}\{\exp(iy) + \exp(-iy)\},\\ \sin y &= \tfrac{1}{2i}\{\exp(iy) - \exp(-iy)\};\end{aligned}$$

if we write herein ix instead of y we have

$$\begin{aligned}\cos ix &= \tfrac{1}{2}\{\exp x + \exp(-x)\},\\ \sin ix &= \tfrac{1}{2i}\{\exp x - \exp(-x)\},\end{aligned}$$

viz. these values are

$$\begin{aligned}\cos ix &= 1 + \frac{x^2}{1\cdot 2} + \frac{x^4}{1\cdot 2\cdot 3\cdot 4} + \dots,\\ \sin ix &= \frac{x}{1} + \frac{x^3}{1\cdot 2\cdot 3} + \dots,\end{aligned}$$

each of them real when x is real. The functions in question $1+\frac{x^2}{1\cdot 2}+\dots$ and $x + \frac{x^3}{1\cdot 2\cdot 3} + \dots$, regarded as functions of x, are termed the hyperbolic cosine and sine, and are represented by the notations $\cosh x$ and $\sinh x$ respectively; and similarly we have the hyperbolic tangent $\tanh x$, &c.: although it is easy to remember that $\cos ix$, $\frac{1}{i}\sin ix$ are, in fact, real functions of x, and to understand accordingly the formulae wherein they occur, yet the use of these notations of the hyperbolic functions is often convenient.

§ 6.

Writing $u = \exp v$, then v is conversely a function of u which is called the logarithm (hyperbolic logarithm, to distinguish it from the tabular or Briggian logarithm), and we write $v = \log u$, or what is the same thing, we have $u = \exp(\log u)$: and it is clear that if u be real and positive there is always a real and positive value of $\log u$, in particular the real logarithm of e is $= 1$; it is however to be observed that the logarithm is not a one-valued function, but has an infinity of values corresponding to the different integer values of a constant s; in fact, if $\log u$ be any one of its values, then $\log u + 2s\pi i$ is also a value, for we have $\exp(\log u + 2s\pi i) = \exp\log u \cdot \exp 2s\pi i$, or since $\exp 2s\pi i$ is $= 1$, this is $= u$; that is, $\log u + 2s\pi i$ is a value of the logarithm of u.

We have

$$uv = \exp(\log uv) = \exp\log u \cdot \exp\log v,$$

and hence the equation which is commonly written

$$\log uv = \log u + \log v,$$

but which requires the addition on one side of a term $2s\pi i$. And reverting to the equation $x+iy = r(\cos\theta + i\sin\theta)$, or as it is convenient to write it, $x+iy = r\exp i\theta$, we hence have

$$\log(x+iy) = \log r + i(\theta + 2s\pi),$$

where $\log r$ may be taken to denote the real logarithm of the real positive quantity r, and θ the completely determinate angle defined as already mentioned.

Reverting to the function $u^{\frac{1}{m}}$, we have $u = \exp\log u$, and thence $u^{\frac{1}{m}} = \exp(\frac{1}{m}\log u)$, which, on account of the infinity of values of $\log u$, has in general (as before remarked) an infinity of values; if $u = e$, then $e^{\frac{1}{m}} = \exp(\frac{1}{m}\log e)$, has in general in like manner an infinity of values, but in regarding $e^{\frac{1}{m}}$ as identical with the one-valued function $\exp\frac{1}{m}$, we take $\log e$ to be $=$ its real value, 1.

The inverse functions $\cos^{-1}x$, $\sin^{-1}x$, $\tan^{-1}x$, are in fact logarithmic functions; thus in the equation $\exp ix = \cos x + i\sin x$, writing first $\cos x = u$, the equation becomes $\exp(i\cos^{-1}u) = u + \sqrt{u^2-1}$, or we have $\cos^{-1}u = \frac{1}{i}\log(u+\sqrt{u^2-1})$, and from the same equation, writing secondly $\sin x = u$, we have $\sin^{-1}u = \frac{1}{i}\log(\sqrt{1-u^2}+iu)$. But the formula for $\tan^{-1}u$ is a more elegant one, as not involving the radical $\sqrt{1-u^2}$; we have

$$\tan x = \frac{1}{i}\frac{\exp ix - \exp(-ix)}{\exp ix + \exp(-ix)} = \frac{1}{i}\frac{\exp 2ix - 1}{\exp 2ix + 1},$$

and thence $\exp 2ix = \dfrac{1+i\tan x}{1-i\tan x}$; that is,

$$x = \frac{1}{2i}\log\frac{1+i\tan x}{1-i\tan x},$$

or, if $\tan x = u$, then

$$\tan^{-1}u = \frac{1}{2i}\log\frac{1+iu}{1-iu}.$$

The logarithm (or inverse exponential function) and the inverse circular functions present themselves as the integrals of algebraic functions:

$$\int \frac{dx}{x} = \log x,$$

whence also $\int \frac{dx}{1+x^2} = \frac{1}{2i} \log \frac{1+ix}{1-ix} = \tan^{-1} x$, and $\int \frac{dx}{\sqrt{1-x^2}} = \sin^{-1} x$.

§ 7.

Each of the functions $\exp u$, $\sin u$, $\cos u$, $\tan u$, &c., is a one-valued function of u: in this respect analogous to a rational function of u; and there are further analogies of $\exp u$, $\sin u$, $\cos u$, to a rational and integral function; and of $\tan u$, $\sec u$, &c., to a rational non-integral function.

A rational and integral function has a certain number of roots, or zeros each of a given multiplicity, and is completely determined (except as to a constant factor) when the several roots and the multiplicity of each of them is given; i.e., if $a, b, c, \dots$ be the roots, $p, q, r, \dots$ their multiplicities, then the form is $a(x-a)^p(x-b)^q \dots$; a rational (non-integral) function has a certain number of infinities, or "poles," each of them of a given multiplicity; viz. the infinities are the roots or zeros of the rational and integral function which is its denominator.

The function $\exp u$ has no finite roots, but an infinity of roots each $= \infty$; this appears from the equation $\exp u = \left(1 + \frac{u}{n}\right)^n$, where n is indefinitely large and positive. The function $\sin u$ has the roots $s\pi$, where s is any positive or negative integer, zero included; or, what is the same thing, its roots are 0 and $\pm s\pi$, s now denoting any positive integer from 1 to ∞; each of these is a simple root, and we in fact have

$$\sin u = u \prod_s \left(1 - \frac{u^2}{s^2\pi^2}\right).$$

Similarly the roots of $\cos u$ are $(s + \frac{1}{2})\pi$, s denoting any positive or negative integer, zero included, or, what is the same thing, they are $\pm(s + \frac{1}{2})\pi$, s now denoting any positive integer from 1 to ∞; each root is simple, and we have

$$\cos u = \prod_s \left(1 - \frac{u^2}{(s+\frac{1}{2})^2\pi^2}\right).$$

Obviously $\tan u$, as the quotient $\sin u \div \cos u$, has both roots and infinities, its roots being the roots of $\sin u$, its infinities the roots of $\cos u$; $\sec u$, as the reciprocal of $\cos u$, has infinities only, these being the roots of $\cos u$, &c.

§ 8.

In the foregoing expression $\sin u = u \prod_s (1 - \frac{u^2}{s^2\pi^2})$, the product must be understood to mean the limit of $\prod_{-n}^{n}(1 - \frac{u^2}{s^2\pi^2})$ for an indefinitely large positive integer value of n, viz. the product is first to be formed for the values $s = 1, 2, 3, \ldots$ up to a determinate number n, and then n is to be taken indefinitely large. If, separating the positive and the negative values of s, we consider the product $u \prod_1^m (1 + \frac{u}{s\pi}) \cdot \prod_1^n (1 - \frac{u}{s\pi})$, where in the first product s has all the positive integer values from 1 to m, and in the second product s has all the positive integer values from 1 to n, then by making m and n each of them indefinitely large, the function does not approximate to $\sin u$ unless $m : n$ be a ratio of equality*.

* The value of the function in question $u \prod_1^m (1 + \frac{u}{s\pi}) \cdot \prod_1^n (1 - \frac{u}{s\pi})$, when m, n are each indefinitely large, but $\frac{m}{n}$ not $= 1$, is $= \left(\frac{m}{n}\right)^{\frac{u}{\pi}} \sin u$.

§ 9.

The functions $\sin u$, $\cos u$, are periodic, having the period 2π, $\cos(u + 2\pi) = \cos u$; and the half-period π, $\cos(u + \pi) = -\cos u$; the periodicity may be verified by means of the foregoing fractional forms, but some attention is required; thus writing, as we may do, $\sin u = \frac{u \prod(u \pm s\pi)}{\cdots}$, where s extends from $-n$ to n, n ultimately infinite, if for u we write $u + \pi$, each factor of the numerator is changed into the following one and the numerator is unaltered, save only that there is an introduced factor $u + (n+1)\pi$ at the superior limit, and an omitted factor $u - n\pi$ at the inferior limit; the ratio of these, $(u + n + 1\pi) \div (u - n\pi)$, for n infinite is $= -1$, and we thus have, as we should have, $\sin(u + \pi) = -\sin u$.

The most general periodic function having no infinities, and each root a simple root, and having a given period a, has the form $A \sin \frac{2\pi u}{a} + B \cos \frac{2\pi u}{a}$, or, what is the same thing, $C \sin\left(\frac{2\pi u}{a} + D\right)$.

§ 10.

We come now to the Elliptic Functions. These arose from the consideration of the integral $\int \frac{dx}{\sqrt{X}}$, where R is a rational function of x, and X is the general rational and integral quartic function $\alpha x^4 + \beta x^3 + \gamma x^2 + \delta x + \epsilon$; a form arrived at was

$$\int \frac{dx}{\sqrt{1-x^2 \cdot 1-k^2x^2}} = \int \frac{d\phi}{\sqrt{1-k^2\sin^2\phi}},$$

on putting therein $x = \sin\phi$, and this last integral was represented by $F\phi$, and called the elliptic integral of the first kind. In the particular case $k = 0$, the integral is $= \sin^{-1} x$, and it thus appears that $F\phi$ is of the nature of an inverse function: for passing to the direct functions we write $F\phi = u$, and consider ϕ as hereby determined as a function of u, $\phi =$ amplitude of u, or for shortness u. And the functions $\sin\phi$, $\cos\phi$, and $\sqrt{1-k^2\sin^2\phi}$ were then considered as functions of the amplitude, and written $\sin u$, $\cos u$, Δu: these were afterwards written $\mathrm{sn}u$, $\mathrm{cn}u$, $\mathrm{dn}u$, which may be regarded either as mere abbreviations of the former functional symbols, or (in a different point of view) as functions, no longer of u, but of u itself as the argument of the functions: sn is thus a function in some respects analogous to a sine, and cn and dn functions analogous to a cosine; they have the corresponding property that the three functions of $u+v$ are expressible in terms of the functions of u and of v. The following formulae may be mentioned:

$$\mathrm{cn}^2u = 1 - \mathrm{sn}^2u, \quad \mathrm{dn}^2u = 1 - k^2\mathrm{sn}^2u,$$
$$\mathrm{sn}'u = \mathrm{cn}u\mathrm{dn}u, \quad \mathrm{cn}'u = -\mathrm{sn}u\mathrm{dn}u, \quad \mathrm{dn}'u = -k^2\mathrm{sn}u\mathrm{cn}u,$$

where the accent denotes differentiation in regard to u; and the addition-formulae

$$\begin{aligned}\mathrm{sn}(u+v) &= \mathrm{sn}u\mathrm{cn}v\mathrm{dn}v + \mathrm{sn}v\mathrm{cn}u\mathrm{dn}u,\\ \mathrm{cn}(u+v) &= \mathrm{cn}u\mathrm{cn}v - \mathrm{sn}u\mathrm{dn}u\mathrm{sn}v\mathrm{dn}v,\\ \mathrm{dn}(u+v) &= \mathrm{dn}u\mathrm{dn}v - k^2\mathrm{sn}u\mathrm{cn}u\mathrm{sn}v\mathrm{cn}v,\end{aligned}$$

each of the expressions on the right-hand side being the numerator of a fraction of which

$$\text{Denom.} = 1 - k^2\mathrm{sn}^2u\mathrm{sn}^2v.$$

It may be remarked that any one of the fractional expressions, differentiated in regard to u and to v respectively, gives the same result; such expression is therefore a function of $u + v$, and the addition-formulae can be thus directly verified.

§ 11.

The existence of a denominator in the addition-formulae suggests that sn, cn, dn are not, like the sine and cosine, functions having zeros only, without infinities; they are in fact functions, having each its own zeros, but having a common set of infinities; moreover, the zeros and the infinities are simple zeros and infinities respectively. And this further suggests, what in fact is the case, that the three functions are quotients having each its own numerator but a common denominator, say they are the quotients of four ϑ-functions, each of them having zeros only (and these simple zeros) but no infinities.

The functions sn, cn, dn, but not the ϑ-functions, are moreover doubly periodic; that is, there exist values 2ω, 2ν, $= 4K$ and $4i(K + iK')$ in the ordinary notation, such that the sn, cn, or dn of $u + 2\omega$, and the sn, cn, and dn of $u + 2\nu$ are equal to the sn, cn, and dn respectively of u; or say that $\phi(u + 2\omega) = \phi(u + 2\nu) = \phi u$, where ϕ is any one of the three functions.

As regards this double periodicity, it is to be observed that the equations $\phi(u+2\omega) = \phi u$, $\phi(u+2\nu) = \phi u$, imply $\phi(u+2m\omega+2n\nu) = \phi u$, and hence it easily follows that if ω, ν were commensurable, say if they were multiples of some quantity σ, we should have $\phi(u + 2\sigma) = \phi u$, an equation which would replace the original two equations $\phi(u + 2\omega) = \phi u$, $\phi(u + 2\nu) = \phi u$, or there would in this case be only the single period σ; ω and ν must therefore be incommensurable. And this being so, they cannot have a real ratio, for if they had, the integer values m, n could be taken such as to make $2m\omega + 2n\nu = k$ times a given real or imaginary value, k as small as we please; the ratio $\omega : \nu$ must be therefore imaginary, as is in fact the case when the values are $4K$ and $4i(K + iK')$.

§ 12.

The function sn$u$ has the zero u and the zeros $m\omega + n\nu$, m and n any positive or negative integers whatever; and this suggests that the numerator of sn$u$ is equal to a doubly infinite product, (Cayley, “On the Inverse Elliptic Functions," *Camb. Math. Jour.* t. iv., 1845, [24]; and "Mémoire sur les fonctions doublement périodiques," *Liouville*, t. x., 1845, [25]). The numerator is equal to

$$u \prod_{m,n} \left(1 - \frac{u}{m\omega + n\nu}\right),$$

m and n having any positive or negative integer values whatever, including zero, except that m, n must not be simultaneously $= 0$, these values being taken account of in the factor u outside the product. But until further defined, such a product has no definite value, and consequently no meaning whatever. Imagine m, n to be coordinates, and suppose that we have, surrounding the origin, a closed curve having the origin for its centre, i.e. the curve is such that, if α, β be the coordinates of a point thereof, then $-\alpha, -\beta$ are also the coordinates of a point thereof; suppose further that the form of the curve is given, but that its magnitude depends upon a parameter h, and that the curve is such that, when h is indefinitely large, each point of the curve is at an indefinitely large distance from the origin; for instance, the curve might be a circle or ellipse, or a parallelogram, the origin being in each case the centre. Then if in the double product, taking the value of h as given, we first give to m, n all the positive or negative integer values (the simultaneous values $0, 0$ excluded) which correspond to points within the curve, and then make h indefinitely large, the product will thus have a definite value: but this value will still be dependent on the form of the curve.

Moreover, varying in any manner the form of the curve, the ratio of the two values of the double product will be $= \exp \beta u^2$, where β is a determinate value depending only on the forms of the two curves; or, what is the same thing, if we first give to the curve a certain form, say we take it to be a circle, and then give it any other form, the product in the latter case is equal to its former value multiplied by $\exp \beta u^2$, where β depends only upon the form of the curve in the latter case.

Considering the form of the bounding curve as given, and writing the double product in the form

$$u \prod_{m,n} \left(1 - \frac{u}{m\omega + n\nu}\right),$$

the simultaneous values $m = 0, n = 0$ being now admitted in the numerator, although still excluded from the denominator, then if we write for instance $u+2\omega$ instead of u, each factor in the numerator is changed into a contiguous factor, and the numerator remains unaltered, except that we introduce certain factors which lie outside the bounding curve, and omit certain factors which lie inside the bounding curve; we, in fact, affect the result by a singly infinite series of factors belonging to points adjacent to the bounding curve: and it appears on investigation that we thus introduce a constant factor $\exp\gamma(u+\omega)$. The final result thus is that the product

$$u\prod\left(1-\frac{u}{m\omega+n\nu}\right)$$

does not remain unaltered when u is changed into $u+2\omega$, but that it becomes therefore affected with a constant factor, $\exp\gamma(u+\omega)$. And similarly the function does not remain unaltered when u is changed into $u+2\nu$, but it becomes affected with a factor, $\exp\delta(u+\nu)$. The bounding curve may however be taken such that the function is unaltered when u is changed into $u+2\omega$: this will be the case if the curve is a rectangle such that the length in the direction of the axis of m is infinitely great in comparison of that in the direction of the axis of n; or it may be taken such that the function is unaltered when u is changed into $u+2\nu$: this will be so if the curve be a rectangle such that the length in the direction of the axis of n is indefinitely great in comparison with that in the direction of the axis of m; but the two conditions cannot be satisfied simultaneously.

§ 13.

We have three other like functions, viz. writing for shortness $\bar{m}$, $\bar{n}$ to denote $m+\frac{1}{2}$, $n+\frac{1}{2}$ respectively, and (m,n) to denote $m\omega+n\nu$, then the four functions are

$$\vartheta_1=\nu\prod_{m,n}\left(1-\frac{u}{(m,n)}\right),\quad \vartheta_2=\nu\prod_{m,n}\left(1-\frac{u+\frac{1}{2}\omega}{(m,n)}\right),$$
$$\vartheta_3=\nu\prod_{m,n}\left(1-\frac{u+\frac{1}{2}\nu}{(m,n)}\right),\quad \vartheta_4=\nu\prod_{m,n}\left(1-\frac{u+\frac{1}{2}\omega+\frac{1}{2}\nu}{(m,n)}\right),$$

the bounding curve being in each case the same; and, dividing the first three of these each by the last, we have (except as to constant factors) the three functions $\operatorname{sn}u$, $\operatorname{cn}u$, $\operatorname{dn}u$; writing in each of the four functions $u+2\omega$ or $u+2\nu$ in place of u, the functions acquire each of them the same exponential factor $\exp\gamma(u+\omega)$, or $\exp\delta(u+\nu)$, and the quotient of any two of them, and therefore each of the functions $\mathrm{sn}u$, $\mathrm{cn}u$, $\mathrm{dn}u$, remains unaltered.

It is easily seen that, disregarding constant factors, the four ϑ-functions are in fact one and the same function with different arguments, or they may be written ϑu, $\vartheta(u+\frac{1}{2}\omega)$, $\vartheta(u+\frac{1}{2}\nu)$, $\vartheta(u+\frac{1}{2}\omega+\frac{1}{2}\nu)$; by what precedes, the functions may be so determined that they shall remain unaltered when u is changed into $u+2\omega$, that is, be singly periodic, but that the change u into $u+2\nu$ shall affect them each with the same exponential factor $\exp\delta(u+\nu)$.

§ 14.

Taking the last-mentioned property as a definition of the function ϑ, it appears that ϑu may be expressed as a sum of exponentials

$$\vartheta u = \sum_m A \exp \frac{}{}(\nu m^2 + um),$$

where the summation extends to all positive and negative integer values of m, including zero. In fact, if we first write herein $u+2\omega$ instead of u, then in each term the index of the exponential is altered by $-2\omega m$, $= 2\nu m i$, and the term itself thus remains unaltered; that is, $\vartheta(u+2\omega)=\vartheta u$. But writing $u+2\nu$ in place of u, each term is changed into the succeeding term, multiplied by the factor $\exp \frac{}{}(m+1)\nu$; in fact, making the change in question u into $u+2\nu$, and writing also $m-1$ in place of m, νm^2+um becomes $\nu(m-1)^2+(u+2\nu)(m-1)$, $=\nu m^2+um-u-\nu$, and we thus have

$$\vartheta(u+2\nu) = \exp \frac{}{}(u+\nu)\cdot\vartheta u.$$

In order to the convergency of the series it is necessary that $\exp \frac{}{}\nu m^2$ should vanish for indefinitely large values of m, and this will be so if $\frac{}{}\nu$ be a complex quantity of the form $\alpha+\beta i$, α negative; for instance, this will be the case if ω be real and positive and ν be $= i$ multiplied by a real and positive quantity. The original definition of ϑ as a double product seems to put more clearly in evidence the real nature of this function, but the new definition has the advantage that it admits of extension to the ϑ-functions of two or more variables.

The elliptic functions sn$u$, cn$u$, dn$u$, have thus been expressed each of them as the quotient of two ϑ-functions, but the question arises to express conversely a ϑ-function by means of the elliptic functions; the form is found to be

$$\vartheta u = C \exp \left(Au^2 + B \int \mathrm{sn}^2 u \, du \right),$$

viz. ϑu is expressible as an exponential, the index of which depends on the double integral $\int \mathrm{sn}^2 u \, du$.

The object has been to explain the general nature of the elliptic functions sn$u$, cn$u$, dn$u$, and of the ϑ-functions with which they are thus intimately connected; it would be out of place to go into the theories of the multiplication, division, and transformation of the elliptic functions, or into the theory of the elliptic integrals, and of the application of the ϑ-functions to the representation of the elliptic integrals of the second and third kinds.

§ 15.

The reasoning which shows that for a doubly periodic function the ratio of the two periods 2ω, 2ν is imaginary shows that we cannot have a function of a single variable, which shall be triply periodic, or of any higher order of periodicity. For if the periods of a triply periodic function $\phi(u)$ were 2ω, 2ν, 2χ, then m, n, p being any positive or negative integer values, we should have $\phi(u+2m\omega+2n\nu+2p\chi) = \phi u$; ω, ν, χ must be incommensurable, for if not, the three periods would really reduce themselves to two periods, or to a single period; and being incommensurable, it would be possible to determine the integers m, n, p in such manner that the real part and also the coefficient of i of the expression $m\omega + n\nu + p\chi$ shall be each of them as small as we please, say $< \epsilon$; then $\phi(u+\epsilon) = \phi u$, and thence $\phi(u+k\epsilon) = \phi u$ (k an integer), and $k\epsilon$ as near as we please to any given real or imaginary value whatever. We have thus the nugatory result ϕu = a constant, or at least the function if not a constant is a function of an infinitely and perpetually discontinuous kind, a conception of which can hardly be formed. But a function of two variables may be triply or quadruply periodic: viz. we may have a function $\phi(u, v)$ having for u, v the simultaneous periods 2ω, $2\omega'$; 2ν, $2\nu'$; 2χ, $2\chi'$; or, what is the same thing, it may be such that, m, n, p, q being any integers whatever, we have

$$\phi(u{+}2m\omega{+}2n\nu{+}2p\chi{+}2q\psi,\ v{+}2m\omega'{+}2n\nu'{+}2p\chi'{+}2q\psi') = \phi(u, v);$$

and similarly a function of $2n$ variables may be $2n$-tuply periodic.

It is, in fact, in this manner that we pass from the elliptic functions and the single ϑ-functions to the hyperelliptic or Abelian functions and the multiple ϑ-functions; the case next succeeding the elliptic functions is when we have X, Y the same rational and integral sextic functions of x, y respectively, and then writing

$$\int \frac{dx}{\sqrt{X}} + \int \frac{dy}{\sqrt{Y}} = du, \quad \int \frac{x\,dx}{\sqrt{X}} + \int \frac{y\,dy}{\sqrt{Y}} = dv,$$

we regard certain symmetrical functions of x, y, in fact, the ratios of ($2^4 =$) 16 such symmetrical functions, as functions of (u, v); say we thus have 15 hyperelliptic functions $f(u, v)$, analogous to the 3 elliptic functions sn$u$, cn$u$, dn$u$, and being quadruply periodic. And these are the quotients of 16 double ϑ-functions $\vartheta(u, v)$, the general form being

$$\vartheta(u, v) = \sum\sum A \exp \overline{-}(a, b, b)(m, n)^2 + mu + nv,$$

where the summations extend to all positive and negative integer values of (m, n): and we thus see the form of the ϑ-function for any number of variables whatever. The epithet “hyperelliptic” is used in the case where the differentials are of the form just mentioned $\frac{dx}{\sqrt{X}}$, where X is a rational and integral function of x; the epithet “Abelian” extends to the more general case where the differential involves the irrational function of x, determined by any rational and integral equation $\phi(x, y) = 0$ whatever.

As regards the literature of the subject, it may be noticed that the various memoirs by Riemann, 1851–1866, are republished in the collected edition of his works, Leipsic, 1876; and shortly after his death we have the *Theorie der Abel'schen Functionen*, by Clebsch and Gordan, Leipsic, 1866. Preceding this, we have by MM. Briot and Bouquet, the *Théorie des Fonctions doublement périodiques et en particulier des Fonctions Elliptiques*, Paris, 1859, the results of which are reproduced and developed in their larger work, *Théorie des Fonctions Elliptiques*, 2nd ed., Paris, 1875.

§ 16.

It is proper to mention the gamma (Γ) or Π function, $\Gamma(n+1) = \Pi n = 1\cdot 2\cdot 3\ldots n$, when n is a positive integer. In the case just referred to, n a positive integer, this presents itself almost everywhere in analysis,—for instance, the binomial coefficients, and the coefficients of the exponential series are expressible by means of such functions of a number n. The definition for any real positive value of n is taken to be

$$\Pi n = \int_0^\infty x^n \exp(-x)\, dx;$$

it is then shown that, n being real and positive, $\Gamma(n+1) = n\Gamma n$, and by assuming that this equation holds good for positive or negative real values of n, the definition is extended to real negative values; the equation gives $\Pi 1 = 0\cdot\Pi 0$, that is, $\Pi 0 = \infty$; and similarly $\Pi(-n) = \infty$, where $-n$ is any negative integer. The definition by the definite integral has been or may be extended to imaginary values of n, but the theory is not an established one. A definition extending to all values of n is that of Gauss

$$\Pi n = \text{limit } \frac{k^n\cdot k!}{(n+1)(n+2)(n+3)\ldots(n+k)},$$

the ultimate value of k being $=\infty$; but the function is chiefly considered for real values of the variable.

A formula for the calculation, when x has a large real and positive value, is

$$\Pi x = \sqrt{2\pi}\, x^{x+\frac{1}{2}} \exp(-x + \frac{1}{12x} - \ldots),$$

or as this may also be written, neglecting the negative powers of x,

$$\Pi x = \sqrt{2\pi} \exp\{(x+\tfrac{1}{2})\log x - x\}.$$

Another formula is $\Pi x\cdot\Pi(-x) = \dfrac{\pi x}{\sin\pi x}$: or, as this may also be written,

$$\Pi(x-1)\Pi(-x) = \frac{\pi}{\sin\pi x}.$$

It is to be observed that the function Π serves to express the product of a set of factors in arithmetical progression; we have

$$(x+a)(x+2a)\ldots(x+ma) = a^m\Pi\left(\frac{x}{a}+1\right)\Pi\left(\frac{x}{a}+2\right)\ldots\left(\frac{x}{a}+m\right) = a^m\frac{\Pi(\frac{x}{a}+m)}{\Pi\frac{x}{a}}.$$

We can consequently express by means of it the product of any number of the factors which present themselves in the factorial expression of $\sin u$. Starting from the form

$$\sin u = u \prod_{s} \left(1 - \frac{u^2}{s^2\pi^2}\right),$$

where Π is here as before the sign of a product of factors corresponding to the different integer values of s, this is thus converted into

$$\sin u = u\Pi\left(-\frac{u}{s\pi}\right)\Pi\left(\frac{u}{s\pi}\right),$$

or as this may also be written,

$$\sin u = u\Pi\left(-\frac{u}{\pi}\right)\Pi\left(\frac{u}{\pi}\right)\Pi\left(-\frac{u}{2\pi}\right)\Pi\left(\frac{u}{2\pi}\right)\dots,$$

which, in virtue of

$$\Pi(-\xi)\Pi\xi = \frac{\pi\xi}{\sin \pi\xi},$$

becomes

$$\sin u = u\Pi\left(-\frac{u}{\pi}\right)\Pi\left(\frac{u}{\pi}\right)\Pi\left(-\frac{u}{2\pi}\right)\Pi\left(\frac{u}{2\pi}\right)\dots = u\frac{\pi u}{\sin \pi u}\dots.$$

Here m and n are large and positive; calculating the second factor by means of the formula for Πx, in this case we have the beforementioned formula

$$\sin u = u \prod_{s=1}^{\infty} \left(1 - \frac{u^2}{s^2\pi^2}\right).$$

The gamma or Π function is the so-called second Eulerian integral; the first Eulerian integral

$$\int_0^1 x^{p-1}(1-x)^{q-1}\,dx = \frac{\Gamma p \cdot \Gamma q}{\Gamma(p+q)},$$

is at once expressible in terms of Γ, and is therefore not a new function to be considered.

§ 17.

We have the function defined by its expression as a hypergeometric series

$$F(\alpha,\beta,\gamma,u)=1+\frac{\alpha\cdot\beta}{1\cdot\gamma}u+\frac{\alpha(\alpha+1)\cdot\beta(\beta+1)}{1\cdot 2\cdot\gamma(\gamma+1)}u^2+\dots,$$

i.e. this expression of the function serves as a definition, if the series be finite or if, being infinite, it is convergent. The function may also be defined as a definite integral; in other words, if, in the integral

$$\int_0^1 x^{\alpha'-1}(1-x)^{\beta'-1}(1-ux)^{-\gamma'}\,dx,$$

we expand the factor $(1-ux)^{-\gamma'}$ in powers of ux, and then integrate each term separately by the formula for the second Eulerian integral, the result is

$$\frac{\Gamma\alpha'\cdot\Gamma\beta'}{\Gamma(\alpha'+\beta')}\left\{1+\frac{\alpha'\cdot\gamma'}{\alpha'+\beta'\cdot 1}u+\dots\right\},$$

which is

$$\frac{\Gamma\alpha'\cdot\Gamma\beta'}{\Gamma(\alpha'+\beta')}\left\{1+\frac{\alpha'}{\alpha'+\beta'}\frac{\gamma'}{1}u+\frac{\alpha'(\alpha'+1)}{\alpha'+\beta'\cdot\alpha'+\beta'+1}\frac{\gamma'(\gamma'+1)}{1\cdot 2}u^2+\dots\right\},$$

or writing α', β', $\gamma'=\alpha$, $\gamma-\alpha$, β respectively, this is

$$\frac{\Gamma\alpha\cdot\Gamma(\gamma-\alpha)}{\Gamma\gamma}F(\alpha,\beta,\gamma,u),$$

so that the new definition is

$$F(\alpha,\beta,\gamma,u)=\frac{\Gamma\gamma}{\Gamma\alpha\cdot\Gamma(\gamma-\alpha)}\int_0^1 x^{\alpha-1}(1-x)^{\gamma-\alpha-1}(1-ux)^{-\beta}\,dx,$$

but this is in like manner only a definition under the proper limitations as to the values of α,β,γ,u. It is not here considered how the definition is to be extended so as to give a meaning to the function $F(\alpha,\beta,\gamma,u)$ for all values, say of the parameters α,β,γ, and of the variable u. There are included a large number of special forms which are either algebraic or circular or exponential, for instance $F(\alpha,\beta,\beta,u)=(1-u)^{-\alpha}$, &c.; or which are special transcendents which have been separately studied, for instance, Bessel's functions, the Legendrian functions X_n presently referred to, series occurring in the development of the reciprocal of the distance between two planets, &c.

§ 18.

There is a class of functions depending upon a variable or variables $x, y, \ldots$ and a parameter n, say the function for the parameter n is X_n, such that the product of two functions having the same variables, multiplied it may be by a given function of the variables, and integrated between given limits, gives a result $= 0$ or not $= 0$, according as the parameters are unequal or equal; $\int \mathfrak{A} \cdot X_m X_n \, dx = 0$, but $\int \mathfrak{A} \cdot X_n^2 \, dx$ not $= 0$; the admissible values of the parameters being either any integer values, or it may be the roots of a determinate algebraical or transcendental equation; and the functions $\mathfrak{A}_n$ may be either algebraical or transcendental. For instance, such a function is $\cos nx$; m and n being integers, we have $\int_0^\pi \cos mx \cos nx \, dx = 0$, but $\int_0^\pi \cos^2 nx \, dx = \frac{1}{2}\pi$.

Assuming the existence of the expansion of a function fx, in a series of multiple cosines, we thus obtain at once the well-known Fourier series, wherein the coefficient of $\cos mx$ is $= \frac{2}{\pi} \int_0^\pi \cos mx \cdot fx \, dx$. The question whether the process is applicable is elaborately discussed in Riemann's memoir (1854), *Ueber die Darstellbarkeit einer Function durch eine trigonometrische Reihe*, No. xii. in the collected works. And again we have the Legendrian functions, which present themselves as the coefficients of the successive powers of a in the development of $(1-2ax+a^2)^{-\frac{1}{2}}$, $X_0 = 1$, $X_1 = x$, $X_2 = \frac{1}{2}(3x^2-1)$, &c.: here m, n being any positive integers, $\int_{-1}^1 X_m X_n \, dx = 0$, but $\int_{-1}^1 X_n^2 \, dx = \frac{2}{2n+1}$: and we have also Laplace's functions, &c.

Functions in General.

§ 19.

In what precedes, a review has been given, not by any means an exhaustive one, but embracing the most important kinds of known functions; but there are questions to be considered in regard to functions in general.

A function of $x + iy$ has been built up by means of analytical operations performed upon $x + iy$, $(x + iy)^2 = x^2 - y^2 + i \cdot 2xy$, &c., and the question next referred to has not arisen. But observe that, knowing $x + iy$, we know x and y, and therefore any two given functions $\phi(x, y)$, $\psi(x, y)$ of x and y: we therefore also know $\phi(x, y) + i\psi(x, y)$, and the question is, whether such a function of x, y (being known when $x + iy$ is known) is to be regarded as a function of $x + iy$; and if not, what is the condition to be satisfied in order that $\phi(x, y) + i\psi(x, y)$ may be a function of $x + iy$. Cauchy at one time considered that the general form was to be regarded as a function of $x + iy$, and he introduced the expression "*fonction monogène*," monogenous function, to denote the more restricted form which is the proper function of $x + iy$.

Consider for a moment the above general form, say $x' + iy' = \phi(x, y) + i\psi(x, y)$, where ϕ, ψ are any real functions of the real variables (x, y); or what is the same thing, assume $x' = \phi(x, y)$, $y' = \psi(x, y)$: if these functions have each or either of them more than one value, we attend only to one value of each of them. We may then as before take x, y to be the coordinates of a point P in a plane Π, and x', y' to be the coordinates of a point P' in a plane Π'. If, for any given values of x, y, the increments of $\phi(x, y)$, $\psi(x, y)$ corresponding to the indefinitely small real increments h, k of x, y be $Ah+Bk$, $Ch+Dk$, A, B, C, D being functions of x, y, then if the new coordinates of P are $x + h$, $y + k$, the new coordinates of P' will be $x'+Ah+Bk$, $y'+Ch+Dk$; or P, P' will respectively describe the indefinitely small straight paths at the inclinations $\tan^{-1}\frac{k}{h}$, $\tan^{-1}\frac{Ch+Dk}{Ah+Bk}$ to the axes of x, x' respectively; calling these angles θ, θ', we have therefore $\tan\theta' = \frac{C+D\tan\theta}{A+B\tan\theta}$. And if $x'+iy'$ be $= f(x+iy)$, a function of $x+iy$, the condition to be satisfied is that the increment of $x'+iy'$ shall be proportional to the increment $h+ik$ of $x+iy$, or say that it shall be $= (\lambda+i\mu)(h+ik)$, λ, μ being functions of x, y, but independent of h, k; we must therefore have $Ah + Bk$, $Ch + Dk = \lambda h - \mu k$, $\mu h+\lambda k$ respectively, that is $A, B, C, D = \lambda, -\mu, \mu, \lambda$ respectively, and the equation for $\tan\theta'$ thus becomes $\tan\theta' = \frac{\mu+\lambda\tan\theta}{\lambda-\mu\tan\theta}$; hence writing $\frac{\mu}{\lambda} = \tan\alpha$, where α is a function of x, y, but independent of h, k, we have $\tan\theta' = \frac{\tan\alpha+\tan\theta}{1-\tan\alpha\tan\theta}$; that is, $\theta' = \alpha + \theta$; or for the given points (x, y), (x', y'), the path of P being at any inclination whatever θ to the axis of x, the path of P' is at the inclination $\theta+$constant angle α to the axis of x; also $(\lambda h - \mu k)^2 + (\mu h + \lambda k)^2 = (\lambda^2 + \mu^2)(h^2 + k^2)$, i.e., the lengths of the paths are in a constant ratio.

The condition may be written $\delta(x'+iy') = (\lambda+i\mu)(\delta x+i\delta y)$; or what is the same thing

$$\frac{dx'}{dx} + i\frac{dy'}{dx} = (\lambda + i\mu)\left(1 + i\frac{dy}{dx}\right),$$

that is,

$$\frac{dx'}{dx} = \lambda - \mu\frac{dy}{dx}, \quad \frac{dy'}{dx} = \mu + \lambda\frac{dy}{dx};$$

consequently

$$\frac{dx'}{dy} + i\frac{dy'}{dy} = (\lambda + i\mu)\left(\frac{dx}{dy} + i\right),$$

that is,

$$\frac{dx'}{dy} = \lambda\frac{dx}{dy} - \mu, \quad \frac{dy'}{dy} = \mu\frac{dx}{dy} + \lambda;$$

as the analytical conditions in order that $x' + iy'$ may be a function of $x + iy$. They obviously imply $\frac{d^2x'}{dx^2} + \frac{d^2x'}{dy^2} = 0$, $\frac{d^2y'}{dx^2} + \frac{d^2y'}{dy^2} = 0$; and if x' be a function of x, y, satisfying the first of these conditions, then $\frac{dx'}{dx}\,dx + \frac{dx'}{dy}\,dy$ is a complete differential, and $\int\left(\frac{dx'}{dx}\,dx + \frac{dx'}{dy}\,dy\right)$ is a function of $x + iy$.

§ 20.

We have, in what just precedes, the ordinary behaviour of a function $\phi(x + iy)$ in the neighbourhood of the value $x + iy$ of the argument or point $x + iy$; or say the behaviour in regard to a point $x + iy$ such that the function is in the neighbourhood of this point a continuous function of $x+iy$ (or that the point is not a point of discontinuity): the correlative definition of continuity will be that the function $\phi(x+iy)$, assumed to have at the given point $x + iy$ a single finite value, is continuous in the neighbourhood of this point, when the point $x + iy$ describing continuously a straight infinitesimal element $h + ik$, the point $\phi(x+iy)$ describes continuously a straight infinitesimal element $(\lambda + i\mu)(h + ik)$: or what is really the same thing, when the function $\phi(x + iy)$ has at the point $x + iy$ a differential coefficient.

§ 21.

It would doubtless be possible to give for the continuity of a function $\phi(x + iy)$ a less stringent definition not implying the existence of a differential coefficient; but we have this theory only in regard to the functions of a real variable in memoirs by Riemann, Hankel, du Bois Reymond, Schwarz, Gilbert, Klein, and Darboux. The last-mentioned geometer, in his "Mémoire sur les fonctions discontinues," *Jour. de l'École Normale*, t. iv. (1875), pp. 57–112, gives (after Bonnet) the following definition of a continuous function (observe that we are now dealing with real quantities only):

The function $f(x)$ is continuous for the value $x = x_0$, when, h and k being positive quantities as small as we please and δ any positive quantity at pleasure between 0 and 1, we have, for all the values of δ, $f(x_0 \pm \delta h) - f(x_0)$ less in absolute magnitude than ϵ; and moreover $f(x)$ is continuous through the interval x_0, x_1 ($x_1 > x_0$, that is, nearer $+\infty$) when $f(x)$ is continuous for every value of x between x_0 and x_1, and, h tending to zero through positive values, $f(x_0 + h)$ and $f(x_1 - h)$ tend to the limits $f(x_0)$, $f(x_1)$ respectively. It is possible, consistently with this definition, to form continuous functions not having in any proper sense a differential coefficient, and having other anomalous properties; thus if $a_1, a_2, a_3, \ldots$ be an infinite series of real positive or negative quantities, such that the series Σa_n is absolutely convergent (i.e. the sum $\Sigma \pm a_n$, each term being made positive, is convergent), then the function $\Sigma a_n(\sin n\pi x)^2$ is a continuous function actually calculable for any assumed value of x: but it is shown in the memoir that, taking $x =$ any commensurable value $\frac{m}{n}$ whatever, and then writing $x = \frac{m}{n} + h$, h indefinitely small, the increment of the function is of the form $(k + \epsilon)h^\mu$, k finite, ϵ an indefinitely small quantity vanishing with h; there is thus no term varying with h, nor consequently any differential coefficient. See also Riemann's Memoir, *Ueber die Darstellbarkeit*, &c. (No. xii. in the collected works), already referred to.

§ 22.

It was necessary to allude to the foregoing theory of (as they may be termed) infinitely discontinuous functions; but the ordinary and most important functions of analysis are those which are continuous, except for a finite number (or it may be an infinite number) of points of discontinuity. It is to be observed that a point at which the function becomes infinite is *ipso facto* a point of discontinuity; a value of the variable for which the function becomes infinite is, as already mentioned, said to be an "infinity" (or a "pole") of the function; thus, in the case of a rational function expressed as a fraction in its least terms, if the denominator contains a factor $(x - a)^m$, a a real or imaginary value, m a positive integer, then a is said to be an infinity of the mth order (and in the particular case $m = 1$, it is said to be a simple infinity). The circular functions $\tan x$, $\sec x$ are instances of a function having an infinite number of simple infinities.

A rational function is a one-valued function, and in regard to a rational function the infinities are the only points of discontinuity; but a one-valued function may have points of discontinuity of a character quite distinct from an infinity: for instance, in the exponential function $\exp \frac{1}{u-a}$, where a is real or imaginary, the value $u(=x+iy)=a$, is a point of discontinuity but not an infinity; taking $u=a+\rho e^{i\alpha}$, where ρ is an indefinitely small real positive quantity, the value of the function is $\exp \frac{1}{\rho e^{i\alpha}} = \exp \frac{1}{\rho}(\cos\alpha - i\sin\alpha)$, which is indefinitely large or indefinitely small according as $\cos\alpha$ is positive or negative, and in the separating case $\cos\alpha = 0$, and therefore $\sin\alpha = \pm 1$, it is $= \cos\frac{1}{\rho} \mp i\sin\frac{1}{\rho}$, which is indeterminate. If, instead of $\exp \frac{1}{u-a}$ we consider a linear function

$$\{A + B\exp \tfrac{1}{u-a}\} \cdot \{C + D\exp \tfrac{1}{u-a}\},$$

then writing as before $u=a+\rho e^{i\alpha}$, the value is $=A+C$, or $=B+D$, according as $\cos\alpha$ is negative or positive. As regards the theory of one-valued functions in general, the memoir by Weierstrass, "Zur Theorie der eindeutigen analytischen Functionen," *Berl. Abh.* 1876, pp. 11–60, may be referred to.

§ 23.

A one-valued function *ex vi termini* cannot have a point of discontinuity of the kind next referred to; if the representative point P, moving in any manner whatever, returns to its original position, the corresponding point P' cannot but return to its original position. But consider a many-valued function, say an n-valued function $x'+iy'$, of $x+iy$: to each position of P there correspond n positions, in general all of them different, of P'. But the point P may be such that (to take the most simple case) two of the corresponding points P' coincide with each other, say such a position of P is at V, then (using for greater distinctness a different letter W instead of V) corresponding thereto we have two coincident points (W), and $n-2$ other points W; V is then a branch-point (*Verzweigungspunkt*). Taking for P any point which is not a branch-point, then in the neighbourhood of this value each of the n functions $x'+iy'$ is a continuous function of $x+iy$, and by what precedes, if P describing an infinitely small closed curve (or oval) return to its original position, then each of the corresponding points P' describing a corresponding indefinitely small oval will return to its original position. But imagine the oval described by P to be gradually enlarged, so that it comes to pass through a branch-point V; the ovals described by two of the corresponding points P' will gradually approach each other, and will come to unite at the point (W), each oval then sharpening itself out so that the two form together a figure of eight. And if we imagine the oval described by P to be still further enlarged so as to include within it the point V, then the figure of eight, losing the crossing, will be at first an hour-glass form, or twice-indented oval, and ultimately in form an ordinary oval, but having the character of a twofold oval; i.e. to the oval described by P (and which surrounds the branch-point V) there will correspond this twofold oval, and $n-2$ onefold ovals, in such wise that to a given position of P on its oval there correspond two points, say P'_1, P'_2, on the twofold oval, and $n-2$ points $P'_3, \ldots, P'_n$, each on its own onefold oval. And then as P describing its oval returns to its original position, the point P'_1 describing a portion only of the twofold oval, will pass to the original position of P'_2, while the point P'_2 describing the remaining portion of the twofold oval will pass to the original position of P'_1; the other points $P'_3, \ldots, P'_n$, describing each of them its own onefold oval, will return each of them to its original position. And it is easy to understand how, when the oval described by P surrounds two or more of the branch-points V, the corresponding curves for P' may be a system of manifold ovals, such that the sum of the manifoldness is always $= n$, and to conceive in a general way the behaviour of the corresponding points P and P'.

Writing for a moment $x + iy = u$, $x' + iy' = v$, the branch-points are the points of contact of parallel tangents to the curve $\phi(u, v) = 0$, a line through a cusp (but not a line through a node), being reckoned as a tangent: that is, if this be a curve of the order n and class m, with δ nodes and κ cusps, the number of branch-points is $= m + \kappa$, that is, it is $= n^2 - 2\delta - 2\kappa$, or if $p = \frac{1}{2}(n-1)(n-2) - \delta - \kappa$, be the deficiency, then the number is $= 2n - 2 + 2p$.

To illustrate the theory of the n-valued algebraical function $x' + iy'$ of the complex variable $x + iy$, Riemann introduces the notion of a surface composed of n coincident planes or sheets, such that the transition from one sheet to another is made at the branch-points, and that the n sheets form together a multiply-connected surface, which can by cross-cuts (*Querschnitte*) be converted into a simply-connected surface: the n-valued function $x' + iy'$ becomes thus a one-valued function of $x + iy$, considered as belonging to a point on some determinate sheet of the surface: and upon such considerations he founds the whole theory of the functions which arise from the integration of the differential expressions depending on the n-valued algebraical function (that is, any irrational algebraical function whatever) of the independent variable, establishing as part of the investigation the theory of the multiple ϑ-functions. But it would be difficult to give a further account of these investigations.

The Calculus of Functions.

§ 24.

The so-called Calculus of Functions, as considered chiefly by Herschel and Babbage and De Morgan, is not so much a theory of functions as a theory of the solution of functional equations; or, as perhaps should rather be said, the solution of functional equations by means of known functions, or symbols,—the epithet known being here used in reference to the actual state of analysis. Thus for a functional equation $\phi x + \phi y = \phi(xy)$, taking the logarithm as a known function, the solution is $\phi x = c \log x$; or if the logarithm is not taken to be a known function, then a solution may be obtained by means of the sign of integration $\phi x = c \int \frac{dx}{x}$; but the establishment of the properties of the function logarithm (assumed to be previously unknown) would not be considered as coming within the theory. A class of equations specially considered is where αx, βx, … being given functions of x, the unknown function ϕ is to be determined by means of a given relation between x, ϕx, $\phi\alpha x$, $\phi\beta x, \ldots$; in particular the given relation may be between x, ϕx, $\phi\alpha x$; this can be at once reduced to equations of finite differences; for writing $x = u_n$, $\alpha x = u_{n+1}$, we have $u_{n+1} = \alpha u_n$, giving u_n, and therefore also x, each of them as a function of n; and then writing $\phi x = v_n$, $\phi\alpha x$ will be the same function of $n+1$, $= v_{n+1}$, and the given relation is again an equation of finite differences in v_{n+1}, v_n, and n; we have thus $v_n = \phi x$ as a function of n, that is, of x. As regards the equation $u_{n+1} = \alpha u_n$, considered in itself apart from what precedes, observe that this is satisfied by writing $u_n = \alpha^n(x)$, or the question of solving this equation of finite differences is, in fact, identical with that of finding the nth function $\alpha^n(x)$, where $\alpha(x)$ is a given function of x. It of course depends on the form of $\alpha(x)$ whether this question admits of solution in any proper sense: thus, for a function such as $\log x$, the nth logarithm is expressible in its original form $\log^n x$ $(= \log\log\ldots x)$, and not in any other form. But there are forms, for instance $\alpha x = \frac{A+Bx}{C+Dx}$, where the nth function $\alpha^n x$ is a function of the like form $\alpha^n x = \frac{A_n+B_n x}{C_n+D_n x}$, in which the actual value can be expressed as a function of n; if α be such a form, then $\phi\alpha\phi^{-1}$, whatever ϕ may be, is a like form, for we obviously have $(\phi\alpha\phi^{-1})^n = \phi\alpha^n\phi^{-1}$. The determination of the nth function is, in fact, a leading question in the calculus of functions.

It is to be observed that considering the case of two variables, if for instance $\phi(x,y)$ denote a given function of x, y, the notation $\phi^n(x,y)$ is altogether meaningless: in order to generalize the question, we require an extended notation wherein a single functional symbol is used to denote two functions of the two variables. Thus $\phi(x,y) = \alpha(x,y)$, $\beta(x,y)$, α and β given functions; writing for shortness $x_1 = \alpha(x,y)$, $y_1 = \beta(x,y)$, then $\phi^2(x,y)$ will denote $\phi(x_1,y_1)$, that is, two functions $\alpha(x_1,y_1)$, $\beta(x_1,y_1)$, say these are x_2, y_2; $\phi^3(x,y)$ will denote $\phi(x_2,y_2)$, and so on, so that $\phi^n(x,y)$ will have a determinate meaning. And the like is obviously the case in regard to any number of variables, the single functional symbol denoting in each case a set of functions equal in number to the variables.

§ 788. GALOIS.

[From the *Encyclopædia Britannica*, Ninth Edition, vol. x (1879), p. 48.]

Galois, Evariste (1811–1832), an eminently original and profound French mathematician, born 26th October 1811, killed in a duel May 1832. A necrological notice by his friend M. Auguste Chevalier appeared in the *Revue Encyclopédique*, September 1832, p. 744; and his collected works are published, *Liouville*, t. xi. (1846), pp. 381–444, about fifty of these pages being occupied by researches on the resolubility of algebraic equations by radicals. But these researches, crowning as it were the previous labours of Lagrange, Gauss, and Abel, have in a signal manner advanced the theory, and it is not too much to say that they are the foundation of all that has since been done, or is doing, in the subject. The fundamental notion consists in the establishment of a group of permutations of the roots of an equation, such that every function of the roots invariable by the substitutions of the group is rationally known, and reciprocally that every rationally determinable function of the roots is invariable by the substitutions of the group; some further explanation of the theorem, and in connexion with it an explanation of the notion of an adjoint radical, is given under Equation, No. 32, [786]. As part of the theory (but the investigation has a very high independent value as regards the Theory of Numbers, to which it properly belongs), Galois introduces the notion of the imaginary roots of an irreducible congruence of a degree superior to unity; i.e., such a congruence, $F(x) \equiv 0$ (mod. a prime number p), has no integer root; but what is done is to introduce a quantity i subjected to the condition of verifying the congruence in question, $F(i) \equiv 0$ (mod. p), which quantity i is an imaginary of an entirely new kind, occupying in the theory of numbers a position analogous to that of $\sqrt{-1}$ in algebra.

§ 789. GAUSS.

[From the *Encyclopædia Britannica*, Ninth Edition, vol. x (1879), p. 116.]

Gauss, Carl Friedrich (1777–1855), an eminent German mathematician, was born of humble parents at Brunswick, April 23, 1777, and was indebted for a liberal education to the notice which his talents procured him from the reigning duke. His name became widely known by the publication, in his twenty-fifth year (1801), of the *Disquisitiones Arithmeticæ*. In 1807 he was appointed director of the Göttingen observatory, an office which he retained to his death: it is said that he never slept away from under the roof of his observatory, except on one occasion, when he accepted an invitation from Humboldt to attend a meeting of natural philosophers at Berlin. In 1809 he published at Hamburg his *Theoria Motus Corporum Coelestium*, a work which gave a powerful impulse to the true methods of astronomical observation; and his astronomical workings, observations, calculations of orbits of planets and comets, &c., are very numerous and valuable. He continued his labours in the theory of numbers and other analytical subjects, and communicated a long series of memoirs to the Royal Society of Sciences at Göttingen. His first memoir on the theory of magnetism, *Intensitas vis magneticæ* terrestris ad mensuram absolutam revocata, was published in 1833, and he shortly afterwards proceeded, in conjunction with Professor Wilhelm Weber, to invent new apparatus for observing the earth's magnetism and its changes; the instruments devised by them were the declination instrument and the bifilar magnetometer. With Weber's assistance he erected in 1833 at Göttingen a magnetic observatory free from iron (as Humboldt and Arago had previously done on a smaller scale), where he made magnetic observations, and from this same observatory he sent telegraphic signals to the neighbouring town, thus showing the practicability of an electromagnetic telegraph. He further instituted an association (Magnetische Verein), composed at first almost entirely of Germans, whose continuous observations on fixed term-days extended from Holland to Sicily. The volumes of their publication, *Resultate aus den Beobachtungen des magnetischen Vereins*, extend from 1836 to 1839; and in those for 1838 and 1839 are contained the two important memoirs by Gauss, *Allgemeine Theorie des Erdmagnetismus*, and the *Allgemeine Lehrsätze*—on the theory of forces attracting according to the inverse square of the distance. The instruments and methods thus due to him are substantially those employed in the magnetic observatories throughout the world. He co-operated in the Danish and Hanoverian measurements of an arc and trigonometrical operations (1821–48), and wrote (1843, 1846) the two memoirs *Ueber Gegenstände der höhern Geodäsie*. He co-operated in the Danish and Hanoverian measurements of an arc and trigonometrical operations (1821–48), and wrote (1843, 1846) the two memoirs *Ueber Gegenstände der höhern Geodäsie*. Connected with observations in general we have (1812–26) the memoir *Theoria combinationis observationum erroribus minimis obnoxiæ*, with a second part and a supplement. Another memoir of applied mathematics is the *Dioptrische Untersuchungen*, 1840. Gauss was well versed in general literature and the chief languages of modern Europe, and was a member of nearly all the leading scientific societies in Europe. He died at Göttingen early in the spring of 1855. The centenary of his birth was celebrated (1877) at his native place, Brunswick.

Gauss's collected works have been recently published by the Royal Society of Göttingen, in 7 vols. 4to, Gott., 1863–71, edited by E. J. Schering,—(1) the *Disquisitiones Arithmeticæ*, (2) Theory of Numbers, (3) Analysis, (4) Geometry and Method of Least Squares, (5) Mathematical Physics, (6) Astronomy, and (7) the *Theoria Motus Corporum Coelestium*. They include, besides his various works and memoirs, notices by him of many of these, and of works of other authors in the Göttingische gelehrte Anzeigen, and a considerable amount of previously unpublished matter, *Nachlass*. Of the memoirs in pure mathematics, comprised for the most part in vols. II., III., and IV. (but to these must be added those on Attractions in vol. v.), it may be safely said there is not one which has not signally contributed to the progress of the branch of mathematics to which it belongs, or which would not require to be carefully analysed in a history of the subject. Running through these volumes in order, we have in the second the memoir, *Summatio quarundam, serierum singularium*, the memoirs on the theory of biquadratic residues, in which the notion of complex numbers of the form $a + bi$ was first introduced into the theory of numbers: and included in the *Nachlass* are some valuable tables. That for the conversion of a fraction into decimals (giving the complete period for all the prime numbers up to 997) is a specimen of the extraordinary love which Gauss had for long arithmetical calculations; and the amount of work gone through in the construction of the table of the number of the classes of binary quadratic forms must also have been tremendous. In vol. iii. we have memoirs relating to the proof of the theorem that every numerical equation has a real or imaginary root, the memoirs on the Hypergeometric Series, that on Interpolation, and the memoir *Determinatio Attractionis* in which a planetary mass is considered as distributed over its orbit according to the time in which each portion of the orbit is described, and the question (having an implied reference to the theory of secular perturbations) is to find the attraction of such a ring. In the solution the value of an elliptic function is found by means of the arithmetico-geometrical mean. The *Nachlass* contains further researches on this subject, and also researches (unfortunately very fragmentary) on the lemniscate-function, &c., showing that Gauss was, even before 1800, in possession of many of the discoveries which have made the names of Abel and Jacobi illustrious. In vol. iv. we have the memoir *Allgemeine Auflösung...*, on the graphical representation of one surface upon another, and the *Disquisitiones generales circa superficies curvas.* And in vol. v. we have a memoir On the Attraction of Homogeneous Ellipsoids, and the already mentioned memoir *Allgemeine Lehrsätze...*, on the theory of forces attracting according to the inverse square of the distance.

§ 790. GEOMETRY (ANALYTICAL).

[From the *Encyclopædia Britannica*, Ninth Edition, vol. x (1879), pp. 408–420.]

This will be here treated as a method. The science is Geometry; and it would be possible, analytically, or by the method of coordinates, to develope the truths of geometry in a systematic course. But it is proposed not in any way to attempt this, but simply to explain the method, giving such examples, interesting (it may be) in themselves, as are suitable for showing how the method is employed in the demonstration and solution of theorems or problems.

Geometry is one-, two-, or three-dimensional, or, what is the same thing, it is lineal, plane, or solid, according as the space dealt with is the line, the plane, or ordinary (three-dimensional) space. No more general view of the subject need here be taken: but in a certain sense one-dimensional geometry does not exist, inasmuch as the geometrical constructions for points in a line can only be performed by travelling out of the line into other parts of a plane which contains it, and conformably to the usual practice Analytical Geometry will be treated under the two divisions, Plane and Solid.

It is proposed to consider Cartesian coordinates almost exclusively; for the proper development of the science homogeneous coordinates (three and four in plane and solid geometry respectively) are required: and it is moreover necessary to have the correlative line- and plane-coordinates: and in solid geometry to have the six coordinates of the line. The most comprehensive English works are those of Dr Salmon, *Conic Sections* (5th edition, 1869), *Higher Plane Curves* (2nd edition, 1873), and *Geometry of Three Dimensions* (3rd edition, 1874): we have also, on plane geometry, Clebsch's *Vorlesungen über Geometrie*, posthumous, edited by Dr F. Lindemann, Leipsic, 1875, not yet complete.

I. Plane Analytical Geometry (§§1–25).

§ 1.

It is assumed that the points, lines, and figures considered exist in one and the same plane, which plane, therefore, need not be in any way referred to. The position of a point is determined by means of its (Cartesian) coordinates: i.e. as explained under the article Curve,

[785], we take the two lines $x'Ox$ and $y'Oy$, called the *axes* of x and y respectively, intersecting in a point O called the *origin*, and determine the position of any other point P by means of its coordinates $x = OM$ (or NP), and $y = MP$ (or ON). The two axes are usually (as in fig. 1) at right angles to each other, and the lines PM, PN are then at right angles to the axes of x and y respectively. Assuming a scale at pleasure, the coordinates x, y of a point have numerical values.

It is necessary to attend to the signs: x has opposite signs according as the point is on one side or the other of the axis of y, and similarly y has opposite signs according as the point is on the one side or the other of the axis of x. Using the letters N., E., S., W. as in a map, and considering the plane as divided into four *quadrants* by the axes, the signs are usually taken to be

x	y	quad.
$+$	$+$	N.E.
$+$	$-$	S.E.
$-$	$+$	N.W.
$-$	$-$	S.W.

A point is said to have the coordinates (a, b), and is referred to as *the point* (a, b), when its coordinates are $x = a$, $y = b$; the coordinates x, y of a variable point, or of a point which is for the time being regarded as variable, are said to be *current coordinates*.

§ 2.

It is sometimes convenient to use oblique coordinates; the only difference is that the axes are not at right angles to each other; the lines PM, PN are drawn parallel to the axes of y and x respectively, and the figure $OMPN$ is thus a parallelogram. But in all that follows, the Cartesian coordinates are taken to be rectangular; polar coordinates and other systems will be briefly referred to in the sequel.

§ 3.

If the coordinates (x, y) of a point are not given, but only a relation between them $f(x, y) = 0$, then we have a *curve*. For, if we consider x as a real quantity varying continuously from $-\infty$ to $+\infty$, then, for any given value of x, y has a value or values. If these are all imaginary, there is not any real point; but if one or more of them be real, we have a real point or points, which (as the assumed value of x varies continuously) varies or vary continuously therewith; and the locus of all these real points is *a curve.* The equation completely defines the curve: to trace the curve directly from the equation, nothing else being known, we obtain as above a series of points sufficiently near to each other, and draw the curve through them. For instance, let this be done in a simple case. Suppose $y = 2x - 1$; it is quite easy to obtain and lay down a series of points as near to each other as we please, and the application of a ruler would show that these were in a line: that the curve is a line depends upon something more than the equation itself, viz. the theorem that every equation of the form $y = ax + b$ represents a line: supposing this known, it will be at once understood how the process of tracing the curve may be abbreviated; we have $x = 0$, $y = -1$, and $x = \frac{1}{2}$, $y = 0$; the curve is thus the line passing through these two points. But in the foregoing example the notion of a line is taken to be a known one, and such notion of a line does in fact precede the consideration of any equation of a curve whatever, since the notion of the coordinates themselves rests upon that of a line. In other cases it may very well be that the equation is the definition of the curve; the points laid down, although (as finite in number) they do not actually determine the curve, determine it to any degree of accuracy; and the equation thus enables us to construct the curve.

A curve may be determined in another way; viz. the coordinates x, y may be given each of them as a function of the same variable *parameter*; $x, y = f(\theta)$, $\phi(\theta)$ respectively. Here, giving to θ any number of values in succession, these equations determine the values of x, y, that is, the positions of a series of points on the curve. The ordinary form $y = \phi(x)$, where y is given explicitly as a function of x, is a particular case of each of the other two forms: we have $f(x, y) = y - \phi(x) = 0$; and $x = \theta$, $y = \phi(\theta)$.

§ 4.

As remarked under Curve, [785], it is a useful exercise to trace a considerable number of curves, first taking equations which are purely numerical, and then equations which contain literal constants (representing numbers): the equations most easily dealt with are those wherein one coordinate is given as an explicit function of the other, say $y = \phi(x)$ as above. A few examples are here given, with such explanations as seem proper.

(i) $y = 2x - 1$, as before; it is at once seen that this is a line; and taking it to be so, any two points, for instance, $(0, -1)$ and $(\frac{1}{2}, 0)$, determine the line.

(ii) $y = x^2$. The equation shows that x may be positive or negative, but that y is always positive, and has the same values for equal positive and negative values of x: the curve passes through the origin, and through the points $(\pm 1, 1)$. It is already known that the curve lies wholly above the axis of x. To find its form in the neighbourhood of the origin, give x a small value, $x = \pm 0.1$ or ± 0.01, then y is very much smaller, $= 0.01$ and 0.0001 in the two cases respectively: this shows that the curve touches the axis of x at the origin. Moreover, x may be as large as we please, but when it is large, y is much larger; for instance, $x = 10$, $y = 100$. The curve is a *parabola* (fig. 2).

(iii) $y = x^3$. Here x being positive y is positive, but x being negative y is also negative: the curve passes through the origin, and also through the points $(1, 1)$ and $(-1, -1)$. Moreover, when x is small, $= 0.1$ for example, then not only is y, $= 0.001$, very much smaller than x, but it is also very much smaller than y was for the last-mentioned curve $y = x^2$, that is, in the neighbourhood of the origin the present curve approaches more closely the axis of x. The axis of x is a tangent at the origin, but it is a tangent of a peculiar kind (a stationary or inflexional tangent), cutting the curve at the origin, which is an *inflexion*. The curve is the *cubical parabola* (fig. 3).

(iv) $y^2 = x - 1 \cdot x - 3 \cdot x - 4$. Here $y = 0$ for $x = 1, = 3, = 4$. Whenever $x - 1 \cdot x - 3 \cdot x - 4$ is positive, y has two equal and opposite values: but when $x - 1 \cdot x - 3 \cdot x - 4$ is negative, then y is imaginary. In particular, for x less than 1, or between 3 and 4, y is imaginary, but for x between 1 and 3, or greater than 4, y has two values. It is clear that for x somewhere between 1 and 3, y will attain a maximum: the values of x and y may be found approximately by trial. The curve will consist of an oval and infinite branch, and it is easy to see that, as shown in fig. 4, the curve where it cuts the axis of x cuts it at right angles. It may be further remarked that, as x increases from 4, the value of y will increase more and more rapidly: for instance, $x = 6$, $y^2 = 8$, $x = 10$, $y^2 = 378$, &c., and it is easy to see that this implies that the curve has on the infinite branch two inflexions as shown.

(v) $y^2 = x - c \cdot x - b \cdot x - a$, where $a > b > c$ (that is, a nearer to $+\infty$, c to $-\infty$). The curve has the same general form as in the last figure, the oval extending between the limits $x = c$, $x = b$, the infinite branch commencing at the point $x = a$.

(vi) $y^2 = (x-c)^2(x-a)$. Suppose that in the last-mentioned curve, $y^2 = x - c \cdot x - b \cdot x - a$, b gradually diminishes, and becomes ultimately $= c$. The infinite branch (see fig. 5) changes its form, but not in a very marked manner, and it retains the two inflexions. The oval lies always between the values $x = c$, $x = b$, and therefore its length continually diminishes: it is easy to see that its breadth will also continually diminish: ultimately it shrinks up into a mere point. The curve has thus a *conjugate or isolated point*, or *acnode*. For a direct verification observe that $x = c$, $y = 0$, so that $(c, 0)$ is a point of the curve, but if x is either less than c, or between c and a, y^2 is negative, and y is imaginary.

(vii) $y^2 = (x-c)(x-a)^2$. If in the same curve b gradually increases and becomes ultimately $= a$, the oval and the infinite branch change each of them its form, the oval extending always between the values $x = c$, $x = b$, and thus continually approaching the infinite branch, which begins at $x = a$. The consideration of a few numerical examples, with careful drawing, would show that the oval and the infinite branch as they approach sharpen out each towards the other, the two inflexions on the infinite branch coming always nearer to the point $(a, 0)$,—so that finally, when b becomes $= a$, the curve has the form shown in fig. 6, there being now a double point or node (crunode) at A, and the inflexions on the infinite branch having disappeared.

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790. GEOMETRY. [Continued]

§ 5. Cubical curves (concluded)

In the last four examples, the curve is one of the cubical curves called the
divergent parabolas: (iv) is a mere numerical example of (v), and (vi), (vii), (viii) are in Newton's language the parabola cum ovali, punctata, and nodata respectively. When a, b, c are all equal, or the form is $y^2 = (x-c)^3$, we have a cuspidal form, Newton's parabola cuspidata, otherwise the semicubical parabola.

(viii) As an example of a curve given by an implicit equation, suppose the equation is

$$x^2 + y^2 - 3xy = 0;$$

this is a nodal cubic curve, the node at the origin, and the axes touching the two branches respectively (fig. 7). An easy mode of tracing it is to express x, y each of them in terms of a variable θ,

$$x = \frac{3\theta}{1+\theta^3}, \qquad y = \frac{3\theta^2}{1+\theta^3},$$

but it is instructive to trace the curve directly from its equation.

§ 5. Tangent by algebraical process

5. It may be remarked that the purely algebraical process, which is, in fact, that employed in finding a differential coefficient $\frac{dy}{dx}$, if applied directly to the equation of the curve, determines the point consecutive to any given point of the curve, that is, the direction of the curve at such given point, or, what is the same thing, the direction of the tangent at that point. In fact, if α, β are the coordinates of any point on a curve $f(x,y)=0$, then writing in the equation of the curve $x=\alpha+h$, $y=\beta+k$, and in the resulting equation $f(\alpha+h,\beta+k)=0$, developed in powers of h and k, omitting the term $f(\alpha,\beta)$, which vanishes, and the terms containing the second and higher powers of h, k, we have a linear equation $Ah+Bk=0$, which determines the ratio of the increments h, k. Of course, in the analytical development of the theory, we translate this into the notation of the differential calculus; but the question presents itself, and is thus seen to be solvable, as soon as it is attempted to trace a curve from its equation.

Descriptive and Metrical Geometry

6. A geometrical proposition is either *descriptive* or *metrical*: in the former case it is altogether independent of the idea of magnitude (length, inclination, &c.); in the latter case it has reference to this idea. It is to be noticed that, although the method of coordinates seems to be by its inception essentially metrical, and we can hardly, except by metrical considerations, connect an equation with the curve which it represents (for instance, even assuming it to be known that an equation $Ax + By + C = 0$ represents a line, yet if it be asked what line, the only form of answer is, that it is the line cutting the axes at distances from the origin $-C \div A$, $-C \div B$ respectively), yet in dealing by this method with descriptive propositions, we are, in fact, eminently free from all metrical considerations.

7. It is worth while to illustrate this by the instance of the well-known theorem of the radical centre of three circles. The theorem is that, given any three circles A, B, C (fig. 8), the common chords $\alpha\alpha'$, $\beta\beta'$, $\gamma\gamma'$ of the three pairs of circles meet in a point.

The geometrical proof is metrical throughout:

Take the point of intersection of $\alpha\alpha'$, $\beta\beta'$, and joining this with γ', suppose that $\gamma' O$ does not pass through γ, but that it meets the circles A, B in two distinct points γ_1, γ_2 respectively. We have then the known metrical property of intersecting chords of a circle; viz. in the circle C, where $\alpha\alpha'$, $\beta\beta'$ are chords meeting at a point O,

$$O\alpha \cdot O\alpha' = O\beta \cdot O\beta',$$

where, as well as in what immediately follows $O\alpha$, &c., denote, of course, lengths or distances.

Similarly in the circle A,

$$O\beta \cdot O\beta' = O\gamma_1 \cdot O\gamma',$$

and in the circle B,

$$O\alpha \cdot O\alpha' = O\gamma_2 \cdot O\gamma'.$$

Consequently $O\gamma_1 \cdot O\gamma' = O\gamma_2 \cdot O\gamma'$, that is, $O\gamma_1 = O\gamma_2$, or the points γ_1 and γ_2 coincide; that is, they each coincide with γ.

PAGE 553. We contrast this with the analytical method.

Here it only requires to be known that an equation $Ax + By + C = 0$ represents a line, and an equation $x^2 + y^2 + Ax + By + C = 0$ represents a circle. A, B, C have, in the two cases respectively, metrical significations but these we are not concerned with. Using S to denote the function $x^2+y^2+Ax+By+C$, the equation of a circle is $S = 0$, where S stands for its value; more briefly, we say the equation is $S = x^2+y^2+Ax+By+C = 0$. Let the equation of any other circle be $S' = x^2 + y^2 + A'x + B'y + C' = 0$; the equation $S - S' = 0$ is a linear equation: $S-S'$ is, in fact, $= (A-A')x+(B-B')y+C-C'$: and it thus represents a line; this equation is satisfied by the coordinates of each of the points of intersection of the two circles (for at each of these points $S = 0$ and $S' = 0$, therefore also $S - S' = 0$); hence the equation $S - S' = 0$ is that of the line joining the two points of intersection of the two circles, or say it is the equation of the common chord of the two circles. Considering then a third circle $S'' = x^2 + y^2 + A''x + B''y + C'' = 0$, the equations of the common chords are $S - S' = 0$, $S - S'' = 0$, $S' - S'' = 0$ (each of these a linear equation); at the intersection of the first and second of these lines $S = S'$ and $S = S''$, therefore also $S' = S''$, or the equation of the third line is satisfied by the coordinates of the point in question; that is, the three chords intersect in a point O, the coordinates of which are determined by the equations $S = S' = S''$.

It further appears that, if the two circles $S = 0$, $S' = 0$ do not intersect in any real points, they must be regarded as intersecting in two imaginary points, such that the line joining them is the real line represented by the equation $S - S' = 0$; or that two circles, whether their intersections be real or imaginary, have always a real common chord (or radical axis), and that for any three circles the common chords intersect in a point (of course real) which is the radical centre. And by this very theorem, given two circles with imaginary intersections, we can, by drawing circles which meet each of them in real points, construct the radical axis of the first-mentioned two circles.

8. The principle employed in showing that the equation of the common chord of two circles is $S - S' = 0$ is one of very extensive application, and some more illustrations of it may be given.

Suppose $S = 0$, $S' = 0$ are lines, that is, let S, S' now denote linear functions $Ax + By + C$, $A'x + B'y + C'$, then $S - kS' = 0$ (k an arbitrary constant) is the equation of any line passing through the point of intersection of the two given lines. Such a line may be made to pass through any given point, say the point (x_0, y_0); i.e., if S_0, S'_0 are what S, S' respectively become on writing for (x, y) the values (x_0, y_0), then the value of k is $k = S_0 \div S'_0$. The equation in fact is $SS'_0 - S_0S' = 0$; and starting from this equation we at once verify it *a posteriori*; the equation is a linear equation satisfied by the values of (x, y) which make $S = 0$, $S' = 0$; and satisfied also by the values (x_0, y_0); and it is thus the equation of the line in question.

If, as before, $S = 0$, $S' = 0$ represent circles, then (k being arbitrary) $S - kS' = 0$ is the equation of any circle passing through the two points of intersection of the two circles; and to make this pass through a given point (x_0, y_0) we have again $k = S_0 \div S'_0$. In the particular case $k = 1$, the circle becomes the common chord; more accurately, it becomes the common chord together with the line infinity, but this is a question which is not here gone into.

If S denote the general quadric function,

$$S = ax^2 + 2hxy + by^2 + 2fy + 2gx + c = (a, b, c, f, g, h(x, y, 1))^2,$$

then the equation $S = 0$ represents a conic; assuming this, then, if $S' = 0$ represents another conic, the equation $S - kS' = 0$ represents any conic through the four points of intersection of the two conics.

Returning to the equation

$$Ax + By + C = 0$$

of a line, if this pass through two given points (x_1, y_1), (x_2, y_2), then we must have $Ax_1 + By_1 + C = 0$, $Ax_2 + By_2 + C = 0$, equations which determine the ratios $A : B : C$, and it thus appears that the equation of the line through the two given points is

$$x(y_1 - y_2) - y(x_1 - x_2) + x_1y_2 - x_2y_1 = 0;$$

or, what is the same thing,

$$\begin{vmatrix} x & y & 1 \\ x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \end{vmatrix} = 0.$$

PAGE 555. **9.** The object still being to illustrate the mode of working with coordinates, we consider the theorem of the polar of a point in regard to a circle. Given a circle and a point (fig. 9), we draw through any two lines meeting the circle in the points A, A' and B, B' respectively, and then taking Q as the intersection of the lines AB' and $A'B$, the theorem is that the locus of the point Q is a right line depending only upon O and the circle, but independent of the particular lines OAA' and OBB'.

Taking O as the origin, and for the axes any two lines through O at right angles to each other, the equation of the circle will be

$$x^2 + y^2 + 2Ax + 2By + C = 0;$$

and if the equation of the line OAA' is taken to be $y = mx$, then the points A, A' are found as the intersections of the straight line with the circle or to determine x we have

$$x^2(1 + m^2) + 2x(A + Bm) + C = 0.$$

PAGE 555 (CONTINUED). If (x_1, y_1) are the coordinates of A, and (x_2, y_2) of A', then the roots of this equation are x_1, x_2, whence easily

$$\frac{1}{x_1} + \frac{1}{x_2} = \frac{2(A + Bm)}{C}.$$

And similarly, if the equation of the line OBB' is taken to be $y = m'x$, and the coordinates of B, B' to be (x_3, y_3) and (x_4, y_4) respectively, then

$$\frac{1}{x_3} + \frac{1}{x_4} = \frac{2(A + Bm')}{C}.$$

We have then

$$\begin{aligned} x(y_1 - y_4) - y(x_1 - x_4) + x_1y_4 - x_4y_1 = 0, \\ x(y_2 - y_3) - y(x_2 - x_3) + x_2y_3 - x_3y_2 = 0, \end{aligned}$$

as the equations of the lines AB' and $A'B$ respectively; for the first of these equations, being satisfied if we write therein (x_1, y_1) or (x_4, y_4) for (x, y), is the equation of the line AB': and similarly the second equation is that of the line $A'B$. Reducing by means of the relations

$y_1 = mx_1$, $y_2 = mx_2$, $y_3 = m'x_3$, $y_4 = m'x_4$, the two equations become

$$\begin{aligned}
x(mx_4 - m'x_1) - y(x_1 - x_4) + (m' - m)x_1x_4 = 0,\\
x(mx_3 - m'x_2) - y(x_2 - x_3) + (m' - m)x_2x_3 = 0;
\end{aligned}$$

and if we divide the first of these equations by mx_1x_4, and the second by $m'mx_2x_3$, and then add, we obtain

$$\left(\frac{1}{x_1} + \frac{1}{x_2}\right) - \left(\frac{1}{x_3} + \frac{1}{x_4}\right)y - \left[\left(\frac{1}{x_1} + \frac{1}{x_2}\right) - \left(\frac{1}{x_3} + \frac{1}{x_4}\right)\right]mx = 0,$$

or, what is the same thing,

$$\frac{y - m'x}{x_1x_2} - \frac{y - mx}{x_3x_4} = 0,$$

which by what precedes is the equation of a line through the point Q. Substituting herein for $\dfrac{1}{x_1} + \dfrac{1}{x_2}$, $\dfrac{1}{x_3} + \dfrac{1}{x_4}$ their foregoing values, the equation becomes

$$(A + Bm)(y - m'x) + (A + Bm')(y - mx) + m' - m = 0;$$

that is,

$$(m - m')(Ax + By + C) = 0;$$

or finally it is $Ax + By + C = 0$, showing that the point Q lies in a line the position of which is independent of the particular lines OAA', OBB' used in the construction.

It is proper to notice that there is no correspondence to each other of the points A, A' and B, B'; the grouping might as well have been A, A' and B', B; and it thence appears that the line $Ax+By+C = 0$ just obtained is in fact the line joining the point Q with the point R which is the intersection of AB and $A'B'$.

PAGE 556. **10.** The equation $Ax + By + C = 0$ of a line contains in appearance 3, but really only 2 constants (for one of the constants can be divided out), and a line depends accordingly upon 2 parameters, or can be made to satisfy 2 conditions. Similarly, the equation $(a, b, c, f, g, h(x, y, 1))^2 = 0$ of a conic contains really 5 constants, and the equation $(*(x, y, 1))^2 = 0$ of a cubic contains really 9 constants. It thus appears that a cubic can be made to pass through 9 given points, and that the cubic so passing through 9 given points is completely determined. There is, however, a remarkable exception. Considering two given cubic curves $S = 0$, $S' = 0$, these intersect in 9 points, and through these 9 points we have the whole series of cubics $S - kS' = 0$, where k is an arbitrary constant: k may be determined so that the cubic shall pass through a given tenth point, viz. $k = S_0 \div S_0'$, if the coordinates are (x_0, y_0), and S_0, S_0' denote the corresponding values of S, S'. The resulting curve $SS_0' - S'S_0 = 0$ may be regarded as the cubic determined by the conditions of passing through 8 of the 9 points and through the given point (x_0, y_0); and from the equation it thence appears that the curve passes through the remaining one of the 9 points. In other words, we thus have the theorem, any cubic curve which passes through 8 of the 9 intersections of two given cubic curves passes through the 9th intersection.

The applications of this theorem are very numerous; for instance, we derive from it Pascal's theorem of the inscribed hexagon. Consider a hexagon inscribed in a conic. The three alternate sides constitute a cubic, and the other three alternate sides another cubic. The cubics intersect in 9 points, being the 6 vertices of the hexagon, and the 3 Pascalian points, or intersections of the pairs of opposite sides of the hexagon. Drawing a line through two of the Pascalian points, the conic and this line constitute a cubic passing through 8 of the 9 points of intersection, and it therefore passes through the remaining point of intersection; that is, the third Pascalian point; and since obviously this does not lie on the conic, it must lie on the line; that is, we have the theorem that the three Pascalian points (or points of intersection of the pairs of opposite sides) lie on a line.

Metrical Theory

11. The foundation of the metrical theory consists in the simple theorem that if a finite line PQ (fig. 10) be projected upon any other line OO' by lines perpendicular to OO', then the length of the projection

$P'Q'$ is equal to the length of PQ multiplied by the cosine of its inclination to $P'Q'$; or, what is the same thing, that the perpendicular distance PQ' of any two parallel lines is equal to the inclined distance PQ multiplied by the cosine of the inclination. It at once follows that the algebraical sum of the projections of the sides of a closed polygon upon any line is $= 0$; or, reversing the signs of certain sides, and considering the polygon as consisting of two broken lines, each extending from the same initial to the same terminal point, the sum of the projections of the lines of the first set upon any line is equal to the sum of the projections of the lines of the second set. Observe that, if any line be perpendicular to the line on which the projection is made, then its projection is $= 0$.

Thus, if we have a right-angled triangle PQR (fig. 11), where QR, RP, QP are $= \xi$, η, p respectively, and whereof the base-angle is $= \alpha$, then projecting successively on the three sides, we have

$$\xi = p\cos\alpha, \qquad \eta = p\sin\alpha, \qquad p = \xi\cos\alpha + \eta\sin\alpha$$

and we thence obtain

$$p^2 = \xi^2 + \eta^2, \qquad \cos^2\alpha + \sin^2\alpha = 1.$$

And again, by projecting on a line Qx', inclined at the angle α' to QR, we have

$$p\cos(\alpha - \alpha') = \xi\cos\alpha' + \eta\sin\alpha'$$

and by substituting for ξ, η their foregoing values,

$$\cos(\alpha - \alpha') = \cos\alpha\cos\alpha' + \sin\alpha\sin\alpha'.$$

It is to be remarked that, assuming only the theory of similar triangles, we have herein a proof of Euclid, Book I., Prop. 47; in fact, the same as is given Book VI., Prop. 31; and also a proof of the trigonometrical formula for $\cos(\alpha - \alpha')$. The formulae for $\cos(\alpha + \alpha')$ and $\sin(\alpha + \alpha')$ could be obtained in the same manner.

Draw PT at right angles to Qx', and suppose QT, TP $= \xi_1$, η_1 respectively, so that we have now the quadrilateral $QRPTQ$, or, what is the same thing, the two broken lines QRP and QTP, each extending from Q to P. Projecting on the four sides successively, we have

$$\begin{aligned}
\xi &= \xi_1 \cos \alpha' + \eta_1 \sin \alpha', \\
\eta &= \xi_1 \sin \alpha' + \eta_1 \cos \alpha', \\
\xi_1 &= \xi \cos \alpha' + \eta \sin \alpha', \\
\eta_1 &= -\xi \sin \alpha' + \eta \cos \alpha',
\end{aligned}$$

where the third equation is that previously written $p \cos(\alpha - \alpha') = \xi \cos \alpha' + \eta \sin \alpha$.

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Equations of Right Line and Circle. Transformation of Coordinates

12. The required formulae are really contained in the foregoing results. For, in fig. 11, supposing that the axis of x is parallel to QR, and taking a, b for the coordinates of Q, and (x, y) for those of P, then we have ξ, $\eta = x - a$, $y - b$ respectively and therefore

$$x - a = p\cos\alpha, \qquad y - b = p\sin\alpha,$$
$$p^2 = (x-a)^2 + (y-b)^2.$$

Writing the first two of these in the form

$$\frac{x-a}{\cos\alpha} = \frac{y-b}{\sin\alpha}(= p),$$

we may regard Q as a fixed point, but P as a point moving in the direction Q to P, so that α remains constant, and then, omitting the equation $(= p)$, we have a relation between the coordinates x, y of the point thus moving in a right line, that is, we have the equation of the line through the given point (a, b) at a given inclination α to the axis of x. And, moreover, if, using this equation $(= p)$, we write $x = a + p\cos\alpha$, $y = b + p\sin\alpha$, then we have expressions for the coordinates x, y of a point of this line, in terms of the variable parameter p.

Again, take the point T to be fixed, but consider the point P as moving in the line TP at right angles to QT. If instead of ξ_1 we take p for the distance QT, then the equation $\xi_1 = \xi\cos\alpha' + \eta\sin\alpha'$ will be

$$(x-a)\cos\alpha' + (y-b)\sin\alpha' = p;$$

that is, this will be the equation of a line such that its perpendicular distance from the point (a, b) is $= p$, and that the inclination of this distance to the axis of x is $= \alpha'$.

From either form it appears that the equation of a line is, in fact, a linear equation of the form $Ax + By + C = 0$. It is important to notice that, starting from this equation, we can determine conversely the α but not the (a, b) of the form of equation which contains these quantities; and in like manner the α' but not the (a, b) or p of the other form of equation. The reason is obvious. In each case (a, b) denote the coordinates of a point, fixed indeed, but which is in the first form any point of the line, and in the second form any point whatever. Thus, in the second form the point from which the perpendicular is let fall may be the origin. Here $(a, b) = (0, 0)$, and the equation is $x \cos \alpha' + y \sin \alpha' - p = 0$. Comparing this with $Ax + By + C = 0$, we have the values of $\cos \alpha'$, $\sin \alpha'$, and p.

13. The equation

$$p^2 = (x - a)^2 + (y - b)^2$$

is an expression for the squared distance of the two points (a, b) and (x, y). Taking as before the point Q, coordinates (a, b), as a fixed point, and writing c in the place of p, the equation

$$(x - a)^2 + (y - b)^2 = c^2$$

expresses that the point (x, y) is always at a given distance c from the given point (a, b); viz. this is the equation of a circle, having (a, b) for the coordinates of its centre, and c for its radius.

The equation is of the form

$$x^2 + y^2 + 2Ax + 2By + C = 0,$$

and here, the number of constants being the same, we can identify the two equations; we find $a = -A$, $b = -B$, $c^2 = A^2 + B^2 - C$, or the last equation is that of a circle having $-A$, $-B$ for the coordinates of its centre, and $\sqrt{A^2 + B^2 - C}$ for its radius.

PAGE 559. **14.** Drawing (fig. 11) Qy_1 at right angles at Qx_1, and taking Qx_1, Qy_1 as a new set of rectangular axes, if instead of ξ_1, η_1 we write x_1, y_1, we have x_1, y_1 as the new coordinates of the point P; and writing also α in place of α', α now denoting the inclination of the axes Qx_1 and Ox, we have the formulas for transformation between two sets of rectangular axes. These are

$$\begin{aligned} x - a &= x_1 \cos\alpha - y_1 \sin\alpha, & x_1 &= (x-a)\cos\alpha + (y-b)\sin\alpha, \\ y - b &= x_1 \sin\alpha + y_1 \cos\alpha, & y_1 &= -(x-a)\sin\alpha + (y-b)\cos\alpha, \end{aligned}$$

each set being obviously at once deducible from the other one. In these formulae (a, b) are the xy-coordinates of the new origin Q, and α is the inclination of Qx_1 to Ox. It is to be noticed that Qx_1, Qy_1 are so placed that, by moving to Q, and then turning the axes Ox_1, Oy_1 round Q (through an angle α measured in the sense Ox to Oy), the original axes Ox, Oy will come to coincide with Qx_1, Qy_1 respectively. This could not have been done if Qy_1 had been drawn (at right angles always to Qx_1) in the reverse direction: we should then have had in the formulae $-y_1$ instead of $+y_1$. The new formulae which would be thus obtained are of an essentially distinct form: the analytical test is that in the formulae as written down we can, by giving to α a proper value (in fact, $\alpha = 0$), make the $(x - a)$ and $(y - b)$ equal to x_1 and y_1 respectively; in the other system we could only make them equal to x_1, $-y_1$, or $-x_1$, y_1 respectively. But for the very reason that the second system can be so easily derived from the first, it is proper to attend exclusively to the first system, that is, always to take the new axes so that the two sets admit of being brought into coincidence.

In the foregoing system of two pairs of equations, the first pair give the original coordinates x, y in terms of the new coordinates x_1, y_1; the second pair the new coordinates x_1, y_1 in terms of the original coordinates x, y. The formulae involve (a, b), the original coordinates of the new origin; it would be easy, instead of these, to introduce (a_1, b_1), the new coordinates of the origin. Writing $(a, b) = (0, 0)$, we have, of course, the formulae for transformation between two sets of rectangular axes having the same origin, and it is as well to write the formulae in this more simple form; the subsequent transformation to a new origin, but with axes parallel to the original axes, can then be effected without any difficulty.

PAGE 560. **15.** All questions in regard to the line may be solved by means of one or other of the foregoing forms

$$Ax + By + C = 0,$$
$$y = Ax + B,$$
$$\frac{x-a}{\cos\alpha} = \frac{y-b}{\sin\alpha},$$
$$(x-a)\cos\alpha' + (y-b)\sin\alpha' - p = 0;$$

or it may be by a comparison of these different forms: thus, using the first form, it has been already shown that the equation of the line through two given points (x_1, y_1), (x_2, y_2) is

$$x(y_1 - y_2) - y(x_1 - x_2) + x_1 y_2 - x_2 y_1 = 0,$$

or, as this may be written,

$$\begin{vmatrix} x & y & 1 \\ x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \end{vmatrix} = 0;$$

a particular case is the equation

$$\frac{x}{a} + \frac{y}{b} = 1$$

representing the line through the points $(a, 0)$ and $(0, b)$, or, what is the same thing, the line meeting the axes of x and y at the distances from the origin a and b respectively. It may be noticed that, in the form $Ax + By + C = 0$, $-B \div A$ denotes the tangent of the inclination to the axis of x, or we may say that $B \div \sqrt{A^2 + B^2}$ and $-A \div \sqrt{A^2 + B^2}$ denote respectively the cosine and the sine of the inclination to the axis of x. A better form is this: $A \div \sqrt{A^2 + B^2}$ and $B \div \sqrt{A^2 + B^2}$ denote respectively the cosine and the sine of inclination to the axis of x of the perpendicular upon the line. So, of course, in regard to the form $y = Ax + B$, A is here the tangent of the inclination to the axis of x; $1 \div \sqrt{A^2 + 1}$ and $A \div \sqrt{A^2 + 1}$ are the cosine and sine of this inclination, &c. It thus appears that the condition, in order that the lines $Ax + By + C = 0$ and $A'x + B'y + C' = 0$ may meet at right angles, is $AA' + BB' = 0$; so when the equations are $y = Ax + B$, $y = A'x + B'$, the condition is $AA' + 1 = 0$, or say the value of A' is

$$A' = -\frac{1}{A}.$$

The perpendicular distance of the point (a,b) from the line $Ax + By + C = 0$ is $(Aa + Bb + C) \div \sqrt{A^2 + B^2}$. In all the formulae involving $\sqrt{A^2 + B^2}$ or $\sqrt{A^2 + 1}$, the radical should be written with the sign $\pm$, which is essentially indeterminate: the like indeterminateness of sign presents itself in the expression for the distance of two points $p = \pm\sqrt{(x-a)^2 + (y-b)^2}$; if, as before, the points are Q, P, and the indefinite line through these is $z'QPz$, then it is the same thing whether we measure off from Q along this line, considered as drawn from z' towards z, a positive distance k, or along the line considered as drawn reversely from z towards z', the equal negative distance $-k$, and the expression for the distance p is thus properly of the form $\pm k$.

It is interesting to compare expressions which do not involve a radical: thus, in seeking for the expression for the perpendicular distance of the point (a,b) from a given line, let the equation of the given line be taken in the form,

$$x \cos \alpha + y \sin \alpha - p = 0,$$

p being the perpendicular distance from the origin, α its inclination to the axis of x: the equation of the line may also be written $(x-a)\cos\alpha + (y-b)\sin\alpha - p_1 = 0$, and we have thence $p_1 = p - a\cos\alpha - b\sin\alpha$, the required expression for the distance p_1: it is here assumed that p_1 is drawn from (a,b) in the same sense as p is drawn from the origin, and the indeterminateness of sign is thus removed.

PAGE 561. **16.** As an instance of the mode of using the formulae, take the problem of finding the locus of a point such that its distance from a given point is in a given ratio to its distance from a given line.

We take (a, b) as the coordinates of the given point, and it is convenient to take (x, y) as the coordinates of the variable point, the locus of which is required: it thus becomes necessary to use other letters, say (X, Y), for current coordinates in the equation of the given line. Suppose this is a line such that its perpendicular distance from the origin is $= p$, and that the inclination of p to the axis of x is $= \alpha$; the equation is $X \cos\alpha + Y \sin\alpha - p = 0$. In the result obtained in § 15, writing (x, y) in place of (a, b), it appears that the perpendicular distance of this line from the point (x, y) is

$$= p - x\cos\alpha - y\sin\alpha;$$

hence the equation of the locus is

$$\sqrt{(x-a)^2 + (y-b)^2} = e(p - x\cos\alpha - y\sin\alpha),$$

or say

$$(x-a)^2 + (y-b)^2 - e^2(x\cos\alpha + y\sin\alpha - p)^2 = 0,$$

an equation of the second order.

The Conics (Parabola, Ellipse, Hyperbola)

17. The conics or, as they were called, conic sections were originally defined as the sections of a right circular cone; but Apollonius substituted a definition, which is, in fact, that of the last example: the curve is the locus of a point such that its distance from a given point (called the *focus*) is in a given ratio to its distance from a given line (called the *directrix*); and taking the ratio as $e : 1$, then e is called the *eccentricity.*

Take FD for the perpendicular from the focus F upon the directrix, and the given ratio being that of $e : 1$ ($e >, =,$ or < 1, but positive), and let the distance FD be divided at O in the given ratio, say we have $OD = m$, $OF = em$, where m is positive; then the origin may be taken at O, the axis Ox being in the direction OF (that is, from O to F), and the axis Oy at right angles to it. The distance of the point (x, y) from F is $= \sqrt{(x-em)^2 + y^2}$, its distance from the directrix is $= x + m$; the equation therefore is

$$(x-em)^2 + y^2 = e^2(x+m)^2;$$

or, what is the same thing, it is

$$(1-e^2)x^2 - 2me(1+e)x + y^2 = 0.$$

If $e^2 = 1$, or, since e is taken to be positive, if $e = 1$, this is

$$y^2 - 4mx = 0,$$

which is the parabola.

PAGE 562. If e^2 not $= 1$, then the equation may be written

$$(1-e^2)\left\{x - \frac{me}{1-e}\right\}^2 + y^2 = m^2e^2(1-e^2).$$

Supposing e positive and < 1, then, writing $a = \dfrac{me}{1-e^2}$, the equation becomes

$$(1-e^2)(x-a)^2 + y^2 = a^2(1-e^2)^2,$$

that is,

$$\frac{(x-a)^2}{a^2} + \frac{y^2}{a^2(1-e^2)} = 1;$$

or, changing the origin and writing $b^2 = a^2(1-e^2)$, this is

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1,$$

which is the ellipse.

And similarly if e be positive and > 1, then writing $a = \dfrac{me}{e^2-1}$, the equation becomes

$$(1-e^2)(x+a)^2 + y^2 = a^2(1-e^2)^2,$$

that is,

$$\frac{(x+a)^2}{a^2} + \frac{y^2}{a^2(1-e^2)} = 1;$$

or changing the origin and writing $b^2 = a^2(e^2-1)$, this is

$$\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1,$$

which is the hyperbola.

18. The general equation $ax^2 + 2hxy + by^2 + 2fy + 2gx + c = 0$, or as it is written $(a, b, c, f, g, h(x, y, 1))^2 = 0$, may be such that the quadric function breaks up into factors, $= (\alpha x+\beta y+\gamma)(\alpha' x+\beta' y+\gamma')$; and in this case the equation represents a pair of lines, or (it may be) two coincident lines. When it does not so break up, the function can be put in the form $\lambda\{(x-a')^2 + (y-b')^2 - e^2(x\cos\alpha + y\sin\alpha - p)^2\}$, or, equating the two expressions, there will be six equations for the determination of λ, a', b', e, p, α; and by what precedes, if a, b', e, p, α are real, the curve is either a parabola, ellipse, or hyperbola.

The original coefficients (a, b, c, f, g, h) may be such as not to give any system of real values for a, b', e, p, α; but when this is so the equation $(a, b, c, f, g, h(x, y, 1))^2 = 0$ does not represent a real curve[13]; the imaginary curve which it represents is, however, regarded as a conic. Disregarding the special cases of the pair of lines and the twice repeated line, it thus appears that the only real curves represented by the general equation $(a, b, c, f, g, h(x, y, 1))^2 = 0$ are the parabola, the ellipse, and the hyperbola. The circle is considered as a particular case of the ellipse.

The same result is obtained by transforming the equation $(a, b, c, f, g, h)(x, y, 1)^2 = 0$ to new axes. If in the first place the origin be unaltered, then the directions of the new (rectangular) axes Ox_1, Oy_1 can be found so that h_1 (the coefficient of the term x_1y_1) shall be $= 0$; when this is done, then either one of the coefficients of x_1^2, y_1^2 is $= 0$, and the curve is then a parabola, or neither of these coefficients is $= 0$, and the curve is then an ellipse or hyperbola, according as the two coefficients are of the same sign or of opposite signs.

[13]It is proper to remark that, when $(a, b, c, f, g, h)(x, y, 1)^2 = 0$ does represent a real curve, there are, in fact, four systems of values of a', b', e, p, α, two real, the other two imaginary; we have thus two real equations and two imaginary equations, each of them of the form $(x - a')^2 + (y - b')^2 = e^2(x \cos\alpha + y \cos\beta - p)^2$, representing each of them one and the same real curve. This is consistent with the assertion of the text that the real curve is in every case represented by a real equation of this form.

PAGE 563. **19.** The curves can be at once traced from their equations:

$$y^2 = 4mx, \qquad \text{for the parabola (fig. 13),}$$

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1, \qquad \text{for the ellipse (fig. 14),}$$

$$\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1, \qquad \text{for the hyperbola (fig. 15);}$$

and it will be noticed how the form of the last equation puts in evidence the two asymptotes

$$\frac{x}{a} \pm \frac{y}{b} = 0$$

of the hyperbola. Referred to the asymptotes (as a set of oblique axes) the equation of the hyperbola takes the form $xy = c$; and in particular, if in this equation the axes are at right angles, then the equation represents the rectangular hyperbola referred to its asymptotes as axes.

Tangent, Normal, Circle and Radius of Curvature, &c.

20. There is great convenience in using the language and notation of the infinitesimal analysis; thus we consider on a curve a point with coordinates (x, y), and a consecutive point the coordinates of which are $(x + dx, y + dy)$, or again a second consecutive point with coordinates $(x + dx + \frac{1}{2}d^2x, y + dy + \frac{1}{2}d^2y)$, &c.; and in the final results the ratios of the infinitesimals must be replaced by differential coefficients in the proper manner; thus, if x, y are considered as given functions of a parameter θ, then dx, dy have in fact the values $\frac{dx}{d\theta}d\theta$, $\frac{dy}{d\theta}d\theta$, and (only the ratio being really material) they may in the result be replaced by $\frac{dx}{d\theta}$, $\frac{dy}{d\theta}$. This includes the case where the equation of the curve is given in the form $y = \phi(x)$; θ is here $= x$, and the increments dx, dy are in the result to be replaced by 1, $\frac{dy}{dx}$. So also with the infinitesimals of the higher orders d^2x, &c.

21. The tangent at the point (x, y) is the line through this point and the consecutive point $(x+dx, y+dy)$; hence, taking ξ, η as current coordinates, the equation is

$$\frac{\xi - x}{dx} = \frac{\eta - y}{dy},$$

an equation which is satisfied on writing therein ξ, η = (x, y) or $= (x + dx, y + dy)$. The equation may be written

$$\eta - y = \frac{dy}{dx}(\xi - x),$$

$\frac{dy}{dx}$ being now the differential coefficient of y in regard to x; and this form is applicable whether y is given directly as a function of x, or in whatever way y is in effect given as a function of x: if as before x, y are given each of them as a function of θ, then the value of $\frac{dy}{dx}$ is $= \left(\frac{dy}{d\theta}\right) \div \left(\frac{dx}{d\theta}\right)$, which is the result obtained from the original form on writing therein $\frac{dx}{d\theta}$, $\frac{dy}{d\theta}$, for dx, dy respectively.

So again, when the curve is given by an equation $u = 0$ between the coordinates (x, y), then $\frac{dy}{dx}$ is obtained from the equation $\frac{du}{dx} + \frac{du}{dy}\frac{dy}{dx} = 0$. But here it is more elegant, using the original form, to eliminate dx, dy by the formula $\frac{du}{dx}dx + \frac{du}{dy}dy$; we thus obtain the equation of the tangent in the form

$$(\xi - x)\frac{du}{dx} + (\eta - y)\frac{du}{dy} = 0.$$

For example, in the case of the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$, the equation is

$$\frac{x}{a^2}(\xi - x) + \frac{y}{b^2}(\eta - y) = 0;$$

or reducing by means of the equation of the curve the equation of the tangent is

$$\frac{x\xi}{a^2} + \frac{y\eta}{b^2} = 1.$$

The normal is a line through the point at right angles to the tangent; the equation therefore is

$$(\xi - x)\,dx + (\eta - y)\,dy = 0,$$

where dx, dy are to be replaced by their proportional values as before.

PAGE 565. **22.** The circle of curvature is the circle through the point and two consecutive points of the curve. Taking the equation to be

$$(\xi-\alpha)^2+(\eta-\beta)^2=\rho^2,$$

the values of α, β are given by

$$\alpha=x-\frac{dy(dx^2+dy^2)}{dx\,d^2y-dy\,d^2x},\qquad \beta=y+\frac{dx(dx^2+dy^2)}{dx\,d^2y-dy\,d^2x},$$

and we then have

$$\rho^2=(x-\alpha)^2+(y-\beta)^2=\frac{(dx^2+dy^2)^3}{(dx\,d^2y-dy\,d^2x)^2}.$$

In the case where y is given directly as a function of x, then, writing for shortness $p=\frac{dy}{dx}$, $q=\frac{d^2y}{dx^2}$, this is $\rho^2=\frac{(1+p^2)^3}{q^2}$, or, as the equation is usually written, $\rho=\frac{(1+p^2)^{\frac{3}{2}}}{q}$, the radius of curvature, considered to be positive or negative according as the curve is concave or convex to the axis of x.

It may be added that the centre of curvature is the intersection of the normal by the consecutive normal.

The locus of the centre of curvature is the *evolute*. If from the expressions of α, β regarded as functions of x we eliminate x, we have thus an equation between (α,β), which is the equation of the evolute.

Polar Coordinates

23. The position of a point may be determined by means of its distance from a fixed point and the inclination of this distance to a fixed line through the fixed point. Say we have r the distance from the origin, and θ the inclination of r to the axis of x; r and θ are then the polar coordinates of the point, r the radius vector, and θ the inclination. These are immediately connected with the Cartesian coordinates x, y by the formulae $x=r\cos\theta$, $y=r\sin\theta$; and the transition from either set of coordinates to the other can thus be made without difficulty. But the use of polar coordinates is very convenient,—as well in reference to certain classes of questions relating to curves of any kind (for instance, in the dynamics of central forces) as in relation to curves having in regard to the origin the symmetry of the regular polygon (curves such as that represented by the equation $r = \cos m\theta$), and also in regard to the class of curves called spirals, where the radius vector r is given as an algebraical or exponential function of the inclination θ.

Trilinear Coordinates

24. Consider a fixed triangle ABC, and (regarding the sides as indefinite lines) suppose for a moment that p, q, r denote the distances of a point P from the sides BC, CA, AB respectively, these distances being measured either perpendicularly to the several sides, or each of them in a given direction. To fix the ideas each distance may be considered as positive for a point inside the triangle, and the sign is thus fixed for any point whatever. There is then an identical relation between p, q, r: if a, b, c are the lengths of the sides, and the distances are measured perpendicularly thereto, the relation is $ap + bq + cr =$ twice the area of triangle. But taking x, y, z proportional to p, q, r, or if we please proportional to given multiples of p, q, r, then only the ratios of x, y, z are determined; their absolute values remain arbitrary. But the ratios of p, q, r, and consequently also the ratios of x, y, z determine, and that uniquely, the point; and it being understood that only the ratios are attended to, we say that (x, y, z) are the coordinates of the point. The equation of a line has thus the form $ax + by + cz = 0$, and generally that of a curve of the nth order is a homogeneous equation of this order between the coordinates, $(*(x, y, z))^n = 0$. The advantage over Cartesian coordinates is in the greater symmetry of the analytical forms, and in the more convenient treatment of the line infinity and of points at infinity. The method includes that of Cartesian coordinates, the homogeneous equation in x, y, z is, in fact, an equation in $\frac{x}{z}$, $\frac{y}{z}$ which two quantities may be regarded as denoting Cartesian coordinates; or, what is the same thing, we may in the equation write $z = 1$. It may be added that, if the trilinear coordinates (x, y, z) are regarded as the Cartesian coordinates of a point of space, then the equation is that of a cone having the origin for its vertex; and conversely that such equation of a cone may be regarded as the equation in trilinear coordinates of a plane curve.

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General Point-Coordinates. Line-Coordinates

25. All the coordinates considered thus far are point-coordinates. More generally, any two quantities (or the ratios of three quantities) serving to determine the position of a point in the plane may be regarded as the coordinates of the point; or, if instead of a single point they determine a system of two or more points, then as the coordinates of the system of points. But, as noticed under Curve, [785], there are also line-coordinates serving to determine the position of a line; the ordinary case is when the line is determined by means of the ratios of three quantities ξ, η, ζ (correlative to the trilinear coordinates x, y, z). A linear equation $a\xi + b\eta + c\zeta = 0$ represents then the system of lines such that the coordinates of each of them satisfy this relation, in fact, all the lines which pass through a given point; and it is thus regarded as the line-equation of this point; and generally a homogeneous equation $(*(\xi,\eta,\zeta))^n = 0$ represents the curve which is the envelope of all the lines the coordinates of which satisfy this equation, and it is thus regarded as the line-equation of this curve.

II. Solid Analytical Geometry (§§ 26–40)

26. We are here concerned with points in space, the position of a point being determined by its three coordinates x, y, z. We consider three coordinate planes, at right angles to each other, dividing the whole of space into eight portions called octants, the coordinates of a point being the perpendicular distances of the point from the three planes respectively, each distance being considered as positive or negative according as it lies on the one or the other side of the plane. Thus the coordinates in the eight octants have respectively the signs

	I	II	III	IV	V	VI	VII	VIII
x	$+$	$-$	$-$	$+$	$+$	$-$	$-$	$+$
y	$+$	$+$	$-$	$-$	$+$	$+$	$-$	$-$
z	$+$	$+$	$+$	$+$	$-$	$-$	$-$	$-$

The positive parts of the axes are usually drawn as in fig. 16, which represents a point P, the coordinates of which have the positive values OM, MN, NP.

PAGE 568. **27.** It may be remarked, as regards the delineation of such solid figures, that if we have in space three lines at right angles to each other, say Oa, Ob, Oc, of equal lengths, then it is possible to project these by parallel lines upon a plane in such wise that the projections Oa', Ob', Oc' shall be at given inclinations to each other, and that these lengths shall be to each other in given ratios: in particular, the two lines Oa', Oc' may be at right angles to each other, and their lengths equal, the direction of Ob', and its proportion to the two equal lengths Oa', Oc', being arbitrary. It thus appears that we may as in the figure draw Ox, Oz at right angles to each other, and Oy in an arbitrary direction; and moreover represent the coordinates x, z on equal scales, and the remaining coordinate y on an arbitrary scale (which may be that of the other two coordinates x, z, but is in practice usually smaller). The advantage, of course, is that a figure in one of the coordinate planes xz is represented in its proper form without distortion; but it may be in some cases preferable to employ the isometrical projection, wherein the three axes are represented by lines inclined to each other at angles of 120°, and the scales for the coordinates are equal (fig. 17).

For the delineation of a surface of a tolerably simple form, it is frequently sufficient to draw (according to the foregoing projection) the sections by the coordinate planes; and in particular, when the surface is symmetrical in regard to the coordinate planes, it is sufficient to draw the quarter-sections belonging to a single octant of the surface; thus fig. 18 is a convenient representation of an octant of the wave surface. Or a surface may be delineated by means of a series of parallel sections, or (taking these to be the sections by a series of horizontal planes) say by a series of contour lines. Of course, other sections may be drawn or indicated, if necessary. For the delineation of a curve, a convenient method is to represent, as above, a series of the points thereof, each point being accompanied by the ordinate PN, which serves to refer the point to the plane of xy; this is in effect a representation of each point P of the curve, by means of two points P, N such that the line PN has a fixed direction. Both as regards curves and surfaces, the employment of stereographic representations is very interesting.

28. In plane geometry, reckoning the line as a curve of the first order, we have only the point and the curve. In solid geometry, reckoning a line as a curve of the first order, and the plane as a surface of the first order, we have the point, the curve, and the surface; but the increase of complexity is far greater than would hence at first sight appear. In plane geometry a curve is considered in connexion with lines (its tangents); but in solid geometry the curve is considered in connexion with lines and planes (its tangents and osculating planes), and the surface also in connexion with lines and planes (its tangent lines and tangent planes); there are surfaces arising out of the line—cones, skew surfaces, developables, doubly and triply infinite systems of lines, and whole classes of theories which have nothing analogous to them in plane geometry: it is thus a very small part indeed of the subject which can be even referred to in the present article.

In the case of a surface, we have between the coordinates (x, y, z) a single, or say a onefold relation, which can be represented by a single relation $f(x, y, z) = 0$; or we may consider the coordinates expressed each of them as a given function of two variable parameters p, q; the form $z = f(x, y)$ is a particular case of each of these modes of representation; in other words, we have in the first mode $f(x, y, z) = z - f(x, y)$, and in the second mode $x = p$, $y = q$ for the expression of two of the coordinates in terms of the parameters.

In the case of a curve, we have between the coordinates (x, y, z) a twofold relation: two equations $f(x, y, z) = 0$, $\phi(x, y, z) = 0$ give such a relation; i.e., the curve is here considered as the intersection of two surfaces (but the curve is not always the complete intersection of two surfaces, and there are hence difficulties); or, again, the coordinates may be given each of them as a function of a single variable parameter. The form $y = \phi x$, $z = \psi x$, where two of the coordinates are given in terms of the third, is a particular case of each of these modes of representation.

29. The remarks under plane geometry as to descriptive and metrical propositions, and as to the non-metrical character of the method of coordinates when used for the proof of a descriptive proposition, apply also to solid geometry; and they might be illustrated in like manner by the instance of the theorem of the radical centre of four spheres. The proof is obtained from the consideration that S and S' being each of them a function of the form $x^2+y^2+z^2+ax+by+cz+d$, the difference $S - S'$ is a mere linear function of the coordinates, and consequently that $S - S' = 0$ is the equation of the plane containing the circle of intersection of the two spheres $S = 0$ and $S' = 0$.

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Metrical Theory

30. The foundation in solid geometry of the metrical theory is, in fact, the before-mentioned theorem that, if a finite right line PQ be projected upon any other line OO' by lines perpendicular to OO', then the length of the projection $P'Q'$ is equal to the length of PQ multiplied by the cosine of its inclination to $P'Q'$; or (in the form in which it is now convenient to state the theorem) the perpendicular distance PQ' of two parallel planes is equal to the inclined distance PQ multiplied by the cosine of the inclination. Hence also the algebraical sum of the projections of the sides of a closed polygon upon any line is $= 0$; or, reversing the signs of certain sides and considering the polygon as made up of two broken lines each extending from the same initial to the same terminal point, the sum of the projections of the one set of lines upon any line is equal to the sum of the projections of the other set of lines upon the same line. When any of the lines are at right angles to the given line (or, what is the same thing, in a plane at right angles to the given line), the projections of these lines severally vanish.

31. Consider the skew quadrilateral $QMNP$, the sides QM, MN, NP being respectively parallel to the three rectangular axes Ox, Oy, Oz; let the lengths of these sides be ξ, η, ζ, and that of the side QP be $= p$; and let the cosines of the inclinations (or say the cosine-inclinations) of p to the three axes be α, β, γ; then projecting successively on the three sides and on QP, we have

$$\xi = p\alpha, \quad \eta = p\beta, \quad \zeta = p\gamma,$$

and

$$p = \xi\alpha + \eta\beta + \zeta\gamma,$$

whence $p^2 = \xi^2 + \eta^2 + \zeta^2$, which is the relation between a distance p and its projections ξ, η, ζ upon three rectangular axes. And from the same equations we obtain $\alpha^2 + \beta^2 + \gamma^2 = 1$, which is a relation connecting the cosine-inclinations of a line to three rectangular axes.

Suppose we have through Q any other line QT, and let the cosine-inclinations of this to the axes be α', β', γ', and θ be its cosine-inclination to QP; also let ϖ be the length of the projection of QP upon QT; then projecting on QT, we have

$$\varpi = \alpha'\xi + \beta'\eta + \gamma'\zeta = p\theta.$$

And in the last equation substituting for ξ, η, ζ their values $p\alpha$, $p\beta$, $p\gamma$, we find

$$\theta = \alpha\alpha' + \beta\beta' + \gamma\gamma',$$

which is an expression for the mutual cosine-inclination of two lines, the cosine-inclinations of which to the axes are α, β, γ and α', β', γ' respectively. We have of course $\alpha^2+\beta^2+\gamma^2 = 1$, and $\alpha'^2+\beta'^2+\gamma'^2 = 1$, and hence also

$$1 - \theta^2 = (\beta\gamma' - \beta'\gamma)^2 + (\gamma\alpha' - \gamma'\alpha)^2 + (\alpha\beta' - \alpha'\beta)^2,$$

so that the sine of the inclination can only be expressed as a square root. These formulae are the foundation of spherical trigonometry.

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The Line, Plane, and Sphere

32. The foregoing formulae give at once the equations of these loci.

For first, taking Q to be a fixed point, coordinates (a, b, c) and the cosine-inclinations (α, β, γ) to be constant, then P will be a point in the line through Q in the direction thus determined; or, taking (x, y, z) for its coordinates, these will be the current coordinates of a point in the line. The values of ξ, η, ζ then are $x-a$, $y-b$, $z-c$, and we thus have

$$\frac{x-a}{\alpha} = \frac{y-b}{\beta} = \frac{z-c}{\gamma} (= p),$$

which (omitting the last equation, $= p$) are the equations of the line through the given point (a, b, c), the cosine-inclinations to the axes being α, β, γ, where $\alpha^2+\beta^2+\gamma^2 = 1$. The equations may be written in a form not involving the cosine-inclinations; viz. they may be written

$$\frac{x-a}{l} = \frac{y-b}{m} = \frac{z-c}{n},$$

and then l, m, n, instead of being equal, will only be proportional to the cosine-inclinations. Using the last equation, and writing

$$x, y, z = a + \alpha p,\ \ b + \beta p,\ \ c + \gamma p,$$

these are expressions for the current coordinates in terms of a parameter p, which is in fact the distance from the fixed point (a, b, c).

It is easy to see that, if the coordinates (x, y, z) are connected by any two linear equations, these equations can always be brought into the foregoing form, and hence that the two linear equations represent a line.

Secondly, taking for greater simplicity the point Q to be coincident with the origin, and α', β', γ', p to be constant, then p is the perpendicular distance of a plane from the origin, and α', β', γ' are the cosine-inclinations of this distance to the axes $(\alpha'^2 + \beta'^2 + \gamma'^2 = 1)$. Now P is any point in this plane; taking its coordinates to be (x, y, z), then (ξ, η, ζ) are $= (x, y, z)$, and the foregoing equation $p = \alpha'\xi + \beta'\eta + \gamma'\zeta$ becomes

$$\alpha' x + \beta' y + \gamma' z = p,$$

which is the equation of the plane in question.

If, more generally, Q is not coincident with the origin, then, taking its coordinates to be (a, b, c), and writing p_1 instead of p, the equation is

$$\alpha'(x-a) + \beta'(y-b) + \gamma'(z-c) = p_1;$$

and we thence have $p_1 = p - (a\alpha' + b\beta' + c\gamma')$, which is an expression for the perpendicular distance of the point (a, b, c) from the plane in question.

It is obvious that any linear equation $Ax + By + Cz + D = 0$ between the coordinates can always be brought into the foregoing form, and hence that such equation represents a plane.

PAGE 572. Thirdly, supposing Q to be a fixed point, coordinates (a, b, c) and the distance QP, $= p$, to be constant, say this is $= d$, then, as before, the values of ξ, η, ζ are $x - a$, $y - b$, $z - c$, and the equation $\xi^2 + \eta^2 + \zeta^2 = p^2$ becomes

$$(x-a)^2 + (y-b)^2 + (z-c)^2 = d^2,$$

which is the equation of the sphere, coordinates of the centre $= (a, b, c)$ and radius $= d$.

A quadric equation wherein the terms of the second order are $x^2 + y^2 + z^2$, viz. an equation

$$x^2 + y^2 + z^2 + Ax + By + Cz + D = 0,$$

can always, it is clear, be brought into the foregoing form; and it thus appears that this is the equation of a sphere, coordinates of the centre $= -\frac{1}{2}A$, $-\frac{1}{2}B$, $-\frac{1}{2}C$, and squared radius $= \frac{1}{4}(A^2 + B^2 + C^2) - D$.

Cylinders, Cones, Ruled Surfaces

33. A singly infinite system of lines, or a system of lines depending upon one variable parameter, forms a surface; and the equation of the surface is obtained by eliminating the parameter between the two equations of the line.

If the lines all pass through a given point, then the surface is a cone; and, in particular, if the lines are all parallel to a given line, then the surface is a cylinder.

Beginning with this last case, suppose the lines are parallel to the line $x = mz$, $y = nz$, the equations of a line of the system are $x = mz + a$, $y = nz + b$, where a, b are supposed to be functions of the variable parameter, or, what is the same thing, there is between them a relation $f(a, b) = 0$: we have $a = x - mz$, $b = y - nz$, and the result of the elimination of the parameter therefore is $f(x - mz, y - nz) = 0$, which is thus the general equation of the cylinder the generating lines whereof are parallel to the line $x = mz$, $y = nz$. The equation of the section by the plane $z = 0$ is $f(x, y) = 0$, and conversely if the cylinder be determined by means of its curve of intersection with the plane $z = 0$, then, taking the equation of this curve to be $f(x, y) = 0$, the equation of the cylinder is $f(x - mz, y - nz) = 0$. Thus, if the curve of intersection be the circle $(x - \alpha)^2 + (y - \beta)^2 = \gamma^2$, we have $(x - mz - \alpha)^2 + (y - nz - \beta)^2 = \gamma^2$ as the equation of an oblique cylinder on this base, and thus also $(x-\alpha)^2+(y-\beta)^2=\gamma^2$ as the equation of the right cylinder.

If the lines all pass through a given point (a,b,c), then the equations of a line are $x-a=\alpha(z-c)$, $y-b=\beta(z-c)$, where α, β are functions of the variable parameter, or, what is the same thing, there exists between them an equation $f(\alpha,\beta)=0$; the elimination of the parameter gives, therefore, $f\left(\frac{x-a}{z-c},\frac{y-b}{z-c}\right)=0$; and this equation, or, what is the same thing, any homogeneous equation $f(x-a,y-b,z-c)=0$, or, taking f to be a rational and integral function of the order n, say $(*(x-a,y-b,z-c))^n=0$, is the general equation of the cone having the point (a,b,c) for its vertex. Taking the vertex to be at the origin, the equation is $(*(x,y,z))^n=0$; and, in particular, $(*(x,y,z))^2=0$ is the equation of a cone of the second order, or quadricone, having the origin for its vertex.

PAGE 573. **34.** In the general case of a singly infinite system of lines, the locus is a ruled surface (or *regulus*). If the system be such that a line does not intersect the consecutive line, then the surface is a *skew surface*, or *scroll*; but if it be such that each line intersects the consecutive line, then it is a *developable*, or *torse*.

Suppose, for instance, that the equations of a line (depending on the variable parameter θ) are

$$\frac{x}{a} = \frac{y-\theta b}{\theta b} = \frac{z-c}{-c}, \quad \text{or say} \quad \frac{x}{a} = \frac{y-\theta b}{\theta b} = \frac{z-c}{-c},$$

then, eliminating θ, we have $\dfrac{x^2}{a^2}+\dfrac{y^2}{b^2}-\dfrac{z^2}{c^2}=1$, if $z \neq c$, or say $x^2/a^2+y^2/b^2-z^2/c^2=1$, the equation of a quadric surface, afterwards called the hyperboloid of one sheet; this surface is consequently a scroll. It is to be remarked that we have upon the surface a second singly infinite series of lines; the equations of a line of this second system (depending on the variable parameter ϕ) are

$$\frac{x}{a}-\frac{z}{c}=\lambda\left(1-\frac{y}{b}\right), \quad \frac{x}{a}+\frac{z}{c}=\frac{1}{\lambda}\left(1+\frac{y}{b}\right).$$

It is easily shown that any line of the one system intersects every line of the other system.

Considering any curve (of double curvature) whatever, the tangent lines of the curve form a singly infinite system of lines, each line intersecting the consecutive line of the system, that is, they form a developable, or torse; the curve and torse are thus inseparably connected together, forming a single geometrical figure. A plane through three consecutive points of the curve (or osculating plane of the curve) contains two consecutive tangents, that is, two consecutive lines of the torse, and is thus a tangent plane of the torse along a generating line.

Transformation of Coordinates

35. There is no difficulty in changing the origin, and it is for brevity assumed that the origin remains unaltered. We have, then, two sets of rectangular axes, Ox, Oy, Oz, and Ox_1, Oy_1, Oz_1, the mutual cosine-inclinations being shown by the diagram

	x	y	z
x_1	α	β	γ
y_1	α'	β'	γ'
z_1	α''	β''	γ''

that is, α, β, γ are the cosine-inclinations of Ox_1 to Ox, Oy, Oz; α', β', γ' those of Oy_1, &c.

PAGE 574. And this diagram gives also the linear expressions of the coordinates (x_1, y_1, z_1) or (x, y, z) of either set in terms of those of the other set; we thus have

$$\begin{aligned} x_1 &= \alpha x + \beta y + \gamma z, & x &= \alpha x_1 + \alpha' y_1 + \alpha'' z_1, \\ y_1 &= \alpha' x + \beta' y + \gamma' z, & y &= \beta x_1 + \beta' y_1 + \beta'' z_1, \\ z_1 &= \alpha'' x + \beta'' y + \gamma'' z, & z &= \gamma x_1 + \gamma' y_1 + \gamma'' z_1, \end{aligned}$$

which are obtained by projection, as above explained. Each of these equations is, in fact, nothing else than the before-mentioned equation $p = \alpha'\xi + \beta'\eta + \gamma'\zeta$, adapted to the problem in hand.

But we have to consider the relations between the nine coefficients. By what precedes, or by the consideration that we must have identically $x^2 + y^2 + z^2 = x_1^2 + y_1^2 + z_1^2$, it appears that these satisfy the relations

$$\alpha^2 + \beta^2 + \gamma^2 = 1, \quad \alpha'^2 + \beta'^2 + \gamma'^2 = 1, \quad \alpha''^2 + \beta''^2 + \gamma''^2 = 1,$$
$$\alpha'\alpha'' + \beta'\beta'' + \gamma'\gamma'' = 0, \quad \alpha''\alpha + \beta''\beta + \gamma''\gamma = 0, \quad \alpha\alpha' + \beta\beta' + \gamma\gamma' = 0,$$

either set of six equations being implied in the other set.

It follows that the square of the determinant

$$\begin{vmatrix} \alpha & \beta & \gamma \\ \alpha' & \beta' & \gamma' \\ \alpha'' & \beta'' & \gamma'' \end{vmatrix}$$

is $= 1$; and hence that the determinant itself is $= \pm 1$. The distinction of the two cases is an important one: if the determinant is $= +1$, then the axes Ox_1, Oy_1, Oz_1 are such that they can by a rotation about O be brought to coincide with Ox, Oy, Oz respectively; if it is $= -1$, then they cannot. But in the latter case, by measuring x_1, y_1, z_1 in the opposite directions we change the signs of all the coefficients and so make the determinant to be $= +1$; hence this case need alone be considered, and it is accordingly assumed that the determinant is $= +1$. This being so, it is found that we have a further set of nine equations, $\alpha = \beta'\gamma'' - \beta''\gamma'$, &c.; that is, the coefficients arranged as in the diagram have the values

	x	y	z
x_1	$\beta'\gamma'' - \beta''\gamma'$	$\gamma'\alpha'' - \gamma''\alpha'$	$\alpha'\beta'' - \alpha''\beta'$
y_1	$\beta''\gamma - \beta\gamma''$	$\gamma''\alpha - \gamma\alpha''$	$\alpha''\beta - \alpha\beta''$
z_1	$\beta\gamma' - \beta'\gamma$	$\gamma\alpha' - \gamma'\alpha$	$\alpha\beta' - \alpha'\beta$

PAGE 575. **36.** It is important to express the nine coefficients in terms of three independent quantities. A solution which, although unsymmetrical, is very convenient in Astronomy and Dynamics is to use for the purpose the three angles θ, ϕ, τ of fig. 19; say $\theta =$ longitude of the node; $\phi =$ inclination; and $\tau =$ longitude of x_1 (from node, Υ).

The diagram of transformation then is

	x	y	z
x_1	$\cos\tau\cos\theta - \sin\tau\sin\theta\cos\phi$	$\cos\tau\sin\theta + \sin\tau\cos\theta\cos\phi$	$\sin\tau\sin\phi$
y_1	$-\sin\tau\cos\theta - \cos\tau\sin\theta\cos\phi$	$-\sin\tau\sin\theta + \cos\tau\cos\theta\cos\phi$	$\cos\tau\sin\phi$
z_1	$\sin\theta\sin\phi$	$-\cos\theta\sin\phi$	$\cos\phi$

But a more elegant solution (due to Rodrigues) is that contained in the diagram

	x	y	z
x_1	$1+\lambda^2-\mu^2-\nu^2$	$2(\lambda\mu-\nu)$	$2(\lambda\nu+\mu)$
y_1	$2(\lambda\mu+\nu)$	$1-\lambda^2+\mu^2-\nu^2$	$2(\mu\nu-\lambda)$
z_1	$2(\nu\lambda-\mu)$	$2(\mu\nu+\lambda)$	$1-\lambda^2-\mu^2+\nu^2$

$\div(1+\lambda^2+\mu^2+\nu^2)$.

The nine coefficients of transformation are the nine functions of the diagram, each divided by $1+\lambda^2+\mu^2+\nu^2$; the expressions contain as they should do the three arbitrary quantities λ, μ, ν; and the identity $x_1^2+y_1^2+z_1^2 = x^2+y^2+z^2$ can be at once verified.

It may be added that the transformation can be expressed in the quaternion form

$$ix_1 + jy_1 + kz_1 = (1+\Lambda)(ix+jy+kz)(1+\Lambda)^{-1},$$

where Λ denotes the vector $i\lambda + j\mu + k\nu$.

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Quadric Surfaces (Paraboloids, Ellipsoid, Hyperboloids)

37. It appears, by a discussion of the general equation of the second order $(a, \ldots (x, y, z, 1))^2 = 0$, that the proper quadric surfaces[14] represented by such an equation are the following five surfaces (a and b positive):

$$
\begin{array}{lll}
(1) & \dfrac{x^2}{a^2} + \dfrac{y^2}{b^2} = 2z, & \text{elliptic paraboloid,} \\
(2) & \dfrac{x^2}{a^2} - \dfrac{y^2}{b^2} = 2z, & \text{hyperbolic paraboloid,} \\
(3) & \dfrac{x^2}{a^2} + \dfrac{y^2}{b^2} + \dfrac{z^2}{c^2} = 1, & \text{ellipsoid,} \\
(4) & \dfrac{x^2}{a^2} + \dfrac{y^2}{b^2} - \dfrac{z^2}{c^2} = 1, & \text{hyperboloid of one sheet,} \\
(5) & \dfrac{x^2}{a^2} + \dfrac{y^2}{b^2} - \dfrac{z^2}{c^2} = -1, & \text{hyperboloid of two sheets.}
\end{array}
$$

It is at once seen that these are distinct surfaces; and the equations also show very readily the general form and mode of generation of the several surfaces.

In the elliptic paraboloid (fig. 20), the sections by the planes of zx and zy are the parabolas

$$\frac{x^2}{2a^2} = z, \quad \frac{y^2}{2b^2} = z$$

having the common axis Oz; and the section by any plane $z = \gamma$ parallel to that of xy is the ellipse

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 2\gamma,$$

so that the surface is generated by a variable ellipse moving parallel to itself along the parabolas as directrices.

[14]The improper quadric surfaces represented by the general equation of the second order are (1) the pair of planes or plane-pair, including as a special case the twice repeated plane, and (2) the cone, including as a special case the cylinder. There is but one form of cone; but the cylinder may be parabolic, elliptic, or hyperbolic.

PAGE 577. In the hyperbolic paraboloid (fig. 21), the sections by the planes of zx, zy are the parabolas

$$\frac{x^2}{2a^2} = z, \quad -\frac{y^2}{2b^2} = z$$

having the opposite axes Oz, Oz'; and the section by a plane $z = \gamma$ parallel to that of xy is the hyperbola

$$\frac{x^2}{a^2} - \frac{y^2}{b^2} = 2\gamma,$$

which has its transverse axis parallel to Ox or Oy according as γ is positive or negative. The surface is thus generated by a variable hyperbola moving parallel to itself along the parabolas as directrices. The form is best seen from fig. 22, which represents the sections by planes parallel to the plane of xy, or say the contour lines; the continuous lines are the sections above the plane of xy, and the dotted lines the sections below this plane. The form is, in fact, that of a saddle.

In the ellipsoid (fig. 23), the sections by the planes of zx, zy, and xy are each of them an ellipse, and the section by any parallel plane is also an ellipse. The surface may be considered as generated by an ellipse moving parallel to itself along two ellipses as directrices.

In the hyperboloid of one sheet (fig. 24), the sections by the planes of zx, zy are the hyperbolas

$$\frac{x^2}{a^2} - \frac{z^2}{c^2} = 1, \quad \frac{y^2}{b^2} - \frac{z^2}{c^2} = 1$$

having a common conjugate axis zOz'; the section by the plane of xy, and that by any parallel plane, is an ellipse; and the surface may be considered as generated by a variable ellipse moving parallel to itself along the two hyperbolas as directrices.

In the hyperboloid of two sheets (fig. 25), the sections by the planes of zx and zy are the hyperbolas

$$\frac{x^2}{a^2} - \frac{z^2}{c^2} = 1, \quad \frac{y^2}{b^2} - \frac{z^2}{c^2} = 1$$

having the common transverse axis zOz'; the section by any plane $z = \pm\gamma$ parallel to that of xy, γ being in absolute magnitude $> c$, is the ellipse

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = \frac{\gamma^2}{c^2} - 1,$$

and the surface, consisting of two distinct portions or sheets, may be considered as generated by a variable ellipse moving parallel to itself along the hyperbolas as directrices.

The hyperbolic paraboloid is such (and it is easy from the figure to understand how this may be the case) that there exist upon it two singly infinite series of right lines. The same is the case with the hyperboloid of one sheet (ruled or skew hyperboloid, as with reference to this property it is termed). If we imagine two equal and parallel circular disks, their points connected by strings of equal length, so that these are the generating lines of a right circular cylinder, then by turning one of the disks about its centre through the same angle in one or the other direction, the strings will in each case generate one and the same hyperboloid, and will in regard to it be the two systems of lines on the surface, or say the two systems of generating lines; and the general configuration is the same when instead of circles we have ellipses. It has been already shown analytically that the equation $\dfrac{x^2}{a^2} + \dfrac{y^2}{b^2} - \dfrac{z^2}{c^2} = 1$ is satisfied by each of two pairs of linear relations between the coordinates.

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Curves; Tangent, Osculating Plane, Curvature, &c.

38. It will be convenient to consider the coordinates (x, y, z) of the point on the curve as given in terms of a parameter θ, so that dx, dy, dz, d^2x, &c., will be proportional to $\dfrac{dx}{d\theta}$, $\dfrac{dy}{d\theta}$, $\dfrac{dz}{d\theta}$, $\dfrac{d^2x}{d\theta^2}$, &c.; and in them ξ, η, ζ are used as current coordinates.

The tangent is the line through the point (x, y, z) and the consecutive point $(x + dx, y + dy, z + dz)$; its equations therefore are

$$\frac{\xi - x}{dx} = \frac{\eta - y}{dy} = \frac{\zeta - z}{dz}.$$

The osculating plane is the plane through the point and two consecutive points, and contains therefore the tangent; its equation is

$$\begin{vmatrix} \xi - x & \eta - y & \zeta - z \\ dx & dy & dz \\ d^2x & d^2y & d^2z \end{vmatrix} = 0,$$

or, what is the same thing,

$$(\xi - x)(dy\,d^2z - dz\,d^2y) + (\eta - y)(dz\,d^2x - dx\,d^2z) \\ + (\zeta - z)(dx\,d^2y - dy\,d^2x) = 0. \quad (1)$$

The normal plane is the plane through the point at right angles to the tangent. It meets the osculating plane in a line called the *principal normal*; and drawing through the point a line at right angles to the osculating plane, this is called the *binormal*. We have thus at the point a set of three rectangular axes—the tangent, the principal normal, and the binormal.

We have through the point and three consecutive points a sphere of spherical curvature, the centre and radius thereof being the centre, and radius, of spherical curvature. The sphere is met by the osculating plane in the circle of absolute curvature, the centre and radius thereof being the centre, and radius, of absolute curvature. The centre of absolute curvature is also the intersection of the principal normal by the normal plane at the consecutive point.

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Surfaces; Tangent Lines and Plane, Curvature, &c.

39. It will be convenient to consider the surface as given by an equation $f(x, y, z) = 0$ between the coordinates; taking (x, y, z) for the coordinates of a given point, and $(x + dx, y + dy, z + dz)$ for those of a consecutive point, the increments dx, dy, dz satisfy the condition

$$\frac{df}{dx}dx + \frac{df}{dy}dy + \frac{df}{dz}dz = 0;$$

but the ratio of two of the increments, suppose $dx : dy$, may be regarded as arbitrary.

Only a part of the analytical formulae will be given; in them ξ, η, ζ are used as current coordinates.

We have through the point a singly infinite series of right lines, each meeting the surface in a consecutive point, or say having each of them two-point intersection with the surface. These lines lie all of them in a plane which is the tangent plane; its equation is

$$(\xi - x)\frac{df}{dx} + (\eta - y)\frac{df}{dy} + (\zeta - z)\frac{df}{dz} = 0,$$

as is at once verified by observing that this equation is satisfied (irrespectively of the value of $dx : dy$) on writing therein ξ, η, $\zeta = x + dx$, $y + dy$, $z + dz$.

Page 581. The line through the point at right angles to the tangent plane is called the *normal*; its equations are

$$\frac{\xi - x}{\frac{df}{dx}} = \frac{\eta - y}{\frac{df}{dy}} = \frac{\zeta - z}{\frac{df}{dz}}.$$

In the series of tangent lines there are in general two (real or imaginary) lines, each of which meets the surface in a second consecutive point, or say it has three-point intersection with the surface; these are called the *chief-tangents* (Haupt-tangenten). The tangent-plane cuts the surface in a curve, having at the point of contact a node (double point), the tangents to the two branches being the chief-tangents.

In the case of a quadric surface the curve of intersection, qua curve of the second order, can only have a node by breaking up into a pair of lines; that is, every tangent-plane meets the surface in a pair of lines, or we have on the surface two singly infinite systems of lines; these are real for the hyperbolic paraboloid and the hyperboloid of one sheet, imaginary in other cases.

At each point of a surface the chief-tangents determine two directions; and passing along one of them to a consecutive point, and thence (without abrupt change of direction) along the new chief-tangent to a consecutive point, and so on, we have on the surface a chief-tangent curve; and there are, it is clear, two singly infinite series of such curves. In the case of a quadric surface, the curves are the right lines on the surface.

40. If at the point we draw in the tangent-plane two lines bisecting the angles between the chief-tangents, these lines (which are at right angles to each other) are called the *principal tangents*[15]. We have thus at each point of the surface a set of rectangular axes, the normal and the two principal tangents.

Proceeding from the point along a principal tangent to a consecutive point on the surface, and thence (without abrupt change of direction) along the new principal tangent to a consecutive point, and so on, we have on the surface a *curve of curvature*; there are, it is clear, two singly infinite series of such curves, cutting each other at right angles at each point of the surface.

[15]The point on the surface may be such that the directions of the principal tangents become arbitrary; the point is then an *umbilicus*. It is in the text assumed that the point on the surface is not an umbilicus.

Passing from the given point in an arbitrary direction to a consecutive point on the surface, the normal at the given point is not intersected by the normal at the consecutive point; but passing to the consecutive point along a curve of curvature (or, what is the same thing, along a principal tangent) the normal at the given point is intersected by the normal at the consecutive point; we have thus on the normal two centres of curvature, and the distances of these from the point on the surface are the two *principal radii of curvature* of the surface at that point; these are also the radii of curvature of the sections of the surface by planes through the normal and the two principal tangents respectively; or say they are the radii of curvature of the normal sections through the two principal tangents respectively. Take at the point the axis of z in the direction of the normal, and those of x and y in the directions of the principal tangents respectively, then, if the radii of curvature be a, b (the signs being such that the coordinates of the two centres of curvature are $z = a$ and $z = b$ respectively), the surface has in the neighbourhood of the point the form of the paraboloid

$$z = \frac{x^2}{2a} + \frac{y^2}{2b},$$

and the chief-tangents are determined by the equation $0 = \dfrac{x^2}{2a} + \dfrac{y^2}{2b}$. The two centres of curvature may be on the same side of the point or on opposite sides; in the former case a and b have the same sign, the paraboloid is elliptic, and the chief-tangents are imaginary; in the latter case a and b have opposite signs, the paraboloid is hyperbolic, and the chief-tangents are real.

The normal sections of the surface and the paraboloid by the same plane have the same radius of curvature; and it thence readily follows that the radius of curvature of a normal section of the surface by a plane inclined at an angle θ to that of zx is given by the equation

$$\frac{1}{\rho} = \frac{\cos^2\theta}{a} + \frac{\sin^2\theta}{b}.$$

The section in question is that by a plane through the normal and a line in the tangent plane inclined at an angle θ to the principal tangent along the axis of x. To complete the theory, consider the section by a plane having the same trace upon the tangent plane, but inclined to the normal at an angle ϕ; then it is shown without difficulty (Meunier's theorem) that the radius of curvature of this inclined section of the surface is $= \rho \cos \phi$.

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791. LANDEN.

[From the *Encyclopædia Britannica*, Ninth Edition, vol. xiv. (1882), p. 271.]

LANDEN, JOHN, a distinguished mathematician of the 18th century, was born at Peakirk near Peterborough in Northamptonshire in 1719, and died 15th January 1790 at Milton in the same county. Most of his time was spent in the pursuits of active life, but he early showed a strong talent for mathematical study, which he eagerly cultivated in his leisure hours. In 1762 he was appointed agent to the Earl Fitzwilliam, and held that office to within two years of his death. He lived a very retired life, and saw little or nothing of society; when he did mingle in it, his dogmatism and pugnacity caused him to be generally shunned. He was first known as a mathematician by his essays in the *Ladies' Diary* for 1744. In 1766 he was elected a Fellow of the Royal Society. He was well acquainted and *au courant* with the works of the mathematicians of his own time, and has been called the English D'Alembert. In his Discourse on the "Residual Analysis," in which he proposes to substitute for the method of fluxions a purely algebraical method, he says, "It is by means of the following theorem, viz.

$$\frac{X^n - Y^n}{X^m - Y^m} = \frac{X^{n-1} + X^{n-2}Y + \cdots + Y^{n-1}}{X^{m-1} + X^{m-2}Y + \cdots + Y^{m-1}} \quad (n \text{ and } m \text{ terms}),$$

(where m and n are integers), that we are able to perform all the principal operations in our said analysis; and I am not a little surprised that a theorem so obvious, and of such vast use, should so long escape the notice of algebraists." The idea is of course a perfectly legitimate one, and may be compared with that of Lagrange's *Calcul des Fonctions*. His memoir (1775) on the rotatory motion of a body contains (as the author was aware) conclusions at variance with those arrived at by D'Alembert and Euler in their researches on the same subject. He reproduces and further develops and defends his own views in his *Mathematical Memoirs*, and in his paper in the *Philosophical Transactions* for 1785. But Landen's capital discovery is that of the theorem known by his name (obtained in its complete form in the memoir of 1775, and reproduced in the first volume of the

Mathematical Memoirs) for the expression of the arc of an hyperbola in terms of two elliptic arcs. To find this, he integrates a differential equation derived from the equation

$$\frac{\dot{x}}{\dot{y}} = \frac{y\sqrt{1+g\,t^2}}{t\sqrt{1+g'\,t^2}},$$

interpreting geometrically in an ingenious and elegant manner three integrals which present themselves. If in the foregoing equation we write $m = 1$, $g = k^2$, and instead of t consider the new variable $y = t\sqrt{1-k^2}$, then

$$\frac{dy}{\sqrt{(1-y^2)(1-k'^2y^2)}} = \frac{dx}{\sqrt{(1-x^2)(1-k^2x^2)}},$$

which is the form known as Landen's transformation in the theory of elliptic functions: but his investigation does not lead him to obtain the equivalent of the resulting differential equation

$$\sqrt{1-y^4}\,\frac{dy}{\sqrt{1-y^4}} = \cdots,$$

due it would appear to Legendre, and which (over and above Landen's own beautiful result) gives importance to the theorem as leading directly to the quadric transformation of an elliptic integral in regard to the modulus.

The list of his writings is as follows: *Ladies' Diary*, various communications, 1744—1760; papers in the *Phil. Trans.*, 1754, 1760, 1768, 1771, 1775, 1777, 1785; *Mathematical Lucubrations*, 1755; *Discourse concerning the Residual Analysis*, 1758; *The Residual Analysis*, book i., 1764; *Animadversions on Dr Stewart's Method of computing the Sun's Distance from the Earth*, 1771; *Mathematical Memoirs*, 1780, 1789.

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792. LOCUS.

[From the *Encyclopædia Britannica*, Ninth Edition, vol. xiv. (1882), pp. 764, 765.]

LOCUS, in Greek $\tau\acute{o}\pi o\varsigma$, a geometrical term, the invention of the notion of which is attributed to Plato. It occurs in such statements as these: the locus of the points which are at the same distance from a fixed point, or of a point which moves so as to be always at the same distance from a fixed point, is a circle; conversely a circle is the locus of the points at the same distance from a fixed point, or of a point moving so as to be always at the same distance from a fixed point; and so, in general, a curve of any given kind is the locus of the points which satisfy, or of a point moving so as always to satisfy, a given condition. The theory of loci is thus identical with that of curves; and it is in fact in this very point of view that a curve is considered in the article Curve, [785]; see that article, and also Geometry (Analytical), [790]. It is only necessary to add that the notion of a locus is useful as regards determinate problems or theorems; thus, to find the centre of the circle circumscribed about a given triangle ABC, we see that the circumscribed circle must pass through the two vertices A, B, and the locus of the centres of the circles which pass through these two points is the straight line at right angles to the side AB at its mid-point; similarly the circumscribed circle must pass through A, C, and the locus of the centres of the circles through these two points is the line at right angles to the side AC at its mid-point; thus we get the ordinary construction, and also the theorem that the lines at right angles to the sides, at their mid-points respectively, meet in a point. The notion of a locus applies, of course, not only to plane but also to solid geometry. Here the locus of the points satisfying a single (or onefold) condition is a surface; the locus of the points satisfying two conditions (or a twofold condition) is a curve in space, which is in general a twisted curve or curve of double curvature.

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793. MONGE.

[From the *Encyclopædia Britannica*, Ninth Edition, vol. xvi. (1883), pp. 738, 739.]

MONGE, GASPARD (1746–1818), a French mathematician, the inventor of descriptive geometry, was born at Beaune on the 10th May 1746. He was educated first at the college of the Oratorians at Beaune, and then in their college at Lyons, where, at sixteen, the year after he had been learning physics, he was made a teacher of it. Returning to Beaune for a vacation, he made, on a large scale, a plan of the town, inventing the methods of observation and constructing the necessary instruments; the plan was presented to the town, and preserved in their library. An officer of engineers seeing it wrote to recommend Monge to the commandant of the military school at Mézières, and he was received as draftsman and pupil in the practical school attached to that institution; the school itself was of too aristocratic a character to allow of his admission to it. His manual skill was duly appreciated: "I was a thousand times tempted," he said long afterwards, "to tear up my drawings in disgust at the esteem in which they were held, as if I had been good for nothing better." An opportunity, however, presented itself: being required to work out from data supplied to him the "defilement" of a proposed fortress (an operation then only performed by a long arithmetical process), Monge, substituting for this a geometrical method, obtained the result so quickly that the commandant at first refused to receive it—the time necessary for the work had not been taken; but upon examination the value of the discovery was recognized, and the method was adopted. And Monge, continuing his researches, arrived at that general method of the application of geometry to the arts of construction which is now called descriptive geometry. But such was the system in France before the Revolution that the officers instructed in the method were strictly forbidden to communicate it even to those engaged in other branches of the public service; and it was not until many years afterwards that an account of it was published. The method consists, as is well known, in the use of the two halves of a sheet of paper to represent say the planes of xy and xz at right angles to each other, and the consequent representation of points, lines, and figures in space by means of their plan and elevation, placed in a determinate relative position.

In 1768 Monge became professor of mathematics, and in 1771 professor of physics, at Mézières; in 1778 he married Madame Horbon, a young widow whom he had previously defended in a very spirited manner from an unfounded charge; in 1780 he was appointed to a chair of hydraulics at the Lyceum in Paris (held by him together with his appointments at Mézières), and was received as a member of the Academy; his intimate friendship with Berthollet began at this time. In 1783, quitting Mézières, he was, on the death of Bezout, appointed examiner of naval candidates. Although pressed by the minister to prepare for them a complete course of mathematics, he declined to do so, on the ground that it would deprive Madame Bezout of her only income, arising from the sale of the works of her late husband; he wrote, however (1786), his *Traité élémentaire de la Statique*.

Monge contributed (1770–1790) to the *Memoirs of the Academy of Turin*, the *Mémoires des Savants Étrangers* of the Academy of Paris, the *Mémoires* of the same Academy, and the *Annales de Chimie*, various mathematical and physical papers. Among these may be noticed the memoir "Sur la théorie des déblais et des remblais" (Mém. de l'Acad. de Paris, 1781), which, while giving a remarkably elegant investigation in regard to the problem of earthwork referred to in the title, establishes in connexion with it his capital discovery of the curves of curvature of a surface. Euler, in his paper on curvature in the Berlin Memoirs for 1760, had considered, not the normals of the surface, but the normals of the plane sections through a particular normal, so that the question of the intersection of successive normals of the surface had never presented itself to him. Monge's memoir just referred to gives the ordinary differential equation of the curves of curvature, and establishes the general theory in a very satisfactory manner; but the application to the interesting particular case of the ellipsoid was first made by him in a later paper in 1795. A memoir in the volume for 1783 relates to the production of water by the combustion of hydrogen; but Monge's results in this matter had been anticipated by Watts and Cavendish.

In 1792, on the creation by the Legislative Assembly of an executive council, Monge accepted the office of minister of the marine, but retained it only until April 1793. When the Committee of Public Safety made an appeal to the savants to assist in producing the matériel required for the defence of the republic, he applied himself wholly to these operations, and distinguished himself by his indefatigable activity therein; he wrote at this time his *Description de l'art de fabriquer les canons*, and his *Avis aux ouvriers en fer sur la fabrication de l'acier.* He took a very active part in the measures for the establishment of the Normal School (which existed only during the first four months of the year 1795), and of the School for Public Works, afterwards the Polytechnic School, and was at each of them professor of descriptive geometry; his methods in that science were first published in the form in which the shorthand writers took down his lessons given at the Normal School in 1795, and again in 1798–99. In 1796 Monge was sent into Italy with Berthollet and some artists to receive the pictures and statues levied from several Italian towns, and made there the acquaintance of General Bonaparte.

PAGE 588. Two years afterwards he was sent to Rome on a political mission, which terminated in the establishment, under Massena, of the shortlived Roman republic; and he thence joined the expedition to Egypt, taking part with his friend Berthollet as well in various operations of the war as in the scientific labours of the Egyptian Institute of Sciences and Arts; they accompanied Bonaparte to Syria, and returned with him in 1798 to France. Monge was appointed president of the Egyptian commission, and he resumed his connexion with the Polytechnic School. His later mathematical papers are published (1794–1816) in the *Journal* and the *Correspondance* of the Polytechnic School. On the formation of the Senate he was appointed a member of that body, with an ample provision and the title of count of Pelusium; but on the fall of Napoleon he was deprived of all his honours, and even excluded from the list of members of the reconstituted Institute. He died at Paris on the 28th July 1818.

For further information see B. Brisson, *Notice historique sur Gaspard Monge*; Dupin, *Essai historique sur les services et les travaux scientifiques de Gaspard Monge*, Paris, 1819, which contains (pp. 162–166) a list of Monge's memoirs and works; and the biography by Arago (*Œuvres*, t. II., 1854).

Monge's various mathematical papers are to a considerable extent reproduced in the *Application de l'Analyse à la Géométrie*, 4th edition (last revised by the author), Paris, 1819; the pure text of this is reproduced in the 5th edition (revue, corrigée et annotée par M. Liouville), Paris, 1850, which contains also Gauss's Memoir, "Disquisitiones generales circa superficies curvas," and some valuable notes by the editor. The other principal separate works are *Traité élémentaire de la Statique*, 8^e Edition, conformée à la précédente, par M. Hachette, et suivie d'une Note etc., par M. Cauchy, Paris, 1846; and the *Géométrie Descriptive* (originating, as mentioned above, in the lessons given at the Normal School). The 4th edition, published shortly after the author's death, seems to have been substantially the same as the 7th (*Géométrie Descriptive*, par G. Monge, suivie d'une théorie des Ombres et de la Perspective, extraite des papiers de l'auteur, par M. Brisson, Paris, 1847).

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794. NUMBERS, PARTITION OF.

[From the *Encyclopædia Britannica*, Ninth Edition, vol. xvii. (1884), p. 614.]

This subject, created by Euler, though relating essentially to positive integer numbers, is scarcely regarded as a part of the Theory of Numbers. We consider in it a number as made up by the addition of other numbers: thus the partitions of the successive numbers 1, 2, 3, 4, 5, 6, &c., are as follows:

$$\begin{gathered}1;\\ 2,\ 11;\\ 3,\ 21,\ 111;\\ 4,\ 31,\ 22,\ 211,\ 1111;\\ 5,\ 41,\ 32,\ 311,\ 221,\ 2111,\ 11111;\\ 6,\ 51,\ 42,\ 411,\ 33,\ 321,\ 3111,\ 222,\ 2211,\ 21111,\ 111111.\end{gathered}$$

These are formed each from the preceding ones; thus, to form the partitions of 6 we take first 6; secondly, 5 prefixed to each of the partitions of 1 (that is, 51); thirdly, 4 prefixed to each of the partitions of 2 (that is, 42, 411); fourthly, 3 prefixed to each of the partitions of 3 (that is, 33, 321, 3111); fifthly, 2 prefixed, not to each of the partitions of 4, but only to those partitions which begin with a number not exceeding 2 (that is, 222, 2211, 21111); and lastly, 1 prefixed to all the partitions of 5 which begin with a number not exceeding 1 (that is, 111111); and so in other cases.

The method gives all the partitions of a number, but we may consider different classes of partitions: the partitions into a given number of parts, or into not more than a given number of parts; or the partitions into given parts, either with repetitions or without repetitions, &c. It is possible, for any particular class of partitions, to obtain methods more or less easy for the formation of the partitions either of a given number or of the successive numbers 1, 2, 3, &c. And of course in any case, having obtained the partitions, we can count them and so obtain the number ofpartitions.

PAGE 590. Another method is by Arbogast's rule of the last and the last but one; in fact, taking the value of a to be unity, and, understanding this letter in each term, the rule gives b; c, b^2; d, bc, b^3; e, bd, c^2, b^2c, b^4, &c., which, if b, c, d, e, &c., denote 1, 2, 3, 4, &c., respectively, are the partitions of 1, 2, 3, 4, &c., respectively.

An important notion is that of *conjugate partitions*. Thus a partition of 6 is 42; writing this in the form

$$\begin{array}{ll} ** & ** \\ * & \\ * & \end{array}$$

and summing the columns instead of the lines, we obtain the conjugate partition 2211; evidently, starting from 2211, the conjugate partition is 42. If we form all the partitions of 6 into not more than three parts, these are

$$6,\ 51,\ 42,\ 33,\ 411,\ 321,\ 222,$$

and the conjugates are

$$111111,\ 21111,\ 2211,\ 222,\ 3111,\ 321,\ 33,$$

where no part is greater than 3; and so, in general, we have the theorem, the number of partitions of n into not more than k parts is equal to the number of partitions of n with no part greater than k.

We have for the number of partitions an analytical theory depending on generating functions; thus for the partitions of a number n with the parts 1, 2, 3, 4, 5, &c., without repetitions, writing down the product

$$(1+x)(1+x^2)(1+x^3)(1+x^4)(1+x^5)\ldots = 1+x+x^2+2x^3+\ldots+Nx^n+\ldots,$$

it is clear that, if x^α, x^β, x^γ,... are terms of the series x, x^2, x^3,... for which $\alpha+\beta+\gamma+\ldots = n$, then we have in the development of the product a term x^n, and hence that, in the term Nx^n of the product, the coefficient N is equal to the number of partitions of n with the parts 1, 2, 3,..., without repetitions; or say that the product is the generating function (G. F.) for the number of such partitions. And so in other cases we obtain a generating function.

Thus for the function

$$\frac{1}{(1-x)(1-x^2)(1-x^3)(1-x^4)\ldots} = 1+x+2x^2+\ldots+Nx^n+\ldots,$$

observing that any factor $1/(1-x^k)$ is $= 1 + x^k + x^{2k} + \ldots$, we see that, in the term Nx^n, the coefficient N is equal to the number of partitions of n, with the parts 1, 2, 3,…, with repetitions.

Introducing another letter z, and considering the function

$$(1+xz)(1+x^2z)(1+x^3z)\ldots = 1+z(x+x^2+\ldots)+\ldots+Nx^nz^k+\ldots$$

we see that, in the term Nx^nz^k of the development, the coefficient N is equal to the number of partitions of n into k parts, with the parts 1, 2, 3, 4,…, without repetitions. And similarly, considering the function

$$\frac{1}{(1-xz)(1-x^2z)(1-x^3z)\ldots} = 1+z(x+x^2+\ldots)+\ldots+Nx^nz^k+\ldots$$

we see that, in the term Nx^nz^k of the development, the coefficient N is equal to the number of partitions of n into k parts, with the parts 1, 2, 3, 4,…, with repetitions.

PAGE 591. We have such analytical formulae as

$$\frac{1}{(1-xz)(1-x^2z)(1-x^3z)\dots} = 1 + \frac{zx}{1-x} + \frac{z^2x^2}{(1-x)(1-x^2)} + \dots,$$

which lead to theorems in the Partition of Numbers. A remarkable theorem is

$$(1-x)(1-x^2)(1-x^3)(1-x^4)\dots = 1-x-x^2+x^5+x^7-x^{12}-x^{15}+\dots,$$

where the only terms are those with an exponent $\frac{1}{2}(3n^2 \pm n)$ and for each such pair of terms the coefficient is $(-)^n$. The formula shows that, except for numbers of the form $\frac{1}{2}(3n^2 \pm n)$, the number of partitions without repetitions into an odd number of parts is equal to the number of partitions without repetitions into an even number of parts, whereas for the excepted numbers these numbers differ by unity. Thus for the number 11, which is not an excepted number, the two sets of partitions are

$$\begin{gathered} 11,\ 821,\ 731,\ 641,\ 632,\ 542, \\ 10.1,\ 92,\ 83,\ 74,\ 65,\ 5321, \end{gathered}$$

in each set 6.

We have

$$(1-x)(1+x)(1+x^2)(1+x^3)(1+x^4)\dots = 1;$$

or, as this may be written,

$$(1+x)(1+x^2)(1+x^3)(1+x^4)\dots = \frac{1}{1-x} = 1 + x + x^2 + x^3 + \dots,$$

showing that a number n can always be made up, and in one way only, with the parts 1, 2, 4, 8,.... The product on the left-hand side may be taken to k terms only: thus if $k = 4$, we have

$$(1+x)(1+x^2)(1+x^4)(1+x^8) = \frac{1-x^{16}}{1-x} = 1 + x + x^2 + \dots + x^{15},$$

that is, any number from 1 to 15 can be made up, and in one way only, with the parts 1, 2, 4, 8; and similarly any number from 1 to $2^k - 1$ can be made up, and in one way only, with the parts 1, 2, 4,..., 2^{k-1}. A like formula is

$$\frac{(1-x^3)(1-x^9)(1-x^{27})(1-x^{81})}{x(1-x)(x^2-x^3)(x^5-x^9)(x^{14}-x^{27})(x^{41}-x^{81})} = \dots,$$

that is,

$$x^{-40} + 1 + x \cdot x^{-39} + 1 + x^2 \cdot \ldots \cdot x^{-1} + 1 + x + x^2 + \ldots + x^{39} + x^{40},$$

showing that any number from -40 to $+40$ can be made up, and that in one way only, with the parts 1, 3, 9, 27 taken positively or negatively; and so in general any number from $-\frac{1}{2}(3^k - 1)$ to $+\frac{1}{2}(3^k - 1)$ can be made up, and that in one way only, with the parts 1, 3, 9,..., 3^{k-1} taken positively or negatively.

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795. NUMBERS, THEORY OF.

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The Theory of Numbers, or higher arithmetic, otherwise arithmology, is a subject which, originating with Euclid, has in modern times, in the hands of Legendre, Gauss, Lejeune-Dirichlet, Kummer, Kronecker, and others, been developed into a most extensive and interesting branch of mathematics. We distinguish between the ordinary (or say the simplex) theory and the various complex theories.

In the ordinary theory we have, in the first instance, positive integer numbers, the unit or unity 1, and the other numbers 2, 3, 4, 5, &c. We introduce the zero 0, which is a number *sui generis*, and the negative numbers -1, -2, -3, -4, &c., and we have thus the more general notion of integer numbers, 0, ± 1, ± 2, ± 3, &c.; $+1$ and -1 are units or unities. The sum of any two or more numbers is a number; conversely, any number is a sum of two or more parts; but even when the parts are positive a number cannot be, in a determinate manner, represented as a sum of parts. The product of two or more numbers is a number; but (disregarding the unities $+1$, -1, which may be introduced as factors at pleasure) it is not conversely true that every number is a product of numbers. A number such as 2, 3, 5, 7, 11, &c., which is not a product of numbers, is said to be a *prime number*; and a number which is not prime is said to be *composite*. A number other than zero is thus either prime or composite; and we have the theorem that every composite number is, in a determinate way, a product of prime factors.

We have complex theories in which all the foregoing notions (integer, unity, zero, prime, composite) occur; that which first presented itself was the theory with the unit i ($i^2 = -1$); we have here complex numbers, $a+bi$, where a and b are in the before-mentioned (ordinary) sense positive or negative integers, not excluding zero; we have the zero 0, $= 0+0i$, and the four units 1, -1, i, $-i$. A number other than zero is here either prime or else composite; for instance, 3, 7, 11, are prime numbers, and $5 = (2+i)(2-i)$, $9 = 3 \cdot 3$, $13 = (3+2i)(3-2i)$, are composite numbers (generally any positive real prime of the form $4n+3$ is prime, but any positive real prime of the form $4n+1$ is a sum of two squares, and is thus composite). And disregarding unit factors we have, as in the ordinary theory, the theorem that every composite number is, in a determinate way, a product of prime factors.

There is, in like manner, a complex theory involving the cube roots of unity—if ω be an imaginary cube root of unity ($\omega^2+\omega+1=0$), then the integers of this theory are $a+b\omega$ (a and b real positive or negative integers, including zero); a complex theory with the fifth roots of unity—if ω be an imaginary fifth root of unity ($\omega^4+\omega^3+\omega^2+\omega+1=0$), then the integers of the theory are $a+b\omega+c\omega^2+d\omega^3$ (a, b, c, d, real positive or negative integers, including zero); and so on for the roots of the orders 7, 11, 13, 17, 19. In all these theories, or at any rate for the orders 3, 5, 7 (see No. 37, post), we have the foregoing theorem: disregarding unit factors, a number other than zero is either prime or composite, and every composite number is, in a determinate way, a product of prime factors. But coming to the 23rd roots of unity the theorem ceases to be true. Observe that it is a particular case of the theorem that, if N be a prime number, any integer power of N has for factors only the lower powers of N,—for instance, $N^3 = N \cdot N^2$; there is no other decomposition $N^3 = A \cdot B$. This is obviously true in the ordinary theory, and it is true in the complex theories preceding those for the 3rd, 5th, and 7th roots of unity, and probably in those for the other roots preceding the 23rd roots; but it is not true in the theory for the 23rd roots of unity. We have, for instance, 47, a number not decomposable into factors, but $47^2 = AB$, is a product of two numbers each of the form $a + b\omega + \ldots + l\omega^{22}$ (ω a 23rd root). The theorem recovers its validity by the introduction into the theory of Kummer's notion of an ideal number.

The complex theories above referred to would be more accurately described as theories for the complex numbers involving the periods of the roots of unity: the units are the roots either of the equation $x^{p-1} + x^{p-2} + \ldots + x + 1 = 0$ (p a prime number) or of any equation $x^{\frac{p-1}{q}} + \ldots + 1 = 0$ belonging to a factor of the function of the order $p-1$: in particular, this may be the quadric equation for the periods each of $\frac{1}{2}(p-1)$ roots; they are the theories which were first and have been most completely considered, and which led to the notion of an ideal number. But a yet higher generalization which has been made is to consider the complex theory, the units whereof are the roots of any given irreducible equation which has integer numbers for its coefficients.

There is another complex theory the relation of which to the foregoing is not very obvious, viz. Galois's theory of the numbers composed with the imaginary roots of an irreducible congruence, $F(x) = 0$ (modulus a prime number p); the nature of this will be indicated in the sequel.

In any theory, ordinary or complex, we have a first part, which has been termed (but the name seems hardly wide enough) the theory of congruences; a second part, the theory of homogeneous forms: this includes in particular the theory of the binary quadratic forms $(a, b, c)(x, y)^2$; and a third part, comprising those miscellaneous investigations which do not come properly under either of the foregoing heads.

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Ordinary Theory, First Part

1. We are concerned with the integer numbers 0, ±1, ±2, ±3, &c., or in the first place with the positive integer numbers 1, 2, 3, 4, 5, 6, &c. Some of these, 1, 2, 3, 5, 7, &c., are prime, others, $4 = 2^2$, $6 = 2 \cdot 3$, &c., are composite; and we have the fundamental theorem that a composite number is expressible, and that in one way only, as a product of prime factors, $N = a^\alpha b^\beta c^\gamma \ldots$ (a, b, c, $\ldots$ primes other than 1; α, β, γ, $\ldots$ positive integers).

Gauss makes the proof to depend on the following steps: (i) the product of two numbers each smaller than a given prime number is not divisible by this number; (ii) if neither of two numbers is divisible by a given prime number the product is not so divisible; (iii) the like as regards three or more numbers; (iv) a composite number cannot be resolved into factors in more than one way.

2. Proofs will in general be only indicated or be altogether omitted, but, as a specimen of the reasoning in regard to whole numbers, the proofs of these fundamental propositions are given at length. (i) Let p be the prime number, a a number less than p, and if possible let there be a number b less than p, and such that ab is divisible by p; it is further assumed that b is the only number, or, if there is more than one, then that b is the least number having the property in question; b is greater than 1, for a being less than p is not divisible by p. Now p as a prime number is not divisible by b, but must lie between two consecutive multiples of b, say mb and $(m+1)b$; if so, $p-mb$, which is less than b, and since neither p nor mb is divisible by b, $p-mb$ is not divisible by b; but ab and mb are each of them divisible by b, therefore their difference $ab - mb = a(p-mb) + m(ab-p)$, since $ab - p$ is divisible by b, must also be divisible by p, or, writing $p - mb = b'$, we have ab' divisible by p, where b' is less than b: so that b is not the least number for which ab is divisible by p. (ii) If a and b are neither of them divisible by p, then a divided by p leaves a remainder α which is less than p, say we have $a = mp + \alpha$; and similarly b divided by p leaves a remainder β which is less than p, say we have $b = np + \beta$; then

$$ab = (mp+\alpha)(np+\beta) = (mnp + n\alpha + m\beta)p + \alpha\beta,$$

and $\alpha\beta$ is not divisible by p, therefore ab is not divisible by p. (iii) The like proof applies to the product of three or more factors a, b, c, $\ldots$ (iv) Suppose that the number $N = a^\alpha b^\beta c^\gamma \ldots$ (a, b, c, $\ldots$ prime numbers other than 1), is decomposable in some other way into prime factors; we can have no prime factor p, other than a, b, c, $\ldots$, for no such number can divide $a^\alpha b^\beta c^\gamma \ldots$; and we must have each of the numbers a, b, c, $\ldots$, for if any one of them, suppose a, were wanting, the number N would not be divisible by a. Hence the new decomposition if it exists must be a decomposition $N = a^{\alpha'} b^{\beta'} c^{\gamma'} \ldots$; and here, if any two corresponding indices, say α, α', are different from each other, then one of them, suppose α, is the greater, and we have $N \div a^\alpha = b^\beta c^\gamma \ldots = a^{\alpha'-\alpha} b^{\beta'} c^{\gamma'} \ldots$. That is, we have the number $N \div a^\alpha$ expressed in two different ways as a product, the number a being a factor in the one case, but not a factor in the other case. Thus the two exponents cannot be unequal, that is, we must in fact have $\alpha' = \alpha$, and similarly we have $\beta' = \beta$, $\gamma' = \gamma$, $\ldots$; that is, there is only the one decomposition $N = a^\alpha b^\beta c^\gamma \ldots$.

PAGE 595. **3.** The only numbers divisible by a number $N = a^\alpha b^\beta c^\gamma \ldots$ are the numbers $a^{\alpha'} b^{\beta'} c^{\gamma'} \ldots$, where each exponent α' is equal to or greater than the corresponding exponent α. And conversely the only numbers which divide N are those of the form $a^{\alpha'} b^{\beta'} c^{\gamma'} \ldots$, where each index α' is at most equal to the corresponding index α; and in particular each or any of the indices α' may be $= 0$. Again, the least common multiple of two numbers $N = a^\alpha b^\beta c^\gamma \ldots$ and $N' = a^{\alpha'} b^{\beta'} c^{\gamma'} \ldots$ is $a^{\alpha''} b^{\beta''} c^{\gamma''} \ldots$, where each index α'' is equal to the largest of the corresponding indices α, α';—observe that any one or more of the indices α, β, $\gamma, \ldots$, α', β', $\gamma', \ldots$ may be $= 0$, so that the theorem extends to the case where either of the numbers N, N', has prime factors which are not factors of the other number. And so the greatest common measure of two numbers $N = a^\alpha b^\beta c^\gamma \ldots$ and $N' = a^{\alpha'} b^{\beta'} c^{\gamma'} \ldots$ is $a^{\alpha''} b^{\beta''} c^{\gamma''} \ldots$, where each index α'' is equal to the least of the corresponding indices α and α'.

4. The divisors of $N = a^\alpha b^\beta c^\gamma \ldots$ are the several terms of the product

$$(1 + a + \ldots + a^\alpha)(1 + b + \ldots + b^\beta)(1 + c + \ldots + c^\gamma) \ldots,$$

where unity and the number itself are reckoned each of them as a divisor. Hence the number of divisors is $= (\alpha + 1)(\beta + 1)(\gamma + 1) \ldots$, and the sum of the divisors is

$$\frac{(a^{\alpha+1} - 1)(b^{\beta+1} - 1)(c^{\gamma+1} - 1) \ldots}{(a - 1)(b - 1)(c - 1) \ldots}.$$

5. In $N = a^\alpha b^\beta c^\gamma \ldots$ the number of integers less than N and prime to it is

$$\phi(N) = N \left(1 - \frac{1}{a}\right) \left(1 - \frac{1}{b}\right) \left(1 - \frac{1}{c}\right) \ldots$$

To find the numbers in question write down the series of numbers $1, 2, 3, \ldots, N$; strike out all the numbers divisible by a, then those divisible by b, then those divisible by c, and so on; there will remain only the numbers prime to N. For actually finding the numbers we may of course in striking out those divisible by b disregard the numbers already struck out as divisible by a, and in striking out with respect to c disregard the numbers already struck out as divisible by a or b, and so on; but in order to count the remaining numbers it is more convenient to ignore the previous strikings out. Suppose, for a moment, there are only two prime factors a and b, then the number of terms struck out as divisible by a is $= N \div a$, and the number of terms struck out as divisible by b is $= N \div b$; but then each term divisible by ab will have been twice struck out; the number of these is $= N \div ab$, and thus the number of the remaining terms is $N(1 - 1/a - 1/b + 1/ab) = N(1 - 1/a)(1 - 1/b)$. By treating in like manner the case of three or more prime factors a, b, c,... we arrive at the general theorem. The formula gives $\phi(1) = 1$: viz. when $N = 1$, there is no factor 1; and it is necessary to consider $\phi(1)$ as being $= 1$. The explanation is that $\phi(N)$ properly denotes the number of integers not greater than N and prime to it; so that, when $N = 1$, we have 1 an integer not greater than N and prime to it; but in every other case the two definitions agree.

PAGE 596. **6.** If N, N', are numbers prime to each other, then $\phi(NN') = \phi(N)\phi(N')$, and so also for any number of numbers having no common divisor; in particular,

$$\phi(a^\alpha b^\beta c^\gamma \ldots) = \phi(a^\alpha)\phi(b^\beta)\phi(c^\gamma)\ldots; \quad \phi(a^\alpha) = a^\alpha\left(1 - \frac{1}{a}\right).$$

and the theorem is at once verified. We have $N = \sum \phi(N')$, where the summation extends to all the divisors N' of N, unity and the number itself being included; thus $15 = \phi(15)+\phi(5)+\phi(3)+\phi(1) = 8+4+2+1$.

7. The prime factor of the binomial function $x^N - 1$ is

$$[x^N - 1] = \frac{(x^N - 1)(x^{N/pq} - 1)\ldots}{(x^{N/p} - 1)(x^{N/q} - 1)\ldots},$$

a rational and integral function of the degree $\phi(N)$; say this is called $[x^N - 1]$, and we have $x^N - 1 = \prod[x^{N'} - 1]$, where the product extends to all the divisors N' of N, unity and the number included. For instance

$$[x^6 - 1] = \frac{(x^6 - 1)(x - 1)}{(x^3 - 1)(x^2 - 1)},$$

and we have

$$x^6-1 = [x^6-1][x^3-1][x^2-1][x-1] = (x^2-x+1)(x^2+x+1)(x+1)(x-1).$$

8. Congruence to a given modulus. A number x is congruent to 0, to the modulus N, $x = 0$ (mod. N), when x is divisible by N; two numbers x, y are congruent to the modulus N, $x = y$ (mod. N), when their difference $x - y$ divides by N, or, what is the same thing, if $x - y = 0$ (mod. N). Observe that, if $xy = 0$ (mod. N), and x be prime to N, then $y = 0$ (mod. N).

9. Residues to a given modulus. For a given modulus N we can always find, and that in an infinity of ways, a set of numbers, say residues, such that every number whatever is, to the modulus N, congruent to one and only one of these residues. For instance, the residues may be 0, 1, 2, 3,..., $N - 1$ (the residue of a given number is here simply the positive remainder of the number when divided by N); or, N being odd, the system may be 0, ± 1, ± 2,..., $\pm\frac{1}{2}(N - 1)$, and, N even, 0, ± 1, ± 2,..., $\pm\frac{1}{2}(N - 2)$, $+\frac{1}{2}N$.

10. Prime residues to a given modulus. Considering only the numbers which are prime to a given modulus N, we have here a set of $\phi(N)$ numbers, say $\phi(N)$ prime residues, such that every number prime to N is, to the modulus N, congruent to one and only one of these prime residues. For instance, the prime residues may be the numbers less than N and prime to it. In particular, if N is a prime number p, then the residues may be the $p-1$ numbers, 1, 2, 3,..., $p-1$.

In all that follows, the letter p, in the absence of any statement to the contrary, will be used to denote an odd prime other than unity. A theorem for p may hold good for the even prime 2, but it is in general easy to see whether this is so or not.

PAGE 597. **11. Fermat's theorem,** $a^{p-1} - 1 = 0$ (mod. p). The generalized theorem is $a^{\phi(N)} - 1 = 0$ (mod. N). The proof of the generalized theorem is as easy as that of the original theorem. Consider the series of the $\phi(N)$ numbers a, b, c, ..., each less than N and prime to it; let x be any number prime to N, then each of the numbers xa, xb, xc,..., is prime to N, and no two of them are congruent to the modulus N, that is, we cannot have $x(a-b) = 0$ (mod. N); in fact, x is prime to N, and the difference $a - b$ of two positive numbers each less than N will be less than N. Hence the numbers xa, xb, xc, ..., are in a different order congruent to the numbers a, b, c, ...; and multiplying together the numbers of each set we have $x^{\phi(N)}abc\ldots = abc\ldots$ (mod. N), or, since a, b, c, ..., are each prime to N, and therefore also the product $abc\ldots$ is prime to N, we have $x^{\phi(N)} = 1$, or say $x^{\phi(N)} - 1 = 0$ (mod. N).

In particular, if N be a prime number $= p$, then $\phi(N)$ is $= p - 1$, and the theorem is $x^{p-1} - 1 = 0$ (mod. p), x being now any number not divisible by p.

12. The general congruence $f(x) = 0$ (mod. p). $f(x)$ is written to denote a rational and integral function with integer coefficients which may without loss of generality be taken to be each of them less than p; it is assumed that the coefficient A of the highest power of x is not $= 0$. If there is for x an integer value α such that $f(\alpha) = 0$ (mod. p, throughout), then α is said to be a *root* of the congruence $f(x) = 0$; we may, it is clear, for α substitute any value whatever $\alpha' = \alpha + kp$, or say any value α' which is $\equiv \alpha$, but such value α' is considered not as a different root but as the same root of the congruence. We have thus $f(\alpha) = 0$; and therefore $f(x) = f(x) - f(\alpha), = (x - \alpha)f_1(x)$, where $f_1(x)$ is a function of like form with $f(x)$, that is, with integer coefficients, but of the next inferior order $n - 1$. Suppose there is another root b of the congruence, that is, an integer value b such that $f(b) = 0$; we have then $(b - \alpha)f_1(b) = 0$, and $b - \alpha$ is not $= 0$ (for then b would be the same root as α). Hence $f_1(b) = 0$, and $f(x) = (x - \alpha)[f_1(x) - f_1(b)] = (x - \alpha)(x - b)f_2(x)$, where $f_2(x)$ is an integral function such as $f(x)$, but of the order $n - 2$; and so on, that is, if there exist n different (non-congruent) roots of the congruence $f(x) = 0$, then $f(x) = A(x - \alpha)(x - b)\ldots(x - k)$, and the congruence may be written $A(x - \alpha)(x - b)\ldots(x - k) = 0$. And this cannot be satisfied by any other value l; for if so we should have $A(l - \alpha)(l - b)\ldots(l - k) = 0$, that is, some one of the congruences $(l - \alpha) = 0$, &c., would have to be satisfied, and l would be the same as one of the roots a, b, c, ..., k. That is, a congruence of the order n cannot have more than n roots, and if it have precisely n roots a, b, c, ..., k, then the form is $f(x) = A(x-a)(x-b)\dots(x-k) = 0$.

Observe that a congruence may have equal roots, viz. if the form be $f(x) = A(x-a)^{\alpha}(x-b)^{\beta}\dots = 0$, then the roots a, b, ... are to be counted α times, β times, ... respectively; but clearly the whole number of roots $\alpha + \beta + \dots$ is at most $= n$.

It is hardly necessary to remark that this theory of a congruence of the order n is precisely analogous to that of an equation of the order n, when only real roots are attended to. The theory of the imaginary roots of a congruence will be considered further on (see No. 41).

PAGE 598. **13. The linear congruence** $ax = c$ (mod. b). This is equivalent to the indeterminate equation $ax+by = c$; if a and b are not prime to each other, but have a greatest common measure q, this must also divide c; supposing the division performed, the equation becomes $a'x + b'y = c'$, where a' and b' are prime to each other, or, what is the same thing, we have the congruence $a'x = c'$ (mod. b'). This can always be solved, for, if we consider the b' numbers 0, 1, 2,. . . , $b' - 1$, one and only one of these will be $\equiv c'$ (mod. b'). Multiplying these by any number a' prime to b', and taking the remainders in regard to b', we reproduce in a different order the same series of numbers 0, 1, 2,. . . , $b' - 1$; that is, in the series a', $2a'$, . . . , $(b' - 1)a'$ there will be one and only one term $\equiv c'$ (mod. b'), or, calling the term in question α, we have $x = \alpha$ as the solution of the congruence $a'x = c'$ (mod. b'); $a'\alpha - c'$ is then a multiple of b', say it is $= -b'\beta$, and the corresponding value of y is $y = \beta$. We may for α write $\alpha + mb'$, m being any positive or negative integer, not excluding zero (but, as already remarked, this is not considered as a distinct solution of the congruence); the corresponding value of y is clearly $\beta - ma'$.

The value of x can be found by a process similar to that for finding the greatest common measure of the two numbers a' and c'; this is what is really done in the apparently tentative process which at once presents itself for small numbers, thus $6x = 9$ (mod. 35), we have $36x = 54$, or, rejecting multiples of 35, $x = 12$, or, if we please, $x = 35m + 12$.

In particular, we can always find a number ξ such that $a'\xi = 1$ (mod. b'); and we have then $x = c'\xi$ as the solution of the congruence $a'x = c'$ (mod. b'). The value of ξ may be written $\xi = \frac{1}{a'}$ (mod. b'), where $\frac{1}{a'}$ stands for that integer value ξ which satisfies the original congruence $a'\xi = 1$ (mod. b'); and the value of x may then be written $x \equiv c' \cdot \frac{1}{a'}$ (mod. b').

Another solution of the linear congruence is given in No. 21.

14. Wilson's theorem, $1\cdot2\cdot3\ldots(p-1)+1 = 0$ (mod. p). It has been seen that, for any prime number p, the congruence $x^{p-1} - 1 = 0$ (mod. p) of the order $p-1$ has the $p-1$ roots 1, 2,. . . , $p-1$; we have therefore

$$x^{p-1} - 1 = (x - 1)(x - 2) \ldots (x - p + 1),$$

or, comparing the terms independent of x, it appears that $1 \cdot 2 \cdot 3 \ldots (p - 1) = -1$, that is,

$$1 \cdot 2 \cdot 3 \ldots (p - 1) + 1 \equiv 0 \pmod p,$$

—the required theorem. For instance, where $p = 5$, then $1\cdot2\cdot3\cdot4+1 \equiv 0$ (mod. 5), and where $p = 7$, then $1 \cdot 2 \cdot 3 \cdot 4 \cdot 5 \cdot 6 + 1 \equiv 0$ (mod. 7).

PAGE 599. **15.** A proof on wholly different principles may be given. Suppose, to fix the ideas, $p = 7$; consider on a circle 7 points, the summits of a regular heptagon, and join these in any manner so as to form a heptagon (7-gon), the number of regular heptagons (convex or stellated) is $= \frac{1}{2}(7-1), = 3$, while the number of remaining heptagons must be divisible by 7, for by making any one rotation we have 6 other heptagons successively, each of the 6 sets of 7 being heptagons which can be obtained from it by a rotation of $\frac{1}{6}$ of $360°$, i.e. $1 \cdot 2 \cdot 3 \cdot 4 \cdot 5 \cdot 6 - \frac{1}{2}(7-1) = 0$ (mod. 7), whence $1 \cdot 2 \cdot 3 \cdot 4 \cdot 5 \cdot 6 - 7 + 1 = 0$, or finally $1 \cdot 2 \cdot 3 \cdot 4 \cdot 5 \cdot 6 + 1 \equiv 0$ (mod. 7). It is clear that the proof applies without alteration to the case of any prime number p.

If p is not a prime number, then $1 \cdot 2 \cdot 3 \ldots (p-1) \not\equiv -1$ (mod. p); hence the theorem shows directly whether a number p is or is not a prime number; but it is not of any practical utility for this purpose.

16. Prime roots of a prime number—application to the binomial equation $x^p - 1 = 0$. Take, for instance, $p = 7$. By what precedes, we have

$$x^7-1 = [x^6-1][x^3-1][x^2-1][x-1] = (x^2-x+1)(x^2+x+1)(x+1)(x-1);$$

and we have

$$x^7 - 1 = (x-1)(x-2)(x-3)(x-4)(x-5)(x-6) \quad (\text{mod } 7),$$

whence also

$$(x^2-x+1)(x^2+x+1)(x+1)(x-1) = (x-1)(x-2)(x-3)(x-4)(x-5)(x-6).$$

These two decompositions must agree together, and we in fact have

$$x^2 - x + 1 = (x-3)(x-5),$$
$$x^2 + x + 1 = (x-2)(x-4), \quad x + 1 = x - 6, \quad x - 1 = x - 1.$$

In particular, we thus have 3, 5, as the roots of the congruence $x^2 - x + 1 = 0$, that is, $[x^6 - 1] = 0$, and these roots 3, 5, are not the roots of any other of the congruences $[x^3 - 1] = 0$, $[x^2 - 1] = 0$, $[x - 1] = 0$; that is, writing $a = 3$ or 5 in the series of numbers a, a^2, a^3, a^4, a^5, a^6, we have a^k as the first term which is $\equiv 1$ (mod. 7); the series in fact are

$$3,\ 9,\ 27,\ 81,\ 243,\ 729 \equiv 3,\ 2,\ 6,\ 4,\ 5,\ 1,$$
$$5,\ 25,\ 125,\ 625,\ 3125,\ 15625 \equiv 5,\ 4,\ 6,\ 2,\ 3,\ 1.$$

And so, in general, the congruence $x^{p-1} - 1 = 0$ (mod. p) has the $p-1$ real roots 1, 2, 3,..., $p-1$; hence the congruence $[x^{p-1} - 1] = 0$, which is of the order $\phi(p-1)$, has this number $\phi(p-1)$ of real roots; and, calling any one of these g, then in the series of powers g, g^2, g^3, ..., g^{p-1}, the first term which is $\equiv 1$ (mod. p) is g^{p-1}, that is, we have g, g^2, g^3, ..., $g^{p-1} \equiv 1, 2, 3, \ldots, p-1$ in a different order. Any such number g is said to be a *prime root* of p, and the number of prime roots is $\phi(p-1)$, the number of integers less than and prime to $p-1$.

The notion of a prime root was applied by Gauss to the solution of the binomial equation $x^p - 1 = 0$, or, what is the same thing, to the question of the division of the circle (Kreistheilung), see Equation, Nos. 30 and 31, [786]; and, as remarked in the introduction to the present article, the roots or periods of roots of this equation present themselves as the units of a complex theory in the Theory of Numbers.

PAGE 600. **17.** Any number x less than p is $= g^m$, and, if m is not prime to $p-1$, but has with it a greatest common measure e, suppose $m = ke$, $p-1 = e \cdot f$, then

$$x^f = g^{ke \cdot f} = g^{k(p-1)} \equiv 1;$$

that is, $x^f \equiv 1$; and it is easily seen that in the series of powers x, $x^2, \ldots, x^f$, we have x^f as the first term which is $\equiv 1 \pmod{p}$. A number $= g^m$, where m is not prime to $p-1$, is thus not a prime root; and it further appears that, g being any particular prime root, the $\phi(p-1)$ prime roots are the numbers g^m, where m is any number less than $p-1$ and prime to it. Thus in the foregoing example $p = 7$, where the prime roots were 3 and 5, the integers less than 6 and prime to it are 1, 5; and we, in fact, have $5 \equiv 3^5$ and $3 \equiv 5^5 \pmod{7}$.

18. Integers belonging to a given exponent; index of a number. If, as before, $p-1 = ef$, that is, if f be a submultiple of $p-1$, then any integer x such that x^f is the lowest power of x which is $\equiv 1 \pmod{p}$ is said to belong to the exponent f. The number of residues, or terms of the series $1, 2, 3, \ldots, p-1$, which belong to the exponent f is $\phi(f)$, the number of integers less than f and prime to it; these are the roots of the congruence $[x^f - 1] = 0$ of the order $\phi(f)$. It is hardly necessary to remark that the prime roots belong to the exponent $p-1$.

A number $x = g^m$ is said to have the index m; observe the distinction between the two terms exponent and index; and, further, that the index is dependent on the selected prime root g.

19. Special forms of composite modulus. If instead of a prime modulus p we have a modulus p^m which is the power of an odd prime, or a modulus $2p$ or $2p^m$ which is twice an odd prime or a power of an odd prime, then there is a theory analogous to that of prime roots, viz. the numbers less than the modulus and prime to it are congruent to successive powers of a prime root g; thus, if $p^m = 3^2$, we have

$$2,\ 4,\ 8,\ 16,\ 32,\ 64 \equiv 2,\ 4,\ 8,\ 7,\ 5,\ 1 \pmod{9},$$

and if $2p^m = 2 \cdot 3^2$, we have

$$5,\ 25,\ 125,\ 625,\ 3125,\ 15625 \equiv 5,\ 7,\ 11,\ 13,\ 17,\ 1 \pmod{18}.$$

As regards the even prime 2 and its powers: for the modulus 2 or 4 the theory of prime roots does not come into existence, and for the higher powers it is not applicable; thus with modulus $= 8$ the numbers less than 8 and prime to it are 1, 3, 5, 7; and we have $3^2 = 5^2 = 7^2 = 1$ (mod. 8).

20. Composite modulus $N = a^\alpha b^\beta c^\gamma \dots$ no prime roots; irregularity. In the general case of a composite modulus it has been seen that, if x is any number less than N and prime to it, then $x^{\phi(N)} - 1 \equiv 0$ (mod. N). But, except in the above-mentioned cases p^m, $2p^m$, 2 or 4, there is not any number a such that $a^{\phi(N)}$ is the first power of a which is $\equiv 1$; there is always some submultiple $l = \phi(N)/\sigma$ such that a^l is the first

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790. GEOMETRY.

[From the *Encyclopaedia Britannica*, Ninth Edition, vol. x. (1879), pp. 436–451.]

[The article commences on an earlier page. The present portion begins at § 37, speaking of the paraboloid.]

37. *The Elliptic and Hyperbolic Paraboloids; the Ellipsoid, Hyperboloid of One Sheet, and Hyperboloid of Two Sheets.*

In the elliptic paraboloid (fig. 20), the sections by the planes of zx, zy are the parabolas

$$x^2 = 2az, \qquad y^2 = 2bz,$$

having the common axis Oz; and the section by any plane $z = \gamma$ parallel to that of xy is the ellipse

$$\frac{x^2}{2a\gamma} + \frac{y^2}{2b\gamma} = 1,$$

so that the surface is generated by a variable ellipse moving parallel to itself along the parabolas as directrices.

In the hyperbolic paraboloid (fig. 21), the sections by the planes of zx, zy are the parabolas

$$\frac{x^2}{2a} = z, \qquad -\frac{y^2}{2b} = z,$$

having the opposite axes Oz, Oz'; and the section by a plane $z = \gamma$ parallel to that of xy is the hyperbola

$$\frac{x^2}{2a\gamma} - \frac{y^2}{2b\gamma} = 1,$$

which has its transverse axis parallel to Ox or Oy according as γ is positive or negative. The surface is thus generated by a variable hyperbola moving parallel to itself along the parabolas as directrices. The form is best seen from fig. 22, which represents the sections by planes parallel to the plane of xy, or say the contour lines; the continuous lines are the sections above the plane of xy, and the dotted lines the sections below this plane. The form is, in fact, that of a saddle.

In the ellipsoid (fig. 23), the sections by the planes of zx, zy, and xy are each of them an ellipse, and the section by any parallel plane is also an ellipse. The surface may be considered as generated by an ellipse moving parallel to itself along two ellipses as directrices.

In the hyperboloid of one sheet (fig. 24), the sections by the planes of zx, zy are the hyperbolas

$$\frac{x^2}{a^2} - \frac{z^2}{c^2} = -1,$$

having a common conjugate axis zOz'; the section by the plane of xy, and that by any parallel plane, is an ellipse; and the surface may be considered as generated by a variable ellipse moving parallel to itself along the two hyperbolas as directrices.

In the hyperboloid of two sheets (fig. 25), the sections by the planes of zx and zy are the hyperbolas

$$\frac{x^2}{a^2} - \frac{z^2}{c^2} = 1,$$

having the common transverse axis zOz'; the section by any plane $z = \pm\gamma$ parallel to that of xy, γ being in absolute magnitude $> c$, is the ellipse

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = \frac{\gamma^2}{c^2} - 1,$$

and the surface, consisting of two distinct portions or sheets, may be considered as generated by a variable ellipse moving parallel to itself along the hyperbolas as directrices.

The hyperbolic paraboloid is such (and it is easy from the figure to understand how this may be the case) that there exist upon it two singly infinite series of right lines. The same is the case with the hyperboloid of one sheet (ruled or skew hyperboloid, as with reference to this property it is termed). If we imagine two equal and parallel circular disks, their points connected by strings of equal length, so that these are the generating lines of a right circular cylinder, then by turning one of the disks about its centre through the same angle in one or the other direction, the strings will in each case generate one and the same hyperboloid, and will in regard to it be the two systems of lines on the surface, or say the two systems of generating lines; and the general configuration is the same when instead of circles we have ellipses. It has been already shown analytically that the equation

$$\frac{x^2}{a^2}+\frac{y^2}{b^2}-\frac{z^2}{c^2}=1$$

is satisfied by each of two pairs of linear relations between the coordinates.

Curves; Tangent, Osculating Plane, Curvature, &c.

38. It will be convenient to consider the coordinates (x, y, z) of the point on the curve as given in terms of a parameter θ, so that dx, dy, dz, d^2x, &c., will be proportional to $\frac{dx}{d\theta}$, $\frac{dy}{d\theta}$, $\frac{dz}{d\theta}$, $\frac{d^2x}{d\theta^2}$, &c. Only a part of the analytical formulae will be given; in them ξ, η, ζ are used as current coordinates.

The tangent is the line through the point (x, y, z) and the consecutive point $(x+dx, y+dy, z+dz)$; its equations therefore are

$$\frac{\xi-x}{dx}=\frac{\eta-y}{dy}=\frac{\zeta-z}{dz}.$$

The osculating plane is the plane through the point and two consecutive points, and contains therefore the tangent; its equation is

$$\begin{vmatrix} \xi-x, & \eta-y, & \zeta-z \\ dx, & dy, & dz \\ d^2x, & d^2y, & d^2z \end{vmatrix}=0,$$

or, what is the same thing,

$$(\xi-x)(dy\,d^2z-dz\,d^2y)+(\eta-y)(dz\,d^2x-dx\,d^2z)+(\zeta-z)(dx\,d^2y-dy\,d^2x)=0.$$

The normal plane is the plane through the point at right angles to the tangent. It meets the osculating plane in a line called the principal normal; and drawing through the point a line at right angles to the osculating plane, this is called the binormal. We have thus at the point a set of three rectangular axes—the tangent, the principal normal, and the binormal.

We have through the point and three consecutive points a sphere of spherical curvature,—the centre and radius thereof being the centre, and radius, of spherical curvature. The sphere is met by the osculating plane in the circle of absolute curvature,—the centre and radius thereof being the centre, and radius, of absolute curvature. The centre of absolute curvature is also the intersection of the principal normal by the normal plane at the consecutive point.

Surfaces; Tangent Lines and Plane, Curvature, &c.

39. It will be convenient to consider the surface as given by an equation $f(x, y, z) = 0$ between the coordinates; taking (x, y, z) for the coordinates of a given point, and $(x+dx, y+dy, z+dz)$ for those of a consecutive point, the increments dx, dy, dz satisfy the condition

$$\frac{df}{dx}\,dx + \frac{df}{dy}\,dy + \frac{df}{dz}\,dz = 0,$$

but the ratio of two of the increments, suppose $dx : dy$, may be regarded as arbitrary. Only a part of the analytical formulae will be given; in them ξ, η, ζ are used as current coordinates.

We have through the point a singly infinite series of right lines, each meeting the surface in a consecutive point, or say having each of them two-point intersection with the surface. These lines lie all of them in a plane which is the tangent plane; its equation is

$$(\xi - x)\frac{df}{dx} + (\eta - y)\frac{df}{dy} + (\zeta - z)\frac{df}{dz} = 0,$$

as is at once verified by observing that this equation is satisfied (irrespectively of the value of $dx : dy$) on writing therein ξ, η, $\zeta = x + dx$, $y + dy$, $z + dz$.

The line through the point at right angles to the tangent plane is called the normal; its equations are

$$\frac{\xi - x}{\dfrac{df}{dx}} = \frac{\eta - y}{\dfrac{df}{dy}} = \frac{\zeta - z}{\dfrac{df}{dz}}.$$

In the series of tangent lines there are in general two (real or imaginary) lines, each of which meets the surface in a second consecutive point, or say it has three-point intersection with the surface; these are called the chief-tangents (Haupt-tangenten). The tangent-plane cuts the surface in a curve, having at the point of contact a node (double point), the tangents to the two branches being the chief-tangents.

In the case of a quadric surface the curve of intersection, qua curve of the second order, can only have a node by breaking up into a pair of lines; that is, every tangent-plane meets the surface in a pair of lines, or we have on the surface two singly infinite systems of lines; these are real for the hyperbolic paraboloid and the hyperboloid of one sheet, imaginary in other cases.

At each point of a surface the chief-tangents determine two directions; and passing along one of them to a consecutive point, and thence (without abrupt change of direction) along the new chief-tangent to a consecutive point, and so on, we have on the surface a chief-tangent curve; and there are, it is clear, two singly infinite series of such curves. In the case of a quadric surface, the curves are the right lines on the surface.

40. If at the point we draw in the tangent-plane two lines bisecting the angles between the chief-tangents, these lines (which are at right angles to each other) are called the principal tangents. We have thus at each point of the surface a set of rectangular axes, the normal and the two principal tangents.

Proceeding from the point along a principal tangent to a consecutive point on the surface, and thence (without abrupt change of direction) along the new principal tangent to a consecutive point, and so on, we have on the surface a curve of curvature; there are, it is clear, two singly infinite series of such curves, cutting each other at right angles at each point of the surface.

Passing from the given point in an arbitrary direction to a consecutive point on the surface, the normal at the given point is not intersected by the normal at the consecutive point; but passing to the consecutive point along a curve of curvature (or, what is the same thing, along a principal tangent) the normal at the given point is intersected by the normal at the consecutive point; we have thus on the normal two centres of curvature, and the distances of these from the point on the surface are the two principal radii of curvature of the surface at that point; these are also the radii of curvature of the sections of the surface by planes through the normal and the two principal tangents respectively; or say they are the radii of curvature of the normal sections through the two principal tangents respectively.

Take at the point the axis of z in the direction of the normal, and those of x and y in the directions of the principal tangents respectively, then, if the radii of curvature be a, b (the signs being such that the coordinates of the two centres of curvature are $z = a$ and $z = b$ respectively), the surface has in the neighbourhood of the point the form of the paraboloid

$$z = \frac{x^2}{2a} + \frac{y^2}{2b},$$

and the chief-tangents are determined by the equation

$$\frac{x^2}{2a} + \frac{y^2}{2b} = 0.$$

The two centres of curvature may be on the same side of the point or on opposite sides; in the former case a and b have the same sign, the paraboloid is elliptic, and the chief-tangents are imaginary; in the latter case a and b have opposite signs, the paraboloid is hyperbolic, and the chief-tangents are real.

The normal sections of the surface and the paraboloid by the same plane have the same radius of curvature; and it thence readily follows that the radius of curvature of a normal section of the surface by a plane inclined at an angle θ to that of zx is given by the equation

$$\frac{1}{\rho} = \frac{\cos^2\theta}{a} + \frac{\sin^2\theta}{b}.$$

The section in question is that by a plane through the normal and a line in the tangent plane inclined at an angle θ to the principal tangent along the axis of x.

To complete the theory, consider the section by a plane having the same trace upon the tangent plane, but inclined to the normal at an angle ϕ; then it is shown without difficulty (Meunier's theorem) that the radius of curvature of this inclined section of the surface is $= \rho\cos\phi$.

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791. LANDEN.

[From the *Encyclopaedia Britannica*, Ninth Edition, vol. xiv. (1882), p. 271.]

Landen, John, a distinguished mathematician of the 18th century, was born at Peakirk near Peterborough in Northamptonshire in 1719, and died 15th January 1790 at Milton in the same county. Most of his time was spent in the pursuits of active life, but he early showed a strong talent for mathematical study, which he eagerly cultivated in his leisure hours. In 1762 he was appointed agent to the Earl Fitzwilliam, and held that office to within two years of his death. He lived a very retired life, and saw little or nothing of society; when he did mingle in it, his dogmatism and pugnacity caused him to be generally shunned. He was first known as a mathematician by his essays in the *Ladies' Diary* for 1744. In 1766 he was elected a Fellow of the Royal Society. He was well acquainted and *au courant* with the works of the mathematicians of his own time, and has been called the English D'Alembert.

In his Discourse on the "Residual Analysis," in which he proposes to substitute for the method of fluxions a purely algebraical method, he says, "It is by means of the following theorem, viz.

$$\frac{x^{m/n} - v^{m/n}}{x - v} = \frac{x^{m/n}}{v} \cdot \frac{1 + \frac{v}{x} + \frac{v^2}{x^2} + \dots \ (m \text{ terms})}{1 + \frac{v^{m/n}}{x^{m/n}} + \frac{v^{2m/n}}{x^{2m/n}} + \dots \ (n \text{ terms})},$$

that we are able to perform all the principal operations in our said analysis; and I am not a little surprised that a theorem so obvious, and of such vast use, should so long escape the notice of algebraists." The idea is of course a perfectly legitimate one, and may be compared with that of Lagrange's *Calcul des Fonctions.*

His memoir (1775) on the rotatory motion of a body contains (as the author was aware) conclusions at variance with those arrived at by D'Alembert and Euler in their researches on the same subject. He reproduces and further develops and defends his own views in his *Mathematical Memoirs*, and in his paper in the *Philosophical Transactions* for 1785.

But Landen's capital discovery is that of the theorem known by his name (obtained in its complete form in the memoir of 1775, and reproduced in the first volume of the *Mathematical Memoirs*) for the expression of the arc of a hyperbola in terms of two elliptic arcs. To find this, he integrates a differential equation derived from the equation

$$\frac{dx}{\sqrt{1-x^4}} = \frac{dy}{\sqrt{1-y^4}} = \frac{dt}{\sqrt{1+t^4}},$$

interpreting geometrically in an ingenious and elegant manner three integrals which present themselves. If in the foregoing equation we write $m = 1$, $g = k^2$, and instead of t consider the new variable $y = t\sqrt{1-k^2}$, then

$$\frac{dy}{\sqrt{1-y^2}\,\sqrt{1-\lambda^2y^2}} = \frac{2}{1+k'}\cdot\frac{dx}{\sqrt{1-x^2}\,\sqrt{1-k^2x^2}}, \quad \text{where} \quad \lambda = \frac{1-k'}{1+k'},$$

which is the form known as Landen's transformation in the theory of elliptic functions: but his investigation does not lead him to obtain the equivalent of the resulting differential equation

$$\frac{dy}{\sqrt{1-y^2}\,\sqrt{1-\lambda^2y^2}} = \frac{(1+k')\,dx}{\sqrt{1-x^2}\,\sqrt{1-k^2x^2}}, \quad \text{where} \quad \lambda = \frac{1-k'}{1+k'},$$

due it would appear to Legendre, and which (over and above Landen's own beautiful result) gives importance to the theorem as leading directly to the quadric transformation of an elliptic integral in regard to the modulus.

The list of his writings is as follows: *Ladies' Diary*, various communications, 1744–1760; papers in the *Phil. Trans.*, 1754, 1760, 1768, 1771, 1775, 1777, 1785; *Mathematical Lucubrations*, 1755; *A Discourse concerning the Residual Analysis*, 1758; *The Residual Analysis*, book i., 1764; *Animadversions on Dr Stewart's Method of computing the Sun's Distance from the Earth*, 1771; *Mathematical Memoirs*, 1780, 1789.

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792. LOCUS.

[From the *Encyclopaedia Britannica*, Ninth Edition, vol. xiv. (1882), pp. 764, 765.]

LOCUS, in Greek *topos*, a geometrical term, the invention of the notion of which is attributed to Plato. It occurs in such statements as these:—the locus of the points which are at the same distance from a fixed point, or of a point which moves so as to be always at the same distance from a fixed point, is a circle; conversely a circle is the locus of the points at the same distance from a fixed point, or of a point moving so as to be always at the same distance from a fixed point; and so, in general, a curve of any given kind is the locus of the points which satisfy, or of a point moving so as always to satisfy, a given condition. The theory of loci is thus identical with that of curves; and it is in fact in this very point of view that a curve is considered in the article *Curve*, [785]; see that article, and also *Geometry* (Analytical), [790]. It is only necessary to add that the notion of a locus is useful as regards determinate problems or theorems: thus, to find the centre of the circle circumscribed about a given triangle ABC, we see that the circumscribed circle must pass through the two vertices A, B, and the locus of the centres of the circles which pass through these two points is the straight line at right angles to the side AB at its mid-point; similarly the circumscribed circle must pass through A, C, and the locus of the centres of the circles through these two points is the line at right angles to the side AC at its mid-point; thus we get the ordinary construction, and also the theorem that the lines at right angles to the sides, at their mid-points respectively, meet in a point.

The notion of a locus applies, of course, not only to plane but also to solid geometry. Here the locus of the points satisfying a single (or onefold) condition is a surface; the locus of the points satisfying two conditions (or a twofold condition) is a curve in space, which is in general a twisted curve or curve of double curvature.

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793. MONGE.

[From the *Encyclopaedia Britannica*, Ninth Edition, vol. xvi. (1883), pp. 738, 739.]

Monge, Gaspard (1746–1818), a French mathematician, the inventor of descriptive geometry, was born at Beaune on the 10th May 1746. He was educated first at the college of the Oratorians at Beaune, and then in their college at Lyons,—where, at sixteen, the year after he had been learning physics, he was made a teacher of it. Returning to Beaune for a vacation, he made, on a large scale, a plan of the town, inventing the methods of observation and constructing the necessary instruments; the plan was presented to the town, and preserved in their library. An officer of engineers seeing it wrote to recommend Monge to the commandant of the military school at Mézières, and he was received as draftsman and pupil in the practical school attached to that institution; the school itself was of too aristocratic a character to allow of his admission to it. His manual skill was duly appreciated: "I was a thousand times tempted," he said long afterwards, "to tear up my drawings in disgust at the esteem in which they were held, as if I had been good for nothing better." An opportunity, however, presented itself: being required to work out from data supplied to him the "defilement" of a proposed fortress (an operation then only performed by a long arithmetical process), Monge, substituting for this a geometrical method, obtained the result so quickly that the commandant at first refused to receive it—the time necessary for the work had not been taken; but upon examination the value of the discovery was recognized, and the method was adopted. And Monge, continuing his researches, arrived at that general method of the application of geometry to the arts of construction which is now called descriptive geometry. But such was the system in France before the Revolution that the officers instructed in the method were strictly forbidden to communicate it even to those engaged in other branches of the public service; and it was not until many years afterwards that an account of it was published. The method consists, as is well known, in the use of the two halves of a sheet of paper to represent say the planes of xy and xz at right angles to each other, and the consequent representation of points, lines, and figures in space by means of their plan and elevation, placed in a determinate relative position.

In 1768 Monge became professor of mathematics, and in 1771 professor of physics, at Mézières; in 1778 he married Madame Horbon, a young widow whom he had previously defended in a very spirited manner from an unfounded charge; in 1780 he was appointed to a chair of hydraulics at the Lyceum in Paris (held by him together with his appointments at Mézières), and was received as a member of the Academy; his intimate friendship with Berthollet began at this time. In 1783, quitting Mézières, he was, on the death of Bézout, appointed examiner of naval candidates. Although pressed by the minister to prepare for them a complete course of mathematics, he declined to do so, on the ground that it would deprive Madame Bézout of her only income, arising from the sale of the works of her late husband; he wrote, however (1786), his *Traité élémentaire de la Statique*. Monge contributed (1770–1790) to the *Memoirs of the Academy of Turin*, the *Mémoires des Savants Étrangers* of the Academy of Paris, the *Mémoires* of the same Academy, and the *Annales de Chimie*, various mathematical and physical papers. Among these may be noticed the memoir "Sur la théorie des déblais et des remblais" (Mém. de l'Acad.

de Paris, 1781), which, while giving a remarkably elegant investigation in regard to the problem of earthwork referred to in the title, establishes in connexion with it his capital discovery of the curves of curvature of a surface. Euler, in his paper on curvature in the Berlin *Memoirs* for 1760, had considered, not the normals of the surface, but the normals of the plane sections through a particular normal, so that the question of the intersection of successive normals of the surface had never presented itself to him. Monge's memoir just referred to gives the ordinary differential equation of the curves of curvature, and establishes the general theory in a very satisfactory manner; but the application to the interesting particular case of the ellipsoid was first made by him in a later paper in 1795. A memoir in the volume for 1783 relates to the production of water by the combustion of hydrogen; but Monge's results in this matter had been anticipated by Watts and Cavendish.

In 1792, on the creation by the Legislative Assembly of an executive council, Monge accepted the office of minister of the marine, but retained it only until April 1793. When the Committee of Public Safety made an appeal to the savants to assist in producing the materiel required for the defence of the republic, he applied himself wholly to these operations, and distinguished himself by his indefatigable activity therein; he wrote at this time his *Description de l'art de fabriquer les canons*, and his *Avis aux ouvriers en fer sur la fabrication de l'acier*. He took a very active part in the measures for the establishment of the Normal School (which existed only during the first four months of the year 1795), and of the School for Public Works, afterwards the Polytechnic School, and was at each of them professor of descriptive geometry; his methods in that science were first published in the form in which the shorthand writers took down his lessons given at the Normal School in 1795, and again in 1798–99. In 1796 Monge was sent into Italy with Berthollet and some artists to receive the pictures and statues levied from several Italian towns, and made there the acquaintance of General Bonaparte. Two years afterwards he was sent to Rome on a political mission, which terminated in the establishment, under Masséna, of the short-lived Roman republic; and he thence joined the expedition to Egypt, taking part with his friend Berthollet as well in various operations of the war as in the scientific labours of the Egyptian Institute of Sciences and Arts; they accompanied Bonaparte to Syria, and returned with him in 1798 to France. Monge was appointed president of the Egyptian commission, and he resumed his connexion with the Polytechnic School. His later mathematical papers are published (1794–1816) in the *Journal* and the *Correspondance* of the Polytechnic School.

On the formation of the Senate he was appointed a member of that body, with an ample provision and the title of count of Pelusium; but on the fall of Napoleon he was deprived of all his honours, and even excluded from the list of members of the reconstituted Institute. He died at Paris on the 28th July 1818.

For further information see B. Brisson, *Notice historique sur Gaspard Monge*; Dupin, *Essai historique sur les services et les travaux scientifiques de Gaspard Monge*, Paris, 1819, which contains (pp. 162–166) a list of Monge's memoirs and works; and the biography by Arago (*Œuvres*, t. II., 1854). Monge's various mathematical papers are to a considerable extent reproduced in the *Application de l'Analyse à la Géométrie*, 4th edition (last revised by the author), Paris, 1819—the pure text of this is reproduced in the 5th edition (revue, corrigée et annotée par M. Liouville), Paris, 1850, which contains also Gauss's Memoir, "Disquisitiones generales circa superficies curvas," and some valuable notes by the editor. The other principal separate works are *Traité élémentaire de la Statique*, 8ᵉ Édition, conformée à la précédente, par M. Hachette, et suivie d'une Note etc., par M. Cauchy, Paris, 1846; and the *Géométrie Descriptive* (originating, as mentioned above, in the lessons given at the Normal School). The 4th edition, published shortly after the author's death, seems to have been substantially the same as the 7th (*Géométrie Descriptive, par G. Monge, suivie d'une théorie des Ombres et de la Perspective, extraite des papiers de l'auteur, par M. Brisson*, Paris, 1847).

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794. NUMBERS, PARTITION OF.

[From the *Encyclopaedia Britannica*, Ninth Edition, vol. xvii. (1884), p. 614.]

This subject, created by Euler, though relating essentially to positive integer numbers, is scarcely regarded as a part of the Theory of Numbers. We consider in it a number as made up by the addition of other numbers: thus the partitions of the successive numbers 1, 2, 3, 4, 5, 6, &c., are as follows:

1;

2, 11;

3, 21, 111;

4, 31, 22, 211, 1111;

5, 41, 32, 311, 221, 2111, 11111;

6, 51, 42, 411, 33, 321, 3111, 222, 2211, 21111, 111111.

These are formed each from the preceding ones; thus, to form the partitions of 6 we take first 6; secondly, 5 prefixed to each of the partitions of 1 (that is, 51); thirdly, 4 prefixed to each of the partitions of 2 (that is, 42, 411); fourthly, 3 prefixed to each of the partitions of 3 (that is, 33, 321, 3111); fifthly, 2 prefixed, not to each of the partitions of 4, but only to those partitions which begin with a number not exceeding 2 (that is, 222, 2211, 21111); and lastly, 1 prefixed to all the partitions of 5 which begin with a number not exceeding 1 (that is, 111111); and so in other cases. The method gives all the partitions of a number, but we may consider different classes of partitions: the partitions into a given number of parts, or into not more than a given number of parts; or the partitions into given parts, either with repetitions or without repetitions, &c. It is possible, for any particular class of partitions, to obtain methods more or less easy for the formation of the partitions either of a given number or of the successive numbers 1, 2, 3, &c. And of course in any case, having obtained the partitions, we can count them and so obtain the number of partitions.

Another method is by Arbogast's rule of the last and the last but one; in fact, taking the value of a to be unity, and, understanding this letter in each term, the rule gives

$$b; \quad c,\ b^2; \quad d,\ bc,\ b^3; \quad e,\ bd,\ c^2,\ b^2c,\ b^4, \quad \&c.,$$

which, if b, c, d, e, &c., denote 1, 2, 3, 4, &c., respectively, are the partitions of 1, 2, 3, 4, &c., respectively.

An important notion is that of conjugate partitions. Thus a partition of 6 is 42; writing this in the form

$$\begin{array}{cc} \bullet & \bullet \\ \bullet & \bullet \\ \bullet & \\ \bullet & \end{array}$$

and summing the columns instead of the lines, we obtain the conjugate partition 2211; evidently, starting from 2211, the conjugate partition is 42. If we form all the partitions of 6 into not more than three parts, these are 6, 51, 42, 33, 411, 321, 222, and the conjugates are 111111, 21111, 2211, 222, 3111, 321, 33, where no part is greater than 3; and so, in general, we have the theorem, the number of partitions of n into not more than k parts is equal to the number of partitions of n with no part greater than k.

We have for the number of partitions an analytical theory depending on generating functions; thus for the partitions of a number n with the parts 1, 2, 3, 4, 5, &c., without repetitions, writing down the product

$$(1+x)(1+x^2)(1+x^3)(1+x^4)\ldots = 1+x+x^2+2x^3+\ldots+Nx^n+\ldots,$$

it is clear that, if α, β, γ, $\ldots$ are terms of the series x, x^2, x^3, $\ldots$ for which $\alpha+\beta+\gamma+\ldots = n$, then we have in the development of the product a term x^n, and hence that, in the term Nx^n of the product, the coefficient N is equal to the number of partitions of n with the parts 1, 2, 3,..., without repetitions; or say that the product is the generating function (G. F.) for the number of such partitions. And so in other cases we obtain a generating function. Thus for the function

$$\frac{1}{(1-x)(1-x^2)(1-x^3)\ldots} = 1+x+2x^2+\ldots+Nx^n+\ldots,$$

observing that any factor $1/(1-x^k)$ is $= 1+x^k+x^{2k}+\ldots$, we see that, in the term Nx^n, the coefficient is equal to the number of partitions of n, with the parts 1, 2, 3,..., with repetitions.

Introducing another letter z, and considering the function

$$(1+xz)(1+x^2z)(1+x^3z)\ldots = 1+z(x+x^2+\ldots)+\ldots+Nx^nz^k+\ldots,$$

we see that, in the term Nx^nz^k of the development, the coefficient N is equal to the number of partitions of n into k parts, with the parts 1, 2, 3, 4,..., without repetitions. And similarly, considering the function

$$\frac{1}{(1-xz)(1-x^2z)(1-x^3z)\ldots} = 1+z(x+x^2+\ldots)+\ldots+Nx^nz^k+\ldots,$$

we see that, in the term Nx^nz^k of the development, the coefficient N is equal to the number of partitions of n into k parts, with the parts 1, 2, 3, 4,..., with repetitions.

We have such analytical formulae as

$$\frac{1}{1-xz}\cdot\frac{1}{1-x^2z}\cdot\frac{1}{1-x^3z}\cdots=\frac{1}{1-x}\cdot\frac{1}{1-x^2}\cdots,$$

which lead to theorems in the Partition of Numbers.

A remarkable theorem is

$$(1-x)(1-x^2)(1-x^3)(1-x^4)\ldots=1-x-x^2+x^5+x^7-x^{12}-x^{15}+\ldots,$$

where the only terms are those with an exponent $\frac{1}{2}(3m^2\pm m)$ and for each such pair of terms the coefficient is $(-)^m$. The formula shows that, except for numbers of the form $\frac{1}{2}(3m^2\pm m)$, the number of partitions without repetitions into an odd number of parts is equal to the number of partitions without repetitions into an even number of parts, whereas for the excepted numbers these numbers differ by unity. Thus for the number 11, which is not an excepted number, the two sets of partitions are

$$\begin{array}{llllll} 11, & 821, & 731, & 641, & 632, & 542, \\ 10.1, & 92, & 83, & 74, & 65, & 5321, \end{array}$$

in each set 6.

We have

$$(1-x)(1+x)(1+x^2)(1+x^4)(1+x^8)\ldots=1;$$

or, as this may be written,

$$(1+x)(1+x^2)(1+x^4)(1+x^8)\ldots=\frac{1}{1-x}=1+x+x^2+x^3+\ldots,$$

showing that a number n can always be made up, and in one way only, with the parts 1, 2, 4, 8,.... The product on the left-hand side may be taken to k terms only: thus if $k=4$, we have

$$(1+x)(1+x^2)(1+x^4)(1+x^8)=\frac{1-x^{16}}{1-x}=1+x+x^2+\ldots+x^{15},$$

that is, any number from 1 to 15 can be made up, and in one way only, with the parts 1, 2, 4, 8; and similarly any number from 1 to 2^k-1 can be made up, and in one way only, with the parts 1, 2, 4,..., 2^{k-1}.

A like formula is

$$\frac{1-x^3}{1-x}\cdot\frac{1-x^9}{1-x^3}\cdot\frac{1-x^{27}}{1-x^9}\cdot\frac{1-x^{81}}{1-x^{27}}\cdot\frac{1-x^{243}}{1-x^{81}}=x^{-40}+x^{-39}+\ldots+1+x+\ldots+x^{40},$$

that is, any number from -40 to $+40$ can be made up, and that in one way only, with the parts 1, 3, 9, 27 taken positively or negatively; and so in general any number from $-\frac{1}{2}(3^k-1)$ to $+\frac{1}{2}(3^k-1)$ can be made up, and that in one way only, with the parts 1, 3, 9,..., 3^{k-1} taken positively or negatively.

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795. NUMBERS, THEORY OF.

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The Theory of Numbers, or higher arithmetic, otherwise arithmology, is a subject which, originating with Euclid, has in modern times, in the hands of Legendre, Gauss, Lejeune-Dirichlet, Kummer, Kronecker, and others, been developed into a most extensive and interesting branch of mathematics. We distinguish between the ordinary (or say the simplex) theory and the various complex theories.

In the ordinary theory we have, in the first instance, positive integer numbers, the unit or unity 1, and the other numbers 2, 3, 4, 5, &c. We introduce the zero 0, which is a number *sui generis*, and the negative numbers -1, -2, -3, -4, &c., and we have thus the more general notion of integer numbers, 0, ±1, ±2, ±3, &c.; $+1$ and -1 are units or unities. The sum of any two or more numbers is a number; conversely, any number is a sum of two or more parts; but even when the parts are positive a number cannot be, in a determinate manner, represented as a sum of parts. The product of two or more numbers is a number; but (disregarding the unities $+1$, -1, which may be introduced as factors at pleasure) it is not conversely true that every number is a product of numbers. A number such as 2, 3, 5, 7, 11, &c., which is not a product of numbers, is said to be a prime number; and a number which is not prime is said to be composite. A number other than zero is thus either prime or composite; and we have the theorem that every composite number is, in a determinate way, a product of prime factors.

We have complex theories in which all the foregoing notions (integer, unity, zero, prime, composite) occur; that which first presented itself was the theory with the unit i ($i^2 = -1$); we have here complex numbers, $a+bi$, where a and b are in the before-mentioned (ordinary) sense positive or negative integers, not excluding zero; we have the zero 0, $= 0+0i$, and the four units $1, -1, i, -i$. A number other than zero is here either prime or else composite; for instance, 3, 7, 11, are prime numbers, and $5 = (2+i)(2-i)$, $9 = 3 \cdot 3$, $13 = (3+2i)(3-2i)$, are composite numbers (generally any positive real prime of the form $4n+3$ is prime, but any positive real prime of the form $4n+1$ is a sum of two squares, and is thus composite). And disregarding unit factors we have, as in the ordinary theory, the theorem that every composite number is, in a determinate way, a product of prime factors.

There is, in like manner, a complex theory involving the cube roots of unity—if ω be an imaginary cube root of unity ($\omega^2+\omega+1 = 0$), then the integers of this theory are $a+b\omega$ (a and b real positive or negative integers, including zero); a complex theory with the fifth roots of unity—if α be an imaginary fifth root of unity ($\alpha^4+\alpha^3+\alpha^2+\alpha+1 = 0$), then the integers of the theory are $a + b\alpha + c\alpha^2 + d\alpha^3$ (a, b, c, d, real positive or negative integers, including zero); and so on for the roots of the orders 7, 11, 13, 17, 19. In all these theories, or at any rate for the orders 3, 5, 7 (see No. 37, post), we have the foregoing theorem: disregarding unit factors, a number other than zero is either prime or composite, and every composite number is, in a determinate way, a product of prime factors.

But coming to the 23rd roots of unity the theorem ceases to be true. Observe that it is a particular case of the theorem that, if N be a prime number, any integer power of N has for factors only the lower powers of N,—for instance, $N^2 = N \cdot N$; there is no other decomposition $N^2 = AB$. This is obviously true in the ordinary theory, and it is true in the complex theories preceding those for the 3rd, 5th, and 7th roots of unity, and probably in those for the other roots preceding the 23rd roots; but it is not true in the theory for the 23rd roots of unity. We have, for instance, 47, a number not decomposable into factors, but $47^2 = AB$, is a product of two numbers each of the form $a + b\alpha + \ldots + l\alpha^{22}$ (α a 23rd root). The theorem recovers its validity by the introduction into the theory of Kummer's notion of an ideal number.

The complex theories above referred to would be more accurately described as theories for the complex numbers involving the periods of the roots of unity: the units are the roots either of the equation $x^{p-1}+x^{p-2}+\ldots+x+1=0$ (p a prime number) or of any equation $x^{\frac{p-1}{q}}+\ldots+1=0$ belonging to a factor of the function of the order $p-1$: in particular, this may be the quadric equation for the periods each of $\frac{1}{q}(p-1)$ roots; they are the theories which were first and have been most completely considered, and which led to the notion of an ideal number. But a yet higher generalization which has been made is to consider the complex theory, the units whereof are the roots of any given irreducible equation which has integer numbers for its coefficients. There is another complex theory the relation of which to the foregoing is not very obvious, viz. Galois's theory of the numbers composed with the imaginary roots of an irreducible congruence, $F(x)=0$ (modulus a prime number p); the nature of this will be indicated in the sequel.

In any theory, ordinary or complex, we have a first part, which has been termed (but the name seems hardly wide enough) the theory of congruences; a second part, the theory of homogeneous forms: this includes in particular the theory of the binary quadratic forms $(a,b,c)(x,y)^2$; and a third part, comprising those miscellaneous investigations which do not come properly under either of the foregoing heads.

Ordinary Theory, First Part.

1. We are concerned with the integer numbers 0, ±1, ±2, ±3, &c., or in the first place with the positive integer numbers 1, 2, 3, 4, 5, 6, &c. Some of these, 1, 2, 3, 5, 7, &c., are prime, others, $4=2^2$, $6=2\cdot3$, &c., are composite; and we have the fundamental theorem that a composite number is expressible, and that in one way only, as a product of prime factors,

$$N=a^{\alpha}b^{\beta}c^{\gamma}\ldots$$

(a, b, c, … primes other than 1; α, β, γ, … positive integers). Gauss makes the proof to depend on the following steps: (i) the product of two numbers each smaller than a given prime number is not divisible by this number; (ii) if neither of two numbers is divisible by a given prime number the product is not so divisible; (iii) the like as regards three or more numbers; (iv) a composite number cannot be resolved into factors in more than one way.

2. Proofs will in general be only indicated or be altogether omitted, but, as a specimen of the reasoning in regard to whole numbers, the proofs of these fundamental propositions are given at length.

(i) Let p be the prime number, a a number less than p, and if possible let there be a number b less than p, and such that ab is divisible by p; it is further assumed that b is the only number, or, if there is more than one, then that b is the least number having the property in question; b is greater than 1, for a being less than p is not divisible by p. Now p as a prime number is not divisible by b, but must lie between two consecutive multiples mb and $(m+1)b$ of b. Hence, ab being divisible by p, mab is also divisible by p; moreover, ap is divisible by p, and hence the difference of these numbers, $= a(p - mb)$, must also be divisible by p, or, writing $p-mb = b'$, we have ab' divisible by p, where b' is less than b; so that b is not the least number for which ab is divisible by p.

(ii) If a and b are neither of them divisible by p, then a divided by p leaves a remainder α which is less than p, say we have $a = mp + \alpha$; and similarly b divided by p leaves a remainder β which is less than p, say we have $b = np + \beta$; then

$$ab = (mp + \alpha)(np + \beta) = (mnp + n\alpha + m\beta)p + \alpha\beta,$$

and $\alpha\beta$ is not divisible by p, therefore ab is not divisible by p.

(iii) The like proof applies to the product of three or more factors a, b, c, ...

(iv) Suppose that the number $N = a^\alpha b^\beta c^\gamma \ldots$ (a, b, c, ... prime numbers other than 1), is decomposable in some other way into prime factors; we can have no prime factor p, other than a, b, c, ..., for no such number can divide $a^\alpha b^\beta c^\gamma \ldots$; and we must have each of the numbers a, b, c, ..., for if any one of them, suppose a, were wanting, the number N would not be divisible by a. Hence the new decomposition if it exists must be a decomposition $N = a^{\alpha'} b^{\beta'} c^{\gamma'} \ldots$; and here, if any two corresponding indices, say α, α', are different from each other, then one of them, suppose α, is the greater, and we have $N \div a^{\alpha'} = b^\beta c^\gamma \ldots = a^{\alpha-\alpha'} b^{\beta'} c^{\gamma'} \ldots$. That is, we have the number $N \div a^{\alpha'}$ expressed in two different ways as a product, the number a being a factor in the one case, but not a factor in the other case. Thus the two exponents cannot be unequal, that is, we must have $\alpha = \alpha'$, and similarly we have $\beta = \beta'$, $\gamma = \gamma'$, ...; that is, there is only the original decomposition $N = a^\alpha b^\beta c^\gamma \ldots$.

3. The only numbers divisible by a number $N = a^\alpha b^\beta c^\gamma \ldots$ are the numbers $a^{\alpha'} b^{\beta'} c^{\gamma'} \ldots$, where each exponent α' is equal to or greater than the corresponding exponent α. And conversely the only numbers which divide N are those of the form $a^{\alpha'} b^{\beta'} c^{\gamma'} \ldots$, where each index α' is at most equal to the corresponding index α; and in particular each or any of the indices α' may be $= 0$. Again, the least common multiple of two numbers $N = a^\alpha b^\beta c^\gamma \ldots$ and $N' = a^{\alpha'} b^{\beta'} c^{\gamma'} \ldots$ is $a^{\alpha''} b^{\beta''} c^{\gamma''} \ldots$, where each index α'' is equal to the largest of the corresponding indices α, α';—observe that any one or more of the indices α, β, γ, $\ldots$, α', β', γ', $\ldots$, may be $= 0$, so that the theorem extends to the case where either of the numbers N, N', has prime factors which are not factors of the other number. And so the greatest common measure of two numbers $N = a^\alpha b^\beta c^\gamma \ldots$ and $N' = a^{\alpha'} b^{\beta'} c^{\gamma'} \ldots$ is $a^{\alpha''} b^{\beta''} c^{\gamma''} \ldots$, where each index α'' is equal to the least of the corresponding indices α and α'.

4. The divisors of $N = a^\alpha b^\beta c^\gamma \ldots$ are the several terms of the product

$$(1 + a + \ldots + a^\alpha)(1 + b + \ldots + b^\beta)(1 + c + \ldots + c^\gamma),$$

where unity and the number N itself are reckoned each of them as a divisor. Hence the number of divisors is $= (\alpha + 1)(\beta + 1)(\gamma + 1) \ldots$, and the sum of the divisors is

$$\frac{(a^{\alpha+1} - 1)(b^{\beta+1} - 1)(c^{\gamma+1} - 1) \ldots}{(a - 1)(b - 1)(c - 1) \ldots}.$$

5. In $N = a^\alpha b^\beta c^\gamma \ldots$ the number of integers less than N and prime to it is

$$\phi(N) = N \left(1 - \frac{1}{a}\right) \left(1 - \frac{1}{b}\right) \left(1 - \frac{1}{c}\right) \ldots$$

To find the numbers in question write down the series of numbers $1, 2, 3, \ldots, N$; strike out all the numbers divisible by a, then those divisible by b, then those divisible by c, and so on; there will remain only the numbers prime to N. For actually finding the numbers we may of course in striking out those divisible by b disregard the numbers already struck out as divisible by a, and in striking out with respect to c disregard the numbers already struck out as divisible by a or b, and so on; but in order to count the remaining numbers it is more convenient to ignore the previous strikings out. Suppose, for a moment, there are only two prime factors a and b, then the number of terms struck out as divisible by a is $= N \cdot \frac{1}{a}$, and the number of terms struck out as divisible by b is $= N \cdot \frac{1}{b}$; but then each term divisible by ab will have been twice struck out; the number of these is $= N \cdot \frac{1}{ab}$, and thus the number of the remaining terms is $N(1 - \frac{1}{a} - \frac{1}{b} + \frac{1}{ab}) = N(1 - \frac{1}{a})(1 - \frac{1}{b})$. By treating in like manner the case of three or more prime factors a, b, c, ... we arrive at the general theorem. The formula gives $\phi(1) = 1$; viz. when $N = 1$, there is no factor $1 - \frac{1}{a}$; and it is necessary to consider $\phi(1)$ as being $= 1$. The explanation is that $\phi(N)$ properly denotes the number of integers not greater than N and prime to it; so that, when $N = 1$, we have 1 an integer not greater than N and prime to it; but in every other case the two definitions agree.

6. If N, N' are numbers prime to each other, then $\phi(NN') = \phi(N)\phi(N')$, and so also for any number of numbers having no common divisor; in particular,

$$\phi(a^\alpha b^\beta c^\gamma \ldots) = \phi(a^\alpha)\phi(b^\beta)\phi(c^\gamma)\ldots;$$

$\phi(a^\alpha) = a^\alpha(1 - \frac{1}{a})$, and the theorem is at once verified. We have $N = \sum \phi(N')$, where the summation extends to all the divisors N' of N, unity and the number N itself being included; thus $15 = \phi(15) + \phi(5) + \phi(3) + \phi(1) = 8 + 4 + 2 + 1$.

7. The prime factor of the binomial function $x^N - 1$ is

$$[x^N - 1] = \frac{(x^N - 1)(x^{N/pq} - 1)\ldots}{(x^{N/p} - 1)(x^{N/q} - 1)\ldots},$$

a rational and integral function of the degree $\phi(N)$; say this is called $[x^N - 1]$, and we have

$$x^N - 1 = \prod [x^{N'} - 1],$$

where the product extends to all the divisors N' of N, unity and the number N included. For instance

$$x^{30} - 1 = [x^{30} - 1][x^{15} - 1][x^{10} - 1][x^6 - 1][x^5 - 1][x^3 - 1][x^2 - 1][x - 1],$$

and we have

$$x^{30} - 1 = [x^{30} - 1][x^{15} - 1][x^{10} - 1][x^6 - 1][x^5 - 1][x^3 - 1][x^2 - 1][x - 1].$$

8. *Congruence to a given modulus.* A number x is congruent to 0, to the modulus N, $x = 0 \pmod{N}$, when x is divisible by N; two numbers x, y are congruent to the modulus N, $x \equiv y \pmod{N}$, when their difference $x - y$ divides by N, or, what is the same thing, if $x - y \equiv 0 \pmod{N}$. Observe that, if $xy \equiv 0 \pmod{N}$, and x be prime to N, then $y \equiv 0 \pmod{N}$.

9. *Residues to a given modulus.* For a given modulus N we can always find, and that in an infinity of ways, a set of N numbers, say N residues, such that every number whatever is, to the modulus N, congruent to one and only one of these residues. For instance, the residues may be $0, 1, 2, 3, \ldots, N-1$ (the residue of a given number is here simply the positive remainder of the number when divided by N); or, N being odd, the system may be $0, \pm 1, \pm 2, \ldots, \pm\frac{1}{2}(N-1)$, and N even, $0, \pm 1, \pm 2, \ldots, \pm\frac{1}{2}(N-2), +\frac{1}{2}N$.

10. *Prime residues to a given modulus.* Considering only the numbers which are prime to a given modulus N, we have here a set of $\phi(N)$ numbers, say $\phi(N)$ prime residues, such that every number prime to N is, to the modulus N, congruent to one and only one of these prime residues. For instance, the prime residues may be the numbers less than N and prime to it. In particular, if N is a prime number p, then the residues may be the $p-1$ numbers, $1, 2, 3, \ldots, p-1$. In all that follows, the letter p, in the absence of any statement to the contrary, will be used to denote an odd prime other than unity. A theorem for p may hold good for the even prime 2, but it is in general easy to see whether this is so or not.

11. Fermat's theorem, $a^{p-1} - 1 \equiv 0 \pmod{p}$. The generalized theorem is $a^{\phi(N)} - 1 \equiv 0 \pmod{N}$. The proof of the generalized theorem is as easy as that of the original theorem. Consider the series of the $\phi(N)$ numbers $a, b, c, \ldots$, each less than N and prime to it; let x be any number prime to N, then each of the numbers xa, xb, xc, $\ldots$, is prime to N, and no two of them are congruent to the modulus N, that is, we cannot have $x(a-b) \equiv 0 \pmod{N}$; in fact, x is prime to N, and the difference $a-b$ of two positive numbers each less than N will be less than N. Hence the numbers xa, xb, xc, $\ldots$, are in a different order congruent to the numbers a, b, c, $\ldots$; we thus have

$$xa \cdot xb \cdot xc \ldots \equiv a \cdot b \cdot c \ldots \pmod{N},$$

that is,

$$x^{\phi(N)} abc \ldots \equiv abc \ldots \pmod{N};$$

and $abc\ldots$, being a product of numbers each prime to N, is itself prime to N, whence $x^{\phi(N)} \equiv 1$ (mod. N).

12. If p is a prime, and a any number whatever, then $a^p \equiv a$ (mod. p). This is the same theorem as $a^{p-1} - 1 \equiv 0$ (mod. p), if a is prime to p; but if a is divisible by p, then the theorem $a^p \equiv a$ (mod. p) is obviously true.

13. *The linear congruence* $ax \equiv c$ (mod. b). This is equivalent to the indeterminate equation $ax + by = c$; if a and b are not prime to each other, but have a greatest common measure q, this must also divide c; supposing the division performed, the equation becomes $a'x + b'y = c'$, where a' and b' are prime to each other, or, what is the same thing, we have the congruence $a'x \equiv c'$ (mod. b'). This can always be solved, for, if we consider the b' numbers $0, 1, 2, \ldots, b'-1$, one and only one of these will be $\equiv c'$ (mod. b'). Multiplying these by any number a' prime to b', and taking the remainders in regard to b', we reproduce in a different order the same series of numbers $0, 1, 2, \ldots, b'-1$; that is, in the series $a', 2a', \ldots, (b'-1)a'$ there will be one and only one term $\equiv c'$ (mod. b'), or, calling the term in question α, we have $x = \alpha$ as the solution of the congruence $a'x \equiv c'$ (mod. b'); $a'\alpha - c'$ is then a multiple of b', say it is $= -b'\beta$, and the corresponding value of y is $y = \beta$. We may for α write $\alpha + mb'$, m being any positive or negative integer, not excluding zero (but, as already remarked, this is not considered as a distinct solution of the congruence); the corresponding value of y is clearly $= \beta - m\alpha'$. The value of x can be found by a process similar to that for finding the greatest common measure of the two numbers a' and c'; this is what is really done in the apparently tentative process which at once presents itself for small numbers, thus $6x \equiv 9$ (mod. 35), we have $36x = 54$, or, rejecting multiples of 35, $x = 19$, or, if we please, $x = 35m+19$. In particular, we can always find a number ξ such that $a'\xi \equiv 1$ (mod. b'); and we have then $x = c'\xi$ as the solution of the congruence $a'x \equiv c'$. The value of ξ may be written $\xi = \frac{1}{a'}$ (mod. b'), where $\frac{1}{a'}$ stands for that integer value ξ which satisfies the original congruence $a'\xi \equiv 1$ (mod. b'); and the value of x may then be written $x \equiv \frac{c'}{a'}$ (mod. b'). Another solution of the linear congruence is given in No. 21.

14. *Wilson's theorem,* $1 \cdot 2 \cdot 3 \ldots (p-1) + 1 \equiv 0$ (mod. p). It has been seen that, for any prime number p, the congruence $x^{p-1} - 1 \equiv 0$ (mod. p) of the order $p-1$ has the $p-1$ roots $1, 2, \ldots, p-1$; we have therefore $x^{p-1} - 1 \equiv (x-1)(x-2)\ldots(x-p+1)$, or, comparing the terms independent of x, it appears that $1 \cdot 2 \cdot 3 \ldots (p-1) - 1 \equiv -1$,

that is,

$$1 \cdot 2 \cdot 3 \ldots (p-1) + 1 \equiv 0 \pmod p,$$

—the required theorem. For instance, where $p = 5$, then $1{\cdot}2{\cdot}3{\cdot}4{+}1 \equiv 0$ (mod. 5), and where $p = 7$, then $1 \cdot 2 \cdot 3 \cdot 4 \cdot 5 \cdot 6 + 1 \equiv 0$ (mod. 7).

15. A proof on wholly different principles may be given. Suppose, to fix the ideas, $p = 7$; consider on a circle 7 points, the summits of a regular heptagon, and join these in any manner so as to form a heptagon; the whole number of heptagons is $\frac{1}{7} \cdot 1 \cdot 2 \cdot 3 \cdot 4 \cdot 5 \cdot 6$. Now of these we have $\frac{1}{7}(7-1) = 3$, which are regular heptagons (convex or stellated); the number of remaining heptagons must be divisible by 7, for with any one such heptagon we connect the 6 heptagons which can be obtained from it by making it rotate through successive angles of $\frac{1}{7}360°$. That is, $\frac{1}{7} \cdot 1 \cdot 2 \cdot 3 \cdot 4 \cdot 5 \cdot 6 - \frac{1}{7}(7-1) \equiv 0$ (mod. 7), whence $1 \cdot 2 \cdot 3 \cdot 4 \cdot 5 \cdot 6 - 7 + 1 \equiv 0$, or finally $1 \cdot 2 \cdot 3 \cdot 4 \cdot 5 \cdot 6 + 1 \equiv 0$ (mod. 7). It is clear that the proof applies without alteration to the case of any prime number p. If p is not a prime number, then $1{\cdot}2{\cdot}3 \ldots (p-1) + 1 \not\equiv 0$ (mod. p); hence the theorem shows directly whether a number p is or is not a prime number; but it is not of any practical utility for this purpose.

16. *Prime roots of a prime number—application to the binomial equation* $x^p - 1 = 0$. Take, for instance, $p = 7$. By what precedes, we have

$$x^7 - 1 = [x^7 - 1][x^3 - 1][x^2 - 1][x - 1];$$

and we have

$$x^7 - 1 \equiv (x-1)(x-2)(x-3)(x-4)(x-5)(x-6) \pmod 7;$$

whence also

$$(x^2-x+1)(x^2+x+1)(x+1)(x-1) \equiv (x-1)(x-2)(x-3)(x-4)(x-5)(x-6).$$

These two decompositions must agree together, and we in fact have

$$\begin{aligned} x^2 - x + 1 &\equiv (x-3)(x-5), \\ x^2 + x + 1 &\equiv (x-2)(x-4), \\ x + 1 &\equiv x - 6, \\ x - 1 &\equiv x - 1. \end{aligned}$$

In particular, we thus have 3, 5, as the roots of the congruence $x^2 - x + 1 = 0$, that is, $[x^3 - 1] = 0$, and these roots 3, 5, are not the roots of any other of the congruences $[x^2 - 1] = 0$, $[x^1 - 1] = 0$; that is, writing $a = 3$ or 5 in the series of numbers a, a^2, a^3, a^4, a^5, a^6, we have a^6 as the first term which is $\equiv 1$ (mod. 7); the series in fact are

$$3,\ 9,\ 27,\ 81,\ 243,\ 729 \equiv 3,\ 2,\ 6,\ 4,\ 5,\ 1,$$
$$5,\ 25,\ 125,\ 625,\ 3125,\ 15625 \equiv 5,\ 4,\ 6,\ 2,\ 3,\ 1.$$

And so, in general, the congruence $x^{p-1} - 1 \equiv 0$ (mod. p) has the $p-1$ real roots $1, 2, 3, \ldots, p-1$; hence the congruence $[x^{p-1}-1] = 0$, which is of the order $\phi(p-1)$, has this number $\phi(p-1)$ of real roots; and, calling any one of these g, then in the series g, g^2, g^3, $\ldots$, g^{p-1}, we have g^{p-1} as the first term which is $\equiv 1$ (mod. p); the number g is said to be a prime root of the prime number p.

17. Any number x less than p is $\equiv g^m$, and, if m is not prime to $p-1$, but has with it a greatest common measure e, suppose $m = ke$, $p-1 = ef$, then $x^f \equiv g^{kef} = g^{k(p-1)} \equiv 1$; and it is easily seen that in the series of powers x, x^2, $\ldots$, x^f, we have x^f as the first term which is $\equiv 1$ (mod. p). A number $\equiv g^m$, where m is not prime to $p-1$, is thus not a prime root; and it further appears that, g being any particular prime root, the $\phi(p-1)$ prime roots are the numbers g^m, where m is any number less than $p-1$ and prime to it. Thus in the foregoing example $p = 7$, where the prime roots were 3 and 5, the integers less than 6 and prime to it are 1, 5; and we, in fact, have $5 \equiv 3^5$ and $3 \equiv 5^5$ (mod. 7).

18. *Integers belonging to a given exponent; index of a number.* If, as before, $p-1 = ef$, that is, if f be a submultiple of $p-1$, then any integer x such that x^f is the lowest power of x which is $\equiv 1$ (mod. p) is said to belong to the exponent f. The number of residues, or terms of the series $1, 2, 3, \ldots, p-1$, which belong to the exponent f is $\phi(f)$, the number of integers less than f and prime to it; these are the roots of the congruence $[x^f - 1] \equiv 0$ of the order $\phi(f)$. It is hardly necessary to remark that the prime roots belong to the exponent $p-1$. A number $x = g^m$ is said to have the index m; observe the distinction between the two terms exponent and index; and, further, that the index is dependent on the selected prime root g.

19. *Special forms of composite modulus.* If instead of a prime modulus p we have a modulus p^m which is the power of an odd prime, or a modulus $2p^m$ or 2^2p^m which is twice an odd prime or a power of an odd prime, then there is a theory analogous to that of prime roots, viz. the numbers less than the modulus and prime to it are congruent to successive powers of a prime root g; thus, if $p^m = 3^2$, we have 2, 4, 8, 16, 32, $64 \equiv 2, 4, 8, 7, 5, 1$ (mod. 9), and if $2p^m = 2 \cdot 3^2$ we have 5, 25, 125, 625, 3125, $15625 \equiv 5, 7, 11, 13, 17, 1$ (mod. 18). As regards the even prime 2 and its powers—for the modulus 2 or 4 the theory of prime roots does not come into existence, and for the higher powers it is not applicable; thus with modulus $= 8$ the numbers less than 8 and prime to it are 1, 3, 5, 7; and we have $3^2 = 5^2 = 7^2 = 1$ (mod. 8).

20. *Composite modulus $N = a^\alpha b^\beta c^\gamma \dots$ —no prime roots—irregularity.* In the general case of a composite modulus it has been seen that, if x is any number less than N and prime to it, then $x^{\phi(N)} - 1 \equiv 0$ (mod. N). But, except in the above-mentioned cases p^m, $2p^m$, 2 or 4, there is not any number a such that $a^{\phi(N)}$ is the first power of a which is $\equiv 1$; there is always some submultiple $f = \frac{\phi(N)}{\sigma}$ such that a^f is the first power which is $\equiv 1$. For instance, say $N = 24$, $\phi(N) = 8$, then the numbers less than 24 and prime to it are 1, 5, 7, 11, 13, 17, 19, 23; and we have $1^1 = 1$, $5^2 = 7^2 = 11^2 = 13^2 = 17^2 = 19^2 = 23^2 = 1$ (mod. 24), that is, 1 has the exponent 1, but all the other numbers have the exponent 2. So again where $N = 48$, the 16 numbers less than 48 and prime to it have, 1 the exponent 1, and 7, 13, 17, 23, 25, 31, 35, 41, 47 each the exponent 2, and the remaining numbers 5, 11, 19, 29, 37, 43 each the exponent 4. We cannot in this case by means of any single root or of any two roots express all the numbers, but we can by means of three roots, for instance, 5, 7, 13, express all the numbers less than 48 and prime to it; the numbers are in fact $\equiv 5^\alpha 7^\beta 13^\gamma$, where $\alpha = 0$, 1, 2, or 3, and β and γ each $= 0$ or 1. Comparing with the theorem for a prime number p, where the several numbers $1, 2, 3, \dots, p-1$, are expressed by means of a single prime root, $\equiv g^m$, where $m = 0$, $1, 2, \dots, p-2$, we have the analogue of a case presenting itself in the theory of quadratic forms, the "irregularity" of a determinant (post. No. 31); the difference is that here (the law being known, $N =$ a composite number) the case is not regarded as an irregular one, while the irregular determinants do not present themselves according to any apparent law.

21. *Maximum indicator—application to solution of a linear congruence.* In the case $N = 48$ it was seen that the exponents were 1, 2, 4, the largest exponent 4 being divisible by each of the others, and this property is a general one, viz. if $N = a^\alpha b^\beta c^\gamma \dots$ in the series of exponents (or, as Cauchy calls them, indicators) of the numbers less than N and prime to it, the largest exponent f is a multiple of each of the other exponents, and this largest exponent Cauchy calls the maximum indicator; the maximum indicator f is thus a submultiple of $\phi(N)$, and it is the smallest number such that for every number x less than N and prime to it we have $x^f - 1 \equiv 0$ (mod. N). The values of f have been tabulated from $N = 2$ to $N = 1000$. Reverting to the linear congruence $ax \equiv c$ (mod. b), where a and b are prime to each other, then, if f is the maximum indicator for the modulus b, we have $a^f \equiv 1$, and hence it at once appears that the solution of the congruence is $x \equiv ca^{f-1}$.

22. *Residues of powers for an odd prime modulus.* For the modulus p, if g be a prime root, then every number not divisible by p is $\equiv$ one of the series of numbers $g, g^2, \ldots, g^{p-1}$; and, if k be any positive number prime to $p-1$, then raising each of these to the power k we reproduce in a different order the same series of numbers $g, g^2, \ldots, g^{p-1}$, which numbers are in a different order $\equiv 1, 2, \ldots, p-1$, that is, the residue of a kth power may be any number whatever of the series $1, 2, \ldots, p-1$. But, if k is not prime to $p-1$, say their greatest common measure is e, and that we have $p-1 = ef$, $k = me$, then for any number not divisible by p the kth power is $\equiv$ one of the series of f numbers $g^e, g^{2e}, \ldots, g^{fe}$; there are thus only $f = \frac{1}{e}(p-1)$, out of the $p-1$ numbers $1, 2, 3, \ldots, p-1$, which are residues of a kth power.

23. *Quadratic residues for an odd prime modulus.* In particular, if $k = 2$, then $e = 2$, $f = \frac{1}{2}(p-1)$, and the square of every number not divisible by p is $\equiv$ one of the $\frac{1}{2}(p-1)$ numbers $g^e, g^{2e}, \ldots, g^{(p-1)e}$; that is, there are only $\frac{1}{2}(p-1)$ numbers out of the series $1, 2, 3, \ldots, p-1$ which are residues of a square number, or say quadratic residues, and the remaining $\frac{1}{2}(p-1)$ numbers are said to be quadratic non-residues of the modulus p,—we may say simply, residues and non-residues. But this result can be obtained more easily without the aid of the theory of prime roots. Every number not divisible by p is, to the modulus p, $\equiv$ one of the series of numbers $\pm 1, \pm 2, \pm 3, \ldots, \pm\frac{1}{2}(p-1)$, hence every square number is $\equiv$ one of the series of numbers $1^2, 2^2, 3^2, \ldots, \{\frac{1}{2}(p-1)\}^2$; and thus the $p-1$ numbers $1, 2, 3, \ldots, p-1$, are one-half of them residues and the other half non-residues of p. Thus, in the case $p = 11$, every number not divisible by 11 is, to this modulus, $\equiv$ one of the series $\pm 1, \pm 2, \pm 3, \pm 4, \pm 5$; whence the square of any such number is $\equiv$ one of the series 1, 4, 9, 16, 25, or say the series 1, 4, 9, 5, 3 (mod. 11); and hence 1, 3, 4, 5, 9 are the five residues and 2, 6, 7, 8, 10 are the five non-residues.

24. *Legendre's symbol* $\left(\frac{a}{p}\right)$. This denotes $+1$ if a is a quadratic residue of p, and -1 if a is a non-residue; we have $\left(\frac{a}{p}\right) \equiv a^{\frac{1}{2}(p-1)}$ (mod. p); and this at once gives the theorem $\left(\frac{ab}{p}\right) = \left(\frac{a}{p}\right)\left(\frac{b}{p}\right)$.

25. *Gauss's theorem,* the quadratic reciprocity of two odd primes. If p and q are any two odd primes, then

$$\left(\frac{p}{q}\right)\left(\frac{q}{p}\right) = (-1)^{\frac{1}{2}(p-1)\cdot\frac{1}{2}(q-1)}.$$

And supplementarily $\left(\frac{-1}{p}\right) = (-1)^{\frac{1}{2}(p-1)}$, and $\left(\frac{2}{p}\right) = (-1)^{\frac{1}{8}(p^2-1)}$.

The demonstration by Gauss of the theorem of reciprocity was first given in the *Disquisitiones Arithmeticae* (1801). It was subsequently expressed in a much simpler form by third and fifth demonstrations (1818), and by the second demonstration of Jacobi; and Cauchy's demonstration (1840) is also very simple. But no proof has yet been obtained which is really easy; and, if the theorem itself is very remarkable, this is still more the case with the theorem as extended to the complex theories (which is not here entered upon).

26. *The congruence* $x^2 \equiv a$ (mod. p). If a is a non-residue of p, the congruence has no root; if a is a residue of p, then the congruence has two roots; and if a is divisible by p, then the congruence has the one root $x \equiv 0$.

27. *Binary quadratic forms.* A binary quadratic form is $ax^2 + 2bxy + cy^2$, which is for shortness written $(a, b, c)(x, y)^2$, or simply (a, b, c); it is assumed that a, b, c are integer numbers, and that these may be either all positive or all negative; the determinant of the form is $D = ac - b^2$, or sometimes the determinant is taken to be $\Delta = b^2 - ac = -D$; but it is best to take it as $D = ac - b^2$, a positive number. The form is said to be "properly primitive" if a, $2b$, c have no common divisor; and "improperly primitive" if a, $2b$, c have the common divisor 2, but no larger common divisor.

28. *Equivalence of forms.* Two forms (a, b, c) and (a', b', c') are said to be "equivalent" if, by the linear substitutions

$$x = \alpha x' + \beta y', \qquad y = \gamma x' + \delta y',$$

where α, β, γ, δ are integer numbers such that $\alpha\delta - \beta\gamma = 1$, the one form is transformed into the other; the condition $\alpha\delta - \beta\gamma = 1$ secures that the two forms have the same determinant; equivalent forms represent the same series of numbers. The number of classes of forms of a given determinant is finite.

29. *Reduction of forms.* A form (a, b, c) is said to be "reduced" if $2|b| \leq a \leq c$, and if in the case of equality $2|b| = a$ we have $b \geq 0$, and in the case $a = c$ we have $b \geq 0$. Every class contains a reduced form, and in general only one; the reduced forms being determined by the inequalities just stated, it is easy to obtain for a given determinant the complete set of reduced forms, and thence the number of classes.

30. The principal form $(1, 0, D)$, compounded with any other form (a, b, c), gives rise to this same form (a, b, c); the principal form is on this account denoted by 1, viz. denoting the other form by ϕ, and expressing composition in like manner with multiplication, we have $1 \cdot \phi = \phi$. The form ϕ may be compounded with itself, giving a form denoted by ϕ^2; compounding this again with ϕ, we have a form denoted by ϕ^3; and so on. Since the whole number of forms is finite, we must in this manner arrive at the principal form, say we have $\phi^n = 1$, n being the least exponent for which this equation is satisfied. In particular, if the form ϕ belong to the principal genus, then the forms ϕ^2, ϕ^3, $\ldots$, ϕ^{n-1} will all belong to the principal genus, or the principal genus will include the forms 1, ϕ, ϕ^2, $\ldots$, ϕ^{n-1}, the powers of a form ϕ having the exponent n.

31. *Regular and irregular determinants.* The principal genus may consist of such a series of forms, and the determinant is then said to be regular; in particular, for a negative determinant $D, = -1$ to -1000, the determinant is always regular except in the thirteen cases $-D = 243$, 307, 339, 459, 576, 580, 675, 755, 820, 884, 891, 900, 974 (and, Perott, in Crelle, vol. xcv., 1883, except also for $-D = 468$, 931); the determinant is here said to be irregular. Thus for each of the values $-D = 576$, 580, 820, 900, the principal genus consists of four forms, not 1, ϕ, ϕ^2, ϕ^3, where $\phi^4 = 1$, but 1, ϕ, ϕ_1, $\phi^2\phi_1$, where $\phi^2 = 1$, $\phi_1^2 = 1$, and therefore also $(\phi\phi_1)^2 = 1$.

Compounding together any two forms, we have a form with the characters compounded of the characters of the two forms; and in particular, combining a form with itself, we have a form with the characters of the principal form. Or, what is the same thing, any two genera compounded together give rise to a determinate genus, viz. the genus having the characters compounded of the characters of the two genera; and any genus compounded with itself gives rise to the principal genus. Considering any regular determinant, suppose that there is more than one genus, and that the number of forms in each genus is $= n$; then, except in the case $n = 2$, it can be shown that there are always forms having the exponent $2n$. For instance, in the case $D = -35$, we have two genera each of three forms; there will be a form g having the exponent 6, or $g^6 = 1$; and the forms are 1, g, g^2, g^3, g^4, g^5, where 1, g^2, g^4, belong to the principal genus, and g, g^3, g^5, to the other genus. The characters refer to and the forms are

$++$	$--$
$(1, 0, 35)$	$(3, -1, 12)$
$(4, 1, 9)\quad g^2$	$(5, 0, 7)\quad g^3$
$(4, -1, 9)\quad g^4$	$(3, 1, 12)\quad g^5$

An instance of the case $n = 1$ is $D = -21$, there are here four genera each of a single form 1, c, c_1, cc_1, where $c^2 = 1$, $c_1^2 = 1$; an instance of the case $n = 2$ is $D = -88$, there are here two genera each of two forms 1, c, and c_1, cc_1, where $c^2 = 1$, $c_1^2 = 1$, thus there is here no form having the exponent $2n = 4$. (See Cayley, Tables, &c., in Crelle, t. LX., 1862, pp. 357–372, [335].) We may have 2^{k+1} genera, each of n forms, viz. such a system may be represented by

$$(1, \phi, \ldots, \phi^{2n-1}; \quad \phi, \phi^2, \ldots, \phi^{2n-1})(1, c)(1, c_1) \ldots (1, c_{k-1}),$$

where $\phi^{2n} = 1$, $c^2 = 1$, $c_1^2 = 1$, $\ldots$, $c_{k-1}^2 = 1$; there is no peculiarity in the form ϕ: we may instead of it take any form such as $c\phi$, $c_1\phi$, &c., for each of these is like ϕ, a form belonging to the exponent $2n$, and such that the even powers give the principal genus.

32. *Ternary and higher quadratic forms—cubic forms, &c.* The theory of the ternary quadratic forms $(a, b, c, a', b', c')(x, y, z)^2 = ax^2 + by^2 + cz^2 + 2a'yz + 2b'zx + 2c'xy$, or when only the coefficients are attended to, (a, b, c, a', b', c'), has been studied in a very complete manner: and those of the quaternary and higher quadratic forms have also been studied; in particular, the forms $x^2 + y^2 + z^2$, $x^2 + y^2 + z^2 + w^2$ composed of three or four squares; and the like forms with five, six, seven, and eight squares. The binary cubic forms $(a, b, c, d)(x, y)^3 = ax^3 + 3bx^2y + 3cxy^2 + dy^3$, or when only the coefficients are attended to, (a, b, c, d), have also been considered, though the higher binary forms have been scarcely considered at all. The special ternary cubic forms $ax^3 + by^3 + cz^3 + 6lxyz$ have been considered. Special forms of the degree 4 have also been studied.

Ordinary Theory, Third Part.—Miscellaneous Investigations.

33. *The solution in integers of a linear equation.* The equation is $ax+by+cz+\ldots=t$, where a, b, c, … t are given integers, positive or negative. The solution depends on the greatest common measure of a, b, c, …: if this does not divide t, the equation is insoluble; if it does divide t, then dividing the equation by this common measure, we have an equation wherein the coefficients of the variables have no common divisor, and the equation is then always soluble. Writing $ax+by=t$, if a and b have a common divisor q, then q must divide t; supposing this so, and writing $a=qa'$, $b=qb'$, $t=qt'$, the equation is $a'x+b'y=t'$, where a' and b' have no common divisor; and the solution is as given above, No. 13. For three variables, the equation is $ax+by+cz=t$; putting $ax+by=t'$, this can be solved as above, giving x, y in terms of t' and an arbitrary integer; then $t'+cz=t$, or $cz=t-t'$, can be solved, giving z and t' in terms of an arbitrary integer; and substituting, we obtain x, y, z each in terms of two arbitrary integers. And so for any number of variables.

34. *Fermat's theorem $x^n+y^n=z^n$.* The equation is not solvable in integers for any value of n greater than 2. For $n=4$, the proof is easy. For n an odd prime, the proof is very difficult; Kummer proved it for all regular primes, and more generally for all primes except those which are irregular in a certain sense. The theorem has been verified for large values of n, but the general proof remains one of the most celebrated unsolved problems in mathematics.

35. *Representation of numbers by quadratic forms.* The question whether a given number can be represented by a given quadratic form, and in how many ways, has been much studied. For binary quadratic forms, the theory is closely connected with that of complex multiplication and elliptic functions. For ternary quadratic forms, the representation of a number as a sum of three squares is a celebrated problem; a number is so representable if and only if it is not of the form $4^\alpha(8\beta+7)$. The representation as a sum of four squares was proved by Lagrange to be always possible; every positive integer is a sum of four squares.

36. *Continued fractions.* The theory of continued fractions has important applications in the Theory of Numbers; in particular, to the solution of Pell's equation $x^2-Dy^2=1$, and to the approximation of irrational numbers by rational fractions. If D is a positive non-square integer, then the equation $x^2 - Dy^2 = 1$ has always an infinity of solutions, which are given by the convergents of the continued fraction for $\sqrt{D}$.

37. *Complex theories.* The complex theories referred to in the introduction have been much developed. The theory for the cube roots of unity (Eisenstein's theory) is especially interesting; the integers are $a + b\omega$, where $\omega^2 + \omega + 1 = 0$, and the theory of cubic reciprocity has been established. For the fifth roots of unity, Kummer established the theory, and introduced his ideal numbers to restore the fundamental theorem of unique decomposition into prime factors. The theory has been extended by Kronecker and others to very general classes of complex numbers.

38. *Galois's theory of congruences.* Galois considered the imaginary roots of an irreducible congruence $F(x) \equiv 0$ (mod. p), where p is a prime number. If the congruence is of the order n, there are n roots, say j, j_1, ..., j_{n-1}; and these can be combined by addition and multiplication in the same manner as ordinary complex numbers. The theory leads to important results in regard to the solution of congruences and the theory of groups.

39. *Transcendental theory.* This includes the application of transcendental methods to the Theory of Numbers. Lejeune-Dirichlet's celebrated theorem, that in any arithmetic progression a, $a + b$, $a + 2b$, ..., where a and b are prime to each other, there are infinitely many prime numbers, was proved by transcendental methods. Riemann's memoir "Ueber die Anzahl der Primzahlen unter einer gegebenen Grösse" (1859) opened up a new field of research, connecting the distribution of prime numbers with the properties of the zeta-function $\zeta(s) = \sum n^{-s}$.

40. *The distribution of prime numbers.* Legendre conjectured that the number of primes less than x is approximately $\frac{x}{\log x - A}$, where A is a constant nearly equal to 1. Tchebycheff proved that the number of primes between n and $2n$ is at least $\frac{n}{5 \log n}$ and at most $\frac{5n}{4 \log n}$. Riemann's formula is

$$\Pi(x) = \mathrm{li}(x) - \sum \mathrm{li}(x^\rho) + \ldots,$$

where the summation extends over the complex roots ρ of the equation $\zeta(\rho) = 0$.

41. *Congruences of a higher degree—imaginary roots.* The theorem that a congruence of the degree n cannot have more than n roots was established above (No. 12) for the case of real roots. For imaginary roots, the theorem is: if p is a prime number, then a congruence $f(x) \equiv 0$ (mod. p) of the degree n has always n roots, real or imaginary; and these roots can be exhibited as symbols j, j_1, $\ldots$, j_{n-1}, which are combined according to the ordinary laws of algebra, but are such that $f(j) \equiv 0$ (mod. p), and similarly for j_1, $\ldots$, j_{n-1}.

42. *Theory of equations in regard to prime moduli.* The theory of algebraic equations in regard to a prime modulus p has many analogies with the theory of algebraic equations in ordinary algebra. A congruence $f(x) \equiv 0$ (mod. p) can be solved by trial for small values of p; and for larger values, the methods of solution are analogous to those for ordinary equations. The theory of symmetric functions of the roots, and of the discriminant, can be developed in a similar manner.

43. *The arithmetic theory of forms.* This includes the theory of the higher congruences, and of the forms which arise from them. The theory has been developed by Kronecker, Dedekind, and others; and it has important applications to the theory of algebraic numbers and functions.

44. *Fermat's equation $x^n + y^n = z^n$.* The equation $x^\lambda + y^\lambda = z^\lambda$, where λ is any positive integer greater than 2, is not resoluble in whole numbers (a theorem of Fermat's). The general proof depends on the theory of the complex numbers composed of the λth roots of unity, and presents very great difficulty; in particular, distinctions arise according as the number λ does or does not divide certain of Bernoulli's numbers.

45. Lejeune-Dirichlet employs, for the determination of the number of quadratic forms of a given positive or negative determinant, a remarkable method depending on the summation of a series $\sum n^{-s}$, where the index s is greater than but indefinitely near to unity.

46. Very remarkable formulae have been given by Legendre, Tchebycheff, and Riemann for the approximate determination of the number of prime numbers less than a given large number x. Factor tables have been formed for the first nine million numbers, and the number of primes counted for successive intervals of 50,000; and these are found to agree very closely with the numbers calculated from the approximate formulae. Legendre's expression is of the form $\frac{x}{\log x - A}$, where A is a constant not very different from unity; Tchebycheff's depends on the logarithm-integral $\mathrm{li}(x)$; and Riemann's, which is the most accurate, but is of a much more complicated form, contains a series of terms depending on the same integral.

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796. SERIES.

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A SERIES is a set of terms considered as arranged in order. Usually the terms are or represent numerical magnitudes, and we are concerned with the sum of the series. The number of terms may be limited or without limit; and we have thus the two theories, finite series and infinite series. The notions of convergency and divergency present themselves only in the latter theory.

Finite Series.

1. Taking the terms to be numerical magnitudes, or say numbers, if there be a definite number of terms, then the sum of the series is nothing else than the number obtained by the addition of the terms; e.g. $4+9+10=23$, $1+2+4+8=15$. In the first example there is no apparent law for the successive terms; in the second example there is an apparent law. But it is important to notice that in neither case is there a determinate law: we can in an infinity of ways form series beginning with the apparently irregular succession of terms 4, 9, 10, or with the apparently regular succession of terms 1, 2, 4, 8. For instance, in the latter case we may have a series with the general term 2^n, when for $n=0, 1, 2, 3, 4, 5, \ldots$ the series will be 1, 2, 4, 8, 16, 32,...; or a series with the general term $\frac{1}{2}(n^2+5n+6)$, where for the same values of n the series will be 1, 2, 4, 8, 15, 26,... The series may contain negative terms, and in forming the sum each term is of course to be taken with the proper sign.

2. But we may have a given law, such as either of those just mentioned, and the question then arises, to find the sum of an indefinite number of terms, or say of n terms (n standing for any positive integer number at pleasure) of the series. The expression for the sum cannot in this case be obtained by actual addition; the formation by addition of the sum of two terms, of three terms, &c., will, it may be, suggest (but it cannot do more than suggest) the expression for the sum of n terms of the series. For instance, for the series of odd numbers $1 + 3 + 5 + 7 + \ldots$, we have $1 = 1$, $1 + 3 = 4$, $1 + 3 + 5 = 9$, &c. These results at once suggest the law,

$$1 + 3 + 5 + \ldots + (2n - 1) = n^2,$$

which is in fact the true expression for the sum of n terms of the series; and this general expression, once obtained, can afterwards be verified.

3. We have here the theory of finite series: the general problem is, u_n being a given function of the positive integer n, to determine as a function of n the sum $u_0 + u_1 + u_2 + \ldots + u_n$, or, in order to have n instead of $n + 1$ terms, say the sum $u_0 + u_1 + u_2 + \ldots + u_{n-1}$. Simple cases are the three which follow.

(i) *The arithmetic series,*

$$a + (a + b) + (a + 2b) + \ldots + (a + \overline{n-1}\, b);$$

writing here the terms in the reverse order, it at once appears that twice the sum is $= 2a + \overline{n-1}\, b$ taken n times: that is, the sum $= na + \frac{1}{2}n(n - 1)b$. In particular, we have an expression for the sum of the natural numbers $1 + 2 + 3 + \ldots + n = \frac{1}{2}n(n + 1)$.

(ii) *The geometric series,*

$$1 + x + x^2 + \ldots + x^{n-1} = \frac{1 - x^n}{1 - x};$$

if x is positive and less than 1, this is $= \frac{1}{1-x}$ less the positive quantity $\frac{x^n}{1-x}$; if x is negative and greater than -1, the sum is $= \frac{1}{1-x}$ plus or minus the quantity $\frac{x^n}{1-x}$, according as n is even or odd. And if instead of x we write $-x$, the formula gives $1 - x + x^2 - \ldots = \frac{1-(-x)^n}{1+x}$.

(iii) *The series of powers,*

$$1 + 2^m + 3^m + \ldots + n^m,$$

where m is any positive integer. For $m = 1$, the sum is $\frac{1}{2}n(n + 1)$; for $m = 2$, the sum is $\frac{1}{6}n(n + 1)(2n + 1)$; for $m = 3$, the sum is $\{\frac{1}{2}n(n+1)\}^2$; and so on. The general formula involves the Bernoullian numbers.

4. The general problem of finite series is solved by means of the calculus of finite differences. If u_n is a given function of n, and we can find a function v_n such that $v_{n+1} - v_n = u_n$, then the sum $u_0 +$

$u_1 + \ldots + u_{n-1} = v_n - v_0$. The function v_n is called the integral of u_n; and the theory is analogous to that of integration in the infinitesimal calculus.

Infinite Series.

5. We have an infinite series when the number of terms is without limit. The series $u_0 + u_1 + u_2 + \ldots$, where the terms are given by a determinate law, may or may not have a finite sum. If the sum of the first n terms, say $S_n = u_0 + u_1 + \ldots + u_{n-1}$, approaches a finite limit S as n increases indefinitely, then the series is said to be *convergent*, and S is called the sum of the series. If S_n does not approach a finite limit, the series is said to be *divergent.*

6. The geometric series $1 + x + x^2 + \ldots$, where x is a real or complex quantity, is convergent if the modulus of x is less than 1, and its sum is then $\frac{1}{1-x}$; it is divergent if the modulus of x is equal to or greater than 1.

7. The harmonic series $1 + \frac{1}{2} + \frac{1}{3} + \frac{1}{4} + \ldots$ is divergent; its sum increases without limit, though very slowly. More generally, the series $\sum \frac{1}{n^s}$ is convergent if $s > 1$, and divergent if $s \leq 1$.

8. A series of positive terms is convergent if each term is less than the corresponding term of a known convergent series, and it is divergent if each term is greater than the corresponding term of a known divergent series. This is the comparison test.

9. Cauchy's test: the series $\sum u_n$ of positive terms is convergent if the limit of $u_n^{1/n}$ is less than 1, and divergent if it is greater than 1. D'Alembert's test: the series is convergent if the limit of $\frac{u_{n+1}}{u_n}$ is less than 1, and divergent if it is greater than 1. If the limit is $= 1$, the tests fail.

10. For a series of positive and negative terms, we have the notion of *absolute convergence*: the series is absolutely convergent if the series of the moduli of the terms is convergent. An absolutely convergent series is convergent; and the sum is the same in whatever order the terms are taken. A series which is convergent but not absolutely convergent is said to be *semi-convergent* or *conditionally convergent*; the terms of such a series can be rearranged so as to make the sum equal to any assigned value.

11. The exponential series $1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \ldots$ is absolutely convergent for all values of x, real or complex; its sum is denoted by e^x or $\exp(x)$. The logarithmic series $x - \frac{x^2}{2} + \frac{x^3}{3} - \ldots$ is convergent if the modulus of x is not greater than 1 (except for $x = -1$, when it is divergent); its sum is $\log(1+x)$. The binomial series $1 + mx + \frac{m(m-1)}{2!}x^2 + \ldots$ is absolutely convergent if the modulus of x is less than 1; its sum is $(1+x)^m$.

12. Fourier's theorem: a function $f(x)$, subject to certain conditions, can be expressed in the form

$$f(x) = a_0 + a_1 \cos x + a_2 \cos 2x + \ldots + b_1 \sin x + b_2 \sin 2x + \ldots,$$

where the coefficients are given by definite integrals involving $f(x)$. This is the foundation of the theory of trigonometrical series, which has important applications in mathematical physics.

13. A series of functions $u_0(x) + u_1(x) + u_2(x) + \ldots$ is said to be *uniformly convergent* in a given interval if, for any assigned small quantity ϵ, a value of n can be found such that the sum of all terms after the nth is less than ϵ for all values of x in the interval. The sum of a uniformly convergent series of continuous functions is itself a continuous function; and such a series can be integrated term by term.

14. The product of two series: if $\sum a_n$ and $\sum b_n$ are absolutely convergent, then the product series $\sum c_n$, where $c_n = a_0 b_n + a_1 b_{n-1} + \ldots + a_n b_0$, is absolutely convergent, and its sum is the product of the sums of the two given series.

15. The reversion of series: given $y = a_1 x + a_2 x^2 + a_3 x^3 + \ldots$, to express x as a series in powers of y. If $a_1 \neq 0$, we can always find $x = b_1 y + b_2 y^2 + b_3 y^3 + \ldots$, where the coefficients b_n can be determined in terms of a_1, a_2, $\ldots$ by successive substitution.

797. SURFACE, CONGRUENCE, COMPLEX.

[From the *Encyclopaedia Britannica*, Ninth Edition, vol. xxii. (1887), pp. 668–672.]

In the article *Curve* [785], the subject was treated from an historical point of view for the purpose of showing how the leading ideas of the theory were successively arrived at. These leading ideas apply to surfaces, but the ideas peculiar to surfaces are scarcely of the like fundamental nature, being rather developments of the former set in their application to a more advanced portion of geometry; there is consequently less occasion for the historical mode of treatment. Curves in space were briefly considered in the same article, and they will not be discussed here; but it is proper to refer to them in connexion with the other notions of solid geometry. In plane geometry the elementary figures are the point and the line; and we then have the curve, which may be regarded as a singly infinite system of points, and also as a singly infinite system of lines. In solid geometry the elementary figures are the point, the line, and the plane; we have, moreover, first, that which under one aspect is the curve and under another aspect the developable (or torse), and which may be regarded as a singly infinite system of points, of lines, or of planes; and secondly, the surface, which may be regarded as a doubly infinite system of points or of planes, and also as a special triply infinite system of lines. (The tangent lines of a surface are a special complex.) As distinct particular cases of the first figure, we have the plane curve and the cone; and as a particular case of the second figure, the ruled surface, regulus, or singly infinite system of lines; we have, besides, the congruence or doubly infinite system of lines, and the complex or triply infinite system of lines. And thus crowds of theories arise which have hardly any analogues in plane geometry; the relation of a curve to the various surfaces which can be drawn through it, and that of a surface to the various curves which can be drawn upon it, are different in kind from those which in plane geometry most nearly correspond to them—the relation of a system of points to the different curves through them and that of a curve to the systems of points upon it. In particular, there is nothing in plane geometry to correspond to the theory of the curves of curvature of a surface. Again, to the single theorem of plane geometry, that a line is the shortest distance between two points, there correspond in solid geometry two extensive and difficult theories, that of the geodesic lines on a surface and that of the minimal surface, or surface of minimum area, for a given boundary. And it would be easy to say more in illustration of the great extent and complexity of the subject.

Surfaces in General; Torses, &c.

1. A surface may be regarded as the locus of a doubly infinite system of points—that is, the locus of the system of points determined by a single equation $U = (\,*\,x, y, z, 1)^n = 0$, between the Cartesian coordinates (to fix the ideas, say rectangular coordinates) x, y, z; or, if we please, by a single homogeneous relation $U = (\,*\,x, y, z, w)^n = 0$, between the quadriplanar coordinates x, y, z, w. The degree n of the equation is the order of the surface; and this definition of the order agrees with the geometrical one, that the order of the surface is equal to the number of the intersections of the surface by an arbitrary line. Starting from the foregoing point definition of the surface, we might develop the notions of the tangent line and the tangent plane; but it will be more convenient to consider the surface *ab initio* from the more general point of view in its relation to the point, the line, and the plane.

2. Mention has been made of the plane curve and the cone; it is proper to recall that the order of a plane curve is equal to the number of its intersections by an arbitrary line (in the plane of the curve), and that its class is equal to the number of tangents to the curve which pass through an arbitrary point (in the plane of the curve). The cone is a figure correlative to the plane curve: corresponding to the plane of the curve we have the vertex of the cone, to its tangents the generating lines of the cone, and to its points the tangent planes of the cone. But from a different point of view, we may consider the generating lines of the cone as corresponding to the points of the curve and its tangent planes as corresponding to the tangents of the curve. From this point of view, we define the order of the cone as equal to the number of its intersections (generating lines) by an arbitrary plane through the vertex, and its class as equal to the number of the tangent planes which pass through an arbitrary line through the vertex. And in the same way that a plane curve has singularities (singular points and singular tangents), so a cone has singularities (singular generating lines and singular tangent planes).

3. Consider now a surface in connexion with an arbitrary line. The line meets the surface in a certain number of points, and, as already mentioned, the order of the surface is equal to the number of these intersections. We have through the line a certain number of tangent planes of the surface, and the class of the surface is equal to the number of these tangent planes. But, further, through the line imagine a plane; this meets the surface in a curve the order of which is equal (as is at once seen) to the order of the surface. Again, on the line imagine a point; this is the vertex of a cone circumscribing the surface, and the class of this cone is equal (as is at once seen) to the class of the surface. The tangent lines of the surface, which lie in the plane, are nothing else than the tangents of the plane section, and thus form a singly infinite series of lines; similarly, the tangent lines of the surface, which pass through the point, are nothing else than the generating lines of the circumscribed cone, and thus form a singly infinite series of lines. But, if we consider those tangent lines of the surface which are at once in the plane and through the point, we see that they are finite in number; and we define the rank of a surface as equal to the number of tangent lines which lie in a given plane and pass through a given point in that plane. It at once follows that the class of the plane section and the order of the circumscribed cone are each equal to the rank of the surface, and are thus equal to each other. It may be noticed that for a general surface $(*x, y, z, w)^n = 0$, of order n without point singularities the rank is $a = n(n-1)$, and the class is $n_1 = n(n-1)^2$; this implies (what is, in fact, the case) that the circumscribed cone has line singularities, for otherwise its class, that is, the class of the surface, would be $a(a-1)$, which is not $= n(n-1)^2$.

4. In the last preceding number, the notions of the tangent line and the tangent plane have been assumed as known, but they require to be further explained in reference to the original point definition of the surface. Speaking generally, we may say that the points of the surface consecutive to a given point on it lie in a plane which is the tangent plane at the given point, and conversely the given point is the point of contact. But the consecutive points of the surface lie also on the tangent lines through the point of contact; and we have thus on the tangent plane a singly infinite series of tangent lines, forming what is called the tangent plane. The tangent plane at any point of a surface $f(x, y, z) = 0$ is given by the equation

$$(\xi - x)\frac{df}{dx} + (\eta - y)\frac{df}{dy} + (\zeta - z)\frac{df}{dz} = 0,$$

where ξ, η, ζ are current coordinates.

The surface has upon it two systems of curves of curvature. These are defined as follows: at any point of the surface there are two principal directions, at right angles to each other; proceeding along one of these to a consecutive point, and thence along the new principal direction to another consecutive point, and so on, we obtain a curve of curvature. The curves of curvature form two systems, cutting each other at right angles at every point of the surface. The normals to the surface at points along a curve of curvature intersect consecutively; and the distances from the surface to the points of intersection are the principal radii of curvature. The product of the principal radii of curvature is the measure of curvature of the surface at the point; and the locus of the centres of curvature is the surface of centres.

5. A surface may be represented by an equation between three parameters, or say by an equation $F(\alpha, \beta, \gamma) = 0$, where α, β, γ are functions of the coordinates x, y, z; and in particular, a plane may be represented by an equation $l\alpha + m\beta + n\gamma + p\delta = 0$, where α, β, γ, δ are linear functions of the coordinates. In the theory of surfaces, the notion of the plane is fundamental; and we have the theorem that any surface may be regarded as the envelope of a doubly infinite system of planes. The equation of the tangent plane at any point of a surface may be written in the form $l\xi + m\eta + n\zeta = p$, where (l, m, n) are the direction-cosines of the normal and p is the perpendicular distance from the origin; and the coordinates (l, m, n, p) of the tangent plane may be regarded as the coordinates of the point in a dual space.

6. The torse, or developable surface, is a particular kind of surface. It may be defined as the envelope of a singly infinite system of planes; or as the locus of the tangents to a curve; or as the surface generated by a line moving so as to have consecutive positions intersecting each other. The intersection of two consecutive planes of the torse is a generating line, and the envelope of these generating lines is the edge of regression, or cuspidal edge, of the torse. The torse is a unilateral surface, and its Gaussian curvature is zero at every point. Examples of torses are the tangent surface of a curve, the cone, and the cylinder.

Congruences and Complexes.

7. A congruence is a doubly infinite system of lines. Any line may be represented by its six coordinates $(l, m, n, \lambda, \mu, \nu)$, where l, m, n are the direction-cosines of the line and λ, μ, ν are the moments about the coordinate axes; these coordinates satisfy the relation $l\lambda + m\mu + n\nu = 0$. A congruence is represented by two equations between these coordinates, $F(l, m, n, \lambda, \mu, \nu) = 0$, $G(l, m, n, \lambda, \mu, \nu) = 0$; and the order of the congruence is the number of lines of the congruence which pass through an arbitrary point, while the class is the number of lines which lie in an arbitrary plane. A congruence has in general a focal surface, upon which the lines of the congruence touch; the focal surface consists of two sheets, and the lines of the congruence are common tangents to these two sheets.

8. A complex is a triply infinite system of lines. It is represented by a single equation $F(l, m, n, \lambda, \mu, \nu) = 0$ between the six coordinates of a line. The degree of the complex is the number of lines of the complex which pass through an arbitrary point and lie in an arbitrary plane through that point; or, what is the same thing, it is the number of lines of the complex which meet an arbitrary line. The complex is a natural generalization of the congruence, and includes as particular cases the linear complex (represented by a linear equation between the six coordinates) and the quadratic complex. The linear complex is of fundamental importance in the geometry of lines; it consists of all the lines which meet a given line (the directrix) and also all the lines which are parallel to a given plane, the combination of these two conditions giving the general linear complex.

9. The theory of complexes is intimately connected with the theory of surfaces. The tangent lines of a surface form a special complex; and conversely, a complex determines in general a family of surfaces. The singular lines of a complex are lines which belong to the complex together with their consecutive lines; they form a congruence, and the focal surface of this congruence is the singular surface of the complex. The theory of linear complexes is equivalent to the theory of screws, or infinitesimal motions, in mechanics; and it has important applications in the theory of curvature of surfaces and in the geometry of dynamics.

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798. WALLIS.

[From the *Encyclopaedia Britannica*, Ninth Edition, vol. xxiv. (1888), pp. 60–63.]

WALLIS, JOHN (1616–1703), a mathematician and divine, was born at Ashford in Kent on the 23rd of November 1616. He was educated at Felsted school in Essex, and at Emmanuel College, Cambridge, where he was admitted in 1632. He took the degree of M.A. in 1640, and was ordained priest in the same year. During the Civil War he was chaplain to Sir Richard Darley; and his skill in deciphering the royalist despatches having been discovered, he was employed by the parliament in this capacity, and is said to have succeeded in deciphering a letter of the king's which had baffled all other attempts. In 1649 he was appointed Savilian professor of geometry at Oxford, a chair which he held till his death on the 28th of October 1703.

Wallis was one of the most original mathematicians of the 17th century. He was a precocious child; it is related that he learnt the alphabet in a fortnight, and that at fifteen he mastered the "Clavis

Mathematicae" of Oughtred. His mathematical reputation was established by his treatise "*Arithmetica Infinitorum*" (1655), in which he developed the methods of Cavalieri and others, and laid the foundations of the integral calculus. In this work he introduced the symbol ∞ for infinity, and the fraction $\frac{1}{\infty}$ for an infinitesimal; and he made extensive use of infinite series and products. He also wrote on algebra, conic sections, mechanics, and music; and he was one of the first to give a systematic account of the properties of the parabola, ellipse, and hyperbola, considered as sections of a cone.

The works of Wallis are numerous, and relate to a multiplicity of subjects. His *Institutio Logicae*, published in 1687, was very popular, and in his *Grammatica Linguae Anglicanae* we find indications of an acute and philosophic intellect. The mathematical works are published some of them in a small 4to volume, Oxford, 1657, and a complete collection in three thick folio volumes, Oxford, 1695–1699. The third volume includes, however, some theological treatises, and the first part of it is occupied with editions of treatises on harmonics and other works of Greek geometers, some of them first editions from the MSS., and in general with Latin versions and notes (Ptolemy, Porphyrius, Briennius, Archimedes, Eutocius, Aristarchus, and Pappus). The second and third volumes include also two collections of letters to and from Brouncker, Frenicle, Leibnitz, Newton, Oldenburg, Schooten, and others; and there is a tract on trigonometry by Caswell. Excluding all these, the mathematical works contained in the first and second volumes occupy about 1800 pages.

The Arithmetica Infinitorum. The *Arithmetica Infinitorum* relates chiefly to the quadrature of curves by the so-called method of indivisibles established by Cavalieri, 1629, and cultivated in the interval by him, Fermat, Descartes, and Roberval. The method is substantially that of the integral calculus; thus, e.g., for the curve $y = x^2$ to find the area from $x = 0$ to $x = 1$, the base is divided into n equal parts, and the area is obtained as

$$\frac{1}{n^3}(1^2 + 2^2 + \ldots + n^2) = \frac{1}{n^3} \cdot \frac{1}{6} n(n+1)(2n+1),$$

which, taking n indefinitely large, is $= \frac{1}{3}$. The case of the general parabola $y = x^m$ (m a positive integer or fraction), where the area is $\frac{1}{m+1}$, had been previously solved. Wallis made the important remark that the reciprocal of such a power of x could be regarded as a power with a negative exponent ($\frac{1}{x^m} = x^{-m}$), and he was thus enabled to extend the theorem to certain hyperbolic curves, but the case of a negative value of m larger than 1 presented a difficulty which he did not succeed in overcoming. It should be noticed that Wallis, although not using the notation x^m in the case of a positive or negative fractional value, nor indeed in the case of a negative integer value of m, deals continually with such powers, and speaks of the positive or negative integer or fractional value of m as the *index* of the power. The area of a curve, $y =$ sum of a finite number of terms Ax^m, was at once obtained from that for the case of a single term; and Wallis, after thus establishing the several results which would now be written

$$\int_0^1 (x-x^2)\,dx = \tfrac{1}{6}, \quad \int_0^1 (x-x^2)^2\,dx = \tfrac{1}{30}, \quad \int_0^1 (x-x^2)^3\,dx = \tfrac{1}{140}, \quad \int_0^1 (x-x^2)^4\,dx$$

proposed to himself to interpolate from these the value of $\int_0^1 (x - x^2)^{\frac{1}{2}}\,dx$, which is the expression for the area $(= \frac{1}{8}\pi)$ of a semicircle, diameter $= 1$; making a slight transformation, the actual problem was to find the value of $\Pi = \binom{\frac{1}{2}}{\frac{1}{2}}$, the term halfway between 1 and 2, in the series of terms 1, 2, 6, 20, 70, ...; and he thus obtained the remarkable expression

$$\pi = \frac{2 \cdot 2 \cdot 4 \cdot 4 \cdot 6 \cdot 6 \cdot 8 \cdot 8 \ldots}{1 \cdot 3 \cdot 3 \cdot 5 \cdot 5 \cdot 7 \cdot 7 \cdot 9 \ldots},$$

together with a succession of superior and inferior limits for the number π.

In the same work, Wallis obtained the expression which would now be written $ds = dx\sqrt{1 + (\frac{dy}{dx})^2}$ for the length of the element of a curve, thus reducing the problem of rectification to that of quadrature. An application of this formula to an algebraical curve was first made a few years later by W. Neil; the investigation is reproduced in the “Tractatus de Cissoide, &c.” (1659, as above), and Wallis adds the remark that the curve thus rectified is in fact the semicubical parabola.

Other Works. The *Mathesis Universalis* is a more elementary work intended for learners. It contains copious dissertations on fundamental points of algebra, arithmetic, and geometry, and critical remarks.

The *De Algebra Tractatus* contains (chapters 66–69) the idea of the interpretation of imaginary quantities in geometry. This is given somewhat as follows: the distance represented by the square root of a negative quantity cannot be measured in the line backwards or forwards, but can be measured in the same plane above the line, or (as appears elsewhere) at right angles to the line either in the plane, or in the plane at right angles thereto. Considered as a history of algebra, this work is strongly objected to by Montucla, on the ground of its unfairness as against the early Italian algebraists and also Vieta and Descartes, and in favour of Harriot; but De Morgan, while admitting this, attributes to it considerable merit.

The two treatises on the cycloid and on the cissoid, &c., and the *Mechanica* contain many results which were then new and valuable. The latter work contains elaborate investigations in regard to the centre of gravity, and it is remarkable also for the employment of the principle of virtual velocities. The cuno-cuneus is a highly interesting surface; it is a ruled quartic surface, the equation of which may be written $cy = (c - z)^2(a^2 - x^2)$.

Among the letters in volume III., there is one to the editor of the Leipsic Acts, giving the decipherment of two letters in secret characters. The ciphers are different, but on the same principle: the characters in each are either single digits or combinations of two or three digits, standing some of them for letters, others for syllables or words—the number of distinct characters which had to be deciphered being thus very considerable.

For the prolonged conflict between Hobbes and Wallis, see the article Hobbes, [*Encyclopaedia Britannica*, ninth edition,] vol. xii. pp. 36–38.

End of Vol. XI.

Repair TODO

The following page ranges were not present in the source TeX drops and are missing from this reconstruction:

- Pages 451–500